EaJEE NOTES

for

PHYSICAL CHEMISTRY

JEE Main & Advanced

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KHOOBSURAT BAATEIN
from the desk of #GOAT

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Notes

1 Some Basic Concepts of Chemistry

INTRODUCTION

This chapter introduces you to fundamental ideas like the mole concept, atomic mass, and the laws of chemical combination—concepts that have not only shaped the field of chemistry but also revolutionized science as a whole.

MATTER

Matter is anything that occupies space and has mass.

- Examples:

Hot air balloon

Pen

NON-MATTER

- Non-matter refers to things that do not occupy space, have mass, or consist of particles.

Examples of Non-Matter- Energy, forces, time

1. Identify matter and non-matter:

1. Pen	4. Time
2. Battle	5. Paper
3. Book	6. Electron
7. Electricity	17. Human memory
8. Love	18. Heat
9. Sound	19. Rain fall
10. Light	20. Drop of rain
11. Depression	21. Smoke
12. Intelligence	22. Fog
13. Ball	23. Happiness
14. Anger	24. Age
15. Density	25. Thought
16. Brain	

Sol. Matter: Pen, Bottle, Book, Paper, Electron, Ball, Density, Brain, Drop of rain, Smoke, Fog

Non-matter: Time, Electricity, Love, Sound, Light, Depression, Intelligence, Anger, Human memory, heat, Rainfall, Happiness, Age, Thoughts.

yeh sawaal hua ya nhi?

PHYSICAL CLASSIFICATION OF MATTER

Property	Solid	Liquid	Gas
Shape	Fix	not fix	not fix

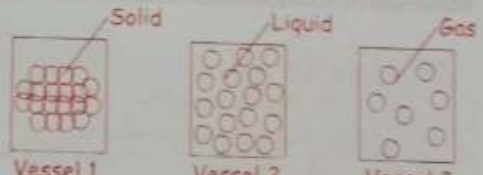
Property	Solid	Liquid	Gas
Volume	Fix	Fix	not fix

Property	Solid	Liquid	Gas
Intermolecular force	Maximum	Moderate	Negligible

Property	Solid	Liquid	Gas
Compressibility	Never	Negligible	Compressible

Property	Solid	Liquid	Gas
Melting Point	M.P > Room temp	M.P < Room temp	M.P < Room temp
Boiling Point	B.P > Room temp	B.P > Room temp	B.P < Room temp

2. Which vessel has the matter having least intermolecular force of attraction?



(A) Vessel 1 (B) Vessel 2
(C) Vessel 3 (D) All have same

Sol. (C) In vessel 3, particles are very far from each other, hence having the least intermolecular force of attraction.

agar yeh nhi hua toh sar ko pakad lo

3. Why is the density of solids generally higher than that of liquids and gases?

Sol.
- In solids, particles are tightly packed with negligible spaces between them, resulting in a higher mass per unit volume.
- In liquids, particles are less tightly packed, and in gases, they are widely spaced, leading to lower density.

Apart from these three states of matter, two other states of matter are:

1. Plasma 2. Bose Einstein condensate

Plasma

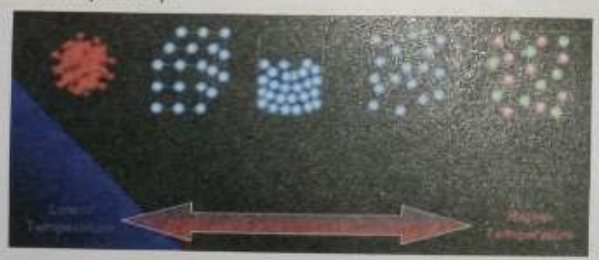
Plasma is a highly ionized state of matter consisting of charged particles: positively charged ions and free electrons. It is often referred to as an "ionized gas."

Examples of Plasma:

- Stars, Lightning, Neon Signs

Bose-Einstein Condensate

- BEC is created at temperatures very close to **absolute zero** (0 K or -273.15°C), typically only a few billionths of a degree above absolute zero. At such low temperatures, atoms lose almost all of their thermal energy and start to behave in a unique way.

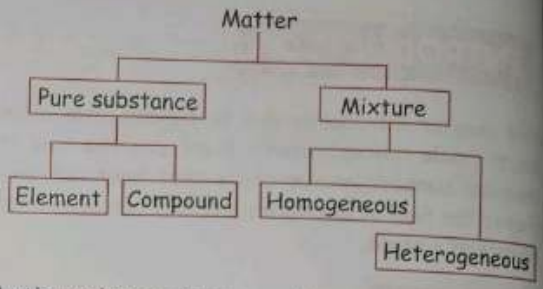


4. Draw a diagram to illustrate the interversion of solids, liquids, gases and plasma states of matter. Mention the processes involved in each step.

Sol. For explanation, refer to QR code given on the last page.

CHEMICAL CLASSIFICATION OF MATTER

Chemical classification of matter can be summarized as:



Let's explain each term one by one in detail.

Pure Substance

- A material containing only one type of substance.
- It can not be separated into simpler substance by physical methods.
- They are of two types → Element, Compound

(a) Element:

- A pure substance containing one type of atom.

Example: Fe, Na, Cs, Mg, O_2, Cl_2, N_2, O_3, P_4, S_8, F_2 etc.

(b) Compound:

- A pure substance containing more than one kind of element which combine together in a fixed mass ratio.
- They can be separated into simple substances by chemical method.
- The property of the compound is entirely different from those of their constituent elements.

Example: HCl, H_2O, KCl, H_2SO_4, $CaSO_4$, NH_3 etc.

Important

- NH_3 is a pure substance, so it is a compound. It has more than one type of atoms.

$$NH_3 \longrightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$$

Property of NH_3 is different than the property of its constituent parts.

- Similarly H_2O (water) is also a compound as it has more than one type of atoms.

$$H_2O \xrightarrow{\text{current}} H_2 + \frac{1}{2}O_2$$

Property of H_2O is different than the property of its constituent parts.

5. Which of the following is/are the characteristics of a compound?
   (A) Two elements combine in the fixed proportion by mass to form compound.
   (B) It can be decomposed into its constituents by chemical method.
   (C) Properties of compounds are same as their constituent elements.
   (D) Compounds can be separated into its constituents by physical method.

Sol. (A,B) Among the given options, the correct ones are A and B.

Socho kyu?

6. Identify the elements and compounds-

1.	H_2	7.	$NH_3(g)$
2.	O_2	8.	$C_6H_{12}O_6(s)$
3.	NaCl	9.	$C_{12}H_{22}O_{11}(s)$
4.	$CaCl_2$	10.	$I_2(s)$
5.	O_3	11.	Dry ice
6.	$H_2O(l)$	12.	Diamond

Sol. Elements: H_2, O_2, O_3, I_2, Diamond

Compounds: NaCl, $CaCl_2$, H_2O, NH_3, $C_6H_{12}O_6$, $C_{12}H_{22}O_{11}$, Dry ice

yeh asaan hai bahut

Mixtures

- A material which consists of more than one type of substance and which are mixed in any ratio by weight.
- The property of the mixture is the property of its compounds.
- These can be separated into their constituents substances by physical method.

- They are of two types → Homogeneous, Heterogeneous

1. Homogeneous Mixtures

A mixture having same composition throughout.

Note

Examples:

(i) Salt + water mixture (ii) Alloys

2. Heterogeneous Mixtures

The mixture in which components are present non-uniformly.

Note

Examples:

(i) Water + sand (ii) Milk

7. Which of the following is a homogeneous mixture?
   (A) Oil and water (B) Air
   (C) Sand and salt (D) Sand and water

Sol. (B) For explanation, refer to QR code given on the last page.

ATOM

Important

Smallest particle of an element which may or may not have independent existence.

Element	Symbol	Exists independently	not exist independently	
Oxygen	O		✓	O_2
Helium	He	✓		
Hydrogen	H		✓	H_2
Neon	Ne	✓		
Phosphorus	P		✓	P_4
Sulphur	S		✓	S_8
Chlorine	Cl		✓	Cl_2
Xenon	Xe	✓		

dhyan lgalo idhar apna

MOLECULE

- Smallest particle of an element or compound which always have independent existence.
- In a molecule two or more atoms connected by chemical bond.
- The smallest particle of the substance that has all of the physical and chemical properties of that substance.

Examples:

H	H_2	2 atoms of Hydrogen
S	S_8	8 atoms of Sulphur

Important

Lets take some examples to understand this:

CO_2 ⟶ 1 atom of C + 2 atoms of oxygen

$C_6H_{12}O_6$ ⟶ 6 carbon atoms + 12 hydrogen atoms + 6 oxygen atoms

$CuSO_4.5H_2O$ ⟶ 1 Cu atom + 1 S atom + 9 oxygen atoms + 10 hydrogen atoms

Types of Molecules

Homoatomic:

A molecule is said to be homoatomic if it consists of atoms of the same element chemically bonded together.

Heteroatomic:

A molecule is said to be heteroatomic if it consists of atoms of different elements chemically bonded together.

8. Classify the given below species as homoatomic or heteroatomic.

O_2,	P_2O_5,	C_4H_{10},	P_4,
NO,	S_2Cl_2,	$C_6H_{12}O_6$,	S_8,
CO_2,	$CaCO_3$,	H_2SO_4,	O_3,
CO,	H_2O,	$KMnO_4$,	NO_2

Sol.

O_2	Homo	P_2O_5	Hetero
NO	Hetero	S_2Cl_2	Hetero
CO_2	Hetero	$CaCO_3$	Hetero
CO	Hetero	H_2O	Hetero
C_4H_{10}	Hetero	P_4	Homo
$C_6H_{12}O_6$	Hetero	S_8	Homo
H_2SO_4	Hetero	O_3	Homo
$KMnO_4$	Hetero	NO_2	Hetero

maza agaya hoga

9. How many O-atoms present in a packet of Glauber's Salt sample?
(Glauber Salt = $Na_2SO_4.10H_2O$)

(A) 4 (B) 14
(C) 140 (D) 1400

Khoobsurat Baat

Sol. (B,C,D) Since the number of molecules of Glauber Salt are not given so any answer which is an integral multiple of 14 will be the correct answer

Ans-B, C, D.

dimag ki batti hui chaalu

Common Mistake

Students may forget to include the oxygen atoms from the 10 water molecules in the hydrated salt, leading to the incorrect conclusion that there are only 4 oxygen atoms (from SO_4^{2-}).

10. If there are 400 oxygen atoms present in a potash alum($K_2SO_4.Al_2(SO_4)_3.24H_2O$ sample, then find out the total H-atoms present in the sample.

(A) 400 (B) 480
(C) 48 (D) None of these

Sol. (B) Potash alum = $K_2SO_4.Al_2(SO_4)_3.24H_2O$
In one molecule of alum → 40 O-atoms
In 10 molecules of alum → 40 × 10 = 400 O-atoms
In one molecules of alum = 48 H-atoms
In 10 molecules of alum = 48 × 10 = 480 H-atoms

socho yeh kaise nikala?

11. How many O-atoms are present in 20 molecules of sucrose?

(A) 22 (B) 220 (C) 11 (D) 110

Sol. (B) For explanation, refer to QR code given on the last page.

ATOMICITY

Atomicity refers to the **number of atoms** present in a single molecule of an element or compound.

Note

Molecule	Atomicity	Reason
He	1	Contains 1 atom (monoatomic)
H_2	2	Contains 2 atoms (diatomic)
O_3	3	Contains 3 atoms (triatomic)
P_4	4	Contains 4 atoms (tetratomic)
S_8	8	Contains 8 atoms (polyatomic)

12. Calculate the atomicity of $C_6H_{12}O_6$ (glucose).

Sol. The molecule $C_6H_{12}O_6$ contains:

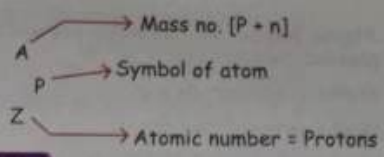
- 6 carbon atoms, 12 hydrogen atoms, 6 oxygen atoms.

Total atoms = 6 + 12 + 6 = 24.

ekdum balle balle sawaal hai

Representation of an Atom

- To represent an atom, first write its symbol, followed by an atomic number (equal to the number of protons) which comes in the left bottom of the symbol. Finally, write the mass number (equals to the sum of the number of protons and neutrons) which comes in the left top of the symbol.
- For symbol P, atomic number Z and mass number number A, general representation is:



Note

Elements along with general representation are given in the table given below:

Element	Symbol	Z	A	Notation
Hydrogen	H	1	1	$_1H^1$
Helium	He	2	4	$_2He^4$
Lithium	Li	3	7	$_3Li^7$
Beryllium	Be	4	9	$_4Be^9$
Boron	B	5	11	$_5B^{11}$
Carbon	C	6	12	$_6C^{12}$
Nitrogen	N	7	14	$_7N^{14}$
Oxygen	O	8	16	$_8O^{16}$
Fluorine	F	9	19	$_9F^{19}$
Neon	Ne	10	20	$_{10}Ne^{20}$
Sodium	Na	11	23	$_{11}Na^{23}$
Magnesium	Mg	12	24	$_{12}Mg^{24}$
Aluminium	Al	13	27	$_{13}Al^{27}$
Silicon	Si	14	28	$_{14}Si^{28}$
Phosphorus	P	15	31	$_{15}P^{31}$
Sulphur	S	16	32	$_{16}S^{32}$
Chlorine	Cl	17	35	$_{17}Cl^{35}$
Argon	Ar	18	40	$_{18}Ar^{40}$
Potassium	K	19	39	$_{19}K^{39}$
Calcium	Ca	20	40	$_{20}Ca^{40}$

is table par dhyan do-ache se

13. If the no. of neutrons in Phosphorus = a and the no. of electrons in Si = b then find out the value of $a^2 + b^2$ = ?

Sol. (452) For explanation, refer to QR code given on the last page.

14. If no. of protons in Potassium = x

no. of neutrons in Ca = y

if $a = x^2 + y^2$, and $a = p^q + q^p$ then find out the value of p and q. (Given p > q)

Sol. Atomic number of potassium = Number of protons = $x = 19$

Atomic number of Ca = 20,

Mass number of Ca = 40

Number of neutrons = 40 – 20 = 20 = y

$a = (19)^2 + (20)^2 = 761$

$761 = 760^{1} + 1^{760}$

$[p = 760, q = 1]$

Pata lga kaise kia

15. Find the total no. of electrons in one molecule of CO_2.

(A) 22 (B) 44

(C) 66 (D) 88

Sol. (A) For explanation, refer to QR code given on the last page.

16. The number of protons, electrons and neutrons in a molecule of heavy water are respectively. [JEE Main 2013 Online]

(A) 8, 10, 11 (B) 10, 10, 10

(C) 10, 11, 10 (D) 11, 10, 10

Sol. (B) For explanation, refer to QR code given on the last page.

CATIONS AND ANIONS

Cations

A cation is a positively charged ion.

Examples of Cations

1. Sodium Ion (Na^+), 2. Calcium Ion (Ca^{2+}),
3. Ammonium Ion (NH_4^+):

Anions

- An anion is a negatively charged ion.

Examples of Anions

1. Chloride Ion (Cl^-), 2. Sulfide Ion (S^{2-}),
3. Nitrate Ion (NO_3^-):

Note

List of Cations [ous = कम charge, ic = ज्यादा charge]					
1.	Hydrogen ion	H^+	11.	Ferric ion	Fe^{+3}
2.	Sodium ion	Na^+	12.	Plumbous ion	Pb^{+2}
3.	Calcium ion	Ca^{+2}	13.	Plumbic ion	Pb^{+4}
4.	Potassium ion	K^+	14.	Stennous ion	Sn^{+2}
5.	Magnesium ion	Mg^{2+}	15.	Stennic ion	Sn^{+4}
6.	Barium ion	Ba^{2+}	16.	Ammonium ion	NH_4^+
7.	Aluminium ion	Al^{+3}	17.	Mercurous ion	Hg^+ or Hg_2^{2+}
8.	Cuprous ion	Cu^{+1}	18.	Mercuric ion	Hg^{2+}
9	Cupric ion	Cu^{+2}	19.	Silver ion	Ag^+
10.	Ferrous ion	Fe^{+2}	20.	Lithium ion	Li^+

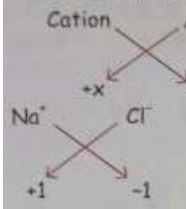
thoda dhyan lagao

List of Anions					
1.	Chloride ion	Cl^-	12.	Sulphite ion	SO_3^{2-}
2.	Bromide ion	Br^-	13.	Phosphate ion	PO_4^{3-}
3.	Iodide ion	I^-	14.	Oxalate ion	$C_2O_4^{2-}$
4.	Fluoride ion	F^-	15.	Nitrate ion	NO_3^-
5.	Hydroxide ion	OH^-	16.	Nitrite ion	NO_2^-
6.	Hydride ion	H^-	17.	Carbonate ion	CO_3^{2-}
7.	Dichromate ion	$Cr_2O_7^{2-}$	18.	Bicarbonate ion	HCO_3^-
8.	Sulphide ion	S^{2-}	19.	Permanganate ion	MnO_4^-
9.	Nitride ion	N^{3-}	20.	Chromate ion	CrO_4^{2-}
10.	Phosphide ion	P^{3-}	21.	Cyanide ion	CN^-
11.	Sulphate ion	SO_4^{2-}	22.	Thiosulfate ion	$S_2O_3^{2-}$

isko bhi toh dekhlo ek baar

How to make a formula of a salt

1. First write the cation along with its charge on the left side.
2. Then write the anion along with its charge on the right side.
3. Cross multiply both charges, the formula obtained will be the formula of salt.



$\Rightarrow (cation)_y (anion)_x$

$\Rightarrow Na_1Cl_1 = NaCl$ Sodium chloride

17. Write the formula for the compound formed between aluminum (Al) and sulfur (S).

Sol. Cation: Aluminum (Al^{3+}) has a charge of +3.

Anion: Sulfur (S^{2-}) has a charge of -2.

To balance the charges, 2 aluminum ions (Al^{3+}) are needed to balance the charge of 3 sulfide ions (S^{2-}).

So, formula formed will be Al_2S_3 (Aluminum sulfide).

charge toh pata hai na dono ke

Common Mistake

Confusing the subscripts and writing ratios that do not reflect charge neutrality, such as Al_3S_3 instead of Al_2S_3.

18. Write down the formula of these given salts-

(i) Silver nitrate (ii) Sodium phosphate
(iii) Magnesium nitride (iv) Calcium phosphide
(v) Ammonium Nitrate (vi) Sodium hydrogen sulphate
(vii) Potassium dichromate (viii) Cuprous oxalate
(ix) Ferric oxalate (x) Aluminium sulphate

Sol. (i) $AgNO_3$ (ii) Na_3PO_4 (iii) Mg_3N_2
(iv) Ca_3P_2 (v) NH_4NO_3 (vi) $NaHSO_4$
(vii) $K_2Cr_2O_7$ (viii) CuC_2O_4 (ix) $Fe_2(C_2O_4)_3$
(x) $Al_2(SO_4)_3$

Page ke kone me karke dekho

LAWS OF CHEMICAL COMBINATION

Law of Conservation of Mass

The **Law of Conservation of Mass** is a fundamental principle of chemistry that states:

"Mass can neither be created nor destroyed in a chemical reaction."

This means the total mass of the reactants in a chemical reaction is always equal to the total mass of the products.

Example: For reaction: $CaCO_3 \xrightarrow{\Delta(heat)} CaO + CO_2$

Here,

Mass of reactant ($CaCO_3$) = 100 g

Mass of product ($CaO + CO_2$) = 100 g

Note

Limitation

This law is not applicable in case of nuclear reactions. In nuclear reactions, mass is converted into energy.

$\Delta E = \Delta mc^2$

19. If 6.3 g of $NaHCO_3$ is added to 15.0 g of CH_3COOH solution, then residue is found to weigh 18.0 g. What is the mass of CO_2 released in the reaction?

Sol. The reaction can be represented as:

$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$

Now, 6.3 g + 15 g = 18 g + Mass of CO_2

Mass of CO_2 = 3.3 g

bilkul hi seedha sawaal hai

20. What mass of silver nitrate will react with 5.85 g of sodium chloride to produce 14.35 g of silver chloride and 8.5 g of sodium nitrate if the Law of conservation of mass is true.

Sol. (17 g) For explanation, refer to QR code given on the last page.

21. A sealed vessel contains 5 g of baking soda and 10 g of acetic acid. After the reaction, the mass of the vessel is measured as 15 g. Prove the law of conservation of mass.

Sol. Since the system is sealed, no mass escapes.
Mass of reactants = 5 + 10 = 15 g
Mass of products = 15 g

Law of Constant or Definite Proportions

The **Law of Constant or Definite Proportions**, also known as **Proust's Law**, states that a given chemical compound always contains exactly the same proportion of elements by mass, regardless of the amount or source of the compound.

For example:

In H_2O, % of H = $\left(\frac{\text{Weight of H}}{\text{Total weight of H}}\right) \times 100\%$

$= (2/18) \times 100\% = 11.11\%$

% of O = (100% - 11.11)% = 88.89%

The mass percentage of H and O in water remains always same, irrespective of the source of water.

Note

Limitations

The law is not applicable if an element exists in different isotopes which may be involved in the formation of compound.

Suppose we have CO_2 molecule

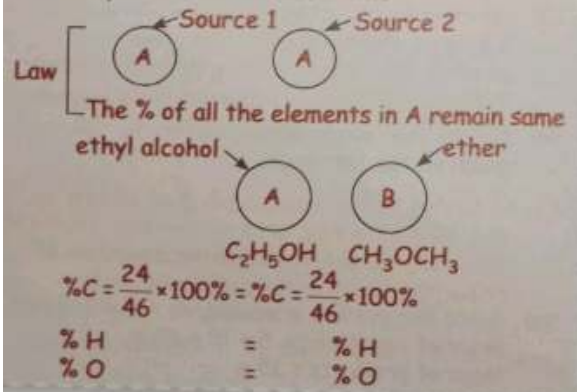
Carbon has two isotopes C^{12} and C^{14}

$$CO_2 \begin{cases} \nearrow {}^{12}CO_2 \rightarrow \left(\frac{12}{12+32}\right) \times 100\% = 2.27\% \\ \searrow {}^{14}CO_2 \rightarrow \left(\frac{14}{14+32}\right) \times 100\% = 30.43\% \end{cases}$$

So, in both isotopes, the percentage of C will be different.

Khoobsurat Baat

The reverse of this law is not true.
"Elements may combine in the same mass ratio but compounds can be different.



22. 1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated in steam gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.

Sol. Metal + O_2 → Metal oxide

1.8 g 3.0 g

Mass of O_2 = 3.0 g - 1.8 g = 1.2 g

Ratio of metal and oxygen = 1.8 g:1.2 g = 3:2

Metal + steam → Metal oxide

1.5 g 2.5 g

Mass of steam = 2.5 g - 1.5 g = 1.0 g

Ratio of metal and oxygen = 1.5 g:1.0 g = 3:2

pagla mat jana

23. 2.16 g of copper metal when treated with nitric acid followed by ignition of the nitrate gave 2.70 g of copper-oxide. In another experiment, 1.15 g of copper oxide upon reduction with hydrogen gave 0.92 g of copper. Show that the above data illustrates the Law of definite proportion.

Sol. (20%) For explanation, refer to QR code given on the last page.

24. 5.250g of lead combined with 2.541 g of oxygen yields lead peroxide PbO_2 which is synthesized from another method by heating lead nitrate. The percentage of oxygen obtained is 32.61. Illustrate the law of Constant proportion?

Sol. Weight of lead peroxide = Weight of lead + weight of oxygen = 5.250 + 2.541 = 7.791 g

- 7.791 g of lead peroxide contains 2.541 g of oxygen
- So 100 g of lead peroxide contains $\frac{2.541}{7.791} \times 100\% = 32.61\%$
- The percentage of oxygen obtained by heating lead nitrate is 32.61 by the second method.
- Since the percentage composition is the same from both the methods, the data illustrates the Law of Constant Proportion.

Ghabra kyu rahe ho?

Law of Multiple Proportions

The Law of Multiple Proportions states that when two elements combine to form more than one compound, the different masses of one element that combine with a fixed mass of the other element are in simple whole number ratios.

For example: CO and CO_2

In these compounds, the mass of C will be fixed, so masses of O in CO and CO_2 will bear a simple whole number ratio.

In CO, mass of O = 16 g

In CO_2, Mass of O = 32 g

Ratio = 16:32 = 1:2

Note

Limitations

The law fails when applied to some organic compounds. Organic compounds have carbon which has a tendency to link itself with other carbon atoms by single, double and triple bonds.

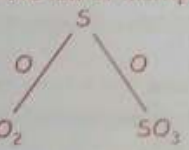
Example:

CH_4	12:4	(12):4
C_2H_6	24:6	(12):3
C_3H_8	36:8	(12):2.67

The wt ratio of hydrogen combining with a fixed mass of carbon (12 g) is not a whole number ratio.

4:3:2.67

25. Prove law of multiple proportion in-



[S = 32, O = 16]

Sol. In SO_2 and SO_3, mass of S will be fixed.

Mass of O in SO_2 = 32 g, Mass of O in SO_3 = 48 g

Ratio = 32:48 = 2:3

As the ratio is in simple whole numbers, this law is verified.

pehle law ko toh samjho tabhi toh hoga

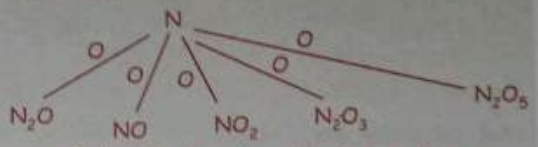
26. Which of the following follows law of multiple proportion?

(A) H_2O, H_2S (B) SO_2, SO_3

(C) N_2O_5, N_2O (D) H_2O, H_2O_2

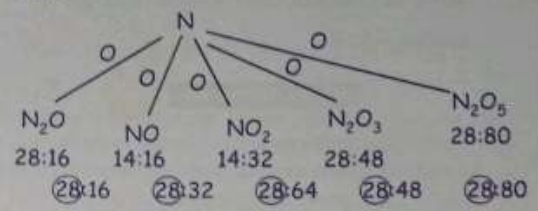
Sol. (B,C,D) For explanation, refer to QR code given on the last page.

27. Prove



follows the law of constant proportions.

Sol.



Mass ratio of O-atom → 16:32:64:48:80

combining with fixed mass (28 g) of N

= 1:2:4:3:5

mushkil nhi hai bhayi, asaan hai

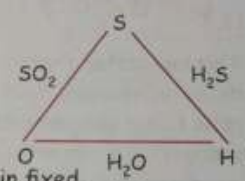
Law of Reciprocal Proportions

Law of reciprocal proportions states that the ratio of the masses of two elements A and B which combine separately with a fixed mass of the third element 'C' is either the same or some simple multiple of the ratio of the masses in which 'A' and 'B' combine directly with each other.

Example:

In SO_2, ratio of masses of S and O = 32:32

In H_2S, ratio of masses of S and H = 32:2



Here, the mass of S will remain fixed.

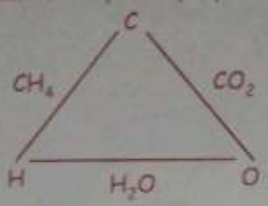
The ratio of H and O combining with 32 g of S (which is fixed) = 2:32 = 1:16

Hence,

H : O
2 : 16
1 : 8

H : O = 1 : 8
H : O = 1 : 16
ratio = 2 : 1

28. Prove law of reciprocal proportions holds for



Sol. CH_4 = 12:4, CO_2 = 12:32

Ratio of H:O = 1:8

Ratio of H and O, combining with 12 gm carbon = 1 : 8

thoda tedha hai

29. Hydrogen combines with chlorine to form HCl, and chlorine combines with oxygen to form Cl_2O. If 1 gram of hydrogen combines with 35.5 grams of chlorine in HCl, and 35.5 grams of chlorine combine with 16 grams of oxygen in Cl_2O, verify the law of reciprocal proportions.

Sol. • Mass ratio of Cl to H in HCl = $\frac{35.5}{1} = 35.5$

• Mass ratio of Cl to O in Cl_2O = $\frac{35.5}{16} = 2.22$

The ratio of H to O

$\frac{1}{35.5} : \frac{16}{35.5} = 1:16$

This matches the observed ratio in water (H_2O), thus, verifies the law.

dekhne se kitna mushkil lag raha hai na

Gay Lussac's Law of Gaseous Volume

This law states that when gases react together, they do so in volumes that bear a simple whole-number ratio to one another and to the volumes of the gaseous products (if gaseous), provided all volumes are measured under the same conditions of temperature and pressure.

30. 10 L of H_2 gas combine with 10 L of Cl_2 gas to produce HCl g. Find out the volume of HCl g formed (under similar conditions).

Sol. (20 L) For explanation, refer to QR code given on the last page.

31. In the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, how much ammonia will be produced if 10 liters of N_2 react with excess H_2?

Sol. From the equation:

1 volume of N_2 produces 2 volumes of NH_3.

Volume of NH_3 = 10 × 2 = 20 L.

ekdum balle balle sawaal hai

32. The formula of a gaseous hydrocarbon which required 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is.

Sol. The combustion reaction in terms of x and y can be written as:

$$C_xH_{y(g)} + \left(x + \frac{y}{4}\right)O_2(g) \rightarrow xCO_2(g) + \frac{y}{2}H_2O(l)$$

V 6V -

- - $V \times x = 4V$

$\Rightarrow x = 4$

Since volume of O_2 is 6 times to the volume of C_xH_y, so,

$\Rightarrow V\left(x + \frac{y}{4}\right) = 6V$

$\Rightarrow \boxed{\left(x + \frac{y}{4}\right) = 6} \Rightarrow 4 + \frac{y}{4} = 6 \Rightarrow \boxed{y = 8}$

dimag ki ghanti bja dega

Avogadro's Law

Equal volume of all gases have equal number of molecules (not atoms) at same temperature and pressure condition.

Mathematically, for ideal gases, $V \propto n$ (constant T & P)

S.T.P. (Standard Temperature and Pressure):

Temperature = 0°C or 273 K

Pressure = 1 atm = 760 mm of Hg

Volume of one mole of an ideal gas = 22.4 litres

Khoobsurat Baat

When two gaseous sample under same pressure, temperature condition with same volume of container are taken then—

- moles of both the samples are always same.
- molecules of both the samples are always same.

Khoobsurat Baat

- masses of both the sample will be same only when molar masses of gases are same.
- atoms of both the samples will be same only when atomicity is same.
- Nucleons of both the samples will be same only when molar masses of both the gases are same.

Under 1 atm pressure and 273 K or 0°C temperature, volume of 1 mole of gas = 22.4 lt.

↓

STP

↓

Standard temp, pressure conditions

Under 1 bar pressure and 273 K or 0°C temperature, volume of 1 mole of gas = 22.7 lt.

1 atm = 1.01325 bar

yeh kamaal ka point hai, chhod mat dena not even by mistake

33. If 10 liters of hydrogen gas react with oxygen to form water, what volume of oxygen is required?

Sol. (5 L) For explanation, refer to QR code given on the last page.

34. 5 g of zinc is treated separately with an excess of [9 Jan, 2020 (Shift-II)]

(A) dilute hydrochloric acid and

(B) aqueous sodium hydroxide.

The ratio of the volumes of H_2 evolved in these two reactions is:

(A) 1:4 (B) 1:1

(C) 1:2 (D) 2:1

Sol. (B) The reaction between Zn and dil. HCl is,

$Zn + 2\ dil.HCl \rightarrow ZnCl_2 + H_2$

The reaction between Zn and aq. NaOH is,

$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

Same amount of H_2 is evolved from 5 g of Zn according to the reaction.

Isko solve karne ke liye bas equation likhni aani chaiye

35. 1.0 L of a mixture of CO and CO_2 is taken. This mixture is passed through a tube containing red hot charcoal. The volume now becomes 1.6 L. The volumes are measured under the same conditions. Find the composition of the mixture by volume. [JEE Advanced]

Sol. (0.6 L) For explanation, refer to QR code given on the last page.

Relative Atomic Mass (RAM)

Relative atomic mass of an element is given by the ratio of mass of one atom of the element and $\frac{1}{12}^{th}$ mass of one carbon-12 atom.

$$= \frac{\text{Mass of total nucleons present in that atom}}{\frac{1}{12} \times \text{mass of 12 nucleons of } C\text{-}12 \text{ atom}}$$

$$= \frac{\text{Total nucleons} \times \text{mass of one nucleon of that substance}}{\frac{1}{12} \times 12 \times \text{mass of one nucleons of } C\text{-}12 \text{ atom}}$$

= Total nucleons

Khoobsurat Baat

Interpretation

Relative atomic mass of any atom = Total nucleons present in that atom

Also,

Number of nucleons = Sum of protons + neutrons

Example:

RAM of oxygen atom = Total nucleons of oxygen atom

= Protons + neutrons

= 8 + 8 = 16

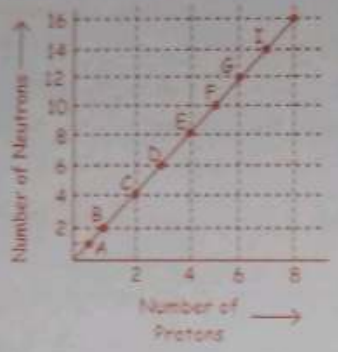
36. The RAM of C, H, N, Na are respectively

(A) 12, 1, 7, 11 (B) 6, 2, 7, 23

(C) 12, 2, 14, 23 (D) 12, 1, 14, 23

Sol. (D) For explanation, refer to QR code given on the last page.

37. In the graph, number of protons are plotted vs number of neutrons from the element A to G and H (hydrogen). Max number of neutrons are present in one formula unit of:



(A) A_2E (B) DE_2

(C) GE (D) DH_4E

Sol. (B) DE_2 = (6 + 8 × 2) = 22 neutrons

Graph par dhyan do bas

ATOMIC MASS UNIT

The atomic mass unit (amu) is a standard unit of mass that quantifies the mass of atoms and molecules.

It is defined as $1/12^{th}$ the mass of one atom of carbon-12 (C-12), which is the isotope of carbon with 6 protons and 6 neutrons.

Khoobsurat Baat

$$1\ amu = 1u = \frac{1}{12} \times \text{mass of one } C-12 \text{ atom}$$

$$= \frac{1}{12} \times \text{mass of 12 nucleon}$$

$$= \frac{1}{\cancel{12}} \times \cancel{12} \times \text{mass of one nucleon}$$

1 u = 1 amu = mass of one nucleon

1 u = 1 amu = 1.66×10^{-24} g

Atomic Weight or Molecular Weight

Atomic weight or Molecular weight is the weight of one atom or molecule. Weight of any atom or molecule is governed by nucleons, so weight of anything is the weight of nucleons present in that entity.

Atomic weight = Total nucleons x wt of one nucleon

Atomic weight = RAM × 1u

Molecular weight = Total nucleons x wt of one nucleon

Note

Atomic weight and Molecular weight

	No. of nucleons	Wt (u)	Wt (gm)
H	1	1 u	$1 \times 1.66 \times 10^{-24}$ g
N	14	14 u	$14 \times 1.66 \times 10^{-24}$ g
C	12	12 u	$12 \times 1.66 \times 10^{-24}$ g
O	16	16 u	$16 \times 1.66 \times 10^{-24}$ g
Na	23	23 u	$23 \times 1.66 \times 10^{-24}$ g

table ko mat chhodo, inme bhut gyaan hai

How to calculate the number of entities (atoms, molecules, ions) and total number of protons, neutrons, electrons and valence electrons in a given sample without using any formula

Lets understand this with the help of a simple example—

If we have a massless (assumed) box and it is filled with apples (100 g each) and we are asked the number of apples present in the box if the total weight of the box is 800 gm.

$$\text{No. of apples} = \frac{\text{Total weight of box}}{\text{weight of one apple}} = \frac{800\,g}{100\,g} = 8 \text{ apples}$$

Similarly if we are asked the number of entities (atoms, molecules and ions) in a sample then we have to divide the total weight of sample from weight of one entity.

38. How many atoms are present in a 1000 u sample of H-atom?

Sol. No. of apples = $\frac{\text{Total wt. of the sample (box)}}{\text{Wt. of one apple (H - atoms)}}$

$= \frac{1000\ \cancel{u}}{1\ \cancel{u}} = 1000$ atoms

simple hi concept use hua hai bas

39. Find out the number of Na atoms present in 920000 u sample of sodium sample.

Sol.

Total apples = $\frac{\text{Total wt. of sample (box)}}{\text{wt. of one apple (Na-atoms)}}$

$= \frac{920000\ \cancel{u}}{23\ \cancel{u}} = 40000$ atoms

40. Find out the total number of atoms present in 20 g of calcium atom sample.

Sol. $\left(\frac{1}{2}N_A\right)$ For explanation, refer to QR code given on the last page.

Khoobsurat Baat

$\frac{1}{1.66 \times 10^{-24}} = 6.022 \times 10^{23}$ = Avogadro's number (N_A)

Valence Electrons

Valence electrons, in terms of the K, L, M, N shells, refer to the electrons in the outermost shell of an atom. These shells correspond to the principal quantum numbers n = 1, 2, 3, and 4 respectively.

Important

Shells and their maximum capacity:

- K shell (n = 1): Maximum 2 electrons.
- L shell (n = 2): Maximum 8 electrons.
- M shell (n = 3): Maximum 18 electrons.
- N shell (n = 4): Maximum 32 electrons.

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The number of valence electrons is determined by how many electrons are present in the outermost shell, not the total number of electrons in the atom.

For Example—

Valence electrons in Carbon atom

Carbon atom has 6 electrons.
So the electronic configuration of carbon = 2, 4

Valence electrons in carbon = 4

Valence electrons in oxygen - atom

Total electrons in oxygen atom = 8

Electronic configuration = 2, 6

Valence electrons in oxygen atom = 6

41. Find out total proton, neutron, electron and valence electron of carbon ($^{12}_{6}C$) atom.

(A) 6, 6, 6, 6 (B) 12, 6, 6, 6
(C) 6, 6, 6, 4 (D) 6, 6, 12, 4

Sol. (C) Atomic number of C = No. of protons = 6
= No. of electrons

Mass number = 12

No. of neutron = 12 - 6 = 6

No of valence electrons : 4

nahi hai na mushkil, bola tha na

42. Find out the total number of protons, neutrons, electrons, nucleon and valence electrons in—

(a) N^{3-} (nitride)

(b) NH_4^+ (ammonium ion)

(c) CO_3^{2-} (carbonate ion)

Sol. (a) For explanation, refer to QR code given on the last page.

AVOGADRO NUMBER

Avogadro's number is a fundamental constant in chemistry and physics that defines the number of particles (atoms, molecules, ions, etc.) in one mole of a substance. The value of Avogadro's number is:

$N_A = 6.022 \times 10^{23}$ particles/mol

This number is used to convert between the atomic or molecular scale and the macroscopic scale.

43. Calculate the total number of atoms present in 50 g $CaCO_3$ sample.

Sol. No of $CaCO_3$ molecules $= \dfrac{\text{Total wt. of sample}}{\text{wt. of one } CaCO_3 \text{ molecule}}$

$$= \frac{50\,g}{100\,u} = \frac{50\,g}{\frac{100}{2} \times 1.66 \times 10^{-24}\,g}$$

$$= \frac{1}{2} \times 6.022 \times 10^{23} = \frac{1}{2} N_A$$

Total atoms present in the sample
= No. of molecules × atomicity

$$= \frac{N_A}{2} \times 5 = \frac{5N_A}{2} \quad (\text{Atomicity} = 1 + 1 + 3 = 5)$$

yeh sawaal toh ho hi jaega

Khoobsurat Baat

Atomicity = No. of atoms present in one molecule

Common Mistake

Students might forget to multiply the no of molecules by atomicity while calculating total atoms which leads to conceptual errors.

44. Find out the total number of protons, neutrons, electrons and total nucleons in 50 g CO_2 sample.

Sol. (25 N_A) For explanation, refer to QR code given on the last page.

Khoobsurat Baat

It implies that if the weight of sample 'x' gm then total nucleons = xN_A.

gram हटाओ, N_A लगाओ, Nucleons ले जाओ

Note

20 g $CaCO_3$ sample = 20 N_A nucleons

20 g H_2SO_4 sample = 20 N_A nucleons

20 g $KMnO_4$ sample = 20 N_A nucleons

Khoobsurat Baat

Irrespective of the name of the sample, if the weights are same, number of nucleons are same.

Jugaad!

Whenever we are asked the number of protons and neutrons in a given sample of 'x' gm substance then:

(i) gm हटाओ, N_A लगाओ, Nucleons ले जाओ

(ii) $\text{Protons} = \dfrac{\text{Total nucleons}}{2}$

(iii) $\text{Neutrons} = \dfrac{\text{Total nucleons}}{2}$

45. Calculate total protons and neutrons in 150 g NO_2 sample.

Sol. (75 N_A) For explanation, refer to QR code given on the last page.

46. Calculate total p, n and e in 88 mg CO_2 sample.

Sol. Method 1 (General)

$$\text{Total } CO_2 \text{ molecules} = \frac{88 \times 10^{-3}\,g}{44\,u} = 2 \times 10^{-3} N_A$$

In one CO_2 → Protons = 22
Neutrons = 22
Electrons = 22

Total protons in sample
$= 22 \times 2 \times 10^{-3} N_A = 44 \times 10^{-3} N_A$
Total neutrons in sample
$= 22 \times 2 \times 10^{-3} N_A = 44 \times 10^{-3} N_A$
Total electrons in sample
$= 22 \times 2 \times 10^{-3} N_A = 44 \times 10^{-3} N_A$

$88 \times 10^{-3} N_A$

Method 2 (Jugaad!)

88 mg $CO_2 = 88 \times 10^{-3}$ gm

gm hatao, N_A lagao = Nucleons le jao (always true)

Total nucleons = $88 \times 10^{-3} N_A$

Total protons = $\frac{88 \times 10^{-3} N_A}{2} = 44 \times 10^{-3} N_A$

Total neutrons = $\frac{88 \times 10^{-3} N_A}{2} = 44 \times 10^{-3} N_A$

hai na mazedaar, ab shant karo dimag ko

Common Mistake

This jugaad is applicable only when we have sample containing only those atoms in which protons = neutrons.

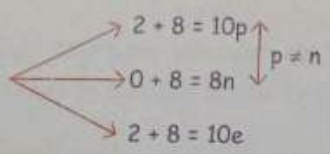
47. Calculate total number of nucleons, protons and neutrons in 72 g of H_2O sample.

Sol. **Method 1 (General)**

No. of water molecules = $\frac{\text{Total wt. of sample}}{\text{wt. of one } H_2O \text{ molecule}}$

$= \frac{72\,gm}{18u} = \frac{72\,gm}{18 \times 1.66 \times 10^{-24}\,gm} = 4 N_A$

One water molecule says



Total protons = $10 \times 4N_A = 40 N_A$
Total neutrons = $8 \times 4N_A = 32 N_A$
Total electrons = $10 \times 4N_A = 40 N_A$

$72 N_A$ nucleons

Method 2 (Jugaad!)

72 gm H_2O sample [p ≠ n]

gm hatao, N_A lagao = Total nucleons le jao

Total nucleons = $72 N_A$ ✓

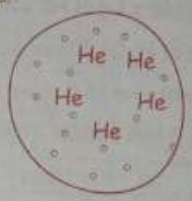
Total protons = $\frac{72 N_A}{2} = 36 N_A$ ✗

Total neutrons = $\frac{72 N_A}{2} = 36 N_A$ ✗

Socho kyu

socho socho, thoda toh dimag lagao

48. Find out the total He atoms present in 52 g sample.



Sol. No. of He atoms = $\frac{\text{wt. of sample}}{\text{wt. of one He atom}}$

$= \frac{52\,g}{4u} = \frac{52\,g}{4 \times 1.66 \times 10^{-24}\,g} = 13 N_A$

49. Calculate the number of C-atoms in 12 g of C-12 sample.

Sol. No. of Carbon atoms = $\frac{\text{Total weight}}{\text{wt. of one } C-12 \text{ atom}}$

$= \frac{12\,g}{12u} = \frac{12\,g}{12 \times 1.66 \times 10^{-24}\,g} = \frac{1}{1.66 \times 10^{-24}}$

$= 6.022 \times 10^{23} = N_A$

dimag ki ghanti baji kya tann tann

50. How many nucleons are present in 1 gm sample of any substance?

Sol. (N_A) For explanation, refer to QR code given on the last page.

MOLE

A mole is a quantity used to express amounts of a chemical substance. It is equivalent to the number of particles (atoms, molecules, ions, etc.) present in exactly 12 grams of carbon-12.

This number is called **Avogadro's number** and is approximately:

$$1 \text{ mol} = 6.022 \times 10^{23} \text{ particles.}$$

Thus, **1 mole of any substance contains 6.022×10^{23}** representative particles, regardless of whether they are atoms, molecules or ions.

Example:

6.022×10^{23} atoms of He = 1 mol He atoms

6.022×10^{24} molecules of H_2O = 10 mol water molecules

3 N_A CO_3^{2-} ions (carbonate ion) = 3 mole CO_3^{2-} ions

5 N_A Sodium atoms = 5 mol sodium atoms

$$\text{Moles} = \frac{\text{no. of entities}}{N_A}$$

Rearrange the above expression for no. of entities:

$$\text{Number of entities} = \text{moles} \times N_A$$

Khoobsurat Baat

$$\frac{\text{wt. of sample}}{\text{wt. of one entity}} = \text{no. of entities}$$

- total (p) → no. of entities × protons in one entity
- total (n) → no. of entities × neutrons in one entity
- total (e) → no. of entities × electrons in one entity
- total valence (e) → no. of entities × valences in one entity

yeh chart bht gyaan wala hai

51. Calculate moles of 1500 gm of $CaCO_3$ sample

Sol. No. of $CaCO_3$ molecule $= \frac{1500 \text{ gm}}{100 \text{ u}}$

$$= \frac{1500 \text{ g}}{100 \times 1.66 \times 10^{-24} \text{ g}} = 15 \, N_A$$

$$\text{Moles} = \frac{\text{Total entities}}{N_A} = \frac{15 \, N_A}{N_A} = 15 \text{ mol}$$

Dekho kitna asaan ho gya ab

52. Calculate no. of $CaCO_3$ molecules in 15 mol sample.

Sol. 1 mole → N_A $CaCO_3$ molecules

15 mole → 15 N_A $CaCO_3$ molecules

GRAM ATOMIC AND GRAM MOLECULAR WEIGHT

Weight of one mole of an atom or one mole of molecule represented in gm are called gm atomic and gm molecular weight respectively.

Khoobsurat Baat

It is also called Molar Mass.

Example:

*gm atomic wt of O is-

wt. of one O-atom = 16 amu = $16 \times 1.66 \times 10^{-24}$ gm

* wt. of one mole of O-atom

$$= 16 \times \underbrace{1.66 \times 10^{-24} \times 6.022 \times 10^{23}}_{1} \text{ gm}$$

= 16 gm

* gm molecular weight of H_2O is-

wt. of one molecule of H_2O = 18 amu

$= 18 \times 1.66 \times 10^{-24}$ gm

wt. of one mole of H_2O molecules

$= 18 \times 1.66 \times 10^{-24} \times 6.022 \times 10^{23}$ gm

= 18 gm

Important

Gram atomic weight of Carbon

Weight of one mole C-12 atom in grams

wt. of one C-12 = 12 u = $12 \times 1.66 \times 10^{-24}$ g

wt. of one mole C-12 atoms =

$$= 12 \times \underbrace{1.66 \times 10^{-24} \times 6.022 \times 10^{23}}_{1} \text{ g}$$

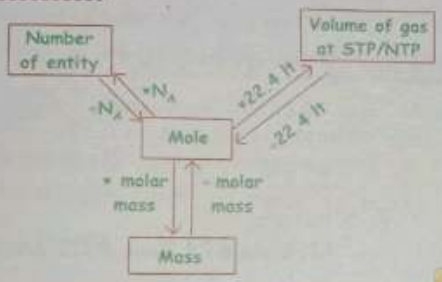
= 12 g

Khoobsurat Baat

$$\text{Mole} = \frac{\text{weight of the sample}}{\text{weight of one entity}} = \text{no. of entities}$$

$$\frac{\text{weight of the sample}}{\text{gm atomic or gm molecular weight}} = \text{no. of moles}$$

Y-MAP



is map ko halke mei mat lo bhulkar bhi

53. The number of atoms in 8 g of sodium is $x \times 10^{23}$ [JEE-Mains 2021]

The value of x (nearest integer)

Given: atomic mass = 23 u, $N_A = 6.02 \times 10^{23}$

Sol. Moles of Na atoms = $\frac{\text{Given mass}}{\text{Molar mass}} = \frac{8}{23}$ mol

gm wala mass

No. of entities (Na atoms)

$$= \text{moles} \times N_A = \frac{8}{23} \times 6.02 \times 10^{23} = 2 \times 10^{23}$$

gm wala mass dekhkar pagla mat jana

54. Find the number of moles of

(i) 3.011×10^{23} molecules of $CaCO_3$

(ii) 1.5055×10^{22} molecules of H_2SO_4

(iii) 30.11×10^{23} atoms of sodium

(iv) 0.5 N_A ions of NH_4^+

Sol. (i) 3.011×10^{23} molecules of $CaCO_3 \div N_A$

$$\frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} = 0.5 \text{ mol}$$

(ii) 1.5055×10^{22} molecules of $H_2SO_4 \div N_A$

$$\frac{1.5055 \times 10^{22}}{6.022 \times 10^{23}} = 0.025 \text{ mol}$$

(iii) 30.11×10^{23} molecules of sodium $\div N_A$

$$\frac{30.11 \times 10^{23}}{6.022 \times 10^{23}} = 5 \text{ mol}$$

(iv) 0.5 N_A ions of $NH_4^+ \div N_A$

$$\frac{0.5\,N_A}{N_A} = 0.5 \text{ mol}$$

are itna assan kaise

55. Which of the following contains minimum no. of molecules?

(A) 11.2 Lt N_2 (0°C, 1 atm) (B) 24 g of O_3

(C) 16 g of O_2 (D) 25 g $CaCO_3$

Sol. (D) 11.2 Lt $N_2 \div 22.4$ Lt $\Rightarrow \frac{11.2 \text{ Lt}}{22.4 \text{ Lt}} = 0.5$ mol

= 0.5 N_A N_2 molecules

24 g of $O_3 \div$ molar mass $\Rightarrow \frac{24 \text{ g}}{48 \text{ g}} = \frac{1}{2}$ mol

= 0.5 N_A O_3 molecules

16 g of $O_2 \div$ molar mass $\Rightarrow \frac{16 \text{ g}}{32 \text{ g}} = \frac{1}{2}$ mol

= 0.5 N_A O_2 molecules

25 g $CaCO_3 \div$ molar mass $\Rightarrow \frac{25 \text{ g}}{100 \text{ g}} = \frac{1}{4}$ mol

$= \frac{1}{4}$ N_A $CaCO_3$ molecules

Note

- If the mass of a sample is given in gm, then we divide it with molar mass (in gm) to get the moles first then multiply it by N_A to get no. of entities.
- If the weight of sample is given in 'u', then divide it by weight of one entity (in u) to get the number of entities.

moles ki aisi ki taisi!

Khoobsurat Baat

moles ↑ entity ↑

If the entity is molecule or ion, then moles ↑ molecules or ions ↑ but no of atoms also depends on atomicity.

56. Which of the following contains maximum number of atoms?

(A) 11.2 Lt $N_{2(g)}$ (0°C, 1 atm)

(B) 24 g of O_3

(C) 16 g of O_2

(D) 25 g $CaCO_3$

Sol. (B) For explanation, refer to QR code given on the last page.

57. The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1:4. The ratio of number of their molecules is-

[JEE Main 2014 Offline]

(A) 1 : 4 (B) 7 : 32

(C) 1 : 8 (D) 3 : 16

Sol. (B) $M_{O_2} = x\,g \rightarrow n_{O_2} = \frac{x}{32} \rightarrow \frac{x}{32} \times N_A$

$M_{N_2} = 4x\,g \rightarrow n_{N_2} = \frac{4x}{28} \rightarrow \frac{4x}{28} \times N_A$

$= 7 : 32$

58. Find no. of NH_3 molecules present in 34 g sample. Also find the number of atoms present in the sample.

Sol. Moles $= \frac{\text{mass}}{\text{molar mass}} = \frac{34g}{(17g/mol)} = 2$ mol

Number of molecules $= 2N_A$

No. of atoms = Molecules × Atomicity

$= 2N_A \times 4 = 8N_A$

59. A metal chloride contains 55.0% of chlorine by weight. 100 mL vapours of the metal chloride at STP weigh 0.57 g. The molecular formula of the metal chloride is (Given: Atomic mass of chlorine is 35.5 u)

[JEE Main-12 April, 2023 Shift-I]

(A) MCl_2 (B) MCl_4 (C) MCl_3 (D) MCl

Sol. (A) First calculate the weight of metal chloride:

$= \frac{0.57}{100} \times 22700 = 129.39$

Weight of Cl = 129.39 × 0.55 = 71.1645

$\therefore$ Mole of $Cl = \frac{71.1645}{35.5} \cong 2$

Hence, the molecular formula of metal chloride is MCl_2.

moles toh nikal hi loge, agar isme bhi galti hui toh bilkul lallu ho

60. Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B. [JEE Main 27 June 2022 Shift-1]

Sol. Given: Molar mass of $A_2B = AB_3$

$\therefore (2A + B) = (A + 3B)$ $\begin{bmatrix} A \rightarrow \text{Atomic wt. of A} \\ B \rightarrow \text{Atomic wt. of B} \end{bmatrix}$

On solving, A = 2B that is the atomic weight of A is two times to the atomic weight of B.

kitne asaan tareeke se kardia humne isko

61. 120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are:

[JEE Main 24 June 2022 Shift-2]

(A) 25 and 75 (B) 40 and 60

(C) 60 and 40 (D) 75 and 25

Sol. (D) Mass of carbon = $n_{CO_2} \times 12$

$= \frac{330}{44} \times 12 = 90g$

$\%\text{ of } C = \frac{90}{120} \times 100 = 75\%$

Mass of hydrogen = $n_{H_2O} \times 2$

$= \frac{270}{18} \times 2 = 30g$

$\%\text{ of hydrogen} = \frac{30}{120} \times 100 = 25\%$

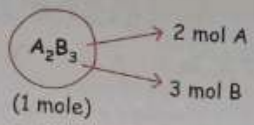
pagla mat jana

CALCULATIONS INVOLVING PERCENTAGE

Important concept

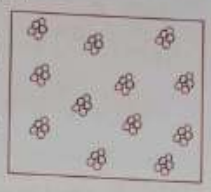
Lets consider a A_2B_3 molecule

atomicity of A = 2
atomicity of B = 3
Atomicity of A_2B_3 = 5



Moles of A = (moles of molecule) × atomicity of A

Moles of B = (moles of molecule) × atomicity of B



62. Find out the weight of C atoms in 200 gm $CaCO_3$.

Sol. Moles of $CaCO_3$ molecules = $\frac{200\,g}{100\,g}$ = 2 mole

Moles of carbon atoms = moles of $CaCO_3$ molecule × atomicity of carbon = 2 × 1 = 2 mole

Weight of carbon atoms = 2 × 12 = 24 g.

Important

1 mol atom	=	1 gram-atom
1 mol molecules	=	1 gram-molecule
1 mol ion	=	1 gram-ion

63. Calculate number of moles in 2 gm-atoms of oxygen

Sol. 2 gm- atoms of oxygen = 2 mol

64. Find out the total weight of C, H and O in 360 g of glucose sample.

Sol. $C_6H_{12}O_6$ = molar mass = 180 g

Moles of glucose sample = $\frac{360\,g}{180\,g} = 2$

$C_6H_{12}O_6$ → 2 × 6 = 12 mol C → 12 × 12 = 144 g
→ 2 × 12 = 24 mol H → 24 × 1 = 24 g
→ 2 × 6 = 12 mol O → 12 × 16 = 192 g

360 g

$$\%C = \frac{\text{wt. of } C}{\text{Total wt.}} \times 100 = \frac{144\,g}{360\,g} \times 100 = 40\%$$

kaafi asaaan hai yeh, padhlo ache se

65. Calculate the mole % of carbon in methane (CH_4).

Sol. mole % = $\frac{\text{mole of } C}{\text{Total moles}} \times 100 = 20\%$

66. What will be the molecular weight of a compound having 3.2% sulphur by weight?

(A) 1000 u (B) 2000 u
(C) 3000 u (D) 4000 u

Khoobsurat Baat

Sol. (A,B,C,D) All the answers will be correct which are integral multiples of 1000 u.

$M \times \frac{3.2}{100} = 32,\ M = 1000$

$M \times \frac{3.2}{100} = 64,\ M = 2000$

$M \times \frac{3.2}{100} = 96,\ M = 3000$

$M \times \frac{3.2}{100} = 128,\ M = 4000$

yeh bhut hi shandaar sawaal hai

67. Find out the molecular wt of insulin which has 3.2% of S and contains 2-sulphur atoms.

Sol. 3.2% of molecular wt = wt of sulphur

$\frac{3.2 \times M}{100} = 64,\ M = 2000$ u

68. An organic compound contains C, H and S. The minimum molecular weight of the compound containing 8% Sulphur is: (atomic weight of S = 32 amu) **(JEE Main 2016 Online)**

(A) 200 g mol^{-1} (B) 400 g mol^{-1}
(C) 600 g mol^{-1} (D) 300 g mol^{-1}

Sol. (B) Let the minimum molecular weight of sulphur be M.

According to the given information,

$\frac{M \times 8}{100} = 32,\ M = 400\ g\ mol^{-1}$

Note

For minimum molecular wt of the molecule, we will consider one atom of the element for which % is given.

69. Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of hemoglobin is: (Given: Atomic mass of Fe is 56 u, N_A, in 6.022×10^{23} mol^{-1})

[JEE Main 26 July 2022 Shift-2]

(A) 1.21×10^{5} (B) 12.0×10^{16}

(C) 1.21×10^{20} (D) 3.4×10^{22}

Sol. (C) Calculate no. of Fe atoms as:

$$\frac{0.34}{100} \times \frac{3.3}{56} \times 6.022 \times 10^{23} = 1.206 \times 10^{20}$$

70. A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is ____ $\times 10^3$ g mol^{-1} [nearest integer]

[JEE Main 25 June 2022 Shift-2]

Sol. Let the minimum molecular weight be M g.

According to the given information,

$$\frac{M \times 0.3}{100} = 75, \quad M = 25000 = 25 \times 10^3$$

araha hoga maza pakka sawaal nikalte hue

71. When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is ____ g mol^{-1} [nearest integer]

[JEE Main-29 Jan, 2023 Shift-II]

Sol. $\underset{0.01 \text{ mole}}{\text{Organic compound}} \xrightarrow[\Delta]{O_2} \underset{4.4 \text{ gm}}{CO_2}$

moles of $CO_2 = \frac{4.4}{44} = 0.1$ mole

moles of C = 0.1 × 1 = 0.1 mol = 1.2 g

wt. of C in one mole org comp = $\frac{1.2}{0.01} = 120$ g

Let the molar mass of org compound = M

$$M \times \frac{60}{100} = 120, \quad M = 200 \text{ g}$$

72. The number of nitrogen atoms in 681 g of $C_7H_5N_3O_6$ is $x \times 10^{21}$. The value of 'x' is-

($N_A = 6.02 \times 10^{23}$) [JEE Mains 2022]

Sol. Moles = $\frac{\text{mass}}{\text{molar mass}} = \frac{681}{(227 \text{ g/mol})} = 3$ mol

Number of nitrogen atoms = $3 \times 3 \times 6.02 \times 10^{23} = 5.418 \times 10^{24} = 5418 \times 10^{21}$

73. In a sample of $CuSO_4.5H_2O$ there are 10 moles of oxygen atom. Find out the moles of H-atoms in the sample.

(A) 10 mol (B) $\frac{10}{9}$ mol

(C) $\frac{100}{9}$ mol (D) $\frac{200}{9}$ mol

Sol. (C) Method 1

If 9 mol oxygen atom then sample would be 1 mole.

10 mol oxygen atom then sample would be $\frac{1}{9} \times 10$ mol

moles of sample = $\frac{10}{9}$ mol

moles of H-atom = moles of sample × $(\text{atomicity})_H$

$= \frac{10}{9} \times 10 = \frac{100}{9}$ mol

Method 2

$$\text{moles of sample} = \frac{\text{moles of atom}}{\text{atomicity of that atom}}$$

moles of sample = $\frac{10}{9}$

moles of H – atom = $\frac{10}{9} \times 10 = \frac{100}{9}$ mol

Method 3

9 mole of oxygen → 10 mole of hydrogen

1 mole oxygen → $\frac{10}{9}$ mole of hydrogen

10 mole oxygen →

$\frac{10}{9} \times 10 = \frac{100}{9}$ mol hydrogen atom

sare methods aane chaiye

74. In a sample of $K_2SO_4.Al_2(SO_4)_3.24H_2O$ we have 96 moles of H-atoms. If all the oxygen atoms of this sample are some how converted to Ozone gas, then find out the volume of ozone formed at STP.

Sol. In $K_2SO_4.Al_2(SO_4)_3.24H_2O$ that is potash alum, 48 moles of H and 40 moles of O are present.

48 mol H → 40 mol O

96 mol H → 80 mol O

1 molecule of O_3 is formed by 3 atoms of oxygen.

moles of O_3 formed by 80 mol of O-atoms

$= \frac{80}{3}$ mol

Volume of O_3 at STP $= \frac{80}{3} \times 22.4$ Lt $= 597.33$ Lt

75. A polystyrene of formula $Br_3C_6H_2(C_8H_8)_n$ was prepared by heating styrene with tribromobenzyl peroxide in the absence of air. It was found to contain 10.46% bromine by mass. The value of n is (Br = 80)

(A) 20 (B) 21 (C) 19 (D) 22

Sol. (C) $\frac{\text{wt. of Bromine}}{\text{Total weight}} \times 100 = 10.46$

Let we have one mole polystyrene

Moles of Br = 1 × (atomicity)$_{Br}$ = 1 × 3 = 3 mol

$10.46 = \%Br = \frac{3 \times 80}{M} \times 100$

$M = 314 + 104\,n$, $n = 19$

76. A compound contains 28% nitrogen and 72% metal by mass. 3 atoms of the metal combine with two atoms of nitrogen. The atomic mass of metal.

(A) 36 (B) 20 (C) 24 (D) 40

Sol. (C) According to the given formula, the formula of metal and nitrogen is M_3N_2.

$\%\text{ metal} = \frac{\text{wt. of metal}}{\text{mol wt.}} \times 100$

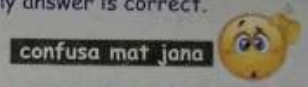
$72 = \frac{3x}{(3x+28)} \times 100$, $x = 24$

77. Find out the wt loss on heating Glauber salt ($Na_2SO_4.10H_2O$) to get anhydrous salt.

(A) 180 g (B) 90 g (C) 45 g (D) 20 g

Sol. (A,B,C,D) $Na_2SO_4.10H_2O \xrightarrow{\Delta} Na_2SO_4 + 10H_2O$

Since we are not provided with the amount of salt so any answer is correct.



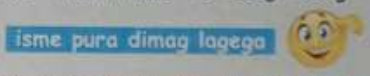
78. Find out the weight loss on heating 0.5 mol of $Na_2SO_4.10H_2O$

(A) 180 g (B) 45 g (C) 90 g (D) 20 g

Sol. (C) 1 mole $Na_2SO_4.10H_2O$ loss → 10 mole of H_2O on heating

0.5 mole $Na_2SO_4.10H_2O$ loss → 5 mole of H_2O on heating

Wt. = mol × molar mass = 5 × 18 g = 90 g



79. The hydrated salt $Na_2SO_4.10H_2O$ undergoes X% wt loss on heating and becomes anhydrous. Value of X is...

(A) 10 (B) 45 (C) 56 (D) 70

Sol. (C) Assuming we have 1 mole Glauber salt

$Na_2SO_4.10H_2O \longrightarrow Na_2SO_4 + 10H_2O$

$\%\text{ loss} = \frac{\text{wt. of } H_2O}{\text{Molar mass}} \times 100$

$= \frac{180}{(142+180)} \times 100 = 56\%$

Empirical Formula and Molecular Formula

Molecular formula: The formula which gives the exact number of atoms of elements present in one molecule of the substance.

Empirical formula: The formula which gives the simplest ratio of atoms of different elements present in one molecule of the substance.

Note

When percentage of atoms are given, then we will calculate the moles of atoms present in a molecule, then will determine the simplest ratio by dividing the smallest value of moles with all the calculated values of moles of other atoms involved.

Example: $C_6H_6 \Rightarrow$ benzene

Molecular formula = C_6H_6

1 molecule of benzene has 6 atoms of C and 6 atoms of H

Simplest formula = (CH) = Empirical formula

$$(\overset{13}{C}H) \times 6 = \overset{=78}{C_6H_6}$$

(Empirical formula) × n = Molecular formula

$$n = \frac{\text{Molecular wt.}}{\text{Empirical formula wt.}}$$

80. An organic compound has 42.1% carbon, 6.4% hydrogen and remainder is oxygen. If molecular weight is 342, then the molecular formula is. **[JEE Mains 2024]**

(A) $C_{11}H_{18}O_2$ (B) $C_{12}H_{20}O_{12}$

(C) $C_{14}H_{20}O_{10}$ (D) $C_{12}H_{22}O_{11}$

Sol. (D)

Element	Symbol	%	At. wt	Mole	Mole ratio	Simple whole no. ratio
Carbon	C	42.1	12	3.5	3.5/3.2 = 1.09	1.09 × 11 = 12
Hydrogen	H	6.4	1	6.4	6.4/3.2 = 2	2 × 11 = 22
Oxygen	O	51.5	16	3.2	3.2/3.2 = 1	1 × 11 = 11

So, empirical formula is $C_{12}H_{22}O_{11}$

Since, the empirical formula weight is equal to molecular weight, as a result, molecular formula is $C_{12}H_{22}O_{11}$.

Tabular form mei acha lagta hai sawaal karna

81. Quantitative analysis of an organic compound (X) shows following % composition.

C: 14.5% Cl: 64.46% H: 1.8%

(Empirical formula mass of the compound (X) is _____ $\times 10^{-1}$

(Given molar mass in g mol^{-1} of C : 12, H : 1, O : 16, Cl : 35.5) **[JEE Mains Shift-I, 2025]**

Sol. [1655]

Element	%	Moles	Ratio	Simplest Ratio
C	14.5	1.2 mol	1	2
H	1.8	1.8 mol	1.5	3
Cl	64.46	1.8 mol	1.5	3
O	19.24	1.2 mol	1	2

∴ Empirical formula is $C_2H_3O_2Cl_3$

Molar mass = 165.5 g mol^{-1} = 1655×10^{-1} g mol^{-1}

∴ Answer is 1655.

ekdum asaan sawaal hai, karne mei maza aaya hoga

82. Compound A contains 8.7% hydrogen, 74% carbon, 17.3% nitrogen. The molecular weight of compound is 162 then the molecular formula of the compound is- **[JEE Mains 2022]**

(A) $C_4H_6N_2$ (B) C_2H_3N

(C) C_5H_7N (D) $C_{10}H_{14}N_2$

Sol. (D)

Element	Symbol	%	At. wt	Moles	Mole ratio
Hydrogen	H	8.7	1	8.7	$\frac{8.7}{1.24} = 7$
Carbon	C	74	12	6.2	$\frac{6.2}{1.24} = 5$
Nitrogen	N	17.3	14	1.24	$\frac{1.24}{1.24} = 1$

Empirical formula = C_5H_7N

Molecular formula = n × Empirical formula

$n = \frac{162}{81} = 2$, M.F = 2 × E.F. = $C_{10}H_{14}N_2$

83. Number of hydrogen atoms per molecule of hydrocarbon 'A' having 85.8% carbon is- (Molar mass of A is 84 g/mol). **[JEE Mains 2023]**

Sol. Mass of C = 85.8 g

Mass of H = (100 – 85.8) = 14.2 g

Element	Symbol	%	At. wt	Mole ratio	Simple ratio
Carbon	C	85.8	12	7.15	$\frac{7.15}{7.15} = 1$
Hydrogen	H	14.2	1	14.2	$\frac{14.2}{7.15} = 2$

$$n = \frac{\text{Molecular mass}}{\text{Emp. formula mass}} = \frac{84}{14} = 6$$

M.F = n × E.F = $1(CH_2) = C_6H_{12}$

No. of H-atoms per molecular of hydrocarbon = 12

isme dimag lagega

AVERAGE ATOMIC MASS

The average atomic mass of an element is the weighted average of the atomic masses of all its naturally occurring isotopes, based on their relative abundances.

$$\text{avg at. wt} = \frac{\%\text{ of isotope 1} \times \text{at wt. of isotope 1} + \%\text{ of isotope 2} \times \text{of wt. of isotope 2}}{100}$$

84. Naturally occurring chlorine is 75% Cl^{35} which has an atomic mass of 35 amu and 25% Cl^{37} which has a mass of 37 amu. Calculate the average atomic mass of chlorine.

Sol. Avg. atomic. wt = $\frac{75 \times 35 + 25 \times 37}{100}$ = 35.5u

Seedha formula lagana hai bas

85. If the avg. atomic wt. of Cl = 35.5 u then find out the % existence or % abundance of two isotopes of Cl. (Cl^{35} and Cl^{37})

Sol. (75%) For explanation, refer to QR code given on the last page.

Note

Statement 1: If the % abundances of different isotopes of an element are given, then we can find out average at. weight. (Ⓣ/F)

Statement 2: If the average at. wt. is given, then we can always find out the % abundances of different isotope of that element. (T/Ⓕ)

Tum lallu ho! This is true only if we have two isotopes of an element.

86. Given that the abundance of isotopes ^{54}Fe, ^{56}Fe and ^{57}Fe is 5%, 90% and 5%. The atomic mass of Fe is [IIT JEE]

(A) 55.85 (B) 55.95

(C) 55.75 (D) 55.05

Sol. (B) Avg at wt

$$= \frac{(5 \times 54) + (90 \times 56) + (5 \times 57)}{100} = 55.95$$

bas formula lagana hai

Common Mistake

Avg. at. wt. = (54 + 56 + 57)/3

This is possible only when all the isotopes are in same percentage.

87. The average molar mass of chlorine is 35.5 g mol^{-1}. The ratio of ^{35}Cl to ^{37}Cl in naturally occurring chlorine is close to: [6 Sept, 2020, shift II]

Sol.

	^{35}Cl	^{37}Cl
Molar ratio be	x	1 - x

Average molar mass = $\frac{n_1M_1 + n_2M_2}{n_1 + n_2}$

$2x = 1.5$, $x = 0.75$; $1 - x = 0.25$

So, ratio of ^{35}Cl: ^{37}Cl = $\frac{0.75}{0.25} = 3:1$

direct formula based hai, values uthai or lagadi, kaam ban gaya apna

AVERAGE MOLECULAR WEIGHT

Average molecular weight is a concept used to represent the weighted average of the molecular weights of different molecules in a mixture.

Suppose there are three species A, B, and C having n_A, n_B and n_C moles respectively.

Let M_A, M_B and M_C be their molecular weights respectively.

$$\text{Moles} = \frac{\text{mass}}{\text{mol.wt.}}, \quad \text{Mol wt} = \frac{\text{mass}}{\text{moles}}$$

$$\text{Avg molecular wt} = \frac{\text{Total weight}}{\text{Total moles}} = \frac{n_A M_A + n_B M_B + n_C M_C}{n_A + n_B + n_C}$$

$$\text{Avg mol. wt} = \frac{M_A + M_B + M_C}{3} \text{ (Only when } n_A = n_B = n_C)$$

88. 2 moles of H_2 and 3 moles of He are present in the container. Find out the average mol. wt.

Sol. (3.2) For explanation, refer to QR code given on the last page.

89. Find out the average molecular weight of gaseous mixture which contains 2 moles O_2, 3 moles N_2 and 5 moles of He.

Sol. Average molecular weight

$$= \frac{(\text{moles} \times \text{molar mass of } O_2) + (\text{moles} \times \text{molar mass of } N_2) + (\text{moles} \times \text{molar mass of He})}{(\text{moles of } O_2 + N_2 + H_e)}$$

$$= \frac{(2 \times 32) + (3 \times 28) + (5 \times 4)}{2 + 3 + 5} = 16.8 \text{ g/mol}$$

ghabra kyu rahe ho paglo, mushkil nahi hai, samjho

90. A gaseous mixture contains 2 g of H_2 and 32 g of O_2. Calculate the average molecular weight of mixture.

(Molar mass of H_2 = 2 g/mol, O_2 = 32 g/mol)

Sol. (17 g/mol) For explanation, refer to QR code given on the last page.

91. Air contains 78% N_2, 21% O_2, and 1% Ar by volume. Calculate the average molecular weight.

(Molar masses: N_2 = 28 g/mol, O_2 = 32 g/mol, Ar = 40)

Sol. $M_{avg} = (0.78 \times 28) + (0.21 \times 32) + (0.01 \times 40)$

$= 21.84 + 6.72 + 0.4 = 28.96$ g/mol.

ABSOLUTE DENSITY AND RELATIVE DENSITY

Absolute Density: Absolute density, commonly referred to simply as density, is the mass of a substance per unit volume. It is an intrinsic property of a material and is expressed using the formula:

$$\text{Density}(\rho) = \frac{\text{Mass}(m)}{\text{Volume}(V)}$$

Unit: Commonly in kg/m^3 or g/cm^3.

Note

For an ideal gas:

$$PV = nRT,\ PV = \frac{W}{M}RT,\ \frac{W}{V} = \frac{PM}{RT} = d_{gas}$$

Relative density: Relative density (also known as specific gravity) is the ratio of the density of a substance to the density of a reference substance, usually water (for liquids and solids) or air (for gases).

$$\text{Relative density} = \frac{\text{Density of the substance}}{\text{Density of the reference substance}}$$

formula toh yaad karna hi padega, chhod kar kaam nahi chalega

Unitless: As it is a ratio, it has no units.

For liquids and solids: Water (density = 1 g/cm^3) is typically used as the reference substance.

For gases: Air or hydrogen is used as the reference.

Some important terms and formulas related to density are:

(i) **Specific gravity:**

$$\text{Specific gravity} = \frac{\text{Density of the substance}}{\text{Density of water at } 4^\circ C}$$

Unit: It is a dimensionless quantity since it is a ratio

Note

Reference substance: Water is used as the reference substance for most practical purposes, with its density taken as 1 g/cm^3 at 4°C.

92. If a density of a solution is 1.2 g/mL, find out its sp. gravity.

Sol. Sp. gr

$$= \frac{\text{Density of the solution}}{\text{Density of } H_2O \text{ at } 4^\circ C} = \frac{1.2 \text{ g/ml.}}{1.0 \text{ g/ml.}} = 1.2$$

93. If the sp gravity of a solution is 1.15 then find out the total number of nucleons present in its 50 ml solution. [Expected]

Sol. Mass of 50 ml solution = density × volume

$$= 1.15 \frac{\text{gm}}{\text{ml}} \times 50 \text{ml} = 57.50 \text{ gm}$$

Total no. of nucleous = 57.50 N_A

thoda lagega tedha lekin hai nahi

(ii) Density of gas 1 wrt gas 2

$$= \frac{\frac{P_1M_1}{RT_1}}{\frac{P_2M_2}{RT_2}} = \frac{P_1M_1T_2}{P_2M_2T_1} = \frac{M_1}{M_2}$$

[When P, T conditions are same $P_1 = P_2$, $T_1 = T_2$]

P_1 = Pressure of gas 1, P_2 = Pressure of gas 2

M_1 = Pressure of gas 1, M_2 = mol. wt of gas 2

T_1 = Temp of gas 1, T_2 = Temp of gas 2

(iii) Vapour density

$$\text{Vapor density} = \frac{\text{Density of gas or vapor}}{\text{Density of } H_2 \text{ gas (under similar condition)}}$$

$$\text{Vapour density} = \frac{M}{M_{H_2}} = \frac{M}{2}$$

Khoobsurat Baat

Always calculate the molecular weight first to calculate the vapour density.

94. Find out the density of SO_3 wrt CH_4.

(A) 4 (B) 3 (C) 5 (D) 1

Sol. (C) For explanation, refer to QR code given on the last page.

95. The density of air at STP is 0.001293 g ml^{-1}. Its vapour density is:

(A) 29.0 (B) 32.0 (C) 14.5 (D) 16.0

Sol. (C)

wt of 22400 ml air = $\underbrace{0.001293 \times 22400 \text{ g}}_{\text{Molar mass}}$

$$V.D = \frac{0.001293 \times 22400}{2} = 14.5$$

dekh kar ho gyi hogi balle balle

96. If a one mole of a sample of a substance contains 2.4×10^{25} nucleons, then find out its V.D. ($N_A = 6 \times 10^{23}$)

(A) 20 (B) 40 (C) 1.0 (D) 30

Sol. (A) **Method-1:** gm हटाओ, N_A लगाओ, nucleons ले जाओ।

No. of nucleons in one mole of sample

$= 2.4 \times 10^{25} = 40\ N_A$

mass of one mole sample = 40 g (mol. wt)

$$V.D = \frac{Mwt}{2} = \frac{40}{2} = 20$$

Method-2: wt of one nucleon = 1.66×10^{-24} g

wt of 2.4×10^{15} nucleons

$$= 1.66 \times 10^{-24} \times 2.4 \times 10^{25} \text{ g} = \underbrace{40 \text{ g}}_{1 \text{ mole}}$$

$$V.D = \frac{Mwt}{2} = \frac{40}{2} = 20$$

Dono methods mazedaar hain

97. 0.05 cm thick coating of silver is deposited on a plate of 0.05 m^2 area. The _____ number of silver atoms deposited on plate are $\times 10^{23}$.

(Ag = 108, d = 7.9 g cm^{-3}) **[JEE Mains 2024]**

Sol. (11×10^{23}) For explanation, refer to QR code given on the last page.

STOICHIOMETRY

It is a quantitative analysis of a chemical reaction.

Lets understand this by taking an example:

$$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2(g)}$$

On heating $CaCO_{3(s)}$, we get CO_2 gas.

—Qualitative analysis

On heating certain amount of $CaCO_{3(s)}$, how much gas is formed?

—Quantitative analysis

Important

Steps involved in stoichiometric calculations

- Write down the chemical reaction.
- Balance that chemical reaction.
- Read that chemical reaction
- Now see the data given in the question.

98. Find the moles of CO_2 formed and O_2 required for complete combustion of 3 moles of CH_4.

Sol. Step 1: Write the chemical equation

$CH_4 + O_2 \rightarrow CO_2 + H_2O$

Step 2: Balance it

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Step 3: Read the chemical reaction

1 mole of CH_4 reacts with 2 moles of O_2 to give 1 mole CO_2 and 2 moles H_2O

Step 4: See the data given in the question

1 mole CH_4 gives 1 mole of CO_2

3 moles CH_4 gives 3 moles of CO_2

steps ka sequence pta hona chaiye

99. 27g of Al will react completely with how many grams of oxygen?

(A) 8g (B) 16g (C) 32g (D) 24g

Sol. (D) For explanation, refer to QR code given on the last page.

100. Xg of benzoic acid on reaction with aq $NaHCO_3$ released CO_2 that occupied 11.2 L volume at STP. X is_____ g.

[JEE Mains Shift-I, 2025]

Sol. $C_6H_5COOH + NaHCO_3 \rightarrow C_6H_5COONa + H_2O + CO_2$

At STP, 1 mole of CO_2 occupies 22.4 L.

The volume of CO_2 released = 11.2 L at STP

$$\text{Moles of } CO_2 = \frac{11.2}{22.4} = 0.5 \text{ moles}$$

Molar mass of benzoic acid $(C_6H_5COOH) = 122\ g/mol$

Mass of benzoic acid reacted:

Mass = Moles × Molar mass = 0.5 × 122 = 61 g

101. Complete combustion of 3g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is _____ (round off to the nearest integer)

[Use: $N_A = 6.023 \times 10^{23}$; Atomic masses in u: C 12.0, O : 16.0; H : 1.0]

[JEE Mains 17 Sept, 2021, Shift-I]

Sol. (18) For explanation, refer to QR code given on the last page.

102. The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is _____

[JEE Mains 05 Sept, 2020, Shift-I]

Sol. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

[O_2 consumed by 1 mole C_3H_8 = 5 moles]

$$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$$

[O_2 consumed by 2 mole C_4H_{10} = 13 moles]

Total consumption = 5 + 13 = 18 mole O_2

103. 1g of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01 mol of CO_2. The molar mass of M_2CO_3 is _____ g mol^{-1} (Nearest integer)

[JEE Mains 13 April, 2023 Online Shift-II]

Sol. $M_2CO_3 + 2HCl \rightarrow 2MCl + H_2O + CO_2$

1 mol CO_2 is formed by 1 mol M_2CO_3

0.01 mole CO_2 is formed by 0.01 mole M_2CO_3

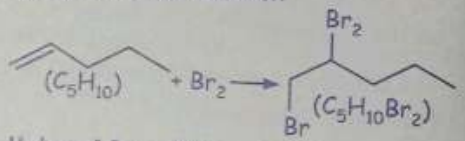
Wt of 0.01 mole M_2CO_3 is 1 gm, then wt of 1 mole $M_2CO_3 = \frac{1}{0.01} = 100g$

reaction ko balance mat karna bhul jana, samajh gaye

104. Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) (Nearest integer)

[JEE Mains 25 June, 2022, Shift-I]

Sol. First write the reaction



Moles of Br_2 = Moles of C_5H_{10}

$$W = \frac{5 \times 160}{70} g = 11.428g = 1143 \times 10^{-2} g$$

105. The volume of hydrogen liberated at STP by treating 2.4 g of magnesium with excess of hydrochloric acid is _____ $\times 10^{-2}$ L.

[JEE Mains 11 April, 2023 Online Shift-II]

Sol. The reaction between Mg and HCl is:

$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$

$$\text{Moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{2.4g}{(24\ g/mol)} = 0.1 \text{ mol}$$

1 mol of gas at STP occupies 22.4 L volume

0.1 mol of gas occupies = 22.4 × 0.1 = 2.24 L

$= 224 \times 10^{-2}$ L

106. $NaClO_3$ is used, even in spacecraft, to produce O_2. The daily consumption of pure O_2 by a person is 492 L at 1 atm, 300 K. How much amount of $NaClO_3$, in grams is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K?

$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$

$R = 0.082\ L\ atm\ mol^{-1}\ k^{-1}$

[JEE Mains 8 Jan, 2020, Shift-II]

Sol. Acc. to the given reaction, moles of $NaClO_3$ = moles of O_2

$$\text{Moles of } O_2 = \frac{1 \times 492}{0.082 \times 300} = 20 \text{ mole} \quad \left(\because n = \frac{PV}{RT}\right)$$

Mass of $NaClO_3 = 20 \times 106.5 = 2130$ g.

sawaal dekh kar dar gaye na

107. Dissolving 1.24 g of white phosphorus in boiling NaOH solution in an inert atmosphere gives gas Q. The amount of $CuSO_4$ (in g) required to completely consume the gas Q is _____.

[JEE Advance, 2022]

[Given: Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63]

Sol. (2.385 g) For explanation, refer to QR code given on the last page.

LIMITING REAGENT

The limiting reagent (or limiting reactant) is the reactant in a chemical reaction that is completely consumed first, thereby limiting the amount of product that can be formed.

Important

How to calculate limiting reagent

$$\left(\frac{\text{Mole}}{\text{Stoichiometric coefficient}}\right) = \text{Limiting reagent (L.R)}$$

Substance for which value comes out to be the smallest that will act as a limiting reagent.

108. $2A + 3B \rightarrow 5C$

Find out the weight of 'C' formed (mol. wt = 60 g/mol) during the reaction of 20 moles of A and 20 moles of B. Also find the moles of excess reagent left.

Sol. A: $\frac{\text{Moles}}{S.C} = \frac{20}{2} = 10$ mol

B: $\frac{\text{Moles}}{S.C} = \frac{20}{3} = 6.6$ mol

Limiting reagent is 'B'

Excess reagent is 'A'

Analysis of C (Product)

3 mol B $\rightarrow$ 5 mol C

20 mol B $\rightarrow \frac{5}{3} \times 20 = \frac{100}{3}$ mol $C = \frac{100}{3} \times 60$

= 2000 g

Analysis of A (Excess Reagent)

3 mol B consumed 2 mol A

20 mol B consumed $= \frac{2}{3} \times 20 = \frac{40}{3}$ mol A

Left amount of $A = A_{initial} - A_{consumed}$

$= 20 - \frac{40}{3} = \frac{20}{3}$ mole

bas sawaal ki language ko samajhna hai

109. SO_2Cl_2 on reaction with excess of water results into acidic mixture

[JEE Mains 25 July, 2022 Shift-I]

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

16 moles of NaOH is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is:

(A) 16 (B) 8 (C) 4 (D) 2

Sol. (C) Let 'x' mole of SO_2Cl_2 be used

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

(x mol) (2x mol)

$HCl + NaOH \rightarrow NaCl + H_2O$

(2x mol) (2x mol)

Total consumption of NaOH = 2x + 2x = 4x mol

$4x = 16 \Rightarrow x = 4$ mol

110. When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is _____ (Nearest integer)

[JEE Mains Shift-II, 2025]

Molar mass of Al is 27.0 g mol^{-1}

Molar mass of O is 16.0 g mol^{-1}

Molar atomic mass (g mol^{-1}): O = 16
Molar atomic mass (g mol^{-1}): C = 12]

[JEE mains, 29 June 2022, shift-I]

(A) 1400 (B) 2200 (C) 3360 (D) 4200

Sol. (C) Moles of $Fe_3O_4 = \frac{4.64 \times 10^3}{232} = 20$

Moles of $CO = \frac{2.52 \times 10^3}{28} = 90$

According to the reaction:

4 moles of CO reacts with 1 mol of Fe_3O_4

$\Rightarrow$ 90 moles of CO reacts with 22.5 mole of Fe_3O_4

Thus, limiting reagent = Fe_3O_4

Thus, moles of Fe formed = 60

Weight of Fe = 60 × 56 = 3360 g

isme acha khasa lagega dimag

SOME IMPORTANT CHEMICAL REACTIONS

Combination of Elements with Oxygen

(i) Heating mercury at its boiling point in air.

$Hg + O_2 \rightarrow 2HgO$ (red mercuric oxide)

(ii) Heating magnesium in air forms mostly magnesium oxide and some magnesium nitride.

$2Mg + O_2 \rightarrow 2MgO$

$3Mg + N_2 \rightarrow Mg_3N_2$

(iii) Calcium behaves similarly,

$2Ca + O_2 \rightarrow 2CaO$

$3Ca + N_2 \rightarrow Ca_3N_2$

(iv) Silver does not combine with oxygen, as Ag_2O is unstable to heat.

(v) Many non-metals burn in O_2 forming their respective oxides.

$2H_2 + O_2 \rightarrow 2H_2O$; $S + O_2 \rightarrow SO_2$

$C + O_2 \rightarrow CO_2$; $P_4 + 5O_2 \rightarrow P_4O_{10}$

$N_2 + O_2 \xrightarrow{3000°C} 2NO$

Cl_2, Br_2 and I_2 do not directly combine with oxygen.

Action of Heat on Certain Oxides

(i) Mercuric oxide and silver oxide are unstable to heat decompose readily.

$2HgO \xrightarrow{\Delta} 2Hg + O_2$; $2Ag_2O \xrightarrow{\Delta} 4Ag + O_2$

(ii) Various higher oxides, dioxides, mixed and peroxides are decomposed to oxygen and a lower oxide.

$3Pb_3O_4 \rightarrow 6PbO + O_2$; $2PbO_2 \rightarrow 2PbO + O_2$

$3MnO_2 \rightarrow Mn_3O_4 + O_2$; $2BaO_2 \rightarrow 2Ba + 2O_2$

Compounds Rich in Oxygen Decompose to give Oxygen

(i) $2KNO_3 \rightarrow 2KNO_2 + O_2\uparrow$

(ii) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2\uparrow$
(purple) (green) (black)

(iii) $4K_2Cr_2O_7 \rightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2\uparrow$
(orange-red) (yellow) (green)

Potassium chlorate when heated just above its melting point decomposes into potassium perchlorate and potassium chloride. This reaction is called disproportionation or auto-oxidation and auto-reduction. On heating further, $KClO_4$ decomposes to KCl and oxygen.

(iv) $4KClO_3 \rightarrow 3KClO_4 + KCl$

$KClO_4 \rightarrow KCl + 2O_2$

yeh karna pdega tabhi sawaal honge reactions wale

Action of Heat on Nitrates

Generally heavy metal nitrates decompose to metal oxide, reddish brown nitrogen dioxide gas and oxygen.

(i) Lead nitrate decomposes to PbO, NO_2 & O_2.

$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2\uparrow + O_2$

litharge or lead (II) oxide

(red when hot and yellow when cold)

(ii) Cupric nitrate decomposes to CuO, NO_2 & O_2.

$2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2\uparrow + O_2$
(green) (black)

(iii) Zinc nitrate decomposes to ZnO, NO_2 & O_2.

$2Zn(NO_3)_2 \rightarrow 2ZnO + 4NO_2\uparrow + O_2$

(zinc oxide, yellow when hot and white when cold)

knowing the colours of some oxides will be useful in qualitative analysis.

(iv) Nitrate of mercury and silver, whose oxides are unstable, decompose into the metal, NO_2 & O_2.

$Hg(NO_3)_2 \rightarrow Hg + 2NO_2 + O_2$

$2AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$

(v) Alkali metal nitrates decompose to give the metal nitrite and O_2 (no reddish brown NO_2 gas).

$2KNO_3 \rightarrow 2KNO_2 + O_2$

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

(vi) Ammonium nitrate on heating leaves no residue and forms nitrous oxide and steam.

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

yaad karna padega or koi chaara nahi hai

Action of Heat on Ammonium Compounds

Generally an ammonium compound decomposes into ammonia and an acid or acidic oxide if the acid is unstable to heat.

(i) $NH_4Cl \rightarrow NH_3 + HCl$

(ii) $(NH_4)_2SO_4 \rightarrow 2NH_3 + H_2SO_4$

(iii) $(NH_4)_3PO_4 \rightarrow 3NH_3 + H_3PO_4$

(iv) $(NH_4)_2CO_3 \rightarrow 2NH_3 + CO_2 + H_2O$

(v) Ammonium compounds which do not give ammonia on heating are ammonium nitrate, ammonium dichromate.

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

$NH_4NO_2 \rightarrow N_2 + 2H_2O$

$(NH_4)_2Cr_2O_7 \rightarrow N_2 + 4H_2O + Cr_2O_3$

Action of Heat on Metallic Carbonates

(i) Generally metallic carbonates decompose to give metal oxide and CO_2.

$CaCO_3 \xrightarrow{900°C} CaO + CO_2\uparrow$

$MgCO_3 \rightarrow MgO + CO_2\uparrow$

$CuCO_3 \rightarrow CuO + CO_2\uparrow$

(pale green) (black)

$PbCO_3 \rightarrow PbO + CO_2\uparrow$

(yellow)

$ZnCO_3 \rightarrow ZnO + CO_2\uparrow$

(ii) Carbonates of strongly electropositive metals (alkali metal except lithium) do not decompose on heating.

(iii) Silver carbonate decomposes to give the metal, CO_2 & O_2.

$2Ag_2CO_3 \rightarrow 2Ag + 2CO_2 + O_2$

(iv) Ammonium carbonate (smelling salt) decomposes to give NH_3, H_2O & CO_2. All the products are in gaseous phase and there is no residue left.

$(NH_4)_2CO_3 \rightarrow 2NH_3 + H_2O + CO_2$

Action of Heat on Metallic Bicarbonates

(i) Only $NaHCO_3$ and $KHCO_3$ are solids; others are known in solution. All bicarbonates decompose to give the metal carbonate, H_2O & CO_2.

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2\uparrow$

$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2\uparrow$

$Mg(HCO_3)_2 \rightarrow MgCO_3 + H_2O + CO_2\uparrow$

yaad karlo baalako

Action of Heat on Certain Hydrated Chlorides

Hydrated halides on heating are converted to oxides, H_2O and halo acids.

(i) $MgCl_2.6H_2O$ does not get completely dehydrated because $MgCl_2$ is hydrolysed by water to give basic $MgCl_2$.

$MgCl_2.6H_2O \rightarrow Mg(OH)Cl + 5H_2O + HCl$

(ii) $Al_2Cl_6.12H_2O \rightarrow Al_2O_3 + 6HCl + 9H_2O$

(iii) $SnCl_2.2H_2O$ undergoes hydrolysis to form basic chloride.

$SnCl_2.2H_2O \rightarrow Sn(OH)Cl + H_2O + HCl$

(iv) On heating, certain halides of metal ions in higher oxidation state changes to halides of lower oxidation state.

$2FeCl_3 \rightarrow 2FeCl_2 + Cl_2$

$2CuCl_2 \rightarrow Cu_2Cl_2 + Cl_2$

Action of Heat on Some Other Compounds

(i) When sodium sulphite is heated, it undergoes disproportionation reaction.

$4Na_2SO_3 \rightarrow 2Na_2SO_4 + Na_2S$

(ii) Sodium thiosulphate $Na_2S_2O_3.5H_2O$ loses water of hydration and becomes anhydrous which on further heating gives a mixture of sodium sulphate, sodium sulphide and Sulphur.

$Na_2S_2O_3.5H_2O \rightarrow Na_2S_2O_3 + 5H_2O$

$4Na_2S_2O_3 \rightarrow 3Na_2SO_4 + Na_2S + 4S$

(iii) When hydrated copper sulphate (blue vitriol) is heated, CuO & CO_2 formed.

$CuSO_4.5H_2O \xrightarrow{100°C} CuSO_4.H_2O + 4H_2O$

$CuSO_4.H_2O \xrightarrow{230°C} CuSO_4 + H_2O$

$CuSO_4 \xrightarrow{750°C} CuO + SO_3$

$2SO_3 \rightarrow 2SO_2 + O_2$

(iv) Gypsum $CaSO_4.2H_2O$, when heated to 120-130° C forms a hemihydrate called Plaster of Paris.

$$CaSO_4.2H_2O \rightarrow CaSO_4.\frac{1}{2}H_2O + \frac{3}{2}H_2O$$

If heated above 200°C, it forms anhydrous calcium sulphate which does not set with water.

(v) Green vitriol, $FeSO_4.7H_2O$, when heated forms Fe_2O_3, SO_2, SO_3 & H_2O.

$2FeSO_4.7H_2O \rightarrow Fe_2O_3 + SO_2 + SO_3 + 14H_2O$

(vi) $MgSO_4.7H_2O \xrightarrow{\Delta} MgSO_4 + 7H_2O$
(Epsom salt)

$MgSO_4 \xrightarrow{\Delta} MgO + SO_3$

(vii) $Al_2(SO_4)_3.18H_2O \xrightarrow{\Delta} Al_2O_3 + 3SO_3 + 18H_2O$
(purple) (white)

(viii) $Cr_2(SO_4)_3.18H_2O \xrightarrow{\Delta} Cr_2O_3 + 3SO_3 + 18H_2O$
(green)

(ix) $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$
(yellow) (blackish brown)

(x) $2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_3 + SO_2$
(light green) (blackish brown)

page par likh likh kar dekho

Some other Useful Reactions

$O_3 + 2KI + H_2O \rightarrow 2KOH + I_2 + O_2$

$BaCO_3 + HCl \rightarrow BaCl_2 + CO_2 + H_2O$

$BaCl_2 + H_2CrO_4 \rightarrow BaCrO_4 + 2HCl$

$2BaCrO_4 + 6KI + 8H_2SO_4 \rightarrow 3I_2 + 2BaSO_4 + 3H_2SO_4 + Cr_2(SO_4)_3 + 8H_2O$

$2CuSO_4 + 4KI \rightarrow Cu_2I_2 + I_2 + 2K_2SO_4$

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$

$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow 2MnSO_4 + 5Fe_2(SO_4)_3 + K_2SO_4 + 8H_2O$

$Mn_3O_4 + 2FeSO_4 + 4H_2SO_4 \rightarrow 3MnSO_4 + Fe_2(SO_4)_3 + 4H_2O$

$2NH_2OH + 4Fe^{3+} \rightarrow N_2O + H_2O + 4Fe^{2+} + 4H^+$

$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

$KMnO_4 + 5KI + 4H_2SO_4 \rightarrow 3K_2SO_4 + MnSO_4 + \frac{5}{2}I_2 + 4H_2O$

$K_2Cr_2O_7 + 6KI + 7H_2SO_4 \rightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 3I_2 + 7H_2O$

reactions bhut important hai
yaad karni pdengi

CONCENTRATION TERMS

There are some terms to define the concentration of solutions, named as concentration terms.

Solution A homogeneous mixture of two or more components is called solution

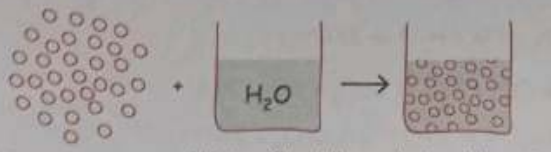
Example: Mixture of sugar in water.

Solution = solute (sugar) + solvent (water)

Important

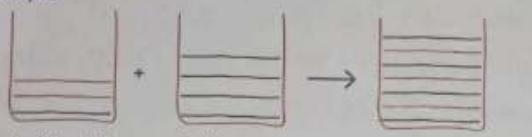
- Solvent decides the final phase (gas, liquid solid) of the solution.
- If both the components are in same phase then
  The component in lesser quantity = solute
  The component in larger quantity = solvent

Example:

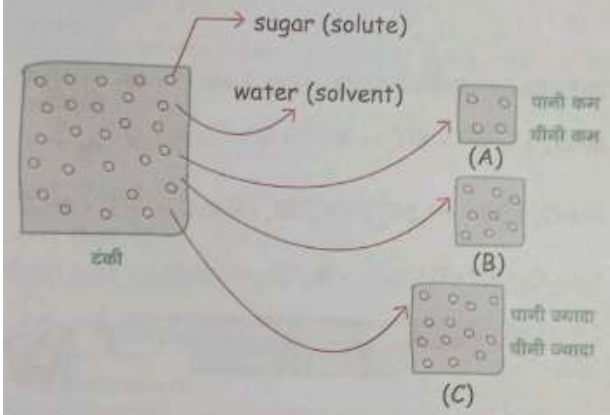


Sugar (solid) (solute) | Water (liquid) (solvent) | solution (liquid)

Example:



A (liquid) (solute) | B (liquid) (solvent) | Solution (liquid)



$$(\text{concentration terms})_A = (\text{concentration terms})_B = (\text{concentration terms})_C$$

aaya samajh?

Molarity

- Molarity (M) is defined as the number of moles of solute per litre of solution. The formula for molarity is:

$$M = \frac{\text{moles of solute}}{\text{volume of solution in liters}}$$

formulae par dhyan lagao

- Rearrange the above equation for moles of solute:
  Moles of solute = M × Volume of solution in L
  In millimoles:
  Milimoles of solute = Molarity × volume of solution in mL
- Molarity depends on temperature because it depends on volume and volume is temperature dependent.
- Suppose 'a' M aqueous solution of NaCl is given, then its molarity can be described as:

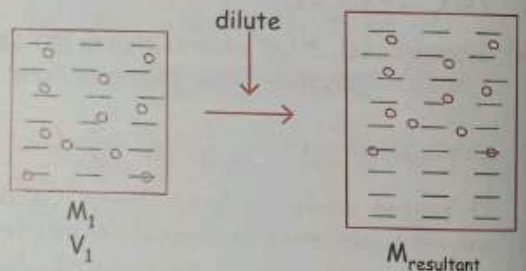
Ek line angrezi ki

a mole of NaCl is dissolved in 1 L or 1000 mL of solution.

1 lt = 1000 ml = 1000 cc = 1 dm^3 = 10^{-3} m^3

Molarity on Dilution

Molarity on Dilution refers to the change in concentration of a solution when more solvent is added. **Although the volume of the solution changes, the total moles of solute remain constant.**



The relationship for dilution is given by the formula:

Initial moles of solute = $M_1 \times V_1$

Final moles of solute = $M_{resultant} \times V_2$

$M_1V_1 = M_{resultant} \times V_2$

$$M_{resultant} = \frac{M_1V_1}{V_2}$$

This formula arises because the number of moles of solute before and after dilution remains the same.

Resultant Molarity during Mixing

When two or more solutions of the same solute are mixed, the **resultant molarity** can be calculated using the principle of conservation of moles of solute.

Note

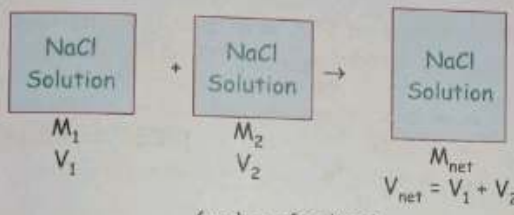
The formula for the resultant molarity (M_r) for two solutions being mixed:

$$M_r = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

Where:

- M_1, M_2: Molarities of the two solutions
- V_1, V_2: Volumes of the two solutions (in liters)
- M_r: Resultant molarity after mixing

Let two solutions of NaCl having molarities M_1 and M_2 are mixed,



So, for this, $M_{net} = \frac{(\text{moles of solute})_{net}}{(\text{vol of solution})_{net}}$

$$M_{net} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

samajh aaya formula, confusa toh nahi gaye

$(\text{no. of moles of solute})_{net} = M_1V_1 + M_2V_2$

$V_{net} = V_1 + V_2$

128. Find out the resultant molarity of solution if we add 100 ml of 0.1 M HCl with 200 ml of 0.05 M HCl and 500 ml of water.

Sol. $M_{net} = \frac{\text{Resultant moles of solute}}{\text{Resultant volume of solution}}$

$$= \frac{(100 \times 0.1) + (200 \times 0.05)}{(100 + 200 + 500)} = \frac{20}{800} = \frac{1}{40} M$$

129. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is:

[JEE Main-2019]

(A) 4 g. (B) 2 g (C) 8 g (D) 10 g

Sol. (A) $H_2C_2O_4 + 2NaOH \rightarrow Na_2C_2O_4 + 2H_2O$

Moles of $H_2C_2O_4 = \frac{0.5 \times 50}{1000} = 0.025$ moles

Moles of NaOH consumed

$= 0.025 \times 2 = 0.05$ moles

Wt of NaOH in 25 ml of its solution

$= 0.05 \times 40 = 2g$

50 ml of its solution $= 2 \times 2 = 4g$

bhayi asaan sa hai yeh

130. When 200 ml of 0.2 M acetic acid is shaken with 0.6 g of wood charcoal, the final concentration of acetic after adsorption is 0.1 M. The mass of acetic acid adsorbed per gram of carbon is _____ g.

[JEE Main 24 June 2022 Shift-2]

Sol. Mass of acetic acid adsorbed

$$= \frac{M_1V_1 - M_2V_2}{1000} \times 60$$

$$= \frac{0.2 \times 200 - 0.1 \times 200}{1000} \times 60 = 1.2\ g$$

Now, mass of acetic acid adsorbed per gram of $C = 1.2/0.6 = 2$

131. 1 L aqueous solution of H_2SO_4 contains 0.02 mmol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 mmol of H_2SO_4 are added. Total mmols of H_2SO_4 in the final solution is _____ $\times 10^3$ m moles.

[25 June, 2022 (Shift-I)]

Sol. [0] 50% of 0.02mmol in 1 L of H_2SO_4

$= 0.01$ mmol

Total moles of H_2SO_4 after adding 0.01 mol of H_2SO_4

$= 0.01 + 0.01 = 0.02$ mmol $= 0.00002 \times 10^3$ mmol

132. The $NaNO_3$ weighed out to make 50 mL of an aqueous solution containing 70.0 mg Na^+ per mL is_____g. (Rounded off to the nearest integer) **[26 Feb, 2021 (Shift-II)]**

Sol. [13] Mass of Na^+ in 50 mL = 70×50 = 3500 mg

23000 mg of Na^+ = 85000 mg of $NaNO_3$

(1 mole $NaNO_3$ contains 1 mole Na^+)

$\therefore$ 3500 mg Na^+ will be present in $\frac{85000}{23000} \times 3500$

$= 12934.78$ mg $= 12.93478$ gm $\cong 13$

133. Find the molarity of water. Given: $\rho = 1000\ kg/m^3$ **[IIT JEE-2003]**

Sol. 1 litre of water = 1 kg.

(Since, Density is 1000 kg/m^3)

Thus, 1000ml of water = 1 kg

(Since, 1 litre = 1000 ml)

Now, Mass of Water = 1 kg = 1000 g

$$\text{No. of moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{1000}{18} = 55.56 \text{ moles}$$

$$\text{Molarity} = \frac{55.56}{1000} \times 1000 = 55.56\ M$$

baaye hath ka khel hai isko karna

Molality

Molality (molal or m):

- This term is also used to define the concentration of a solution.
- It is defined as the no. of moles of solute dissolved in 1 kg or 1000 gm solvent

$$\text{Molality} = \frac{\text{moles of solute}}{\text{wt of solvent in kg}}$$

- Moles of solute = molality × wt. of solvent in kg
- Molality is temperature independent
- 1.2 molal(m) aq urea solution means

↓ Ek line angrezi ki

1.2 moles of urea dissolved in 1000 gm solvent (H_2O)

moles of solute ⟷ moles of solvent

134. Density of 3 M NaCl solution is 1.25 g/mL. The molality of the solution is: **[JEE Mains Shift-II, 2025]**

(A) 2 m (B) 2.79 m

(C) 1.79 m (D) 3 m

Sol. (B) $\text{Molality} = \frac{M \times 1000}{1000\,d - M \times M_w}$

$$= \frac{3000}{1250 - 175.5} = 2.79\ m$$

135. It is given that the molarity of H_2SO_4 solution is 18 M. Find out molality. (Density=1.8 g/mL)

Sol. 18 moles of H_2SO_4 dissolved in 1000 ml solution

wt of solution = 1000 ml × 1.8 g/ml = 1800 gm

wt of solute = 18 × 98 = 1764 gm

wt of solvent = wt of solution − wt of solute

= 1800 − 1764 = 36 g

$$\text{Molality} = \frac{\text{moles of solute}}{\text{wt of solvent in kg}} = \frac{18}{\frac{36}{1000}} = 500\ m$$

dimag ki jali batti

136. A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3}. The molarity of the solution is _____ mol dm^{-3}. (Round off to the nearest integer).

Sol. (9) For explanation, refer to QR code given on the last page.

137. A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g mL^{-1}. Assuming no change in volume upon dissolution, the molality of a 3.2 molar solution is **[JEE Adv.-2014]**

Sol. $M = 3.2 = \frac{n_B}{V_{solution}}$

$W_{Solvent} = 0.4 \times 1000 = 400g = W_A$

3.2 moles of H_2X in 1000 ml of Solution

$$\text{Molality} = \frac{n_B}{W_A} \times 1000 = \frac{3.2 \times 1000}{400} = 8\ m$$

138. Calculate the molality of 1.0 L solution of 93% H_2SO_4 (weight/volume). The density of the solution is 1.84 g/mL. **[IIT JEE-1990]**

Sol. 93% H_2SO_4 w/V means 93 g of H_2SO_4 is present in 100 ml of solution.

Since volume is 1000 ml, amount of H_2SO_4 is 930 g/ litre. Also density is given 1.84 g/ml

M = density × volume = 1.84 × 1000 = 1840 g

wt of solvent = 1840 − 930 = 910 g

$$\because \text{Molality} = \frac{\text{moles of solute}}{\text{wt of solvent in kg}} = \frac{930}{98} \times \frac{1000}{910}$$

= 10.43 mol/kg

agar sare formulae pata hai toh ho jaega

Mole Fraction

- It is defined as the ratio of the number of moles of a specific component to the total number of moles of all components in the mixture.

$$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{n_1}{n_1 + n_2}$$

$$X_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}} = \frac{n_2}{n_1 + n_2}$$

$$X_{solute} + X_{solvent} = \frac{n_1}{n_1 + n_2} + \frac{n_2}{n_1 + n_2} = 1$$

(Here, n_1 = Moles of solute, n_2 = moles of solvent)

Example:

In a solution of aqueous NaCl, the $X_{NaCl} = 0.1$

$$X_{NaCl} = 0.1 = \frac{1}{10} = \frac{1}{1+9}$$

(n_{NaCl} and n_{H_2O})

$$\frac{n_{NaCl}}{n_{NaCl} + n_{H_2O}} \quad \text{compare}$$

139. Find out the molality of an aqueous solution of urea in which mole fraction of urea = $\frac{1}{31}$

Sol. $\left(\frac{50}{27}m\right)$ For explanation, refer to QR code given on the last page.

140. The mole fraction of a solute in a 100 molal aqueous solution _____ $\times 10^{-2}$ (Round off to the Nearest Integer)

[JEE Main 17 March 2021 Shift-1]

Sol. Mole fraction of solute = $\frac{n_{solute}}{n_{solute} + n_{solvent}}$

100m means 100 moles of solute are present in 1000 g or 1 kg of solvent. Substitute these values in the above expression:

$$= \frac{100}{100 + \frac{1000}{18}} = \frac{1800}{2800} = 0.6428 = 64.28 \times 10^{-2}$$

141. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same its molality. Density of this solution at 298 K is 2.0 g/ml. The ratio of the molecular weights of the solute and solvent, is

[JEE Adv.-2016]

Sol. $\left(\frac{9}{1}\right)$ For explanation, refer to QR code given on the last page.

142. 100 g of propane is completely reacted with 1000 g of oxygen. The mole fraction of carbon dioxide in the resulting mixture is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)

[Atomic weight: H = 1.008; C = 12.00; O = 16.00] **[27August 2021 (Shift-II)]**

Sol. $n_{C_3H_8} = \frac{100}{44} = 2.27$ mol, $n_{O_2} = \frac{1000}{32} = 31.25$ mol

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

At t = 0;	2.27 mol	31.25 mol	0	0
At t = ∞;	0	19.9 mol	6.81mol	9.08 mol

$$x_{CO_2} = \frac{6.81}{19.9 + 6.81 + 9.08}$$

$$= 0.1902 = 19.02 \times 10^{-2} \approx 19 \times 10^{-2}$$

reaction likho lekin balance karna bhul gye toh beda bilkul garg

143. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is ____.

(Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol, respectively)

[JEE Adv.-2019]

Sol. $X_{urea} = 0.05 = \frac{n}{n+50}$, $19n = 50$, $n = 2.6315$

$$V_{sol} = \frac{(2.6315 \times 60 + 900)}{1.2} = 881.5789 \text{ ml}$$

$$\text{Molarity} = \frac{2.6315 \times 1000}{881.5789} = 2.9849$$

144. The density of a 3 M sodium thiosulphate solution ($Na_2S_2O_3$) is 1.25 g per mL. Calculate (i) the percentage by weight of sodium thiosulphate (ii) the mole fraction of sodium thiosulphate and (iii) the molalities of Na^+ and $S_2O_3^{-2}$ ions. **[IIT JEE-1983]**

Sol. (i) 37.92, (ii) 0.065, (iii) 3.86 m

For explanation, refer to QR code given on the last page.

%w/w

- It is defined as the amount of solute dissolved (in gm) in 100 gm of solution

$$\%w/w = \frac{\text{wt of solute in gm}}{\text{wt of solution in gm}} \times 100$$

- Temperature independent
- 8% w/w HCl solution or 8% by mass HCl

↓ Ek line angrezi ki

8 gm of HCl dissolved in 100 gm solution

wt of solute ⟷ wt of solution

145. Find out the X_{urea} and molality of 10% w/w aq. urea solution.

Sol. $\left(\frac{50}{27}\text{ molal}\right)$ For explanation, refer to QR code given on the last page.

146. Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

[JEE Mains Shift-II, 2025]

Given: Density of nitric acid solution is 1.25 g/mL

(A) 45 (B) 55 (C) 32 (D) 40

Sol. (C) Since 75% of the solution is nitric acid by mass, 30 g is 75% of the total mass.

Let the total mass of the solution be M.

$0.75 \times M = 30g,\ M = \frac{30}{0.75} = 40g$

$V = \frac{\text{Mass}}{\text{Density}} = \frac{40g}{1.25g/mL} = 32mL$

147. 29.2%(w/w) HCl stock solution has density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5 g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution 0.4 M HCl is **[JEE Adv. 2012]**

Sol. First calculate moles of HCl:

Moles = $0.4 \times 0.2 = 0.08$ mol

Let x be the volume of original HCl solution, then mass = 1.25 x g

mass of HCl = (1.25x × 0.292) g

mole of HCl $= \frac{1.25x \times 0.292}{36.5} = 0.08$

So, $x = \frac{36.5 \times 0.08}{0.29 \times 1.25} = 8$ mL

adv mei aaya hai, ignore karne ki galti mat karna

%w/V

- %w/V is defined as the mass of solute (in gm) dissolved in 100 ml solution

$$\%w/V = \frac{\text{wt of solute in gm}}{\text{volume of solution in ml}} \times 100$$

- Temperature dependent
- 10% w/V aqueous sucrose ($C_{12}H_{22}O_{11}$ = 342 g) solution (चीनी)

↓ ek line angrezi kil

10 gm of sucrose dissolved in 100 ml solution

wt of solute ⟷ volume of solution

148. Calculate the molarity of 10% w/V aqueous sucrose solution.

Sol. 10% w/V ⇒ 10 g sucrose dissolved in 100 ml solution

$$\text{Molarity} = \frac{\left(\frac{10}{342}\right)}{\frac{100}{100}} = \frac{100}{342}M$$

149. A solution contains 10% w/V NaCl. Calculate its molarity.

(Molar mass of NaCl = 58.5 g/mol; density of solution = 1 g/mL)

Sol. Mass of NaCl in 1 L = 10% × 1000 = 100 g.

$\text{Moles of NaCl} = \frac{\text{Mass}}{\text{Molar mass}} = \frac{100}{58.5} = 1.71$ mol

$M = \frac{\text{Moles}}{\text{Volume}} = 1.71\ M$

bhayi sawaal ki bhasha samjho bas

%V/V

- %V/V is defined as the volume of solute (in ml) dissolved in 100 ml solution

$$\%V/V = \frac{\text{volume of solute in ml}}{\text{volume of solution in ml}} \times 100$$

- Temperature independent
- 5 % V/V C_2H_5OH solution

↓ Ek line angrezi ki

5 ml ethanol (C_2H_5OH) dissolved in 100 ml solution

volume of solute ⟷ volume of solution

150. How much alcohol (in mL) is present in 250 mL of a 40% V/V alcohol solution

Sol. $\text{Volume of solute} = \frac{\%V/V \times \text{Volume of solution}}{100}$

$$\text{Volume of solute} = \frac{40 \times 250}{100} = 100 \text{ mL}.$$

151. 100 mL of a 50% V/V ethanol solution is diluted to 500 mL. What is the new % V/V?

Sol. $\text{New } \%V/V = \frac{\text{Original } \%V/V \times \text{Initial volume}}{\text{Final volume}}$

$$= \frac{50 \times 100}{500} = 10\%$$

isme bhi ab pagla mat jana, samjhe?

ppm

- ppm is defined as the no. of gm of solute present in 10^6 gm of solution.
- This concentration term is defined for the solutions having very less amount of solute.
- mass of solution $\simeq$ mass of solvent
- Unit $\Rightarrow$ ppm
- It is temperature independent

$$\text{ppm} = \frac{\text{mass of solute (gm)}}{\text{mass of solution (gm)}} \times 10^6$$

152. 10.30 mg of O_2 is dissolved into a litre of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is _____

[JEE Main Jan 2020 Shift-2]

Sol. mass of $O_2 = 10.3 \times 10^{-3}$ gm

$$\text{mass of solution} = 1000 \text{ ml} \times 1.03 \frac{\text{gm}}{\text{ml}} = 1030 \text{ gm}$$

$$\text{Ppm conc} = \frac{\text{mass of } O_2}{\text{mass of sea water}} \times 10^6$$

$$= \frac{10.3 \times 10^{-3} \times 10^6}{1030} = 10 \text{ ppm}$$

153. A water sample contains 3 mg of dissolved oxygen per 1000 g of sample. Calculate concentration of oxygen in parts per million.

Sol. (3 ppm) For explanation, refer to QR code given on the last page.

ppb

- ppb is defined as the no. of gm of solute dissolved in 10^9 gm of solution.

$$\text{ppb} = \frac{\text{mass of solute (gm)}}{\text{mass of solution (gm)}} \times 10^9$$

- mass of solution = mass of solvent
- This concentration term is used for highly diluted solutions
- Unit $\Rightarrow$ ppb
- Temperature independent

154. Which urea solution is more concentrated?

10% w/w, (A) 10 ppm, (B) 10 ppb (C)

Sol. 10% w/w means 10 g of solute is present in 100 g of solution.

10 ppm means 10 g of solute is present in 10^6 g of solution

10 ppb means 10 g of solute is present in 10^9 g of solution.

So, the correct order is: A > B > C

Formality

- Formality is defined as the no. of gram formula masses of the ionic solute dissolved per lt of solution.
- It is defined for ionic solids.

$$\text{Formality} = \frac{\text{no. of gm formula mass of solute}}{\text{volume of solution in lt}}$$

$$= \frac{\text{mass of solute}}{\text{formula mass of solute} \times \text{vol of solution in lt}}$$

- unit = formal (F)
- Temperature dependent
- defined for ionic solids
- For ionic solids, molarity is termed as formality

isko formality mat samajh lena, properly karke jana

155. 5.85 gm of NaCl dissolved in 2 lt of solution. Find the formality of NaCl solution.

Sol. gm formula mass = $\frac{\text{given mass}}{\text{formula mass}} = \frac{5.85}{58.5} = 0.1$

Formality = $\frac{0.1}{2} = 0.05$ F

156. Calculate the formality of sodium chloride (NaCl) solution, 5.85 g of which have been dissolved to form 250 mL of the given solution.

Sol. Formality = $\frac{\text{Given weight of solute}}{\text{Molecular weight of solute} \times \text{Volume of solution in litre}}$

Formality = $\frac{5.85\ g}{58.5\ g\ mol^{-1} \times 0.250\ L} = 0.4$ F

seedha seedha formula laga hai bas

Strength of Solution

- It is mass of solute (in gm) present in 1 lt of solution

$$\text{strength} = \frac{\text{mass of solute in gm}}{\text{volume of solution in lt}}$$

- 1.2 g/L aq. urea solution means 1.2 gm of urea dissolved in 1000 ml or 1 lt solution
- 1.2 g/L density of aq urea solution means wt. of 1000 ml or 1 lt aq urea solution is 1.2 gm

Important

Relationship between strength and Molarity

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution(lt)}}$$

$$= \frac{\text{mass of solute in gm}}{\text{molecular wt} \times \text{volume of solution in lt}}$$ (strength)

$$\text{Molarity} = \frac{\text{Strength}}{\text{mol wt. of solute}}$$

157. Find out the strength of aq. urea solution having molarity $\frac{11}{6}$ M.

Sol. Method-1: Molarity = $\frac{\text{strength}}{\text{mol wt. of solute}}$

Strength = molarity × mol wt. of solute

= $\frac{11}{6} \times 60 = 110$ g/L

Method-2:

$\frac{11}{6}$ moles of urea present in 1 lt solution.

$\frac{11}{6} \times 60$ gm of urea present in 1 lt solution

110 gm in 1 lt ⇒ strength = 110 g/L

dono methods seekhlo, kaam aaenge

158. Find out % w/w of 110 g/L strength of aq urea solution having density 1.1 gm/ml.

Sol. (10%) For explanation, refer to QR code given on the last page.

159. 20 mL of sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____ M. (Nearest Integer value)

(Given : Na = 23, I = 127, Ag = 108, N = 14, O = 16 g mol^{-1}) **[08 April, 2025 (Shift-II)]**

Sol. (1) For explanation, refer to QR code given on the last page.

160. $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Consider the above reaction, what mass of $CaCl_2$ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$?

(Given: Molar mass of Ca, C, O, H and Cl are 40, 12, 16, 1 and 35.5 g mol^{-1}, respectively)

[02 April, 2025 (Shift-I)]

(A) 3.908 g (B) 2.636 g

(C) 10.545 g (D) 5.272 g

Sol. (C) For explanation, refer to QR code given on the last page.

Scan the QR code to get the explanations of unsolved questions.

2 Structure of Atom

To access one shot Revision Video Scan this QR Code

INTRODUCTION

The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.) who were of the view that atoms are the fundamental building blocks of matter.

According to them, the continued subdivisions of matter would ultimately yield atoms which would not be further divisible.

DISCOVERY OF SUB-ATOMIC PARTICLES

- Atoms are divided into subatomic particles like electron, neutron, proton are present in the atom.
- Matter is made up of molecules and molecules are made up of atoms.
- Electrons, protons and neutrons are the fundamental particles of atom.

DISCOVERY OF ELECTRON

Electron is that fundamental particle that carries one unit negative charge and has mass nearly equal to 1/1837th of that of hydrogen atom.

Electrons were discovered using cathode ray discharge tube experiment.

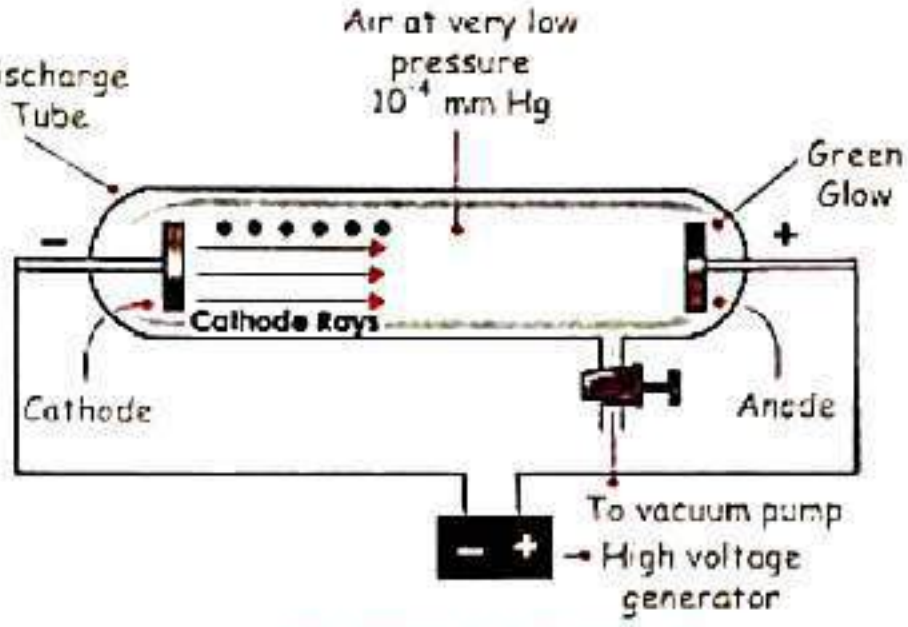
Cathode Ray Experiment

The Cathode Ray Experiment, also known as the **Crookes Tube Experiment**, was a ground breaking study conducted by William Crookes in 1869 and later by J.J. Thomson in 1897. The experiment involved the study of cathode rays, which are streams of negatively charged particles emitted from the cathode (negative electrode) of a vacuum tube.

Kuch pada palle

Note

"The origin of cathode ray is cathode (metal) surface."



Production of Cathode Ray

$Metal \rightarrow e_m^-$

Cathode

$A - A \xrightarrow{e_m^-} A + A$

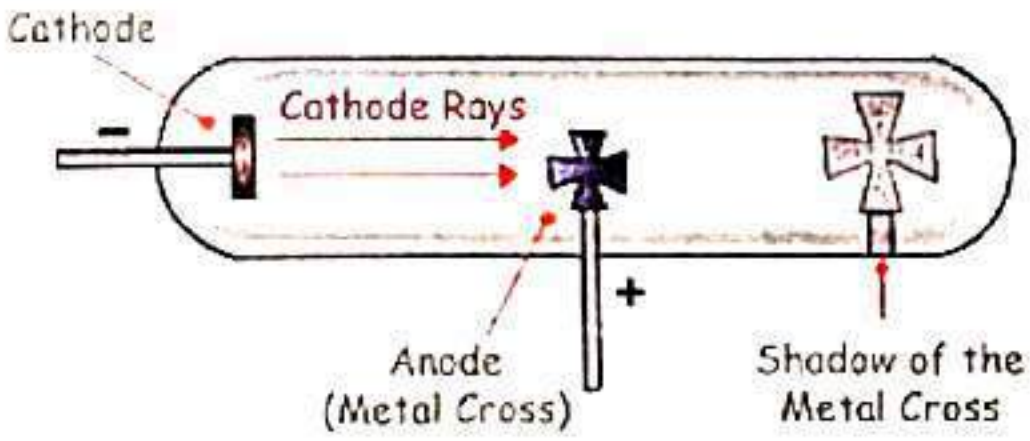
$A \xrightarrow{e_m^-} A^+ + e_g^-$

$(e_m^- + e_g^-)$ = Cathode Rays

Ab padhte hai cathode experiment ki kuch properties

1. **Cathode rays cast shadows of the objects placed in their path**

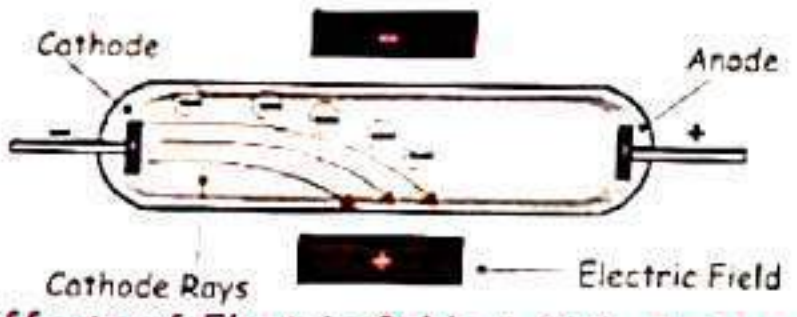
Conclusion: Cathode rays travel in a straight line.



Cathode Rays cast shadows of the objects placed in their path

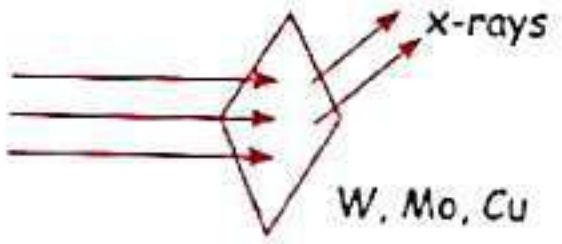
2. Cathode rays deflect towards positive plate

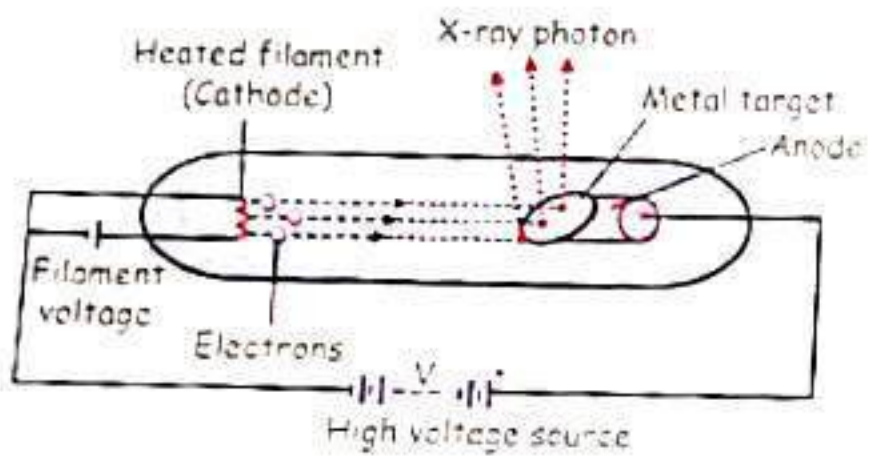
Conclusion: They are made up of negatively charged particles.



Effects of Electric field on Cathode Rays

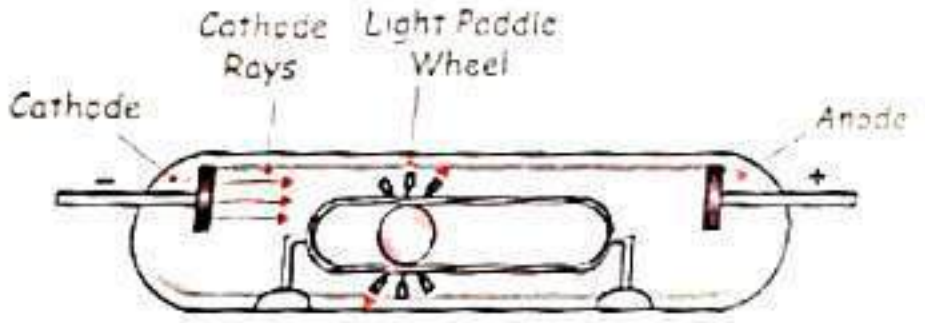
3. When strike on metal of high atomic mass, they produce X-rays





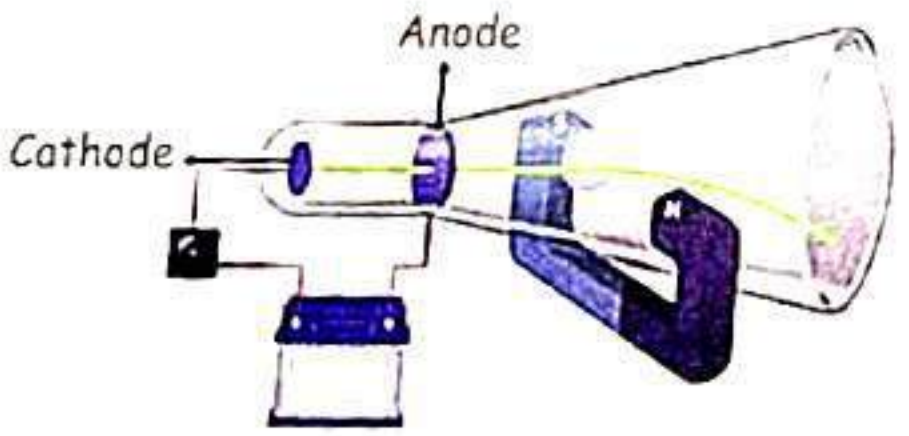
4. Rotates mica paddle wheel placed in their path

Conclusion: They are made up of particles having mass also.



Cathode Rays can Rotate a High Paddle Wheel placed in their path

5. Cathode rays are deflected by magnetic field



6. Cathode rays can ionize gases

$M \rightarrow e_M^-$

Cathode

$A \xleftarrow{e_M^-} A \rightarrow A_{(g)} + A_{(g)}$

$A_{(g)} + e_M^- \rightarrow A_{(g)}^{\cdot} + \underbrace{e_g^- + e_M^-}_{\text{Cathode rays}}$

Laga na mazedaar?

Important

- Specific charge of cathode Ray

$$= \frac{\text{Charge}}{\text{Mass}} = \frac{-1.6 \times 10^{-19} C}{9.1 \times 10^{-31} kg} = -1.76 \times 10^{11} C/Kg$$

- Charge = electronic charge = -1.6×10^{-19} C
- Mass of electron = 9.1×10^{-31} Kg
- Specific charge of cathode rays remains constant and Universal. It will not change on changing the gas.

$M \rightarrow e_M^-$ (Cathode)

$A \xleftarrow{e_M^-} A \rightarrow A_{(g)} + A_{(g)}$

$A_{(g)} + e_M^- \rightarrow A_{(g)}^{\cdot} + \underbrace{e_g^- + e_M^-}_{\text{Cathode rays}}$

$M \rightarrow e_M^-$ (Cathode)

$B \xleftarrow{e_M^-} B \rightarrow B_{(g)} + B_{(g)}$

$B_{(g)} + e_M^- \rightarrow B_{(g)}^{\cdot} + \underbrace{e_g^- + e_M^-}_{\text{Cathode rays}}$

yeh padhlo dhyan se

7. Cathode rays penetrate through th sheets of aluminium and metals.

8. They affect the photographic plates.

1. The specific charge of cathode rays:
   (A) depends on the nature of gas.
   (B) depends on material of discharge tube.
   (C) depends on potential difference.
   (D) is a universal constant.

Sol. (D) The specific charge of the cathode ray is a universal constant.

Are yeh bhi nhi hua toh tum lallu ho

Common Mistake

Specific charge of cathode rays depends on the nature of gas.

2. i. Calculate the number of electrons which will together weigh 1 gm.

   ii. Calculate the mass and charge of one mole of electrons.

Sol. (i) 1.098×10^{2} , (ii) $9.65 \times 10^{4} C$

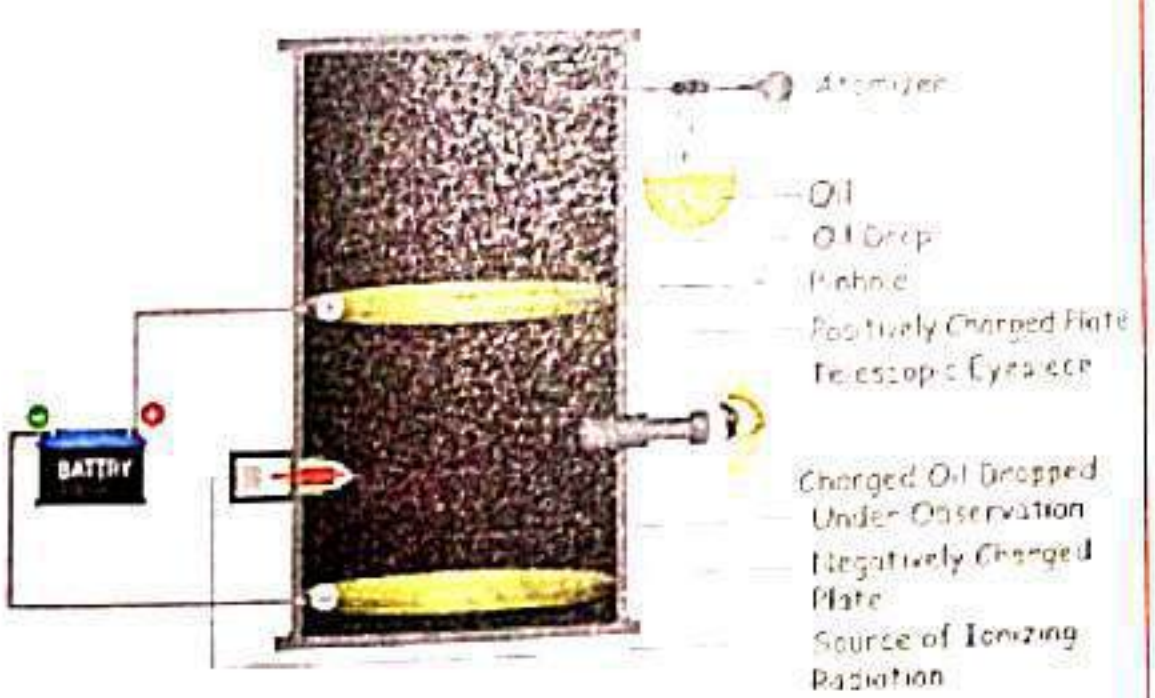
For explanation, refer to QR code given on the last page.

Millikan's Oil Experiment

- Millikan determined the charge of the electron by an **oil drop experiment**. Millikan concluded that the magnitude of electrical charge, q, on the droplets is always an integral multiple of the electrical charge, e, that is $q = ne$, where $n = 1, 2, 3 \ldots\ldots$
- Charge on the oil drops was always an integral multiple of $1.60 \times 10^{-19} C$.
- The mass on an electron was determined by combining the results of Thomson's experiment and Millikan's oil drop experiment.

$$m_e = \frac{e}{e/m_e} = \frac{1.60 \times 10^{-19} C}{1.758820 \times 10^{11} C kg^{-1}}$$

$$= 9.1094 \times 10^{-31} Kg$$



khatarnaak lagra hoga bahut

Note

- Atomizer converts oil into very tiny droplets.
- Drop of oil falls from the tiny hole done on positive plate.
- Beam of electrons or X-rays ionize the air present inside the chamber and released electrons are absorbed by oil drops.
- Potential difference is applied such that the net force acting on the oil drop becomes zero.

$mg = qE$

$q = \frac{mg}{E}$

3. Assuming that in oil drop experiment, the charge on oil drops was found to be -

   9×10^{-18} unit, 6×10^{-18} unit,

   15×10^{-18} unit, 3×10^{-18} unit,

   4.5×10^{-18} unit, then the charge of e -

   (A) 3×10^{-18} unit (B) 9.0×10^{-18} unit

   (C) 1.5×10^{-18} unit (D) 1.6×10^{-19} unit

Sol. According to the given data, 1.5×10^{-18} must be the answer.

$1.5 \times 10^{-18} \times 2 = 3 \times 10^{-18}$,

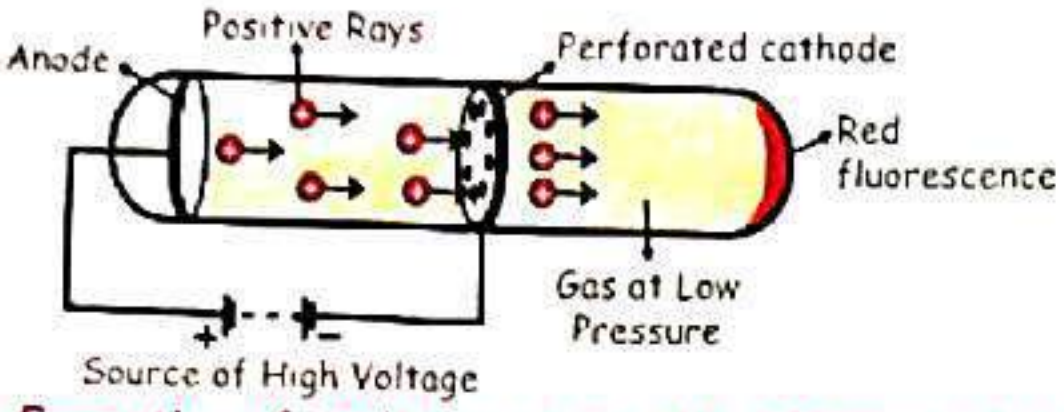
$1.5 \times 10^{-18} \times 3 = 4.5 \times 10^{-18}$

$1.5 \times 10^{-18} \times 4 = 6 \times 10^{-18}$ and so on

POSITIVE RAYS-DISCOVERY OF PROTON

The existence of positively charged particles in an atom was shown by E. Goldstein in 1886. He repeated the same discharge tube experiments by using a perforated cathode. It was observed that when a high potential difference was applied between the electrodes, not only cathode rays were produced but also a new type of rays were produced simultaneously

from anode moving towards cathode and passed through the holes or canal of the cathode. These rays are termed as canal ray or anode ray.



Formation of positive rays in the immersion tube

dimag ghum toh nahi gaya?

Note

$M \rightarrow e_M^-$ (Cathode)

$A \xleftarrow{e_M^-} A \rightarrow A_{(g)} + A_{(g)}$

$A_{(g)} + e_M^- \rightarrow A_{(g)}^{\bullet} + e_g^- + e_M^-$

($A_{(g)}^{\bullet}$: Anode rays; $e_g^- + e_M^-$: Cathode rays)

$M \rightarrow e_M^-$ (Cathode)

$B \xleftarrow{e_M^-} B \rightarrow B_{(g)} + B_{(g)}$

$B_{(g)} + e_M^- \rightarrow B_{(g)}^{\bullet} + e_g^- + e_M^-$

($B_{(g)}^{\bullet}$: Anode rays; $e_g^- + e_M^-$: Cathode rays)

$H \xleftarrow{e_M^-} H \rightarrow H_{(g)}^{\bullet} + e_g^- + e_M^-$

For this, $\frac{e}{M}$ ratio is maximum and we name this species as proton

Anode Rays Characteristics

(i) The e/m ratio for these rays is smaller than that of electrons.

(ii) Unlike cathode rays, their e/m value is dependent upon the nature of the gas taken in the tube. For different gases used in the discharge tube, the charge to mass ratio (e/m) of the positive particles constituting the positive rays is different. When hydrogen gas is taken in the discharge tube, the e/m value obtained for the positive rays is found to be maximum. Since the value of charge (e) on the positive particle obtained from different gases is the same, the value of m must be minimum for the positive particles obtained from hydrogen gas. Thus, the positive particle obtained from hydrogen gas is the lightest among all the positive particles obtained from different gases. This particle is called the proton.

(iii) They are capable to produce ionisation in gases.

(iv) They can produce physical and chemical changes.

language mushkil lag rahi hogi but hai bilkul asaan

4. Which of the following statement is incorrect?
   (A) Cathode rays are emitted out from the surface of cathode.
   (B) Anode rays are heavier than cathode rays.
   (C) Anode rays are emitted out from the surface of anode.
   (D) Cathode rays travel in a straight line

Khoobsurat Baat

Sol. (C) Anode rays are not emitted from the surface of anode but are produced in the space present between the anode and the cathode.

Common Mistake

Anode rays are emitted out from the surface of anode.

5. The e/m ratio is maximum for
   (A) Al^{+3} (B) Na^{+}
   (C) H^{+} (D) He^{++}

Sol. (C) When hydrogen gas is taken in the discharge tube, the e/m value obtained for the positive rays is found to be maximum.

padhlo toh padega hi bacho

6. e/m ratio or specific charge of anode rays depends on the
   (A) Nature of anode material
   (B) Nature of cathode material
   (C) Nature of gas filled in the tube
   (D) Potential difference applied.

Sol. (C) Specific charge of anode rays depends on the nature of gas filled in the tube.

7. The ratio of the charges on two particles A and B are in 4:9 and their mass ratio is 2:3, then find out their specific charge ratio.

(A) 1 : 1 (B) 2 : 3

(C) 4 : 9 (D) 3 : 2

Sol. (B) Charge ratio: $q_A/q_B = 4/9$

Mass ratio: $m_A/m_B = 2/3$

The specific charge ratio is given by:

$(q_A/q_B) / (m_A/m_B) = (4/9) \times (3/2) = 2 : 3$

simple hai bilkull

Discovery of Neutrons

Later, a need was felt for the presence of electrically neutral particles as one of the constituent of atom. These particles were discovered by Chadwick in 1932 by bombarding a thin sheet of Beryllium with α-particles, when electrically neutral particles having a mass slightly greater than that of protons were emitted. He named these particles as neutrons.

$$^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{12}_{6}C + ^{1}_{0}n$$

Properties of Charge

1. Charge is always quantised

$$Q = ne, e = 1.6 \times 10^{-19} \text{ coulumb}$$

8. i A body can have a charge of 20 es (T/F)

ii. A body can have a charge of 20 mole of es (T/F)

iii. A body can have a charge of 6.32 mole of es (T/F)

iv. A body can have a charge of 6.32 es (T/F)

Sol. (i) T, (ii) T, (iii) T, (iv) F

2. Charge is of two types → Positive ($+1.6 \times 10^{-19}$ C)
Negative (-1.6×10^{-19} C)

3. When two charge bodies are separated by distance r, then there can exist force of attraction or repulsion and this force is given by:

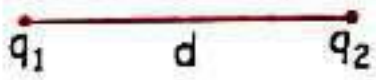
$$F = K\frac{q_1 q_2}{r^2}$$

$K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$

Khoobsurat Baat

Opposite charges attract each other while similar charges repel each other.

4. When two bodies are separated by distance r, then the magnitude of potential energy (P.E.) is given by:



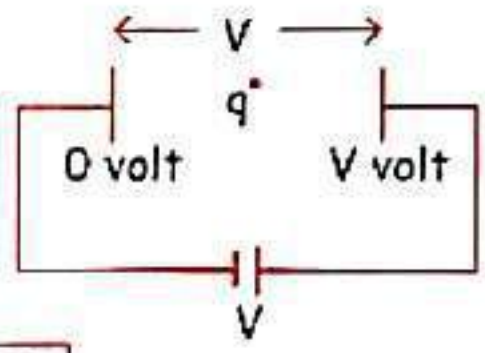
$$|PE| = K\frac{q_1 q_2}{d}$$

Note

PE = (+) due to repulsive forces.

PE =(-) due to attractive force.

5. When a charge body is placed on the surface of potential difference V volts, then the potential difference of this charge body is qV.



$$qV = \frac{1}{2}mv^2 \qquad v = \sqrt{\frac{2qV}{m}}$$

prabhu ko yaad karlena

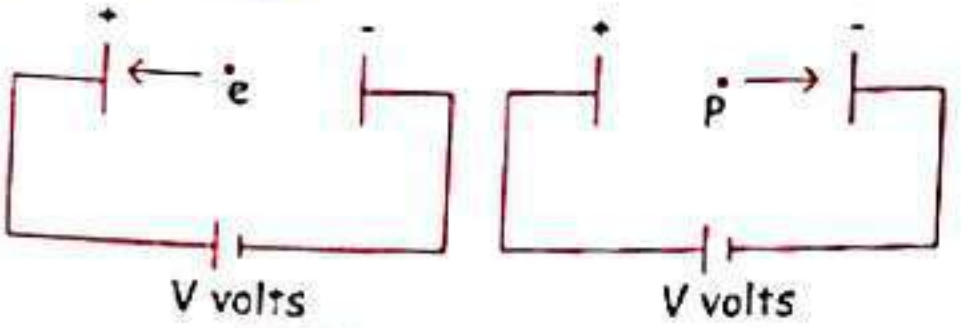
Important

(+)Ve charge → high to low potential

(-) Ve charge → low to high potential

The velocity gained by a charge body of charge q, when accelerated through potential difference V volt is given by $V = \sqrt{\frac{2qV}{m}}$

Important



$$v_e = \sqrt{\frac{2q_eV}{m_e}} \; , \; v_p = \sqrt{\frac{2q_pV}{m_p}}$$

9. 1. If a proton and electron both are accelerated through same potential difference then which will move faster?

2. Also, calculate the ratio of potentials through which they are accelerated to gain same velocity

Sol. 1. $q_p = q_e = 1.6 \times 10^{-19} C$

$$\frac{v_e = \sqrt{\frac{2q_eV}{m_e}}}{v_p = \sqrt{\frac{2q_eV}{m_e}}} = \sqrt{\frac{m_p}{m_e}} \; , \; m_p = m_n, \; m_p = 1837 \, m_e$$

2. For $v_e = v_p$

$$\frac{v_e = \sqrt{\frac{2q_eV_e}{m_e}}}{v_p = \sqrt{\frac{2q_pV_p}{m_p}}} \Rightarrow \frac{V_e}{m_e} = \frac{V_p}{m_p}$$

$$\frac{V_e}{V_p} = \frac{m_e}{m_p} = \frac{9.1 \times 10^{-31} kg}{1.66 \times 10^{-27} kg} = \frac{1}{1837}$$

$$V_p = 1837 V_e$$

are mushkil nahi hai bhayi

Common Mistake

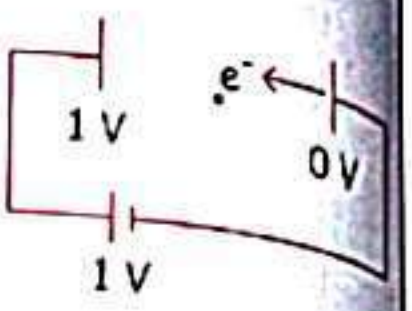
It is important to know the values of mass of electron as well as mass of proton. If values are not known, students might face difficulty to solve problem.

One Electron Volt

Energy gained by an electron when it is accelerated from rest through a potential difference of 1 volt.

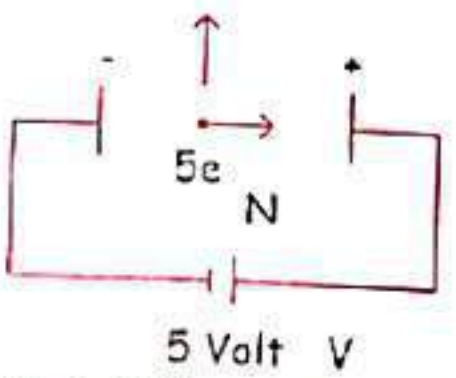
Note

Positive charge always moves from high potential to low potential and -ve charge always moves from low potential to high potential if set free.



$$1eV = 1.6 \times 10^{-19} J$$

10. A particle has a mass of 1.73×10^{-20} gm. Find out the velocity.



Sol. $PE = qV = (5e)(5V) = 25eV = 25 \times 1.6 \times 10^{-19} J$

$$25 \times 1.6 \times 10^{-19} = \frac{1}{2}mv^2 = \frac{1}{2} \times 1.73 \times 10^{-23} v^2$$

$v = 680$ m/s

11. Find out the velocity of a charged particle moving with K.E 20 eV. ($m = 1.67 \times 10^{-23}$ gm)

Sol. (1.96×10^4 m/s) For explanation, refer to QR code given on the last page.

Common Mistake

Don't forget to convert the mass into kg otherwise it will lead to conceptual error.

12. Calculate the velocity of electron when it is accelerated from rest through the potential difference of 20 V.

Sol. $KE = eV = 1.602 \times 10^{-19} C \times 20V = 3.204 \times 10^{-18} J$

$KE = (1/2)mv^2$

$$v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times 3.204 \times 10^{-18} J}{9.109 \times 10^{-31} kg}} \approx 8.38 \times 10^6 \, m/s$$

13. An electron is moving with velocity 8.38×10^6 m/s. Find out the stopping potential.

Sol. (20 V) For explanation, refer to QR code given on the last page.

CLOSEST DISTANCE OF APPROACH

The closest distance of approach (r_0) is the minimum distance between a charged particle and a nucleus of an atom.

$(TE)_B = (TE)_A$

$(KE)_B + (PE)_B = (KE)_A + (PE)_A$

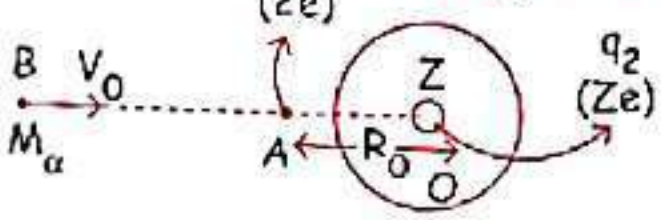
$$\frac{1}{2}m_\alpha V_0^2 + 0 = 0 + k\frac{(2e)(Ze)}{R_0}$$

Z = Protons

He$^+$ = α - particle

OA = R_0 = CDA

$$R_0 = \frac{4KZe^2}{m_\alpha V_0^2}$$



For proton,

$$\frac{1}{2}m_p V_0^2 = K\frac{(e)(Ze)}{R_0} \quad (\text{Proton} = e, \text{Mass} = m_p)$$

$$R_0 = \frac{2KZe^2}{m_p V_0^2}$$

yeh closest distance ka halke me mat lena

Important

For α - particle

$$R_0 = \frac{4KZe^2}{m_\alpha V_0^2}$$

For proton,

$$R_0 = \frac{2KZe^2}{m_p V_0^2}$$

14. Find the distance of closest approach when an α particle is projected towards the nucleus of silver atom having speed v. (mass of a particle = m_α, atomic number of Ag = 47)

Sol. Closest distance of approach

$$R_0 = \frac{4KZe^2}{m_\alpha V_0^2} = \frac{188Ke^2}{m_\alpha V^2}$$

PARTICLE NATURE OF ELECTROMAGNETIC RADIATION

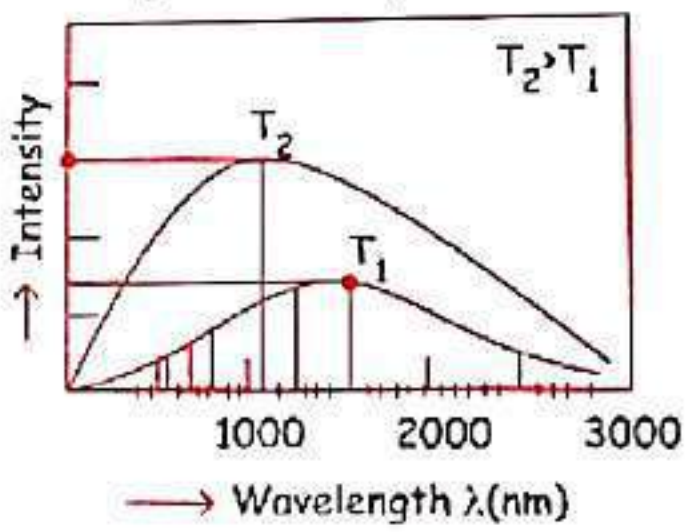
Some of the experimental phenomenon such as diffraction and interference can be explained by the wave nature of the electromagnetic radiation. However, following are some of the observations which could not be explained

(i) The nature of emission of radiation from hot bodies (black - body radiation)

(ii) Ejection of electrons from metal surface when radiation strikes it (photoelectric effect)

Black Body Radiation

When solids are heated they emit radiation over a wide range of wavelengths. The ideal body, which emits and absorbs all frequencies, is called a black body and the radiation emitted by such a body is called black body radiation. The exact frequency distribution of the emitted radiation (i.e., intensity versus frequency curve of the radiation) from a black body depends only on its temperature.



black body radiation bhi kamaal ki hai

Important

The above experimental results cannot be explained satisfactorily on the basis of the wave theory of light. Planck suggested that atoms and molecules could emit (or absorb) energy only in discrete quantities and not in a continuous manner.

15. The number of incorrect statement/s about the black body from the following is

[10 April, 2023 (Shift-I)]

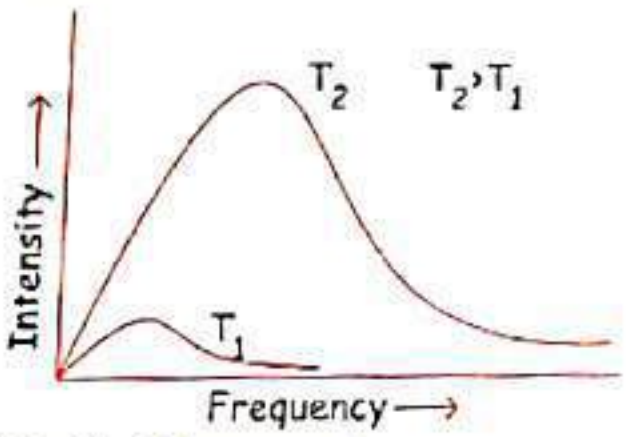
(1) Emit or absorb energy in the form of electromagnetic radiation

(2) Frequency distribution of the emitted radiation depends on temperature

(3) At a given temperature, intensity vs frequency curve passes through a maximum value

(4) The maximum of the intensity vs frequency curve is at a higher frequency at higher temperature compared to that at lower temperature

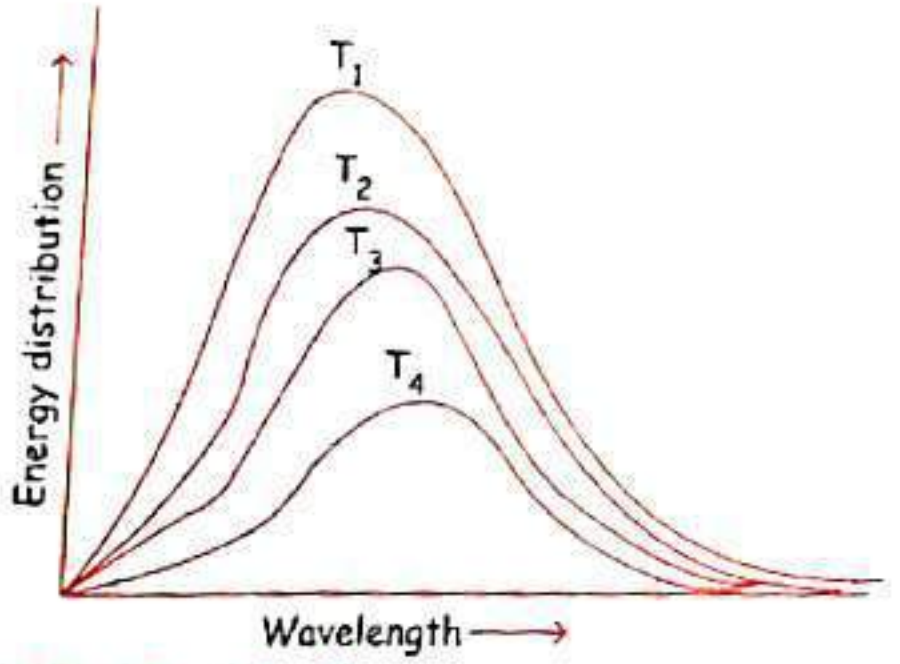
Sol. [0] A blackbody can emit and absorb all the wavelengths in electromagnetic spectrum ⇒ (1) is correct.



⇒ (2), (3), (4) correct.

16. Following figure shows spectrum of an ideal black body at four different temperatures. The number of correct statement/s from the following is ____.

[24 Jan, 2023 (Shift-II)]



(1) $T_4 > T_3 > T_2 > T_1$

(2) The black body consists of particles performing simple harmonic motion

(3) The peak of the spectrum shifts to shorter wavelength as temperature increases

(4) $\frac{T_1}{V_1} = \frac{T_2}{V_2} = \frac{T_3}{V_3} \neq$ constant

(5) The given spectrum could be explained using quantisation of energy

Sol. [2] The spectrum of Black body radiation is explained by the phenomenon quantization of energy. With increase in temperature, peak of spectrum get shifted to shorter wavelength or higher frequency. For above graph; the correct order is → $T_1 > T_2 > T_3 > T_4$.

kyu ghabra rahe ho

QUANTUM THEORY OF LIGHT

The smallest quantity of energy that can be emitted or absorbed in the form of EMR is called quantum of light or quanta or photon

$$E = h\nu = \frac{hc}{\lambda} \qquad (c = \nu\lambda)$$

h = Planck's const = 6.626×10^{-34} J sec

c = Speed of light = 3×10^{8} m/sec

λ = wavelength of EMR

Khoobsurat Baat

$$\lambda_{\text{Å}} = \frac{12400}{\Delta E_{eV}}$$

$\lambda_{\text{Å}}$: Wavelength of photon in Å; ΔE_{eV}: Energy of photon in eV

17. Find out the wavelength of photon having energy 5 eV in Å?

Sol. Method 1

$$E = \frac{hc}{\lambda}$$

J ← E; hc → J-sec, m/sec.; λ → m

$$5\times1.6\times10^{-19}=\frac{6.626\times10^{-34}\times3\times10^{8}}{\lambda}$$

$\lambda = 2480\ \text{Å}$

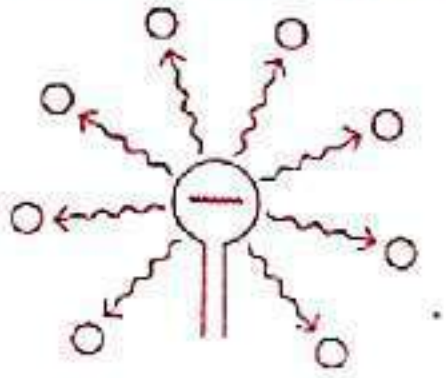
Method 2

$$\lambda_{\text{Å}}=\frac{12400}{E_{eV}}=\frac{12400}{5}=2480\ \text{Å}$$

paper mei ek se hi karke aana budhu

18. A 100 watt bulb emits monochromatic light of wavelength 400 nm. Calculate number of photons emitted per second by this bulb.

Sol.



100W = 100 J/sec

Emits 100 J energy in one second.

$\frac{hc}{\lambda}$ × no. of photons = 100 J

$$\text{No of photons}=\frac{100}{\frac{6.626\times10^{-34}\times3\times10^{8}}{400\times10^{-9}}}=10^{20}$$

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No of photon × energy of one photon = Total energy emitted by the source.

$h\nu$ or $\frac{hc}{\lambda}$

19. Bond energy of Br_2 is 194 kJ/mol. The wave number of photon required to break this bond is

Sol. ($1.62\times10^{6}\ m^{-1}$) For explanation, refer to QR code given on the last page.

20. The energy of one mole of photons of radiation of wavelength 300 nm is

(Given : $h = 6.63\times10^{-34}$ J s, $N_A = 6.02\times10^{23}\ mol^{-1}$, $c = 3\times10^{8}\ ms^{-1}$)

[JEE Main 2023 Online 24th June Shift-II]

(A) 235 kJ mol^{-1} (B) 325 kJ mol^{-1}

(C) 399 kJ mol^{-1} (D) 435 kJ mol^{-1}

Sol. (C) Energy of one mole of photons = $\frac{hc}{\lambda}\times N_A$

$= 399.13\times10^{3}$ J/mol = 399 kJ/mole

21. Calculate the energy of 100 photons if the wavelength of the light is 200Å.

Sol. Total energy = No of photons × energy of one photon

Energy of one photon = $\frac{hc}{\lambda}$

$$100\times\left(\frac{6.625\times10^{-34}\times3\times10^{8}}{2000\times10^{-10}}\right)=621.1\ \text{eV}$$

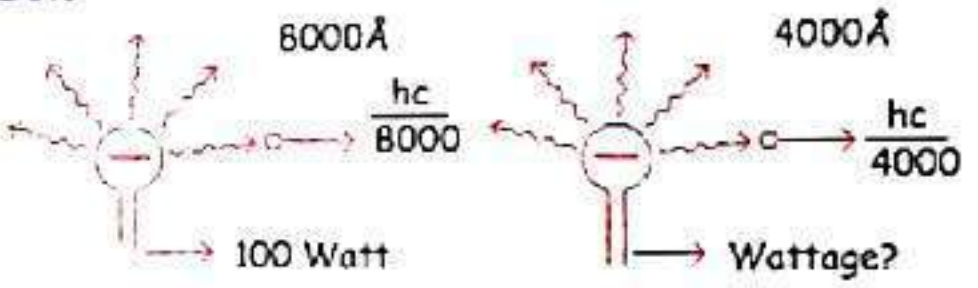
agar unit nahi aati badalni toh beda garg hai

Common Mistake

Students should know the unit conversion.

22. Two bulbs 'A' and 'B' emit red light and yellow light at 8000 Å and 4000 Å respectively. The number of photons emitted by both the bulbs per second is the same. If the red bulb is labelled as 100 watts, find the wattage of the yellow bulb.

Sol.



Let the power of second bulb be W watt.

$$\frac{W}{\left(\frac{hc}{4000}\right)}=\frac{100}{\left(\frac{hc}{8000}\right)}=\text{No. of photons emitted}$$

per second

W = 200 watt

kitna sundar sawaal hai, waah waah

23. How many photons are emitted per second by a 5 mW laser operating at 620 nm ?

Sol. (1.6×10^{16}) For explanation, refer to QR code given on the last page.

24. One quantum is absorbed per gaseous molecule of Br_2 for converting into Br atoms. If light absorbed has wavelength 5000Å, calculate energy required in kJ/Mol.

Sol. Energy required to break one Br - Br $\Rightarrow \frac{hc}{\lambda}$

Per mol dissociation energy = $\left(\frac{hc}{\lambda}\right)N_A$

= 239.3 kJ/mol

Khoobsurat Baat

To calculate per mol, multiply the quantity by N_A.

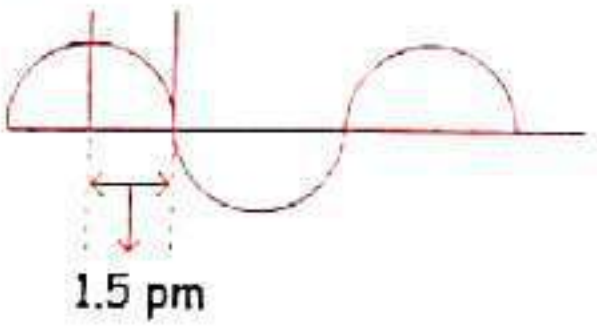
25. The eyes of a certain member of the reptile family pass a visual signal to the brain when the visual recepters are struck by photons of wavelength 850 nm. If a total energy of 3.15×10^{-14} J is required to trip the signal, what is the minimum number of photons that must strike the receptor? ($h = 6.6 \times 10^{-34}$)

Sol. Energy of one photon × no of photons = total energy

$$\frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{850 \times 10^{-9}} \times n = 3.15 \times 10^{-14}$$

$$n = 1.37 \times 10^{5}$$

26. A hypothetical electromagnetic wave is shown below



The frequency of the wave is $x \times 10^{19}$ Hz.

x = _____ (nearest integer)

[08 April, 2024 (Shift-I)]

Sol. $\lambda = 1.5 \times 4$ pm $= 6 \times 10^{-12}$ metre

$\lambda\nu = c$

$$\nu = \frac{3 \times 10^{8}}{6 \times 10^{-12}} = 5 \times 10^{19}\text{Hz}$$

formula hi toh lagaya hai bas

27. Wavenumber for a radiation having 5800 Å wavelength is $x \times 10\ \text{cm}^{-1}$. The value of x is

[08 April, 2024 (Shift-II)]

Sol. $\bar{\nu}$(wave no.) $= \frac{1}{\lambda}$

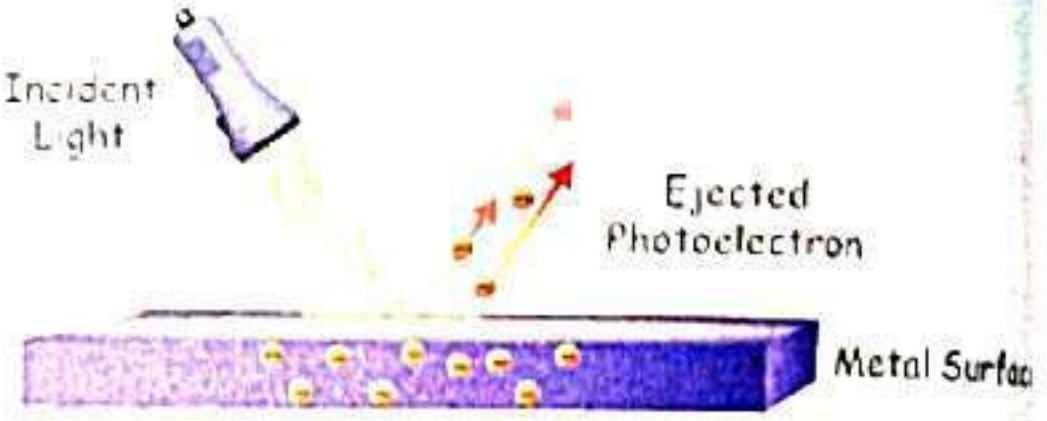
$$\frac{1}{\lambda} = \frac{1}{5800 \times 10^{-8}\text{ cm}} = 17241$$

Common Mistake

Don't forget to convert the wavelength into cm.

PHOTOELECTRIC EFFECT

When certain metals (for example Potassium, Rubidium, Caesium etc.) were exposed to a beam of light electrons were ejected as shown in figure given below:



Work Function

It is defined as the minimum energy required to eject an electron on the surface of metal.

Postulates

(i) The electrons are ejected from the metal surface as soon as the beam of light strikes the surface, i.e., there is no time lag between the striking of light beam and the ejection of electrons from the metal surface.

(ii) The number of electrons ejected is proportional to the intensity or brightness of light. (current)

(iii) For each metal, there is a characteristic minimum frequency, ν_0 (also known as threshold frequency) below which photoelectric effect is not observed. At a frequency $\nu > \nu_0$, the ejected electrons come out with certain kinetic energy. The kinetic energies of these electrons increase with the increase of frequency of the light used.

Important

(a) Current (no of electrons) ∝ Intensity of light (brightness)

(b) K.E of electron (photoelectron) α Energy of incident light.

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$h\nu = W + K.E$

$h\nu$ = energy of incident light photon

W = work funchon

KE = K.E of photoelectron

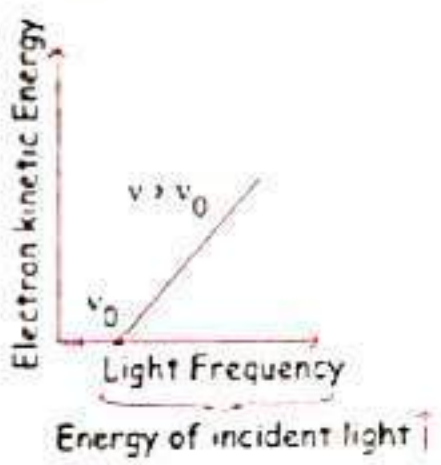
Important

The minimum required frequency of incident Photon is known as threshold frequency, ν_0

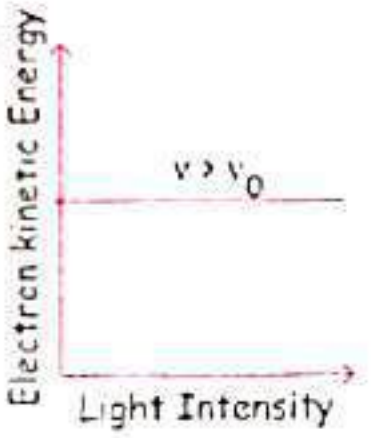
$$h\nu = h\nu_0 + KE$$

(energy of incident photon) (work function)

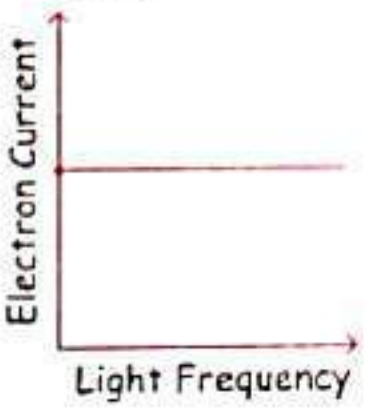
A.



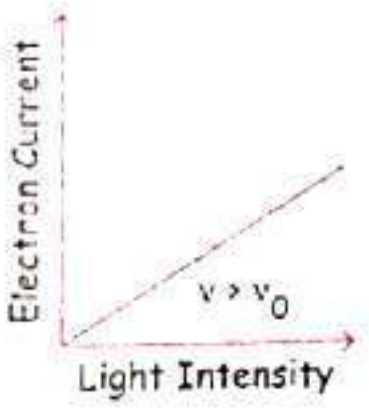
B.



C.



D.



graphs par dhyan lagao ache se

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Photon ↑ no. badhenge toh electron के no. badhenge
(intensity) (current)

Photon ↑ energy badhegi toh electron की energy badhegi
(frequency) (Kinetic energy)

28. The threshold frequency ν_0 for a metal is $6 \times 10^{14}\ s^{-1}$. Calculate the kinetic energy of an electron emitted when radiation of frequency $\nu = 1.1 \times 10^{15}\ s^{-1}$ hits the metal.

Sol. $h\nu = h\nu_0 + KE$, $1eV = 1.6 \times 10^{-19} J$

$6.626 \times 10^{-34} \times 1.1 \times 10^{15} = 6.626 \times 10^{-34} \times 6 \times 10^{14} + KE$

$KE = 33.13 \times 10^{14}\ J$

fomula lagega bas

29. The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is

[Given : The threshold frequency of platinum is $1.3 \times 10^{15}\ s^{-1}$ and $h = 6.6 \times 10^{-34}\ J\ s$.]

[JEE Main 2023 Online 25th June Shift-II]

(A) 3.21×10^{-14} J (B) 6.24×10^{-16} J

(C) 8.58×10^{-19} J (D) 9.76×10^{-20} J

Sol. (C) $W = h\nu_0 = 6.626 \times 10^{-34}\ J - \cancel{Sec} \times 1.3 \times 10^{15}\ \cancel{sec^{-1}}$

$= 8.58 \times 10^{-19}\ J$

30. A light source of wavelength λ illuminates a metal and ejects photoelectrons with $(K.E.)_{max}$ = 1eV. Another light source of wavelength $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(K.E.)_{max}$ = 4eV. Find the value of work function.

(A) 1eV (B) 2 eV

(C) 0.5eV (D) None of these

Sol. (C) Method-1

$$\frac{hc}{\lambda} = W + 1, \quad 3\frac{hc}{\lambda} = W + 4$$

W = 0.5 eV

Method-2

$$\frac{hc}{\lambda} = W + 1, \frac{hc}{\lambda_3} = W + 4 \qquad (1\ \&\ 2)$$

W = 0.5 eV

yeh sawaal mei thoda toh dimag lagega hi

31. If 'p' is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength (λ), then for (1.5p') momentum of the photoelectron, the wavelength of light should be assume KE of ejected photoelectron is very high in comparison to work function)

[JEE Main 2019 Online 08th Apr Shift-II]

(A) $4\lambda/9$ (B) $3\lambda/4$ (C) $2\lambda/3$ (D) $\lambda/2$

Sol. (A) For explanation, refer to QR code given on the last page.

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$K.E = \frac{p^2}{2m}$, p = momentum of body

32. λ_0 is the threshold wavelength of a metal for photoelectron emission. If the metal is exposed to the light wavelength λ, then the velocity of ejected electron will be $\sqrt{\frac{2h}{m}(\lambda_0 - \lambda)K}$ The value of K is:

(A) Speed of light (B) 1

(C) $\frac{c}{\lambda_0\lambda}$ (D) $\frac{1}{\lambda\lambda_0}$

Sol. (C) $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}mv^2$

$\frac{hc}{\lambda\lambda_0}(\lambda_0 - \lambda) = \frac{1}{2}mv^2$, $v = \sqrt{\frac{2hc}{m\lambda\lambda_0}(\lambda_0 - \lambda)}$

3. Values of work function (W_0) for a few metals are given below:

Metal	Li	Na	K	Mg	Cu	Ag
W_0/eV	2.42	2.3	2.25	3.7	4.8	4.3

The number of metals which will show photoelectric effect when light of wavelength 400 nm falls on it is ____

Given: $h = 6.6 \times 10^{-34}$ J s

$c = 3 \times 10^8$ ms^{-1}; $e = 1.6 \times 10^{-19}$ C

[12 April, 2023 (Shift-I)]

$E(eV) = \frac{1240}{400} = 3.1eV$

Mg, Cu, Ag

pehle hi ghabra jate ho yaar

34. If the work function of a metal is 6.63×10^{-19} J, the maximum wavelength of the photon required to remove a photoelectron from the metal is ____ nm. (Nearest integer)

[Given $h = 6.63 \times 10^{-34}$ Js, and $c = 3 \times 10^8$ ms^{-1}]

[28 June, 2022 (Shift-I)]

Sol. Using: $w_0 = h\nu_0 = \frac{hc}{\lambda_0}$ or $\lambda_0 = \frac{hc}{w_0}$

$\lambda_0 = 300 \times 10^{-9}$m = 300 nm

balle balle karo, kitna asaan hai

35. The work function of sodium metal is 4.41×10^{-19}J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be ($h = 6.63 \times 10^{-34}$Js; $c = 3 \times 10^8$ m/s) ____ $\times 10^{-21}$J. **[2 Sept, 2021 (Shift-II)]**

Sol. $(K.E.) = E - E_0 = \frac{hc}{\lambda} - E_0$

$= \frac{6.63 \times 10^{-34}\,J.s \times 3 \times 10^8\,ms^{-1}}{300 \times 10^{-9}\,m} - 4.41 \times 10^{-19}J$

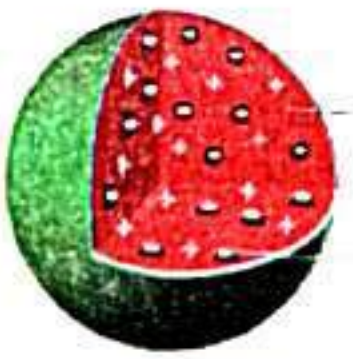
$= 2.22 \times 10^{-19}J = 222 \times 10^{-21}J$

ATOMIC MODELS

Different atomic models were proposed to explain the distributions of these charged particles in an atom.

Thomson Model

J.J. Thomson, in 1898, suggested that an atom has a spherical shape in which the positive charge is uniformly distributed and the embedded number of electrons is sufficient to neutralize the positive charge. According to Thomson, atom is like water melon and electron's are embedded like seeds in watermelon to give most stable electrostatic arrangement.



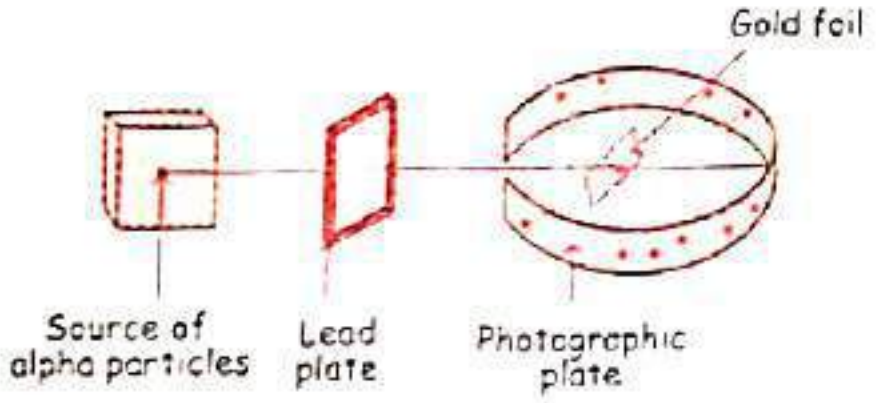
Thomson model of atom

- The model is represented as a pudding or watermelon of positive charge with plums or seeds (electron) embedded into it.
- An important feature of this model is that the mass of the atom is assumed to be uniformly distributed over the atom.
- This model can not explain electrical neutrality of the atom.
- This model also called plum pudding, raisin pudding or watermelon model of atom.

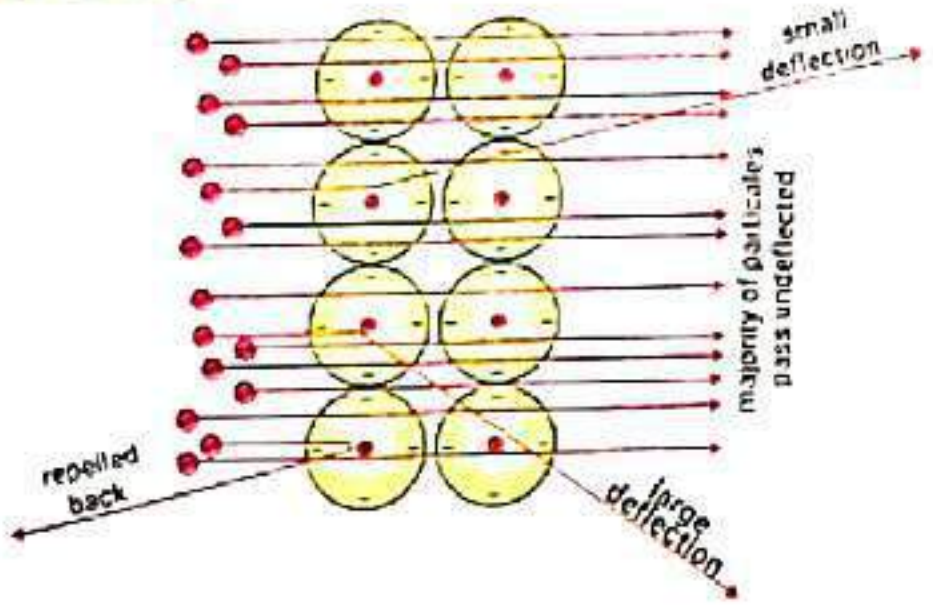
iske sare naam yaad karlo, kuch bhi asakta hai

Rutherford's Nuclear Model of Atom

- Rutherford suggested atomic model based on α-particles scattering experiment.
- A narrow beam of high energy α-particles passed from a thin gold foil or silver foil and it is covered with fluorescent ZnS screen.
- When α-particles struck the screen, then a tiny flash of light was produced at that point.



Note



Schematic molecular view of the gold foil

chakaar khakr mat gir jana

There were certain observation that are observed at time experiment.

Observation

(i) Most of the α-particles passes through the foil goes undeflected.

(ii) A small fraction of α-particles were deflected by angles.

(iii) A very few α-particles deflected back and were deflected by 180°.

Drawbacks of Rutherford's Model

1. This was not according to the classical theory of electromagnetism proposed by Maxwell. According to this theory, every accelerated charged particle must emit radiations in the form of electromagnetic waves and loses it total energy. Since energy of electrons keep on decreasing, so radius of the circular orbits should also decrease and ultimately the electron should fall in nucleus.
2. It could not explain the line spectrum of H-atom.

36. Rutherford's experiment, which established the nuclear model of the atom, used a beam of **[IIT JEE 2002]**

(A) β-particles, which impinged on a metal foil and got absorbed

(B) γ-rays, which impinged on a metal foil and got scattered

(C) helium atoms, which impinged on a metal foil and got scattered

(D) helium nuclei, which impinged on a metal foil and got scattered

Sol. (D) For explanation, refer to QR code given on the last page.

37. Rutherford's alpha particle scattering experiment eventually led to the conclusion that **[IIT JEE 1986]**

(A) mass and energy are related

(B) electrons occupy space around the nucleus

(C) neutrons are buried deep in the nucleus

(D) the point of impact with matter can be precisely determined

Sol. (B) According to the Rutherford's model, the electrons revolve around positively charged core or in other words, negatively charged electrons occupy space around the nucleus and the atom is electrically neutral.

38. Rutherford's scattering experiment is related to the size of the [IIT JEE 1983]

(A) nucleus (B) atom

(C) electron (D) neutron

Sol. (A) Rutherford's discovery of the nucleus of an atom was based on the alpha scattering experiment which means that experiment was also related to the size of the nucleus.

39. Rutherford's on scattering of α-particles showed for the first time that the atom has [IIT JEE 1981]

(A) electrons (B) protons

(C) nucleus (D) neutrons

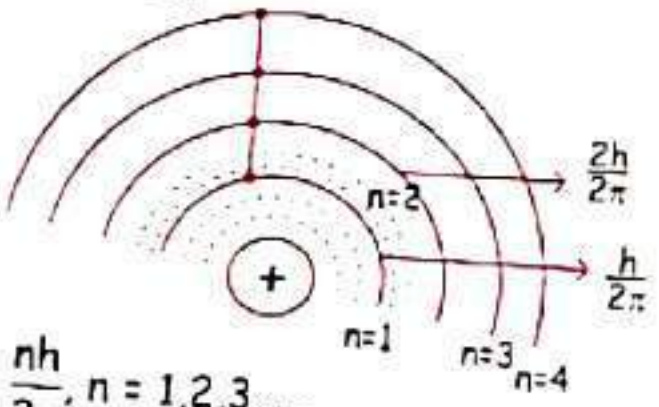
Sol. (C) For explanation, refer to QR code given on the last page.

BOHR'S ATOMIC MODEL

Postulates

1. Electron move in circular orbits in which they do not radiate out energy. These orbits are called stationary orbits.

2. Stationary orbits are the orbit where the angular movements of electron is an integral multiple of $\frac{h}{2\pi}$.



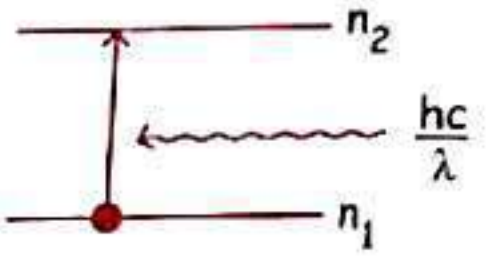
$mvr = \frac{nh}{2\pi}$, n = 1,2,3,.....

for 1st stationary orbit $mvr = \frac{h}{2\pi}$

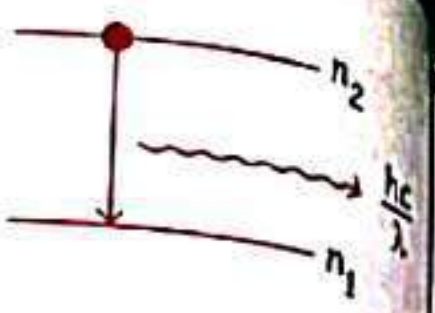
for IInd stationary orbit $mvr = \frac{2h}{2\pi}$

for IIIrd stationary orbit $mvr = \frac{3h}{2\pi}$

On going from lower to higher level, energy is absorbed in the form of photon while coming from higher to lower level, energy is emitted in the form of photon.



$E_{n_2} - E_{n_1} = \Delta E = \frac{hc}{\lambda}$



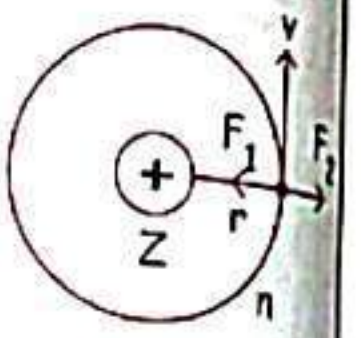
$E_{n_2} - E_{n_1} = \Delta E = \frac{hc}{\lambda}$

Bohr's Theory Parameters

(i) Bohr radius

F_1 = coulombic force $= K\frac{(Ze)(e)}{r^2}$

F_2 = centrifugal force $= \frac{mv^2}{r}$



$K\frac{Ze^2}{r^2} = \frac{mv^2}{r}$ ...(1)

$mvr = \frac{nh}{2\pi}$ ...(2)

$v = \frac{nh}{2\pi mr}$

$K\frac{Ze^2}{r^2} = \frac{mv^2}{r}$

$K\frac{Ze^2}{r} = m\left[\frac{n^2h^2}{4\pi^2m^2r^2}\right]$

$$r = \frac{n^2h^2}{4\pi^2mKZe^2}$$

dar gye kya?

Chalo ab khush ho jao

$h = 6.626 \times 10^{-34}$ J sec, $\pi = 3.14$, $m = 9.1 \times 10^{-31}$ kg

$K = 9 \times 10^9$ Nm^2/C^2, $e = 1.6 \times 10^{-19}$ C

yeh sari values ko jab substitute krenge upar wale formulae mei, dekho kya nikal kar aata hai

$$r_n = 0.529\frac{n^2}{Z}\text{ Å}$$

ab dekho, lagra hai na asaan

i) Velocity of an electron

$$v = \frac{nh}{2\pi mr} = \frac{nh}{2\pi m\left(\frac{n^2h^2}{4\pi^2 mZe^2}\right)} = \frac{2\pi KZe^2}{nh}$$

$$\boxed{v_n = 2.18\times10^6\frac{Z}{n}\,m/sec}$$

ii) Bohr Energy

$$KE = \frac{1}{2}mv^2 \qquad \left(\frac{KZe^2}{r^2} = \frac{mv^2}{r},\ mv^2 = \frac{KZe^2}{r}\right)$$

$$PE = -\frac{K(Ze)(e)}{r} = -K\frac{Ze^2}{r}$$

$$TE = KE + PE$$

$$= \frac{1}{2}mv^2 + \left(-\frac{KZe^2}{r}\right) = \left(\frac{1}{2}\frac{KZe^2}{r}\right) + \left(-\frac{KZe^2}{r}\right)$$

$$T.E = -\frac{1}{2}\frac{KZe^2}{r}$$

1. $KE = |T.E.|$ 2. $2T.E = P.E.$

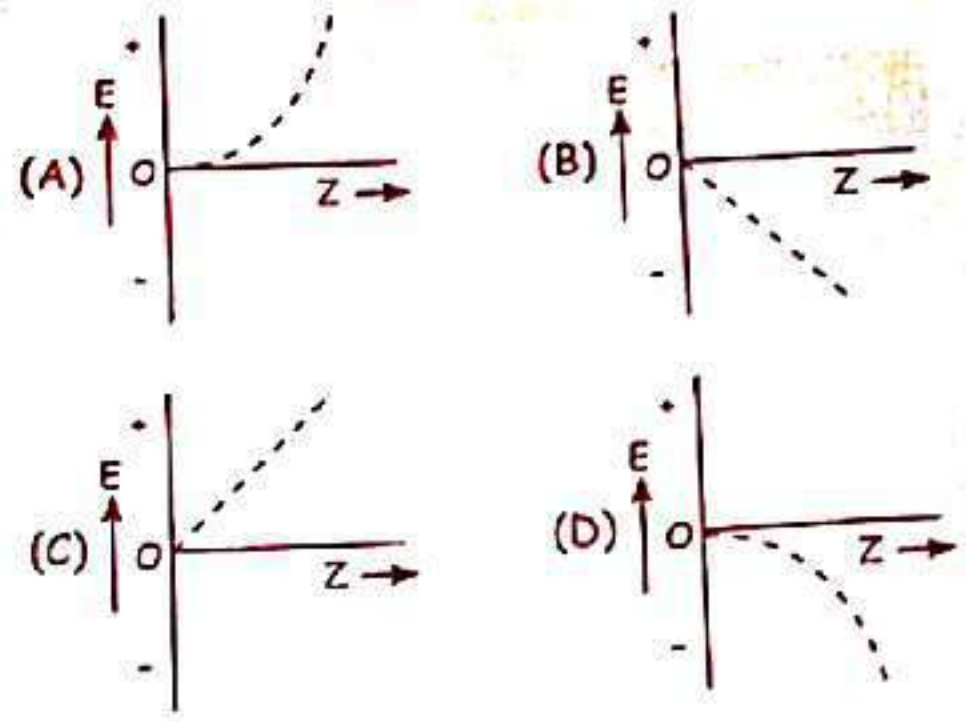
$$T.E. = -\frac{1}{2}\frac{KZe^2}{r}$$

$$= -\frac{1}{2}K\frac{Ze^2}{\frac{n^2h^2}{4\pi^2 mKZe^2}} = \frac{-2\pi^2K^2me^4Z^2}{n^2/h^2}$$

$$\boxed{T.E. = -2.18\times10^{-18}\frac{Z^2}{n^2}J/atom = -13.6\frac{Z^2}{n^2}eV/atom}$$

40. For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n? **[JEE Mains Shift-II, 2025]**

[E: Energy of the stationary state, Z: atomic number, n = principal quantum number]



Sol. (D) The energy of a hydrogen-like species follows $E_n = -\frac{13.6Z^2}{n^2}$, leading to a parabolic downward curve as Z increases. Option (D) correctly represents this trend.

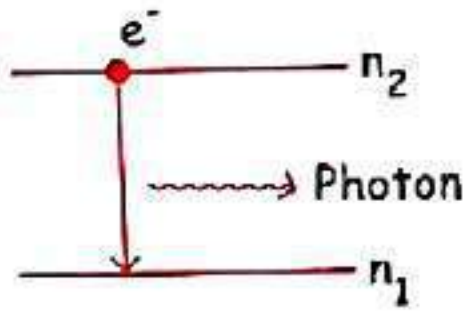
$$r_n = 0.529\frac{n^2}{Z}\text{Å}$$

$$v_n = 2.18\times10^6\frac{Z}{n}m/sec$$

$$T.E. = 13.6\frac{Z^2}{n^2}\ eV/atom$$

Important

$$\Delta E = \frac{hc}{\lambda} = E_{n_2} - E_{n_1}$$



$$E_{n_2} = -\frac{2\pi^2mK^2e^4Z^2}{h^2n_2^2}$$

$$E_{n_1} = -\frac{2\pi^2mK^2e^4Z^2}{h^2n_1^2}$$

$$E_{n_2} - E_{n_1} = \frac{hc}{\lambda} = \frac{2\pi^2mK^2e^4}{h^2}Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda} = \left(\frac{2\pi^2mK^2e^4}{h^3c}\right)Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Rydberg constant(R_H) = 109677 cm^{-1}

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$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

λ = wavelength of a photon emitted when an electron jumps from $n_2 \rightarrow n_1$ in a single electron system having atomic number 'Z'

Yeh formula yaad rakho, sawaal jaldi solve hoga

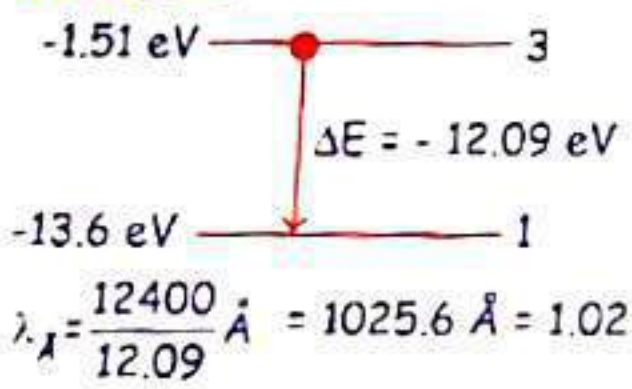
41. Find out the W.L of a photon emitted during the transition of an electron from $3 \rightarrow 1$ in H - atom

Sol. Method-1 $\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$$\frac{1}{\lambda} = 109677\,cm^{-1} \times 1^2\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = 97491\,cm^{-1}$$

$\lambda = 1.02 \times 10^{-5}$ cm

Method-2



$$\lambda_A = \frac{12400}{12.09}\,\text{Å} = 1025.6\,\text{Å} = 1.02 \times 10^{-5}\,cm$$

sara khel samajhne ka hai sawaal ko

42. Find out the difference in PE of electron moving in 2nd and 1st orbit of H - atom

Sol. $(PE)_2 = 6.8$ eV, $(PE)_1 = -27.2$ eV

$(PE)_2 - (PE)_1 = -6.8 - (-27.2) = 20.4$ eV

(iv) Time period calculation

Time taken in one revolution (T)

$$T = \frac{2\pi r}{v} = \frac{2\pi\left(0.529\frac{n^2}{Z} \times 10^{-10}\,m\right)}{2.18 \times 10^6 \frac{Z}{n}\,m/sec}$$

$$= \frac{2\pi \times 0.529 \times 10^{-10}}{2.18 \times 10^6}\left(\frac{n^3}{Z^2}\right) sec$$

$$\boxed{T \propto \frac{n^3}{Z^2}}$$

43. If the time period of an electron moving in 3rd orbit of He$^+$ is 'T' then find out the time period of an electron moving in 2nd orbit of Li^{2+} ion

Sol. $T \propto \frac{n^3}{Z^2}$

$$\frac{(T_{He^+})_3}{(T_{Li^{2+}})_2} = \frac{\frac{(n_{He^+})^3}{(Z_{He^+})^2}}{\frac{(n_{Li^{2+}})^3}{(Z_{Li^{2+}})_2}} \quad \frac{3^3 3^2}{2^2 \cdot 2^3} = \frac{3^5}{2^5}, \quad (T_{Li^{2+}})_2 = \frac{2^5 T}{3^5}$$

Kaahe ghabra rahe ho bacho

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$$\frac{T_1}{T_2} = \frac{n_1^3 \times Z_2^2}{n_2^3 \times Z_1^2}$$

(v) Frequency

$$f = \frac{1}{T} = \frac{v}{2\pi r} = \frac{2.18 \times 10^6 \frac{Z}{n}\,m/sec}{2\pi\left(0.529\frac{n^2}{Z} \times 10^{-10}\,m\right)}$$

$$\boxed{f \propto Z^2/n^3}$$

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$$\frac{f_1}{f_2} = \frac{Z_1^2 \cdot n_2^3}{Z_2^2 \cdot n_1^3}$$

Isko baitthalo dimag ke kisi kone mei

44. $He^+ \rightarrow 2^{nd}$ orbit $\rightarrow f = \sigma$

$H \rightarrow 4^{th}$ orbit $\rightarrow f = ?$

Sol. $f \propto \frac{Z^2}{n^3}$

$$\frac{f_1}{f_2} = \frac{Z_1^2 \cdot n_2^3}{Z_2^2 n_1^3} = \frac{2^2 \cdot 4^3}{1^2 \cdot 2^3}, \quad f_2 = \frac{\sigma}{32}$$

(vi) Acceleration

$$a = \frac{V}{T},\ a \propto \frac{Z^3}{n^4}$$

45. $Li^{+2} \rightarrow 1^{st}$ orbit $\rightarrow$ a

$He^{+} \rightarrow 1^{st}$ orbit $\rightarrow$ acel^n = ?

Sol. $\frac{a}{a'} = \frac{3^3 1^4}{2^3 \cdot 1^4}, a' = \frac{8a}{27}$

simple hi toh hai

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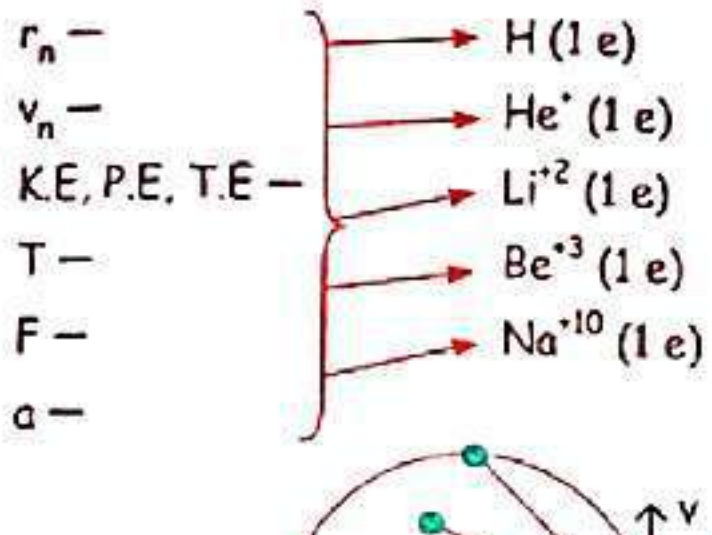
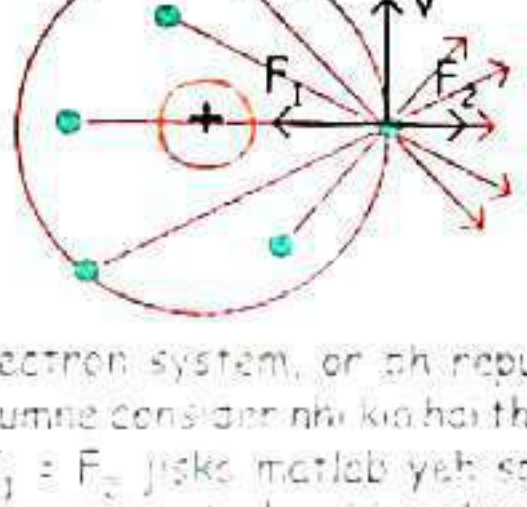
Sab Kuch is valid for single electron system or H- like species.





For a mutielectron system, or bh repulsive forces honge jinko humne consider nhi kia hai that is we have considered $F_1 = F_2$ jiske matlab yeh sare formulae single electron system ke liye inapplicable honge.

ENERGY LEVEL DIAGRAM

According to Bohr's atomic model, electrons occupy specific energy levels, or shells, around the nucleus. The energy level diagram illustrates these discrete energy levels, with the level (n = 1) being the ground state and higher levels (n > 1) being excited states. As electrons jump from one energy level to another, they either absorb or emit photons, resulting in characteristic spectral lines. The energy differences between levels determine the frequencies of these spectral lines, providing a key experimental verification of Bohr's theory.

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$$E_n = -13.6\frac{Z^2}{n^2}\text{ eV}$$

0 eV ..... n = ∞

.....

- 0.28 Z^2eV ..... n = 7
- 0.38 Z^2eV ..... n = 6
- 0.544 Z^2eV ..... n = 5
- 0.85 Z^2eV ..... n = 4
- 1.51 Z^2eV ..... n = 3
- 3.4 Z^2eV ..... n = 2
- 13.6 Z^2eV ..... n = 1

yeh yaad karna hai tum soch bhi nhi iska fayda yaad karne ka

46. Calculate the wavelength in Angstroms of the photon that is emitted when an electron in the Bohr orbit n = 2, returns to the orbit, n = 1, in the hydrogen atom. The ionization potential of the ground state hydrogen atom is 2.17×10^{-11} erg per atom. [IIT-JEE 1982]

Sol. (1220 Å) For explanation, refer to QR code given on the last page.

Common Mistake

Don't forget to calculate the value of wavelength in angstroms.

47. Radius of the first excited state of Helium ion is given as: $a_0 \rightarrow$ radius of first stationary state of hydrogen atom.

(A) $r = 2a_0$ (B) $r = 4a_0$

(C) $r = \frac{a_0}{2}$ (D) $r = \frac{a_0}{4}$

Sol. (A) For explanation, refer to QR code given on the last page.

48. In a certain electronic transition in the hydrogen atom from an initial state 'i' to a final state 'f' the difference in the orbit radius $(r_i - r_f)$ is seven times the Bohr radius. Identify the transition.

(A) $4 \to 1$ (B) $4 \to 2$ (C) $4 \to 3$ (D) $3 \to 1$

Sol. (C) Radius of 1st orbit of H-atom = $0.529\frac{1^2}{1}$ Å

$r_i - r_f = 7 \times 0.529$

$$\frac{0.529n_i^2}{1} - \frac{0.529n_f^2}{1} = 7 \times 0.529$$

$$(n_i^2 - n_f^2) = 7, \quad (n_i + n_f)(n_i - n_f) = 7$$

So, $n_i = 4, n_f = 3$

Isme hit and trial khelna hai bss or kuch nhi karna

49. Which state of the triply ionized Beryllium (Be^{3+}) has the same orbit radius as that of the ground state of hydrogen atom?

Sol. The expression for the radius of Hydrogen and H-like atoms is:

$$r_n = \frac{n^2h^2}{4\pi^2Kme^2} \text{ and } r_n = \frac{1}{Z}\frac{n^2h^2}{4\pi^2Kme^2}$$

For $Z = 1$, $r_1 = r_n$, $\frac{h^2}{4\pi^2Kme^2} = \frac{1}{4}\frac{n^2h^2}{4\pi^2Kme^2}$

or $n^2 = 4$, $\therefore n = 2$

50. If the velocity of the electron in first orbit of H atom is 2.18×10^6 m/s, what is its value in third orbit?

Sol. Velocity = $2.18 \times 10^6 \, Z/n$

$= 2.18 \times 10^6 (1/3) = 0.727 \times 10^6 = 7.27 \times 10^5$ m/s

51. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius]: **[IIT JEE 2012]**

(A) $\frac{h^2}{4\pi^2ma_0^2}$ (B) $\frac{h^2}{16\pi^2ma_0^2}$

(C) $\frac{h^2}{32\pi^2ma_0^2}$ (D) $\frac{h^2}{64\pi^2ma_0^2}$

Sol. (C) $mvr = \frac{nh}{2\pi}$, $v = \frac{nh}{2\pi mr}$

$$K.E = \frac{1}{2}mv^2 = \frac{1}{2}m\left[\frac{n^2h^2}{(4\pi^2m^2r^2)}\right]$$

$$K.E = \frac{n^2h^2}{8\pi^2mr^2} = \frac{h^2}{32\pi^2ma_0^2}$$

Time lagega but ajaega samajh, tension not

52. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity of electron in 3rd orbit of Li^{+2}

(A) V (B) V/3 (C) 3V (D) 9V

Sol. (A) Velocity $\propto Z/n$

$$\frac{V_{Li^{2+}}}{V_H} = \frac{Z_1/n_1}{Z_2/n_2} = \frac{3/3}{1/1} = 1$$

53. Match the following.

(a) Energy of ground state of He^+ (i) +6.04 eV

(b) Potential energy of I orbit of H-atom (ii) -27.2 eV

(c) Kinetic energy of II excited state of He^+ (iii) 54.4 eV

(d) Ionisation potential of He^+ (iv) -54.4 eV

(A) a - (i), b - (ii), c - (iii), d - (iv)

(B) a - (iv), b - (iii), c - (ii), d - (i)

(C) a - (iv), b - (ii), c - (i), d - (iii)

(D) a - (ii), b - (iii), c - (i), d - (iv)

Sol. (A) For explanation, refer to QR code given on the last page.

54. S_1: Potential energy of the two opposite charge system increases with the decrease in distance.

S_2: When an electron make transition from higher orbit to lower orbit it's kinetic energy increases.

S_3: When an electron make transition from lower energy to higher energy state its potential energy increases.

S_4: 1 eV photon can free an electron from the 1st excited state of He^+ ion.

(A) T T T T (B) F T T F

(C) T F F T (D) F F F F

Sol. (B) Among the given statements. S_1 and S_4 are incorrect.

Page ke kone me karke dekho

55. Choose the correct relations on the basis of Bohr's theory

(A) Velocity of electron $\propto \frac{1}{n}$

(B) Frequency of revolution $\propto \frac{1}{n^3}$

(C) Radius of orbit $\propto n^2Z$

(D) Electrostatic force on electron $\propto \frac{1}{n^4}$

Sol. (C) Radius $\propto \frac{n^2}{Z}$

Common Mistake

To solve this problem, students should know the formulas of all terms that come under Bohr's theory.

56. Find out the WL of a photon emitted during the transition of an electron from n = 5 to n = 2 in H-atom.

Sol. The formula to calculate wavelength is:

$$\frac{1}{\lambda} = 109,677\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$\lambda = 4.34 \times 10^{-5}$ cm

Kch bhi nahi kia bas formula lga dia seedha

57. The ionization energy of sodium in kJ mol^{-1}. If electromagnetic radiation of wavelength 242 nm is just sufficient to ionize sodium atom is _____ [31 Jan, 2024 (Shift-1)]

Sol. Given: Wavelength (in nm) = 242 nm

$$E = \frac{hc}{\lambda} = \frac{4.1356 \times 10^{-15}\ eVs \times 3 \times 10^{8}\ ms^{-1}}{242 \times 10^{-9}\ m}$$

$= 5.1239 \times 1.6 \times 10^{-19}$ J/atom

(Since, 1 eV = 1.6×10^{-19} J/atom

$= 8.198 \times 10^{-19}$ J/atom = 494 KJ/mol

58. Number of spectral lines obtained in He$^+$ spectra When an electron makes transition from fifth excited state to first excited state will be _____

[30 Jan, 2024 (Shift-II)]

Sol. For 5th excited state $\Rightarrow n_1 = 6$

For 1st excited state $\Rightarrow n_2 = 2$

$\Delta n = n_1 - n_2 = 6 - 2 = 4$

Formula for maximum number of spectral lines

$$= \frac{\Delta n(\Delta n + 1)}{2} = \frac{4(4+1)}{2} = 10$$

Common Mistake

It is important to know the formula to calculate maximum number of spectral lines.

59. The value of Rydberg constant (R_H) is 2.18×10^{-18} J. The velocity of electron having mass 9.1×10^{-31} kg in Bohr's first orbit of hydrogen atom = ____ $\times 10^5$ ms^{-1} (nearest integer) [05 April, 2024 (Shift-I)]

Sol. [22] For H-atom, n = 1, Z = 1

$$V = 2.18 \times 10^6 \times \frac{Z}{n} = 21.8 \times 10^5 \times \frac{1}{2} \approx 22 \times 10^5$$

60. For hydrogen atom, energy of an electron in first excited state is -3.4eV. K.E. of the same electron of hydrogen atom is xeV. Value of x is _____ $\times 10^{-1}$ eV (Nearest integer).

[06 April, 2024 (Shift-II)]

Sol. Given: $E_2 = -3.4$ eV

K.E = $-E_2 = 3.4$ eV = 34×10^{-1} eV

Hence, Value of x = 34×10^{-1} eV

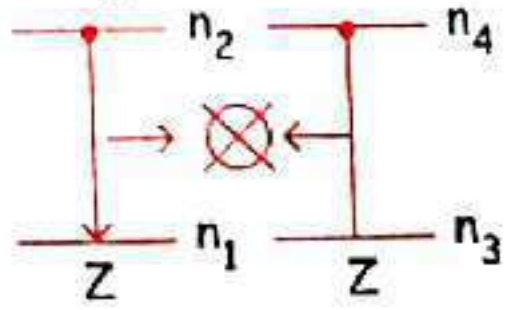
Kyu dimag ko ghuma rahe ho

EQUI-ENERGY TRANSITION

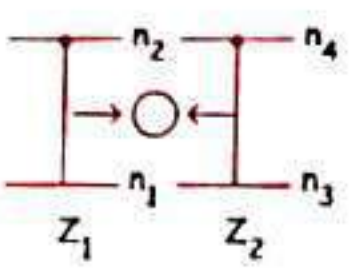
An equi-energy transition refers to a transition between two energy levels that have the same energy difference as the energy of the incident photon. This results in the emission or absorption of a photon, where the energy of the photon matches the energy difference the two levels.

Important

For a given same H-like sample, there will be not any equi-energy transition of an electron.



For a same H-like sample, electrons ke 2 alag alag transitions same energy ke nahi ho sakte lekin 2 alag alag H-like samples ke andar different transitions ki energy same ho sakti hai



$$\left(\frac{-13.6Z_1^2}{n_2^2}\right)-\left(\frac{-13.6Z_1^2}{n_1^2}\right) = \text{energy of photon emitted}$$

$$\left(-\frac{13.6Z_2^2}{n_4^2}\right)-\left(\frac{-13.6Z_2^2}{n_3^2}\right) = \text{energy of photon emitted}$$

Note

For equi-energy

$$-\frac{13.6Z_1^2}{n_2^2}+\frac{136Z_1^2}{n_1^2}=\frac{-13.6Z_2^2}{n_4^2}+\frac{136Z_2^2}{n_3^2}$$

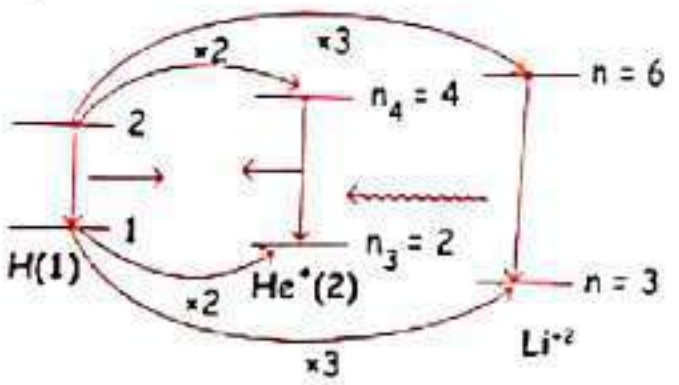
$$\frac{Z_1^2}{n_1^2}-\frac{Z_1^2}{n_2^2}=\frac{Z_2^2}{n_3^2}-\frac{Z_2^2}{n_4^2}$$

$$\frac{Z_1}{n_1}=\frac{Z_2}{n_3} \quad \text{and} \quad \frac{Z_1}{n_2}=\frac{Z_2}{n_4}$$

$$\boxed{n_3=\left(\frac{Z_2}{Z_1}\right)n_1 \quad \& \quad n_4=\left(\frac{Z_2}{Z_1}\right)n_2}$$

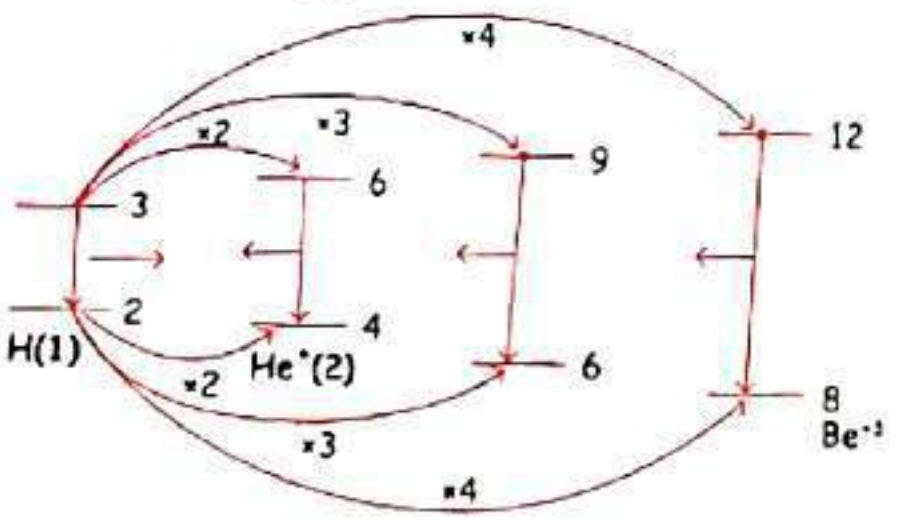
Example:

(1)



2 - 1 transition of H is equi-energy with 4 - 2 transition of He^+ in equi-energy transition with Li^{2+}.

2)



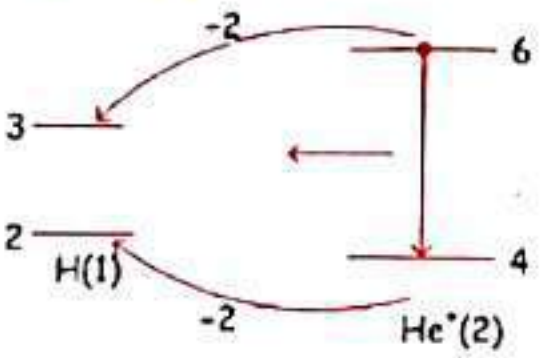
$= 3 \rightarrow 2$, $He^+ = 6 \rightarrow 4$, $Li^{+2} = 9 \rightarrow 6$

Khoobsurat Baat

Hydrogen emission spectrum is always a subset of any H-like species emission spectrum.

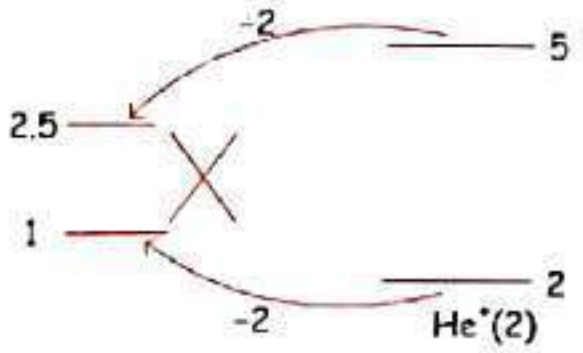
Important

(1)



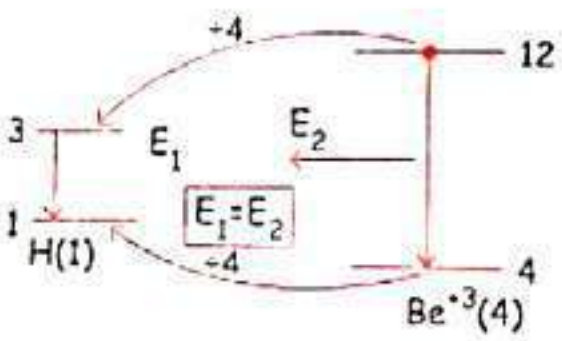
Transition of He^+ $(6 \rightarrow 4)$ is equi-energy with transition of H $(3 \rightarrow 2)$

(2)

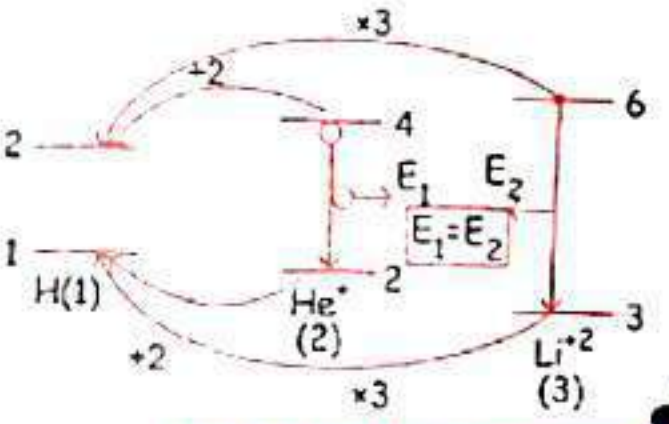


Integer/integer= Integer or Fraction He^+ mei 5 se 2 ka transition H mei exist nhi krega

(3)

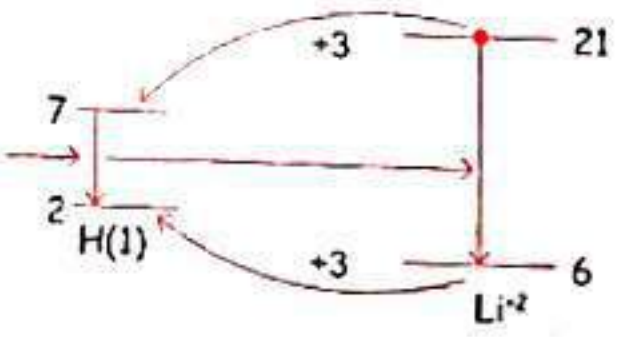


(4)



61. Find out WL of photon emitted during a transition of an electron from $21 \rightarrow 6$ in Li^{+2} ion sample.

Sol.



Energy at n = 7 is 0.28 eV and at n = 2 is 3.40 eV.

$\Delta E = (3.40 - 0.28)\ eV = 3.12\ eV$

Now, $\lambda = \frac{12400}{3.12} = 3974.3 Å$

dekha kitna asaan ho gaya energy level diagram ko yaad karna

62. Find out the wavelength of a photon emitted during a transition of an electron from n = 4 to n = 2 in He^+ ion.

(A) 1216 Å (B) 2359 Å

(C) 1240 Å (D) 12160 Å

Sol. (A) $He^+_{(4 \to 2)} = H_{(2 \to 1)}$

Energy at n = 2 is -3.4 eV and for n = 1 is -13.6 eV.

$\Delta E = (-3.4 + 13.6)\ eV = 10.2\ eV$

$\lambda_\lambda = \frac{12400}{3.12} = 1216 Å$

63. Find out the potential energy of an electron moving in 28th orbit of Be^{3+} ion.

[JEE Mains - 2024]

(A) -0.56 eV (B) 0.56 eV

(C) -0.28 eV (D) 0.28 eV

Sol. (A) For Be^{3+}, Z = 4, n = 28(given)

$E_4 = -13.6 \times \frac{4^2}{28^2}\ eV = 0.28\ eV$

$TE = Be^{+3}_{28} = H_7 = -0.28\ eV$

$PE = 2TE = -0.28 \times 2 = -0.56\ eV$

aankh band karke prabhu ko yaad karo pehle

DEFINITION VALID FOR SINGLE ELECTRON SYSTEM

(i) Ground State: (n = 1)

Lowest energy state of any atom or ion is called ground state of the atom it is n = 1.

Ground state energy of H-atom = -13.6 eV

Ground state energy of He^+ ion = -54.4 eV

Important

$(E_n) = -13.6 \frac{Z^2}{n^2}\ eV$

For ground level n = 1, $E_1 = -13.6\ Z^2\ eV$

(ii) Excited State

States of atom other than the ground state are called excited states :

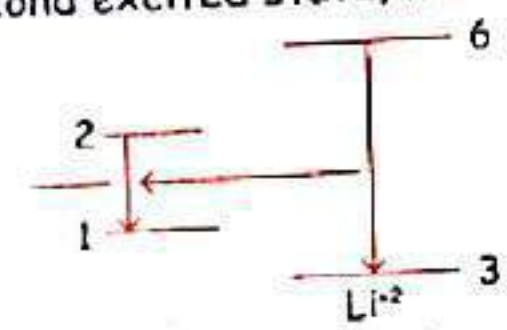
n = 2	first excited state
n = 3	second exited state
n = 4	third excited state
n = n + 1	n^{th} excited state

Common Mistake

n = 1 is not an excited state.

64. Find out the energy of photon excited during the transition of electron from 6th level to 2nd excited state in Li^{+2} ion sample.

Sol. For second excited state, n = 3



Energy at n = 2 is -3.4 eV and at n = 1 is -13.6 eV.

$\Delta E = -3.4 + 13.6 = 10.2\ eV$

dekho kitna asaan bna dia humne isko

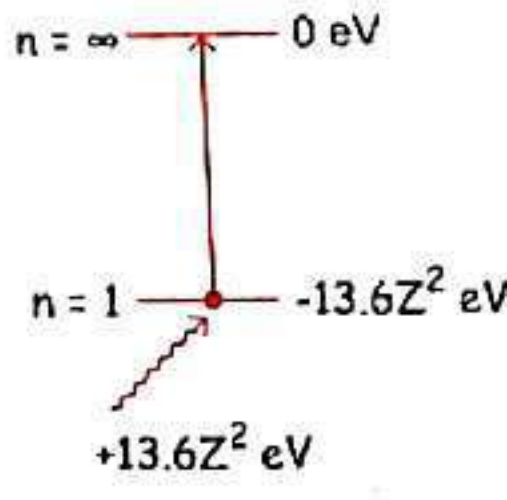
(iii) Ionisation Energy (IE)

Minimum energy required to move an electron from ground state to n = ∞ is called ionisation energy of the atom or ion.

Ionisation energy of H-atom = 13.6 eV

Ionisation energy of He^+ ion = 54.4 eV

Ionisation energy of Li^{+2} ion = 122.4 eV



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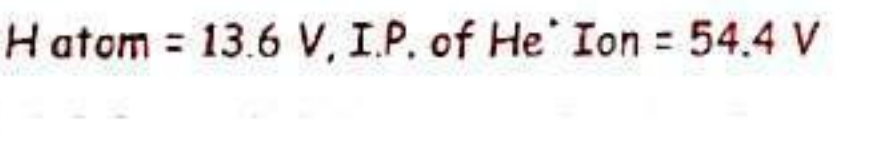
I.E. = $13.6Z^2$ eV

65. Find out the stopping potential of an electron ejected from He^+ ion when irradiated with 60.4 eV photon.

Sol. First calculate energy:

$E_1 = -13.6\ Z^2$ eV $= -13.6 \times 2^2$ eV

$= -54.4$ eV



The KE of e^- = 6 eV

So, the stopping potential = 6 eV

Thoda sa dimag lagega bas

(iv) Ionisation Potential (I.P.)

Potential difference through which a free electron must be accelerated from rest, such that its kinetic energy becomes equal to ionization energy of the atom is called ionization potential of the atom

Khoobsurat Baat

I.P. of H atom = 13.6 V, I.P. of He^+ Ion = 54.4 V

66. If the I.P of a single electron system = 122.9 V then find out the P.E of on electron moving in its 14^{th} excited level?

Sol. For 14^{th} excited state, n = 15

I.P = $13.6\ Z^2$ volt, $13.6\ Z^2 = 122.9$

$Z = 3$ (It is a Li^{2+} ion)

T.E. = $H_5 = -0.544$ eV

$PE = -2 \times 0.544 = -1.08$ eV

ghabrana mat bilkul bhi, okay bacho

Khoobsurat Baat

I.E = $+13.6Z^2$ eV, I.P = $13.6\ Z^2$ V

Common Mistake

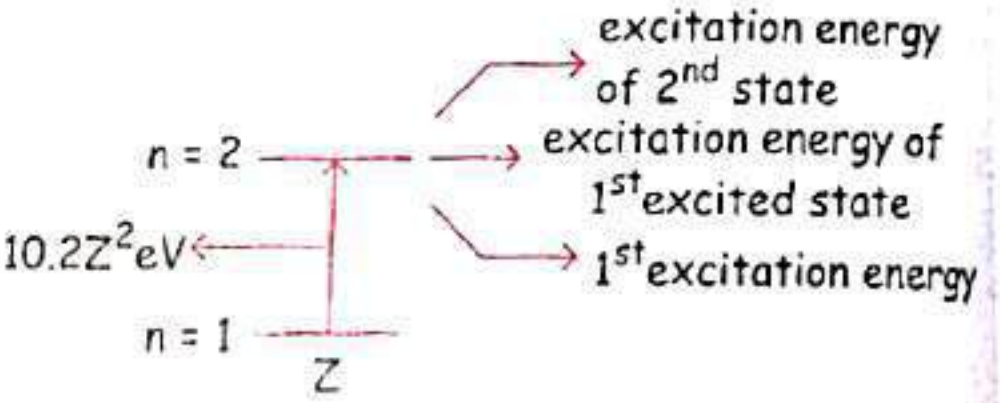
For 14^{th} excited state, students might take the value of n as 14.

(v) Excitation Energy

Energy required to move an electron from ground state of the atom to any other state of the atom is called excitation energy of that state.

Important

Excitation energy of 2^{nd} state = excitation energy of 1^{st} excited state = 1^{st} excitation energy = 10.2 eV.



67. The excitation energy of first excited state of a hydrogen like atom is 40.8 eV. Find the energy needed to remove the electron to form the ion.

Sol. First excited state means $n_1 = 2$, $n_2 = 1$

Energy at n = 2 is -3.4 eV and at n = 1 is -13.6 eV.

$\Delta E = -3.4 + 13.6 = 10.2$ eV

Now, $10.2\ Z^2 = 40.8$

On solving, Z = 2 which means ion will be He^+ ion

IE = $13.6\ Z^2 = 54.4$ eV

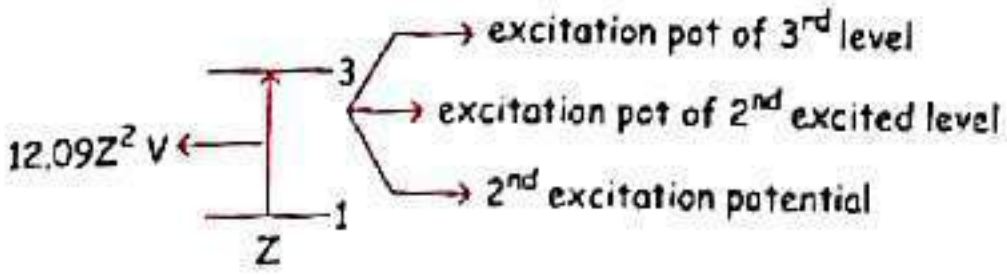
sawaal ko padho, samjho, ho jaega

(vi) Excitation Potential

Potential difference through which an electron must be accelerated from rest to so that its kinetic energy become equal to excitation energy of any state is called excitation potential of that state.

Excitation potential of third state = excitation potential of second excited state = second excitation potential = 12.09 V.

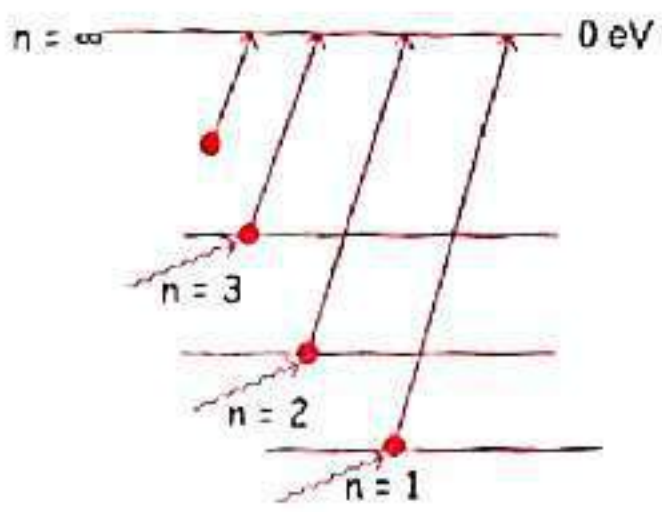
Note



(vii) Binding Energy 'or' Separation Energy

Energy required to move an electron from any state to n = ∞ is called binding energy of that state.

Binding energy of ground state = I.E. of atom or Ion.



Khaobsurat Baat

B.E of n = 1 = 13.6 Z^2 eV (I. E). B.E of n = 2 = 3.4 Z^2 eV

B.E of n = 3 = 1.51 Z^2 eV B.E of n = 4 = 0.85 Z^2 eV

68. The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is (are)

[IIT-JEE 1998]

(A) - 3.4 eV (B) - 4.2 eV

(C) - 6.8 eV (D) -1.51 eV

Sol. (A) $E_n = \frac{-13.6}{n^2} Z^2 eV$

For n = 2, $E = \frac{-13.6}{2^2} 1^2 eV = -3.4eV$

69. If the an electron jumps from an orbit having B.E = 1.51 eV (H = 3) to an orbit where B.E = 13.6 eV in a H-like system whose ionization energy = 122.9 eV, find out the change in orbit angular momentum when electron makes this transition.

Sol. $\frac{9h}{2\pi} - \frac{3h}{2\pi} \Rightarrow \frac{6h}{2\pi} = \frac{3h}{\pi}$

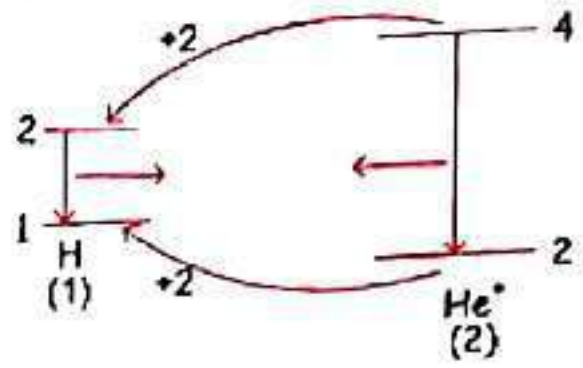
70. Which transition in the hydrogen spectrum would have the same wavelength as of the Balmer type transition from n = 4 to n = 2 of He^+ spectrum?

[JEE Main 2023 Online 31st Jan Shift-I]

(A) n = 3 to n = 4 (B) n = 2 to n = 1

(C) n = 1 to n = 2 (D) n = 1 to n = 3

Sol. (C)



pagla toh nhi gaye

71. Which transition in Li^{2+} would have the same wavelength as the 2 → 4 transition in He^+ ion?

(A) 4 → 2 (B) 2 → 4

(C) 3 → 6 (D) 6 → 2

Sol. (C) For explanation, refer to QR code given on the last page.

72. For He^+, transition takes place from the orbit of radius 105.8 pm to the orbit of radius 26.45 pm. The wavelength (in nm) of the emitted photon during the transition is _____.

[JEE Adv. 2023 Paper-2 Online]

[Use: Bohr radius, a = 52.9 pm

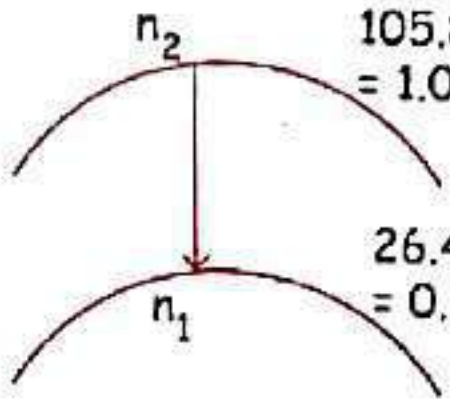
Rydberg constant, $R_H = 2.2 \times 10^{-18}$]

Sol. $r_n = 0.529 \frac{n^2}{Z} Å$

$n_2 = 2$

$0.2645 = \frac{0.529 n_1^2}{2}$

$n_1 = 1$

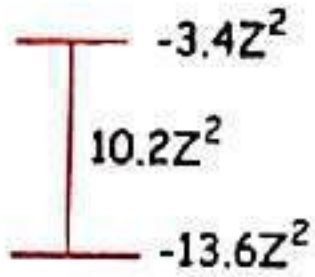


In terms of eV:

$10.2\ Z^2 = 10.2 \times 2^2$

$= 40.8$ eV

$\lambda_{nm} = \frac{1240}{40.8} = 300nm$



Thoda hai mushkil lekin practice karte raho

73. In an atom, two electrons move round the nucleus in circular orbits of radius R and 4R. The ratio of the time taken by them to complete one revolution is (Consider Bohr model to be valid)

(A) 1:4 (B) 4:1

(C) 1:8 (D) 8:1

Sol. (C) $\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{R}{4R}\right)^3 = \frac{1}{64}$ or $\frac{T_1}{T_2} = \frac{1}{8}$

bilkul seedha sawaal hai

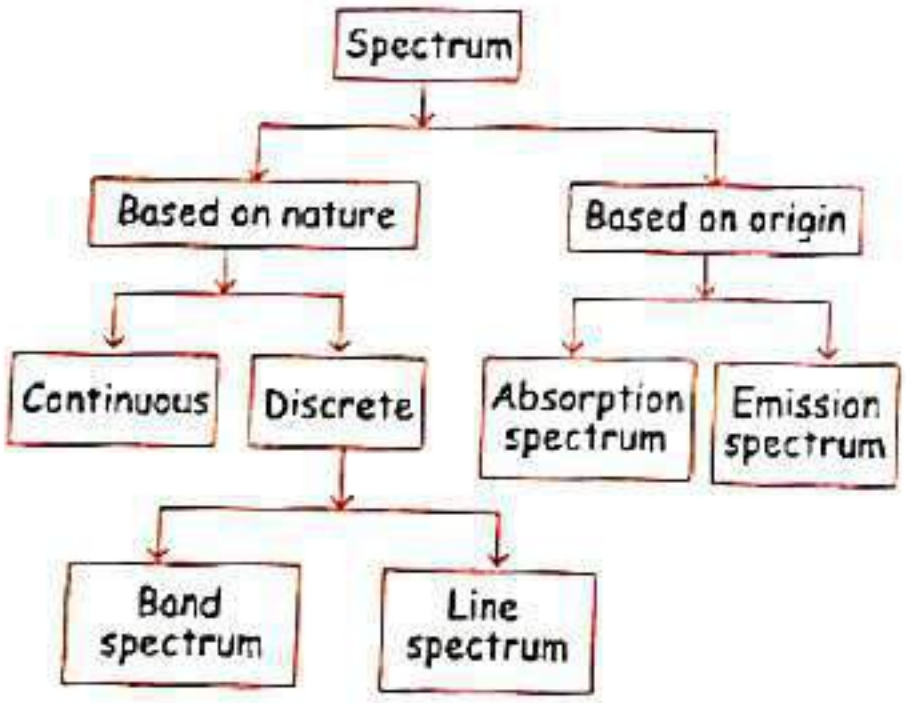
74. The angular momentum of an electron in a given orbit is J, its kinetic energy will be

(A) $\frac{1}{2}\frac{J^2}{mr^2}$ (B) $\frac{Jv}{r}$ (C) $\frac{J^2}{2m}$ (D) $\frac{J^2}{2\pi}$

Sol. (A) $mvr = J,\ v = \frac{J}{mr}, \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{J}{mr}\right)^2 = \frac{1}{2}\frac{J^2}{mr^2}$

SPECTRUM

Note

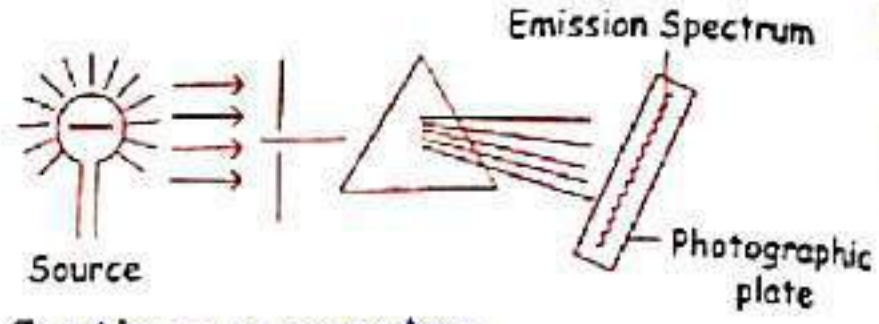


yeh ek baar dekhlo dhyan se kaafi rahega

Emission Spectra

When the radiation emitted from some source e.g. from the sun or by passing electric discharge through a gas at low pressure or by heating some substance to high temperature etc, is passed directly through the prism and then received on the photographic plate, the spectrum obtained is called 'Emission spectrum'.

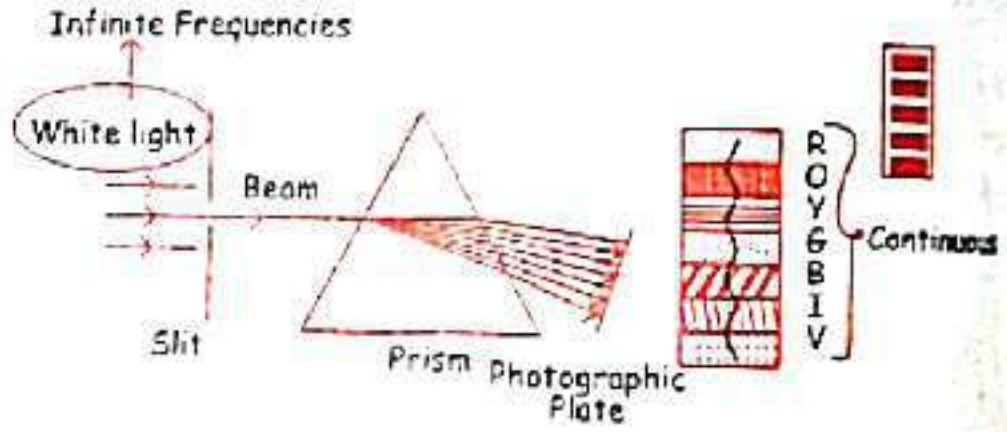
It is of two types- continuous spectra & discrete spectra



(a) Continuous spectra

When white light from any source such as sun, a bulb or any hot glowing body is analysed by passing through a prism it is observed that it splits up into seven different wide band of colours from violet to red. These colours are so continuous that each of them merges into the next. Hence the spectrum is called continuous spectrum.

Important

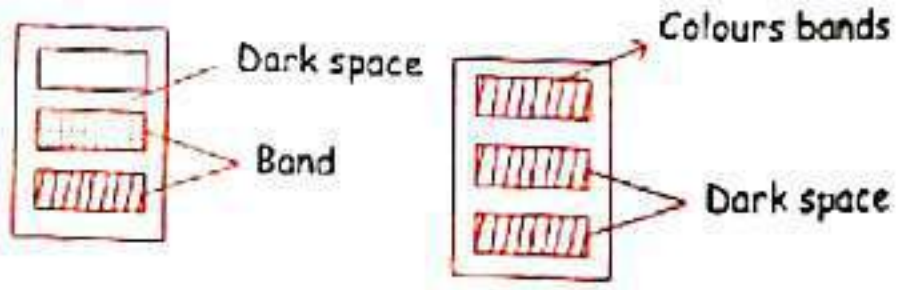


Khoobsurat Baat

White light has infinite frequencies.

(b) Discrete spectra: It is of two types

(i) Band spectrum



Band spectrum contains colourful continuous bands separated by some dark space.

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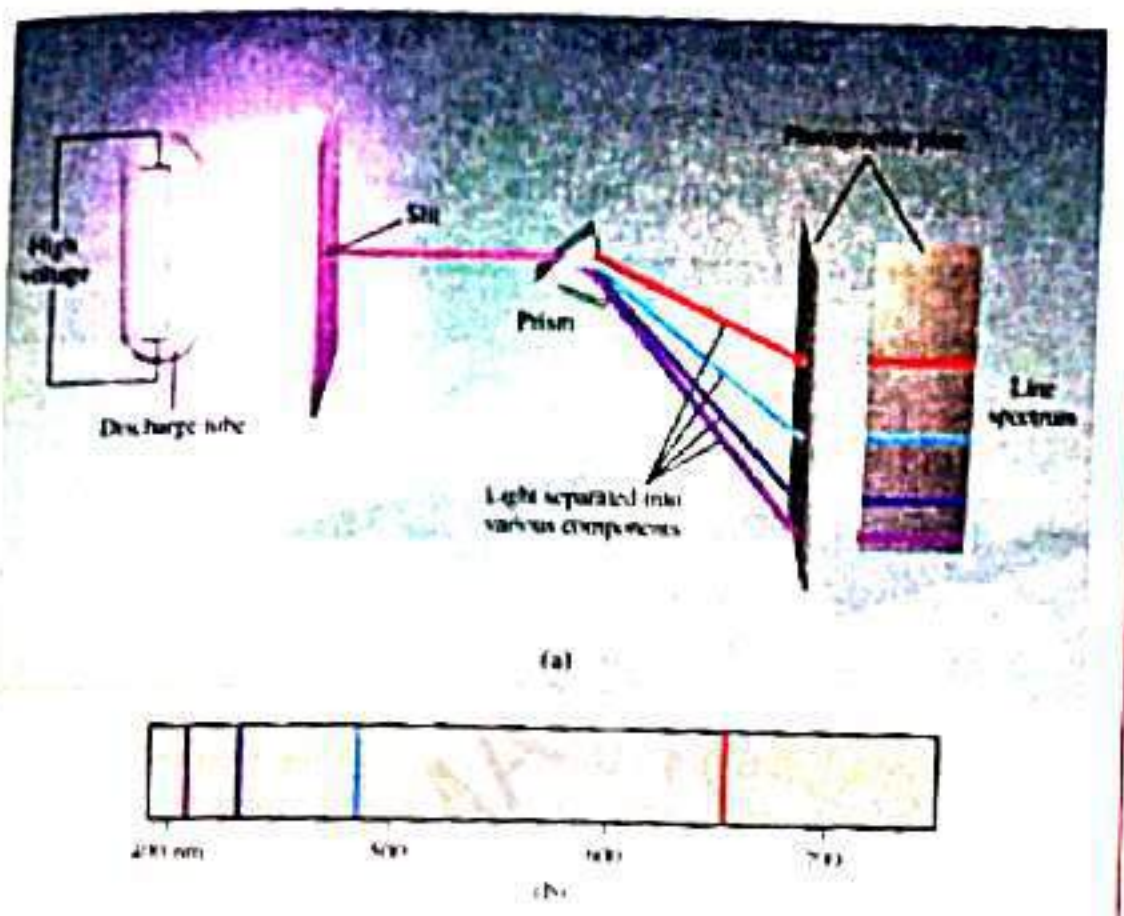
Generally molecular spectrum are band spectrum.

(ii) Line spectrum



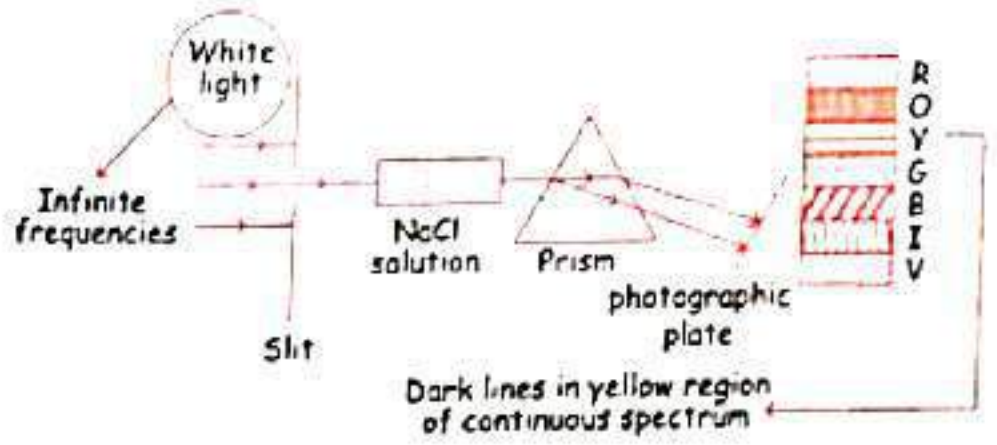
This is the ordered arrangement of lines of particular wavelength separated by dark space eg. hydrogen spectrum.

Line spectrum can be obtained from atoms.

Absorption Spectra

When white light from any source is first passed through the solution or vapours of a chemical substance and then analysed by the spectroscope, it is observed that some dark lines are obtained in the continuous spectrum. These dark lines are supposed to result from the fact that when white light (containing radiations of many wavelengths) is passed through the chemical substance, radiations of certain wavelengths are absorbed, depending upon the nature of the element.

Note



75. The number of following statement/s which is/are incorrect is **[8 April, 2023 (Shift-1)]**

(1) Line emission spectra are used to study the electronic structure

(2) The emission spectra of atoms in the gas phase show a continuous spread of wavelength from red to violet

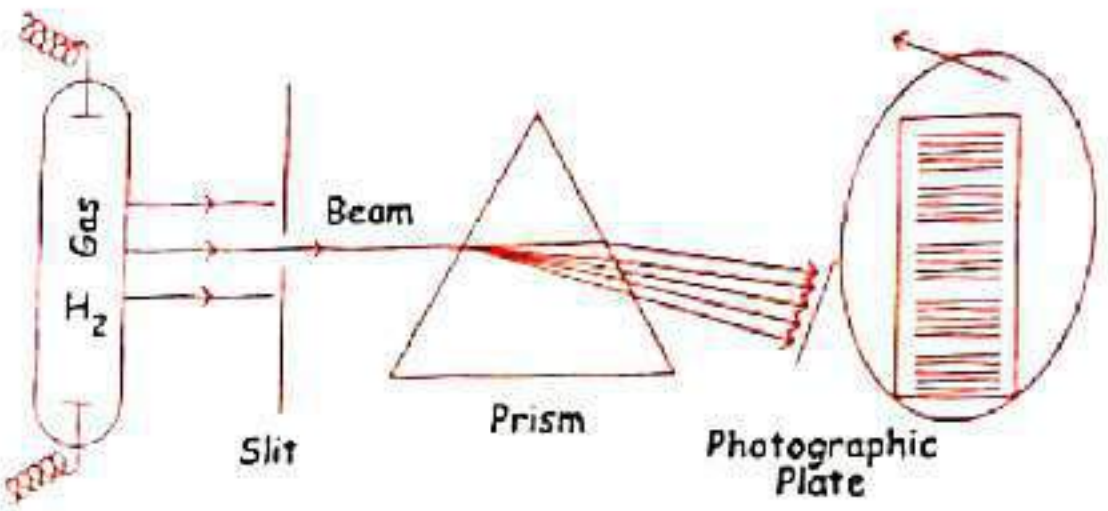
(3) An absorption spectrum is like the photographic negative of an emission spectrum

(4) The element helium was discovered in the sun by spectroscopic method

Sol. [1] The emission spectra of atoms in the gas phase, on the other hand, do not shows a continuous as all of wavelengths of the visible light which means statement (2) is incorrect.

socho kyu?

EMISSION SPECTRUM OF HYDROGEN



When hydrogen gas at low pressure is taken in the discharge tube and the light emitted on passing electric discharge is examined with a spectroscope, the spectrum obtained is called the emission spectrum of hydrogen.

Lets study about different types of series one by one:

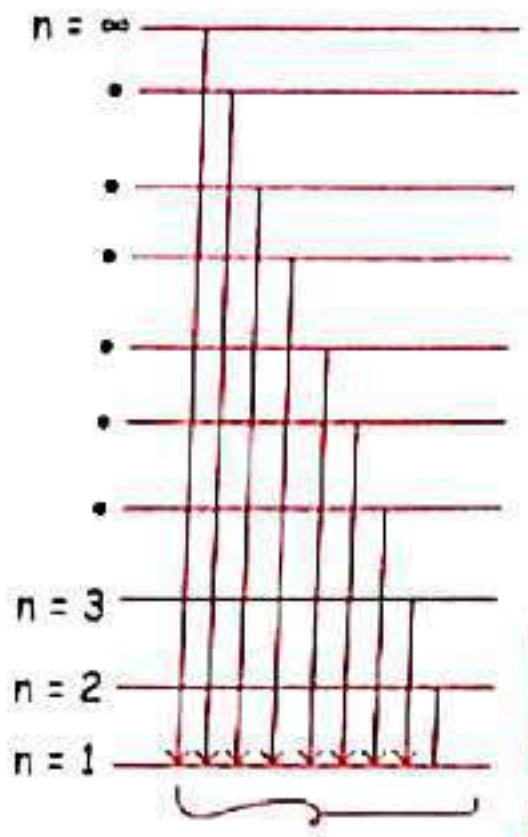
(1) Lyman series

The Lyman series is a series of spectral lines in the ultraviolet region of the electromagnetic spectrum, corresponding to transitions of electrons from higher energy levels to the lowest energy level (n = 1) in hydrogen atoms.

Lyman Series. $E = \frac{hc}{\lambda}$

($n_1 = 1$,

$n_2 = 2 ____ \infty$)



(a) $E_{max} = (\infty \rightarrow 1)$

$= 13.6\ eV$

$E_{max} = (2 \rightarrow 1)$

$= 10.2\ eV$

(b) $\lambda_{max} = \frac{12400}{10.2}\ \text{Å}$,

$\lambda_{min} = \frac{12400}{13.6}\ \text{Å}$

Important

$\lambda_{max}\ (2 \rightarrow 1)$

$$\bar{v} = \frac{1}{\lambda_{max}} = R(1)^2\left[\frac{1}{1^2} - \frac{1}{Z^2}\right]$$

$$\lambda_{max} = \frac{4}{3R} = \frac{12400}{10.2}\ \text{Å}$$

$\lambda_{min}\ (\infty \rightarrow 1)$

$$\bar{v} = \frac{1}{\lambda_{min}} = R(1)^2\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] \Rightarrow \lambda_{min} = \frac{1}{R} = \frac{12400}{13.6}\ \text{Å}$$

Note

Lyman series ki shortest line kaunsi honi chaiye?

Shortest line → $\infty \rightarrow 1$ ✓

Shortest line → $2 \rightarrow 1$ ✗

It refers to wavelength

Shortest line ka matlab hota hai shortest wavelength

Khoobsurat Baat

Shortest line → shortest W.L → $(\infty \rightarrow 1)$

Longest line → longest W.L → $(2 \rightarrow 1)$

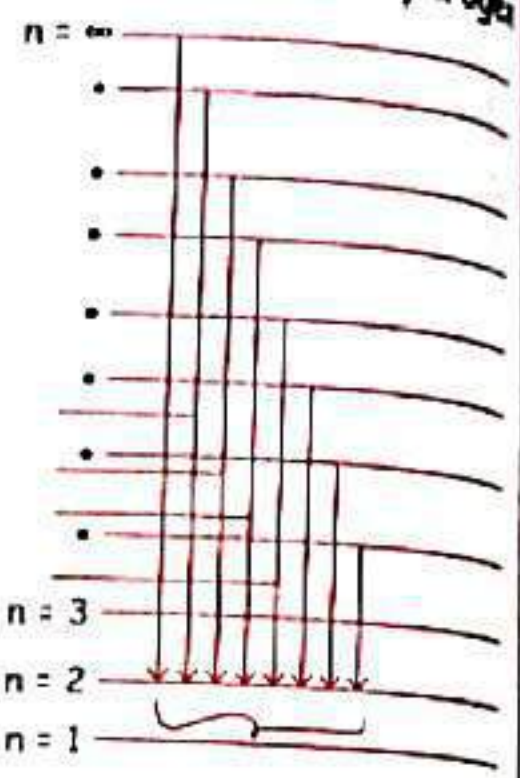
- First line ⟶ 2 ⟶ 1
- Second line ⟶ 3 ⟶ 1
- Third line ⟶ 4 ⟶ 1

- Photons belong to lyman series lies in U.V region

Lyman sabse easy series mei aata hai

(2) Balmer Series

The Balmer Series is a series of spectral lines in the visible region of the electromagnetic spectrum, corresponding to transition of electrons from higher energy levels to the second-lowest energy level (n = 2) in hydrogen atoms.



Balmer series

$n_1 = 2, n_2 = 3 \ldots\ldots \infty$

(a) $E_{max} = (\infty \rightarrow 2)$

$= 3.4\ eV$

$E_{max} = (3 \rightarrow 2)$

$= 1.89\ eV$

(b) $\lambda_{max} \Rightarrow (E_{min}) \rightarrow \frac{12400}{1.89}\ \text{Å}$

$\lambda_{min} \Rightarrow (E_{max}) \rightarrow \frac{12400}{3.4}\ \text{Å}$

$$\frac{1}{\lambda_{max}} = R(1)^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$\lambda_{max} = \frac{36}{5R} = \frac{12400}{1.89}\ \text{Å}$$

$$\lambda_{min} = R(1)^2\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right]$$

$$\lambda_{min} = \frac{4}{R} = \frac{12400}{3.4}\ \text{Å}$$

(c) Photons belong to Balmer series lies in visible region.

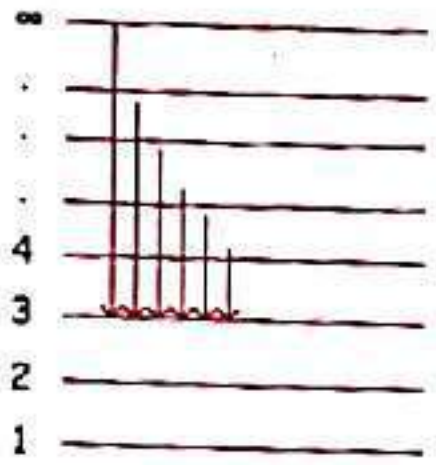
Khoobsurat Baat

Shortest line → shortest W.L → $E_{max} \rightarrow (\infty \rightarrow 2)$

Longest line → longest W.L → $E_{min} \rightarrow (3 \rightarrow 2)$

(3) Paschen series

The Paschen series is a series of spectral lines in the infrared region of the electromagnetic spectrumcorresponding to transitions of electrons from higher energy levels to the third-lowest energy level (n = 3) in hydrogen atoms.



$n_1 = 3, n_2 = 4.....\infty$

$E_{max} = 1.51$ eV, $E_{min} = 0.66$ eV

Photon → infrared region

(4) Brackett series

The Brackett series is a series of spectral lines in the infrared region of the electromagnetic spectrum corresponding to transitions of electrons from higher energy levels to the fourth-lowest energy level (n = 4) in hydrogen atoms.

$n_1 = 4, n_2 = 5.....\infty$

$E_{max} = 0.85$ eV, $E_{min} = 0.31$ eV

Iska halke mei mat lo, samjhe

(5) Pfund series

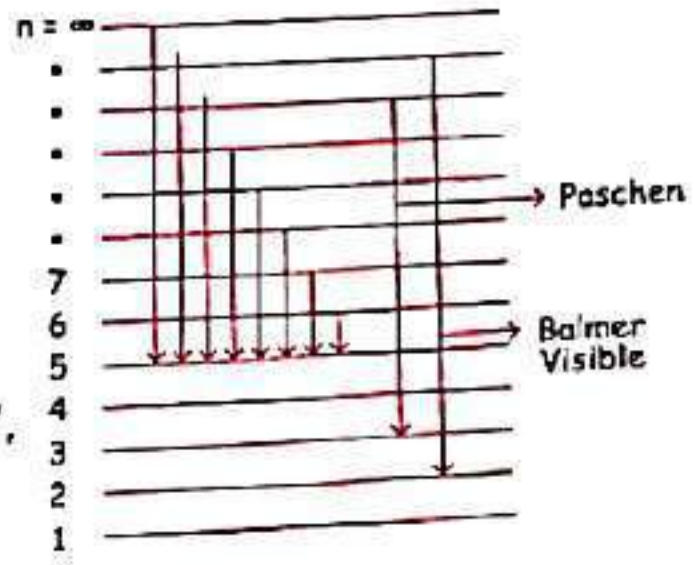
The Pfund series is a series of spectral lines in the infrared region of the electromagnetic spectrum, corresponding to transitions of electrons from higher energy levels to the fifth-lowest energy level (n = 5) in hydrogen atoms.

$n_1 = 5,$

$n_2 = 6.....\infty$

$E_{max} = 0.544$ eV,

$E_{min} = 0.16$ eV



(6) Humphrey series

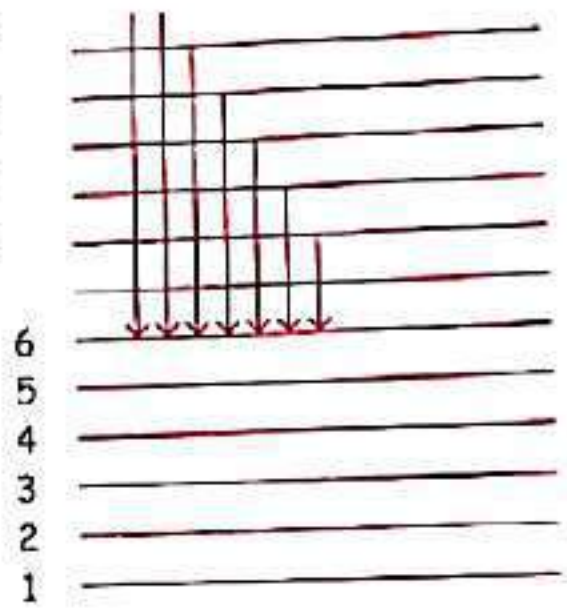
The Humphreys series is a series of lines in the infrared region of the electromagnetic spectrum, corresponding to transition of electrons from higher energy levels to the sixth-lowest energy level (n = 6) in hydrogen atoms.

Humphrey series:

$n_1 = 6, ... n_2 = 7.....\infty$

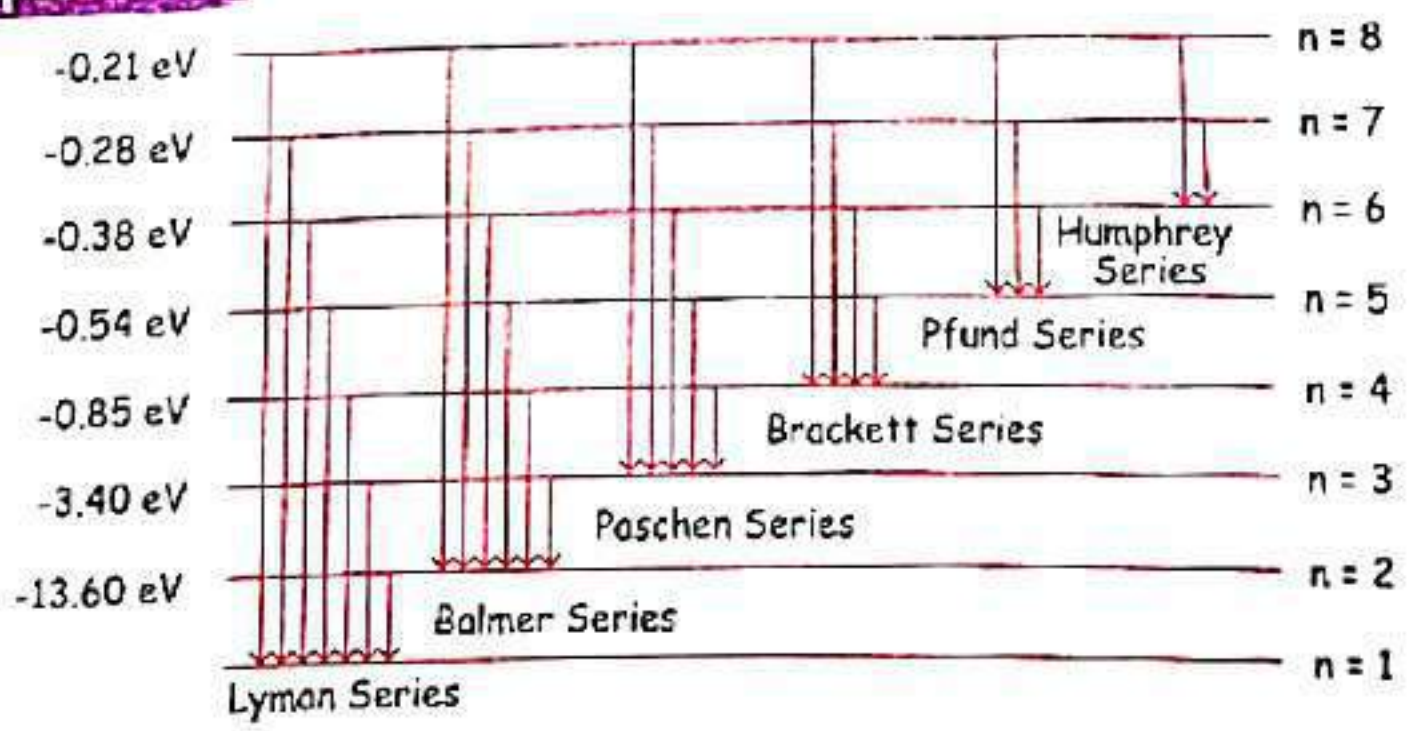
$E_{max} = 0.38$ eV,

$E_{min} = 0.1$ eV



yeh bhi koi kam nahi hai

Important



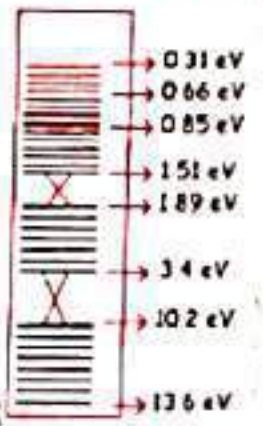
Khoobsurat Baat

10.2 ≤ energy of photon ≤ 13.6 lyman $n_1 = 1$

1.89 ≤ energy of photon ≤ 3.4 Balmar $n_1 = 2$

3.4 ≤ energy of photon ≤ 10.2 Sannata 1

0.66 ≤ energy of photon ≤ 1.51 paschen $n_1 = 3$

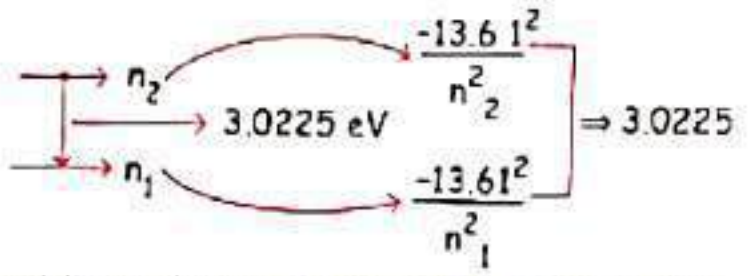


1.51 ≤ energy of photon ≤ 1.89 Sannata 2

yeh bhut mazedaar hai, dhyan lagao

76. Find out the transition of an electron which emits a photon of energy 3.0225 eV in H- emission spectrum.

Sol. Method-1



Using Hit and trial method we get $n_2 = 6$; $n_1 = 2$

Method-2

According to above khoobsurat baat, photon of energy 3.0225 eV belongs to Balmer series, so, $n_1 = 2$, now we can easily find out $n_2 = 6$

77. Find out the transition which emits a photon of energy 6.723 eV

(A) 3 → 1 (B) 7 → 2

(C) 8 → 4 (D) not possible

Sol. (D) According to above khoobsurat baat, 6.723 eV energy ka koi photon emit ho hi nhi sakta hydrogen ke emission spectrum mei

are dimag toh hai na thikane?

78. The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^+ is

[29 Jan, 2023 (Shift-1)]

(A) $\frac{5}{9\lambda}$ (B) $\frac{9\lambda}{5}$ (C) $\frac{36\lambda}{5}$ (D) $\frac{5\lambda}{9}$

Sol. (B) For explanation, refer to QR code given on the last page.

79. If wavelength of the first line of the Paschen series of hydrogen atom is 720 nm, then the wavelength of the second line of this series is _____ nm. (Nearest integer)

[24 Jan, 2023 (Shift-1)]

Sol. (492 nm) For explanation, refer to QR code given on the last page.

NO. OF PHOTONS EMITTED BY A SAMPLE OF H ATOM

Important

No of unique photons emitted during transition of different energies or different W.L.

No of spectral lines observed = no of unique Photons

If sample contains only one atom

Max = n - 1

Min = 1

Sample having enough no of atoms and de-excitation is fill ground level

$$\text{Min} = 1, \quad \text{Max} = {}^{n}C_2 = \left(\frac{n(n-1)}{2}\right)$$

Applicable only when
- (i) enough no. of atom in sample
- (ii) de-excitation is till ground level (n = 1)

If electrons in a sample of enough number of atoms are in any higher state n = n and make a transition to ground state, then total no. of different photons emitted is equal to $\frac{n(n-1)}{2}$.

If electrons are in any higher state $n = n_2$ and make a transition to another excited state $n = n_1$ then,

$$\text{total no. of different photons emitted} = \frac{\Delta n(\Delta n+1)}{2}$$

where $\Delta n = n_2 - n_1$

In case of single isolated atom if electron makes transition from n^{th} state to the ground state then max. number of spectral lines observed = (n – 1)

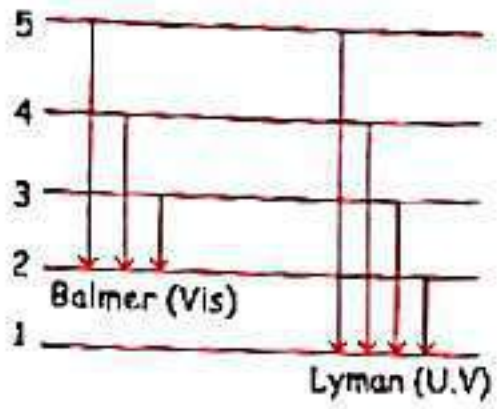
80. No. of visible lines when an electron returns from 5^{th} orbit upto ground state in H spectrum:

(A) 5 (B) 4 (C) 3 (D) 10

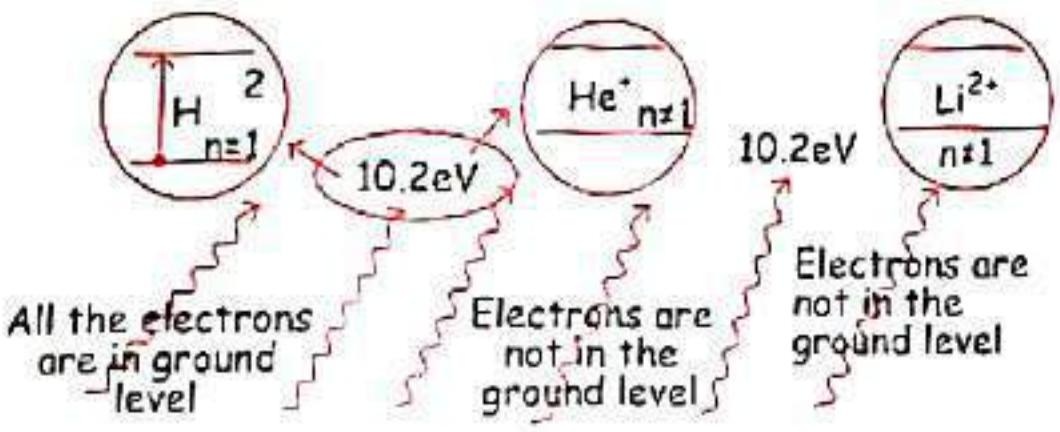
Sol. (D) Max unique photon

$$= \frac{5(5-1)}{2} = 10$$

$$\begin{pmatrix} 4+3+2+1 \\ | \quad | \quad | \quad | \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ L \quad B \quad P \quad B \end{pmatrix}$$



81. Find out the total number of unique photons emitted during the de excitation of the samples till ground level if these samples are excited with 10.2eV photons. (assuming all three samples are accepting bombarded photons)



Sol. (20) For explanation, refer to QR code given on the last page.

Now, lets study study the different cases:

Case-1: Enough sample but de excitation is not till ground level

Khoobsurat Baat

$$\text{Max unique photon} = \frac{\Delta n(\Delta n+1)}{2} \quad [\Delta n = n_2 - n_1]$$

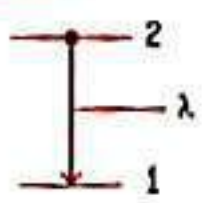
Note

For ground level $\Rightarrow n_1 - n_2 = (n-1)$

$$\text{max} \Rightarrow \frac{n(n-1)}{2}$$

Case-2: Not enough atoms

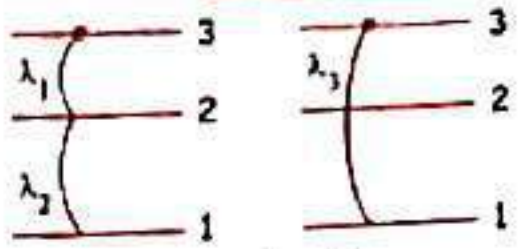
n = 2



$$\frac{n(n-1)}{2}$$

$$= \frac{2(2-1)}{2} = 1$$

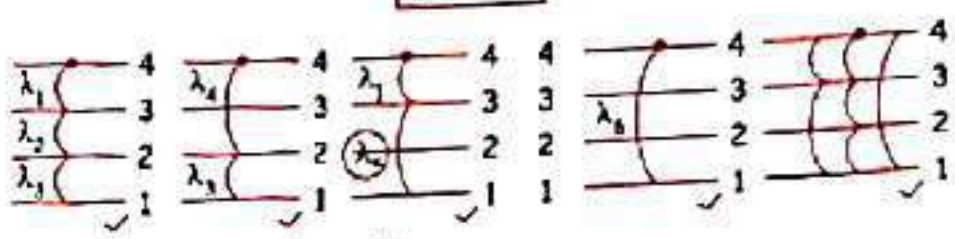
n = 3



$$= \frac{n(n-1)}{2}$$

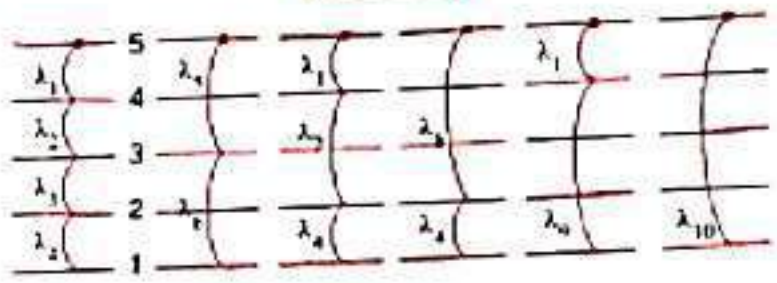
$$= \frac{3(3-1)}{2} = 3$$

n = 4



$$= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

n = 5



$$= \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$$

Bada hi dardnaak lagra hoga

Khoobsurat Baat

What is enough?

For n = 2	enough = 1
For n = 3	enough = 2
For n = 4	enough = 4
For n = 5	enough = 6
For n = 6	enough = 9

DUAL NATURE OF ELECTRON (DE-BROGLIE HYPOTHESIS)

iske do do nature hai, kahin bhi kai bhi dikha sakta hai

(a) Einstein had suggested that light can behave as a wave as well as like a particle i.e. it has dual character.

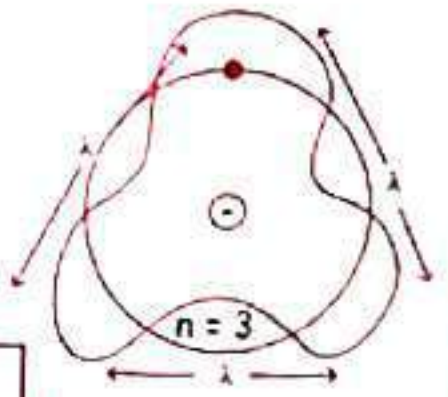
(b) In 1924, de-Broglie proposed that an electron behaves both as a material particle and as a wave.

(c) This proposed a new theory wave mechanical theory of matter. According to this theory, the electrons, protons and even atom when in motion possess wave properties.

$E = \frac{hc}{\lambda}$ $\quad (E = mc^2)$

$\frac{hc}{\lambda} = mc^2, \ \lambda = \frac{h}{mc}$.

Velocity = v



$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

- $\lambda_{de\text{-}broglie} = \frac{h}{p} = \frac{h}{mv}$

No. of orbit

- $2\pi r = 3\lambda, \ 2\pi r = n\lambda$

$$\lambda = \frac{2\pi r}{n}$$

Here, n = no. of orbit.

Quantization of Angular Momentum

$\lambda = \frac{h}{mv}, \lambda = \frac{2\pi r}{n}, \frac{h}{mv} = \frac{2\pi r}{n}$

$$mvr = \frac{nh}{2\pi}$$

quantization of angular momentum

Important

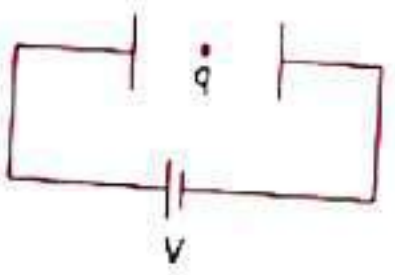
- $\lambda_{DB} = \frac{h}{mv}$
- $\lambda_{DB} = \frac{2\pi r}{n}$
- $\lambda_{DB} = \frac{h}{mv} = \frac{h}{\sqrt{2m(kE)}}$
- $\lambda_{D_B} = \frac{h}{\sqrt{2mqV}}$

$KE = \frac{1}{2}mv^2$

$v = \sqrt{\frac{2K \cdot E}{m}}$

$mv = P = \sqrt{2m(kE)}$

= voltage through which a harge particle (q) is accl^{ed}.



or Proton $\lambda_{DB} = \frac{0.286}{\sqrt{V}}$ Å

For α particle $\lambda_{DB} = \frac{0.101}{\sqrt{V}}$ Å

For electron $\lambda_{DB} = \frac{12.3}{\sqrt{V}}$ Å

Formulae ko yaad karlo copy mei likh likh kar

82. An electron is revolving in 12th orbit of Li^{+2}. Find out its de-Broglie W.L ?

Sol. For 12th - orbit:

T.E = - 0.85 eV

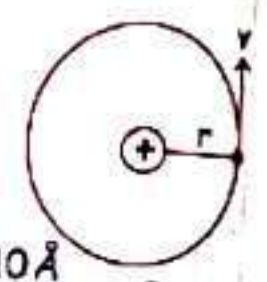
↓

KE = + 0.85 eV

$\lambda = \sqrt{\frac{150}{0.85}}$ Å = 13.28 Å

83. Find out the de-Broglie W.L of an electron moving in n = 3 in H-atom

Sol. Method - 1



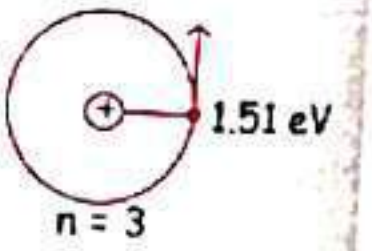
$$\frac{h}{mv} = \lambda_{DB}$$

$$= \frac{6.626 \times 10^{-34} \text{ J-sec}}{9.1 \times 10^{-31} \times \left(2.18 \times 10^6 \times \frac{1}{3} \text{ m/sec}\right)} = 10 Å$$

Method - 2

$$\frac{2\pi r}{n} = \lambda_{DB} \Rightarrow \frac{2 \times 3.14 \times \left(0.529 \times \frac{3^2}{1} Å\right)}{3} = 10 Å$$

Method - 3



$$\frac{h}{\sqrt{2m(KE)}} = \lambda_{DB}$$

$$\Rightarrow \frac{6.626 \times 10^{-34} \text{ J.sec}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.51 \times 1.6 \times 10^{-19}}} = 10 Å$$

Method - 4 $\frac{h}{\sqrt{2mqV}} = \lambda_{DB}$

$$= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 1.51}} = 10 Å$$

are waah ek hi swaal 4 tarike se kar dia humne

ab bina pen uthaye kaise krna hai dekhte hain

$$(\lambda_H)_1 = \frac{h}{\sqrt{2m(K.E)}} = \frac{h}{\sqrt{2m(13.6)}} = x$$

$$(\lambda_H)_2 = \frac{h}{\sqrt{2m(KE)}} = \frac{h}{\sqrt{2m(3.4)}} = 2x$$

$$(\lambda_H)_3 = \frac{h}{\sqrt{2m(K.E)}} = \frac{h}{\sqrt{2m(1.51)}} = 3x$$

Agr x ki value pta chal jae, sare answers pta pta chal jaenge or x ki value = $\frac{10}{3}$ Å

For 3rd orbit: $3 \times \frac{10}{3} = 10$ Å

Khoobsurat Baat

$\lambda_{DB} = \sqrt{\frac{150}{V}}$ Å $= \frac{12.3}{\sqrt{V}}$ Å

Applicable for an electron accelerated through voltage V volts

Important

Hydrogen	He^+	Li^{+2}	Be^{+3}
$\frac{10}{3}$ Å	$\frac{5}{3}$ Å	$\frac{10}{9}$ Å	$\frac{10}{12}$ Å
$\frac{20}{3}$ Å	$\frac{10}{3}$ Å	$\frac{20}{9}$ Å	$\frac{20}{12}$ Å
$\frac{30}{3}$ Å	$\frac{15}{3}$ Å	$\frac{10}{3}$ Å	$\frac{13}{12}$ Å
$\frac{40}{3}$ Å	$\frac{20}{3}$ Å	$\frac{40}{9}$ Å	$\frac{10}{3}$ Å
$\frac{50}{3}$ Å	$\frac{25}{3}$ Å	...	...

yeh bhut gazab ka hai isko chhod mat dena

84. $\frac{(\lambda_{He^+})_8 - (\lambda_H)_4}{(\lambda_{Li^{+2}})_3} = ?$

Sol. $\lambda_{(He^+)_8} = 40/3$

$\lambda_{(H)_4} = 40/3$, $\lambda_{(Li^{2+})_3} = 30/9$

Substitute the values in the above expression, we will get 0 as the final answer.

85. $\frac{(\lambda_{He^+})_6 - (\lambda_{Li^{+2}})_7}{(\lambda_H)_2} = ?$

Sol. $(\frac{1}{3})$ For explanation, refer to QR code given on the last page.

86. Calculate ratio of wavelength for an α particle and proton accelerated through same potential difference.

Sol. $m_\alpha = 4m_p$, $q_\alpha = 2q_p$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{\frac{h}{\sqrt{2m_\alpha q_\alpha V}}}{\frac{h}{\sqrt{2m_p q_p V}}} = \sqrt{\frac{m_p q_p}{m_\alpha q_\alpha}} = \sqrt{\frac{1}{8}} = \frac{1}{2\sqrt{2}}$$

direct formula laga hai bas

87. Hydrogen sample is in Ist excited state A photon of energy 8 eV is used to excite the hydrogen sample. Find the de-brogile wavelength of the electron.

Sol. $\left(\lambda = \sqrt{\frac{150}{4.6}} A^\circ\right)$ For explanation, refer to QR code given on the last page.

88. If the radius of the first orbit of hydrogen atom is a_0, then de Broglie's wavelength of electron in 3rd orbit is

[JEE Main 2023 Online 06th April Shift-II]

(A) $\frac{\pi a^3}{6}$ (B) $3\pi a_0$ (C) $6\pi a_0$ (D) $\frac{\pi a^3}{3}$

Sol. (C) $(r_3)_H = \frac{a_0 n^2}{7} = a_0 \times 3^2 = 9a_0$

Now, $2\pi r = n\lambda$, $2\pi \times 9a_0 = 3\lambda$, $\lambda = 6\pi a_0$

waah waah kitna asaan hai

89. If an electron having kinetic energy 2 eV is accelerated through the potential difference of 2 Volt. Then calculate the wavelength associated with the electron.

Sol. (6.15Å) For explanation, refer to QR code given on the last page.

90. An electron, practically at rest, is initially accelerated through a potential difference of 100 volts. It then has a de-Broglie wavelength = λ_1Å. It then get retarded through 19 volts and then has a wavelength λ_2Å. A further retardation through 32 volts changes the wavelength to λ_3Å. What is $\frac{\lambda_3 - \lambda_2}{\lambda_1}$?

Sol. $\left(\frac{20}{63}\right)$ For explanation, refer to QR code given on the last page.

91. What possibly can be the ratio of the de Broglie wavelengths for two electrons each having zero initial energy and accelerated through 50 volts and 200 volts?

(A) 3:10 (B) 10:3 (C) 1:2 (D) 2:1

Sol. (D) $\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}} = \sqrt{\frac{200}{500}} = \frac{2}{1}$

Kch nhi karna, bas formula lgana hai

92. An electron having initial K.E. of 6 eV is accelerated through a potential difference of 4 V. Find the wavelength of associated with electron.

Sol. (0.3883 nm) For explanation, refer to QR code given on the last page.

93. Photon having energy equivalent to the binding energy of 4th state of He^+ atom is used to eject an electron from the metal surface of work function 1 4 eV If electrons are further accelerated through the potential difference of 4 V then what is the minimum value of de-Broglie wavelength associated with the electron?

Sol. $\lambda = \frac{h}{\sqrt{2me(KE)}} = \frac{6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times6\times1.6\times10^{-19}}}$

$\lambda = 5Å$

94. De-Broglie wavelength of electron in second orbit of Li^{+2} ion will be equal to de-Broglie of wavelength of electron in

(A) n = 3 of H-atom (B) n = 4 of C^{5+} ion

(C) n = 6 of Be^{2+} ion (D) n = 3 of He^+ ion

Sol. (B) Wavelength is related to radius as:

$\lambda \propto \frac{r}{n}$ or $\frac{n}{Z}$

The de-Broglie of electron in second orbit of Li^{2+} ion will be equal to de-Broglie wavelength of electron in n = 4 of C^{5+} ion.

95. The wavelength of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12}$ m. The value of x is ____ (Nearest Integer)

[20 July, 2021 (Shift-II)]

Given: Mass of electron = 9.1×10^{-31}kg

Charge on an electron = 1.6×10^{-19}C

Planck's constant = 6.63×10^{-34}Js

Sol. [6]

$\lambda = \frac{h}{\sqrt{2\times m\times q\times V}}$

$\therefore \lambda = 0.614 \times 10^{-11}$m = 6.14×10^{-12}m

[Nearest integer = 6]

96. The de-Broglie's wavelength of an electron in the 4th orbit is ____ πa_0 (a_0 = Bohr's radius)

[04 April, 2024 (Shift-II)]

Sol. [8] $2\pi a_0 \frac{n^2}{Z} = n\lambda_d$

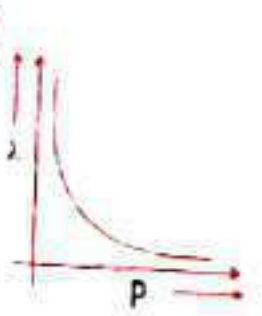
For H-atom, Z = 1, $2\pi a_0 \frac{4^2}{1} = 4\lambda_d$

On solving, $\lambda = 8\pi a_0$.

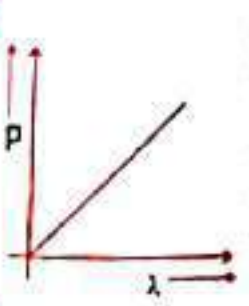
97. According to the wave-particle duality of matter by de-Broglie, which of the following graph plot presents most appropriate relationship between wavelength of electron (λ) and momentum of electron (p)?

[01 Feb, 2024 (Shift-I)]

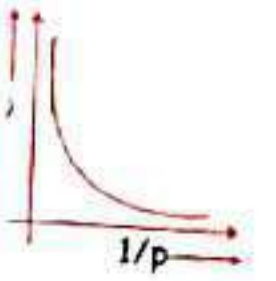
(A)



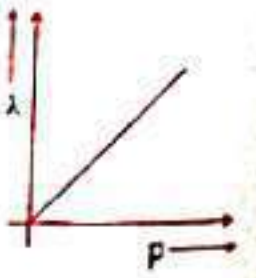
(B)



(C)



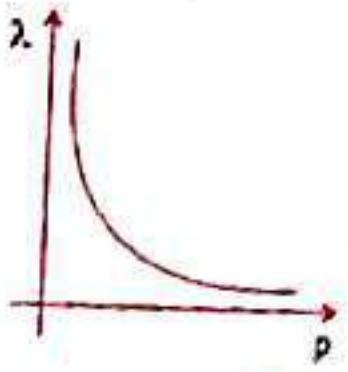
(D)



Sol. (A) $\lambda = \frac{h}{p}\left[\lambda \propto \frac{1}{p}\right]$

$\Rightarrow \lambda p = h$ (constant)

Hence, the plot can be represented as a rectangular hyperbola.



graph ko dekh kar toh ho hi jaega sawaal

98. If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom? [n : any integer] **[JEE Mains Shift-I, 2025]**

(A) $\frac{2a_0}{n\pi}$ (B) $\frac{4\pi a_0}{n}$ (C) $\frac{8\pi a_0}{n}$ (D) $\frac{4n}{\pi a_0}$

Sol. (C) $r_n = n^2 a_0$

For the second orbit (n = 2):

$r_2 = 2^2 a_0 = 4a_0$, $n\lambda = 2\pi r_n$

$\lambda = \frac{2\pi r_n}{n}$, $\lambda = \frac{2\pi(4a_0)}{n} = \frac{8\pi a_0}{n}$

HEISENBERG'S UNCERTAINTY PRINCIPLE

The exact position and momentum of a fast moving particle cannot be calculated precisely at the same moment of time. If Δx is the error in the measurement of position of the particle and if Δp is the error in the measurement of momentum of the particle, then:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \text{ or } \Delta x.(m\Delta v) \geq \frac{h}{4\pi} \text{ or } \Delta E.\Delta t \geq \frac{h}{4\pi}$$

where Δx = uncertainty in position,

Δp = uncertainty in momentum,

Δv = uncertainty in velocity

ΔE = uncertainty in energy

ΔT = uncertainty in time,

If the position of a particle is measured precisely, i.e. $\Delta x \to 0$ then $\Delta p \to \infty$.

If the momentum of the particle is measured precisely, i.e. $\Delta p \to 0$ then $\Delta x \to \infty$.

This is because of a principle of optics that if a light of wavelength 'λ' is used to locate the position of a particle then maximum error in the position measurement will be $\pm\lambda$..

i.e. $\Delta x = \pm\lambda$

If $\Delta x \to 0; \lambda \to 0$

But, $p = \frac{h}{\lambda} \Rightarrow p \to \infty$

So, to make $\Delta x \to 0$, $\lambda \to 0$ a photon of very high energy is used to locate it.

$\therefore$ When this photon will collide with the electron then momentum of electron will get changed by a large amount.

muskil nahi hai bada hi asaan or mazedaar hai

99. Given below are two statements: **[JEE Mains Shift-II, 2025]**

Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II): If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\frac{1}{\sqrt{\pi}}\sqrt{\frac{h}{2m}}$

Sol. (A) Both Statement I and Statement II are true.

100. A golf ball has a mass of 40 g , and a speed of 45 m/s. If the speed can be measured within accuracy of 2%, calculate the uncertainty in the position

Sol. (0.9 m/s) For explanation, refer to QR code given on the last page.

101. Uncertainty in position is twice the Uncertainty in momentum. Uncertainty in velocity is

(A) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ (B) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (C) $\frac{1}{2m}\sqrt{\frac{h}{2\pi}}$ (D) $\frac{h}{4\pi m}$

Sol. (C) $\Delta x = 2\Delta p$, $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

$2(\Delta p)^2 \geq \frac{h}{4\pi}$, $2(m\Delta v)^2 \geq \frac{h}{4\pi}$

$\Delta v \geq \frac{1}{2m}\sqrt{\frac{h}{2\pi}}$

sochlo thoda sa ki hua kya hai

102. Based on Heisenberg's uncertainty principle, the uncertainty in the velocity of the electron to be found within an atomic nucleus of diameter 10^{-15}m is ____ $\times 10^9$ ms^{-1} (nearest integer)

[Given: mass of electron = 9.1×10^{-31} kg, Planck's constant (h) = 6.626×10^{-34} Js]
(Value of $\pi = 3.14$)

[09 April, 2024 (Shift-II)]

Sol. $m\Delta v \cdot \Delta x = \frac{h}{4\pi}$

$$\Delta v = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^{-15} \times 4 \times 3.14}$$

On solving, $\Delta x = 57.97 \times 10^9$ m/sec

103. The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is ____ . (Nearest Integer)
(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

[26 July, 2022 (Shift-I)]

Sol. |1758| $X_e = X_n$, $\Rightarrow M_eV_e = M_nV_n$

$$\Rightarrow \frac{V_e}{V} - \frac{M_n}{M} = \frac{1.6 \times 10^{-27}}{9.1 \times 10^{-31}} \approx 1758.24$$

THE QUANTUM WAVE MECHANICAL MODEL

- Quantum mechanics is a theoretical science that deals with the study of motions of the microscopic objects that have both observable wave-like and particle-like properties.

It specifies the laws of motion that these objects obey.

When quantum mechanics is applied to macroscopic objects (for which wave like properties are insignificant) the results are the same as those from the classical mechanics.

- Quantum mechanical model of atom is the picture of structure of the atom, which emerges from the application of the Schrödinger equation to atoms.
- For a particle whose energy does not vary with time, the Schrodinger equation may be written as

$$\hat{H}\Psi = E\Psi$$

Where $\hat{H}$ = Hamiltonian or Total energy operator,
Ψ = wave function

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m(E-V)}{h^2}\psi = 0$$

where x, y and z are Cartesian Co-ordinates,
m = mass of the electron,
E = Total energy of the electron, V = P.E. of the electron,
Ψ = Wave function of the electron.

- For a given system for which V and m are known, we can use the Schrödinger equation to obtain value of E (the allowed energies of the particle) & ψ (the wave function).
- For H-atom, P.E. (V) = $-\frac{Ke^2}{r}$
- Schrodinger's wave equation becomes

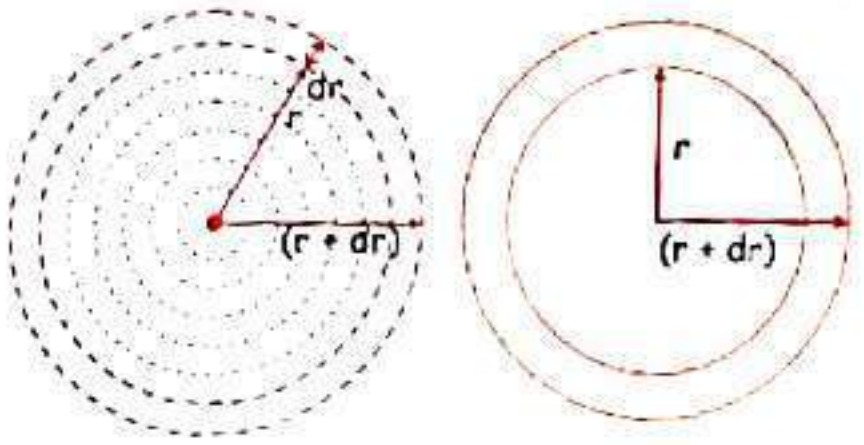
$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m\left(E - \frac{Ke^2}{r}\right)}{h^2}\psi = 0$$

- The wave function itself has no physical meaning, but Ψ^2 is a probability density and for this to be the case, Ψ must have certain properties:
  - ★ It must be finite for all values of x;
  - ★ It can only have one value for any value of x;
  - ★ And $\frac{d\psi}{dx}$ must vary continuously as x varies.

- Ψ must be continuous, have a continuous slope, be single-valued, & be square integrable.
- The results (solutions) of wave equation are
  - ★ The wave function is a solution of the Schrodinger equation and describes the behavior of an electron in region of space called the atomic orbital.

RADIAL PROBABILITY FUNCTION

- Radial probability distribution function $=R^2\,dV$. (where R = radial wave function)
- In order to find the value of dV, the entire space of the electron orbital around the nucleus is supposed to be divided into small imaginary concentric shells of very small thickness dr, each.



RADIAL WAVE FUNCTION FOR H-LIKE ATOMS

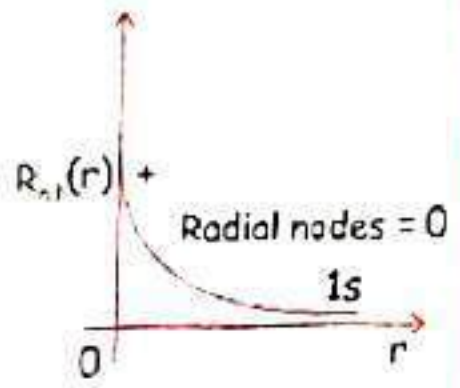
yaha se dhyan bilkul nahi hatana ab

- $R_{n,l}$ = Constant × Polynomial × Exponential.

(I) $R_{n,l}(r)$ v/s plots:

(i) **1s-orbital:**

$$R_{1,0} = 2\left(\frac{Z}{a_0}\right)^{3/2} \times 1 \times e^{-Zr/a_0}$$



- 1s-orbital has a finite value of $R_{n,l}(r)$ at nucleus.

(ii) **2s-orbital:**

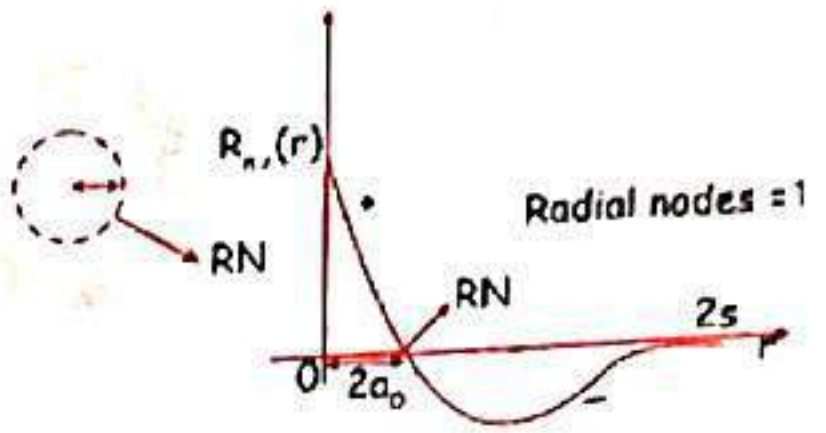
$$R_{2,0} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{1}{2\sqrt{2}} \times \left(2 - \frac{Zr}{a_0}\right) \times e^{-Zr/2a_0}$$

(a) If $2 > \frac{Zr}{a_0}$, then $R_{2,0}$ is +ve.

(b) If $\frac{Zr}{a_0} > 2$, then $R_{2,0}$ is -ve.

(c) If $\frac{Zr}{a_0} = 2$, then $R_{2,0}$ is 0.

At Radial node, $r = \frac{2a_0}{Z}$

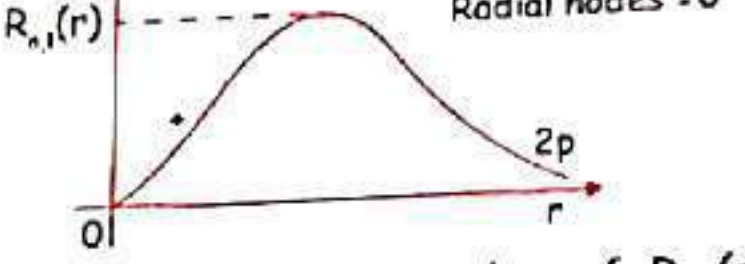


2s-orbital has a finite value of $R_{n,l}(r)$ at nucleus.

2s-orbital wave function also decays exponentially but its decay is slower than 1s-orbital.

(iii) **2p-subshell:**

$$R_{2,1} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{1}{2\sqrt{6}} \times \left(\frac{Zr}{a_0}\right) \times e^{-Zr/2a_0}$$

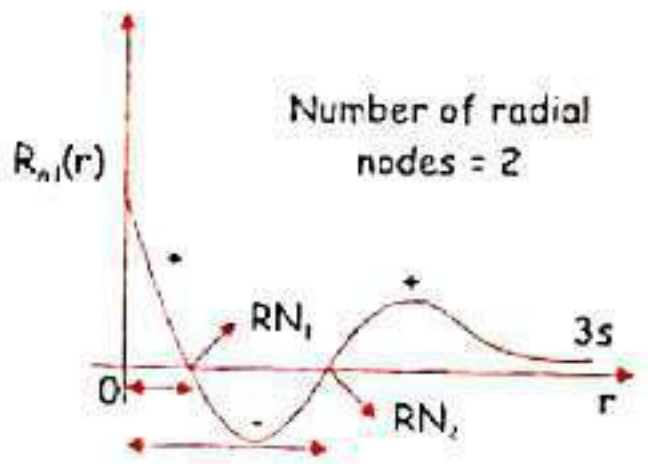


2p-orbital has zero value of $R_{n,l}(r)$ at nucleus.

2p-orbital has a maxima in the $R_{n,l}(r)$ v/s r plot.

(iv) **3s-orbital:**

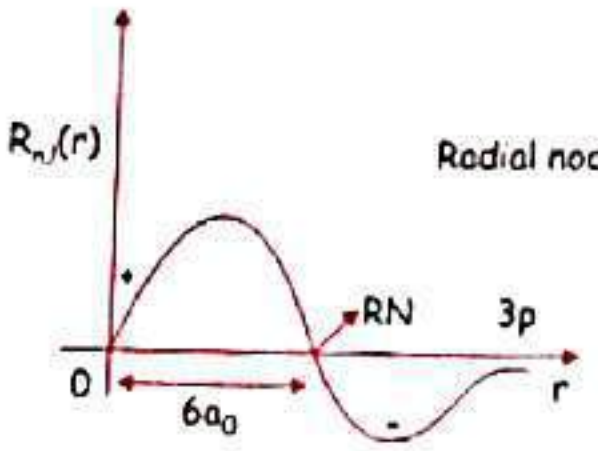
$$R_{3,0} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{2}{81\sqrt{3}} \times \left(27 - 18\frac{Zr}{a_0} + \frac{2Z^2r^2}{a_0^2}\right) \times e^{-Zr/3a_0}$$



3s-orbital has a finite value $R_{n,l}(r)$ at nucleus.

(v) **3p-subshell:**

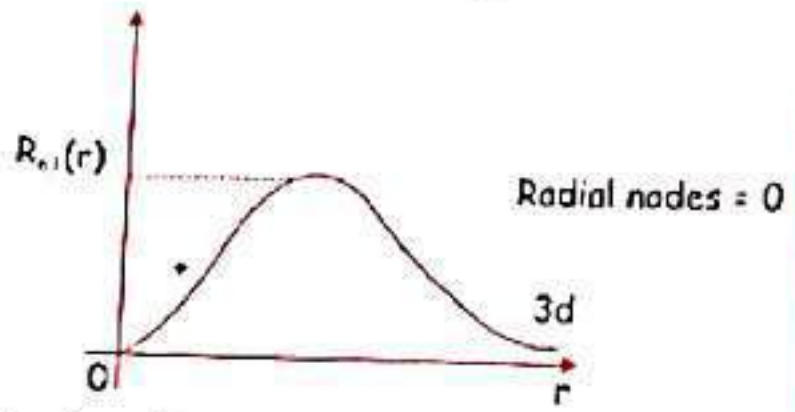
$$R_{3,1} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{4}{81\sqrt{6}} \times \left(6\frac{Zr}{a_0} - \frac{Z^2r^2}{a_0^2}\right) \times e^{-Zr/3a_0}$$



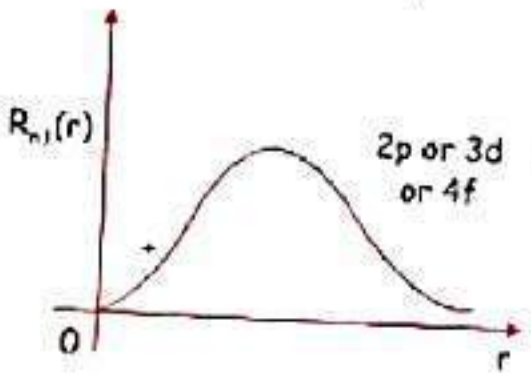
3p-orbital has zero value $R_{n,l}(r)$ at nucleus.

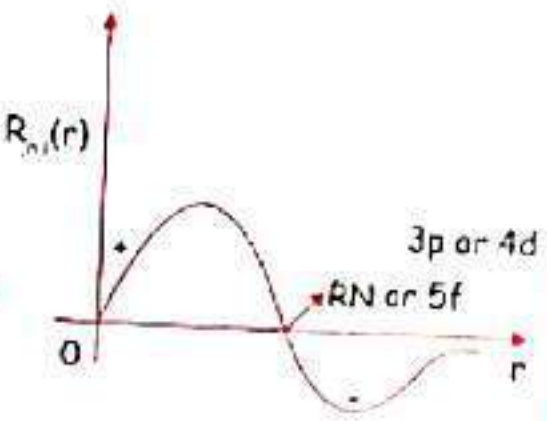
(vi) 3d-subshell:

$$R_{3,2} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{4}{81\sqrt{30}} \times \frac{Z^2r^2}{a_0^2} \times e^{-Zr/3a_0}$$



3d-orbital has zero value of $R_n(r)$ at the nucleus.





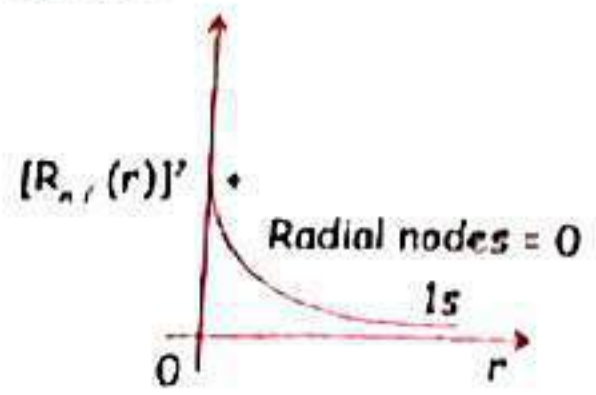
itné sare graphs, tauba tauba

- For all orbitals other than s-orbital, $R_{n,l}(r) = 0$ at the nucleus.
- For all 1 s - orbitals, $R_{n,l}(r)$ is always positive.
- For the first orbital of other types (i.e. 2p, 3d, 4f etc), $R_{n,l}(r)$ is +ve everywhere except at the origin.
- For the second orbital of a given type (i.e. 2s, 3p, 4d, 5f etc.), $R_{n,l}(r)$ may be +ve or -ve but the radial wave function has only 1 sign change, i.e. they have 1 radial node.
- For the third orbital of a given type (i.e. 3s, 4p, 5d, 6f etc.), $R_{n,l}(r)$ may have 2 sign changes i.e. they have 2 radial nodes.

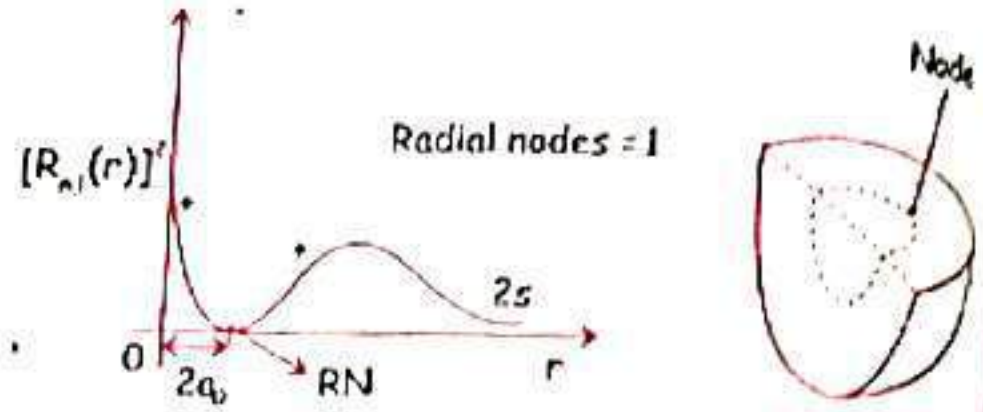
) $[R_{n,l}(r)]^2$ v/s r plots:

$[R_{n,l}(r)]^2$ would always be +ve, irrespective of the sign of $R_{n,l}(r)$ for a given value of r. R^2 represents the probability density of findin electron at a particular point in space.

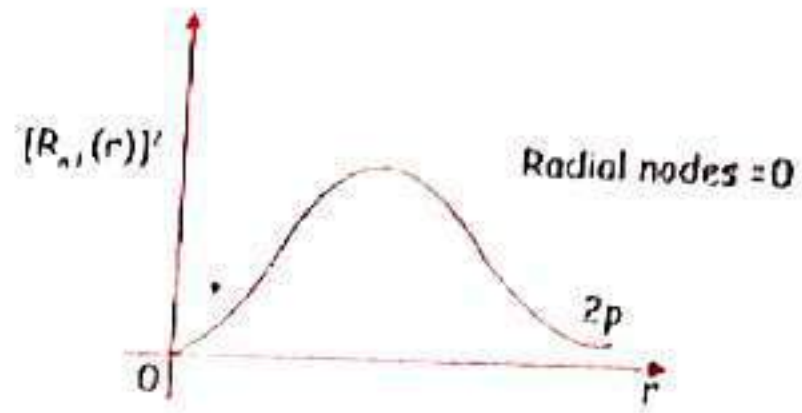
(i) 1s - orbital:



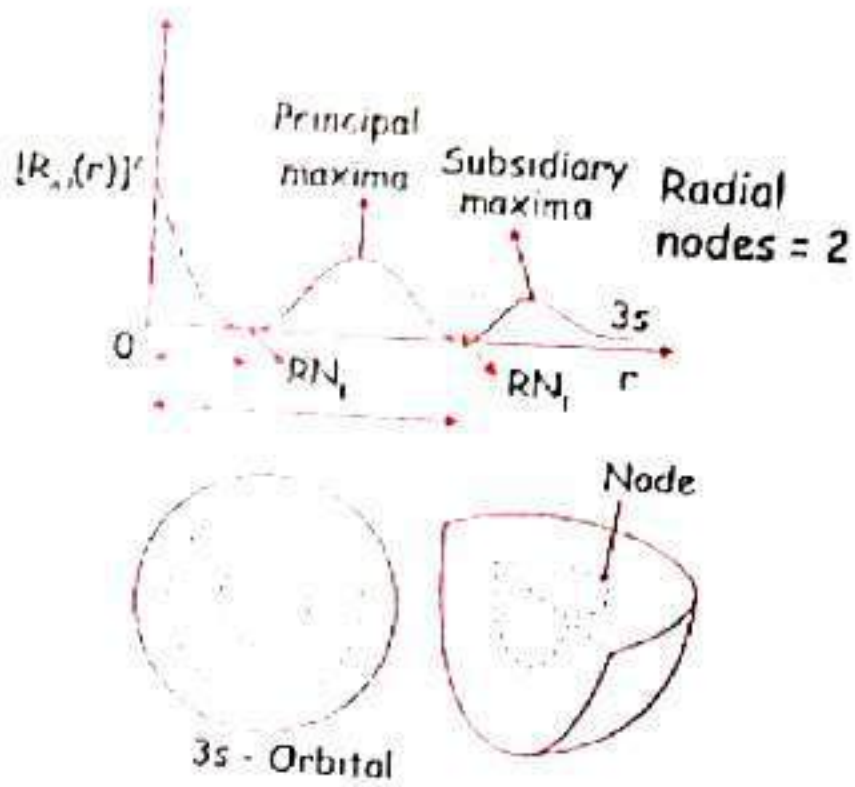
(ii) 2s - orbital:



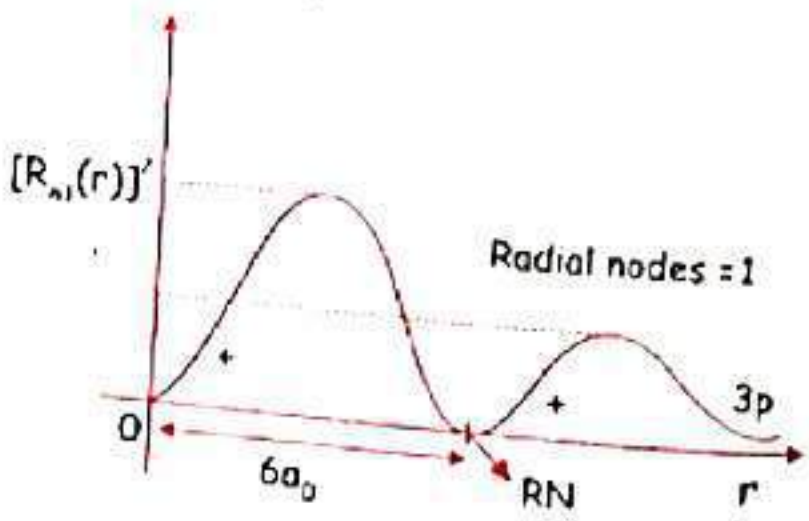
(iii) 2p - subshell:

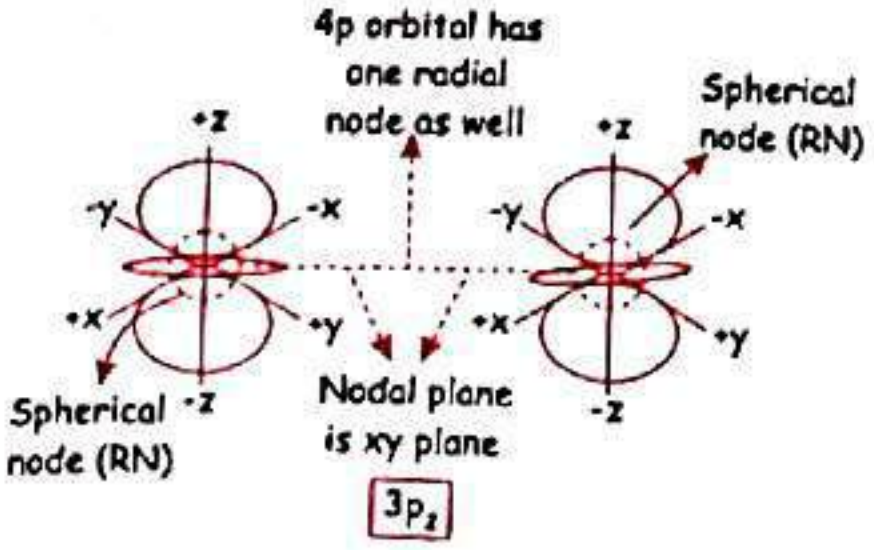


(iv) 3s-orbital:

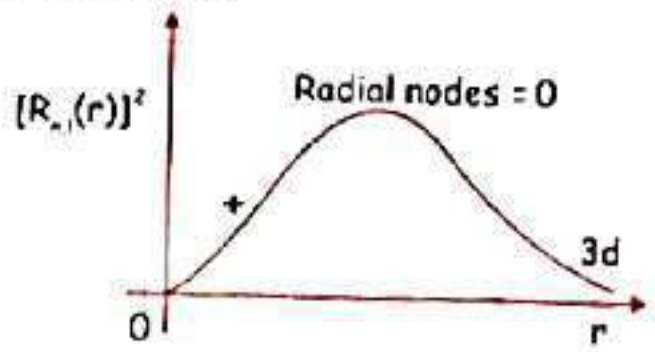


(v) 3p - subshell:





(vi) 3d-subshell:



Note

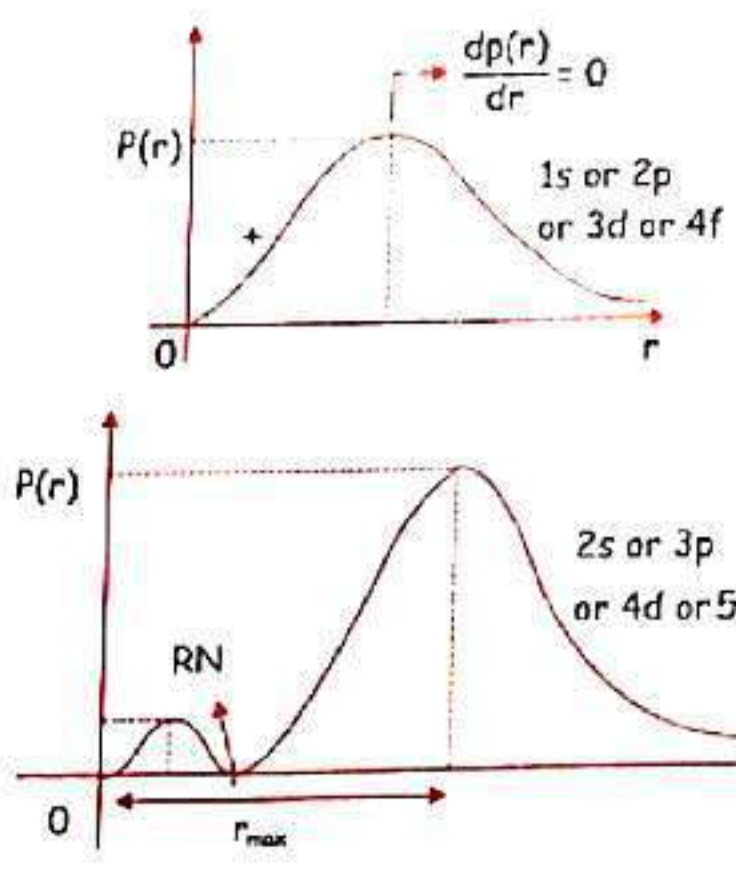
- First maxima is high… second is higher t…
- $R_{n,l}$ (r) v/s r p… r plot are s… '4f' orbit…

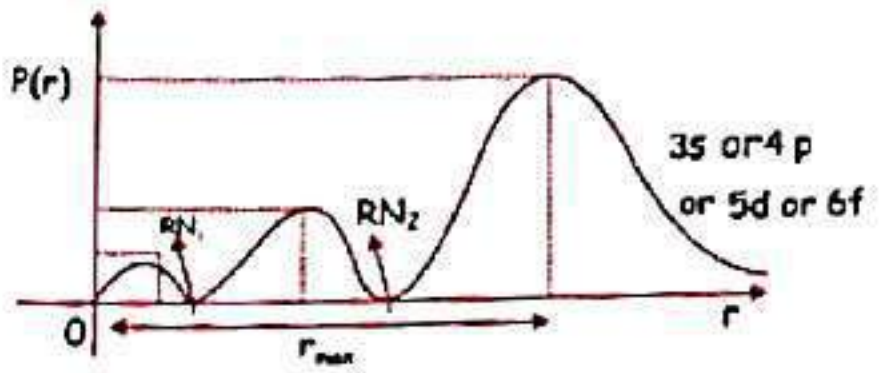
) $4\pi r^2 [R_{n,}$…

Radial dis… distributio…

= p(r) dr = 4…

The radial distri… of choice to deterr… find the electron for a g… of its direction).





arey phirse graphs agaye, haaye ram

104. The maximum radial probability in 1st-orbital occurs at a distance when: [r_0 = Bohr radius]

(A) $r = r_0$ (B) $r = 2r_0$

(C) $r = \frac{r_0}{2}$ (D) $2r = \frac{r_0}{2}$

…st orbital, the distance at which …bability occurs is r_0.

…ments:

…ne in $3d_{x^2-y^2}$

…in $3d_{z^2}$

…s?

… in x and y

…ial.

…unction ('ψ') of 2s is given by

$$\psi_{2s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

At $r = r_0$, radial node is formed. Thus, r_0 in terms of a_0 **[30 Jan, 2023 (Shift-II)]**

(A) $r_0 = a_0$ (B) $r_0 = 4a_0$

(C) $r_0 = \frac{a_0}{2}$ (D) $r_0 = 2a_0$

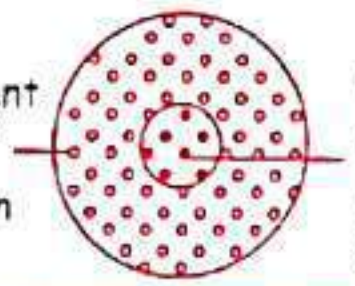
Sol. (D) At node $\psi_{2s} = 0$

$\therefore 2 - \frac{r_0}{a_0} = 0$, $\therefore r_0 = 2a_0$

kitna asaan hai, balle baale ho gayi

ORBITAL

An orbital may be defined as the region of space around the nucleus where the probability of finding an electron is maximum (90% to 95%) Orbitals do define a definite path for the electron, rather they define only the probability of the electron being in various regions of space around the nudes.

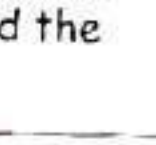


Less dots represent low probability of finding an electron

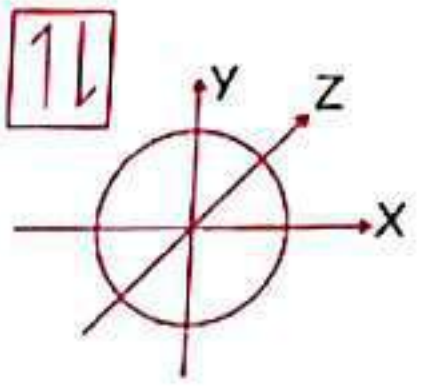
Maximum dots represent high probability of finding an electron

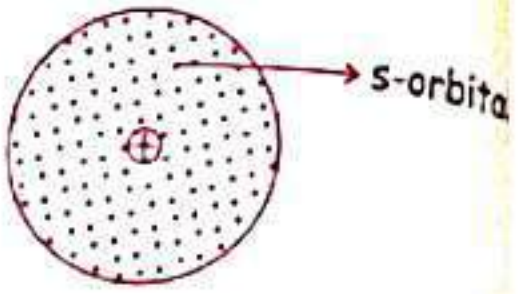
Orbits	Orbitals
1. It is well defined circular path followed by revolving electrons around the nucleus	1. It is well region of space around the nucleus where electron is most likely to be found
2. It represents planar motion of electron	2. It represent 3-dimensional motion of an electron around the nucleus.
3. The maximum no. of electron in an orbitals is $2n^2$ where n stands for no. of orbital.	3. Orbitals can not accommodate more than 2 electrons.
4. Orbits are circular in shape	4. Orbitals have different e.g. s-orbital is spherical, p-orbital is dumb - bell shaped.
. Orbit is non irectional in haracter. Hence, ey cannot explain ape of molecules	5. Orbitals (except s-orbital) have direction character. Hence, they can account for the shape of molecules.
Concept of well fined orbit is ainst Heisenberg's certainty principle.	6. Concept of orbitals is in accordance with Heisenberg's principle

table dekhlo, dhyan se or yaad karlo



S-orbital: **Shape spherical**





spherical - non direction

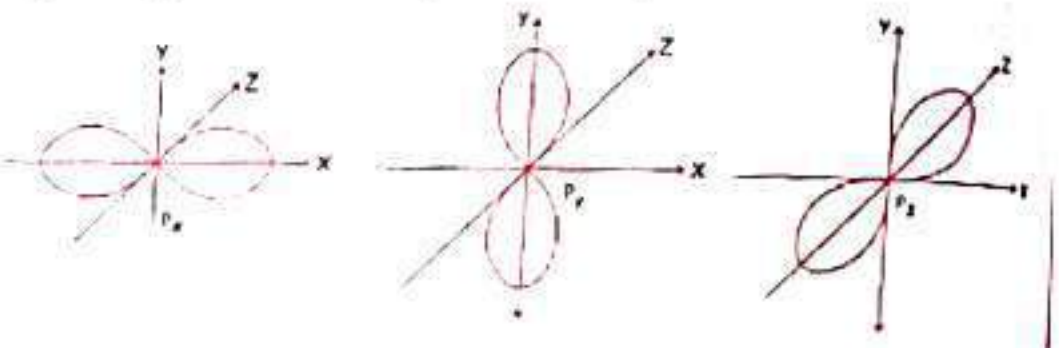
S-orbital is non directional and it is closest to th nucleus, having lowest energy.

S-orbital can accommodate maximum no. of tw electrons.

yeh gol hai bole toh

p-orbital: Shape → dumb-bell

Dumb-bell consists of two loops which are separated by a region of zero probability called node.



p-orbitals can accommodate maximum no. of six electrons.

p-subshell → p_x p_y p_z

yeh bhi koi kam nahi hai, dumb-bell hai

107. Given below are two statements: [JEE Mains Shift-II, 2025]

Statement I: A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.

Statement II: $2p_x$ and $2p_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both statement I and II are true.

(B) Statement I is true but II is false.

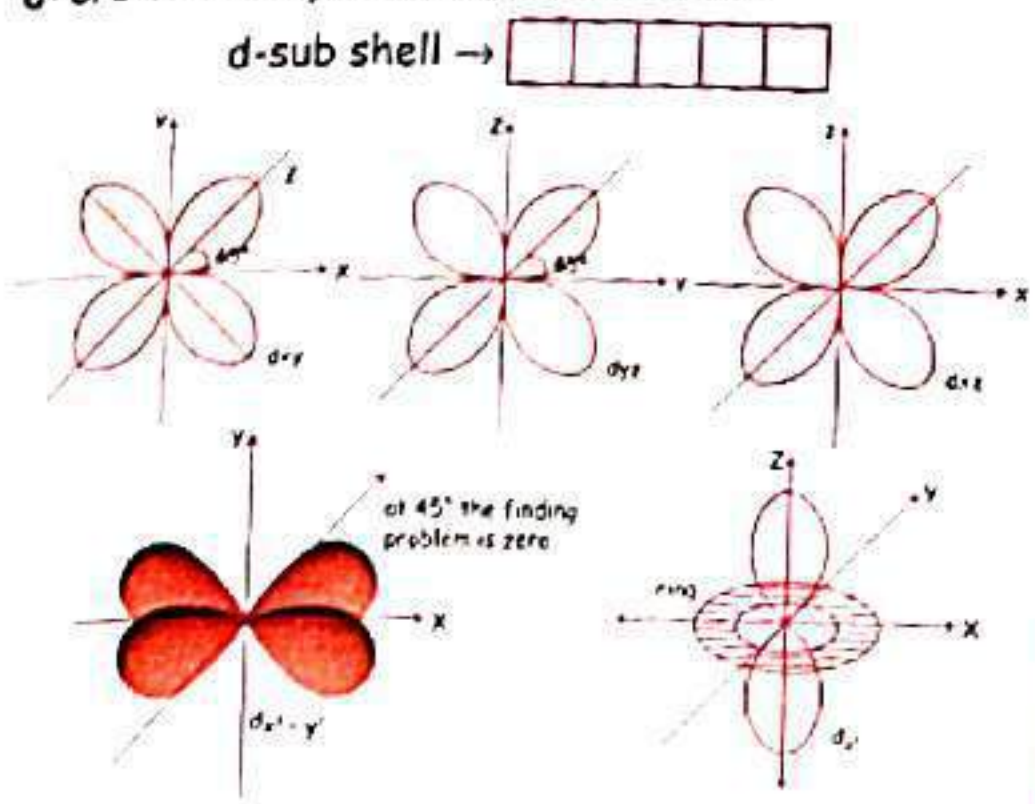
(C) Both statement I and II are false.

(D) Statement I is false but II is true.

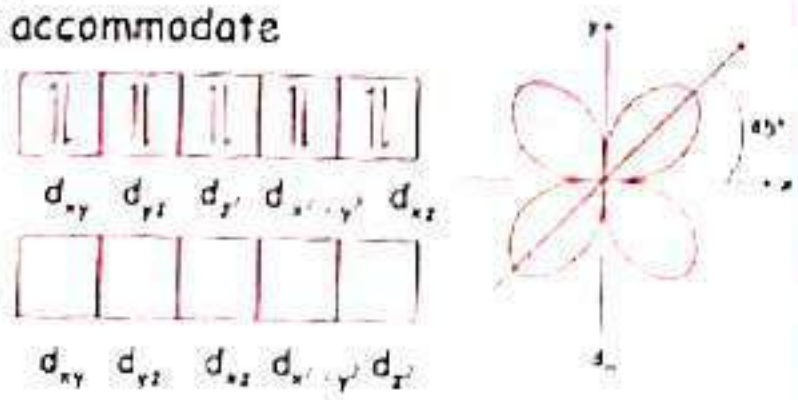
Sol. (D) As $2p_x$ and $2p_y$ are degenerate orbitals, so no spectral line will be observed for $2p_x$ to $2p_y$ transition.

d-orbital: Shape → double dumb-bell

d-sub shell →



d-orbital can accommodate



Along the axis finding probability = 0

At 45° from the axis the finding probability is maximum.

Isko toh bilkul hi halka mat lena, yeh toh double dumb-bell hai

QUANTUM NUMBERS

The set of four numbers required to define an electron completely in an atom are called quantum numbers. The first three have been derived from Schrodinger wave equation.

(I) Principal Quantum Number [n]

It describes the size of the electron wave and the total energy of the electron. It has integral values 1, 2, 3, 4.... etc. and is denoted by K, L, M, N, ... etc

Number of subshell present in n^{th} shell = n

n	subshell
1	s
2	s, p
3	s, p, d
4	s, p, d, f

Number of orbitals present in n^{th} shell = n^2

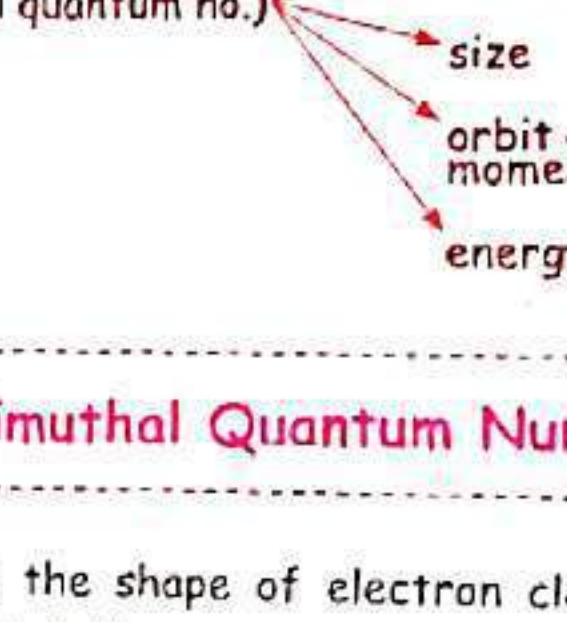
- The maximum number of electrons which can be present in a principal energy shell is equal to $2n^2$.
- Angular momentum of any orbit = $\frac{nh}{2\pi}$

$$mvr = \frac{nh}{2\pi}$$

Note

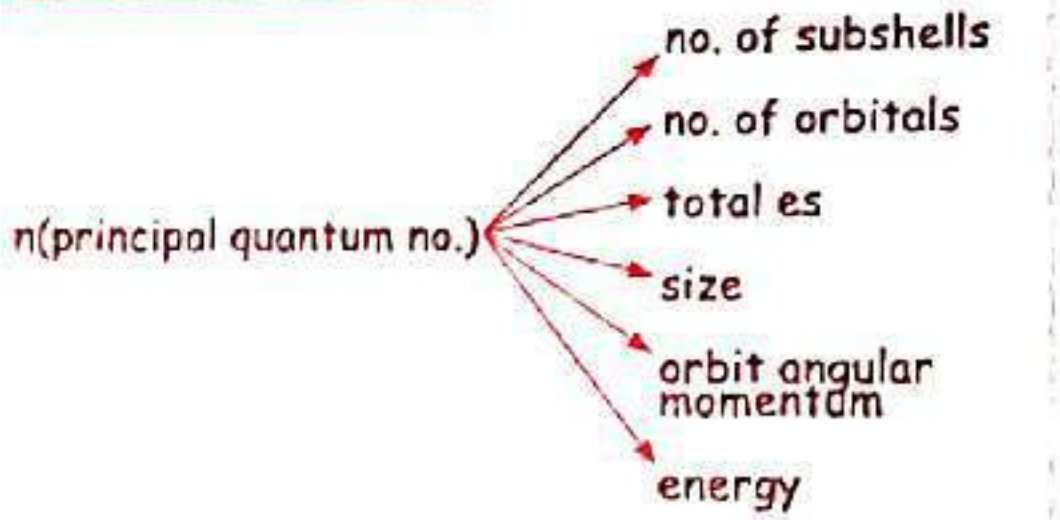
if n = 1 total orbitals = $1^2 = 1$

if n = 2 total orbitals = $2^2 = 4$



sare formulae bhut hi imp hai, yaad karlo

Khoobsurat Baat



(II) Azimuthal Quantum Number (ℓ)

It describes the shape of electron cloud and the number of subshells in a shell

- It can have values from 0 to [n - 1]
- Value of *l* subshell

0	⟶	s
1	⟶	p
2	⟶	d
3	⟶	f

l = 0 ..... (n - 1)

if n = 2 → *l* = 0, 1, n = 3, *l* = 1 ⇒ 3p

- Number of orbitals in a subshell = $2\ell + 1$
- Maximum number of electrons in particular subshell = $2 \times (2\ell + 1)$
- Orbital angular momentum,

$$L = \frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \bar{h}\sqrt{\ell(\ell+1)} \quad \left[\bar{h} = \frac{h}{2\pi}\right]$$

i.e. Orbital angular momentum of s orbital = 0,

Orbital angular momentum of p orbital = $\sqrt{2}\frac{h}{2\pi}$,

Orbital angular momentum of d orbital = $\sqrt{3}\frac{h}{2\pi}$

Orbital angular momentum $\Rightarrow mvr = \frac{nh}{2\pi}$

yeh bhi bht badhiya hai,
sare sawaal kraega

Important

Orbital angular momentum $\Rightarrow \frac{h}{2\pi}\sqrt{\ell(l+1)}$

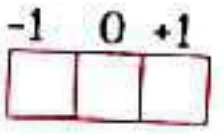
$\ell = 0$ (s) $\Rightarrow 0$

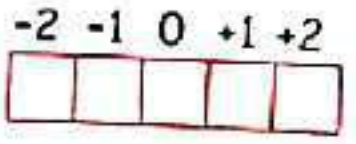
$\ell = 1$ (p) $\Rightarrow \sqrt{2}\frac{h}{2\pi}$, $\ell = 2$ (d) $\Rightarrow \frac{\sqrt{6}h}{2\pi}$

(III) Magnetic Quantum Number (m)

It describes the orientations of the subshells. It can have values from $-\ell$ to $+\ell$ including zero, i.e., total $(2\ell + 1)$ values. Each value corresponds to an orbital, s-subshell has one orbital, p-subshell three orbitals (p_x, p_y and p_z), d-subshell five orbitals (d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$, d_{z^2}) and f-subshell has seven orbitals. The total number of orbitals present in a main energy level is 'n^2'.

$s \rightarrow \ell = 0, m = 0$

$p \rightarrow \ell = 1, m = -1, 0, +1$ 

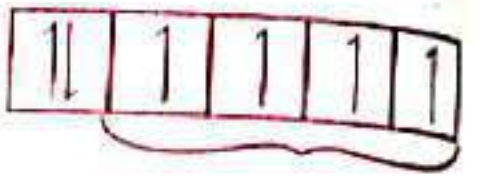
$d \rightarrow \ell = 2, m = -2, -1, 0, +1, +2$ 

(IV) Spin Quantum Number (s)

It describes the spin of the electron. It has value +1/2 and - 1/2. (+) signifies clockwise spinning and (- signifies anticlockwise spinning.

- Spin magnetic moment $\mu = \sqrt{n(n+2)}$ B.M. (n = n of unpaired electrons, B.M. = Bohr magneton)
- It represents the value of spin angular momentum which is equal to $\frac{h}{2\pi}\sqrt{s(s+1)}$
- Maximum spin of atom = $\frac{1}{2}$ × No. of unpaired electron.

Example:



Spin angular momentum

$$= \frac{h}{2\pi}\sqrt{s(s+1)}$$

s = total spin $\Rightarrow \frac{1}{2}\times$ unpaired es

$$\Rightarrow \frac{1}{2} \times 4 = 2 = \frac{h}{2\pi}\sqrt{2(2+1)} = \frac{\sqrt{6}h}{2\pi}$$

yeh bhi badhiya hai quantum number,
isko mat ignore kardena

Khoobsurat Baat

Orbit angular momentum $\Rightarrow \frac{nh}{2\pi} = mvr$

Orbital angular momentum $\Rightarrow \frac{h}{2\pi}\sqrt{\ell(\ell+1)}$

Spin angular momentum $\Rightarrow \frac{h}{2\pi}\sqrt{s(s+1)}$

↓

Total spin

108. In a multielectron atom, which of the following orbitals described by three quantum numbers will have same energy in absence of electric and magnetic fields?

[JEE Mains Shift-I, 2025]

A. $n = 1, l = 0, m_l = 0$
B. $n = 2, l = 0, m_l = 0$
C. $n = 2, l = 1, m_l = 1$
D. $n = 3, l = 2, m_l = 1$
E. $n = 3, l = 2, m_l = 0$

Choose the correct answer from the options given below:

(A) A and B Only (B) D and E Only

(C) B and C Only (D) C and D Only

Sol. (B) In a multi-electron atom, the orbitals (D) n = 3, l = 2, m = 1 and (E) n = 3, l = 2, m = 0 described by the three quantum numbers will have the same energy in the absence of magnetic and electric fields.

109. The number of electrons present in all the completely filled subshells having n = 4 and $s = +\frac{1}{2}$ is_____.

(Where n = principal quantum number and s = spin quantum number)

[27 Jan, 2024 (Shift-1)]

Sol. [16] For explanation, refer to QR code given on the last page.

110. The correct set of four quantum numbers for the valence electron of rubidium atom (Z = 37) is: **[29 Jan, 2024 (Shift-I)]**

(A) $5, 0, 0, +\frac{1}{2}$ (B) $5, 0, 1, +\frac{1}{2}$

(C) $5, 1, 0, +\frac{1}{2}$ (D) $5, 1, 1, +\frac{1}{2}$

Sol. (A) Electronic configuration of Rb = $[Kr]5s^1$

n = 5, l = 0, m = 0, s = +1/2 or - 1/2

111. Given below are two statements:

Statement-I: The orbitals having same energy are called as degenerate orbitals.

Statement-II: In hydrogen atom, 3p and 3d orbitals are not degenrrate orbitals.

In the light of the above statements, choose the most appropriate answer from the options given **[30 Jan, 2024 (Shift-I)]**

(A) Statement-I is true but Statement-II is false

(B) Both Statement-I and Statement-II are true.

(C) Both Statement-I and Statement-II are false

(D) Statement-I is false but Statement-II is true

Sol. (A) In hydrogen 3s, 3p and 3d orbitals have same energy. So, statement I is true and II is false.

padha tha na upar, agr nahi toh gaye kaam se

112. The four quantum numbers for the electron in the outer most orbital of potassium (atomic no. 19) are **[31 Jan, 2024 (Shift-II)]**

(A) $n = 4, \ell = 2, m = -1, s = +\frac{1}{2}$

(B) $n = 4, \ell = 0, m = 0, s = +\frac{1}{2}$

(C) $n = 3, \ell = 0, m = 1, s = +\frac{1}{2}$

(D) $n = 2, \ell = 0, m = 0, s = +\frac{1}{2}$

Sol. (B) $_{19}K \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$

The outermost orbital of potassium is the 4s orbital.

$n = 4, \ell = 0, m_\ell = 0, s = \pm 1/2$

113. The number of radial node/s for 3p orbital is: **[01 Feb, 2024 (Shift-II)]**

(A) 1 (B) 4

(C) 2 (D) 3

Sol. (A) Number of radial node = $n - \ell - 1 = 1$

graphs dekh liye the na lallu

114. The maximum number of orbitals which can be identified with n = 4 and $m_l = 0$ is_____

[04 April, 2024 (Shift-II)]

Sol.

n = 4,	□	□□□	□□□□□	□□□□□□□
	4s	4p	4d	4f
$m_l \Rightarrow$	1	1	1	1

So, Maximum number of orbitals are 4.

115. In an atom, total number of electrons having quantum numbers n = 4, $[m_l] = 1$ and $m_s = -1/2$ is **[05 April, 2024 (Shift-II)]**

Sol. (6) For explanation, refer to QR code given on the last page.

116. Compare the energies of following sets of quantum numbers for multielectron system.

[09 April, 2024 (Shift-I)]

(a) n = 4, l = 1 (b) n = 4, l = 2

(c) n = 3, l = 1 (d) n = 3, l = 2

(e) n = 4, l = 0

Choose the correct answer from the options given below

(A) (b) > (a) > (c) > (e) > (d)

(B) (e) > (c) < (d) < (a) < (b)

(C) (e) > (c) > (a) > (d) > (b)

(D) (c) < (e) < (d) < (a) < (b)

Sol. (D) The energy level can be determined by comparing (n + l) values. More the (n + l) value, more will the energy

Hence, the correct order is:

(b) > (a) > (d) > (e) > (c)

117. The number of given orbitals which have electron density along the axis is_____

P_x, P_y, P_z, d_{xy}, d_{xz}, d_{yz}, d_{z^2}, $d_{x^2-y^2}$

[25 Jan, 2023 (Shift-II)]

Sol. (5.00) P_x, P_y, P_z, d_{z^2}, $d_{x^2-y^2}$ are axial orbitals.

shukar karo prabhu ka

118. The number of atomic orbitals from the following having 5 radial nodes is_____

7s, 7p, 6s, 8p, 8d [8 April, 2023 (Shift-II)]

Sol. [3] Radial node is calculated by the formula n - l -1.

For 7p, 6s and 8d the value of radial node is same i.e. 5.

119. The number of correct statements from the following_____

[11 April, 2023 (Shift-II)]

(1) For 1s orbital, the probability density is maximum at the nucleus.

(2) For 2s orbital, the probability density first increases to maximum and then decreases sharply to zero.

(3) Boundary surface diagrams of the orbitals encloses a region of 100% probability of finding the electron.

(4) p and d-orbitals have 1 and 2 angular nodes respectively.

(5) Probability density of p-orbital is zero at the nucleus.

Sol. (3) For explanation, refer to QR code given on the last page.

120. Given below are two statements:

[JEE Mains Shift-II, 2025]

Statement (I): For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II): For any subshell, the spatial orientation of the orbitals is given by -l to +l values including zero.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Sol. (B) For explanation, refer to QR code given on the last page.

ELECTRONIC CONFIGURATION

Pauli's Exclusion Principle

No two electrons in an atom can have the same set of all the four quantum numbers, i.e., an orbital cannot have more than 2 electrons because three quantum numbers (principle, azimuthal and magnetic) at the most may be same but the fourth must be different, i.e., spins must be in opposite directions.

Aufbau Principle

Aufbau is a German word meaning building up. The electrons are filled in various orbitals in order of their increasing energies. An orbital of lowest energy is filled first. The sequence of orbitals in order of their increasing energy is:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d....

The energy of the orbitals is governed by $(n + \ell)$ rule.

Note

'n + l' Rule

The relative order of energies of various sub-shell in a multi electron atom can be predicted with the help of '$n + \ell$' rule

If two sub-shell has same value of $(n + \ell)$ then the sub-shell with lower value of n has lower energy and it should be filled first.

(n + l) rule ko samajhlo, isse sawaal asakta hai

121. For hydrogen atom, the orbital/s with lowest energy is/are : **[JEE Mains Shift-II, 2025]**

(a) 4s (b) $3p_x$ (c) $3d_{x^2-y^2}$

(d) $3d_{z^2}$ (e) $4p_z$

Choose the correct answer from the options given below :

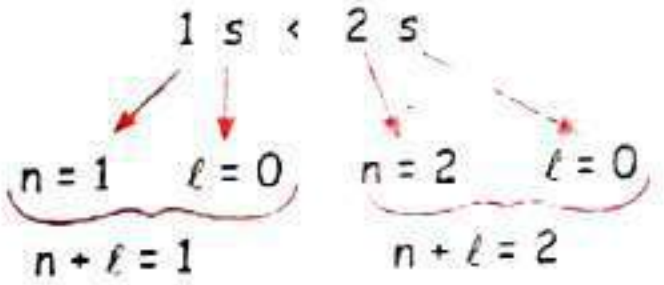
(A) (b) only (B) (a) only

(C) (a) and (e) only (D) (b), (c) and (d) only

Sol. (D) For explanation, refer to QR code given on the last page.

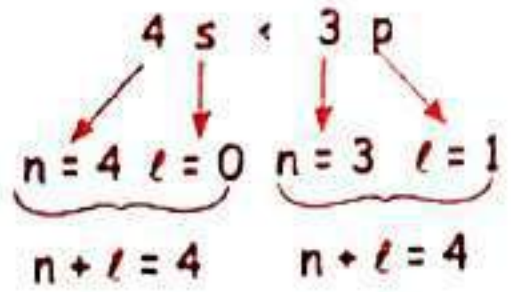
Important

More the $(n + l)$ value more will be the energy.



Khoobsurat Baat

In case of same $(n + l)$ values lesser the 'n' value lesser will be the energy

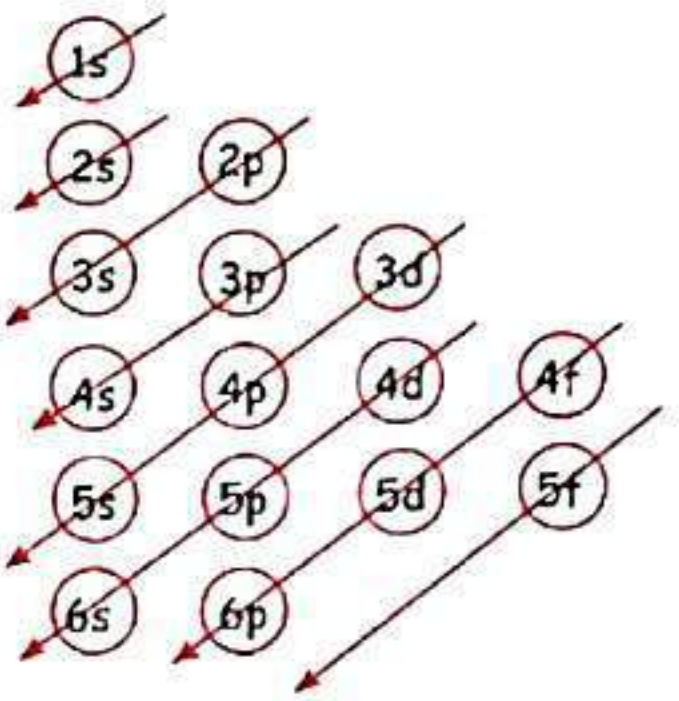


Important

Memory Map

$1s < 2s < 2p < 3s < 3p < 4s < 3d < .....$

$\{Ca(20) = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2\}$



yeh toh yaad karna hi padega varna dikkat aaegi

122. Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) $1s < 2p < 3d < 4s$

(B) $1s < 2s = 2p < 3s = 3p$

(C) $1s < 2s < 2p < 3s < 3p$

(D) $1s < 2s < 4s < 3d$

(A) (A) and (C) only (B) (A) and (B) only

(C) (B) and (D) only (D) (C) and (D) only

Sol. (D) For explanation, refer to QR code given on the last page.

Hund's Rule

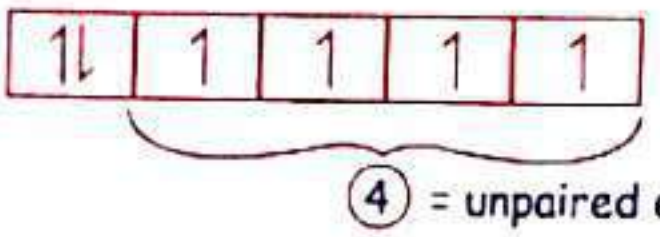
No electron pairing takes place in the orbitals in sub-shell until each orbital is occupied by one electro with parallel spin. Exactly half filled and fully fille orbitals make the atoms more stable, i.e., p^3, p^6, c d^{10}, f^7 and f^{14} configuration are most stable.

123. Find out the total unpaired e^s in Ca(20)?

Sol. $Ca(20) = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ (unpaired = 0)

124. Calculate the spin angular momentum of Fe.

Sol. Fe (26) = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$



$\mu = \sqrt{n(n+2)}$ BM $= \sqrt{24}$ BM

Spin angular momentum $= \frac{h}{2\pi}\sqrt{s(s+1)}$

$s = 4 \times \frac{1}{2} = 2$

Spin angular momentum $= \frac{2.44h}{2\pi}$

dekh toh lo ho kya raha hai yahn

125. Which of the following postulate of Bohr's model of hydrogen atom is not in agreement with quantum mechanical model of an atom?

[03 April, 2025 (Shift-I)]

(A) An atom in a stationary state does not emit electromagnetic radiation as long as it stays in the same state.

(B) An atom can take only certain distinct energies E_1, E_2, E_3, etc. These allowed states of constant energy are called the stationary states of atom.

(C) When an electron makes a transition from a higher energy stationary state to a lower energy stationary state, then it emits a photon of light.

(D) The electron in a H atom's stationary state moves in a circle around the nucleus.

(D) The electron in a hydrogen atom's stationary state follows a circular orbit.

Important

Stability of Completely filled and half filled subshells

- The completely filled and exactly half filled subshells are stable due to the following reason:

1. **Symmetrical Distribution of electrons:** As symmetry leads to stability, the completely filled or half filled subshells have symmetrical distribution of electrons in them and they are therefore more stable.
2. **Exchange Energy:** The number of exchanges which can take place is maximum when the subshell is either half filled or completely filled. As a result the exchange energy is maximum and so is the stability.

sar ko pakad lo ab

126. The extra stability of half-filled subshell is due to:

(a) Symmetrical distribution of electrons

(b) Smaller coulombic repulsion energy

(c) The presence of electrons with the same spin in non degenerate orbitals

(d) Larger exchange energy

(e) Relatively smaller shielding of electrons by one another

Identify the correct statements

[07 April, 2025 (Shift-II)]

(A) (b), (d) and (e) only

(B) (a), (b), (d) and (e) only

(C) (b), (c) and (d) only

(D) (a), (b) and (d) only

Sol. (B) Half-filled subshells are extra stable due to:

Symmetrical electron distribution (a),

Large exchange energy (d),

Smaller coulombic repulsion (b),

Smaller shielding of electrons by one another (e).

Scan the QR code to get the explanations of unsolved questions.

3 Thermodynamics

To access one shot Revision Video Scan this QR Code

INTRODUCTION

Thermodynamics is the branch of chemistry that deals with the study of energy changes, particularly the transfer of energy in the form of heat and work, during chemical reactions and physical processes. It helps us to understand how and why certain processes occur and how energy flows within systems.

System and the Surroundings

System: A system is defined as the specific region or collection of matter where a thermodynamic process takes place.

Surrounding: Surroundings refer to everything outside the system — the rest of the universe that is not part of the system. This is where energy exchanges with the system occur.

Boundary: A boundary is an imaginary or real line that separates the system from the surroundings. The boundary can be fixed or movable, depending on the nature of the system.

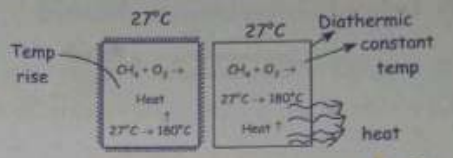
- **Rigid Boundary:** A boundary that does not change its shape or volume. Example: A closed, rigid container.
- **Diathermic Boundary:** A boundary that allows the transfer of heat between the system and surroundings. Example: A metal lid on a beaker that allows heat to flow through.
- **Adiabatic Boundary:** A boundary that does not allow the transfer of heat. Example: A perfectly insulated container.

1. A gas is contained in a piston-cylinder arrangement. The gas expands, doing work on the piston. Identify the system, surroundings, and boundary in this process.

Sol. For explanation, refer to QR code given on the last page.

2. What happens when methane undergoes combustion in system A and B respectively?
[13th April 2023]

Sol. In system A, temperature will rise and in system B, temperature remains constant as heat evolved comes out.



yeh sawaal acha ai, dekhlo ek baar

Types of System

Open system: It can exchange both matter and energy with its surroundings.

Example: A boiling pot of water (heat and steam can escape into the surroundings).

Closed System: It can exchange energy (heat or work) but not matter with its surroundings.

Example: A sealed jar of gas — energy can enter or leave, but gas cannot escape.

Isolated System: It can exchange neither matter nor energy with the surroundings.

Example: An insulated thermos flask — no heat enters or exits, and nothing can enter or leave.

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Perfect example of an isolated system = Universe

Because there is nothing beyond the universe, so heat and mass can not go out or come in to the universe.

Path Functions

- Quantities which are dependent on path where the system has achieved a particular state.
- These quantities are defined when there is a process going on.
- These can not have a particular value in any particular state of system.

Examples:

1. Work (w) 2. Heat (q) 3. Heat capacity(c)

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Small changes of path function is represented as δ. Large changes of a path function is not represented as Δ.

9. Which of the following is not a state function?
(A) Enthalpy (H) (B) Heat (q)
(C) Internal energy (U) (D) Volume

Sol. (B) All the forms of energy = State function
Heat is a path function.

Ialiu ho kya

10. Which of the following is a state function?
(A) Heat (q) (B) Free energy (G)
(C) Internal energy (U) (D) Work (w)

Sol. (B, C) For explanation, refer to QR code given on the last page.

11. Among the following the set of parameters that represent path function
[JEE Mains-2019]
(A) q + w (B) w (C) q (D) H-TS

Sol. (B), (C) Among the given options, q and w are path functions. Hence, B and C are correct options.

Isko galat mat kar dena

THERMODYNAMIC PROCESSES

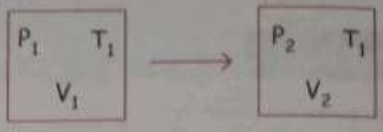
The operation by which a system changes from one state to another state is called a process.

Isothermal Process

When the operation is carried out at constant temperature, the process is said to be isothermal.

For isothermal processes, dT = 0 Where dT is the change in temperature.

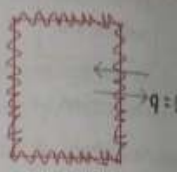
Examples:



Boiling Liq ⇌ vap ΔT = 0
Melting Solid ⇌ liq ΔT = 0

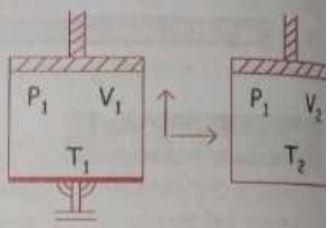
Adiabatic Process

It is a process in which no transfer of heat between the system and surroundings takes place that is q = 0.



Isobaric Process

When the process is carried out at constant pressure, it is said to be isobaric. i.e. dP = 0



ΔP = 0, ΔV ≠ 0, ΔT ≠ 0

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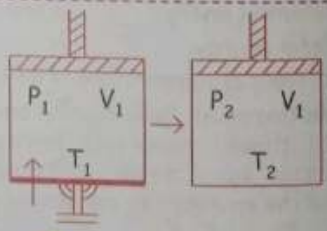
Heat supplied to the system at constant pressure is equal to change in enthalpy.

That is $q_p = \Delta H$

Isochoric (Isovolumetric) Process

A process when carried out at constant volume, it is known as isochoric in nature. dV = 0



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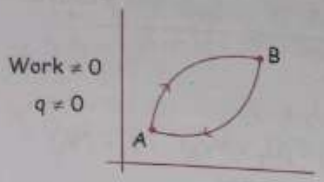
At constant volume process, heat supplied is equal to change in internal energy.

$q_V = \Delta U$

Cyclic Process

When a system undergoes a number of different states by different processes and finally returns to its initial state, it is said to have completed a cycle and the process is called a cyclic process, i.e. $\Delta E = 0$ and also $\Delta H = 0$



$\Delta E = \Delta U = 0$
$\Delta H = 0$
$\Delta S = 0$
$\Delta G = 0$

jaha se start hua wahi akar khatam ho gaya, samjhe

Reversible Process

The process which occurs infinitesimally slowly i.e. opposing force is infinitesimally smaller than driving force and infinitesimal increase in the opposing force can reverse the process.

Some important points related to this process are:

- The process that can be reversed by a very small change is known as reversible process.
- Reversible process is carried out in such a manner so that the system is always in thermodynamic equilibrium at every stage of the process.
- Reversible process is carried out such that the difference in driving force and opposing force is infinitesimally small so that process takes place at infinitesimally slow rate.

  $F_{driving} - F_{opposing} = dF$ and $dF \to 0$
- An ideal reversible process will take infinite time to get completed.
- Strictly speaking there is no ideal reversible process in universe.

Shabdo ko samjho

Note

P_{ext} = variable, $P_{ext} = P_{internal} = P_{gas}$

Irreversible Process

The process goes from initial to final state in a single step or finite number of steps in finite time and cannot be reversed exactly by small changes in P, V, T etc.

Some important points related to this process are:

- The process, which can not be reversed by a small change is known as irreversible process.
- If a process is carried out in such a manner so that the system is in thermodynamic equilibrium
  I. Only at initial & final state of the process but not at the intermediate stages.
  II. System may be in thermodynamic equilibrium state at some finite number of intermediate stages only **for example:** irreversible expansion of a gas.
- If during the process there is a finite difference in driving force and opposing force so that process takes place with a finite rate.
- Irreversible processes will get completed in finite time.
- At intermediate stages of the irreversible process, different state function such as Pressure, temperature etc. are not defined.
- All real process are irreversible.

Note

P_{ext} = const, $P_{ext} \neq P_{gas}$

THERMODYNAMIC VARIABLES

Internal Energy (E or U)

It is the energy associated with a system by virtue of its molecular constitution and the motion of its molecules. The contribution of energy due to molecular constitution is known as internal potential energy and the contribution of energy due to the motion of molecules is called internal kinetic energy. Internal energy of a system is given by the sum of these two types of energy.

Every substance possesses a definite amount of energy which depends upon factors such as composition of the substance, temperature and pressure. This is called intrinsic or internal energy (E).

$$U = E_T + E_R + E_V + E_B + E_E \ldots\ldots\ldots\ldots$$

Where E_T = Translational energy, E_R = Rotational energy, E_V = Vibrational Energy, E_B = Bonding energy and E_E = Electronic energy.

Internal energy of a system is a state function and its absolute value can not be calculated but its change can be determined. When the system undergoes a change in its state, the internal energy change is given by

$$\Delta U = U_2 - U_1$$

Where U_1 = Internal energy in the initial state and U_2 = Internal energy in the final state.

In a chemical reaction,

Internal energy arises due to the existence of the system on its own due to which gravitation energy potential is not included. It is represented as:

$$U_1 = KE_1 + P.E_1 + V.E_1 + \ldots\ldots\ldots$$
$$U_2 = KE_2 + P.E_2 + V.E_2 + \ldots\ldots\ldots$$
$$U_2 - U_1 = \Delta U = (KE_2 - KE_1) + (PE_2 - PE_1) + \ldots\ldots\ldots$$

Internal energy cannot be calculated but change in internal energy can be calculated.

Calculation of internal energy of an ideal gas

$U = KE + \cancel{PE}^{0} + \cancel{VE}^{0} + \ldots\ldots\ldots^{0}$

$U_{ideal\ gas} = K.E$ [overall internal energy = KE]

$$\boxed{U_{ideal\ gas} = K.E = \frac{3}{2}nRT} \qquad \boxed{\Delta U_{ideal} = K.E = \frac{3}{2}nR(\Delta T)}$$

n = no. of moles — change in internal energy

R = gas constant, T = absolute temp (K)

12. Find out the internal energy of an ideal gas (1 mol) at 127°C.

Sol. $U_{ideal\ gas} = \frac{3}{2}nRT = \frac{3}{2} \times 1 \times R \times 400 = 600R$

Bas formula lagega or kuch nhi karna

13. Find out the change in internal energy of 3 moles of an ideal gas on heating it from 27°C → 127°C.

Sol. $\Delta U_{ideal\ gas} = \frac{3}{2}nR\Delta T$

$= \frac{3}{2} \times 3 \times R(400 - 300) = 450R$

14. Internal energy does not include:

(A) Vibration energy

(B) Rotational energy

(C) Energy arising by gravitational pull

(D) Nuclear energy

Sol. (C) Internal energy does not include energy arising by gravitational pull because it is an external energy.

internal energy ko halke mei mat lena

Enthalpy

When we deal certain process in open vessels (at constant pressure), it becomes essential to introduce in place of internal energy, a new thermodynamic function called heat enthalpy. This new function is denoted by H.

$$\boxed{H = E + PV}$$
$$\boxed{H = U + PV}$$

Case 1: At constant P,

State 1 → State 2

H_1, U_1, P, V_1 — H_2, U_2, P, V_2

$H_1 = U_1 + PV_1$, $H_2 = U_2 + PV_2$

Now, $H_2 - H_1 = (\Delta H) = (U_2 + PV_2) - (U_1 + PV_1)$

$= (U_2 - U_1) + P(V_2 - V_1)$

$\Delta H = \Delta U + P\,\Delta V$ ...(i)

An ideal gas equation is:

$PV = nRT$

At constant T, $P\Delta V = \Delta n_g RT$

So, for any chemical reaction at constant temperature

$$\boxed{\Delta H = \Delta U + \Delta n_g RT}$$

Case 2: When P is not constant

State 1 → State 2

H_1, U_1, P_1, V_1 — H_2, U_2, P_2, V_2

$H_1 = U_1 + P_1V_1$, $H_2 = U_2 + P_2V_2$

$H_2 - H_1 (\Delta H) = (U_2 + P_2V_2) - (U_1 + P_1V_1)$

$= (U_2 - U_1) + (P_2V_2 - P_1V_1)$

$$\boxed{\Delta H = \Delta U + \Delta(PV)}$$

15. The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane (l) is carried out at a temperature T, is equal to

[JEE Mains, 10 April 2019 Shift-II]

(A) 3RT (B) -3RT (C) -4RT (D) 4RT

Sol. (C) For explanation, refer to QR code given on the last page.

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$\Delta H = \Delta U + P\Delta V$ (P constant)

$\Delta H = \Delta U + \Delta n_g RT$ (for a chemical reaction at const T)

$\Delta H = \Delta U + \Delta(PV)$ (pressure is variable)

16. For water at 100°C and 1 bar
$\Delta_{vap}H - \Delta_{vap}U =$ _____ $\times 10^2$ J mol^{-1}
(Round off to the Nearest integer)
[Use : R = 8.31 J mol^{-1} K^{-1}]
[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ can be treated as an ideal gas]
[JEE Mains, 27 July 2021 Shift-I]

Sol. (3100 J/mol or 31 × 10^2 J/mol) For explanation, refer to QR code given on the last page.

17. One mole of a non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) → (4.0 atm, 5.0 L, 245 K) with a change in internal energy, ΔU = 30.0 L-atm. The change in enthalpy (ΔH) of the process in L-atm is
[IIT JEE Adv.]
(A) 40.0 (B) 42.3 (C) 44.0
(D) Not defined, because pressure is not constant

Sol. (C) For explanation, refer to QR code given on the last page.

18. The difference between heats of reaction at constant pressure and at constant volume for the reaction at 25°C in kJ is-
$2C_6H_{6(l)} + 15O_{2(g)} \rightarrow 12CO_{2(g)} + 6H_2O_{(l)}$

Sol. $\Delta H = \Delta E + \Delta n_g RT$
Here, $\Delta n_g = 12 - 15 = -3$
$\therefore \Delta H - \Delta E = -3 \times 8.314 \times 298 = -7.43$ kJ

19. A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36g. When it is subjected to cooking at 100°C, then the internal energy for vaporization in kJ mol^{-1} is _____ (Nearest integer)
[Assume steam to be an ideal gas. Given $\Delta_{vap}H^\circ$ for water at 373 K and 1 bar is 41.1 kJ mol^{-1}, R = 8.31 JK^{-1} mol^{-1}]
[26 June, 2022 (Shift-II)]

Sol. Mole of $H_2O = \frac{36}{18} = 2$ mol
$\Delta U = \Delta H - \Delta n_g RT$, $H_2O(l) \rightarrow H_2O(g)$
$= 41.1 - \frac{1 \times 8.31 \times 373}{1000} = 38$ kJ/mol

yeh agar nhi hua toh tum lallu ho

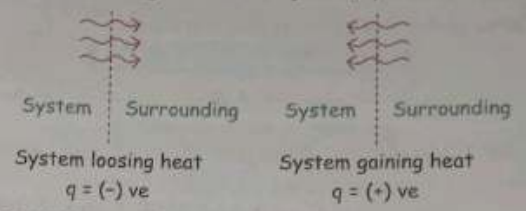
Heat

Heat may be defined as the quantity of energy which flows between a system and its surroundings on account of temperature difference. Heat always flows from high temperature to low temperature.

According to IUPAC system:

(i) Heat flowing into the system, q = +ve.

(ii) Heat flowing out of the system, q = -ve



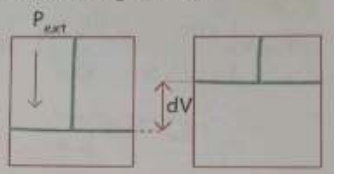
Work

By definition, the work involved is given by:

w = -(External force) (Distance through which piston moves)

$w = -P_{ext}(\Delta V)$

(ΔV = Area × distance)



where ΔV is the change in volume of the system. If the piston moves by an infinitesimal amount, the work involved is given by

$$dw = -P_{ext}dV$$

$$\int dw = w = -\int_{V_1}^{V_2} P_{ext}dV$$

P_{ext} = variable → P_{ext} = const

Case-1

P_{ext} = constant

$dw = -P_{ext}\,dV$

$w = -P_{ext}\int_{V_1}^{V_2} dV$

$= -P_{ext}(V_2 - V_1)$

Case-2

$P_{ext} \neq$ constant (a function of volume)

$dw = -P_{ext}dV$

$w = -\int_{V_1}^{V_2} P_{ext}dV$

Important

$dw = -P_{ext}dV$

In case of compression ⇒ dV = (-) ve

$dw = \underbrace{-P_{ext}}_{(-)ve}\underbrace{dV}_{(-)ve} = (+ve)$

In case of expansion of gas ⇒ $dw = \underbrace{-P_{ext}}_{(-)ve}\underbrace{dV}_{(+)ve} = (-)$ ve

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Work done on system = (+) ve

Work done by system = (-) ve

Area under P-V curve is also work done.

yaad karlo isko, chhodna mat

20. Calculate the work done by the system when it expands from 1 Lt to 10 Lt against a constant pressure of 1 atm. **[JEE Mains-2019]**

Sol. $w = -\int P_{ext} dV = -P_{ext} \int_1^{10} dV$

$= -1 (10 - 1) = -9$ atm. Lt $= -911.7$ J

1 atm - lt = 101.3 J

1 bar - lt = 100 J

minus, plus sign ko mat bhul jana, samjhe!

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* Pressure (Pascals) × volume (m^3) → work (J)

1 atm = 1.013×10^5 Pascals

* Pressure (atm) × volume (Lt) → work (atm-Lt)

1 atm-Lt = 101.3 J

* Pressure (atm) × volume (m^3) → work (atm-m^3)

1 m^3 = 1000 Lt

Note

The sign convention

- w is taken as negative if work is done by the system
- w is taken as positive if work is done on the system.
- When heat is given by the system to surrounding, it is given as negative sign.
- When heat is absorbed by the system from the surrounding, then it has a positive sign.

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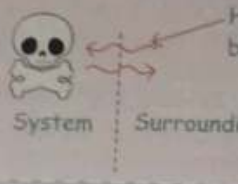
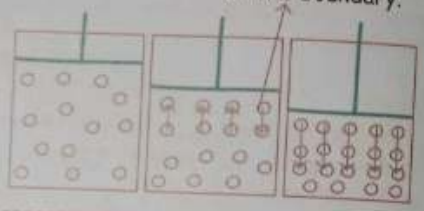
Heat and work both are boundary phenomenon

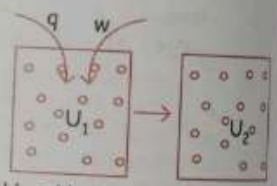


work also first appears at the boundary.



FIRST LAW OF THERMODYNAMICS

- Energy can neither be created nor destroyed, it can be converted from one form to another.
- Universe is an isolated system, hence the sum of all the energies remains constant. So that total energy is constant.



$U_2 = U_1 + q + w$

$U_2 - U_1 = q + w$

$\boxed{\Delta U = q + w}$

21. Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? **[JEE Mains, 8 April 2019 Shift-I]**

(A) Cyclic process : $q = -w$

(B) Isothermal process : $q = -w$

(C) Adiabatic process : $\Delta U = -w$

(D) Isochoric process : $\Delta U = q$

Sol. (C) According to first law:

$\Delta U = q + w$

For adiabatic process, $q = 0$

$\Delta U = w$

22. A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy_____
[25 July 2021]

Sol. (50) For explanation, refer to QR code given on the last page.

23. During compression of spring the work done is 10 kJ and 2 kJ energy is escaped out in the form of heat. The change in internal energy-
[JEE Mains-2019]

Sol. Work done on the system is positive while heat released by the system is negative.
Now, apply first law of thermodynamics:
$\Delta U = -2 + 10 = 8$ kJ or 8000 J

24. A liquid when kept inside a thermally insulated closed vessel at 25°C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters? [22 Jan, Shift-I, 2025]

(A) $\Delta U = 0, q < 0, w > 0$
(B) $\Delta U > 0, q = 0, w > 0$
(C) $\Delta U < 0, q = 0, w > 0$
(D) $\Delta U = 0, q = 0, w = 0$

Sol. (B) For explanation, refer to QR code given on the last page.

(i) Isothermal process: (Boundary wall should be diathermic)

$\Delta U = q + w$, $+q = -w$, $-q = +w$

+q = heat given to the system
−w = work done by the system
+w = work done on the system
−q = heat released by the system

Temperature constant hi rahega

(ii) Adiabatic process: (Boundary should be adiabatic)

$\Delta U = q + w$, $+\Delta U = +w$, $-\Delta U = -w$

+w = work done on the system
$(U_2 - U_1)$ = (+) ve = internal energy (T) is increasing
−w = work done by the system
$(U_2 - U_1)$ = (−) ve = internal energy (T) is decreasing

Heat zero rahegi

(iii) Isochoric process: (Diathermic wall)

$\Delta U = q + w$, $(+)\Delta U = +q$, $(-)\Delta U = -q$

+q = heat supplied to the system
ΔU = (+) ve = internal energy (T) is increasing.
−q = heat is lost by the system
ΔU = (−) ve = internal energy (T) is decreasing.

$$\boxed{q_v = \Delta U}$$

Volume change 0 rahegil

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At constant volume, heat supplied or released by the system is equal to change in internal energy.

(iv) Isobaric process: (Diathermic movable wall)

$\Delta U = q + w$
$q = \Delta U - w = (U_2 - U_1) + P(V_2 - V_1)$
$q = (U_2 + PV_2) - (U_1 + PV_1) = H_2 - H_1$

$$\boxed{q_p = \Delta H}$$

Pressure change zero rahega!

Khoobsurat Baat

Heat exchange at constant pressure is equal to change in enthalpy.

Note

$\Delta H = \Delta U + P\Delta V$, $\Delta H = \Delta U + \Delta n_g RT$
$q_p = q_v + \Delta n_g RT$

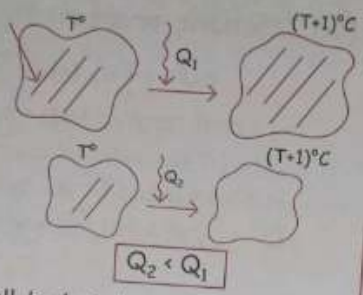
Heat Capacity

It is defined as the quantity of heat required to raise the temperature of the system by one degree.

i.e. $$\boxed{C = \frac{dq}{dT}}$$

Where C = Heat capacity, dq = Amount of heat absorbed by the system and dT = Rise in temperature.
Its unit is J K^{-1} or J/°C.

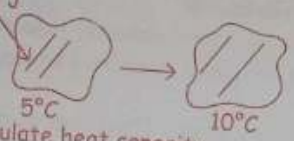
Larger the sample, greater is its heat capacity. It is therefore, common to use specific heat capacity (specific heat) or molar heat capacity.



Heat capacity of small body is less than the heat capacity of bigger body.

dekh toh lo saaf samajh araha hai

25. 100 J



Calculate heat capacity.

Sol. Heat Capacity $(C) = \frac{dq}{dT}$

$$\text{Heat Capacity} = \frac{\text{Heat supplied}}{\text{change in temp}}$$

$$= \frac{100\,J}{(10-5)^\circ C} = 20\,J/^\circ C$$

Specific Heat Capacity

Heat required to raise the temp of 1 gm system by 1°.

$s = \frac{dq}{mdT}$, $dq = msdT$ $\boxed{q = ms(T_2 - T_1)}$

26. Find out the heat required to raise the temp of 50 gm water from 50°C to 80°C. Specific heat capacity = 4.18 J/g°C

Sol. $q = ms\Delta T = 50 \times 4.18 \times 30\,J = 6270\,J = 6.27\,kJ$

Simple formula hi toh lagaya hai bas

27. A sample of oxygen gas expands its volume from 3 litre to 5 litre against a constant pressure of 3 atm. If the work done during expansion be used to heat 10 mole of water initially present at 290 K, its final temperature will be (Specific heat capacity of water = 4.18 $J\,K^{-1}\,g^{-1}$)

(A) 292.0 K (B) 290.8 K (C) 298.0 K (D) 293.7 K

Sol. (B) For explanation, refer to QR code given on the last page.

Molar Heat Capacity

Molar heat capacity is defined as the heat required to raise the temp of 1 mol of the system by 1°C.

$\boxed{C = \frac{dq}{ndT}}$ Unit = J/mol K

C_V ↑

For an ideal gas system → To raise the temp by 1° at constant volume

For an ideal gas system → To raise the temp by 1° at constant P ↓ C_p

C_V = molar heat capacity at constant volume

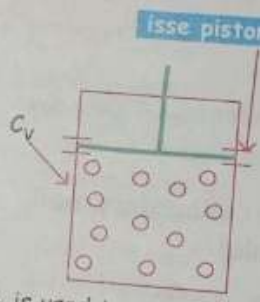
↳ Heat supplied will directly changes its temperature (work done is not permitted $\Delta V = 0$)

C_p = molar heat capacity at constant pressure

↳ Heat supplied will be used in (i) increasing temp (ii) work done

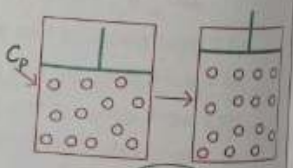
$\boxed{C_p > C_V}$

Isse piston kahin nhi jaega



C_V is used to raise the temp of the gas only. No heat is wasted in doing work ($\Delta V = 0$)

C_p is used to raise the temp of gas and some amount is wasted in doing some work.



$C_V = \left(\frac{dq}{dT}\right)_V = \frac{dU}{dT}$

↓ molar heat capacity at constant volume

$dU = C_V dT$ For one mole

$dU = nCVdT$ For n moles of gas

Important

if C_V is constant

$$\Delta U = \int dV = nc_V \int_{T_1}^{T_2} dT = nC_V(T_2 - T_1)$$

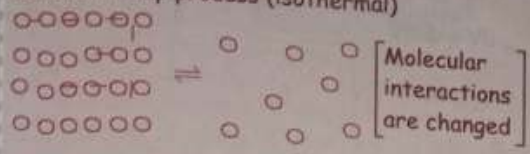
if C_V is temp dependent $\Delta U = \int dU = n\int_{T_1}^{T_2} C_V dT$

Khoobsurat Baat

For any isothermal ($\Delta T = 0$) process, $\Delta U = 0$ is not always correct. As $\Delta U = nC_v\Delta T$ is applicable only when molecular interactions are not changed. For example-

$H_2O_{(l)} \rightleftharpoons H_2O_{(v)}$ [T = B.P (100°C)]

Constant temp process (isothermal)



The above process is isothermal process still $\Delta U \neq 0$, as molecular interactions are not same.

28. At constant volume 4 mole of ideal gas when heated from 300 K to 500 K changes its internal energy by 5000 J. Find C_V, m.....

[Jan-2020]

Sol. (6.25) For explanation, refer to QR code given on the last page.

$$C_p = \left(\frac{dq}{dT}\right)_p = \frac{dH}{dT}$$

$dH = C_pdT$ for one mole, $dH = nC_pdT$ for n mole

Important

If C_p is constant

$$\Delta H = n\int_{T_1}^{T_2} C_pdT$$

If C_p is function of temperature

$$\Delta H = nC_p\Delta T = nC_p(T_2 - T_1)$$

Note

$dT = 0$, $dH = 0$ only when molecular interactions remain same.

$H_2O_{(s)} \xrightleftharpoons{\text{const temp (0°C)}} H_2O_{(l)}$: $\Delta H \neq 0$ ($\Delta H = \Delta H_{fusion}$)

Khoobsurat Baat

At constant temperature, $dH = 0$ only when molecular interactions remain same.

29. 2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is '-x' J. The value of 'x' is _____ [nearest integer) (Given atomic mass of N = 14 g mol^{-1} and of O = 16g mol^{-1}. Molar heat capacity of N_2O is 100 J K^{-1} mol^{-1})

[JEE Mains, 30 Jan 2023 Shift-I]

Sol. (w = +5J, ΔU = -195 J)

For explanation, refer to QR code given on the last page.

Relationship between C_p and C_v:

$H = U + PV$

$H = U + nRT$ (For 'n' moles)

$H = U + RT$ (For one mole)

$$\left(\frac{dH}{dT}\right) = \left(\frac{dU}{dT}\right) + R$$

$C_p = C_V + R$ $\quad\boxed{C_p - C_V = R}$

30. For one mole of an ideal gas, which of these statements must be true?

(a) U and H each depends only on temperature

(b) Compressibility factor Z is not equal to 1

(c) $C_{p,m} - C_{V,m} = R$

(d) $dU = C_vdT$ for any process

(A) (a), (c) and (d) (B) (b), (c) and (d)

(C) (a) and (c) (D) (c) and (d)

Sol. (A) Except statement (b), all are correct.

For an ideal gas, compressibility factor Z is equal to 1.

Poisson's ratio (γ): It is defined as the ratio of C_p to C_v.

$$\boxed{\gamma = C_p/C_V}$$

For monoatomic gases	$C_v = \frac{3}{2}R$	$C_p = \frac{5}{2}R$	$\gamma = 1.66$
For diatomic gases	$C_v = \frac{5}{2}R$	$C_p = \frac{7}{2}R$	$\gamma = 1.4$
For Triatomic gases	$C_v = \frac{6}{2}R$	$C_p = \frac{8}{2}R$	$\gamma = 1.33$

31. 5 moles of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature becomes 200 K. If C_v = 28 J K^{-1} mol^{-1}, calculate ΔU and $\Delta(pV)$ for this process. (R = 8.0 J K^{-1} mol^{-1}) **[JEE Mains-2019]**

(A) ΔU = 14 kJ; $\Delta(pV)$ = 4 kJ

(B) ΔU = 14 kJ; $\Delta(pV)$ = 18 kJ

(C) ΔU = 2.8 kJ; $\Delta(pV)$ = 0.8 kJ

(D) ΔU = 14 kJ; $\Delta(pV)$ = 0.8 kJ

Sol. (A) For explanation, refer to QR code given on the last page.

32. For silver, $C_p(K^{-1} mol^{-1})$ = 23 + 0.01 T. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be close to:

(A) 21 kJ (B) 62 kJ (C) 16 kJ (D) 13 kJ

Sol. (B) $\Delta H = nC_p\Delta T$

$$\int dH = n\int_{T_1}^{T_2} C_p dT$$

$$\Delta H = 3\int_{300}^{1000}(23+0.01T)dT = 3\left[23T + 0.01\frac{T^2}{2}\right]_{300}^{1000}$$

On solving, ΔH = 61.95 kJ/mol ≅ 62 kJ/mol

integration seekhlo bhayi, varna beda garg

33. An ideal gas undergoes isothermal compression from 5 m^3 to 1 m^3 against a constant external pressure of 4 Nm^{-2}. Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is 24 J mol^{-1} K^{-1}, the temperature of Al is increased by **[JEE Mains, 10 Jan 2019 Shift-II]**

(A) 3/2 K (B) 2 K (C) 2/3 K (D) 1 K

Sol. (C) For explanation, refer to QR code given on the last page.

34. 500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: **[29 Jan, Shift-I, 2025]**

Given: R = 8.3 J K^{-1} mol^{-1}

(A) 378 K and 300 J (B) 348 K and 300 J

(C) 368 K and 500 J (D) 378 K and 500 J

Sol. (B) For explanation, refer to QR code given on the last page.

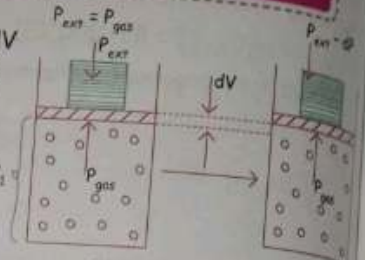
WORK DONE IN AN ISOTHERMAL REVERSIBLE PROCESS

$dw = -(P_{ext} - dP)\,dV$

$= -P_{ext}\,dV + dP\,dV$ ($dP\,dV \to 0$)

$\boxed{dw = -P_{ext}\,dV}$

$dw = -P_{gas}\,dV$



$$\int dw = -nRT\int_{V_1}^{V_2}\frac{dV}{V} = -nRT\ln\frac{V_2}{V_1} = -2.303\,nRT\log\left(\frac{P_1}{P_2}\right)$$

Agr formula yaad hai toh samjhlo kaam ban gaya

Note

$P_1V_1 = P_2V_2$ ($T_{const.}$, n_{const}), $\frac{V_2}{V_1} = \frac{P_1}{P_2}$

Important

Work done calculation against a pressure which is variable and equal to the pressure of the gas-

→ Reversible process

35. When '5' mole of He gas expands isothermally and reversibly at 300 K from 10 L to 20 L the magnitude of the maximum work obtained is _____ (R = 8.314 J/K mole, log 2 = 0.3010) **[27th June-2022]**

Sol. (-8645.7 J) For explanation, refer to QR code given on the last page.

36. Consider the figure provided. 1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at 18°C. If the piston is moved to position B, keeping the temperature unchanged, then 'x' L-atm work is done in this reversible process.

x = _____ L atm. (nearest integer)

[Given: Absolute temperature = °C + 273.15, R = 0.08206 L-atm mol^{-1} K^{-1}] **[08 April, 2024 (Shift-1)]**

Sol. [55] Given, $V_2 = 100L$, $V_1 = 10L$

$T = 18 + 273.15 = 291.15\ K$

$$w = -nRT\ln\left(\frac{V_2}{V_1}\right)$$

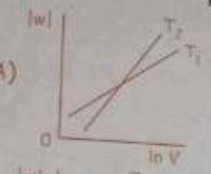
$$= -1 \times 0.08206 \times 291.15 \ln\left(\frac{100}{10}\right) = -55.0128$$

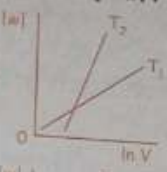
Work done by system ≈ 55 L-atm.

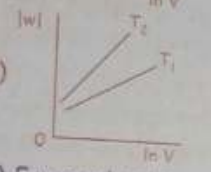
yeh dimag ko ghumaega pura

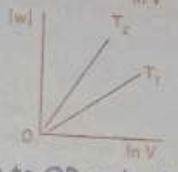
37. Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperature T_1 & T_2 ($T_1 < T_2$). The graphical depiction of the dependence of work done (w) on the final volume (V) is

[9 Jan, 2019 (Shift-I)]

(A) 

(B) 

(C) 

(D) 

Sol. (B) For explanation, refer to QR code given on the last page.

38. 2 moles of helium is compressed from 1 atm to 2 atm pressure at 27°C in reversible path. Calculate
1. Work done
2. Heat transfer
3. Change in internal energy

Sol. (1. 3500 J, 2. -3500 J, 3.0) For explanation, refer to QR code given on the last page.

39. 16 g of oxygen gas expands isothermally and reversibly at 300 K from 10 dm^3 to 100 dm^3. The work done is: (in J)

(A) Zero (B) -2875 J (C) +2875 J (D) Infinite

Sol. (B) $w = -2.303nRT\log\frac{V_2}{V_1}$

$$= -2.303 \times \frac{16}{32} \times 8.314 \times 300 \log\frac{100}{10} \approx -2875\ J$$

asaani se ho jaega

40. If 5 moles of an ideal gas expands from 10 L to a volume of 100 L at 300 K under isothermal and reversible condition then work, w, is -x J. The value of x is _____.

(Given R = 8.314 J K^{-1} mol^{-1})

[31 Jan, 2024 (Shift-II)]

Sol. (-28721J) For explanation, refer to QR code given on the last page.

41. Calculate the work done when 1 mole of an ideal gas is compressed reversibly from 1 bar to 4 bar at a constant temperature of 300 K.

(A) 4.01 kJ (B) - 8.02 kJ

(C) 18.02 kJ (D) 3.458 kJ

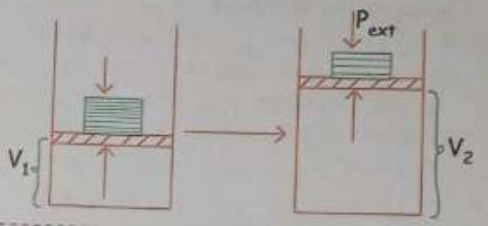
Sol. (D) Work done = $-2.303nRT\log\frac{P_1}{P_2}$

$$= -2.303 \times 1 \times 8.314 \times 300 \log\frac{1}{4}$$

$= 3458.3\ J$ or $3.458\ kJ$

formula pta nhi kya

WORK DONE IN AN ISOTHERMAL IRREVERSIBLE EXPANSION PROCESS



Irreversible Isothermal

$$w_{irreversible} = -\int P_{ext}\,dV = -P_{ext}(V_2 - V_1)$$

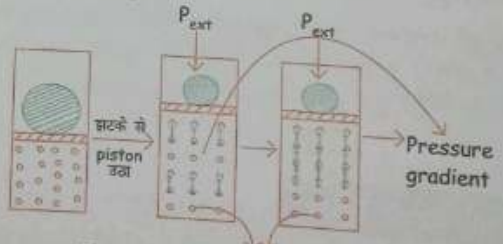
similar expressions

Generally $P_{ext} = P_2$

$$w_{irr} = -P_2\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) = nRT\left(\frac{P_2}{P_1} - 1\right) = nRT\left(\frac{V_1}{V_2} - 1\right)$$

Note

The pressure against which gas is performing work is constant and much less than internal pressure (P_{gas}).



Throughout the pressure is different in the container so gas pressure is not defined.

42. An ideal gas expands in volume from $1 \times 10^{-3}\ m^3$ to $1 \times 10^{-2}\ m^3$ at 300 K against a constant pressure of $1 \times 10^5\ Nm^{-2}$. The work done is

(A) 270 kJ (B) -900 kJ

(C) -900 J (D) 900 kJ

Sol. (B) For explanation, refer to QR code given on the last page.

43. At 25°C, 50 g of iron reacts with HCl to form $FeCl_2$. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is _____ J. (Round off to the nearest integer).

[Given: $R = 8.314\ J\ mol^{-1}\ K^{-1}$. Assume hydrogen is an ideal gas]

[Atomic mass of Fe is 55.85u]

[16 March, 2021 (Shift-II)]

Sol. (2218 J) For explanation, refer to QR code given on the last page.

44. 4.4 g of an ideal gas (mol. wt = 44) at 300K and 10 atm is allowed to expand isothermally against a external pressure of 1 atm. Taking 1 L-atm = 100 J, heat absorbed by the gas is:

(A) 200.5 J (B) 212.5 J

(C) 221.4 J (D) 242.4 J

Sol. (C) For explanation, refer to QR code given on the last page.

Comparison of the work done by the gas in reversible and irreversible process

1. For expansion

Method 1

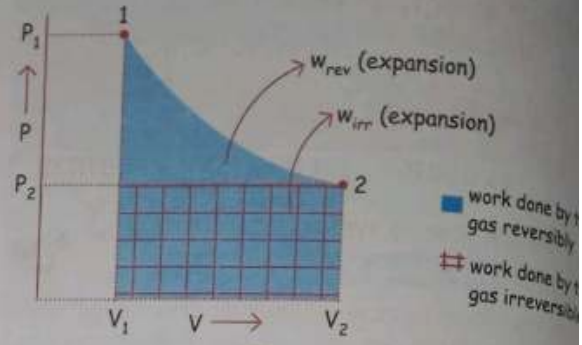
In case of work done by the gas reversibly, work is done against a pressure which is equal to the internal pressure (P_{gas}). ($P_{ext} = P_{gas}$)

In case of irreversible expansion, work done by the gas is against a pressure which is much less than internal pressure (P_{gas}). ($P_{ext} << P_{gas}$)

Khoobsurat Baat

$W_{rev} > W_{irrev}$

Method 2



Khoobsurat Baat

$W_{rev} > W_{irrev}$

2. For contraction

Method 1

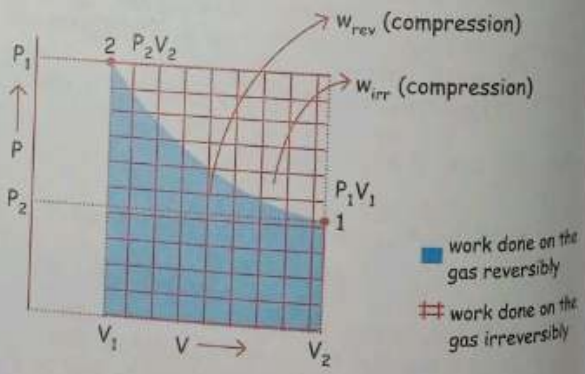
In case of reversible compression the external pressure which is doing work is equal to internal pressure (P_{gas}). ($P_{ext} = P_{gas}$)

In case of irreversible compression the pressure which is doing work is much greater than internal pressure (P_{gas}). ($P_{ext} >> P_{gas}$)

Khoobsurat Baat

$W_{irrev} > W_{rev}$

Method 2

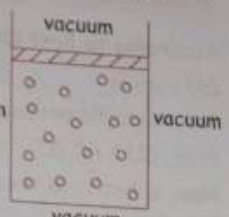


Khoobsurat Baat

$W_{irrev} > W_{rev}$

Free Expansion

Free expansion is a thermodynamic process in which a gas expands into a vacuum (a region with no external pressure) without performing any work and without any heat exchange with the surroundings.



Expansion against $P_{ext} = 0$

$w = -\int P_{ext} dV = 0$

$w = 0, q = 0, \Delta U = 0$

yeh toh sabse acha concept hai

45. When 4 lt of ideal gas is allowed to expand isothermally into vacuum to a total volume of 20 lt, the amount of heat absorbed in kJ is _____. [June-2022]

Sol. As process is taking place in vacuum, so, $w = 0$; $\Delta U = 0$, $\Delta U = q + w$, $q = 0$

46. '5' moles of an ideal gas at 1 bar and 298 K expanded into vacuum to double the volume work done is [Sep-2020]

(A) $C_V (T_2 - T_1)$ (B) $-RT \ln \frac{V_2}{V_1}$

(C) $\frac{RTV_2}{V_1}$ (D) Zero

Sol. (D) As the process is taking place in vacuum, so, $w = 0$.

vacuum shabd ko dekho bas, kch nahi karna

47. When 2 litre of ideal gas expands isothermally into vacuum to a total volume of 6 litre, the change in internal energy is J. (Nearest integer) [JEE Mains, 30 Jan 2023 Shift-1]

Sol. According to the given information, process is taking place in a vacuum, so, change in internal energy will be equal to 0.

WORK DONE IN A REVERSIBLE ADIABATIC PROCESS

Adiabatic Process

- Walls should be adiabatic
- Heat exchange = 0
- Work done on the system will increase the internal energy of the system that is $\Delta U > 0$

  $T_2 > T_1$ (T_2 = final temp, T_1 = initial temp)
- Work done by the system will decrease the internal energy of the system that is $\Delta U < 0$

  $T_2 < T_1$ (T_2 = final temp, T_1 = initial temp)

Apply first law of thermodynamics:

$\Delta U = q + w$, $q = 0$ (adiabatic process)

$nC_V dT = -P_{ext} dV$

kyunki P_{ext} ko P_{gas} bol dia, toh process reversible maana hai

$nC_V dT = -P_{gas} dV$, $nC_V dT = -nRT \frac{dV}{V}$

$C_V \int_{T_1}^{T_2} \frac{dT}{T} = -\int_{V_1}^{V_2} R \frac{dV}{V}$, $C_V \ln \frac{T_2}{T_1} = R \ln \frac{V_1}{V_2}$

$\ln \frac{T_2}{T_1} = \left(\frac{R}{C_V}\right) \ln \frac{V_1}{V_2}$ $(R/C_v = \gamma - 1)$ socho kyu

$\ln \frac{T_2}{T_1} = (\gamma - 1) \ln \frac{V_1}{V_2}$, $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma - 1}$, $T_1 V_1^{\gamma - 1} = T_2 V_2^{\gamma - 1}$

$TV^{\gamma - 1} = \text{constant}$

Note

$TV^{\gamma-1} = \text{const}$	$TV^{\gamma-1} = \text{const}$	$TV^{\gamma-1} = \text{const}$
$\left(\frac{PV}{nR}\right)(V)^{\gamma-1} = \text{const}$	$T\left(\frac{nRT}{P}\right)^{\gamma-1} = \text{const}$	Reversible Adiabatic
$PV^{\gamma} = \text{const}$	$T^{\gamma}P^{1-\gamma} = \text{const}$	
Reversible Adiabatic	Reversible Adiabatic	

(yeh sare formulae yaad karlo ache se, mza ajaega)

Important

$W_{\text{adiabatic reversible}} = nC_V (T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1}$

$= \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

Note all those relations PV^{γ} = const, $TV^{\gamma - 1}$ = const, $P^{1-\gamma}T^{\gamma}$ = const are applicable

48. An ideal monoatomic gas at 300 K expands adiabatically (reversible) to 8 times its volume. What is the final temperature?
(A) 300 K (B) 340 K (C) 560 K (D) 75 K

Sol. (D) $\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{\gamma-1}$, $T_2=300\left(\frac{V}{8V}\right)^{\frac{5}{3}-1}=75\ K$

Gajab lallu hai

Temperature to 300 K se kam hi hoga kyunki expansion ho raha hai to gas ki internal energy kam hi hogi. So answer will be less than 300 K.

Ek hi option tha!

Khoobsurat Baat

While calculating work done in reversible adiabatic process we need to find out T_2 (final temp),

$T_2=\left(\frac{T_1V_1^{\gamma-1}}{V_2^{\gamma-1}}\right)^{\frac{1}{\gamma-1}}$ is used.

$$T_2=\left(\frac{T_1^{\gamma}P_1^{1-\gamma}}{P_2^{1-\gamma}}\right)^{\frac{1}{\gamma}}$$

49. Adiabatic reversible process $P^{-2/5}T^{7/5}$, gas is
(A) He (B) NH_3 (C) O_2 (D) Ne

Sol. (C) For explanation, refer to QR code given on the last page.

50. In adiabatic reversible process $TV^{1/3}$ = constant. Gas is:
(A) He (B) NH_3 (C) H_2 (D) O_2

Sol. (B) For adiabatic reversible process:
$TV^{\gamma-1}$ = const

$\gamma-1=\frac{1}{3}$, $\gamma=1+\frac{1}{3}=\frac{4}{3}$ (Polyatomic)

Among the given options, NH_3 is a polyatomic gas.

dimag ji baji ghanti

51. 1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. The work done is 3 kJ mol^{-1}. The final temperature of the gas is _____K (nearest integer). Given C_V = 20 J mol^{-1} K^{-1}.

[JEE Mains, 30 Jan 2023 Shift-II]

Sol. $\Delta U = \cancel{q}^{0} + w$

$nC_V(T_2-T_1)=-3000$

$1\times 20(T_2-300)=-3000$

$T_2-300=-150$, $T_2=150\ K$

WORK DONE IN AN IRREVERSIBLE ADIABATIC PROCESS

According to first law of thermodynamics:

$\Delta U = q + w$

For adiabatic process, $q = 0$

Also, $\Delta U = nC_v\Delta T$

Now, substitute these values in the equation of first law of thermodynamics:

$$nC_V(T_2-T_1)=w_{irr}$$

While calculating work done in irreversible adiabatic process, we need to calculate T_2(final temp).

$$nC_V(T_2-T_1)=-P_{ext}(V_2-V_1)$$

P_{ext}

$$nC_V(T_2-T_1)=-P_2\left[\frac{nRT_2}{P_2}-\frac{nRT_1}{P_1}\right]$$

{generally if P_{ext} is not given then $P_{ext}=P_2$}

mast formula hai

Important

$$w_{\text{adiabatic irreversible}}=nC_V\Delta T=nC_V(T_2-T_1)$$

$$=\frac{nR(T_2-T_1)}{\gamma-1}=-P_{ext}(V_2-V_1)$$

Note

T_2 can be calculated from here because PV^{γ} = const is not applicable here.

52. 1 m^3 of neon gas $\left(C_V=\frac{3}{2}R\right)$ at 273.2 K and 10 atm undergoes three different expansions to a final pressure of 1 atm. Calculate the final volume and the work done in three different expansions.
(a) Isothermal reversible expansion
(b) Adiabatic reversible expansion and
(c) Irreversible adiabatic expansion

Sol. (a) -2333.5 kJ (b) T_2 = 108.8 K
(c) w = -547 kJ
For explanation, refer to QR code given on the last page.

ISOTHERMAL AND ADIABATIC SLOPE IN P-V GRAPH

For isothermal process:

$$PV = \text{constant}$$

Now, differentiate this equation:

$$PdV + VdP = 0$$

$$\boxed{\frac{dP}{dV} = -\left(\frac{P}{V}\right)}$$

↓

Slope in isothermal process in P-V graph

For adiabatic process:

$$PV^{\gamma} = \text{const}$$

Now differentiate this equation: $P\gamma dV^{\gamma-1} + V^{\gamma}dP = 0$

$$\boxed{\frac{dP}{dV} = -\gamma\left(\frac{P}{V}\right)}$$

↓

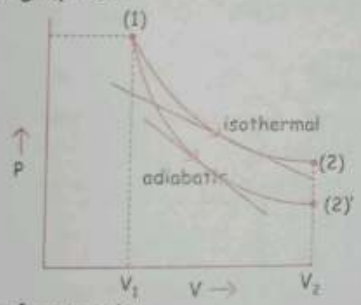
Slope in adiabatic process in P-V graph

Khoobsurat Baat

Slope of adiabatic graph is γ times to the slope of isothermal curve.

Note

Both isothermal and adiabatic processes can be shown in graph as:



In case of expansion,

Slope for curve (1) → (2)' is more as compared to slope for (1) → (2) curve.

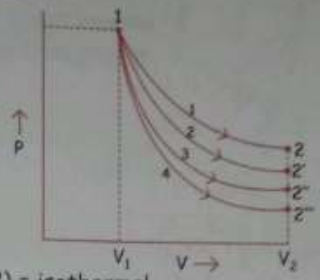
So, (1) → (2)' = adiabatic (1) → (2) = isothermal

Khoobsurat Baat

During expansion, more dhalaan (steeper) wala curve adiabatic curve hoga

Isothermal and Adiabatic Expansion Curves to a Fixed Volume

Note



(1) → (2) = isothermal

(1) → (2)' = adiabatic (polyatomic)

(1) → (2)'' = adiabatic (diatomic)

(1) → (2)''' = adiabatic (monoatomic)

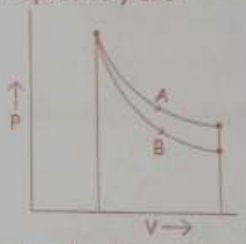
$$(\text{Slope})_{\text{adiabatic}} = -\gamma\left(\frac{P}{V}\right),$$

more γ = more slope

Khoobsurat Baat

Work done by the gas (adiabatic expansion) from V_1 to V_1 It in case of monoatomic gas is less as compared to diatomic and polyatomic gas.

53. From the adiabatic graph given below, gases A and B respectively are:



(A) A = mono, B = dia (B) A = dia, B = poly

(C) A = dia, B = mono (D) A = poly, B = dia

Sol. (C,D) More γ more will be the slope of adiabatic curve.

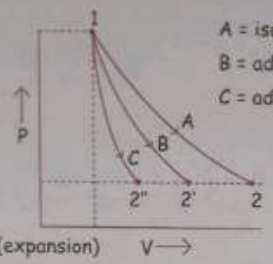
Slope of B is more than slope of A.

$\gamma_B > \gamma_A$

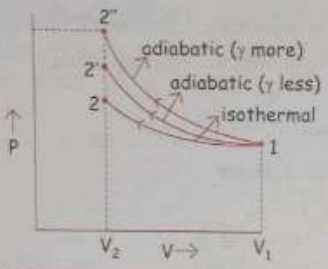
$\gamma_{poly} < \gamma_{dia} < \gamma_{mono}$

tedha hai sawaal, dimag lagega

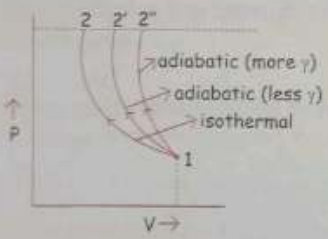
Isothermal and Adiabatic Expansion Curves to a Fixed Pressure



Isothermal and Adiabatic Compression Curve to a Fixed Volume



Isothermal and Adiabatic Compression Curve to a Fixed Pressure



more चढ़ाई will be adiabatic compression curve.

Note

$(Slope)_{adiabatic} = \gamma(slope)_{isothermal}$

Expansion ⇒ more ढलान → adiabatic

Compression ⇒ more चढ़ाई → adiabatic

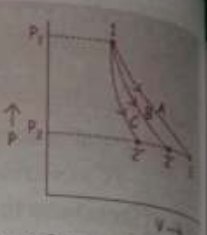
54. Which of the following statements are correct for these isothermal and adiabatic curves for mono and diatomic gases?

(A) Curve B is isothermal curve

(B) Curve A is isothermal curve

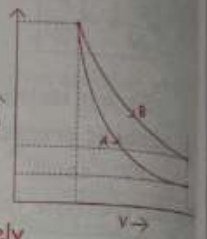
(C) Curve C is adiabatic expansion curve of monoatomic gas

(D) Curve B is adiabatic expansion curve of diatomic gas



Sol. (B,C,D) For explanation, refer to QR code given on the last page.

55. In figure, A and B are two different adiabatic curves for two different gases. Then A and B corresponds to



(A) Ar and He respectively

(B) He and H_2 respectively

(C) O_2 and H_2 respectively

(D) H_2 and He respectively

Sol. (B) For explanation, refer to QR code given on the last page.

56. A monoatomic gas X and a diatomic gas Y, both initially at the same temperature and pressure are compressed adiabatically from a volume V to V/2, which gas will be at higher temperature

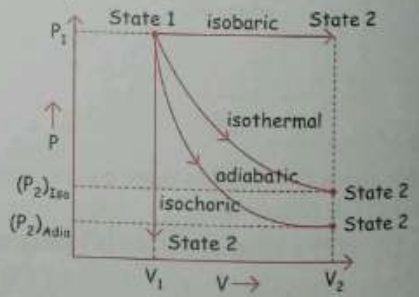
(A) X (B) Y

(C) both the same (D) can't say

Sol. (A) For explanation, refer to QR code given on the last page.

Important

Comparison of work done by the gas in different processes



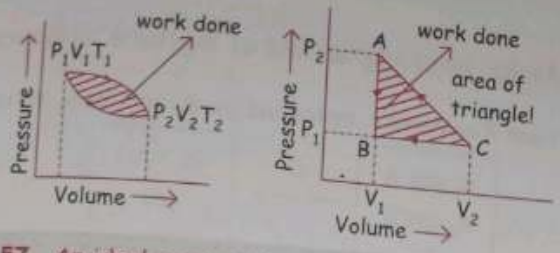
Area under the curve will give the work done:

$W_{isobaric} > W_{isoth} > W_{adiab} > W_{isoch}$

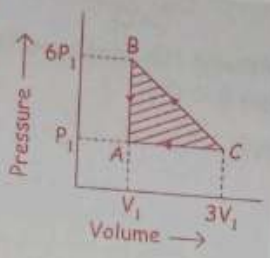
graph ko ache se nihaar lo

Work Done for a Cyclic Process

Work in a cyclic process is given by area enclosed by the cycle.



57. An ideal gas is taken around the cycle ABCA as shown in P-V diagram. The net work done during the cycle is equal to



(A) $12P_1V_1$ (B) $6P_1V_1$

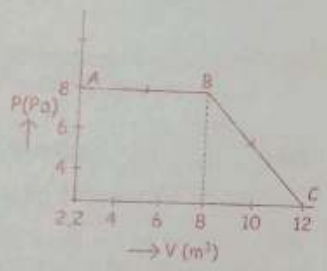
(C) $5P_1V_1$ (D) P_1V_1

Sol. (C) Work done will be given by the area under the curve, so, here, we need to find the area of the triangle.

So, Area $= \frac{1}{2}$(base)(height)

$= \frac{1}{2}(3V_1 - V_1)(6P_1 - P_1) = 5P_1V_1$

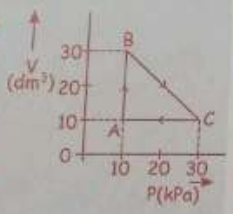
58. The magnitude of the work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is-



[JEE Mains-2020]

Sol. (48 J) For explanation, refer to QR code given on the last page.

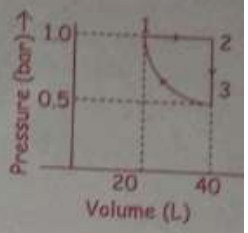
59. An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path $A \rightarrow B \rightarrow C \rightarrow A$ as shown in the diagram. The total work done in the process is _____ J.



[30 Jan, 2024 (Shift-I)]

Sol. (200 Joule) For explanation, refer to QR code given on the last page.

60. One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is _____ J (nearest integer).

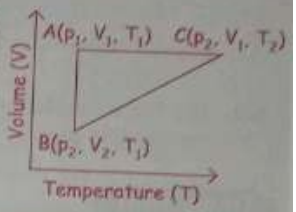


Given: log 2 = 0.3, ln 10 = 2.3

[24 Jan, 2023 (Shift-II)]

Sol. (|W| = 620 J) For explanation, refer to QR code given on the last page.

61. A reversible cyclic process for an ideal gas is shown below. Here, p, V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively.



[JEE Adv. 2018]

The correct option is (are)

(A) $q_{AC} = \Delta U_{BC}$ and $w_{AB} = P_2 (V_2 - V_1)$

(B) $w_{BC} = P_2 (V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$

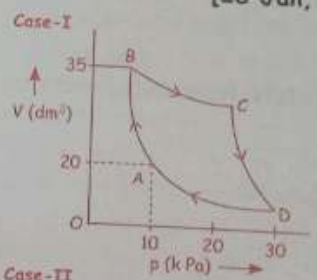
(C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$

(D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} > \Delta U_{CA}$

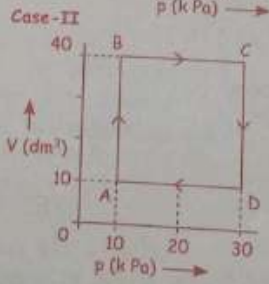
Sol. (B, C) For explanation, refer to QR code given on the last page.

62. [28 Jan, Shift-II, 2025]

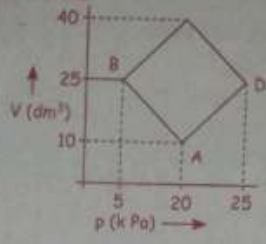
Case-I



Case-II



Case-III



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → D → as shown in the three cases above.

Choose the correct option regarding ΔU:

(A) ΔU (Case -III) > ΔU (Case-II) > ΔU (Case-I)

(B) ΔU (Case -I) = ΔU (Case-II) = ΔU (Case-III)

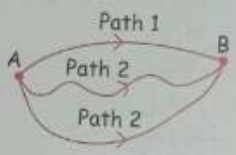
(C) ΔU (Case -I) > ΔU (Case-II) > ΔU (Case-III)

(D) ΔU (Case -I) > ΔU (Case-III) > ΔU (Case-II)

Sol. (B) For explanation, refer to QR code given on the last page.

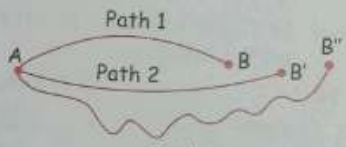
Important

During state change of a gas isothermally, there exist infinite paths joining two states of a gas. Only work done and heat exchange are different in those paths.

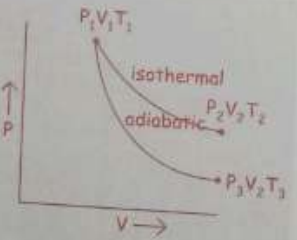


But

For adiabatic process, there exists only one path joining two states of a gas. We can not attain the same state from initial state by two different paths.



63. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in figure. Which of the following statements is/are correct?



[JEE-2012]

(A) $T_1 = T_2$ (B) $T_3 > T_1$

(C) $W_{iso} > W_{adia}$ (D) $\Delta U_{iso} > \Delta U_{adia}$

Sol. (A,C,D) $T_1 = T_2$ (Isothermal process)

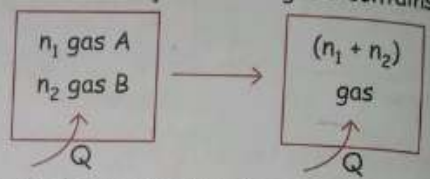
$w_{isothermal} > w_{adiabatic}$ (area under the curve)

$\Delta U_{iso} = 0$ or $\Delta U_{adia} < 0$, so

$\Delta U_{iso} > \Delta U_{adia}$

Molar heat capacities of mixture of gases

Let gas A contains n_1 moles and gas B contains n_2 moles



Assume the heat capacity of gas A is Cv_1 and that of gas B is Cv_2.

$$n_1 C_{v_1} \Delta T + n_2 C_{v_2} \Delta T = (n_1 + n_2) C_{v_{net}} \Delta T$$

$$C_{V_{net}} = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}, \quad C_{P_{net}} = \frac{n_1 C_{P_1} + n_2 C_{P_2}}{n_1 + n_2}$$

Note

$$C_{V_{net}} = \frac{n_1 C_{v_1} + n_2 C_{v_2} + n_3 C_{v_3} + \ldots\ldots}{n_1 + n_2 + n_3 + \ldots\ldots} = \frac{\sum n_i C_{v_i}}{\sum n_i}$$

$$C_{P_{net}} = \frac{n_1 C_{P_1} + n_2 C_{P_2} + n_3 C_{P_3} + \ldots\ldots}{n_1 + n_2 + n_3 + \ldots\ldots} = \frac{\sum n_i C_{P_i}}{\sum n_i}$$

64. Calculate resultant C_v for the mixture of 3 moles H_2 and 2 moles He.

Sol. $\left(\frac{21R}{10}\right)$ For explanation, refer to QR code given on the last page.

65. 3 mol of gas A and 2 mol of He gas are given. If $C_{V_{net}} = 21/10\ R$, identify the nature of gas A.

(A) Monoatomic (B) Diatomic

(C) Triatomic (D) cant say anything

Sol. (B) For explanation, refer to QR code given on the last page.

66. Calculate average molar heat capacity at constant volume of gaseous mixture contained 2 mole of each of two ideal gases.

$A\left(C_{v,m_1} = \frac{3}{2}R\right)$ and $B\left(C_{v,m_2} = \frac{5}{2}R\right)$

(A) R (B) 2R (C) 3R (D) 8R

Sol. (B) For explanation, refer to QR code given on the last page.

Calculation of γ_{mix}

While calculating this, we will use some important formulas that are given below:

$$C_v = \frac{R}{\gamma - 1} \qquad \begin{pmatrix} C_p - C_v = R \\ \frac{C_p}{C_v} = \gamma \end{pmatrix}$$

$$C_{v_{mix}} = \frac{n_1 C_{v_1} + n_2 C_{v_2} + \ldots\ldots}{n_1 + n_2 + \ldots\ldots}$$

$$\frac{R}{\gamma_{mix} - 1} = \frac{\frac{n_1 R}{\gamma_1 - 1} + \frac{n_2 R}{\gamma_2 - 1} + \ldots\ldots}{n_1 + n_2 + n_3 + \ldots\ldots}$$

$$\boxed{\frac{n_1 + n_2 + n_3 + \ldots\ldots}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1} + \ldots\ldots}$$

67. 2 mole of He is mixed with 3 mole of O_2. Find out γ_{mix}

Sol. The expression to calculate γ_{mix} in terms of He and O_2 gas is:

$$\frac{n_{He} + n_{O_2}}{\gamma_{mix} - 1} = \frac{n_{He}}{\gamma_{He} - 1} + \frac{n_{O_2}}{\gamma_{O_2} - 1}$$

$$\frac{2+3}{\gamma_{mix} - 1} = \frac{2}{\frac{5}{3} - 1} + \frac{3}{\frac{7}{5} - 1}$$

$$\gamma_{mix} = \frac{31}{21}$$

68. 2 mole of $_{92}U^{238}$ decay to $_{82}Pb^{206}$ over a certain period of time. If mole of He released are mixed with 3 moles of O_2 and 1 mole of Neon then what would be resultant C_v, C_p and γ of this gaseous mixture?

Sol. $\left(\frac{53}{33}\right)$ For explanation, refer to QR code given on the last page.

69. A monoatomic ideal gas undergoes a process in which the ratio of p to V at any instant is constant and equals to 1. What is the molar heat capacity of the gas? **[IIT JEE-2006]**

(A) $\frac{4R}{2}$ (B) $\frac{3R}{2}$ (C) $\frac{5R}{2}$ (D) 0

Sol. (A) For explanation, refer to QR code given on the last page.

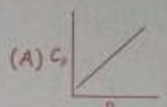
70. Equal volumes of two monoatomic gases, A and B, at same temperature and pressure are mixed. The ratio of specific heats (C_p/C_v) of the mixture will be:

(A) 0.83 (B) 1.50 (C) 3.3 (D) 1.67

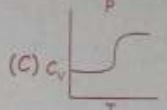
Sol. (D) $\frac{C_p}{C_v} = \frac{\frac{5R}{2}}{\frac{3R}{2}} = \frac{5}{3} = 1.67$

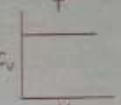
71. For a diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities?

[12 Jan, 2019 (Shift-I)]

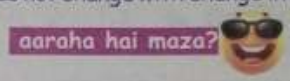
(A) 

(B)

(C) 

(D) 

Sol. (A) C_p does not change with change in pressure.



THERMOCHEMISTRY

Thermochemistry is a branch of chemistry that deals with the relationship between chemical reactions and energy changes, particularly heat transfer. Important concepts include the enthalpy of reaction, formation, combustion, and Hess's Law, which states that the total enthalpy change of a reaction is path-independent.

Lets consider a general reaction:

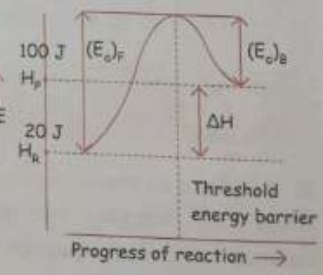
$2A + 3B \rightarrow 4C$; ΔH_R = heat of reaction

If a reaction proceeds by the release of heat, then it is an exothermic reaction.

If a reaction proceeds by the absorption of heat, then it is an endothermic reaction.

Endothermic Reaction

In an endothermic reaction, the energy required to break the bonds of the reactants is greater than the energy released when new bonds are formed in the products.



Lets consider a general reaction:

$$R \underset{(E_a)_b}{\overset{(E_a)_f}{\rightleftharpoons}} P$$

Energy profile diagram for this reaction is:

Heat of reaction can be calculated as:

Heat of Reaction = $H_{product} - H_{reactant}$

= 100 – 20 = +80 J

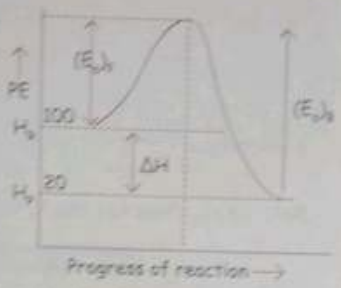
Examples of endothermic reaction are:

* Dissociation reactions, * Fusion reaction (Melting)

* Vapourisation, * Sublimation, * Photochemical reaction

Exothermic Reactions

In an **exothermic reaction**, the energy released during the formation of products is greater than the energy required to break the bonds in the reactants.



Heat of reaction can be calculated as:

$\Delta H = H_p - H_R$ = (–) ve

= 20 – 100 = –80 J

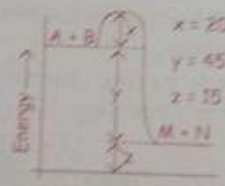
Examples of exothermic reaction are:

* Combustion reaction, * Neutralisation reactIon,

* Respiration process, * Bond formation

72. According to the following figure, the magnitude of the enthalpy change of the reaction A + B → M + N is kJ mol^{-1} is equal to _____ (Integer answer)

[JEE Mains, 31 Aug 2021 Shift-I]



Sol. (45 kJ/mol) For explanation, refer to QR code given on the last page.

HEAT OF REACTION

It is defined as the amount of heat evolved or absorbed when the reacting species, as represented by a balanced chemical equation have completely reacted.

Important

Heat of reaction or enthalpy depends on the type of reaction.

If reaction is formation process — Heat of formation

If reaction is combustion process — Heat of combustion

If reaction is neutralisation process — Heat of neutralisation

If reaction is solution process — Heat of solution

If reaction is isomerisation process — Heat of isomerisation

$\Delta H_R = \sum(\Delta H_f)_p - \sum(\Delta H_f)_R$

Example:

$2A + B + 3C \rightarrow 4D + 3E$

$$\Delta H_R = 4(\Delta H_f)_D + 3(\Delta H_f)_E - \left[2(\Delta H_f)_A + (\Delta H_f)_B + 3(\Delta H_f)_C\right]$$

73. Standard enthalpy of vapourisation for CCl_4 is 30.5 kJ mol^{-1}. Heat required for vapourisation of 284 g of CCl_4 at constant temperature is _____ kJ.

(Given molar mass in g mol^{-1}: C = 12, Cl = 35.5)

[29 Jan, 2024 (Shift-II)]

Sol. (56.247 kJ) For explanation, refer to QR code given on the last page.

74. If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2 kJ mol^{-1}, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

[22 July, 2021 (Shift-II)]

Sol. 1 mole graphite = 12g C

Heat generated on combustion of 1g graphite can be calculated as:

$$= \frac{2.48 \times 10^2}{12} \text{ kJ} = 20.67 \text{ kJ} \approx 21 \text{ kJ}$$

Heat of Formation

The amount of heat evolved or absorbed when one mole of a substance is formed from its constituent elements is called heat of formation.

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$; ΔH = -44 kcal

Thus, enthalpy of formation of HCl is -22 kcal.

Lets consider the formation of HCl from its constituent elements H_2 and Cl_2 as:

$$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightarrow 1HCl(g)$$

(Constituent elements = H and Cl
Elemental states = $H_2(g)$ and $Cl_2(g)$)

The enthalpy content of this reaction will be heat of reaction. Since one mole of HCl is formed from elemental states of its constituent elements so the heat of reaction is named as Heat of formation of HCl.

Note

* 1 mole बनना चाहिए!
* From the constituents in their elemental state.

75. Which of the following equations represents ΔH_f° of the compound?

(A) $C_{(diamond)} + O_2(g) \rightarrow CO_2(g)$

(B) $\frac{1}{2}H_2(g) + \frac{1}{2}Br_2(g) \rightarrow HBr(g)$

(C) $C_{(graphite)} + O_2(g) \rightarrow CO_2(g)$

(D) $\frac{1}{2}H_2(g) + \frac{1}{2}Br_2(l) \rightarrow HBr(g)$

Sol. (C, D) For explanation, refer to QR code given on the last page.

Important

List of elements present in their standard elemental states: ($\Delta H_f = 0$)

$H \rightarrow H_2(g)$, $C \rightarrow C_{(graphite)}$, $Al \rightarrow Al(s)$, $O \rightarrow O_2(g)$, $S \rightarrow S_{(rhombic)}$, $Cl \rightarrow Cl_2(g)$, $P \rightarrow P_{(white)}$, $F \rightarrow F_2(g)$, $N \rightarrow N_2(g)$, $Br \rightarrow Br_2(l)$, $Na \rightarrow Na(s)$, $I \rightarrow I_2(s)$, $Fe \rightarrow Fe(s)$

Example:

$\Delta H_F^\circ(H_2) = 0$	$\Delta H_F^\circ(Br_{(g)}) \neq 0$	$\Delta H_F^\circ(Na_{(s)}) = 0$
$\Delta H_F^\circ(O_2) = 0$	$\Delta H_F^\circ(Br_{(l)}) = 0$	$\Delta H_F^\circ(Al_{(s)}) = 0$
$\Delta H_F^\circ(C_{graphite}) = 0$	$\Delta H_F^\circ(S_{rhombic}) = 0$	
$\Delta H_F^\circ(C_{diamond}) \neq 0$	$\Delta H_F^\circ(P_{white}) = 0$	

Heat of formation of any element present in its standard state = 0

76. $S(g) + \frac{3}{2}O_2(g) \rightarrow SO_3(g) + 2x$ Kcal

$SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g) + y$ Kcal

The heat of formation of $SO_2(g)$ is given by:

[24 Jan, Shift-II, 2025]

(A) 2x + y kcal (B) x + y kcal

(C) y - 2x kcal (D) $\frac{2x}{y}$ kcal

Sol. (C) For explanation, refer to QR code given on the last page.

77. The standard enthalpies of formation of Al_2O_3 and CaO are -1675 kJ mol^{-1} and -635 kJ mol^{-1} respectively. For the reaction

$$3CaO + 2Al \rightarrow 3Ca + Al_2O_3$$

the standard reaction enthalpy ΔH° = _____kJ.
(Round off to the nearest integer)

[JEE Mains, 17 March 2021 Shift-I]

Sol. The reaction is:

$3CaO + Al \rightarrow 3Ca + Al_2O_3$

Calculate standard reaction enthalpy as:

Now, $\Delta_r H^\circ = \Sigma \Delta_f H^\circ_{Products} - \Sigma \Delta_f H^\circ_{Reactants}$

$= [1 \times (-1675) + 3 \times 0] - [3 \times (-635) + 2 \times 0]$

$= +230$ kJ mol^{-1}

bas formula aana chahiye

78. Benzene burns in oxygen according to the following reactions:

$$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 3H_2O(l) + 6CO_2(g)$$

If the standard enthalpies of formation of $C_6H_6(l)$, $H_2O(l)$ and $CO_2(g)$ are 11.7, -68.1 and -94 kcal/mole, respectively, the amount of heat that will liberate by burning 780 g of benzene is-

(A) 7800 kcal (B) 780 kcal

(C) 78 kcal (D) 608.4 kcal

Sol. (A) For explanation, refer to QR code given on the last page.

79. Choose the reaction(s) from the following options, for which the standard enthalpy of reaction is equal to the standard enthalpy of formation. **[JEE Adv, 2019]**

(A) $3/2\ O_2(g) \rightarrow O_3(g)$

(B) $1/8\ S_8(s) + O_2(g) \rightarrow SO_2(g)$

(C) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

(D) $2C(g) + 3H_2(g) \rightarrow C_2H_6(g)$

Sol. (A, B) Standard enthalpy of formation is defined as the change in the standard enthalpy when one mole of the compound is formed from constituent elements present in their standard states. Hence, options A and B are correct.

80. Given:

(A) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$; $\Delta H_1^\circ = x$ kJ/mol

(B) $C_{(graphite)} + O_2(g) \rightarrow 2CO_2(g)$; $\Delta H_2^\circ = y$ kJ/mol

ΔH° for the reaction

$C_{(graphite)} + 1/2O_2(g) \rightarrow CO(g)$ is

[JEE Mains, April, 2023]

Sol. Reverse the first reaction and divide it by 2:

$CO_2(g) \rightarrow CO + 1/2O_2$ $\quad - x/2$

Add above reaction and (B) reaction, the heat of reaction will become:

$$y - \frac{x}{2} = \left(2y - \frac{x}{2}\right)$$

81. At 25°C, the enthalpy of the following processes are given:

[JEE Main 2023 (Online) 1st Feb. Shift-I]

$H_2(g) + O_2(g) \rightarrow 2OH(g)$ $\quad \Delta H^\circ = 78$ kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ $\quad \Delta H^\circ = -242$ kJ mol^{-1}

$H_2(g) \rightarrow 2H(g)$ $\quad \Delta H^\circ = 436$ kJ mol^{-1}

$\frac{1}{2}O_2(g) \rightarrow O(g)$ $\quad \Delta H^\circ = 249$ kJ mol^{-1}

What would be the value of X for the following reaction?_____ (Nearest integer)

$H_2O(g) \rightarrow H(g) + OH(g)$; $\Delta H^\circ = X$ kJ mol^{-1}

Sol. (499 kJ/mol) For explanation, refer to QR code given on the last page.

82. Enthalpies of formation of CCl_4 (g), $H_2O(g)$, $CO_2(g)$ and HCl(g) are -105, -242, -394 and -92 kJ mol^{-1} respectively. The magnitude of enthalpy of the reaction given below is _____ kJ mol^{-1}. (nearest integer)

[JEE Main 2023 (Online) 31st Jan. Shift-II]

$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4HCl(g)$

Sol. $\Delta H_r = \Delta H_f^\circ(CO_2) + 4\Delta H_f^\circ(HCl)$

$- \Delta H_f^\circ(CCl_4) - 2\Delta H_f^\circ(H_2O)$

$= -394 + 4(-92) - (-105) - 2(-242) = -173$ kJ/mol

formula lagao answer nikalo

Enthalpy or Heat of Combustion

The amount of heat evolved or decrease in enthalpy when one mole of a substance is completely oxidized. Organic compounds containing only carbon, hydrogen and oxygen are burnt to give gaseous carbon dioxide and liquid water. For example:

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$;

$\Delta H_R^\circ = -890.36$ kJ/mol

Important

- Combustion of 1 mole substance
- Complete oxidation (reaction)
- The final oxidation state should be maximum

83. Consider the following data:

Heat of formation of $CO_2(g) = -393.5$ kJ mol^{-1}

Heat of formation of $H_2O(l) = -286.0$ kJ mol^{-1}

Heat of combustion of benzene $= -3267.0$ kJ mol^{-1} **[28 Jan, Shift-II, 2025]**

The heat of formation of benzene is _____ kJ mol^{-1}.

(Nearest integer)

Sol. [48] For explanation, refer to QR code given on the last page.

84. Which of the following reactions represent enthalpy of combustion?

(A) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

(B) $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$

(C) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

(D) $2Na(s) + \frac{1}{2}O_2(g) \rightarrow Na_2O(s)$

Sol. (A, B, C) Enthalpy of combustion is defined as when one mole of the substance gets completely oxidized. Among the given options, except (D), all others represent combustion reactions.

Important

Faisal's Rule

- ΔH_R is the enthalpy involved in the reaction and the name of ΔH_R is decided by the nature of reaction.
- $\Delta H_R = \Sigma(\Delta H_f^\circ)_P - \Sigma(\Delta H_f^\circ)_R$,

  $\Delta H_R = \Sigma(\Delta H_C^\circ)_R - \Sigma(\Delta H_C^\circ)_P$
- Heat of combustion of C(s) = Heat of formation of $CO_2(g)$
- Heat of combustion of $H_2(g)$ = Heat of formation of $H_2O(l)$

85. Heat of combustion of C(s), $H_2(g)$, $CH_4(g)$ are a, b and c kcal/mol. Find out heat of formation of CH_4.

Sol. (a+2b-c) For explanation, refer to QR code given on the last page.

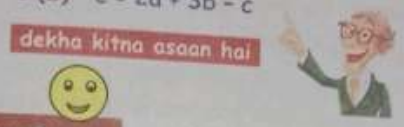
86. Combustion enthalpies of
C(s) = a kcal/mol, $H_2(g)$ = b kcal/mol
$C_2H_6(g)$ = c kcal/mol
Find out $\Delta H_f^\circ(C_2H_6)$

Sol. [2a + 3b - c] First write the enthalpy of formation of C_2H_6:

$2C\ (graphite) + 3H_2(g) \rightarrow C_2H_6\ (l)$

$\Delta H_R^\circ = \Sigma\Delta H_R^\circ$ (reactants) $- \Sigma\Delta H_R^\circ$ (products)

$= 2 \times a + 3\ (b) - c = 2a + 3b - c$



Khoobsurat Baat

$\Delta H_R = \Sigma(\Delta H_f^\circ)_P - \Sigma(\Delta H_f^\circ)_R$

$\Delta H_R = \Sigma(\Delta H_c^\circ)_R - \Sigma(\Delta H_c^\circ)_P$

87. Heat of combustion of C(s), $H_2(g)$, $CH_4(g)$ are -94, -68 and -213 kcal/mol. Then what is ΔH of reaction $C(s) + 2H_2(g) \rightarrow CH_4(g)$?

(A) -17 kcal/mol (B) -111 kcal/mol

(C) -170 kcal/mol (D) -85 kcal/mol

Sol. (A) $\Delta H = \Delta H_c^\circ(C) + 2\Delta H_c^\circ(H_2) - \Delta H_c^\circ(CH_4)$

$= -94 + 2(-68) - (-213) = -17$ kcal/mol

88. At 25°C and 1 atm pressure the enthalpy of combustion of benzene (l) and acetylene (g) are -3268 and -1300 kJ/mol respectively. The change in enthalpy for the reaction-

$3C_2H_2(g) \rightarrow C_6H_6(l)$ **[June Mains -2022]**

Sol. $\Delta H_R = \Sigma(\Delta H_f^\circ)_P - \Sigma(\Delta H_f^\circ)_R = \Sigma(\Delta H_c^\circ)_R - \Sigma(\Delta H_c^\circ)_P$

$\Delta H_R = 3(-1300) - (-3268) = -632$ kJ/mol

89. The standard heat of formation (ΔH°_{298}) of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are -1560, -393.5 and -286 kJ/mol respectively is _____

[JEE Mains, 7 Jan 2020 Shift-II]

Sol. (-192.5 kJ/mole) For explanation, refer to QR code given on the last page.

90. The molar heat of combustion of $C_2H_2(g)$, $C_{(graphite)}$ and $H_2(g)$ are 310.62, 94.05 and 68.32 kcal respectively. Calculate heat of formation of C_2H_2.

Sol. (54.2 kcal) For explanation, refer to QR code given on the last page.

91. The standard enthalpies of combustion at 25°C of H_2, C_6H_{10} and cyclohexane are -241, -3800 and -3920 kJ/mole respectively. Calculate heat of hydrogenation of cyclohexene.

Sol. (-121 kJ) For explanation, refer to QR code given on the last page.

92. From the following data, calculate the enthalpy change for the combustion of cyclopropane at 298 K. The enthalpy of formation of $CO_2(g)$, $H_2O(l)$ and $propene_{(g)}$ are -393.5, -285.8 and 20.42 kJ/mol respectively. The enthalpy of isomerisation of cyclopropane to propene is -33 kJ/mol **[JEE Adv.-1998]**

Sol. First calculate the heat of formation of cyclopropane

$-33 = \Delta H_f^\circ$ (Propene) $- \Delta H_f^\circ$ (Cyclopropane)

ΔH_f° (Cyclopropane) $= 33 + 20.42 = 53.42$ KJ/mol

Now, Cyclopropane $+ \frac{9}{2}O_2 \rightarrow 3CO_2 + 3H_2O$

$\Delta H_R^\circ = 3\Delta H_f^\circ(CO_2) + 3\Delta H_f^\circ(H_2O) - \Delta H_f^\circ$ (Cyclopropane)

$= 3 \times -393.5 + 3(-285.8) - 53.42$

$= -2091.32$ kJ

93. Combustion of 1 mole of benzene is expressed as

$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow CO_2(g) + 3H_2O(l)$

The standard enthalpy of combustion of 2 mol of benzene is - 'x' kJ.

x = _____ **[05 April, 2024 (Shift-II)]**

(1) Standard Enthalpy of formation of 1 mol of $C_6H_6(l)$, for the reaction

$6C_{(graphite)} + 3H_2(g) \rightarrow C_6H_6(l)$ is 48.5 kJ mol^{-1}.

(2) Standard Enthalpy of formation of 1 mol of $CO_2(g)$, for the reaction

$C_{(graphite)} + O_2(g) \rightarrow CO_2(g)$ is -393.5 kJ mol^{-1}.

(3) Standard Enthalpy of formation of $H_2O(l)$, for the reaction

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ is -286 kJ mol^{-1}.

Sol. (-6535 kJ) For explanation, refer to QR code given on the last page.

94. Consider the following data

Heat of combustion of $H_2(g) = -241.8$ kJ mol^{-1}

Heat of combustion of $C(s) = -393.5$ kJ mol^{-1}

Heat of combustion of $C_2H_5OH(l) = -1234.7$ kJ mol^{-1}

The heat of formation of $C_2H_5OH(l)$ is (-) _____ kJ mol^{-1} (Nearest integer)

[6 April, 2023 (Shift-II)]

Sol. $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$

$(\Delta H_f^\circ)_{C_2H_5OH(l)} = \sum(\Delta H_c)_{reactant}$

$(\Delta H_f^\circ)_{C_2H_5OH(l)} = \sum(\Delta H_c)_{reactant} - \sum(\Delta H_c)_{product}$

$= 2 \times (-393.5) + 3(-241.8) - (-1234.7)$.

$= -277.7$ kJ / mol

95. At 25°C and 1 atm pressure, the enthalpies of combustion are as given below:

[24 June, 2022 (Shift-II)]

Substance	H_2	C (graphite)	$C_2H_6(g)$
$\frac{\Delta_c H^\circ}{kJ\ mol^{-1}}$	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is

(A) +54.0 kJ mol^{-1} (B) -68.0 kJ mol^{-1}

(C) -86.0 kJ mol^{-1} (D) +97.0 kJ mol^{-1}

Sol. (C) Enthalpy of formation of ethane can be represented as:

$2C$ (graphite) $+ 3H_2(g) \rightarrow C_2H_6(l)$

Now, $\Delta H_R^\circ = \Sigma \Delta H_C^\circ$ (Reactants) $- \Sigma \Delta H_C^\circ$ (Products)

$= (2 \times -394) + (3 \times -286) - (-1560)$

$= -86$ kJ/mol

Yeh dekho kitna assan tarika hai

96. Enthalpy of combustion of carbon to CO_2 is -393.5 kJ mol^{-1}. What amount of heat will be released upon formation of 35.2 g of CO_2 from carbon and oxygen gas?

(A) 214.8 kJ (B) - 314.8 kJ

(C) 414.8 kJ (D) 514.8 kJ

Sol. (B) For explanation, refer to QR code given on the last page.

97. The heat of formations of CO(g) and CO_2(g) are -26.4 kcal and -94.0 kcal respectively. The heat of combustion of carbon monoxide will be

(A) +26.4 kcal (B) -67.6 kcal

(C) -120.6 kcal (D) +52.8 kcal

Sol. (B) $CO + \frac{1}{2}O_2 \rightarrow CO_2$

$\Delta H = \Delta H_f^\circ(CO_2) - \left[\Delta H_f^\circ(CO) + \frac{1}{2}\Delta H_f^\circ(O_2)\right]$

$= -94.0 - (-26.4) = -67.6$ kcal.

calculation toh kar hi loge ya dikkat aaegi?

Enthalpy or Heat of Solution

The amount of heat evolved or absorbed when one mole of solute is dissolved in excess of solvent so that further addition of solvent makes no heat change. For example,

$H_2SO_4(l)$ + aqueous $\rightarrow H_2SO_4(aq)$: $\Delta H = -20.2$ kcal

$KCl(s)$ + aqueous $\rightarrow KCl(aq)$: $\Delta H = 4.4$ kcal

Heat of an ideal solution is taken as zero.

98. Solute + $1H_2O \rightarrow$ Solute.H_2O; $\Delta H_1 = -69$ kcal in this ΔH_1 a heat of solution of this solute?

(A) Yes (B) No (C) Can't say

Sol. (C) According to the given information, it cannot be said that ΔH_1 is the heat of solution.

99. Solute + $1H_2O \rightarrow$ Solute.H_2O; $\Delta H_1 = -69$ kJ/mol

Solute + $2H_2O \rightarrow$ Solute.$2H_2O$; $\Delta H_2 = -70$ kJ/mol

(A) ΔH_1 = Heat of solution

(B) $\Delta H_1 \neq$ Heat of solution

(C) ΔH_2 = Heat of solution

(D) $\Delta H_2 \neq$ Heat of solution

(E) ΔH_2 may be heat of solution

Sol. (B, E) For explanation, refer to QR code given on the last page.

Enthalpy or Heat of Dilution

The amount of heat evolved or absorbed when solution containing one mole of solute is diluted from one concentration to another. For example.

$HCl(g) + 10H_2O \rightarrow HCl.\ 10H_2O$; $\Delta H_1 = -69.01$ kJ/mol

$HCl(g) + 25H_2O \rightarrow HCl.\ 25H_2O$; $\Delta H_2 = -72.03$ kJ/mol

$HCl(g) + 40H_2O \rightarrow HCl.\ 40H_2O$; $\Delta H_3 = -72.79$ kJ/mol

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$HCl(g) + H_2O \rightarrow HCl.\ \infty H_2O$; $\Delta H_\infty = -74.85$ kJ/mol

Note

According to the above data, heat of solution of HCl is -74.85 kJ/mol

Jab humne 10 mol pani dala tab 69.01 kJ/mol heat evolve hai aur jab humne 25 mole pani dala tab -72.03 kJ/mol heat evolve hai. Iska matlab heat of dilution from 10 mole water to 25 mole water is-

$(\Delta H_{dil})_{10\ mol-25\ mol} = -72.03 - (-69.01) = -3.02$ kJ/mol

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Jiaise-Jaise dilution badhta ja raha hai, heat ka nikalna kam hota ja raha hai.

Important

Heat of solution is a fixed value.

But heat of dilution is dependent on dilution range.

$\Delta H_{solution} = x$

$[\Delta H_{Dil}]_{10\ mol-20\ mol} = y$

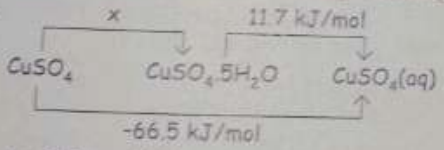
$[\Delta H_{Dil}]_{10\ mol-30\ mol} = z$

$[\Delta H_{Dil}]_{20\ mol-30\ mol} = w$

$x > z > y > w$

100. The enthalpies of solution of anhydrous $CuSO_4$ and hydrated $CuSO_4.5H_2O$ are -66.5 and 11.7 kJ mol^{-1} respectively. Calculate the enthalpy of hydration of $CuSO_4$ to $CuSO_4.5H_2O$.

Sol.



$x + 11.7 = -66.5$

$x = -66.5 - 11.7 = -78.2$ kJ/mol

lag raha hoga muskil lekin hai nahi

101. The heat of solution of anhydrous $CuSO_4$ and $CuSO_4.5H_2O$ are -70 kJ mol^{-1} and +12 kJ mol^{-1} respectively.

The heat of hydration of $CuSO_4$ to $CuSO_4.5H_2O$ is -x kJ. The value of x is_____.

[09 April, 2024 (Shift-I)]

Sol. (x = -82) For explanation, refer to QR code given on the last page.

102. The enthalpy change for the conversion $\frac{1}{2}Cl_2(g)$ to $Cl^-(aq.)$ is (-) _____ kJ mol^{-1} (Nearest integer)

Given, $\Delta_{dis}H^\circ_{Cl_2(g)} = 240$ kJ mol^{-1}

$\Delta_{eg}H^\circ_{Cl(g)} = -350$ kJ mol^{-1}

$\Delta_{hyd}H^\circ_{Cl^-(g)} = -380$ kJ mol^{-1}

[31 Jan, 2023 (Shift-I)]

Sol. $\frac{1}{2}Cl_2(g) \rightarrow Cl(g) \rightarrow Cl^-(g) \rightarrow Cl^-(aq)$

$\Delta H^\circ = \Delta H^\circ_{dis} + \Delta H^\circ_{eg} + \Delta H^\circ_{hyd}$

On substituting values, we get

$\Delta H^\circ = \frac{1}{2} \times 240 + (-350) + (-380) = -610$

Yeh mat sochna mushkil hai

Enthalpy or Heat of Neutralization

The amount of heat evolved when one gram equivalent of an acid is completely neutralized by one gram equivalent of a base in dilute solution is called heat of neutralization.

Acid + Base → Salt + Water + Heat

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When acid and Base react to give one mole of water the heat evolved is known as heat of neutralization.

* $\Delta H^\circ_{neutralization}$ = (-) ve

103. Which $\Delta H^\circ_R = \Delta H^\circ_{neutralization}$?

(A) $2HCl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O$

(B) $HCl + NaOH \rightarrow NaCl + H_2O$

(C) $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

(D) $H_2C_2O_4 + 2NaOH \rightarrow Na_2C_2O_4 + 2H_2O$

Sol. (B, C) Formation of one mole of water should take place. Among the given reactions, B, C and E are correct.

dimag ke ek kone mei socho

(1) For strong acid and strong base

Strong Acid + Strong Base

(a) 13.7

$HCl + NaOH \rightarrow NaCl + H_2O$

$H^+ + Cl^- \quad Na^+ + OH^- \rightarrow Na^+ + Cl^- + H_2O$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(l)$; $\Delta H^\circ_{neutralisation}$ = -13.7 kcal/mole or -57.3 kJ/mole

strong acid, strong base, salt → strong electrolyte → dissociates 100% in water

(b) 13.7

$H_2SO_4 + NaOH \rightarrow NaHSO_4 + H_2O$

$H^+ + HSO_4^- + Na^+ + OH^-_{(aq)} \rightarrow Na^+ + HSO_4^- + H_2O(l)$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(l)$:

$\Delta H^\circ_{neutralisation}$ = -13.7 kcal/mole or -57.3 kJ/mole

(2) For weak acid and strong base

13.7 - q_1 = Heat evolved

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

$CH_3COO^- + H^+_{(aq)} + Na^+ + OH^-_{(aq)} \rightarrow CH_3COO^- Na^+ + H_2O(l)$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(l)$; ΔH = -13.7 kcal/mole

Note

Some heat (q_1) is required to dissociate weak acid completely. This heat is known as **dissociation energy** of that weak acid. If heat of neutralisation of weak acid and strong base or weak base and strong acid is given then we can easily find out the dissociation energy of weak acid or weak base respectively.

(3) For weak acid and weak base

(13.7-q_1 - q_2) = Heat evolved

$CH_3COOH + NH_4OH \rightarrow CH_3COONH_4 + H_2O$

$CH_3COO^- + H^+_{(aq)} + NH_4^+ + OH^-_{(aq)} \rightarrow CH_3COO^-_{(aq)} + NH_4^+_{(aq)} + H_2O(l)$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(l)$; ΔH = -13.7 kcal/mol

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$\Delta H^\circ_{neutralization}$ (strong acid + strong base) = -57.3 kJ/mol

$\Delta H^\circ_{neutralization}$ (weak acid + strong base) = -57.3 kJ/mol + ΔH°_{Diss} (weak acid)

$\Delta H^\circ_{neutralization}$ (weak base + strong acid) = -57.3 kJ/mol + ΔH°_{Diss} (weak base)

$\Delta H^\circ_{neutralization}$ (weak acid + weak base) = -57.3 kJ/mol + ΔH°_{Diss} (weak acid) + ΔH°_{Diss} (weak base)

104. Calculate the heat of dissociation for acetic acid from the following data

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$; ΔH = -13.2 kcal/

$H^+ + OH \rightarrow H_2O$; ΔH = -13.7 kcal

Also calculate heat of dissociation for NH_4OH if

$HCl + NH_4OH \rightarrow NH_4Cl + H_2O$; ΔH = -12.27 kcal

Sol. $(\Delta H^\circ_{Diss})_{Acetic\ Acid}$ = (-13.2) - (-13.7) = 0.5 kcal/mol

$(\Delta H^\circ_{Diss})_{NH_4OH}$ = (-12.27) - (-13.7) = 1.43 kcal/mol

105. Heat of neutralization of oxalic acid is -53.35 kJ mol^{-1} using NaOH. Hence ΔH of

$H_2C_2O_4 \rightleftharpoons C_2O_4^{2-} + 2H^+$ is

(A) 5.88 kJ (B) -5.88 kJ

(C) -13.7 kcal (D) 7.9 kJ

Sol. (7.9 KJ/mol) For explanation, refer to QR code given on the last page.

106. While performing a thermodynamics experiment a student made the following observation.

$HCl + NaOH \rightarrow NaCl + H_2O$ ΔH = -57.3 kJ mol^{-1}

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$ ΔH = -55.3 kJ mol^{-1}

The enthalpy of ionisation of CH_3COOH as calculated by the student is ____ kJ mol^{-1}. (Nearest integer)

[25 July, 2022 (Shift-II)]

Sol. Enthalpy of ionisation is given as:

$\Delta H_{ionization}$ of CH_3COOH = -55.3 - (-57.3) = 2kJ/mole

107. The heat of neutralisation of four acids A, B, C and D when neutralised against a common base are 13.7, 9.4, 11.2 and 12.4 kcal respectively. The weakest among these acids is:

(A) A (B) BO (C) C (D) D

Sol. (B) For explanation, refer to QR code given on the last page.

108. If the heat of neutralisation for a strong acid-base reaction is -57.1 kJ. What would be the heat released when 350 cm^3 of 0.20 M H_2SO_4 is mixed with 650 cm^3 of 0.10 M NaOH?

(A) 37.1 kJ (B) 3.71 kJ

(C) 3.17 kJ (D) 0.317 kJ

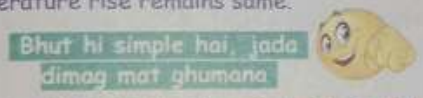
Sol. (B) For explanation, refer to QR code given on the last page.

109. A solution of 500 ml of 0.2 M KOH and 500 ml of 0.2 M HCl is mixed and stirred, the rise in temperature is T_1. The experiment is repeated using 250 ml of each solution, the temperature raised is T_2. Which of the following is true?

(A) $T_1 = T_2$ (B) $T_1 = 2T_2$

(C) $T_1 = 4T_2$ (D) $T_2 = 9T_1$

Sol. (A) Volume and heat evolved is halved, so the temperature rise remains same.



110. Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. If enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol. Calculate $\Delta H_{ionization}$ of H_3PO_3 into its ions:

(A) 50.84 kJ/mol (B) 5 kJ/mol

(C) 2.5 kJ/mol (D) None of these

Sol. (B) $H_3PO_3 \rightarrow 2H^+ + HPO_3^{2-}$; $\Delta_r H = ?$

$2H^+ + 2OH^- \rightarrow 2H_2O$; $\Delta_r H = -55.84 \times 2 = 111.68$

$-106.68 = \Delta_{ion}H - 55.84 \times 2$

$\Delta_{ion}H = 5$ kJ/mol

Enthalpy of Phase Transition

Enthalpy of fusion

It is the enthalpy change involved in the conversion of one mole of a substance from solid state to liquid state at its melting point. For example:

$H_2O(s) \rightarrow H_2O(l)$; $\Delta H = 1.44$ kcal/mol (at 0°C)

Enthalpy of vaporisation

It is the enthalpy change involved in converting one mole of the substance from liquid state to gaseous state at its boiling point. For example:

$H_2O(l) \rightarrow H_2O(g)$; $\Delta H = 10.5$ kcal/mol (at 100°C)

Enthalpy of sublimation

It is the enthalpy change involved in the conversion of one mole of a solid directly into its vapour at a given temperature below its melting point. For example:

$I_2(s) \rightarrow I_2(g)$; $\Delta H = 14.9$ kcal/mol

Heat of atomization

Enthalpy change when one mole of any compound (solid, liquid, gas) is dissociated into its constituent atoms in gaseous state is called heat of atomization.

Examples:

$Br_2(l) \rightarrow 2Br(g)$; $\Delta H_R = \Delta H_{atom}(Br_2(l))$

$I_2(s) \rightarrow 2I(g)$; $\Delta H_R = \Delta H_{atom}(I_2(s))$

$C_2H_6(g) \rightarrow 2C(g) + 6H(g)$; $\Delta H_R = \Delta H_{atom}(C_2H_6)$

$C(s) \rightarrow C(g)$; $\Delta H_R = \Delta H_{atom}C(s)$, $(\Delta H^\circ_{sub}C(s) = \Delta H_{atom}C(s))$

$H_2(g) \rightarrow 2H(g)$; $\Delta H_R = \Delta H_{atom}(H_2(g))$

$O_3(g) \rightarrow 3O(g)$, $\Delta H_R = \Delta H_{atom}(O_3(g))$

111. Define heat of atomisation of Carbon.

Sol. $1C(s) \rightarrow 1C(g)$; $\Delta H^\circ_{atom}(C)$

112. Define heat of atomisation of H.

Sol. $\frac{1}{2}H_2(g) \rightarrow 1H(g)$; $\Delta H^\circ_{atom}(H)$

$H_2(g) \rightarrow 2H(g)$; $\Delta H^\circ_{atom}(H_2)$

113. Define heat of atomisation of $Br_2(l)$.

Sol. $Br_2(l) \rightarrow 2Br(g)$; $\Delta H^\circ_{atom}(Br_2(l))$

114. Define heat of atomisation of $CH_4(g)$.

Sol. $CH_4(g) \rightarrow C(g) + 4H(g)$; $\Delta H^\circ_{atom}(CH_4(g))$

Bond Energies

The amount of energy required to break one mole of a particular type of bond between the atoms in the gaseous state i.e. to separate the atoms in the gaseous state under one atmospheric pressure and the specified temperature is called bond dissociation energy.

e.g. $H\text{-}H(g) \rightarrow 2H(g)$; $\Delta H = 433 kJ\ mol^{-1}$

$H\text{-}I(g) \rightarrow H(g) + I(g)$; $\Delta H = 299\ kJ\ mol^{-1}$

Heat of reaction= (Sum of bond enthalpies of reactants)-(sum of bond enthalpies of products)

Important

- In Bond dissociation energy, reactions and products both are present in gaseous state.
- In atomisation, products are in gaseous state but reactants may or may not be in gaseous state.

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BDE is not always equal to heat of atomisation.

$O_2(g) \rightarrow 2O(g)$

$O=O(g) \rightarrow 2O(g)$ — Bond diss energy $\epsilon_{O=O}$; $\Delta H_{atom}(O_2(g))$ ⟹ $\Delta H_{BDE} = \Delta H_{atom}$

$Br_2(l) \rightarrow 2Br(g)$: $\Delta H_{atom}(Br_2(l))$

$Br_2(g) \rightarrow 2Br(g)$: $\Delta H_{BDE}(Br_2(g))$ ⟹ $\Delta H_{BDE} \neq \Delta H_{atom}$

$I_2(s) \rightarrow 2I(g)$: $\Delta H_{atom}(I_2(s))$

$I_2(g) \rightarrow 2I(g)$: $\Delta H_R = \Delta H_{BDE}$ ⟹ $\Delta H_{BDE} \neq \Delta H_{atom}$

115. If the enthalpy of atomisation for $Br_2(l)$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them is

[Jan-2020 Mains]

(A) $x = y$ (B) $x > y$

(C) does not exist (D) $x < y$

Sol. (B) For explanation, refer to QR code given on the last page.

116. Atomisation of I_2 is a kJ/mol

Bond diss. energy I_2 is b kJ/mol. Then find the relation between a and b.

(A) $a = b$ (B) $a > b$

(C) $a < b$ (D) पता नहीं

Sol. (B) $I_2(s) \rightarrow 2I(g)$; $\Delta H_{atom} = a$

$I_2(g) \rightarrow 2I(g)$; $\Delta H_{BDE} = b$

$a > b$

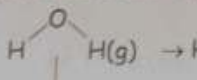
Note

- If in a molecule we have only one type of bond then BDE of that bond is an absolute value.
- If in a molecule there exist more than one of a particular type of bond then BDE of that bond is an average value.

Example:

$H\text{—}H(g) \rightarrow 2H(g)$; $\Delta H_{BDE} = \epsilon_{H\text{-}H} = a$ kJ/mol

(only one particular type of bond)

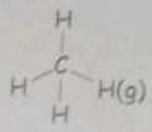


$H\text{-}O(g) \rightarrow H(g) + O(g)$; $\Delta H_2 = y$ kJ/mole

(more than one particular type of bond)

$$\epsilon_{O\text{-}H} = \frac{x+y}{2}\ kJ/mol$$

Example:



$CH_3(g) \rightarrow CH_2(g) + H(g)$; $\Delta H_1 = a$ kcal/mol

$CH_2(g) \rightarrow CH(g) + H(g)$; $\Delta H_2 = b$ kcal/mol

$CH(g) \rightarrow C(g) + H(g)$; $\Delta H_3 = c$ kcal/mol

$C\text{-}H(g) \rightarrow C(g) + H(g)$; $\Delta H_4 = d$ kcal/mol

What is bond diss. energy of C—H bond?

$$\epsilon_{C\text{-}H} = \frac{a+b+c+d}{4}$$

117. If the atomisation energy of CH_4 is x kJ/mole and the bond dissociation energies of 3(C—H) bonds in CH_4 are a, b, c kJ/mole then what would be the bond dissociation energy of fourth C—H bond?

Sol. First write the reaction:

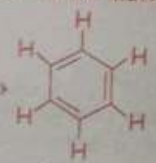
$1CH_4(g) \rightarrow 1C(g) + 4H(g)$; $\Delta H_{atom} = x$

(this energy is used to break 4 mole C—H bonds)

The dissociation energy of fourth C—H bond

$= x-(a+b+c)$ kJ/mole

118. Calculate the heat of formation of C_6H_6

$6C(s) + 3H_2(g) \rightarrow$ 

Sol. For explanation, refer to QR code given on the last page.

119. The bond enthalpies of H – H, Cl – Cl and H – Cl are 435, 243 and 431 kJ mol^{-1}, respectively the enthalpy of formation of HCl(g) will be

(A) 92 kJ mol^{-1} (B) - 92 kJ mol^{-1}

(C) 247 kJ mol^{-1} (D) 770 kJ mol^{-1}

Sol. (B) For explanation, refer to QR code given on the last page.

120. The formation enthalpies, ΔH_f° for $H_{(g)}$ and $O_{(g)}$ are 220.0 and 250.0 kJ mol^{-1} respectively, at 298.15 K, and ΔH_f° for $H_2O_{(g)}$ is -242.0 KJ mol^{-1} at the same temperature. The average bond enthalpy of the O-H bond in water at 298.15 K is _____ kJ mol^{-1} (nearest integer). **[28 Jan, Shift-I, 2025]**

Sol. **Hess's Law Formula**

$\Delta H_f^\circ(H_2O)$ = Bond enthalpies of reactants - Bond enthalpies of products

$$-242.0 = D(H-H) + \frac{1}{2}D(O=O) - 2D(O-H)$$

Substituting Known Values

$$-242.0 = 440 + \frac{1}{2}(500) - 2D(O-H)$$

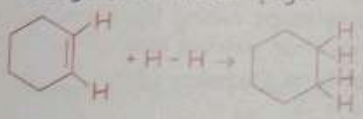
$$D(O-H) = \frac{932}{2} = 466 \text{ kJ/mol}$$

121. Calculate the resonance energy of N_2O from the N_2O is 82 kJ mol^{-1}.

$\varepsilon_{N\equiv N}$ = 946 kJ mol^{-1}, $\varepsilon_{N=N}$ = 418 kJ mol^{-1},

$\varepsilon_{O=O}$ = 498 kJ mol^{-1}, $\varepsilon_{N=O}$ = 607 kJ mol^{-1}

Sol. (-88 kJ/mol) For explanation, refer to QR code given on the last page.

122. 

The bond energies (in kJ/mol) at 25°C are C-C = 346, C-H = 413, H-H = 437, C=C = 611. From these data, the value of ΔH at 25°C for the above reaction is

(A) -289 kJ mol^{-1} (B) -124 kJ mol^{-1}

(C) 124 kJ mol^{-1} (D) 289 kJ mol^{-1}

Sol. According to the reaction,

1 C = C and 1 H-H bond is broken and 1 C-C and 2 C-H bonds are formed.

ΔH = 611 + 437 - 346 - 2(413)

ΔH = -124 kJ/mol

123. Calculate the enthalpy of the following homogeneous gaseous reaction

$CH_3COCH_3 + 2O_2 \rightarrow CH_3COOH + CO_2 + H_2O$

from the following data.

Bond energies (kJ/mol): C-H = 414; C-C = 348

C = O = 580; C-O = 354; O=O = 610; O - H = 462

Magnitude of resonance energies (kJ/mol): COOH = 118; CO_2 = 140

(A) 348 kJ (B) 168 kJ

(C) -168 kJ (D) -348 kJ

Sol. (D) For explanation, refer to QR code given on the last page.

124. Using bond energy data, calculate the heat of formation of isoprene.

$$5C(s) + 4H_2(g) \rightarrow H_2C = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - \underset{\displaystyle H}{\overset{\displaystyle |}{C}} = CH_2$$

Given: C-H = 98.8 kcal

H-H = 104 kcal

C-C = 83 kcal

C=C = 147 kcal

$C(s) \rightarrow C(g)$: 171 kcal

Sol. (20.6 kcal) For explanation, refer to QR code given on the last page.

125. The ΔH°_f values of cyclohexane$_{(l)}$ and benzene$_{(l)}$ at 25°C are -156 and +49 kJ/mol respectively. The standard enthalpy of hydrogenation of cyclohexane$_{(l)}$ at 25°C is -119 kJ/mol. Use these data to estimate the magnitude of resonance energy of benzene. **[JEE Adv.]**

Sol. Reaction of hydrogenation of cyclohexane can be represented as:

$C_6H_{10} + H_2 \rightarrow C_6H_{12}$, ΔH = -119 kJ

$\therefore C_6H_6 + 3H_2 \rightarrow C_6H_{12}$, $\Delta H = 3 \times (-119) = -357$ kJ

Now, Also given $6C + 6H_2 \rightarrow C_6H_{12}(\Delta H = -156)$

Finally, $C_6H_6 + 3H_2 \rightarrow C_6H_{12}$ (l), ΔH = -3576°C

$+ 3H_2 \rightarrow C_6H_6$, ΔH + 201 kJ

Resonance energy = 49-201 = -152 kJ

Socho thoda ki hua kya hai?

126. The enthalpy of formation of ethane (C_2H_6) from ethylene by addition of hydrogen where the bond energies of C - H, C - C, H - H are 414 kJ, 347 kJ, 615 kJ and 435 kJ respectively is - _____ kJ.

[04 April, 2024 (Shift-I)]

Sol. (-125 kJ) For explanation, refer to QR code given on the last page.

127. $A_2 + B_2 \rightarrow 2AB \cdot \Delta H_f^\circ = -200$ kJ mol^{-1} AB, A_2 and B_2 are diatomic molecule. If the bond enthalpies of A_2, B_2 and AB are in the ratio 1:0.5:1, then the bond enthalpy of A_2 is _____ kJmol^{-1} (Nearest integer)

[13 April, 2023 (Shift-I)]

Sol. $A_2 + B_2 \rightarrow 2AB \cdot \Delta H_f^\circ = -200$ kJ mol^{-1}

$\Delta H = (B.E)_{A-A} + (B.E)_{B-B} - 2(B.E)_{A-B}$

If B.E of A-A = x then, B.E of B-B = $\frac{x}{2}$ and that of A-B = x

$\Delta H_f(A-B) = -200$ kJ

$\therefore -200 = x + \frac{x}{2} - 2x = -\frac{x}{2} \Rightarrow x = 400$ kJ/mol

128. The bond dissociation energies for Cl_2, I_2 and ICl are 242.3, 151, 211.3 kJ/mol respectively. The enthalpy of sublimation of iodine is 62.8 kJ/mol. What is the standard enthalpy of formation of ICl (g) nearly equal to?

(A) 16.8 kJ/mole (B) 33.5 kJ/mole

(C) - 14.6 kJ/mole (D) -211.3 kJ/mole

Sol. (A) For explanation, refer to QR code given on the last page.

129. The bond dissociation enthalpy of X_2 ΔH°_{bond} calculated from the given data is _____ kJ mol^{-1}. (Nearest integer)

[23 Jan, Shift-II, 2025]

$M^+X^-(s) \rightarrow M^+(g) + X^-(g)\ \Delta H^\circ_{lattice} = 800$ kJ mol^{-1}

$M(s) \rightarrow M(g)\ \Delta H^\circ_{sub} = 100$ kJ mol^{-1}

$M(g) \rightarrow M^+(g) + e^-(g)\ \Delta H^\circ_i = 500$ kJ mol^{-1}

$X(g) + e^-(g) \rightarrow X^-(g)\ \Delta H^\circ_{eg} = -300$ kJ mol^{-1}

$M(s) + \frac{1}{2}X_2(g) \rightarrow M^+X^-(s)\ \Delta H^\circ_f = -400$ kJ mol^{-1}

[Given: M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 in gaseous state]

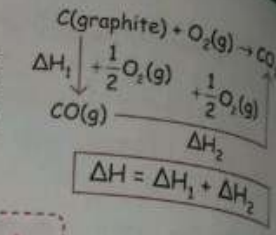
Sol. (200 kJ/mol) For explanation, refer to QR code given on the last page.

Hess's Law of Constant Heat Summation

...re precisely, "the enthalpy change in a chemical or ...ysical process is the same whether the process is ...ried out in one step or in several steps".

For example,

This property is due to the fact that the enthalpy change in a system is independent of the path followed.



$\Delta H = \Delta H_1 + \Delta H_2$

Born-Haber Cycle

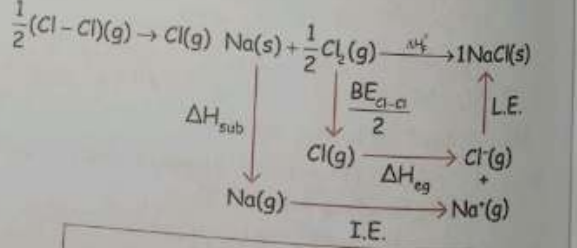
The Born-Haber Cycle is a thermodynamic cycle used to calculate the lattice enthalpy of ionic compounds.

Steps in the Born-Haber Cycle:

For a general ionic compound, MX (e.g., NaCl):

1. **Sublimation of the Metal ($\Delta H_{sublimation}$):**
   $M(s) \rightarrow M(g)$
   Energy is absorbed to convert the metal from solid to gaseous form.
2. **Ionization of the Metal ($\Delta H_{ionization}$):**
   $M(g) \rightarrow M^+(g) + e$
   Energy is absorbed to remove an electron from the gaseous metal atom.
3. **Dissociation of the Non-metal Molecule ($\Delta H_{dissociation}$):**
   $X_2(g) \rightarrow 2X(g)$
   Energy is absorbed to break the bond in the non-metal molecule.
4. **Electron Affinity of the Non-metal (ΔH_{EA}):**
   $X(g) + e \rightarrow X^-(g)$
   Energy is released when the non-metal atom gains an electron.
5. **Formation of the Ionic Solid ($\Delta H_{lattice}$):**
   $M^+(g) + X^-(g) \rightarrow MX(s)$
   Energy is released during the formation of the ionic lattice from gaseous ions.
6. **Formation from Elements ($\Delta H_{formation}$):**
   $M(s) + \frac{1}{2}X_2(g) \rightarrow MX(s)$

Lets understand this by taking an example of NaCl compound.



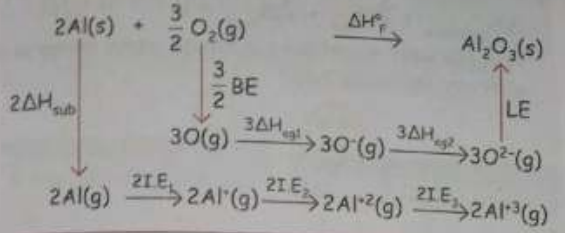
$$\Delta H_f^\circ = \Delta H_{sub} + I.E + \frac{\epsilon_{Cl-Cl}}{2} + \Delta H_{eg} + L.E$$

ΔH_F° = Heat of formation, ΔH_{sub} = Heat of sublimation
I.E = ionisation energy, ϵ_{Cl-Cl} = Bond dissociation
ΔH_{eg} = Electron gain enthalpy

Important

Take another example: Al_2O_3

Born-Haber cycle-$Al_2O_3(s)$



$$\Delta H_F^\circ = 2\Delta H_{sub} + 2(I.E_1 + I.E_2 + I.E_3) + \frac{3}{2}BE_{(O=O)} + 3\Delta H_{eg1} + 3\Delta H_{eg2} + L.E$$

Khoobsurat Baat

The main purpose of Born-Haber cycle is to determine lattice energy of ionic solids.

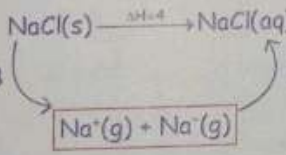
Lattice energy is defined as the amount of energy released during the formation of one mole of solid lattice from its constituents gaseous ions.

130. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is: **[JEE Mains, 5 Spet-2020 Shift-II]**

(A) 784 kJ mol^{-1} (B) -784 kJ mol^{-1}

(C) 780 kJ mol^{-1} (D) -780 kJ mol^{-1}

Sol. (B) According to the given information,



Now,

4 kJ/mol = 788 + ΔH, ΔH = -784 kJ/mol

131. The Born-Haber cycle for KCl is evaluated with the following data :

$\Delta_f H^\circ$ for KCl = - 436.7 kJ mol^{-1};

$\Delta_{sub} H^\circ$ for K = 89.2 kJ mol^{-1};

$\Delta_{ionization} H^\circ$ for K = 419.0 kJ mol^{-1};

$\Delta_{electron\ gain} H^\circ$ for Cl(g) = - 348.6 kJ mol^{-1};

$\Delta_{bond} H^\circ$ for Cl_2 = 243.0 kJ mol^{-1}

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer).

[JEE Main 2021 (Online) 26th Aug Shift-I]

Sol. Apply born-haber cycle concept:

$$\Delta H_F^\circ = \Delta H_{sub}^\circ + I.E + \frac{1}{2}BE + \Delta H_{eg} + L.E$$

-436.7 89.2 419 $\frac{243}{2}$ -348.6 + L.E

On solving, Lattice energy = -717.8 kJ/mol

132. The ionization enthalpy of Na^+ formation from Na(g) is 495.8 kJ mol^{-1}, while the electron gain enthalpy of Br is -325.0 kJ mol^{-1}. Given the lattice enthalpy of NaBr is -728.4 kJ mol^{-1}. The energy for the formation of NaBr ionic solid is (-) _____ $\times 10^{-1}$ kJ mol^{-1}.

[JEE Main 2021 (Online) 25th Feb Shift-I]

Sol. (- 5576 $\times 10^{-1}$ kJ/mol) For explanation, refer to QR code given on the last page.

133. If $C(diamond) \rightarrow C(graphite) + X$ kJ mol^{-1}

$C(diamond) + O_2(g) \rightarrow CO_2(g) + Y$ kJ mol^{-1}

$C(graphite) + O_2(g) \rightarrow CO_2(g) + Z$ kJ mol^{-1}

at constant temperature. Then

[29 Jan, Shift-II, 2025]

(A) X = Y - Z (B) X = Y + Z

(C) -X = Y + Z (D) X = -Y + Z

Sol. (A) For explanation, refer to QR code given on the last page.

SECOND LAW OF THERMODYNAMICS

Statement

- For any irreversible process, entropy of universe is increasing.
- Transfer of heat from cold body to hot can not be done without performing work.
- Any process occurring its own is thermodynamically irreversible.

Limitations of First Law of Thermodynamics

- The first law is clueless about why some processes happen on their own and why some other do not.

- First law can not explain the direction of heat flow, rather it can only calculate the value of q, w, ΔU, ΔH (i.e, energy difference between two states).
- It can not explain the spontaneity of the process.

Spontaneity

Spontaneity refers to the natural tendency of a process to occur without any external intervention once it has been initiated.

Criterea of spontaneity

(i) Heat exchange during the process (ΔH)

Process which are exothermic in nature are only spontanteous process (अपने आप होंगे)

Endothermic processes are non-spontaneous (अपने आप नहीं होंगे)

Lekin ek simple experiment ne is baat ko galat proof kar diya

Khoobsurat Baat

ΔH_{sys} is not the criterea of spontaneity.

(ii) Entropy change of the system (ΔS_{sys})

- Entropy is the degree of randomness or disorderness of a system
- Change in entropy is the capacity for spontaneous change in a system.
- Entropy is a state function.
- Entropy is an extensive property.
- ΔS is a state function.

  $\Delta S_{rev} = \Delta S_{irrev}$
- If entropy is expressed in $JK^{-1}\ mol^{-1}$ then it will be intensive property.

Now comes the new criterea ΔS_{sys}. Process are spontaneous only when he entropy of system increases.

ΔS_{sys} = (+) spontaneous

ΔS_{sys} = (-) non spontaneous

अच्छा पानी इसलिए उड़ गया था क्योंकि vapour phase mein पानी की entropy ज्यादा थी।

Lekin Chai to apne aap thandi hoti hai jabki chai ke thande hone mein uski entropy kam hoti hai

Khoobsurat Baat

So, ΔS_{sys} is not the criterea of spontaneity.

To asli criteria spontaneity ka kya hai?

(iii) $\Delta S_{universe}$ should be positive

$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr} > 0$

Process wahi spontaneous honge jisme universe k entropy increase hogi.

Khoobsurat Baat

$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$

$\Delta S_{universe} > 0$ for spontaneous process

Yahi hai asli criterea for spontaneity

Note

endothermic process → always non spontaneous — False

endothermic process → always spontaneous — False

$\Delta S_{sys} > 0$ → always spontaneous — False

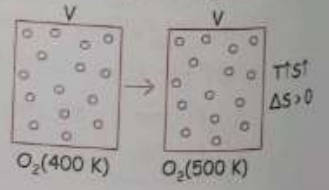
$\Delta S_{sys} < 0$ → always non spontaneous — False

$\Delta S_{universe} > 0$ → always spontaneous — Ture

QUALITATIVE COMPARISON OF ENTROPY

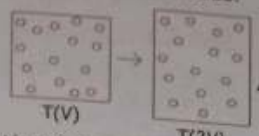
1. solid → liquid $\Delta S > 0$
   liquid → gas $\Delta S > 0$
   gas → liquid $\Delta S < 0$
   liquid → solid $\Delta S < 0$
   solid → gas $\Delta S > 0$
2. If volume is constant and temp changes

   On increasing the temperature, entropy will increase, so, $\Delta S > 0$



3. If temp is constant and volume increases.

Since volume is increasing, entropy will increase.



V↑s↑ $\Delta S > 0$

4. For a chemical reaction that is:

Product gaseous moles - reactant gaseous moles = Δn_g

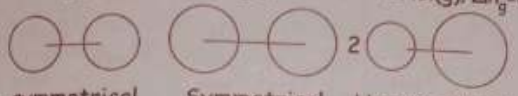
$\Delta n_g > 0$: $\Delta S > 0$, $\Delta n_g < 0$; $\Delta S < 0$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta n_g < 0$; $\Delta S < 0$

Khoobsurat Baat

If $\Delta n_g = 0$

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$; $\Delta n_g = 0$



symmetrical Symmetrical Unsymmetrical

$\Delta S > 0$

Heteronuclear diatomic molecules have more entropy as compare to homonuclear diatomic molecule.

5. For two different gas more the atomicity more will be the randomness.

$SO_3(g) > SO_2(g)$ atomicity↑entropy↑

(atomicity = 4) (atomicity = 3)

6. If atomicity is same then the criteria is molecular wt.

mol wt.↑entropy↑

$NO(g) > N_2(g)$

(mol wt = 30) (mol wt = 28)

Khoobsurat Baat

Spontaneity of a process in terms of total entropy change :

If, ΔS_{total} = +ve ⇒ total spontaneous process

If, ΔS_{total} = -ve ⇒ non spontaneous process

If, $\Delta S_{total} = 0$ ⇒ process is at equilibrium

134. Change in entropy is negative for

(A) $Br_2(l) \rightarrow Br_2(g)$

(B) $C(s) + H_2O(g) \rightarrow CO(g) + H_2(g)$

(C) $N_2(g)$ (10 atm, 298 K) → $N_2(g)$ (1 atm, 298 K)

(D) Fe(s) (400 K) → Fe(s) (300 K)

Sol. (D) A. In option A, l → g, $\Delta S > 0$

B. In option B, $\Delta n_g > 0$, $\Delta S > 0$

C. In option C, volume is increasing, $\Delta S > 0$

D. In option D, temperature is decreasing, $\Delta S < 0$

135. Predict the sign of ΔS

(A) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

(B) $Ni(s) + 4CO(g) \rightarrow Ni(CO)_4(g)$

(C) $2Ca(s) + O_2(g) \rightarrow 2\,CaO(s)$

(D) $H_2(g) + S(s) \rightarrow H_2S(g)$

Sol. (D) For explanation, refer to QR code given on the last page.

Quantitative Estimation of Entropy

$ds_{sys} = \frac{dq_{rev}}{T}$ — dq_{rev}: heat exchange; T: temperature at which this heat is exchanged

$$\Delta S_{sys} = \int ds_{sys} = \int_1^2 \frac{dq_{rev}}{T}$$

Case-1: Temperature of system is changing but temperature of surrounding is constant

Let the temperature of the system be T_1 and that of the surroundings be T_2.

Consider an example of a cup of tea:

T_2 (Temp. of surroundings)

mass of tea = m gm

Temp. of tea = T_1

$T_1 >> T_2$

Entropy change for calculation for a system (tea)

1. If specific heat capacity of tea (s J/kg)is given

$$\Delta S = \int ds_{sys} = \int \frac{dq_{rev}}{T} = \int_{T_1}^{T_2} \frac{msdT}{T} = ms\ln\frac{T_2}{T_1}$$

$$= 2.303 ms \log\frac{T_2}{T_1}$$

2. If molar heat capacity of tea(C_p) is given

$$\Delta S = \int_{T_1}^{T_2} \frac{nC_p dT}{T} = nC_p \ln\frac{T_2}{T_1} = 2.303nC_p \log\frac{T_2}{T_1}$$

Entropy change calculation for surrounding

The entropy change of surroundings

$$\Delta S_{surr} = \frac{-\Delta H_{sys}}{T_{surrounding}} = \frac{-ms(T_2 - T_1)}{T_2}$$

Kya chai apne aap T_1 se T_2 tak aa jaegei?

The answer is 'yes'. This process is

'Spontaneous Process'

because $\Delta S_{univ} > 0$

Reason: Entropy of tea (body at higher temperature) is decreasing and entropy of surroundings (at lower temperature) is increasing but the increase in entropy of surroundings is more than the decrease in entropy of the system as heat exchange from a body at higher temperature does not affect the entropy much as it does in case of lower temperature.

$$\Delta S_{univ} = \underset{\text{-ve}}{\Delta S_{(tea)}} + \underset{\text{+ve}}{\Delta S_{(surrounding)}} > 0$$

Important

iska matlab yeh hua chai thandi hote hote ice cream ban jaegi

Lekin Chai surrounding temperature (T_2) se aur kam temperature per apne aap nahi aa sakti. $\Delta S_{univ} < 0$ ho jaega jiska matlab process non-spontaneous ho jaega.

Khoobsurat Baat

$$\Delta S_{sys} = ms\ln\frac{T_2}{T_1}, \Delta S_{surr} = \frac{-ms\ln(T_2 - T_1)}{T_2}$$

$$\Delta S_{univ} = \left(ms\ln\frac{T_2}{T_1}\right) + \left(\frac{-ms(T_2 - T_1)}{T_2}\right)$$

हर उस सवाल का जवाब है। जहाँ Temperature of system changes but temp of surroundings is constant.

136. Find out the entropy change-

mass = 150 gm
S = 1.9 J/kg K

Sol. The formula to calculate change in entropy is:

$\Delta S = 2.303\ ms \log T_2/T_1$

Here, m = 150 g, s = 1.9 J/kg K, T_2 = 300 K, T_1 = 800 K

Substitute the values in the above expression:

$\Delta S = 2.303 \times 0.15\ kg \times 1.9\ J/kg\ K \times \log(800/300)$

On solving, $\Delta S = 0.279\ J/K$

Case-2: Phase Transformation

solid → liquid	liquid → vapour	solid → vapour
$\Delta S_{sys} = \frac{\Delta H_{fusion}}{T_{M.P}}$	$\Delta S_{sys} = \frac{\Delta H_{vap}}{T_{B.P}}$	$\Delta S_{sys} = \frac{\Delta H_{sub}}{T_{sublimation}}$
$\Delta S_{surrd} = \frac{-\Delta H_{fusion}}{T_{surrd}}$	$\Delta S_{surrd} = \frac{-\Delta H_{vap}}{T_{surrd}}$	$\Delta S_{surrd} = \frac{-\Delta H_{sub}}{T_{surrd}}$
process is spontaneous at T_{surrd} above M.P.	process is spontaneous at temp T_{surrd} above B.P	

137. The entropy S° of the following substances are: **[JEE Mains-2014]**

$CH_4(g) = 186.2\ JK^{-1}\ mol^{-1}$

$O_2(g) = 205.0\ JK^{-1}\ mol^{-1}$

$CO_2(g) = 213 = 69.9\ JK^{-1}\ mol^{-1}$

The entropy change for the reaction

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

Sol. The formula to calculate change in entropy is:

ΔS = Sum of entropies of products- sum of entropies of reactants

= ($\Delta S°$ of CO_2 + 2$\Delta S°$ of H_2O) - ($\Delta S°$ of CH_4 + 2$\Delta S°$ of O_2)

= (213.6 + 2(69.9)) - (186.2 + 2(205.0)) J/K/mol

= 353.4 - 596.2 = -242.8 J/K mol

138. Ice at -5°C is heated to become vapor with temperature of 110°C at atmospheric pressure. The entropy change associated with this process can be obtained from **[23 Jan, Shift-I, 2025]**

(A) $\int_{268K}^{273K} C_{p,m}\,dT + \frac{\Delta H_{m,\text{fusion}}}{T_f} + \frac{\Delta H_{m,\text{vaporisation}}}{T_b} + \int_{273K}^{373K} C_{p,m}\,dT + \int_{373K}^{383K} C_{p,m}\,dT$

(B) $\int_{268K}^{383K} C_p\,dT + \frac{\Delta H_{\text{melting}}}{273} + \frac{\Delta H_{\text{boiling}}}{373}$

(C) $\int_{268K}^{273K} \frac{C_{p,m}}{T}\,dT + \frac{\Delta H_{m,\text{fusion}}}{T_f} + \frac{\Delta H_{m,\text{vaporisation}}}{T_b} + \int_{273K}^{373K} \frac{C_{p,m}\,dT}{T} + \int_{373K}^{383K} \frac{C_{p,m}\,dT}{T}$

(D) $\int_{268K}^{383K} C_p\,dT + \frac{q_{rev}}{T}$

Sol. (C) For explanation, refer to QR code given on the last page.

139. 30.4 kJ of heat is required to melt one mole of sodium chloride and the entropy change at the melting point is 28.4 $JK^{-1}\,mol^{-1}$ at 1 atm. The melting point of sodium chloride in K (Nearest integer)

[JEE Mains, 15 April 2023 Shift-I]

Sol. $\Delta H_{Fusion} = 30.4 \times 10^3$ J/mol

$\Delta S_{sys} = 28.4\ J/K^{-1}\,mol^{-1}$, $\Delta S = \frac{\Delta H_{Fusion}}{T}$

$T = \frac{\Delta H_{Fusion}}{\Delta S} = \frac{30.4 \times 10^3}{28.4}\ K = 1070.4\ K$

are formula hi toh lagaya hai bas

140. The enthalpy change of transition of liquid water to steam is 40.8 kJ mol^{-1} at 373 K. Calculate ΔS for the process.

Sol. The formula to calculate change in entropy for vaporization process is given by:

$$\Delta S = \frac{\Delta H_{vap}}{\Delta T} = \left(\frac{40.8 \times 1000\,J/mol}{373K}\right) = 109.38K$$

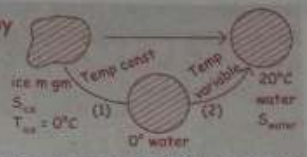
141. Given ΔS = 109.8 $JK^{-1}\,mol^{-1}$ and boiling point of ethanol = 78.5°C. The enthalpy of vaporisation per mole for ethanol is:

(A) 46.581 kJ mol^{-1} (B) 38.595 kJ mol^{-1}

(C) 57.891 kJ mol^{-1} (D) 27.231 kJ mol^{-1}

Sol. (B) For explanation, refer to QR code given on the last page.

142. Calculate entropy change for the transformation given below:



Sol. Process 1 is a fusion process and its entropy is given by:

$$\Delta S_1 = \left(\frac{\Delta H_{Fusion}}{273}\right) n_{ice}$$

For process 2, temperature is changing and its entropy is given by:

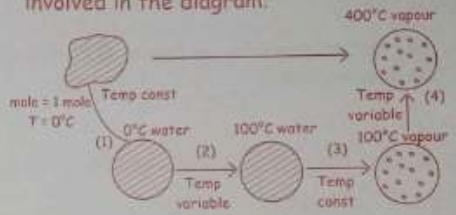
$$\Delta S_2 = mS_{water} \ln\left(\frac{293}{273}\right)$$

Total entropy is given by $\Delta S_1 + \Delta S_2$

$$= \Delta S_1 = \left(\frac{\Delta H_{Fusion}}{273}\right) n_{ice},\ \Delta S_2 = mS_{water} \ln\left(\frac{293}{273}\right)$$

yeh thoda pecheeda hai

143. Calculate net entropy for the processes involved in the diagram.



Sol. ($\Delta S_{net} = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4$)

For explanation, refer to QR code given on the last page.

144. Calculate the total entropy change for the transition at 368 K of 1 mol of suphur from the monoclinic to rhombic solid state, if ΔH = -401.7 J/mole for this transition. Assume the surroundings is ice-water bath at 0°C.

(A) -1.09 JK^{-1} (B) 1.47 JK^{-1}

(C) 0.38 JK^{-1} (D) None of these

Sol. (C) $\Delta S_{sulphur} = \frac{-401.7}{368} = -1.09\ J/K$

$\Delta S_{both} = \frac{401.7}{273} = 1.47\ J/K$

$\Delta S_{total} = \Delta S_{sulphur} + \Delta S_{both} = 0.38\ J/K$

145. The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapour at 383 K is: (specific heat of water liquid and water vapour are 4.2 kJ $K^{-1}\,kg^{-1}$ and 2.0 kJ $K^{-1}\,kg^{-1}$, heat of liquid fusion and

vapourisation of water are 334 kJ kg^{-1} and 2491 kJ kg^{-1}, respectively) (log 273 = 2.436, log 373 = 2.572, log 383 = 2.583)

[JEE Mains, 9 Jan 2020 Shift-II]

(A) 7.90 kJ K^{-1} kg^{-1} (B) 2.64 kJ K^{-1} kg^{-1}

(C) 8.49 kJ K^{-1} kg^{-1} (D) 9.26 kJ K^{-1} kg^{-1}

Sol. (D) For explanation, refer to QR code given on the last page.

146. Two blocks of the same metal having same mass and at temperature T_1 and T_2 respectively, are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS, for this process is

[JEE Mains, 11 Jan 2019 Shift-1]

(A) $C_p \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$ (B) $2C_p \ln\left[\frac{(T_1+T_2)^{1/2}}{T_1T_2}\right]$

(C) $2C_p \ln\left[\frac{(T_1+T_2)}{4T_1T_2}\right]$ (D) $2C_p \ln\left[\frac{(T_1-T_2)}{2T_1T_2}\right]$

Sol. (A) For explanation, refer to QR code given on the last page.

Case-3: An Ideal Gas Undergoing Some Process

$\Delta S_{sys} = \int ds = \int \frac{dq_{rev}}{T}$

Process—कुछ भी
path—reversible

ΔS = const

$dU = dq + dw$, $dq = dU - dw$

$dq = nC_V dT + P_{ext} dV$,

$dq = nC_V dT + P_{gas} dV$

$dq_{rev} = nC_v dT + \frac{nRT}{V} dV$, $\frac{dq_{rev}}{T} = nC_V \frac{dT}{T} + nR\frac{dV}{V}$

$\Delta S_{sys} = \int \frac{dq_{rev}}{T} = nC_V \int_{T_1}^{T_2} \frac{dT}{T} + nR \int_{V_1}^{V_2} \frac{dV}{V}$

$$\Delta S_{sys} = nC_v \ln\frac{T_2}{T_1} + nR \ln\frac{V_2}{V_1}$$

$$\Delta S_{sys} = nR \ln\left(\frac{V_2}{V_1}\right) + nC_v \ln\left(\frac{T_2}{T_1}\right)$$

Volume factor Temp factor

Khoobsurat Baat

To change the entropy of an ideal gaseous system we can-

(i) Change the temperature

(ii) Change the volume

147. * On changing the temperature at constant volume entropy changes. (T/F)

* On changing the volume at constant temperature entropy always changes. (T/F)

* On changing volume and temperature entropy changes always. (T/F)

[Mains + Adv.]

Sol. May be there is a process in which volume increases and temperature decreases (adiabatic process), in that case, both the factors nullify each other.

148. At what temperature will the entropy of one mole of an ideal gas occupying 5 dm^3 at 300 K on expanding to 500 dm^3 increases by 38.294 JK^{-1}? (R = 8.314 JK^{-1} mol^{-1})

(A) 600 K (B) 300 K (C) 250 K (D) 700 K

Sol. (B) For explanation, refer to QR code given on the last page.

Entropy Change for Isothermal Reversible Process

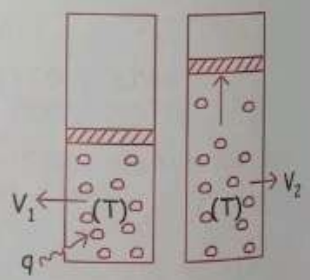
1. Entropy change of system

Let initial volume be V_1 and final volume be V_2 and T be the temperature.

$dU = dq + dw$

$\Delta U = q + w$ ($\Delta U = 0$)

$q = -w$



Change in entropy for system can be calculated as:

$$\Delta S_{system} = nR \ln\frac{V_2}{V_1} + nC_v \ln\frac{T_2}{T_1}$$ ($T_2 = T_1$)

For isothermal process, $T_1 = T_2 = T$

So, final change in entropy can be written as:

$$\Delta S_{system} = nR\ln\frac{V_2}{V_1}$$

Khoobsurat Baat

Expansion → $(V_2 > V_1)$: ΔS_{system} = (+) ve

Compression → $(V_2 < V_1)$: ΔS_{system} = (-) ve

2. Entropy change of surroundings

$$\Delta S_{surrd} = \frac{-\Delta H_{sys}}{T_{surrd}}$$

Work done $= -nRT\ln\frac{V_2}{V_1}$, $\Delta H_{sys} = q = nRT\ln\frac{V_2}{V_1}$ $(q = -w)$

$$\Delta S_{surrounding} = \frac{-nRT\ln\frac{V_2}{V_1}}{T} = -nR\ln\frac{V_2}{V_1}$$

$$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surrd} = nR\ln\frac{V_2}{V_1} + \left(-nR\ln\frac{V_2}{V_1}\right) = 0$$

formula toh yaad karna padega tabhi kaam banega

Important

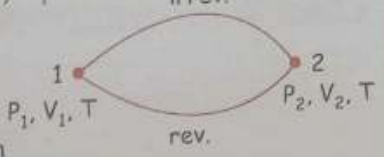
In any reversible process:

$\Delta S_{universe} = 0$

Entropy Change for Irreversible Isothermal Process

1. Entropy change of system

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1} + nC_v\ln\frac{T_2}{T_1}^{\nearrow 0} \text{ (Reversible path)}$$



ΔS = state function

$\Delta S_{rev} = \Delta S_{irrev}$ if states are same

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1}$$

$\Delta S_{surroundings}$ depends on heat exchange between system and surroundings. For any isothermal process, heat exchange is given by the magnitude of work done. Since, work done is a path function that is why heat exchange also depends on path. Since path is irreversible, the heat exchange in this path is different than the heat exchange in reversible path.

$\Delta S_{surroundings\ (rev)}$ is not equal to $\Delta S_{surroundings\ (irreversible)}$

$$(\Delta S_{surrd})_{irrev} = \frac{-\Delta H_{sys}}{T} = \frac{-P_{ext}(V_2 - V_1)}{T}$$

$$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surrounding} = nR\ln\left(\frac{V_2}{V_1}\right) + \left[\frac{-P_{ext}(V_2 - V_1)}{T}\right]$$

149. One mole of an ideal monoatomic gas expands isothermally against constant external pressure 1 atm from initial volume 1 lt to a state where its final pressure becomes equal to external pressure. If initial temp is 300 K then total entropy change of the system in this process-(R = 0.0821 L atm/k-mol or 8.314 J/K mol)

(A) 0 (B) Rln (24.6)

(C) Rln (2490) (D) $\frac{3}{2}$Rln(24.6)

Sol. (B) For explanation, refer to QR code given on the last page.

150. One mole of an ideal gas at 350 K is in a 2.0 L vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from 2.0 L to 3.0 L against a constant pressure of 4 atm. The change in entropy of the surroundings (ΔS) is _____ JK^{-1} (Nearest integer)

Given: R = 8.314 JK^{-1} mol^{-1}

[JEE Mains, 12 April 2023 Shift-I]

Sol. The formula to calculate change in entropy for isothermal process is:

$$\Delta S_{System} = nR\ln\left(\frac{V_2}{V_1}\right)$$

Substitute the values in the above expression:

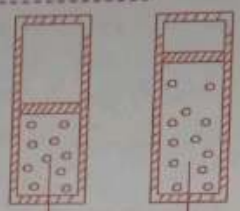
$$= 1 \times 8.314\ln\left(\frac{3}{2}\right)$$

$\Delta S_{System} = 3.37$, $\Delta S_{surr.} = 3.37$

Kuch nahi kia bas formula lgaya hai

Entropy Change for Reversible Adiabatic Process

Let initial temperature be T_1, final temperature be T_2, Initial volume be V_1 and final volume be V_2.



For an adiabatic process,

$$TV^{\gamma-1} = \text{constant}$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

1. Entropy change of system

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1} + nC_V \ln\frac{T_2}{T_1} = nR\ln\frac{V_2}{V_1} + nC_V \ln\left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$= nR\ln\frac{V_2}{V_1} + n(\gamma-1)C_V \ln\frac{V_1}{V_2}$$

$$\boxed{= nR\ln\frac{V_2}{V_1} + nR\ln\frac{V_1}{V_2} = 0}$$

Entropy change of surroundings

$$\boxed{\Delta S_{surrd} = \frac{-\Delta H_{sys}}{T} = 0 \quad [\Delta H_{sys} = q = 0]}$$

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$(\gamma-1)C_V = R, \quad nR\ln\frac{V_1}{V_2} = -nR\ln\frac{V_2}{V_1}$

Adiabatic reversible (Isentropic or isoentropic process)

$\Delta S_{sys} = 0 \quad \Delta S_{surrd} = 0$

$\Delta S_{univ} = \Delta S_{surrd} + \Delta S_{sys}$
$= 0 + 0$
$= 0$

151. One mole of an ideal gas at 25°C is subjected to expand reversibly and adiabatically to ten times of its initial volume. Calculate the change in entropy during expansion in $JK^{-1}\,mol^{-1}$.

(A) 19.15 (B) -19.15 (C) 1.915 (D) zero

Sol. (A) For explanation, refer to QR code given on the last page.

Entropy Change for Adiabatic Irreversible Process

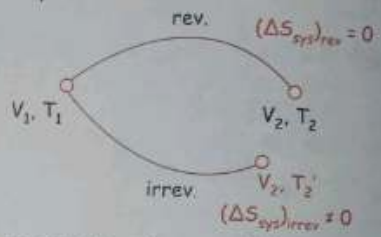
Note

$(\Delta S_{sys})_{irrev} = (\Delta S_{sys})_{rev} = 0$ (adiabatic) ✗

$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surrd} = 0 + 0 = 0$

The above is not true. Let's explain this in detail.

Let V_1, V_2 be the initial and final volume respectively. T_1, T_2 be the initial and final temperature respectively.



Since there exists only one path joining two states of gas adiabatically.

So $(\Delta S_{sys})_{rev} \neq (\Delta S_{sys})_{irrev}$

1. Entropy change of system

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surrd} > 0 \quad (\Delta S_{surrd} = 0)$$

$$\left(\Delta S_{sys}\right)_{rev} = nR\ln\frac{V_2}{V_1} + nC_V \ln\frac{T_2}{T_1} = 0 \ldots..(1)$$

$$\left(\Delta S_{sys}\right)_{irrev} = nR\ln\frac{V_2}{V_1} + nC_V \ln\frac{T_2'}{T_1} > 0 \ldots..(2)$$

2. Entropy change of surroundings

$$\Delta S_{surroundings} = \frac{-\Delta H_{sys}}{T} = 0 \quad [\Delta H_{sys} = 0]$$

The above equation gets modified for different processes involved.

itne sare formulae, dimag thik toh hai na?

Entropy Change for Isochoric Process

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1} + nC_V \ln\frac{T_2}{T_1} \quad (\ln\frac{V_2}{V_1} = 0) \qquad \boxed{\Delta S_{sys} = nC_V \ln\frac{T_2}{T_1}}$$

$$\boxed{\Delta S_{surrd} = \frac{-\Delta H_{sys}}{T_{surrd}} = \frac{-nC_V(T_2 - T_1)}{T_{surrd}}}$$

Entropy Change for Isobaric Process

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1} + nC_v \ln\frac{T_2}{T_1}$$

Both the factors (volume factor and temperature factors affect the entropy change of system)

Q → Expansion → Entropy increases by volume factor and also by temperature factor.

Q → Contraction → Entropy decreases by volume and temperature factor both.

$$\Delta S_{sys} = nR\ln\frac{V_2}{V_1} + nC_v \ln\frac{T_2}{T_1} \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2} = nR\ln\frac{T_2}{T_1} + nC_v \ln\frac{T_2}{T_1}$$

$$\Delta S_{sys} = nC_p \ln\frac{T_2}{T_1}$$

$$\Delta S_{surrd} = \frac{-\Delta H_{sys}}{T_{surrd}} = \frac{-nC_p(T_2 - T_1)}{T_{surrd}}$$

Note

$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surrd}$

$\Delta S_{univ} = 0$ reversible (Eq^{bm})

$\Delta S_{univ} < 0$ Non-spontameous

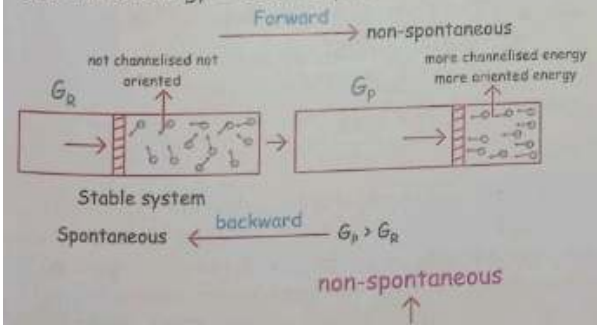
$\Delta S_{univ} > 0$ Spontaneous

GIBBS ENERGY

- Gibb's energy is a thermodynamic quantity which is used to measure the capacity of system to do useful work or Gibb's energy is that part of the total energy of system which can be converted into useful work.

$$\text{System stability} \propto \frac{1}{\text{Free energy of system}}$$

Gibb's free energy ⇒ Used to perform some useful work.



For forward movement of piston, $\Delta G > 0$

For backward movement, $\Delta G < 0$ → Spontaneous

Now,

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surrd} = \Delta S_{sys} + \left(\frac{-\Delta H_{sys}}{T}\right)$$

$$-T\Delta S_{univ} = -T\Delta S_{sys} + \Delta H_{sys}$$

$$\boxed{\Delta G = \Delta H_{sys} - T\Delta S_{sys}}$$

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$\Delta G = -T\Delta S_{univ}$

if $\Delta G < 0$ then $\Delta S_{univ} > 0$ → spontaneous

if $\Delta G > 0$ then $\Delta S_{univ} < 0$ → non-spontaneous

Case-1

$\Delta G = \Delta H - T\Delta S$

$\Delta H_{sys} > 0, \Delta S_{sys} < 0$

ΔG will always be (+) ve at all temperature

Never spontaneous

Case-2

$\Delta H_{sys} < 0 ; \Delta S_{sys} > 0$

ΔG will always be negative

Always Spontaneous at all temperature

Case-3

$\Delta H_{sys} > 0 ; \Delta S_{sys} > 0$

These reaction are spontaneous at higher temperature

The minimum temp above which the process is spontaneous-

$\Delta G = 0$; $\Delta H - T\Delta S = 0$ | above T → spontaneous

$T = \left(\frac{\Delta H}{\Delta S}\right)$ | below T → non-spontaneous

Case-4

$\Delta H_{sys} < 0 ; \Delta S_{sys} < 0$

These reaction are spontaneous at lower temperatures

The minimum temp below which the process is spontaneous-

$\Delta G = 0$; $\Delta H = T\Delta S$ | above T → non-spontaneous

$T = \left(\frac{\Delta H}{\Delta S}\right)$ | below T → spontaneous

dekh liye na sare cases

Lets consider a general reaction

$aA + bB \rightleftarrows cC + dD$

$$\Delta G = \Delta G^\circ + RT\ln\frac{[D]^d[C]^c}{[A]^a[B]^b}$$

At Eq. $\Delta G = 0$

$$\Delta G^\circ = -RT\ln\frac{[D]^d[C]^c}{[A]^a[B]^b} \xrightarrow{K_{eq}} \Delta G^\circ = -RT\ln K_{eq}$$

152. Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option. **[24 Jan, Shift-I, 2025]**

(A) Both ΔH and ΔS are (-ve)

(B) Both ΔH and ΔS are (+ve)

(C) ΔH is (-ve) but ΔS is (+ve)

(D) ΔH is (+ve) but ΔS is (-ve)

Sol. (B) For explanation, refer to QR code given on the last page.

153. For a certain reaction the change in enthalpy and change in entropy are 40.63 kJ mol^{-1} and 100 JK^{-1}. What is the value of ΔG at 27°C and indicate whether the reaction is possible or not?

Sol. $\Delta G = \Delta H - T\Delta S = 40.63 \times 10^3 - 300 \times 100$

$= 40630 - 30000 = 10630$ J

As change in Gibbs free energy is greater than 0, reaction will be non-spontaneous.

shabashi do khud ko

154. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

At 298 K: $\Delta_f H^\circ(SnO_2(s)) = -581.0$ kJ mol^{-1}, $\Delta H^\circ(CO_2(g)) = -394.0$ kJ mol^{-1}

$S^\circ(SnO_2(s)) = 56.0$ J K^{-1} mol^{-1}, $S^\circ(Sn(s))$ 52.0 J K^{-1} mol^{-1}

$S^\circ(C(s)) = 6$ J K^{-1} mol^{-1}, $S^\circ(CO_2(g)) = 210.0$ J K^{-1} mol^{-1}

[JEE Adv.-2020]

Sol. (935 K) For explanation, refer to QR code given on the last page.

155. The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are 55.0 kJ mol^{-1} and 175.0 J/K/mol respectively. The standard free energy change for this reaction at 25°C in J mol^{-1} is _____ (Nearest integer) **[23 Jan, Shift-I, 2025]**

Sol. [2,850] $\Delta G = \Delta H - T\Delta S$

$\therefore \Delta G = 55{,}000 - (298 \times 175) = 2{,}850$ J mol^{-1}

sara khel minus plus se pta chal jaega

156. Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is _____ (integer answer)

[Given: $\Delta_f G^\circ$ ($HC \equiv CH$) = -2.04×10^5 J mol^{-1}; $\Delta_f G^\circ$ (C_6H_6) = -1.24×10^5 J mol^{-1}; R = 8.314 JK^{-1} mol^{-1})]

[JEE Mains, 24 Feb 2021 Shift-II]

Sol. (855) For explanation, refer to QR code given on the last page.

157. For independent process at 300 K.

Process	ΔH/kJ mol^{-1}	$\Delta S/JK^{-1}$
A	-25	-80
B	-22	40
C	25	-50
D	22	20

The number of non-spontaneous process from the following is _____.

[JEE Mains, 24 Jan 2023 Shift-I]

Sol. (C and D) For explanation, refer to QR code given on the last page.

sabhi ko karke dekhna padega, koi or chaara nahi hai

158. The incorrect expression among the following is: **[JEE Mains, 31 Aug 2021 Shift-II]**

(A) $\ln K = \dfrac{\Delta H^\circ - T\Delta S^\circ}{RT}$

(B) For isothermal process,

$$w_{reversible} = -nRT\ln\frac{V_f}{V_i}$$

(C) $\dfrac{\Delta G_{system}}{\Delta S_{Total}} = -T$ (at constant P)

(D) $K = e^{-\Delta G^\circ/RT}$

Sol. (A) The formula to calculate ΔG° is:

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$-RT\ln K = \Delta H^\circ - T.\Delta S^\circ$

$$\ln K = \frac{T\Delta S^\circ - \Delta H^\circ}{RT}$$

159. For the reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0$ JK^{-1} and $\Delta H = -57.8$ kJ mol^{-1}, the magnitude of ΔG at 298 K for the reaction is _____ kJ mol^{-1}. (Nearest integer)

[JEE Mains, 1 sep 2021 Shift-II]

Sol. The formula to calculate ΔG° is:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

Substitute the values in the above equation:

$\Delta G^\circ = -57.8-(298 \times -0.176) = -5.352$ kJ/mol

So, the magnitude will be 5 kJ/mol.

Jab bh magnitude puchta hai toh answer hamesha plus mei hi aaega

160. For a chemical reaction $A + B \rightleftharpoons C + D$ ($\Delta_r H^\circ$ = 80 kJ mol^{-1}) the entropy changes $\Delta_r S^\circ$ depends on the temperature T (in K) as $\Delta_r S^\circ$ = 2T (JK^{-1} mol^{-1}).

Minimum temperature at which it will become spontaneous is _____ K. (Integer).

[JEE Mains, 26 Feb 2021 Shift-1]

Sol. (200 K) For explanation, refer to QR code given on the last page.

161. For a dimerization reaction, $2A(g) \rightarrow A_2(g)$, at 298 K,

ΔU° = -20 kJ mol^{-1}, ΔS° = -30 JK^{-1} mol^{-1}, then the ΔG° will be _____ J.

[JEE Mains, 5 Sept. 2020 Shift-II]

Sol. (-13538 J/mol) For explanation, refer to QR code given on the last page.

162. The reaction

$MgO(s) + C(s) \rightarrow Mg(s) + CO(g)$, for which $\Delta_r H^\circ$ = +491.1 kJ mol^{-1} and $\Delta_r S^\circ$ = 198.0 JK^{-1} is not feasible at 298 K. Temperature above which reaction will be feasible is :

[JEE Mains, 11 Jan 2019 Shift-II]

(A) 2040.5 K (B) 1890.0 K

(C) 2480 K (D) 2380 K

Sol. (C) Feasibility of reaction will be determined by sign of change of Gibbs free energy.

$\Delta H - T\Delta S < 0$

Substitute the values in the above expression:

$$T > \frac{\Delta H}{\Delta S} = \frac{491.1 \times 1000}{198}, \quad T > 2480.3 \text{ K}$$

163. A process that has ΔH = 200 J mol^{-1} and ΔS = 40 J K^{-1}. Out of the value given below, choose the minimum temperature above which the process will be spontaneous

[JEE Mains, 10 Jan 2019 Shift-I]

(A) 20 K (B) 12 K (C) 5 K (D) 4 K

Sol. (C) For explanation, refer to QR code given on the last page.

164. The standard reaction Gibbs energy for a chemical reaction at an absolute temperature T is given by:

$$\Delta_r G^\circ = A - BT$$

Where A and B are non-zero constant. Which of the following is TRUE about this reaction?

[JEE Mains, 11 Jan 2019 Shift-II]

(A) Endothermic if A > 0

(B) Exothermic if A > 0 and B > 0

(C) Endothermic if A < 0 and B > 0

(D) Exothermic if B < 0

Sol. (A) The formula to calculate ΔG° is:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

It is given that A and B are non-zero constants, so, $A = \Delta H^\circ$, $B = \Delta S^\circ$

If ΔH° is positive, the reaction will be endothermic.

165. For the chemical reaction $X \rightleftharpoons Y$, the standard reaction Gibbs energy depends on 3 temperature T (in K) as $\Delta_r G^\circ$ (in kJ mol^{-1})

$$= 120 - \frac{3}{8}T$$

The major component of the reaction mixture at T is: **[JEE Mains, 11 Jan 2019 Shift-I]**

(A) Y if T = 300 K (B) Y if T = 280 K

(C) X if T = 350 K (D) X if T = 315 K

Sol. (A) For explanation, refer to QR code given on the last page.

166. For the reaction $A(l) \rightarrow 2B(g)$

ΔU = 2.1 kcal, ΔS = 20 cal K^{-1} at 300 K

Hence, ΔG in kcal is

[JEE Mains, 7 Jan 2020 Shift-1]

Sol. First calculate ΔH as: $\Delta H = \Delta H + \Delta n_g RT$

And for ΔG is: $\Delta G = \Delta H - T\Delta S$

Substitute the value of ΔG in above equation:

$\Delta G = \Delta U + \Delta n_g RT - T\Delta S$

Now, substitute the values in the above equation:

$$\Delta G = 2.1 + \frac{2 \times 2 \times 300}{1000} - \frac{300 \times 20}{1000}$$

(R = 2 cal K^{-1} mol^{-1})

$= 2.1 + 1.2 - 6 = -2.70$ Kcal/mol

unit ko convert karna mat bhul jana

167. Data given for the following reaction is as follows:

$FeO(s) + C_{(graphite)} \rightarrow Fe(s) + CO(g)$

Substance	$\Delta_f H^\circ$(kJ mol^{-1})	ΔS°(J mol^{-1} K^{-1})
$FeO_{(s)}$	-266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe_{(s)}$	0	27.28
$CO_{(g)}$	-110.5	197.6

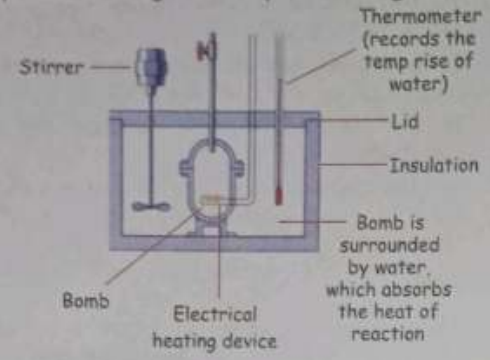
The minimum temperature in K at which the reaction becomes spontaneous is _____

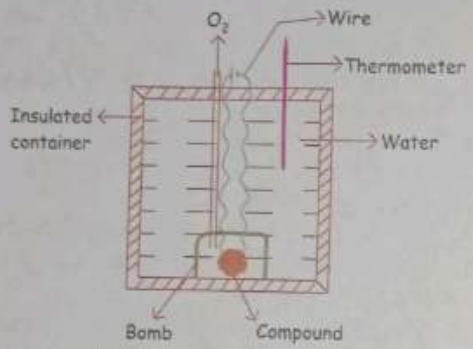
[JEE Mains, 27 Aug 2021 Shift-II]

Sol. (964 K) For explanation, refer to QR code given on the last page.

BOMB CALORIMETER

A **bomb calorimeter** is a device used to measure the heat of combustion of a substance. It provides a controlled environment to determine the energy content of a sample, often a fuel or food, by observing the temperature change in the system during combustion.





- During combustion energy is released.
- This released energy will be absorbed by bomb and water.

$Q = C\Delta T = C(T_2 - T_1)$

T_1 = initial temp before combustion

T_2 = final temperature after combustion

C = heat capacity of colorimeter.

$$Q = n_{Bomb} \cdot C_{Bomb} \Delta T + n_{H_2O} \cdot C_{H_2O} \Delta T$$

$$Q = m_{Bomb} \cdot S_{Bomb} \Delta T + m_{H_2O} \cdot S_{H_2O} \Delta T$$

$\Delta E = \frac{-Q}{n}$, $\Delta H = \Delta E + \Delta n_g RT$

- Overall heat capacity of bomb calorimeter can be calculated as:

$Q = C(T_2 - T_1)$, $\Delta E = \frac{-C}{n}$ → $\frac{wt}{mol\ wt} = n$

= no. of moles of the substance which was combusted

$\Delta H = \Delta E + \Delta n_g RT$

Khoobsurat Baat

First calculate $Q = C\Delta T = C(T_2 - T_1)$ (Heat capacity of calorimeter)

$\Delta E_{rxn} = \frac{-Q}{n}$

$\Delta H = \Delta E + \Delta n_g RT$

168. 12.2 gm solid benzoic acid is combusted in bomb calorimeter at 27°C. Temperature raised from 27°C to 77°C. If the heat capacity of calorimeter is 50 J/°C then-
(i) Calculate change in internal energy in J/mole
(ii) Calculate molar enthalpy of combustion at 27°C

Sol. (i) (25000 J) (ii) (-25000 J)
For explanation, refer to QR code given on the last page.

169. When 1.550 gm of liquid hexane (C_6H_{14}) undergoes combustion in a bomb calorimeter the temperature rises from 25.87°C to 38.13°C. Find $\Delta E_{reaction}$ in kJ/mole.
The heat capacity of calorimeter is 5.730 kJ/°C

Sol. $Q = C\Delta T = 5.730 \times (38.13 - 25.87)$
$= 5.730 \times 12.26$ kJ

$$\Delta E_{rxn} = \frac{-5.730 \times 12.26 \times 86}{1.550} \text{ kJ/mol}$$

$= 3897.7$ kJ/mol

170. The reaction of cyanamide, $NH_2CN(s)$ with oxygen was run in a bomb calorimeter and ΔU was found to be -742.24 kJ mol^{-1}. The magnitude of ΔH_{298} for the reaction $NH_2CN(S) + O_2(g) \rightarrow N_2(g) + O_2(g) + H_2O(l)$ is _____ kJ.

Sol. (-741 kJ/mol)
For explanation, refer to QR code given on the last page.

171. 0.3 g of ethane undergoes combustion at 27°C in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by 0.5°C. The heat evolved during combustion of ethane at constant pressure is _____ kJ mol^{-1}. (Nearest integer)
[Given: The heat capacity of the calorimeter system is 20 kJ K^{-1}, R = 8.3 J K^{-1} mol^{-1}.
Assume ideal gas behaviour.
Atomic mass of C and H are 12 and 1 g mol^{-1} respectively] **[1 Feb, 2023 (Shift-II)]**

Sol. (Bomb calorimeter → const volume)
Calculate heat released by combustion of 1 mole as:

$$C_2H_2(\Delta U) = -\frac{20 \times 0.5}{0.3} \times 30 = -1000 \text{ kJ}$$

$C_2H_2(g) + 7/2O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

$\Delta n_g = 2 - (2 + 7/2) = -7/2$

$= -1000 - 7/2 \times 8.3 \times 300$ kJ

$= -1000 - 6.225 = -1006$ kJ

So, heat released = 1006 kJ mol^{-1}

mushkil hai, time do isko

172. For complete combustion of ethene, $C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$ the amount of heat produced as measured in bomb calorimeter is 1406 kJ mol^{-1} at 300K. The minimum value of $T\Delta S$ needed to reach equilibrium is (-) _____ kJ. (Nearest integer)

Given: R = 8.3 JK^{-1} mol^{-1}

[8 April, 2023 (Shift-II)]

Sol. (-1411 kJ mol^{-1}) For explanation, refer to QR code given on the last page.

173. For complete combustion of methanol

$$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

The amount of heat produced as measured by bomb calorimeter is 726 kJ mol^{-1} at 27°C.

The enthalpy of combustion for the reaction is -x kJ mol^{-1}, where x is_____. (Nearest integer)

(Given: R = 8.3 J K^{-1} mol^{-1})

[26 June, 2022 (Shift-I)]

Sol. $CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$$\Delta H = \Delta U + \Delta n_g RT = -726 - \left(\frac{0.5 \times 8.3}{1000} \times 300\right)$$

$= -726 - 1.24 = -727.24 \approx -727$ kJ/mol

waah waah, kitna asaan hai

174. For combustion of one mole of magnesium in an open container at 300 K and 1 bar pressure, $\Delta_cH^\ominus$ = -601.70kJ mol^{-1}, the magnitude of change in internal energy for the reaction is _____kJ. (Nearest integer)

(Given: R = 8.3 J K^{-1} mol^{-1})

[28 June, 2022 (Shift-II)]

Sol. (600 kJ) For explanation, refer to QR code given on the last page.

175. 2.4 g coal is burnt in a bomb calorimeter in excess of oxygen at 298 K and 1 atm pressure. The temperature of the calorimeter rises from 298 K to 300 K. The enthalpy change during the combustion of coal is -x kJ mol^{-1}. The value of x is _____. (Nearest Integer)

(Given: Heat capacity of bomb calorimeter 20.0 kJ K^{-1}. Assume coal to be pure carbon)

[26 July, 2022 (Shift-I)]

Sol. $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta n_g = 0$

$\Delta T = (300 - 298)K = 2K$

Given, $\Delta H = -x$ kJ/mole

No. of moles of $C = \frac{2.4}{12} = 0.2$ moles

ΔH for 0.2 mole = $0.2 \times (-x)$ kJ = $-0.2 \times$ kJ

As, $q = C\Delta T$

$0.2x = 20 \times 2$ $\quad (\because q = -\Delta H)$

$x = \frac{40}{0.2} = 200$ kJ/$mole^{-1}$

176. "X" gm of ethanal was subjected to combustion in a bomb calorimeter and the heat produced is "Y" Joules. Then:

(A) $\Delta U_{(combustion)} = -XJ$

(B) $\Delta U_{(combustion)} = -YJ$

(C) $\Delta U_{(combustion)} = -44Y/X$ J mol^{-1}

(D) $\Delta H_{(combustion)} = -44Y/X$ J mol^{-1}

Sol. (C) For explanation, refer to QR code given on the last page.

KIRCHOFF'S EQUATION

Enthalpy of reaction also depends on the temperature at which the reaction is carried out. This is due to variation in the heat capacity of the system with temperature.

$\Delta H = H_p - H_R$

$$\left(\frac{d\Delta H}{dT}\right)_p = \left(\frac{dH_p}{dT}\right)_p - \left(\frac{dH_R}{dT}\right)_p = (C_p)_p - (C_p)_R = \Delta C_p$$

$d\Delta H/dT = \Delta C_p$, $d\Delta H = \Delta C_p dT$

$$\int_{\Delta H_1}^{\Delta H_2} d\Delta H = \int_{T_1}^{T_2} \Delta C_p dT \text{ , } \Delta H_2 - \Delta H_1 = \Delta C_p(T_2 - T_1)$$

$$\boxed{\Delta C_p = \frac{\Delta H_2 - \Delta H_1}{T_2 - T_1}}$$

$\Delta U = U_p - U_R$, $\left(\frac{d\Delta U}{dT}\right)_v = \left(\frac{dU_p}{dT}\right)_v - \left(\frac{dU_R}{dT}\right)_v$

$\frac{d\Delta U}{dT} = (C_v)_p - (C_v)_R = \Delta C_v$, $\int_{\Delta U_1}^{\Delta U_2} d\Delta U = \int_{T_1}^{T_2} \Delta C_v dT$

$$\boxed{\Delta C_v = \frac{\Delta U_2 - \Delta U_1}{T_2 - T_1}}$$

177. Consider the reaction,

$$H_2O(g) \rightarrow H_2(g) + \frac{1}{2}O_2(g);$$

$\Delta H^\circ = 241.75$ J at 291.15 K

What would be ΔH° at 298.15 K? Over the small temperature range, the effectively constant C_p values per mole are-

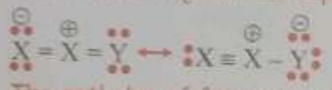
$C_p(H_2O) = 33.56$ J/K mol

$C_p(H_2) = 28.83$ J/K mol

$C_p(O_2) = 29.12$ J/K mol

Sol. (310.56 J) For explanation, refer to QR code given on the last page.

178. Resonance in X_2Y can be represented as



The enthalpy of formation of

$X_2Y\left(X \equiv X(g) + \frac{1}{2}Y = Y(g) \rightarrow X_2Y(g)\right)$ is

80 kJ mol^{-1}. The magnitude of resonance energy of X_2Y is _____ kJ mol^{-1} (nearest integer value)

Given : Bond energies of X = X , X = X, Y = Y and X = Y are 940, 410, 500 and 602 kJ mol^{-1} respectively.

valence X : 3, Y : 2

[08 April, 2025 (Shift-II)]

Sol. (98) $\Delta H_{R.E} = \Delta H_{f(exp))} - \Delta H_{f\,(Theo)}$

$\Delta H_{f(exp)}$ for $X_2Y_{(g)} = 80$ kJ / mole

for ΔH_f (Theo)

$$X_{2(g)} + \frac{1}{2}Y_{2(g)} \rightarrow X_2Y_{(g)}$$

$\Delta H_{f(Theo)} = \left(BE_{X=X} + \frac{1}{2}BE_{Y=Y}\right) - \left(BE_{X=X} + BE_{X=Y}\right)$

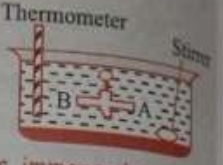
$= \left(940 + \frac{1}{2} \times 500\right) - (410 + 602)$

$= 178$ kJ / mole

$\Delta H_{R.E} = 80 - 178 = -98$ kJ / mol

$|\Delta H_{R.E}| = 98$

179. Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure.



The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true?

[02 April, 2025 (Shift-I)]

(A) $dw \neq 0$ (B) $dq \neq 0$ (C) $dU \neq 0$

(D) The pressure in the vessel B before opening the stopcock is zero.

Sol. (D) This is a free expansion of gas

($P_{ext} = 0$), where, $w = 0$, $q = 0$ and $\Delta U = 0$.

180. For Ag, C_p (JK^{-1} mol^{-1}) is given by $23.43 + 0.00628\,T$. Calculate ΔH if three moles of silver are raised from 25°C to its melting point 961°C under 1 atm pressure.

Sol. $\Delta H = n\int_{T_1}^{T_2} C_p dT = 3\int_{298}^{1234} (23.43 + 0.00628T)dT$

$= 3\left[23.43T + \frac{.00628T^2}{2}\right]_{298}^{1234}$

$= 79.298$ kJ / mol

Khoobsurat Baat

Kirchhoff's equation → We calculate ΔH at T_2 with the help of ΔH_{T_1} and $\Delta C_p[(C_p)_P - (C_p)_R]$ by using this equation-

$$\boxed{\Delta C_p = \frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1}}$$

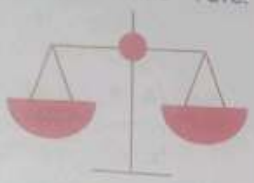
Scan the QR code to get the explanations of unsolved questions.

4 Chemical Equilibrium

To access one shot Revision Video Scan this QR Code

INTRODUCTION

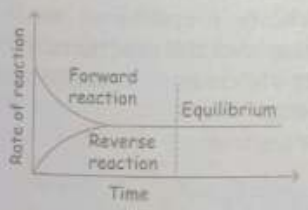
Imagine pouring sugar into a glass of water. Initially, the sugar dissolves quickly, but as you add more, a point comes when no more sugar dissolves, no matter how much you stir. Socho aisa kya hua? This moment marks the establishment of equilibrium - a state where opposing processes, such as dissolving and crystallization, occur at the same rate. Equilibrium is the silent mediator that keeps reactions in balance, whether in the world around us or within the human body.

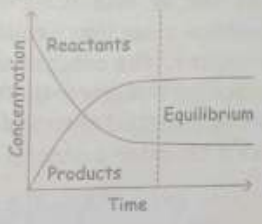


Acha lga hoga naya term sunkar

CHEMICAL EQUILIBRIUM

It occurs in reversible reactions where the forward and backward reactions proceed at equal rates. At this point, the concentrations of reactants and products remain constant, but the reactions themselves don't stop; they continue to be in a dynamic state of balance.





Easy toh hai kitna, kyu ghabra rahe ho?

Let's consider a general reaction

$$A_g + B_g \underset{\text{backward}}{\overset{\text{forward}}{\rightleftharpoons}} C_g + D_g$$

At equilibrium, $(Rate)_{forward} = (Rate)_{backward}$

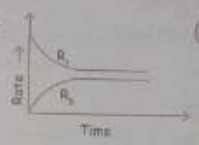
Khoobsurat Baat

Conditions for equilibrium

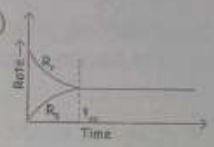
- Rate of forward reaction = rate of backward reaction.
- Concentrations of reactants and products are constant.

1. Which curves define Equilibrium situation ?

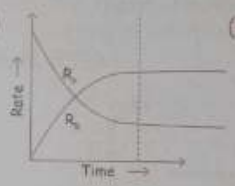
(1)



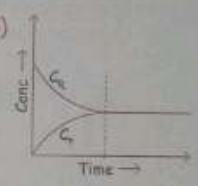
(2)



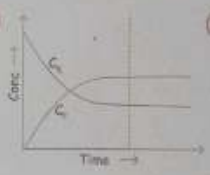
(3)



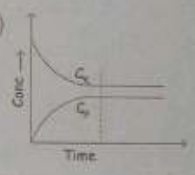
(4)



(5)



(6)



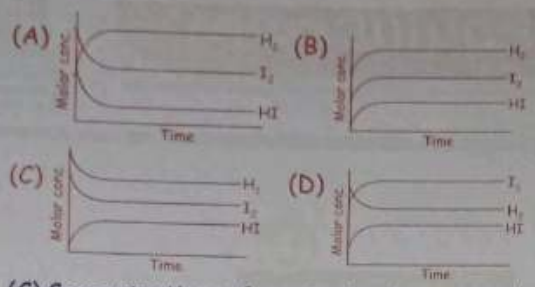
Sol. (2,4,5,6) At equilibrium, the rate of forward reaction becomes equal to the rate of rate of backward direction and concentration becomes constant. According to this, curves given in option 1 and 3 are incorrect.

graphs toh karne hi padenge

2. For the reaction,

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Attainment of equillibrium is predicted correctly by :



Sol. (C) Concentrations of reactants decrease and that of products increase until the equilibrium is reached. At equilibrium, concentrations of reactants and products remain constant.

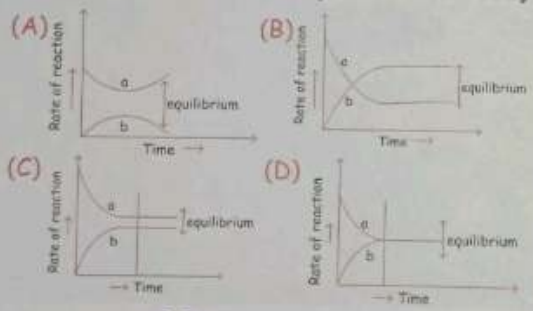
Graph dekh kar hi hoga, pehle wo seekho

Common Mistake

At equilibrium, concentrations of reactants and products become equal.

3. For the equilibrium $A \rightleftharpoons B$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by:

(JEE Mains 2020)



Khoobsurat Baat

Sol. (D) At equilibrium, the rate of forward direction becomes equal to the rate of backward direction.

baar baar aaenge graphs

Important points regarding equilibrium position

- Quantities of reactants and products become constant.
- Chemical equilibrium are dynamic in nature (Reaction not stopped)
- Chemical equilibrium are quasistatic in nature [concentration constant, rate same].
- Equilibrium is the most stable state of system.
- Any disturbance caused by external agent, system tends to achieve its equilibrium state again.

4. The number of correct statements involving equilibrium in physical process from the following is _____

(A) Equilibrium is possible only in closed system at a given temp.

(B) Both the opposing processes occur at same rate.

(C) Value of all the parameters become equal.

(D) For dissolution of solid, solubility is constant at given temp.

Sol. (A, B, D) All the statements except C are correct. At equilibrium, all the parameters become constant.

samajh aaya tabhi kar paoge, lallu

Common Mistake

At equilibrium, all parameters become equal.

LAW OF MASS ACTION

The law of mas action states that the rate of a chemical reaction is directly proportional to the product of the concentrations of the reactants, each raised to the power of its stoichiometric coefficient in the balanced chemical equation.

Let's consider a general reaction-

$$A_g + B_g \underset{k_b}{\overset{k_f}{\rightleftharpoons}} C_g + 2D_g$$

Law of mass action says-

$(\text{rate})_{\text{forward}} \propto [A]^1 [B]^1$

$(\text{rate})_{\text{forward}} = k_f[A][B]$

[k_f = rate constant of forward rx.]

Similarly

$(\text{rate})_{\text{backward}} \propto [C][D]^2$

$(\text{rate})_{\text{backward}} = k_b[C][D]^2$

[k_b = rate constant of backward rx.]

At equilibrium

$(rate)_{forward} = (rate)_{backward}$

$k_f[A][B] = k_b[C][D]^2$, $\frac{k_f}{k_b} = \frac{[C][D]^2}{[A][B]}$

This $\frac{k_f}{k_b}$ is a constant which is known as equilibrium constant (K_{eq}).

$$K_{eq} = \frac{k_f}{k_b} = \frac{[C][D]^2}{[A][B]}$$

Important

To calculate K_{eq}-

(i) $K_{eq} = \frac{k_f}{k_b} = \frac{\text{rate constant of forward reaction}}{\text{rate constant of backward reaction}}$

(ii) $K_{eq} = \frac{[C][D]^2}{[A][B]}$

where [A], [B], [C], [D] are equilibrium concentrations of the species.

Note

$\frac{[C][D]^2}{[A][B]} = Q$ (if these concentration are not equilibrium concentration)

where Q is reaction quotient.

At equilibrium the value of reaction quotient (Q) = K_{eq}

EQUILIBRIUM CONSTANT, K_C

$A_g + B_g \rightleftharpoons C_g + 2D_g$, $K_{eq} = \frac{[C][D]^2}{[A][B]} = K_C$

The equilibrium constant having only concentration terms of those species having variable concentration during reaction till equilibrium reaches is called K_C. While writing the expression of K_C, we do not write the concentration of solid phase and pure liquid phase as their concentration remains constant thought out the reaction.

Example: $A_{(s)} \rightleftharpoons B_{(s)} + 2C_{(g)} + D_{(g)}$, $K_C = [C]^2[D]$

Calculating Equilibrium Concentrations

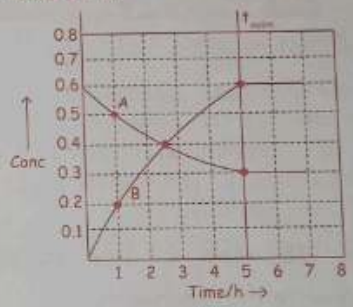
1. Start with the balanced equation for the reaction, indicating the reactants, products, and their stoichiometric coefficients.
2. Record the initial concentrations (C_0) of all species involved. If a species is absent initially, its concentration is 0.
3. Assign a variable (x) to represent the change in concentration for the species involved due to the reaction proceeding toward equilibrium.
4. Express the equilibrium concentration for each species in terms of the initial concentrations and the changes.
5. Substitute the equilibrium concentrations into the equilibrium constant expression.
6. Substitute the value of x into the equilibrium concentration expressions to find concentrations of all species.

yeh bhut jyada important hai

5. In a reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, 4 moles of N_2 and 8 moles of H_2 are taken in 1 Lt container. At eq^{bm}, $NH_3(g)$ has 2 moles. Make a concentration and time curve for all the species.

Sol. For explanation, refer to QR code given on the last page.

6. The progress of reaction
$A \rightleftharpoons n\,B$
with time is represented in Fig. Determine the value of 'n'.



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Sol. According to the plot,

$[A]_{eq} = 0.3$, $[B]_{eq} = 0.6$

1A	⇌	nB	
0.6		0	(0.6 - x = 0.3)
0.6 - x		nx	(x = 0.3)
||		||	
0.3		0.6	

$nx = 0.6$, $n = \frac{0.6}{0.3} = 2$

yeh badhiya sawaal hai, dhayan lgao

7. The equilibrium constant K_c at 298 K for the reaction $A + B \rightleftharpoons C + D$ is 100. Starting with an equimolar 1M, the equilibrium concentration of D is _____ $\times 10^{-2}$ M. (Nearest integer)

Sol. (182) For explanation, refer to QR code given on the last page.

8. 4.5 moles each of hydrogen and iodine is heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is .....

Sol.

	$H_2(g)$	+ $I_2(g)$	$\rightleftharpoons 2HI(g)$
t = 0	4.5	4.5	–
t_{eq}	4.5-x	4.5-x	2x

$2x = 3, x = 1.5$

$[H_2] = 4.5 - x = 4.5 - 1.5 = 3$

$[I_2] = 4.5 - 1.5 = 3$

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(3)^2}{3\times 3} = \frac{9}{9} = 1$$

are waah, kitna asaan hai

9. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

In an experiment, 2.0 mole of NOCl was placed in a non-litre flask and the concentration of NO after equilibrium established was found to be 0.4mol/L. The equilibrium constant at 30° C is __ $\times 10^{-4}$.

(JEE Main –27 June, 2022(shift–I))

Sol. First write the reaction as:

	$2NOCl(g)$	$\rightleftharpoons 2NO(g)$	+ $Cl_2(g)$
t = 0	2	–	–
$t = t_{eq}$	(2–x)M	xM	$\frac{x}{2}$

$\because x = 0.4$ M, $\therefore [NOCl]_{eq} = 1.6$ M

$[NO]_{eq} = 0.4$ M, $[Cl_2]_{eq} = 0.2$ M

$$\Rightarrow K_c = \frac{[NO]^2[Cl_2]}{[NOCl]^2} = \frac{[0.4]^2[0.2]}{[1.6]^2}$$

$K_c = 12.5 \times 10^{-3} = 125 \times 10^{-4}$

kitni asaan hai calculations

EQUILIBRIUM CONSTANT, K_p

The way of writing equilibrium constant having only Partial pressure term of species having variable partial Pressure (Gases) is known as K_p.

Khoobsurat Baat

Solids and pure Liquids (solvent) do not appear in K_c. But gas and solutions have variable concentrations due to which they can appear in the expression of K_c.

10. The equilibrium constant for the reaction

$A(s) \rightleftharpoons M(s) + \frac{1}{2} O_2(g)$

is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. (Round off to the nearest integer) (JEE Main–27 July, 2021)

Sol. According to the given reaction:

$K_p = P_{O_2}^{\frac{1}{2}} = 4, P_{O_2} = 4^2 = 16$ atm

kitna asaan hai, balle balle

Common Mistake

$K_p = P_M \times P O_2^{\frac{1}{2}}$

The above expression to calculate K_p is wrong as species present in solid state do not appear in K_p expression.

11. 4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is [Given R = 0.082 L atm K^{-1} mol^{-1}]

(JEE Main–29 June, 2022 (shift-II))

(A) 2.25 (B) 6.24 (C) 12.13 (D) 15.24

Sol. (A) For explanation, refer to QR code given on the last page.

12. At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_p for the reaction $Cl_2(g) \rightleftharpoons 2Cl(g)$ under the above conditions is $x \times 10^{-1}$. The value of x is ___ (Rounded off to the nearest integer)

(24 Feb, 2021 (Shift-I))

Sol. [5]

	$Cl_2(g)$	$\rightleftharpoons$	$2Cl(g)$
Initially $\rightarrow$	1 mol		–
At eq.	1 – x mol		2x mol

$1 - x = 2x;\ x = \frac{1}{3}$

Moles of Cl_2 at equilibrium $= \frac{2}{3}$

Moles of Cl at equilibrium $= \frac{2}{3}$

Total moles $= \frac{4}{3}$

Now: $P_{Cl_2} = \frac{2}{4} \times 1\ atm = \frac{1}{2} atm$

$P_{Cl} = \frac{2}{4} \times 1\ atm = \frac{1}{2} atm$

$$K_p = \frac{(P_{Cl})^2}{P_{Cl_2}} = \frac{\left(\frac{1}{2}\right)^2}{\frac{1}{2}} = 5 \times 10^{-1}$$

13. A homogeneous ideal gaseous reaction $AB_{2(g)} \rightleftharpoons A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27° C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is ___ (Integer answer)

[R = 0.08206 $dm^3\ atm\ K^{-1}\ mol^{-1}$]

(26 Feb, 2021 (Shift-I))

Sol. (73) For explanation, refer to QR code given on the last page.

14. For a reaction, $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$ in a constant volume container, no products were present initially. The final pressure of the system when 50% of reaction gets completed is [24 Jan, Shift-I, 2025]

(A) 7/4 times of initial pressure
(B) 5/2 times of initial pressure
(C) 5 times of initial pressure
(D) 7/2 times of initial pressure

Sol. (A) $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$

Let the initial pressure of N_2O_5 be P_0.

Initial moles: Only N_2O_5 is present $\rightarrow n_0 = P_0$ (as per ideal gas law).

Let x be the fraction of reaction completed. For 1 mole of N_2O_5 decomposing:

2 moles of NO_2 are formed.

0.5 moles of O_2 are formed.

Given that 50% of the reaction is completed (x = 0.5):

N_2O_5 decomposed: $0.5\ P_0$

NO_2 formed: $2 \times 0.5P_0 = P_0$

O_2 formed: $0.5 \times 0.5P_0 = 0.25P_0$

Remaining N_2O_5: $P_0 - 0.5P_0 = 0.5P_0$

$P_{final} = P_{N_2O_5} + P_{NO_2} + P_{O_2} = \frac{7}{4}P_0$

sar mat pakdo, sawaal par dhyan do

Relationship between K_p and K_c

Lets consider a general reaction:

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + 2D_{(g)}$

$$K_c = \frac{[C][D]^2}{[A][B]} = \frac{\left(\frac{P_C}{RT}\right)\left(\frac{P_D}{RT}\right)^2}{\left(\frac{P_A}{RT}\right)\left(\frac{P_B}{RT}\right)} = \frac{P_C \times P_D^2}{P_A \times P_B}\left(\frac{1}{RT}\right)$$

$$K_c = \frac{P_C \cdot P_D^2}{P_A \cdot P_B}\frac{1}{RT} \qquad \boxed{K_C(RT) = \frac{P_C \cdot P_D^2}{P_A \cdot P_B} = K_p}$$

Now, take another general reaction:

$2A_{(g)} + 2B_{(g)} \rightleftharpoons 3C_{(g)} + 3D_{(g)}$

Q. For the above reaction, find the relation between K_p and K_C for this reaction.

Sol. First write the expression of K_C of the given reaction:

$$K_c = \frac{[C]^3[D]^3}{[A]^2[B]^2} = \frac{\left(\frac{P_C}{R_T}\right)^3\left(\frac{P_D}{R_T}\right)^3}{\left(\frac{P_A}{R_T}\right)^2\left(\frac{P_B}{R_T}\right)^2} = \frac{P_C^3 \times P_D^3}{P_A^2 \times P_B^2} \times \left(\frac{1}{RT}\right)^2$$

$K_c(RT)^2 = \frac{P_C^3 \cdot P_D^3}{P_A^2 \cdot P_B^2} = K_p$, $K_p = K_c(RT)^2$, $K_p = K_c(RT)^{\Delta n_g}$

Pehle equilibrium ke liye $\Delta n_g = 1$ aaya tha toh $(RT)^1$ aaya

ab $\Delta n_g = 2$ aaya, toh $(RT)^2$ aaya, iska matlab hum K_p, K_c ka relation general terms mei aise likh sakte hain:

Δn_g = Gaseous moles of product – gaseous moles of reactants

Note

$K_p = K_c(RT)^{\Delta n_g}$

If $\Delta_{n_g} = 0 \Rightarrow K_p = K_c$, if $RT = 1 \Rightarrow K_p = K_c$

Khoobsurat Baat

For the defined Eq_{bm} constant (K_p)

- There should not be any solution phase.
- There should be atleast one gaseous phase.

isko galti se bhi mat chhod dena, not even by mistake

15. $\log\frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:

(A) $PCl_5 = PCl_3 + Cl_2$ (B) $2SO_2 + O_2 = 2SO_3$
(C) $N_2 + 3H_2 = 2NH_3$ (D) (B) and (C) both

Sol. (B) The relation between K_p and K_c is:

$\log K_p = K_c(RT)^{\Delta n_g}$

Take log on both sides of equation:

$\log K_p = \log K_C = \Delta n_g \log RT$

$\log\frac{K_p}{K_c} = \Delta n_g \log RT$

So, $\Delta n_g = -1$

For reaction given in option (B),

$\Delta n_g = 2 - 3 = -1$

log ko dekh kar paglao mat, asaan hi hai

16. For the reaction $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$, if $K_p = K_C(RT)^x$ where the symbols have usual meaning then the value of x is: (assuming ideality) **(JEE Mains-2014)**

(A) $\frac{1}{2}$ (B) 1 (C) –1 (D) $-\frac{1}{2}$

Sol. Calculate the value of Δn_g as:

Δn_g = Moles of gaseous products–moles of gaseous reactants.

$= 1 - \left(1 + \frac{1}{2}\right) = 1 - \frac{3}{2} = -\frac{1}{2}$

17. In which of the following equilibrium, K_c and K_p are not equal?

(A) $2NO(g) = N_2(g) + O_2(g)$
(B) $SO_2(g) + NO_2(g) = SO_3(g) + NO(g)$
(C) $H_2(g) + I_2(g) = 2HI(g)$
(D) $2C(s) + O_2(g) = 2CO_2(g)$

Sol. (D) For explanation, refer to QR code given on the last page.

18. For which of these following equilibria $K_p = K_c RT^{-1}$?

(A) $2A_{(s)} + 2B_{(g)} \rightleftharpoons C_{(g)} + D_{(s)}$
(B) $A_{(sol)} + B_{(g)} + C_{(g)} \rightleftharpoons D_{(Sol)} + E_{(g)}$
(C) $A_{(g)} + 2B_{(g)} \rightleftharpoons 4C_{(g)}$
(D) $A_{(s)} + B_{(g)} \rightleftharpoons C_{(s)} + D_{(g)}$

Sol. The relation between K_p and K_c is:

$K_p = K_c(RT)^{\Delta n_g}$

According to the expression given in question, the value of Δn_g should be equal to –1.

Among the given reactions, in (A) and (B), value of Δn_g will be –1 but K_p is not defined for (B) as there is a solution phase involved and we cannot write the concentrations of solution in terms of pressure. So, according to this, option A is correct.

is sawaal ko halke mei mat lo, samjhe

Common Mistake

Students might consider that option B is also correct but it should be kept in mind that K_p cannot be defined for those reactions involving solution phase.

19. For reaction: $SO_2(g) + \frac{1}{2}O_2(g) = SO_3(g)$

$K_p = 2 \times 10^{12}$ at 27°C and 1 atm pressure. The K_C for the same reaction is _____ $\times 10^{11}$ (Nearest integer)

(Given R = 0.082 L atm K^{-1} mol^{-1})

(JEE Mains-31 Jan, 2023(Shift-I))

Sol. The relation between K_p and K_c is:

$K_p = K_c \times (RT)^{\Delta n_g}$

Here, $\Delta n_g = 1 - \frac{3}{2} = -\frac{1}{2}$

Now, substitute the values in the above expression:

$2 \times 10^{12} = K_c \times (0.082 \times 300)^{\frac{-1}{2}}$

$K_c = 9.92 \times 10^{12} = 0.992 \times 10^{13} = 1$

20. For the equilibrium $SO_2Cl_2(g) = SO_2(g) + Cl_2(g)$, what is the temperature at which $\frac{K_p(atm)}{K_c(M)} = 3$?

(A) 0.027 K (B) 0.36 K

(C) 36.54 K (D) 273 K

Sol. (C) For explanation, refer to QR code given on the last page.

21. In a study of the gaseous reaction,

$A(g) + 2B(g) = 2C(g) + D(g)$,

A and B are mixed in a reaction vessel kept at 25°C. The initial concentration of B is 1.5 times the initial concentration of A. After the equilibrium has been established, the equilibrium concentrations of A and D were equal. Calculate the equilibrium constant at 25°C.

Sol. (0.32) For explanation, refer to QR code given on the last page.

Units of K_c

Unit of $K_c = (conc)^{\Delta n_g} = \left(\frac{mol}{L}\right)^{\Delta n_g}$

$1A_{(g)} + 2B_{(g)} = C_{(g)}$

$K_c = \frac{[C]}{[A][B]^2} = \frac{\left(\frac{mol}{L}\right)}{\left(\frac{mol}{L}\right)\left(\frac{mol}{L}\right)^2} = \left(\frac{mol}{L}\right)^{-2}$

$\Delta n_g = 1 - 3 = -2$

Unit of $K_c = \left(\frac{mol}{L}\right)^{\Delta n_g} = \left(\frac{mol}{L}\right)^{-2}$

Δn_g bahut jada kamaal ka hai, isko ignore nahi kr sakte

22. In which equilibrium, the unit of $K_c = \left(\frac{mol}{L}\right)^{-1}$?

(A) $2A_{(g)} + B_{(g)} = C_{(g)} + D_{(g)} + E_{(s)}$

(B) $A_{(g)} + B_{(g)} = 3C_{(g)} + D_{(g)}$

(C) $A_{(s)} + 2B_{(g)} = C_{(s)} + D_{(g)}$

(D) $A_{(sol)} + B_{(sol)} = C_{(sol)}$

Sol. (C,D) For explanation, refer to QR code given on the last page.

23. Predict the unit of K_c?

(A) $A_{(g)} + B_{(g)} = C_{(g)}$

(B) $PCl_{5(g)} = PCl_{3(g)} + Cl_{2(g)}$

(C) $N_{2(g)} + 3H_{2(g)} = 2NH_{3(g)}$

(D) $A_{(g)} + B_{(sol)} = C_{(sol)} + 2D_{(g)}$

Sol. (A) $\left(\frac{mol}{L}\right)^{-1}$ (B) $\left(\frac{mol}{L}\right)^{1}$ (C) $\left(\frac{mol}{L}\right)^{-2}$ (D) $\left(\frac{mol}{L}\right)^{1}$

Units of K_p

K_p = expression of equilibrium constant where only partial pressure terms exist.

Unit of $K_p = (atm)^{\Delta n_g}$

24. For which of these equilibria, unit of K_p is $(bar)^{-1}$?

(A) $PCl_{5(g)} = PCl_{3(g)} + Cl_{2(g)}$

(B) $NH_2COONH_{4(s)} = 2NH_{3(g)} + CO_{2(g)}$

(C) $3A_{(g)} = B_{(g)} + C_{(g)}$

(D) $C_{(soln)} + 2D_{(g)} = E_{(soln)} + 1F_{(g)}$

Sol. (C) Here, –1 value indicates the values of Δn_g.

Among the given options, reaction given in option C has –1 value of Δn_g.

arey isme bhi Δn_g, yeh toh sach mei hi kamaal hai

Khoobsurat Baat

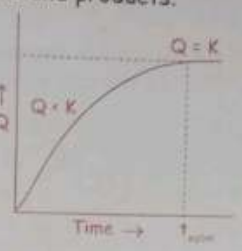
The value of Δn_g gives the power of the unit of K_p that is atm.

REACTION QUOTIENT

The reaction quotient (Q) is a measure of the relative amounts of reactants and products in a chemical reaction at a given time. It is calculated using the concentrations of the reactants and products.

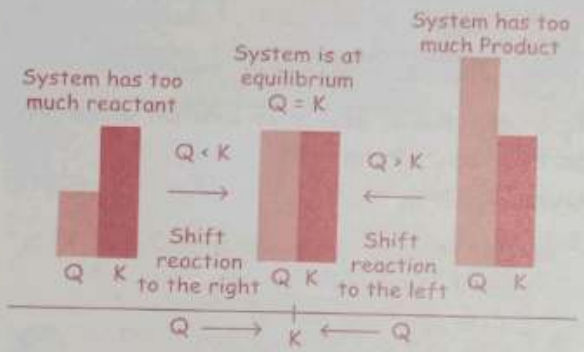
	A	+	B	⇌	C	+	D
t = 0	a		b		0		0

$$\text{Expression} = \frac{[C][D]}{[A][B]} = Q$$



Important

if reaction $Q < K_{eq}$ Forward movement
if reaction $Q = K_{eq}$ Equilibrium reached
if reaction $Q > K_{eq}$ Backward movement



yeh K_c nahi hai, pehle dekh toh lo

Chalo ab padhte hai Types of equilibrium ko

1. HOMOGENEOUS EQUILIBRIUM

- All the species involved are in same phase.

Case-1:

Only reactants are present (vol = 1 lt)

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$; $K_c = 4$

2 mol	2 mol	0	0
2-x	2-x	x	x

$$K_c = \frac{[C][D]}{[A][B]} = \frac{\left(\frac{x}{1}\right)\left(\frac{x}{1}\right)}{\left(\frac{2-x}{1}\right)\left(\frac{2-x}{1}\right)} = 4 \quad \left(Q = \frac{[C][D]}{[A][B]} = 0, Q < K\right)$$

$$\frac{x}{2-x} = 2,\ x = 4 - 2x,\ x = \frac{4}{3}$$

Case-2: Both reactant and product are present initially (vol = 1 lt)

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$; $K_c = 4$

t=0	2 mol	1 mol	1 mole	1 mole
t=t eqbn	2-x	1-x	1+x	1+x

$$K_c = 4 = \frac{\left(\frac{1+x}{1}\right)\left(\frac{1+x}{1}\right)}{\left(\frac{2-x}{1}\right)\left(\frac{1-x}{1}\right)}$$

x = ?

$$Q = \frac{[C][D]}{[A][B]} = \frac{\left(\frac{1}{1}\right)\left(\frac{1}{1}\right)}{\left(\frac{2}{1}\right)\left(\frac{1}{1}\right)}$$

Q < K

Case-3: Both reactant and product present initially

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$; $K_c = 4$

t=0	1 mol	1 mol	4 mole	3 mole
t=t eqbn	1+x	1+x	4-x	3-x

$$K_c = \frac{[C][D]}{[A][B]} = \frac{\left(\frac{4-x}{1}\right)\left(\frac{3-x}{1}\right)}{\left(\frac{1+x}{1}\right)\left(\frac{1+x}{1}\right)} = 4$$

x = ?

$$Q = \frac{[C][D]}{[A][B]} = \frac{\left(\frac{4}{1}\right)\left(\frac{3}{1}\right)}{\left(\frac{1}{1}\right)\left(\frac{1}{1}\right)} = 12 > K$$

Case-4: Both reactant and product are present initially.

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$; $K_c = 4$

t=0	2 mol	2 mol	4 mol	4 mol
t=t eqbn	2 mol	2 mol	4 mol	4 mol

$$Q = \frac{[C][D]}{[A][B]} = \frac{\left(\frac{4}{1}\right)\left(\frac{4}{1}\right)}{\left(\frac{2}{1}\right)\left(\frac{2}{1}\right)} = 4$$

Q = K

Hai na mazedaar

25. For the reaction $NOBr(g) \rightleftharpoons NO(g) + \frac{1}{2}Br_2(g)$, $K_p = 0.15\,atm^{1/2}$ at 90°C. If NOBr, NO and Br_2 are mixed at this temperature having partial pressure 0.5 atm, 0.4 atm and 2.0 respectively, will Br_2 be consumed or formed?

Sol. $$Q = \frac{P_{NO} \cdot P_{Br_2}^{\frac{1}{2}}}{P_{NOBr}} = \frac{0.4(2)^{\frac{1}{2}}}{0.5} = \frac{0.4 \times 1.4}{0.5} > 0.15$$

Q > K

Backward movement, Br_2 will be consumed.

Common Mistake

redict the wrong direction that is in forward novement.

se-5: Equilibrium is already established and we disturbing the existing equilibrium.

26. Already established equilibrium, equilibrium moles are:

$CO_{(g)} + H_2O_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)}$

0.4 0.3 0.6 0.2

How much CO_2 must be added at constant temperature to this existing equilibrium so that at new equilibrium moles of CO would be 0.6?

[JEE Adv.]

Sol.

	$CO_{(g)}$	$H_2O_{(g)}$	$H_{2(g)}$	$CO_{2(g)}$
$(t = t^{eq})_{old}$	0.4	0.3	0.6	0.2
$t = t_{disturb}$	0.4	0.3	0.6	0.2 + x
$(t = t^{eq})_{new}$	0.4 + y	0.3+y	0.6 - y	0.2 + x - y

$$K_c = \frac{\left(\frac{0.6}{V}\right)\left(\frac{0.2}{V}\right)}{\left(\frac{0.4}{V}\right)\left(\frac{0.3}{V}\right)} = 1$$

Moles of CO at new equilibrium are 0.4 + y which is given as 0.6.

so, y = 0.2

$$K_c = \frac{\left(\frac{x}{V}\right)\left(\frac{0.4}{V}\right)}{\left(\frac{0.6}{V}\right)\left(\frac{0.5}{V}\right)} = 1,\ x = \frac{3}{4}\ \text{mol}$$

27.

	$PCl_{5(g)}$	$\rightleftharpoons$	$PCl_{3(g)}$	+	$Cl_{2(g)}$
$(t = t^{eq})_{old}$	2 mole		2 mole		2 mole

How much chlorine should be added so that at constant temperature and pressure, volume is doubled?

Khoobsurat Baat

Sol.

	$PCl_{5(g)}$	$\rightleftharpoons$	$PCl_{3(g)}$	+	$Cl_{2(g)}$
$(t = t^{eq})_{old}$	2		2		2
$(t = t)_{disturb}$	2		2		2 + x
$(t = t^{eq})_{new}$	2 + y		2 - y		2 + x - y

$2 + y + 2 - y + 2 + x - y = 12$ ...(1)

$x - y = 6$

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{2}{V}\right)\left(\frac{2}{V}\right)}{\left(\frac{2}{V}\right)} = \left(\frac{2}{V}\right)$$

$$\left(\frac{2}{V}\right) = \frac{\left(\frac{2+x-y}{2V}\right)\left(\frac{2-y}{2V}\right)}{\frac{2+y}{2V}} \quad \text{...(2)}$$

From equations (1) and (2):

$x = \frac{20}{3}$ mol

yeh sawaal dimag ko ghuma kar rakh dega

28.

	$A_{2(g)}$	+	$B_{2(g)}$	$\rightleftharpoons$	$2AB_{(g)}$
$(t = t^{eq})_{old}$	2 mol		2 mol		2 mol

If some how 1 mole of AB is taken out then find out the new composition of the eq^{bm} mixture at constant temperature. Volume of container = V lt.

Sol. ($[A_2] = 1.67$ M, $[B_2] = 1.67$ M, $[AB] = 1.67$ M)

For explanation, refer to QR code given on the last page.

Case-6: Complex formation

$Ag^+_{(aq)} + 2NH_{3(aq)} \rightleftharpoons [Ag(NH_3)_2]^+_{(aq)}$; K_f = High

t=0	C_1M	C_2M	0
$t=t^{eqbm}$)	(C_1-x)	(C_2-x)	x

$$K_f = \frac{[Ag(NH_3)_2]^+}{[Ag^+][NH_3]^2} = \frac{x}{(C_1 - x)(C_2 - 2x)^2}$$

x = ?

$Fe^{+3}_{(aq)} + 6CN^-_{(aq)} \rightleftharpoons [Fe(CN)_6]^{3-}_{(aq)}$; $K_f \uparrow\uparrow\uparrow$

t = 0	C_1M	C_2M	0
$t = t^{eqbm}$)	C_1-x	C_2-6x	x

$$K_f = \frac{[Fe(CN)_6]^{3-}}{[Fe^{+3}][CN^-]^6} = \frac{x}{(C_1 - x)(C_2 - 6x)^6}$$

x = ?

29. For the reaction $Ag(CN)_2^- \rightleftharpoons Ag^+ + 2CN^-$ the K_c at 25° is 4×10^{-19}. Calculate $[Ag^+]$ in solution which was originally 0.1M in KCN and 0.03M in $AgNO_3$.

Sol. $Ag^+ = 0.034M$, $CN^- = 0.1M$

$$Ag^+_{(aq)} + 2CN^-_{(aq)} \rightleftharpoons [Ag(CN)_2]^-_{(aq)}$$

0.03	0.1	0
0.03 − x	0.1 − 2x	x

$$K_f = \frac{1}{K_{diss.}} = \frac{10^{19}}{4}$$

$$\frac{[Ag(CN)_2]^-}{[Ag^+][CN^-]^2} = \frac{1}{4\times10^{-19}} = \frac{10^{19}}{4}$$

$$\frac{x}{(0.03-x)(0.1-2x)^2} = \frac{10^{19}}{4}$$

(ek toh yeh cubic equation hai which is impossible to solve, isliye beimani karni padegi)
(K ki value bhut badi hai isliye Ag^+ almost khatam ho jaega lekin humein iski concentration bh nikalni hai) toh hum yeh beyimani karenge ki 0.03 - x ko y rakhenge jo 0 jaise hai lekin 0 nahi hai.

$$\frac{0.03}{y(0.1-2\times0.03)} = \frac{10^{19}}{4} \Rightarrow \frac{0.03}{y(0.04)^2} = \frac{10^{19}}{4}$$

$$y = \frac{0.03\times4}{(0.04)^2\times10^{19}} = 7.5\times10^{-18}$$

$$[Ag^+]_{final} = 7.5\times10^{-18}\ M$$

Samjhe ki naahi kuch, ya pagla gaye

30. The number of moles of NH_3, that must be added to 2L of 0.80M $AgNO_3$ in order to reduce the concentration of Ag^+ ions to 5.0×10^{-8}M ($K_{formation}$ for $[Ag(NH_3)_2]^+ = 1.0\times10^8$) is _____ (Nearest integer)

[Assume no volume change on adding NH_3]

[27 Aug, 2021 (Shift - 1)]

Sol. (4 mol) For explanation, refer to QR code given on the last page.

FACTORS AFFECTING EQUILIBRIUM CONSTANT

(i) Stoichiometry of the Reaction

$$A_{(g)} \rightleftharpoons B_{(g)} ; K = \frac{[B]}{[A]}$$

$$2A_{(g)} \rightleftharpoons 2B_{(g)} ; K' = \frac{[B]^2}{[C]^2} = \left[\frac{[B]}{[C]}\right]^2 = K^2$$

$$K' = K^2$$

$$K_{new} = (K_{old})^{\text{multiplying factor}}$$

(ii) Mode of Writing the Chemical Equilibrium

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)} ; K = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)} ; K' = \frac{[PCl_5]}{[PCl_3][Cl_2]} = \frac{1}{K}$$

If we change the mode of writing the chemical Eq^{bn} then the K_{new} is equal to the reciprocal of K_{old}.

31. The value of equilibrium constant of the reaction $HI(g) = \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g)$ is 8.0
The equilibrium constant of the reaction
$H_2(g) + I_2(g) = 2HI(g)$ will be

(A) $\frac{1}{16}$ (B) $\frac{1}{64}$ (C) 16 (D) $\frac{1}{8}$

Sol. (B) For explanation, refer to QR code given on the last page.

32. K_1, K_2 and K_3 are the equilibrium of the following reactions (I), (II) and (III), respectively. **(JEE–Mains (online)-12)**

(I) $N_2 + 2O_2 = 2NO_2$ (II) $2NO_2 = N_2 + 2O_2$

(III) $NO_2 = \frac{1}{2}N_2 + O_2$

The correct relation from the following is:

(A) $K_1 = \sqrt{K_2} = K_3$ (B) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$

(C) $K_1 = \frac{1}{K_2} = K_3$ (D) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

Sol. ③ × 2 = ②; $K_2 = K_3^2$

① ◯ ②: $K_1 = \frac{1}{K_2}$

↓ mode change

$$K_1 = \frac{1}{K_2} = \frac{1}{K_3^2}$$

mushkil nahi hai bacho, lag rha hai bas

33. At 298 K, the equilibrium constant is 2×10^{15} for the reaction:

$$Cu_{(s)} + 2Ag^{+}_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + 2Ag_{(s)}$$

The equilibrium constant for the reaction

$$\frac{1}{2}Cu^{+2}_{(aq)} + Ag_{(s)} \rightleftharpoons \frac{1}{2}Cu_{(s)} + Ag^{+}_{(aq)} \text{ is } x \times 10^{-8}$$

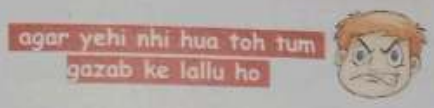
the value of 'x' is _____ (nearest integer)

(July 2022)

Sol. First we will reverse the reaction, then divide the whole reaction by 2:

$$K_{new} = \left(\frac{1}{K}\right)^{\frac{1}{2}} = \left(\frac{1}{2 \times 10^{15}}\right)^{\frac{1}{2}} = 2.2 \times 10^{-8}$$

So, $x = 2.2$



Common Mistake

Students might calculate the wrong value of K_C that is:

$$K'_C = \frac{K_C}{2} = \frac{2 \times 10^{15}}{2} = 1 \times 10^{15}$$

34. At 298 K [29 Jan, 2023(Shift-II)]

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), K_1 = 4 \times 10^5$$

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g), K_2 = 1.6 \times 10^{12}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g), K_3 = 1.0 \times 10^{-13}$$

Based on above equilibria, the equilibrium constant of the reaction,

$$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$$

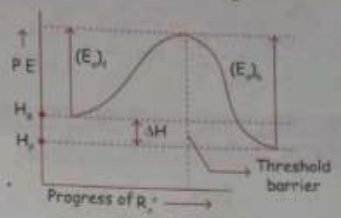
is _____ $\times 10^{-33}$ (Nearest integer)

Sol. (4) For explanation, refer to QR code given on the last page.

(iii) Temperature

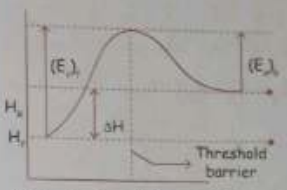
Exothermic

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B ; K = \frac{k_f}{k_b}$$



$\Delta H = (E_a)_f - (E_a)_b = (-)Ve$

Endothermic



$\Delta H = (E_a)_f - (E_a)_b = (+)Ve$

graphs ko samjho, yaad karne ki zarurat nahi hai

Temperature dependence of rate constant can be represented as:

$$k = A e^{\frac{-E_a}{RT}}$$

Effective collisions ← | → Fraction of molecules having energy more than or equal to threshold energy

$$k_F = A_F \cdot e^{\frac{-(E_a)_f}{RT}}, \quad k_b = A_b \cdot e^{\frac{-(E_a)_b}{RT}}$$

$$K_T = \frac{k_f}{k_b} = \frac{A_f \cdot e^{\frac{-(E_a)_f}{RT}}}{A_b e^{\frac{-(E_a)_b}{RT}}} = \frac{A_f}{A_b} \cdot e^{-\frac{[(E_a)_f - (E_a)_b]}{RT}}$$

$$K_T = \frac{A_f}{A_b} \cdot e^{\frac{-\Delta H}{RT}}$$

A_f = Frequency factor for forward rx^n

A_b = Frequency factor for backward rx^n

$(E_a)_f$ = Activation energy for forward rx^n

$(E_a)_b$ = Activation energy for backward rx^n

$T_1 \rightarrow K_{T_1}, \quad T_2 \rightarrow K_{T_2}$

$$K_{T_1} = \frac{A_f}{A_b} \cdot e^{\frac{-\Delta H}{RT_1}}, \quad K_{T_2} = \frac{A_f}{A_b} \cdot e^{\frac{-\Delta H}{RT_2}}$$

$$\frac{K_{T_2}}{K_{T_1}} = \frac{\left(\frac{A_f}{A_b}\right) \cdot e^{-\frac{\Delta H}{RT_2}}}{\left(\frac{A_f}{A_b}\right) \cdot e^{-\frac{\Delta H}{RT_1}}}, \quad \frac{K_{T_2}}{K_{T_1}} = e^{-\frac{\Delta H}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]}$$

$$\ln \frac{K_{T_2}}{K_{T_1}} = \frac{\Delta H}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right], \quad \log \frac{K_{T_2}}{K_{T_1}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

equation yaad karlo, kaam ban jaega lekin sari terms ka matlab pata hona chaiye

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$$\log \frac{K_{T_2}}{K_{T_1}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Case-1 $\Delta H = (+)$ ve $T_2 > T_1$ $K_{T_2} > K_{T_1}$
Case-2 $\Delta H = (+)$ ve $T_2 < T_1$ $K_{T_2} < K_{T_1}$
Case-3 $\Delta H = (-)$ ve $T_2 > T_1$ $K_{T_2} < K_{T_1}$
Case-4 $\Delta H = (-)$ ve $T_2 < T_1$ $K_{T_2} > K_{T_1}$

Equilibrium constant (K) depends on temperature (T) exponentially.

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Agar बढ़ेगा To Bahot बढ़ेगा!
Agar घटेगा To Bahot घटेगा!

DEGREE OF DISSOCIATION

The degree of dissociation (α) is a measure of the extent to which a substance dissociates into its constituent ions or molecules in a given medium. It represents the fraction of the initial amount of a substance that has dissociated.

	$PCl_{5(g)}$	$\rightleftharpoons PCl_{3(g)}$	$+ Cl_{2(g)}$; K_c
$t = 0$	a	0	0
$t = t^{eqbm}$	a − x	x	x

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{x}{V}\right)\left(\frac{x}{V}\right)}{\left(\frac{a-x}{V}\right)}, \quad x = ?$$

$$\alpha = \frac{\text{moles diss at eq}^n}{\text{initial moles}} < 1 \Rightarrow \alpha_{PCl_5} = \frac{x}{a} \frac{(\text{dissociated moles})}{(\text{initial moles})}$$

Formula to dekho pehle ki kya likha hai

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degree of dissociation (α) का मतलब है 1 mole में से कितना टूटा!

	$PCl_{5(g)}$	$\rightleftharpoons 1PCl_{3(g)}$	$+ 1Cl_{2(g)}$; K_c
$t = 0$	a	0	0
$t = t^{eq}$	$a(1-\alpha)$	$a\alpha$	$a\alpha$

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{a\alpha}{V}\right)\left(\frac{a\alpha}{V}\right)}{\frac{a(1-\alpha)}{V}}, \quad \alpha = ?$$

RELATION BETWEEN α, K_p AND P_T

Lets consider a reaction:

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, partial pressure = $P_{total} \times$

Method-1:

	$PCl_{5(g)}$	$\rightleftharpoons PCl_{3(g)}$	$+ Cl_{2(g)}$
$t = 0$	a	0	0
$t = t\ eq^{bm}$	$a(1-\alpha)$	$a\alpha$	$a\alpha$

$$(P_{total})_{eq} = a(1-\alpha) + a\alpha + a\alpha = a(1+\alpha)$$

$$K_p = \frac{P_{PCl_3} P_{Cl_2}}{P_{PCl_5}} = \frac{\left[\frac{a\alpha}{a(1+\alpha)} \cdot P_T\right]\left[\frac{a\alpha}{a(1+\alpha)} \cdot P_T\right]}{\left[\frac{a(1-\alpha)}{a(1+\alpha)} \cdot P_T\right]}$$

$$K_p = \frac{\alpha^2}{1-\alpha^2} \cdot P_T$$

Method-2:

K_c Nikal kar K_p Nikalenge, $K_p = K_c (RT)^{\Delta n_g} = K_c (RT)$

$$K_c = \frac{[PCl_3][Cl_2]}{PCl_5} = \frac{\left(\frac{a\alpha}{V}\right)\left(\frac{a\alpha}{V}\right)}{\frac{a(1-\alpha)}{V}} = \frac{\alpha^2}{1-\alpha}\left(\frac{a}{V}\right)$$

$$K_p = \frac{\alpha^2}{1-\alpha}\left(\frac{a}{V}\right)RT \quad (\Delta n_g = 1)$$

dono methods par dhyan lagao or prabhu ko yaad karo

Note

$P_T V = a(1+\alpha)RT$

$\frac{aRT}{V} = \frac{P_T}{1+\alpha}$, $K_p = \frac{\alpha^2}{(1-\alpha)}\frac{P_T}{(1+\alpha)} = \frac{\alpha^2}{1-\alpha^2}P_T$

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$1A_{(g)} \rightleftharpoons 1B_{(g)} + 1C_{(g)}$, $K_p = \left(\frac{\alpha^2}{1-\alpha^2}\right)P_T$

35. At some temperature and under a pressure of 4 atm, PCl_5 is 10% dissociated. Calculate the pressure at which PCl_5 will be 20% dissociated, temperature remaining same.

Sol. First write the reaction as: $PCl_5 \Leftrightarrow PCl_3 + Cl_2$

Mole before dissociation	1	0	0
Mole after dissociation	$(1-\alpha)$	α	α

$$K_p = \frac{n_{PCl_3} \times n_{Cl_2}}{n_{PCl_5}} \times \left[\frac{P}{\sum n}\right]^{\Delta n} \quad ...(1)$$

At P = 4 atm and $\alpha = 0.1$

$$\therefore K_p = \frac{0.1 \times 0.1}{0.9} \times \left[\frac{4}{1.1}\right]^1 = 0.0404 \text{ atm}$$

At $\alpha = 0.2$,

$$0.0404 = \frac{0.2 \times 0.2}{0.8} \times \left[\frac{P}{1.2}\right]^1$$

$$\therefore P = 0.969 \text{ atm}$$

ghabrane ki baat nahi hai, page ke kone mei karke dekhlo

36. At temperature T, a compound $AB_2(g)$ dissociates according to the reaction

$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$

with degree of dissociation α, which is small as compared to unity. The expression for K_p, in terms of α and the total pressure, P_T is

(A) $\frac{P_T\alpha^3}{2}$ (B) $\frac{P_T\alpha^2}{3}$ (C) $\frac{P_T\alpha^3}{3}$ (D) $\frac{P_T\alpha^2}{2}$

Sol. (A) For explanation, refer to QR code given on the last page.

Common Mistake

Assuming the equilibrium concentration of B_2 be x instead of $\frac{x}{2}$.

37. Consider the reaction

[23 Jan, Shift-II, 2025]

$X_2Y(g) = X_2(g) + \frac{1}{2}Y_2(g)$

The equation representing correct relationship between the degree of dissociation (x) of $X_2Y(g)$ with its equilibrium constant K_p is _____.

Assume x to be very very small.

(A) $x = 3\sqrt{\frac{2Kp^2}{p}}$ (B) $x = 3\sqrt{\frac{Kp}{2p}}$

(C) $x = 3\sqrt{\frac{Kp}{p}}$ (D) $x = 3\sqrt{\frac{2Kp}{p}}$

Sol. (A)

	$X_2Y(g) \rightleftharpoons$	$X_2(g)$	+	$\frac{1}{2}Y_2(g)$
Initially	1			
At eq.	$(1-x)$	x		$\frac{x}{2}$

$$P_{X_2Y} = x_{X_2Y}P = \frac{1-x}{\left(1+\frac{x}{2}\right)} \times P$$

$$P_{X_2} = x_{X_2}P = \frac{x}{\left(1+\frac{x}{2}\right)} \times P, P_{Y_2} = x_{Y_2}P = \frac{x/2}{\left(1+\frac{x}{2}\right)} \times P$$

$$\text{So, } K_p = \frac{P_{X_2} \times (P_{Y_2})^{\frac{1}{2}}}{P_{X_2Y}}$$

$$K_p = \frac{\left(\frac{x}{1+\frac{x}{2}}\right)P \times \left(\left(\frac{x/2}{1+x/2}\right)P\right)^{1/2}}{\frac{1-x}{\left(1+\frac{x}{2}\right)} \times P}$$

$$K_p = \left(\frac{x}{1-x}\right)\left[\left(\frac{x}{2\left(1+\frac{x}{2}\right)}\right)P\right]^{\frac{1}{2}}$$

x is very small, so neglect x in comparison to 1 & 2.

$$K_p = \frac{(x)^{\frac{3}{2}}(P)^{\frac{1}{2}}}{(2)^{\frac{1}{2}}},\ (x)^{\frac{3}{2}} = \frac{(2)^{\frac{1}{2}}K_p}{(P)^{\frac{1}{2}}},\ x = \sqrt[3]{\frac{2K_p^2}{P}}$$

3. For a solution of a weak electrolyte (K_{eq} = equilibrium constant) A_2B_3 of concentration 'c', the degree of dissociation 'α' is

(06 April, 2023 (Shift-I))

(A) $\left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$ (B) $\left(\frac{K_{eq}}{6c^5}\right)^{\frac{1}{5}}$

(C) $\left(\frac{K_{eq}}{5c^4}\right)^{\frac{1}{5}}$ (D) $\left(\frac{K_{eq}}{25c^2}\right)^{\frac{1}{5}}$

ol. (A) For explanation, refer to QR code given on the last page.

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hen $1-\alpha < 0.05$, then $1-\alpha$ can be approximated to 1.

9. $A_{(g)} \rightleftharpoons B_{(g)} + \frac{C(g)}{2}$. The correct relationship between P, K_p, α and equilibrium pressure P is

(31 Jan, 2024(Shift-II))

(A) $K_p = \frac{\alpha^{\frac{1}{2}} P^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}}$ (B) $K_p = \frac{\alpha^{\frac{3}{2}} P^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$

(C) $K_p = \frac{\alpha^{\frac{1}{2}} P^{\frac{3}{2}}}{(2+\alpha)^{\frac{3}{2}}}$ (D) $K_p = \frac{\alpha^{\frac{1}{2}} P^{\frac{1}{2}}}{(2+\alpha)^{\frac{3}{2}}}$

l. (B)

	$A_{(g)}$	$\rightleftharpoons$	$B_{(g)}$	+	$\frac{C}{2}(g)$
$t = t_{eq}$	$(1-\alpha)$		α		$\frac{\alpha}{2}$

$$P_B = \frac{\alpha}{\left(1+\frac{\alpha}{2}\right)}\cdot P,\ P_A = \frac{(1-\alpha)}{\left(1+\frac{\alpha}{2}\right)}\cdot P,\ P_C = \frac{\frac{\alpha}{2}}{\left(1+\frac{\alpha}{2}\right)}\cdot P$$

$$K_p = \frac{P_B \cdot P_C^{\frac{1}{2}}}{P_A} = \frac{(\alpha)^{\frac{3}{2}}(P)^{\frac{1}{2}}}{(1-\alpha)(2+\alpha)^{\frac{1}{2}}}$$

na assan hai na, balle-balle karo

40. $A(g) \rightleftharpoons 2B(g) + C(g)$

For the given reaction, if the initial pressure is 450 mm Hg and the pressure at time t is 720 mm Hg at a constant temperature T and constant volume V. The fraction of A(g) decomposed under these conditions is $x \times 10^{-1}$. The value of x is ____ (nearest integer)

(10 April, 2023(Shift-II))

Sol. $A(g) \rightleftharpoons 2B(g) + C(g)$

At t = 0	450		
At time, t	450 − x	2x	x

Now, $P_T = P_A + P_B + P_C$ (Given: P_T = 720 mm Hg)

$720 = 450 - x + 2x + x$, $x = 135$

Hence, the fraction of A decompose

$\frac{135}{450} = 0.3 = 3 \times 10^{-1}$ So, $x = 3$

Pagla mat jana

41. When 5.1 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 20% of the solid decompose into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is $x \times 10^{-2}$. The value of x is _____.(Integer answer)

(27 Aug, 2021 (Shift-II))

Sol. (6) For explanation, refer to QR code given on the last page.

42. At constant temperature, the equilibrium constant (K_p) for the decomposition reaction, $N_2O_4 \rightleftharpoons 2NO_2$, is expressed by $K_p = \frac{4x^2p}{(1-x^2)}$ where, p = pressure, x = extent of decomposition. Which one of the following statements is true? (IIT JEE 2001)

(A) K_p increases with increase of p

(B) K_p increases with increase of x

(C) K_p increases with decrease of x

(D) K_p remains constant with change in p and x

Sol. (D) The value of the equilibrium constant depends on temperature only. It is not affected by changes in concentration, pressure, extent of dissociation or presence of catalyst.

lallu mat ban jana, question ko padh lo pehle ache se

Common Mistake

Students might think that K_p will increase with increase in pressure. But it is an equilibrium constant which depends only on temperature.

2. HETEROGENEOUS EQUILIBRIUM

- In this equilibrium, species are not present in the same phase.
- Generally decomposition of solid phase in closed container is heterogeneous

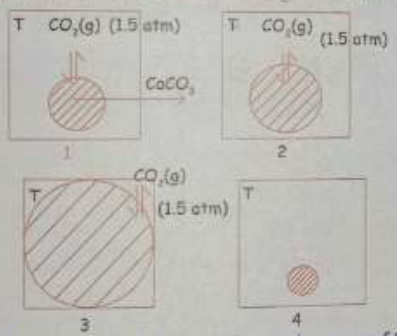
$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)} ; K_p = P_{CO_2} = \frac{xRT}{V}$$

$t = 0$	a	0	0
$t = t_{eq}$	$a - x$	x	x

a = initial moles of $CaCO_{3(s)}$

x = moles of CO_2 at equilibrium which exerts a pressure which is equal to K_p value.

43. $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$; $K_p = 1.5$ atm at T.
Equilibrium pressure of CO_2 = 1.5 atm



Sol. In container 1,2 and 3 we have sufficient. $CaCO_3$ to make the equilibrium. If Equilibrium is formed the pressure of CO_2 should be 1.5 atm only.

$$P_{CO_2} = 1.5$$

The amount of $CaCO_3$ may be less than the amount required to make the equilibrium. So all the $CaCO_3$ will be decomposed to $CO_{2(g)}$

$$P_{CO_2} < 1.5$$

Yeh sawaal hila daalega khopdi ko

44. $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$, $K_p = 1.2$ atm
Find out the minimum amount of $CaCO_3$ taken among the given options to form this eq^{bm} at 500 K and 2 Lt container

(A) 6 gm (B) 5 gm (C) 7gm (D) 8 gm

Sol. (C)

$CaCO_{3(s)}$	$\rightleftharpoons$	$CaO_{(s)}$	$+ CO_{2(g)}$
a		0	0
$a - x$		x	x

$$K_p = 1.2 = P_{CO_2}, P_{CO_2} = \frac{xRT}{V}$$

$$1.2 = \frac{x \times 0.0821 \times 500}{2}, x = \frac{1.2 \times 2}{0.0821 \times 500} = 0.06$$

Minimum amount of $CaCO_3$ required to attain this equilibrium is just greater than 0.06 mol (6 g).

yeh acha sawaal hai, chhod mat dena galti se bhi

45. Find out the amount of $CaCO_{3(s)}$ (8 g) left after heating it in a closed 2 Lt container at 500 K.
$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$; $K_p = 1.2$ atm

Sol. (2 gm) For explanation, refer to QR code given on the last page.

last line toh dekhlo kya hai

46. Solid ammonium carbamate dissociates according to the reaction.

$NH_2COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$; $K_p = 32$ atm^3

Find the total pressure at equilibrium.

Sol. (6 atm) For explanation, refer to QR code given on the last page.

47. For $NH_2COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$

Total pressure at equilibrium = 12 atm. Find K_p value. Also calculate the total pressure at final equilibrium if NH_3 is added so that the new pressure of NH_3 becomes double its initial equilibrium pressure.

Sol. $P_{NH_3} = 2P$, $P_{CO_2} = P$

$(P_T)_{eq} = 3P = 12$, $P = 4$ atm

$K_p = P^2_{NH_3} P_{CO_2} = (8)^2(4) = 256$ atm^3

$K_p = 256 = (P'_{NH_3})(P'_{CO_2})$

$256 = 16^2 P'_{CO_2}$, $P'_{CO_2} = 1$ atm

$(P_T)_{new} = P'_{NH_3} + P_{CO_2} = 16 + 1 = 17$ atm

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For any heterogeneous equilibrium established taking only solid in the container

- if K_p given we can find out equilibrium total pressure.
- If equilibrium total pressure is given then we can find out K_p for the equilibrium.

aaya samajh?

48. $NH_2COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$, $(P_T) = 12$ atm

Find out the total pressure at final equilibrium if CO_2 is added till its new equilibrium pressure becomes double its initial equilibrium pressure.

Sol. $(P_T)_{old} = 12$ atm, $3P = 12$, $P = 4$

$K_p = P^2_{NH_3} P_{CO_2} = 8^2 .4 = 256\ atm^3$

$K_p = P'^2_{NH_3} P'_{CO_2}$, $256 = P'^2_{NH_3} .8$

$P'_{NH_3} = \sqrt{32} = 4\sqrt{2}$ atm

$(P)_{new} = P^2_{NH_3} + P_{CO_2} = \left(4\sqrt{2} + 8\right)$ atm

ganit toh aati hai na ya nahi?

49. 20.0 grams of $CaCO_3(s)$ were placed in a closed vessel, heated and maintained at 727°C under equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO(g)$ and it is found that 75% of $CaCO_3$ was decomposed. What is the value of K_p? The volume of the container was 15 litres.

Sol. (0.833 atm) For explanation, refer to QR code given on the last page.

Common Mistake

Students might calculate the incorrect moles of $CaCO_3$ without considering its decomposition.

0. Solid ammonium carbamate dissociates to give ammonia and carbon dioxide as follows

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

At equilibrium, ammonia is added such that the partial pressure of NH_3 at the new equilibrium equals the original total pressure. Calculate the ratio of total pressure now to the original total pressure.

Sol. $\left(\frac{31}{27}\right)$ For explanation, refer to QR code given on the last page.

51. A vessel at 1000 K contains CO_2 with pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K_p is

[22 Jan, shift-I, 2023]

Sol. First write the reaction as:

$CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$

0.5 atm

0.5 – P 2P

Total pressure = 0.5 – P + 2P = 0.8, P = 0.3

Now, calculate K_p as:

$$K_p = \frac{P^2_{CO}}{P_{CO_2}} = \frac{(2P)^2}{(0.5-P)} = \frac{(0.6)^2}{(0.5-0.3)} = 1.8 atm$$

Multiple Equilibrium

$A \rightleftharpoons B;\ K_1$

$B \rightleftharpoons C;\ K_2$

$A \rightleftharpoons C;\ K_3$

$K_3 = K_1 \times K_2$, $K_1 = \frac{[B]}{[A]}$, $K_2 = \frac{[C]}{[B]}$, $K_3 = \frac{[C]}{[A]}$

$$K_1 \times K_2 = \frac{[B]}{[A]} \times \frac{[C]}{[B]} = \frac{[C]}{[A]} = K_3$$

52. $A \rightleftharpoons B;\ 2$

$C \rightleftharpoons B;\ 4$

$3C \rightleftharpoons 3D;\ 27$

$2A \rightleftharpoons 2D;\ K = ?$

Sol. $A \rightleftharpoons B;\ 2$

$B \rightleftharpoons C;\ \frac{1}{4}$

$C \rightleftharpoons D;\ 27^{\frac{1}{3}} = 3$

$A \rightleftharpoons D;\ 2 \times \frac{1}{4} \times 3 = \left(\frac{3}{2}\right)$

$2A \rightleftharpoons 2D;\ \left(\frac{3}{2}\right)^2 = \left(\frac{9}{4}\right)$

socho hua kya hai, dimag lagao

53. For the following three reactions (i), (ii) and (iii), equilibrium constants are given :

(i) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K_1

(ii) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$; K_2

(iii) $CH_4(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 4H_2(g) + 4H_2(g)$; K_3

Which of the following is correct?

(A) $K_1\sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$

(C) $K_3 = K_1K_2$ (D) $K_3K_2^3 = K_1^2$

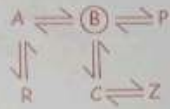
Sol. (C) The final equation is obtained by adding equations (i) and (ii). So, $K_3 = K_1 \times K_2$

Common Mistake

Students might add the values of two equilibrium constants instead of multiplying.

Simultaneous Equilibrium

In Simultaneous equilibrium, multiple equilibria coexist and interact with each other.



$A \rightleftharpoons B$ $B \rightleftharpoons P$ $Z \rightleftharpoons C$ $R \rightleftharpoons A$ $B \rightleftharpoons C$

Banenge to ek sath banenge.!
Disturb honge to ek sath honge !

Example:

$CaCO_{3(s)}$	$\rightleftharpoons$	$CaO_{(s)}$	+ $CO_{2(g)}$	Simultaneous equilibrium
a - x		x	x - y	
$C_{(s)}$	+	$CO_{2(g)}$	$\rightleftharpoons$ $2CO_{(g)}$	
b - y		x - y	2y	

$$K_{P_1} = P_{CO_2},\ K_{P_2} = \frac{P_{CO}^2}{P_{CO_2}} = \frac{P_{CO}^2}{K_{P_1}}$$

54. For these two simultaneous equilibria $K_{P_1} = 2$ atm and $K_{P_2} = 8$ atm. Find out the total pressure of the system when both equilibria are established.

$CaCO_{3(g)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$; $K_{P_1} = 2atm$

$C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$; $K_{P_2} = 8atm$

Sol. $P_{CO_2} = K_{P_1} = 2$ atm

$$\frac{P_{CO}^2}{P_{CO_2}} = K_{P_2} = 8 \quad P^2_{CO} = 16\ atm^2$$

$P_{CO} = 4$ atm, $P_{Total} = P_{CO_2} + P_{CO} = 2 + 4 = 6$ atm

55. Three reactions are at simultaneous equilibrium. Moles of C, E and F are 3, 2 and 4 respectively (at equilibrium). Calculate the K_C values of these equilibrium if initially A, B and D were 8, 8 and 2 moles respectively taken in 1 lt container.

Sol.

	A + 2B	$\rightleftharpoons$ 3C		
	8-x-2z	8-2x-y-3z	3x=3	x = 1
$E_q^{bm\ moles}$	(5)	(2)	(3)	

	B + D	$\rightleftharpoons$ 2E		
	8-2x-y-3z	2-y	2y=2	y = 1
$E_q^{bm\ moles}$	(2)	(1)	(2)	

	3B + 2A	$\rightleftharpoons$ 4F		
	8-2x-y-3z	8-x-2z	4z=4	z = 1
$E_q^{bm\ moles}$	(2)	(5)	(1)	

$$K_{C_1} = \frac{[C]^3}{[A][B]^2} = \frac{\left(\frac{3}{1}\right)^3}{\left(\frac{5}{1}\right)\left(\frac{2}{1}\right)^2} = \frac{27}{20}$$

hai na ekdum khatarnaak sawaal?

56. Two solids dissociate as follows.

$A(s) \rightarrow B(g) + C(g)$; $K_{P_1} = x\ atm^2$

$D(s) \rightarrow C(g) + E(g)$; $K_{P_2} = y\ atm^2$

The total pressure when both the solids dissociate simultaneously is

(A) $\sqrt{x+y}$ atm (B) $\left(2\sqrt{x+y}\right)$ atm

(C) (x+y) atm (D) $x^2 + y^2$ atm

Sol. (B) For explanation, refer to QR code given on the last page.

57. For the given hypothetical reactions, the equilibrium constants are as follows:

$X \rightleftharpoons Y$; $K_1 = 1.0$

$Y \rightleftharpoons Z$; $K_2 = 2.0$

$Z \rightleftharpoons W$; $K_3 = 4.0$

The equilibrium constant for the reaction $X \rightleftharpoons W$ is **(08 April, 2024 (Shift-I))**

(A) 6.0 (B) 12.0 (C) 8.0 (D) 7.0

Sol. (C) $X \rightleftharpoons Y$; $K_1 = 1$

$Y \rightleftharpoons Z$; $K_2 = 2$

$Z \rightleftharpoons W$; $K_3 = 4$

Equilibrium constant for $X \rightleftharpoons W$ will be

$K_1 . K_2 . K_3, K = 1 \times 2 \times 4, K = 8$

yeh bhi kam nahi hai sawaal kisi se

LE-CHATELIER'S PRINCIPLE

If a system at equilibrium is subjected to a change of pressure, temperature, or the number of moles of a substance, there will be a tendency for a net reaction in the direction that tends to reduce the effect of this change.

Any changes or disturbances except temperature only change reaction quotient Q_c or Q_p not K_p.

(A) Effect of Change in Concentration

Concept: $Q > K$ backward

$Q < K$ Forward

$Q = K$ No change in the position of Equilibrium.

Change in the concentration of reactant:

Reactant ↑ Q ↓ Forward

Reactant ↓ Q ↑ backward

Change in the concentration of products:

Product ↑ Q ↑ backward

Product ↓ Q ↓ Forward

Example: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

On adding PCl_5 — Forward

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

On adding Cl_2 — Backward

Note

Effect of change in concentration of reactant or product.

agaya hoga na samjh, asaan sa hai bilkul

Khoobsurat Baat

On increasing the amount of reactant, reaction always goes forward.

$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

Kya yehi sahi hai? socho socho

Since reactant is not in the expression of Q so on changing its amount does not affect the equilibrium.

$Q_c = [CO_2]$, $Q_p = P_{CO_2}$

58. Consider the reaction:

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

On doubling the amount of PCl_5 and tripling the amount of Cl_2 and making the amount of PCl_3 halved. Predict the direction.

Sol. Let a, b, c be the moles of PCl_5, PCl_3, Cl_2 at equilibrium respectively.

Now, according to the given information:

$$Q_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{b}{V}\right)\left(\frac{c}{V}\right)}{\left(\frac{a}{V}\right)} = \frac{bc}{a}\left(\frac{1}{V}\right)$$

$$Q_p = Q_c(RT)^{\Delta n_g} = \frac{bc}{a}\left(\frac{RT}{V}\right)$$

$$(Q_c)_{new} = \frac{\frac{b}{2} \times 3c}{2a}\left(\frac{RT}{V}\right) = \frac{bc}{a}\left(\frac{RT}{V}\right) \times \left(\frac{3}{4}\right) < Q_c$$

So, the direction of the reaction will be forward.

prabhu ke gun gao

Effect of Change in Pressure

1. **Increase in pressure:** When the pressure of a system is increased, the equilibrium shifts to the side with fewer moles of gas. This helps to reduce the pressure.
2. **Decrease in pressure:** When the pressure of a system is decreased, the equilibrium shifts to the side with more moles of gas. This helps to increase the pressure.

(1) For example, consider the reaction:
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

If the pressure is increased, the equilibrium shifts to the right, favoring the production of ammonia (NH_3), which has fewer moles of gas.'

(2) $\underset{a}{PCl_{5(g)}} \rightleftharpoons \underset{b}{PCl_{3(g)}} + \underset{c}{Cl_{2(g)}}$

$$Q = \left(\frac{bc}{a}\right)\left(\frac{RT}{V}\right)$$

Pressure ↑ vol ↓ Q ↑ Backward

Pressure ↓ vol ↑ Q ↓ forward

(3) $2A_{(g)} \rightleftharpoons B_{(g)}$ On increasing the pressure

$$Q_p = \frac{\left(\frac{b}{V}\right)\frac{1}{}}{\left(\frac{a}{V}\right)^2 RT}$$

Pressure ↑ vol ↓ Q ↓ Forward

Pressure ↓ vol ↑ Q ↑ Backward

Kaafi mazedaar hai

Khoobsurat Baat

59. Consider the reaction:

$$A^a_{(g)} + B^b_{(g)} \rightleftharpoons C^c_{(g)} + D^d_{(g)}$$

On increasing the pressure

(A) eq^{bm} does not shift
(B) moves forward
(C) moves backward
(D) conc. of B increases
(E) Moles of B remain some.

Sol. (A,D,E) $Q = \dfrac{\left(\frac{c}{V}\right)\left(\frac{d}{V}\right)}{\left(\frac{a}{V}\right)\left(\frac{b}{V}\right)} = \dfrac{cd}{ab}$

No term of volume in the expression of reaction quotient
P ↑ V ↓ Q remains same
Since volume changes, conc. will slowly change but moles remains same as eq^{bm} does not shift.

So, options A, D and E are correct.

aaya kya samajh, ya dimag ghum gaya pura?

60. Consider the equilibrium **[29 Jan, Shift-II, 2025]**

$CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$

If the pressure applied over the system increases by two fold at constant temperature then

(a) Concentration of reactants and products increases.
(b) Equilibrium will shift in forward direction.
(c) Equilibrium constant increases since concentration of products increase.
(d) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the correct answer from the options given below :

(A) (a) and (b) only (B) (b) and (c) only
(C) (a), (b) and (c) only (D) (a), (b) and (d) only

Sol. (A) Since pressure is doubled at constant temperature, the concentration of gases increases proportionally (PV = nRT). So, both reactants and products increase in concentration.

Since fewer moles of gas are on the right side (products), the equilibrium will shift forward to reduce the total pressure.

61. Which one of the following reactions at equilibrium, with all reactants and products the gaseous phase, would be unaffected by an increase in pressure?

(A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (B) $2CO + O_2 \rightleftharpoons 2CO_2$
(C) $2H_2 + O_2 \rightleftharpoons 2H_2O$ (D) $N_2 + O_2 \rightleftharpoons 2NO$

Sol. (D) In option D, the value of Δn_g is 0, so, equilibrium will remain unaffected by an increase in pressure.

Δn_g ka formula toh aata hai na ya bhul gaye?

62. Consider the following equilibrium in a closed container $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements hold true regarding the equilibrium constant (K_p) and degree of dissociation (α)? **[IIT JEE 2002]**

(A) Neither K_p nor α changes
(B) Both K_p and α change
(C) K_p changes but α does not change
(D) K_p does not change but α changes

Sol. (D) If volume is halved, pressure is doubled, but K_p will not be affected by the pressure. The equilibrium will shift in backward direction, according to Le Chatelier's principle, to nullify the effect of increased pressure. This decreases the degree of dissociation α. Hence, K_p does not change.

badhiya sawaal, karlo ache se, samjhe

63. Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K. If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following
[28 Jan, Shift-I, 2025]

(A) Liquid phase disappears completely.
(B) The amount of ice decreases.
(C) The solid phase (ice) disappears completely.
(D) Volume of system increases

Sol. (C) Ice (solid) has a greater volume than water (liquid). When pressure is increased, the equilibrium will shift towards the phase with the lower volume to minimize the effect of the pressure increase.

Since water has a lower volume than ice, the equilibrium will shift towards the formation of more water.

Isko kar daala toh life jhingalala

Common Mistake

tudents might think that K_p will also change with e change in pressure, but it is important to ep in mind that it changes only with change in mperature.

ddition of Inert Gas

t constant pressure

consider a general reaction:

$$\underset{a}{A_{(g)}} + \underset{b}{B_{(g)}} \rightleftharpoons \underset{c}{C_{(g)}} + \underset{d}{2D_{(g)}}$$

$$Q_c = \frac{[C][D]^2}{[A][B]} = \frac{\left(\frac{c}{V}\right)\left(\frac{d}{V}\right)^2}{\left(\frac{a}{V}\right)\left(\frac{b}{V}\right)} = \frac{cd^2}{ab}\frac{1}{V}$$

$$Q_p = \frac{cd^2}{ab}\left(\frac{RT}{V}\right)$$

Let e moles of gas is added, then,

Total moles = a + b + c + d + e

moles↑volume↑[const P,T]

Forward

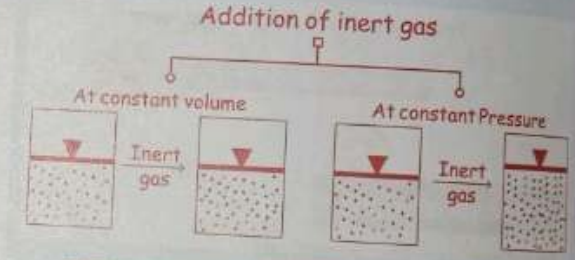
(ii) At constant volume

$$\underset{a}{A_{(g)}} + \underset{b}{B_{(g)}} \rightleftharpoons \underset{c}{C_{(g)}} + \underset{d}{2D_{(g)}}$$

$$Q = \frac{c \cdot d^2}{ab}\left(\frac{RT}{V}\right)$$

On addition of inert gas at constant volume, pressure increases but Q does not change, so, no effect on equilibrium.

Note



lag raha hai na kitna sundar?

Yeh kya ho raha hai, total pressure change ho raha toh partial pressure bh change hona chaiye jiska matlab Q bhi change hona chaiye toh equilibrium bhi toh shift hona chaiye:
Lekin aisa nahi hota hai, aao dekhte hai isko

$$\underset{a}{A_{(g)}} + \underset{b}{B_{(g)}} \rightleftharpoons \underset{c}{C_{(g)}} + \underset{d}{2D_{(g)}}$$

$$Q_{old} = \frac{P_C \cdot P_D^2}{P_A \cdot P_B} = \frac{\left(\frac{c}{a+b+c+d}\cdot P_T\right)\left(\frac{d}{a+b+c+d}\cdot P_T\right)^2}{\left(\frac{a}{a+b+c+d}\cdot P_T\right)\left(\frac{b}{a+b+c+d}\cdot P_T\right)}$$

Now, $P_T \cdot V = (a+b+c+d)RT$

$$P_T(a+b+c+d) = \left(\frac{RT}{V}\right), \quad Q_{old} = \frac{c \cdot d^2}{ab} \cdot \left(\frac{RT}{V}\right)$$

Let e moles of gas is added, so,

$$Q_{new} = \frac{P_C' \cdot P_D'^2}{P_A' \cdot P_B'}$$

$$= \left(\frac{a}{a+b+c+d+e} \cdot P_T'\right)\left(\frac{b}{a+b+c+d+e} \cdot P_T'\right)$$

$$= \left(\frac{c \cdot P_T'}{a+b+c+d+e}\right)\left(\frac{d}{a+b+c+d+e} \cdot P_T'\right)^2$$

$$= \left(\frac{a}{a+b+c+d+e} \cdot P_T'\right)\left(\frac{b}{a+b+c+d+e} \cdot P_T'\right)$$

$P_T' V = (a+b+c+d+e) RT$

$$\frac{P_T'}{a+b+c+d+e} = \left(\frac{RT}{V}\right), \quad Q_{new} = \frac{c \cdot d^2}{ab}\left(\frac{RT}{V}\right)$$

* mole fraction of individual species changes.
* But partial pressure = mole fraction × Total pressure remains same for all the species

Partial Pressure = $\chi_{gas} \times P_{Total}$ = constant

$\chi_{gas} \downarrow \quad P_{Total} \uparrow$

Palle pada kuch

Khoobsurat Baat

On addition of inert gas at constant volume.

* quotient remains same as there is no pressure term
* Equilibrium does not shift
* $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + 2D_{(g)}$

$$Q = \frac{P_C \cdot P_D^2}{P_A \cdot P_B} = \frac{(X_C \cdot P_T)(X_D \cdot P_T)^2}{(X_A \cdot P_T)(X_B \cdot P_T)} = \frac{X_C \cdot X_D^2}{X_A \cdot X_B}(P_T)$$

* Total equilibrium pressure increases.
* Partial pressures of all the species remains same.

64. The effect of addition of helium gas to the following reaction in equilibrium state is:

[1 Feb, 2023 (Shift-II)]

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) The equilibrium will shift in the forward direction and more of Cl_2 and PCl_3 gases will be produced.

(B) The equilibrium will go backward due to suppression of dissociation of PCl_5

(C) Helium will deactivate PCl_5 and reaction will stop.

(D) Addition of helium will not affect the equilibrium.

Sol. (A, D) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Case 1: At constant P, volume will increase due to which reaction will shift in forward direction.

Case 2: At constant volume, no change will take place, so reaction will not shift in any direction.

65. Consider the following reaction:

$N_2O_4(g) \rightleftharpoons 2NO_2(g); \Delta H^0 = +58kJ$

For each of the following cases (a, b) the direction in which the equilibrium shifts is:

[5 Sept, 2020 (Shift-I)]

(i) Temperature is decreased

(ii) Pressure is increased by adding N_2 at constant T.

(A) (i) Towards product, (ii) towards reactant

(B) (i) Towards reactant, (ii) no change

(C) (i) Towards reactant, (ii) towards product

(D) (i) Towards product, (ii) no change

Sol. (B) According to Le-chatelier's principle in endothermic reaction, formation of reactants takes place due to decrease in temperature. Addition of inert gas at constant volume and temperature has no effect on equilibrium because partial pressure remains constant.

sawaal ko dekho samjho phir hi hoga varna beda garg

Common Mistake

Students might think that at constant volume reaction will proceed in forward or backward direction, but at constant volume, no effect on equilibrium as partial pressure remains constant.

66. For the chemical reaction,

$3X(g) + Y(g) \rightleftharpoons X_3Y(g)$

the amount of X_3Y at equilibrium is affected by **[IIT JEE 1999]**

(A) temperature and pressure

(B) temperature only

(C) pressure only

(D) temperature, pressure and catalyst

Sol. (A) For the reaction, $3X(g) + Y(g) \rightleftharpoons X_3Y(g)$

Value of $\Delta n_g = 1 - 4 = -3$

There is change in moles so pressure will change accordingly.

K_p changes with temperature. Catalyst has no effect on K_p. Thus P and T influence the equilibrium concentrations. Hence, increase in pressure will favor forward reaction.

Dimag ki ghanti baji

67. For the reaction,

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$,

at a given temperature, the equilibrium amount of CO_2 can be increased by **[IIT JEE 1998]**

(A) adding suitable catalyst

(B) adding an inert gas

(C) decreasing the volume of the container

(D) increasing the amount of $CO_{(g)}$

Sol. (D) For explanation, refer to QR code given on the last page.

Effect of Catalyst

A catalyst has no effect on the position of the equilibrium since it increases not only the rate of forward reaction but also the rate of backward reaction. However it does help the system to reach the equilibrium faster.

Effect of Change in Temperature

Increasing the temperature of an endothermic reaction favors the forward reaction, while increasing the temperature of an exothermic reaction favors the reverse reaction.

Khoobsurat Baat

Sine K_{eq} depends on temperature exponentially so if on changing temperature K increases, it increase with a large quantity and if decreases, it decreases with large quantity.

अगर बढ़ेंगे तो बहुत बढ़ेंगे
अगर घटेंगे तो बहुत घटेंगे

In case of more than one changes, if one of the changes is temperature also then deciding factor will be temperature only. Because other factors can change the reaction quotient value but temperature factor changes equilibrium constant exponentially.

68. Consider the following equation:

[30 Jan, 2023 (Shift-II)]

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \Delta H = -190kJ$

The number of factors which will increase the yield of SO_3 at equilibrium from the following is_____

(A) Increasing temperature

(B) Increasing pressure

(C) Adding more SO_2

(D) Adding more O_2

(E) Addition of catalyst

Sol. (C) According to Le-Chatelier's Principle, adding more O_2 and SO_2 will increase the yield of SO_3.

Also, by increasing pressure, the reaction will move forward, hence SO_2 yield will get increase.

69. The following reaction occurs in the Blast furnace where iron ore is reduced to iron metal

$Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow Fe_{(l)} + 3CO_{2(g)}$

Using the Le-Chatelier's principle, predict which one of the following will not disturb the equilibrium. **[05 April, 2024 (Shift-I)]**

(A) Addition of Fe_2O_3 (B) Addition of CO_2

(C) Removal of CO (D) Removal of CO_2

Sol. (A) As per Le-Chatelier's principle, when solid is added, there is no effect on equilibrium.

balle balle karo, itna asaan

Physical Equilibrium

$H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$; ΔH = endo = (+)ve

$H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$; ΔH = endo = (+)ve

$C_{(graphite)} \rightleftharpoons C_{(diamond)}$; ΔH = endo = (+)ve

Note

Endothermic	Exothermic
T↑ K↑ Forward	T↑K↓Backward

Khoobsurat Baat

On increasing the pressure of any physical equilibrium, equilibrium shifts forward the compound having more density.

Example: $H_2O_{(s)} \quad H_2O_{(l)}$

ice $\rightleftharpoons$ water ; ΔH = endo

Temperature ↑ forward = more ice melts.
Pressure ↑ forward = more ice melts.

$H_2O_{(l)} \quad H_2O_{(g)}$

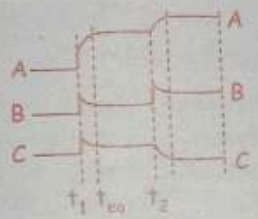
water $\rightleftharpoons$ vapour ; ΔH = endo

Temperature ↑ forward = more vapour formed.
Pressure ↑ Backward = more vapour condensed into liquid.

$C_{(graphite)} \rightleftharpoons C_{(diamond)}$; ΔH = endo

Temperature ↑ forward = more diamond formed
Pressure ↑ forward
= more diamond formed ($d_{diamond} > d_{graphite}$)

70.

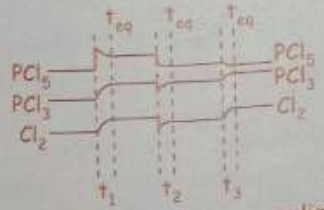


$A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$; ΔH = (+)
a b c

What are the changes applied on t_1 and t_2 time?

Sol. At t_1, conc of all species are increasing, reaction goes backward direction
At t_2, B is increasing, so, reaction goes in backward direction

71.



What are the changes applied at t_1, t_2, t_3 time?

Sol. t_1 = Addition of PCl_5
t_2 = Pressure reduced
t_3 = Temperature increases
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (endo)

Relation between Vapour Density and Degree of Dissociation

Let's consider a reaction:

$$\underset{a}{1A_{n(g)}} \rightleftharpoons \underset{o}{n\,A_{(g)}}$$

$$\underset{a(1-\alpha)}{} \quad \underset{n\alpha}{}$$

moles initial = a

moles at eq^{bm} = $a[1 + (n-1)\alpha]$

$$PV = nRT,\ PV = \frac{w}{M}RT$$

$$\frac{w}{V} = \rho = \frac{PM}{RT}$$

$$M = \frac{\rho RT}{P} = \frac{\rho \cancel{R}\cancel{T}V}{n\cancel{R}\cancel{T}} = \frac{\rho V}{n}$$

$$V.D = \frac{\rho V}{2n},\ V.D \propto \frac{1}{n}$$

Note

$$D = (V.D)_{initial} = (V.D)_{theoretical} \propto \frac{1}{a}$$

$$d = (V \cdot D)_{eq} = (V \cdot D)_{observed} \propto \frac{1}{a(1+(n-1)\alpha]}$$

$$\frac{D}{d} = \frac{(V \cdot D)_{theoretical}}{(V \cdot D)_{observed}} = \frac{a(1+(n-1)\alpha]}{a} = 1 + (n-1)\alpha$$

$$\left(\frac{D}{d} - 1\right) = (n-1)\alpha\ ;\ \alpha = \frac{D-d}{d(n-1)}$$

D = vapour density theoretical

d = vapour density observed

α = degree of dissociation

$$n = \frac{\text{moles of product}}{\text{moles of reactant}}$$

are baapre, itne sare formulae, dilo ko thaam lo

Khoobsurat Baat

$$n = \frac{\text{gaseous moles of product}}{\text{gaseous moles of reactant}}$$

$$\alpha = \frac{(V \cdot D)_{th} - (V \cdot D)_{obs}}{(V \cdot D)_{obs}(n-1)} = \frac{(\text{Mol wt})_{th} - (\text{mol wt})_{obs}}{(\text{mol wt})_{obs}(n-1)}$$

72. $3A_{(g)} \rightleftharpoons 1B_{(g)} + 4C_{(g)}$; $(\text{vap density})_{\text{observed}} = d$

$(D - d)(\text{vap density})_{\text{theoretical}} = D$

Predict the relation between d and D.

(A) $\alpha = \frac{(D-d)}{3d}$ (B) $\alpha = \frac{(D-d)}{4d}$

(C) $\alpha = \frac{3(D-d)}{2d}$ (D) $\alpha = \frac{2(D-d)}{3d}$

Sol. (C) $\alpha = \frac{(D-d)}{d(n-1)}$, $n = \frac{1+4}{3} = \frac{5}{3}$

$$\alpha = \frac{(D-d)}{d\left(\frac{5}{3}-1\right)} = \frac{3(D-d)}{2d}$$

formula lagaya hai bas bacho

73. Derive the relation b/w α, D, d.

$2A_{(g)} \rightleftharpoons 3B_{(g)}$

Sol. $2A_{(g)} \rightleftharpoons 3B_{(g)}$

a 0

$a(1-\alpha)$ $\frac{3}{2}a\alpha$

$$\frac{D}{d} = \frac{\cancel{a}\left[1 + \frac{3}{2}\alpha - \alpha\right]}{\cancel{a}}; \quad \frac{D}{d} = 1 + \alpha\left(\frac{3}{2} - 1\right)$$

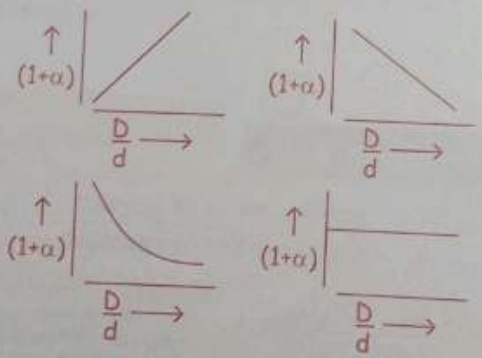
$$\alpha = \frac{D-d}{d\left(\frac{3}{2}-1\right)}$$

74. In the dissociation of $N_2O_{4(g)}$ into $NO_{2(g)}$, $(1 + \alpha)$ varies with the vapour density ratio $\left(\frac{D}{d}\right)$ as

where α = degree of dissociation

D = initial vapour density

d = final vapour density



Sol. $N_2O_4 \rightleftharpoons 2NO_2$

$$\alpha = \frac{D-d}{d(1)}, \; 1 + \alpha = \frac{D-d}{d} + 1, \; (1+\alpha) = \frac{D}{d}$$

75. The vapour Density of an equilibrium mixture of PCl_5, PCl_3 and Cl_2 is 92. Find the degree of dissociation of PCl_5.

Sol. $1PCl_5 \rightleftharpoons 1PCl_3 + 1Cl_2$

$$n = \frac{1+1}{1} = 2, \; d = 92$$

$$D = \frac{\text{Molecular weight}}{2} = \frac{208.5}{2} = 104.25$$

$$\alpha = \frac{104.25 - 92}{92(2-1)} = 0.133$$

simple hai bacho, kyu ghabrate ho itna

76. Consider the following equilibrium,

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

0.1 mol of CO along with a catalyst is present in a 2 dm^3 flask maintained at 500 K. Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of CH_3OH is formed. The K_p is _____ $\times 10^{-3}$ (nearest integer).

Given: $R = 0.08\ dm^3\ bar\ K^{-1}\ mol^{-1}$

Assume only methanol is formed as the product and the system follows ideal gas behaviour. **[02 April, 2025 (Shift-I)]**

Sol. (74) For explanation, refer to QR code given on the last page.

77. Consider the following chemical equilibrium of the gas phase reaction at a constant temperature:

$A(g) \rightleftharpoons B(g) + C(g)$

If p being the total pressure, K_p is the pressure equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium?

[02 April, 2025 (Shift-II)]

(A) If p value is extremely high compared to K_p, $\alpha \approx 1$

(B) When p increases α decreases

(C) If k_p value is extremely high compared to p, α becomes much less than unity

(D) When p increases a increases

Sol. (B) For explanation, refer to QR code given on the last page.

Scan the QR code to get the explanations of unsolved questions.

5 Ionic Equilibrium

To access one shot Revision Video Scan this QR Code

INTRODUCTION

Ionic equilibrium refers to the equilibrium established between ions and molecules in a solution. This concept is crucial in understanding various chemical reactions, especially those involving acids, bases, and salts.

ARRHENIUS CONCEPT

According to Arrhenius theory, acids are substances that dissociates in water to give hydrogen ions [H^+] and bases are substances that produce hydroxyl ions [OH^-].

ARRHENIUS ACIDS

The ionization of an acid HA in its aqueous solution can be represented by the following equation:

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

or

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

Important

- If an acid releases only one H^+ ion per molecule, it is known as monobasic/ monoprotic acid (HA).
  For eg. HCl, HBr, HI, CH_3COOH, HNO_3 etc.
- If an acid releases two H^+ ions per molecule, it is known as dibasic/diprotic(H_2A).
  For e.g. H_2SO_4, H_2SO_3, H_3PO_3, $H_2C_2O_4$, H_2S etc.
- If an acid releases three H^+ ions per molecule, it is known as tribasic/triprotic (H_3A).
  For e.g. H_3PO_4, H_3AsO_4 etc.

Aaya kya samajh

Note

Arrhenius concept of Acid and Base

Acid → H^+ Deta Hai !
- $1H^+$ = monoprotic
- $2H^+$ = Diprotic
- $3H^+$ = Triprotic

Base → OH^- Deta Hai !
- $1OH^-$ = monoacidic
- $2OH^-$ = diacidic
- $3OH^-$ = Triacidic

1. According to Arrhenius concept, which are monobasic acids?
   1. H_2SO_4 2. H_3PO_2 3. H_3BO_3 4. HNO_3
   5. H_3PO_3(Dibasic) 6. H_3PO_4 7. CH_3COOH

Sol. Monobasic acids- Give one H^+ ion on dissociation.

Among the given acids, H_3PO_2, H_3BO_3, HNO_3 and CH_3COOH are monobasic acids.

yeh toh kar hi loge haina!

ARRHENIUS BASES

The ionization of a base BOH in its aqueous solution can be represented by the following equation:

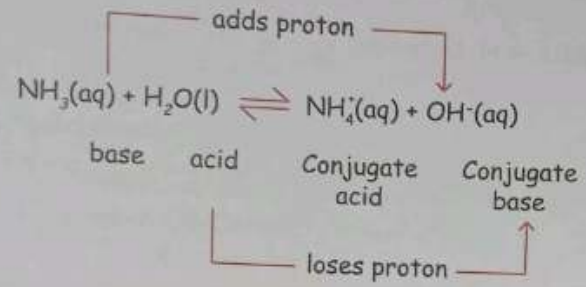
$$BOH(aq) \rightleftharpoons B^+(aq) + OH^-(aq)$$

Important

- If a base releases only one OH^- ion per molecule, it is known as monoacidic base (BOH).
  For e.g. NaOH, KOH, RbOH, CsOH, CsOH, NH_4OH etc.
- If a base releases two OH^- ions per molecule, it is known as diacidic base [$B(OH)_2$].
  For e.g. $Mg(OH)_2$, $Ca(OH)_2$, $Zn(OH)_2$ etc.
- If a base releases three OH^- ions per molecule, it is known as triacidic base [$B(OH)_3$].
  For eg. $Al(OH)_3$, $Fe(OH)_3$ etc.

BRONSTED – LOWRY CONCEPT (THE PROTON – DONOR – ACCEPTOR CONCEPT)

According to Bronsted–Lowry theory, an acid is a substance that is capable of donating a hydrogen ion H^+ and bases are substances capable of accepting a hydrogen ion, H^+. In short, acids are proton donors and bases are proton acceptors.



Samajhne ki koshish toh karo pehle

Note

H^+ deta hai — acid !
H^+ leta hai — base !

$HA \rightarrow A^- + H^+$
Acid — Conjugate base

$NH_3 + H^+ \rightarrow NH_4^+$
Base — conjugate acid

CONJUGATE ACID–BASE PAIR CONCEPT

An acid–base pair that differs only by one proton is called a conjugate acid–base pair.

Note

- To get conjugate acid of a given species add H^+ to it. e.g. conjugate acid of N_2H_4 is $N_2H_5^+$.
- To get conjugate base of any species subtract H^+ from it. e.g. Conjugate base of NH_3 is NH_2^-.
- Stronger a Bronsted acid is, weaker is its conjugate base.
- Stronger a Bronsted base is, weaker is its conjugate acid.

Hai na kitna asaan?

2. The conjugate acid of NH_2^- is (IIT JEE 1988)
(A) NH_3 (B) NH_2OH
(C) NH_4^+ (D) N_2H_4

Sol. (A) The conjugate acid of amide ion (NH_2^-) is ammonia, NH_3.

3. Choose the conjugate acid–base pair –

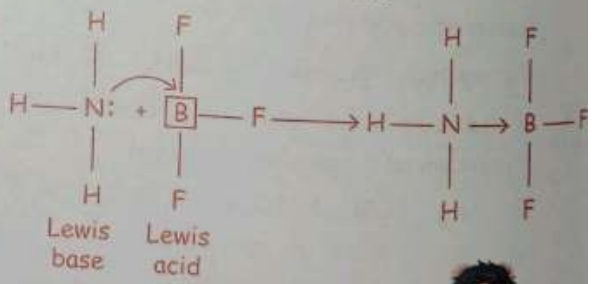
1. H_2S	– HS^-	7. H_2CO_3	– CO_3^{2-}
2. HCl	– Cl^-	8. HCO_3^-	– CO_3^{2-}
3. H_2SO_4	– SO_4^{2-}	9. H_2CO_3	– HCO_3^-
4. H_3PO_4	– $H_2PO_4^-$	10. $H_2PO_4^-$	– HPO_4^{2-}
5. NH_3	– NH_4^+	11. HPO_4^{2-}	– PO_4^{3-}
6. $CH_3COOH-CH_3COO^-H_2$		12. H_2SO_4	– HSO_4^-

Sol. Except 3,5,6,7, all are correct.

LEWIS CONCEPT OF ACID AND BASE

THE LEWIS CONCEPT (ELECTRON DONOR – ACCEPTOR SYSTEM)

According to this theory an acid is any molecule or ion which can accept an electron pair with the formation of a coordinate bond. For example, in BF_3 the boron atom can accept a pair of electrons; so BF_3 is a Lewis acid. A base must therefore be any molecule or ion which has a lone pair of electrons, which it can donate. For example, ammonia molecule has a lone pair of electrons; so it is a Lewis base. The reaction between a Lewis base and a Lewis acid is just the formation of a coordinate bond between them.



sar toh nhi ghum gaya padhkar?

Note

Acid – electron pair acceptor !
Base – electron pair donor !

4. Which of the following compound CANNOT act as a Lewis base?

(17 March, 2021 (Shift-I))

(A) ClF_3 (B) PCl_5 (C) NF_3 (D) SF_4

Sol. (B) PCl_5 cannot act as Lewis base due to non-availability of l.p on P.

IONIC PRODUCT OF WATER

Consider the dissociation of water as:

$$H_2O \rightleftharpoons H^+ + OH^-$$

$$K_a = \frac{[H^+][OH^-]}{[H_2O]}$$

$$K_a[H_2O] = [H^+][OH^-] = K_W \text{ (Ionic product)}$$

At 25°C, $K_W = 10^{-14}$

$$K_a = \frac{K_w}{[H_2O]} = \frac{10^{-14}}{55.55} = 1.8 \times 10^{-16} \ (pK_a = 15.7)$$

Consider the dissociation of H_2O as:

H_2O	$\rightleftharpoons$	H^+	+	OH^-
C		0		0
C(1 - α)		Cα		Cα

$$K_a = \frac{(C\alpha)(C\alpha)}{C(1-\alpha)} \quad [\alpha << 1]$$

$$K_a = C\alpha^2$$

$$\alpha = \sqrt{\frac{K_a}{C}} = (1.8 \times 10^{-16}/55.55)^{1/2} = 1.8 \times 10^{-9}$$

check karo ki correct kiya hai ya nahi

Khoobsurat Baat

At 25°C,

Ionic product of water (K_w) = 10^{-14}

Dissociation constant of water (K_a) = 1.8×10^{-16}

Degree of dissociation of water (α) = 1.8×10^{-9} or 1.8×10^{-7}%

5. In a container at temperature T, $[H^+] = 10^{-7}$ M

$[OH^-] = 10^{-3}$ M

if we increase the $[H^+]$ to 10^{-5} M then what would be the $[OH^-]$ at temperature T?

Sol. At any Temperature

$$K_W = [H^+][OH^-] = 10^{-7} \times 10^{-3} = 10^{-10}$$

At same Temperature

$$K_W = [H^+][OH^-] = 10^{-5} \times [OH^-] = 10^{-10}$$

$$[OH^-] = \frac{10^{-10}}{10^{-5}} = 10^{-5} M$$

are asaan hi hai, kyu ghabra rahe ho?

6. At 90°C, pure water has $[H_3O^+]$ as 10^{-6} molL^{-1}. What is the value of K_W at 90°C ?

(IIT JEE 1981)

(A) 10^{-6} (B) 10^{-12} (C) 10^{-14} (D) 10^{-8}

Sol. (B) For explanation, refer to QR code given on the last page.

pH SCALE

The pH scale is a fundamental concept in chemistry that measures the concentration of hydrogen ions (H^+) in a solution.

pH Scale at 25°C

At 25°C, $K_W = 10^{-14}$

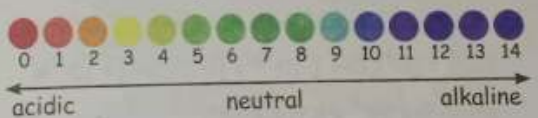
$$K_W = [H^+][OH^-]$$

Since, H_2O is neutral, So, $[H^+] = [OH^-]$

$$[H^+]^2 = 10^{-14}, [H^+] = 10^{-7}$$

pH = 7

pH scale



pH Scale at Temperature > 25°C

$K_W > 10^{-14}$. (Since, T > 25°C, Dissociation of H_2O increases, ionic product also increases)

Suppose $K_W = 10^{-12}$

$$H_2O \rightleftharpoons H^+ + OH^-$$

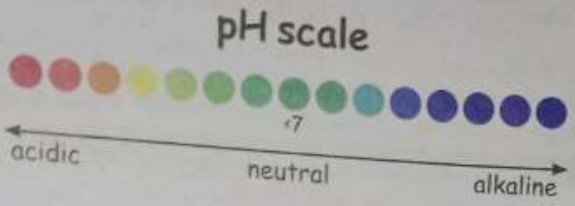
$$K_W = [H^+][OH^-]$$

$$10^{-12} = [H^+][OH^-]$$

$[H^+] = 10^{-6}M$

$pH = -\log [H^+] = -\log 10^{-6} = 6$

At temperature > 25°C, neutral point shifts to (<7)



Khoobsurat Baat

7. At temperature greater than 25° C–
   (A) pH = 7 Neutral point
   (B) pH = 6.8 Neutral point
   (C) pH = 7.1 Neutral point
   (D) pH = 7 is basic point

Sol. (D) $K_w > 10^{-14}$

pH = 7 indicates a basic point.

Yhan neutral point 7 se kam hoga lekin kitna kam hoga yeh humko nahi pata, isliye 6.8 answer galat maana jaega.

pH Scale at Temperature < 25°C

$K_w < 10^{-14}$ (Since, T < 25°C, Dissociation of H_2O decreases, ionic product also decreases)

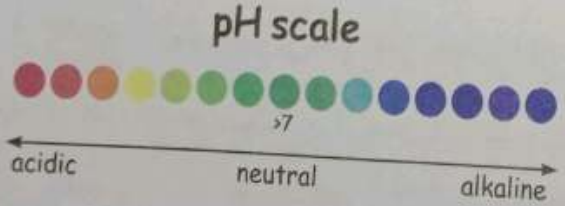
Suppose $K_w = 10^{-16}$

$H_2O \rightleftharpoons H^+ + OH^-$

$K_w = [H^+][OH^-]$

$10^{-16} = [H^+][OH^-]$, $10^{-16} = [H^+]^2$, $[H^+] = 10^{-8}M$

$pH = -\log 10^{-8} = 8$ (Neutral)



Is scale ko halke mei mat lena, samjhe

8. pH of water is 7 at 25°C. If water is heated to 80°C., it's pH will:
   (A) Remains the same
   (B) Decrease
   (C) Increase
   (D) H^+ concentration increases, OH^- concentration decreases

Sol. (B) pH of water is 7 at 25°C.

So, if we increase the temperature, then the pH of water will decrease.

9. At temperature < 25°C-
   (A) pH = 7 neutral point
   (B) pH = 7.1 is neutral point
   (C) pH = 6.9 is neutral point
   (D) pH = 7 is acidic solution

Sol. (D) At temperature < 25°C, Neutral point is greater than 7 which means that at pH = 7, Solution will be acidic.

Yhan neutral point 7 se jada hoga kitna jada hoga yeh humko nahi pata, isliye answer 7.1 galat hoga

Khoobsurat Baat

10. At temperature > 25° C –
    (A) pH = 7 Neutral (B) pH = 7 basic
    (C) pH = 7 may be acidic. (D) pH < 7 acidic
    (E) pH < 7 may be acidic (F) pH < 7 may be basic

Sol. (B),(E),(F)

pH CALCULATION OF STRONG ACID SOLUTION

C M HCl solution

Strong acids are strong electrolytes so, they dissociate completely in aqueous solution.

HCl	→	H^+	+ Cl^-
C		O	O
O		C	C
H_2O	⇌	H^+	+ OH^-

To Calculate the pH of strong acid solution, we need the total $[H^+]$ in the solution.

$[H^+]_{total} = [H^+]_{acid} + [H^+]_{Water}$

$pH = -\log [H^+]_{total}$

in formulae ko baithalo dimag ke kone mei, samjhe

11. Find out the pH of 10^{-4}M, 10^{-5}M and 10^{-6}M HCl solution.

Sol. 10^{-4} M HCl

$[H^+]_{acid} = 10^{-4}M$

$[H^+]_{total} = [H^+]_{acid} + [H^+]_{H_2O}$

$= 10^{-4} + [H^+]_{H_2O}$ $[[H^+]_{H_2O} \ll 10^{-4}M]$

$= 10^{-4}M$

$pH = -\log 10^{-4} = 4$

10^{-5} M HCl

$[H^+]_{acid} = 10^{-5}M$

$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O}$

$= 10^{-5} + [H^+]_{H_2O}$

$\approx 10^{-5}M$ $[[H^+]_{H_2O} \ll 10^{-5}M]$

$pH = -\log 10^{-5} = 5$

10^{-6} M HCl

$[H^+]_{acid} = 10^{-6}M$

$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O}$

$= 10^{-6} + [H^+]_{H_2O} = 10^{-6}M$ $[[H^+]_{H_2O} \ll 10^{-6}M]$

$pH = -\log 10^{-6} = 6$

Khoobsurat Baat

If $[H^+]_{acid} \geq 10^{-6}$ M then we will ignore $[H^+]$ contribution coming from water.

COMMON-ION EFFECT

When a solution contains a common ion, which is an ion that is present in both the reactants and products of a reaction, the equilibrium of the reaction is affected. The presence of the common ion suppresses the formation of the product, causing the reaction to shift in the direction of the reactants.

Example:

Consider the reaction:

$$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$$

If we add Na_2CO_3 (sodium carbonate) to the solution, the CO_3^{2-} ion is common to both the reactant and product. The presence of CO_3^{2-} ions from the sodium carbonate will suppress the formation of CO_3^{2-} ions from the calcium carbonate, causing the reaction to shift to the left.

is effect ko galti se bhi ignore mat kar dena

Khoobsurat Baat

$HCl \rightleftharpoons H^+ + Cl^-$

Common ion effect

$H_2O \rightleftharpoons H^+ + OH^-$

If $[H^+]_{acid} \geq 10^{-6}$ M then these $[H^+]_{acid}$ apply common ion effect on $[H^+]_{H_2O}$ and the equilibrium of dissociation of water shifts backward and $(H^+)_{H_2O}$ reduces further from 10^{-7} M at 25°C.

12. Find out the order of degree of dissociation of water in 10^{-3}M (α_1), 10^{-4}M (α_2), 10^{-5}M (α_3), 10^{-6}M (α_4) HCl solution.

(A) $(\alpha_1) > (\alpha_2) > (\alpha_3) > (\alpha_4)$

(B) $(\alpha_1) < (\alpha_2) < (\alpha_3) < (\alpha_4)$

(C) $(\alpha_1) = (\alpha_2) = (\alpha_3) = (\alpha_4)$

(D) None of these

Khoobsurat Baat

Sol. (B) More the $[H^+]_{acid}$, more will be the common ion effect on H_2O equilibrium so lesser will be the degree of dissociation of water.

bht badhiya sawaal hai

13. Contribution of $[H^+]$ coming from water in:

(a) 10^{-3}M HCl (b) 10^{-4}M HCl

(c) 10^{-5}M HCl (d) 10^{-6}M HCl

is in the order:

Sol. More the $[H^+]$ coming from acid, more will be the common-ion effect, more will be the backward shift, lesser will be the D.O.D and lesser will be the $[H^+]$ coming from water.

(a) < (b) < (c) < (d)

14. Find the pH of 10^{-7} M HCl.

Sol.

Is bachhe ne paani ko ignore kia hai lekin paani ignore nhi ho sakta kyunki $[H^+]_{acid} < 10^{-6}$ M

$pH = -\log 10^{-7}$

$pH = 7$

Is bacche ne paani ko ignore nahi kiya lekin Pani ki baat Sahi Se nahi suni

$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O}$

$[H^+]_{Total} = 10^{-7} + 10^{-7}$

$= 2 \times 10^{-7}$ M

$pH = -\log [H^+]_{Total}$

$= -\log 2 \times 10^{-7}$

$= 7 - \log 2$

$pH = 6.7$

Yeh mera lallu hai

* 10^{-7} M = $[H]_{acid}$
* x = $[H]^+$ coming from water after considering common-ion effect.
* $(x + 10^{-7})$ = total $[H]^+$ in solution after considering common-ion effect.

$[H^+]_{acid} = 10^{-7}$ M

$H_2O \rightleftharpoons H^+ + OH^-$

$x+10^{-7} \quad x$

$[H^+][OH^-] = K_w = 10^{-14}$

$(x + 10^{-7})\, x = 10^{-14}$ $x = ?$

$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O}$

$= 10^{-7} + x$

$pH = -\log(10^{-7} + x)$

(After putting the value of x, we will get the correct answer)

$pH = 6.79$

aaya samajh kab paani ignore hoga kab nahi

15. Find the pH of 10^{-8} M HCl

Sol. (6.98) For explanation, refer to QR code given on the last page.

16. Find the pH of 10^{-10} M HCl

Sol. $[H^+]_{acid} = 10^{-10}$ M

Pani ki bat sunni chahiye

Pani ki hi bat sunni chahiye!

Since the $[H^+]$ coming from acid is very less as compared to $[H^+]$ coming from water , so,

$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O} \approx [H^+]_{H_2O}$

$pH = -\log[H^+] = -\log 10^{-7} = 7$

paani ki baat kyu nhi manre bacho?

Khoobsurat Baat

$[H^+]_{acid} \geq 10^{-6}$ M [Acid will decide the pH]

10^{-8} M $\leq [H^+]_{acid} < 10^{-6}$ M

[Acid and water both collectively decide the pH of solution after considering common-ion effect]

$[H^+]_{acid} < 10^{-8}$ M [water will decide the pH and at 25°C, pH = 7]

17. Calculate pH of 10^{-14} M HCl.

(A) 14 (B) 6.5 (C) 7 (D) 0

Sol. (C) As $[H^+]_{acid} < 10^{-8}$ M

So, only the $[H]^+$ coming from water will be considered,

$[H]^+ = 10^{-7}$, pH = 7

Common Mistake

Students might evaluate pH as 14 without considering that pH of acid cannot be greater than 7.

PH CALCULATION FOR STRONG BASE

Base gives $[OH^-]$ ions.

$[OH^-]_{Total} = [OH^-]_{Base} + [OH^-]_{H_2O}$

if $[OH^-]_{Base} \geq 10^{-6}$ M Paani chup rahega X

$[OH^-]_{Base} < 10^{-6}$ M, Pani ko consider karna padega !

[Common-ion effect considered]

paani bhut zaruri hai, isko ignore karne ka sochna bhi mat

18. Find out the pH of 10^{-3} M in NaOH solution.

Sol. $[OH^-]_{Base} = 10^{-3}$ M

$pOH = -\log [OH^-] = -\log 10^{-3} = 3$

$pH = 14 - 3 = 11$

Khoobsurat Baat

$K_w = [H^+][OH^-] = 10^{-14}$ (At 25° C)

$-\log K_W = -\log[H^+] + -\log[OH^-]$

$-\log 10^{-14} = pH + pOH$

$\boxed{pH + pOH = 14}$

19. What mass of NaOH should be dissolved in sufficient water to get 20 m^3 of an aqueous solution of pH 7.3 at 25°C? [Ignore water]

(A) 0.16 g (B) 1.6×10^{-4} g

(C) 0.4 g (D) 0.12 g

Sol. (A) For explanation, refer to QR code given on the last page.

hai na acha sawaal

20. Millimoles of calcium hydroxide required to produce 100 mL of the aqueous solution of pH = 12 is $x \times 10^{-1}$. The value of x is _____ (Nearest integer).

[29 Jan, 2023 (Shift-I)]

Sol. $pH + pOH = 14$, $pOH = 14 - 12 = 2$

$[OH^-] = 10^{-2}$M

Moles of $OH^- = 10^{-2} \times 100/(1000) = 10^{-3}$ mol

$$\underset{5\cdot10^{-4}}{Ca(OH)_2} \longrightarrow \underset{5\cdot10^{-4}}{Ca^{2+}} + \underset{10^{-3}}{2OH^-}$$

Moles of $Ca(OH)_2$ in 100 mL = 5×10^{-4}

Number of milimoles = 5×10^{-1} mmol

pH OF DIPROTIC OR TRIPROTIC STRONG ACID

pH of CM H_2SO_4 Solution

H_2SO_4	$\rightarrow$	$2H^+$	+	SO_4^{2-}
C		0		0
0		2C		C

Case 1: $2C \geq 10^{-6}$ M means pani chup rahega, isliye,

$\boxed{pH = -\log 2C}$

Case 2: $2C < 10^{-6}$ M, But $\geq 10^{-8}$ M

means paani ki baat sahi se sunni padegi

$\boxed{pH = -\log (x + 2C)}$

ab socho x kaise niklega?

$K_W = (x + 2C)\,x$

$x = ?$

Case 3: $2C < 10^{-8}$ M

means ab paani hi decide krega, isliye

$\boxed{pH = 7}$

yeh mzaedaar hai, padhlo dhyan lagakar

Mixing of Strong Acids

HA	+	H_2B	$\rightarrow$	pH = ?
C_1M, V_1		C_2M, V_2		
(1)		(2)		(3)

$(nH^+)_3 = (nH^+)_1 + (nH^+)_2$ (nH^+ = Moles of H^+)

$(nH^+)_1 = C_1V_1$, $(nH^+)_2 = 2C_2V_2$

$(nH^+)_3 = C_1V_1 + 2C_2V_2$

$[H^+]_3$ = Total moles/total volume $= \dfrac{C_1V_1 + 2C_2V_2}{V_1 + V_2}$

$$\boxed{pH = -\log \frac{C_1V_1 + 2C_2V_2}{V_1 + V_2}}$$

formula ko yaad karlo

21. 200 mL of 0.01 HCl is mixed with 400 mL of 0.01 M H_2SO_4. The pH of the mixture is _____

Given: log 2 = 0.30, log 3 = 0.48, log 5 = 0.70, log 7 = 0.84, log 11 = 1.04

[29 July, 2022 (Shift-I)]

(A) 1.14 (B) 1.78

(C) 2.34 (D) 3.02

Sol. (B) First, calculate the resultant $[H^+]$ as:

$$[H^+] = \frac{(M_1V_1 + M_2V_2)}{(V_1 + V_2)}$$

$$[H^+] = \frac{(0.01 \times 200) + (0.01 \times 2 \times 400)}{600}$$

$$= \frac{2+8}{600} = \frac{10}{600} = \frac{1}{60}$$

$$pH = -\log[H^+]$$

$$= -\log\left[\frac{1}{60}\right] = 1.78$$

question ko samjho toh pehle kehna kya chahta hai

22. Calculate pH of mixture of

$$\left(400\text{ml}, \frac{1}{200}\text{M}H_2SO_4\right) + \left(400\text{ml}, \frac{1}{100}\text{M}HCl\right)$$

+ (200 ml of water)

Sol. (2.1) For explanation, refer to QR code given on the last page.

Common Mistake

Students might consider the $[H^+]$ ions coming from water.

23. Equal volumes of two HCl solutions of pH = 3 and pH = 5 were mixed. What is the pH of resulting solution?

(A) 3.5 (B) 4.0 (C) 4.5 (D) 3.3

Sol. (D) For explanation, refer to QR code given on the last page.

arey mushkil nahi hai

24. 600 mL of 0.01 M HCl is mixed with 400 mL of 0.01 M H_2SO_4. The pH of the mixture is _____ $\times 10^{-2}$. (Nearest integer)

[30 Jan, 2023 (Shift-I)]

[Given log 2 = 0.30, log 3 = 0.48
log 5 = 0.69 log 7 = 0.84
log 11 = 1.04]

Sol. Total milimoles of $H^+ = (600 \times 0.01) + (400 \times 0.01 \times 2) = 14$

$$[H^+] = \frac{\text{Total millimoles}}{\text{total volume(mL)}} = \frac{14}{1000} = 14 \times 10^{-3}\text{M}$$

$$pH = -\log[H]^+ = -\log[14 \times 10^{-3}]$$

$$= 1.86 \text{ or } 186 \times 10^{-2}$$

pH CALCULATION OF STRONG ACID AND STRONG BASE

Case 1: Let's consider HCl and NaOH:

HCl	+	NaOH	→	pH = ?
100 mL, 0.1 M		100 mL, 0.1 M		

HCl	+	NaOH	→	NaCl	+	H_2O
10mmol		10 mmol		0		
0		0		10 mmol		

In the container there is only solution of NaCl which is a salt of SA and SB. pH will be decided by water only that is pH will be equal to 7.

Case 2:

HCl	+	NaOH	→	pH = ?
100 mL, 0.1 M		100 mL, 0.1 M		

HCl	+	NaOH	→	NaCl	+	H_2O
10 mmol		3mmol		0		
7 mmol		0		3 mmol		

In the container we have 3mmol of NaCl and 7mmol of HCl left.

$$[H^+]_{left} = \frac{7}{130}\text{M}, \quad pH = -\log\left(\frac{7}{130}\right)$$

(Volume = 100 + 30 = 130 mL)

25. The pH of a solution obtained by mixing 50 mL of 1 M HCl and 30 mL of 1 M NaOH is $x \times 10^{-4}$. The value of x is_____. (Nearest integer)
[log 2.5 = 0.3979] **31 Aug, 2021 (Shift-II)**

Sol. 50 ml 1M HCl + 30 ml 1M NaOH

HCl	+	NaOH	→	NaCl	+	H_2O
50 mmol		30 mmol		0		
20 mmol		0		30 mmol		

$$[H^+]_{left} = \frac{20}{80} = \frac{1}{4}\text{M}$$

$$pH = -\log\frac{1}{4} = \log 4 = 0.6021 \text{ or } 6021 \times 10^{-4}$$

thoda toh hai tedha sawaal

26. 100 ml 0.1 M HCl + 150 ml 0.1 M NaOH
Calculate pH of the solution.

Sol. (12.3) For explanation, refer to QR code given on the last page.

27. Which of the following solutions will have pH close to 1.0? [IIT JEE 1992]

(A) 100 mL of $\left(\frac{M}{10}\right)$ HCl + 100ml of $\left(\frac{M}{10}\right)$ NaOH

(B) 55 mL of $\left(\frac{M}{10}\right)$ HCl + 45ml of $\left(\frac{M}{10}\right)$ NaOH

(C) 10 mL of $\left(\frac{M}{10}\right)$ HCl + 90ml of $\left(\frac{M}{10}\right)$ NaOH

(D) 75 mL of $\left(\frac{M}{15}\right)$ HCl + 25ml of $\left(\frac{M}{15}\right)$ NaOH

Sol. (D) For explanation, refer to QR code given on the last page.

daro mat bacho, asaan hi hai

28. The pK_a of acetyl salicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2–3 and the pH in the small intestine is about 8. Aspirin will be [IIT JEE 1988]

(A) unionised in the small intestine and in the stomach

(B) completely ionised in the small intestine and in the stomach

(C) ionised in the stomach and almost unionised in the small intestine

(D) ionised in the small intestine and almost unionised in the stomach

Sol. (D) pH is 2–3, i.e. strongly acidic in stomach and aspirin will be not ionised due to common ion effect as aspirin is also acidic in nature. Whereas, pH in small intestine is 8 , i.e., basic, hence as aspirin will be neutralised here.

OSTWALD DILUTION LAW

Generally weak electrolytes do not dissociate 100% in their aqueous solution. But on increasing the dilution of their aqueous solution, the degree of dissociation increases. Infact at infinite dilution, the degree of dissociation approaches to 1.

Note

Dilution ↑ D.O.D ↑

29. Which of CH_3COOH solution will have highest degree of dissociation?

(A) 0.1 M (B) 1.0 M

(C) 0.001 M (D) 0.01 M

Sol. (C) 0.001 M sol^n is least concentrated or most diluted.

pH CALCULATION FOR A SOLUTION OF WEAK ACID

$$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{C\alpha \cdot C\alpha}{C(1-\alpha)}$$

$$= \frac{C\alpha^2}{1-\alpha} \simeq C\alpha^2 [\text{Assuming } \alpha^2 << 1]$$

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{K_a \cdot C}$$

HA	$\rightleftharpoons H^+$	$+ A^-$
C	0	0
$C(1-\alpha)$	$C\alpha$	$C\alpha$
H_2O	$\rightleftharpoons H^+$	$+ OH^-$

$$pH = -\log C\alpha = -\log\sqrt{K_a \cdot C}$$

Important

For numericals in JEE Mains Paper

1. $\alpha << 1$
2. $\alpha = \sqrt{\frac{K_a}{C}}$
3. $C\alpha = [H^+] = \sqrt{K_a \cdot C}$
4. $pH = -\log\sqrt{K_a \cdot C}$
5. $[OH^-] = \frac{K_w}{[H^+]} = \frac{K_w}{C\alpha} = \frac{K_w}{\sqrt{K_a \cdot C}}$

sare formulae par dhyan lagao apna

Actual approach kya honi chaiye?

Step 1:

Sabse pehle copy ke kone mei jakr α ki value check karni hai that is

$$\alpha = \sqrt{\frac{K_a}{C}}$$

If $\alpha < 0.1$ or more precisely 0.05 तो बल्ले बल्ले

$$pH = -\log\sqrt{K_a \times C}$$

Step 2:

If $\alpha > 0.1$, or more precisely > 0.05, toh museebat agayi samjho

Solve this quadratic eq^n in terms of α

$[H^+]_{acid} = C\alpha$

If $[H^+]_{acid} \geq 10^{-6}$ M पानी चुप रहेगा

If $[H^+]_{acid} < 10^{-6}$ M पानी की बात सुननी पड़ेगी

$pH = -\log(x + C\alpha)$

ab socho x kaise niklega?

$(x + C\alpha)x = 10^{-14}$

30. pH of 0.1 M HCOOH acid solution ($K_a = 10^{-5}$)

Sol. $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{0.1}} = 0.01$, Balle balle

$pH = -\log\sqrt{K_a \times C}$

$= -\log\sqrt{10^{-5} \times 10^{-1}} = -\log 10^{-3} = 3$

31. What would be the concentration of CH_3COOH ($\alpha = 0.09$) for which its pH is equal to the pH of 10^{-6} M HCl solution? [IIT JEE]

Sol. p^H of HCl Sol^n $= -\log 10^{-6} = 6$

$10^{-6} = [H^+]_{CH_3COOH} = C(0.09)$

$C = \frac{10^{-6}}{0.09}$ M

32. Calculate pH of 10^{-6} M CH_3COOH ($K_a = 1.8 \times 10^{-5}$)

Sol.

CH_3COOH	$=$	CH_3COO^-	$+$	H^+
C		0		0
$C(1-\alpha)$		$C\alpha$		$C\alpha$

$K_a = \frac{c\alpha^2}{1-\alpha}$

$1.8 \times 10^{-5} = \frac{10^{-6}\alpha^2}{1-\alpha}$

On solving this equation, $\alpha = 0.95$

$[H^+]_{acid} = c\alpha = 10^{-6} \times 0.95$

$[H^+]_{acid} < 10^{-6}$ M पानी बोलेगा

$(x + 0.95 \times 10^{-6})x = 10^{-14}$

On solving, $x = 2.7$

$pH = -\log(x + 0.95 \times 10^{-6})$

$= -\log(2.7 + 0.95 \times 10^{-6}) = 0.43$

adratic equation toh kar loge na solve

33. K_a for butyric acid C_3H_7COOH is 2×10^{-5}. The pH of 0.2 M solution of butyric acid is _____ $\times 10^{-1}$ (Nearest integer)

[Given log 2 = 0.30] [28 July, 2022 (Shift-I)]

Sol. $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2 \times 10^{-5}}{0.2}} = 10^{-1} < 0.1$

$pH = -\log[H^+] = -\log\sqrt{K_a \times C}$

$= -\log\sqrt{2 \times 10^{-5} \times 0.2}$

$= -\log 2 \times 10^{-3}$

$= 3 - \log 2 = 2.7$

34. 0.2 M solution of HCOOH is 3.2% ionized then, find ionization constant of acid.

(A) 4.2×10^{-4} (B) 4.2×10^{-5}

(C) 2.1×10^{-4} (D) 2.1×10^{-5}

Sol. (C) For explanation, refer to QR code given on the last page.

bhayi asaaan hi hai

35. A weak acid HA has degree of dissociation x. Which option gives the correct expression of $(pH - pK_a)$? [JEE Mains-2025]

(A) $\log(1 + 2x)$ (B) 0

(C) $\log\left(\frac{x}{1-x}\right)$ (D) $\log\left(\frac{1-x}{x}\right)$

Sol. (C) We know the dissociation equation for a weak acid HA :

$K_a = \frac{Cx^2}{1-x}$

Taking log on both sides:

$\log K_a = \log C + 2\log x - \log(1 - x)$

Since $pH = -\log[H^+] = -\log(Cx)$, we get:

$pH = -\log C - \log x$

Similarly, $pK_a = -\log K_a$, so:

$pK_a = -\log C - 2\log x + \log(1 - x)$

Now, subtracting:

$pH - pK_a = (-\log C - \log x) - (-\log C - 2\log x + \log(1 - x))$

$= \log x - \log(1 - x)$

$= \log\frac{x}{1-x}$

36. The pH of 0.1 M solution of HQ is 3. Its K_a value is: [JEE Mains-2012]

(A) 1×10^{-7} (B) 3×10^{-7}

(C) 1×10^{-3} (D) 1×10^{-5}

Sol. (D) For explanation, refer to QR code given on the last page.

pH CALCULATION OF A SOLUTION OF WEAK BASE

Let C M be the concentration of BOH.

$BOH \rightleftharpoons$	B^+	+ OH^-
C	0	0
$C(1-\alpha)$	$C\alpha$	$C\alpha$

$[OH^-]_{Base} = C\alpha$

$$K_b = \frac{[B^+][OH^-]}{[BOH]} = \frac{C\alpha \cdot C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{1-\alpha} = C\alpha^2$$

$$\alpha = \sqrt{\frac{K_b}{C}}$$

$$C\alpha = [OH^-] = C\sqrt{\frac{K_b}{C}} = \sqrt{K_b \cdot C}$$

$$pH = -\log\sqrt{K_b \cdot C}$$

$$\boxed{pH = 14 - pOH}$$

37. The pH of 0.02 M ammonia solution which is 5% ionized will be

(A) 2 (B) 11

(C) 5 (D) 7

Sol. (B) For explanation, refer to QR code given on the last page

38. If 1 mM solution of ethylamine produces pH = 9, then the ionization constant (K_b) of ethylamine is 10^{-x}. The value of x is _____ (nearest integer). [23 Jan, 2025 (Shift-I)]

[The degree of ionization of ethylamine can be neglected with respect to unity.]

Sol. The concentration of ethylamine is 1 mM = 10^{-3} M

Given, pH = 9 ⇒ pOH = 5 ⇒ $[OH^-] = 10^{-5}$M

The reaction may be represented as:

$$C_2H_5NH_2(aq) + H_2O \rightleftharpoons C_2H_5NH_3^+ + OH^-$$

$C = 10^{-3}$M	-	-	
$C(1-\alpha)$	$C\alpha$	$C\alpha$	
$\Rightarrow C = 10^{-3}$	$=10^{-5}$	$=10^{-5}$	
$\because 1-\alpha = 1$			

As, we know that,

$$K_b = \frac{[C_2H_5NH_3^+][OH^-]}{[C_2H_5NH_2]} \Rightarrow K_b = \frac{10^{-5} \times 10^{-5}}{10^{-3}} = 10^{-7}$$

Therefore, the value of x = 7.

pH CALCULATION OF MIXTURE OF WEAK ACIDS

HA(C_1M, Dissociation constant- K_{a_1}),
HB(C_2M, dissociation constant- K_{a_2})

HA	$\rightleftharpoons A^- + H^+$	
$C_1(1-\alpha_1)$	$C_1\alpha_1$	$C_1\alpha_1 + C_2\alpha_2$
HB	$\rightleftharpoons B^- + H^+$	
$C_2(1-\alpha_2)$	$C_2\alpha_2$	$C_1\alpha_1 + C_2\alpha_2$

α_1 = D.O.D of HA in presence of HB
α_2 = D.O.D of HB in presence of HA

$[H^+]_{acid} = C_1\alpha_1 + C_2\alpha_2$

$$K_{a_1} = \frac{[H^+][A^-]}{[HA]} = \frac{(C_1\alpha_1 + C_2\alpha_2)\cancel{C_1}\alpha_1}{\cancel{C_1}(1-\alpha_1) = 1} \quad ...(1)$$

$$K_{a_2} = \frac{[H^+][B^-]}{[HB]} = \frac{(C_1\alpha_1 + C_2\alpha_2)\cancel{C_2}\alpha_2}{\cancel{C_2}(1-\alpha_2) = 1} \quad ...(2)$$

$$\boxed{\frac{K_{a_1}}{K_{a_2}} = \frac{\alpha_1}{\alpha_2}}$$

$K_{a_1} = \alpha_1(C_1\alpha_1 + C_2\alpha_2)$ ...(1)'

$K_{a_2} = \alpha_2(C_1\alpha_1 + C_2\alpha_2)$ ...(2)'

$$\alpha_1 = \frac{K_{a_1}}{\sqrt{K_{a_1}C_1 + K_{a_2}C_2}}, \quad \alpha_2 = \frac{K_{a_2}}{\sqrt{K_{a_1}C_1 + K_{a_2}C_2}}$$

$[H^+]_{acids} = C_1\alpha_1 + C_2\alpha_2$

$$= \frac{K_{a_1}C_1}{\sqrt{K_{a_1}C_1 + K_{a_2}C_2}} + \frac{K_{a_2}C_2}{\sqrt{K_{a_1}C_1 + K_{a_2}C_2}}$$

$$\boxed{[H^+]_{acids} = \sqrt{K_{a_1}C_1 + K_{a_2}C_2}}$$

formula toh agaya hoga samajh pakka

Important

The degree of dissociations of both the weak acids are less than their individual values (when they dissociate in a separate container) because H^+ coming from one acid will apply a common-ion effect on other acid.

α_1 = d.o.d of HA in presence of H B

α_2 = d.o.d of HB in presence of H A

$$\alpha_1 < \sqrt{\frac{K_{a_1}}{C_1}}, \quad \alpha_2 < \sqrt{\frac{K_{a_2}}{C_2}}$$

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$$\alpha_1 = \frac{Ka_1}{\sqrt{\sum Ka_i C_i}}$$

$$\alpha_2 = \frac{Ka_2}{\sqrt{\sum K_{a_i} C_i}}$$

$$\alpha_i = \frac{Ka_i}{\sqrt{\sum K_{a_i} C_i}}$$

$$[H^+]_{acid} = \sqrt{K_{a_1} C_1 + K_{a_2} C_2 + \cdots}$$

$$[H^+]_{acids} = \sqrt{\sum K_{a_i} C_i}$$

if $[H^+]_{acids} \geq 10^{-6}$ H_2O चुप

then $pH = -\log \sqrt{\sum K_{a_i} C_i}$

if $[H^+]_{acids} < 10^{-6}$ H_2O बोलेगा

the $pH = -\log \sqrt{\sum K_{a_i} c_i + K_w}$

sare formulae ko yaad karlo, bhut kaam aaenge

39. Calculate pH of solution obtained by mixing equal vol. of 0.02 M HOCl & 0.2M CH_3COOH Solution

given that $K_{a_1}(HOCl) = 2 \times 10^{-4}$ $(\log 2 = 0.3)$

$K_{a2}(CH_3COOH) = 2 \times 10^{-5}$

Also calculate OH^-, OCl^-, CH_3COO^-

Sol. $HA \rightarrow A^- = OCl^-$

$HB \rightarrow B^- = CH_3COO^-$

$[OCl^-] = C_1\alpha_1$, $[CH_3COO^-] = C_2\alpha_2$

$C_1 = 0.01$ M, $C_2 = 0.01$ M

Adding equal volumes leads to the concentration to half.

$$[H^+]_{acid} = \sqrt{K_{a_1} C_1 + K_{a_2} C_2}$$

$$= \sqrt{2 \times 10^{-4} \times 0.01 + 2 \times 10^{-5} \times 0.1}$$

$$= 2 \times 10^{-3} M$$

$$pH = -\log (2 \times 10^{-3}) M = 2.7$$

$$[OH^-] = \frac{K_W}{[H^+]} = \frac{10^{-14}}{2 \times 10^{-3}} = 5 \times 10^{-12} M$$

mushkil lag raha hoga, lekin hai nahi

40. The ratio of dod of HA(C_1 M) and HB(C_2M) when they dissociate separately assuming both the d.o.d are much less than 1.

Sol. $$\frac{\alpha_1 = \sqrt{\frac{K_{a_1}}{C_1}}}{\alpha_2 = \sqrt{\frac{K_{a_2}}{C_2}}} = \sqrt{\frac{K_{a_1} C_2}{K_{a_2} C_1}}$$

41. Find out the ratio of dod of two weak acids (HA & HB) in a common container having concentrations C_1 M and C_2 M respectively. Assuming d.o.d << 1.

Sol. $$\frac{\alpha_1 = \frac{K_{a_1}}{\sqrt{K_{a_1} C_1 + K_{a_2} C_2}}}{\alpha_2 = \frac{K_{a_2}}{\sqrt{K_{a_1} C_1 + K_{a_2} C_2}}} = \frac{K_{a_1}}{K_{a_2}}$$

Irrespective of their conc. the ratio of $\alpha_1 : \alpha_2 = K_{a_1} : K_{a_2}$.

seedha formula hi toh lgaya hai or kuch nahi kia

MIXTURE OF TWO WEAK BASES

For AOH, concentration-C_1M, dissociation constant-K_{b_1}

For BOH, concentration-C_2M, dissociation constant-K_{b_2}

$$\frac{\alpha_1}{\alpha_2} = \frac{K_{b_1}}{K_{b_2}}, \quad \alpha_i = \frac{K_{b_i}}{\sqrt{\sum K_{b_i} C_i}}$$

$$[OH^-]_{base} = \sqrt{\sum K_{b_i} C_i}$$

$$pOH = -\log \sqrt{\sum K_{b_i} C_i}$$

$pH = 14 - pOH$ [at 25°C]

waah waah, kitna asaan hai

pH CALCULATION FOR A WEAK DIPROTIC ACID

$H_2A \xrightarrow{K_{a_1}} \xrightarrow{K_{a_2}}$ $(K_{a_2} \ll K_{a_1})$

H_2A	$=$	HA^-	$+$ H^+ : K_{a_1}	
$C(1-\alpha_1)$		$C\alpha_1(1-\alpha_2)$	$C\alpha_1+C\alpha_1\alpha_2$	
HA^-	$=$	A^{2-}	$+$ H^+ : K_{a_2}	
$C\alpha_1(1-\alpha_2)$		$C\alpha_1\alpha_2$	$C\alpha_1 + C\alpha_1\alpha_2$	

Note

α_2 is very very small because the second ionisation constant(K_{a_2}) is very small and moreover due to common-ion effect from first equilibrium, the d.o.d of HA^- further decreases ($\alpha_2 \lll 1$)

$$K_{a_1} = \frac{[H^+][HA^-]}{[H_2A]} = \frac{(C\alpha_1 + C\alpha_1\alpha_2)C\alpha_1(1-\alpha_2)}{C(1-\alpha_1)}$$

$$= \frac{C\alpha_1(1+\alpha_2)^1 C\alpha_1(1-\alpha_2)^1}{C(1-\alpha_1)}$$

$$K_{a_1} = \frac{C\alpha_1^2}{(1-\alpha_1)} = C\alpha_1^2$$

$$\alpha_1 = \sqrt{\frac{Ka_1}{c}}$$

$$C\alpha_1 = [H^+]_{acid} = C\sqrt{\frac{Ka_1}{C}} = \sqrt{CK_{a_1}}$$

$$\boxed{pH = -\log\sqrt{K_{a_1}C}}$$

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* pH of any polyprotic weak acid is calculated by K_{a_1} value.
* $\alpha_1 = \sqrt{\frac{Ka_1}{C}}$
* $pH = -\log\sqrt{K_{a_1}C}$

$$K_{a_2} = \frac{[A^{2-}][H^+]}{[HA^-]}$$

$$= \frac{(C\alpha_1\alpha_2)(C\alpha_1 + C\alpha_1\alpha_2)}{C\alpha_1(1-\alpha_2)}$$

$$K_{a_2} = \frac{(C\alpha_1\alpha_2)C\alpha_1(1+\alpha_2)}{C\alpha_1(1+\alpha_2)} \quad (\alpha_2 \ll 1)$$

$$K_{a_2} = C\alpha_1\alpha_2$$

$$\boxed{\alpha_2 = \frac{K_{a_2}}{C\alpha_1} = \frac{K_{a_2}}{[H^+]}}$$

$$\boxed{[A^{2-}] = C\alpha_1\alpha_2 = K_{a_2}}$$

sar toh nahi chakra gaya, bacho

Note

$[A^{2-}] = K_{a_2}$ is only when $[H^+]$ are coming from H_2A weak acid only, no external involvement is there.

42. Calculate pH of 0.1M H_2S solution. Given that $K_{a_1} = 10^{-7}$, $K_{a_2} = 10^{-14}$

Sol. $\alpha_1 = \sqrt{\frac{K_{a_1}}{C}} = \sqrt{\frac{10^{-7}}{0.1}} = 10^{-3}$

$H_2S \rightleftharpoons HS^- + H^+$

$HS^- \rightleftharpoons S^{2-} + H^+$

$[H^+] = C\alpha_1 = \sqrt{K_{a_1}C} = \sqrt{10^{-7} \times 0.1} = 10^{-4}M$

$pH = 4$

$\alpha_2 = \frac{K_{a_2}}{[H^+]} = \frac{10^{-14}}{10^{-4}} = 10^{-10}$

$[S^{2-}] = C\alpha_1\alpha_2 = K_{a_2} = 10^{-14}$

43. Calculate $[S^{2-}]$ in a solution original having 0.1 M HCl and 0.1M H_2S, $K_{a_1} = 1.4 \times 10^{-7}$ and $K_{a_2} = 1.0 \times 10^{-14}$.

(A) 0.1 M (B) 2.8×10^{-20} M

(C) 2.8×10^{-22} M (D) 1.4×10^{-20} M

Sol. (D) $K_{a_1} \times K_{a_2} = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$

$1.4 \times 10^{-7} \times 10^{-14} = \frac{(0.1)^2[S^{2-}]}{(0.1)}$

$[S^{2-}] = 1.4 \times 10^{-20}$ M

44. Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature? **[24 Jan, 2025 (Shift-II)]**

(A) 45 mL CH_3COOH and 25 mL NaOH

(B) 50 mL HCl and 20 mL NaOH

(C) 30 mL CH_3COOH and 30 mL NaOH

(D) 30 mL HCl and 30 mL NaOH

Sol. (D) More moles of neutralization → More heat released → Higher temperature rise.

Largest temperature increase occurs when maximum moles are neutralized with a strong acid and strong base. Hence, option (D) is correct.

HYDROLYSIS OF SALTS

Salt hydrolysis is a chemical reaction where the ions of a salt react with water molecules to produce an acidic or basic solution. This occurs when either the cation or anion of the salt, or both, can react with water.

$$AB_{(s)} \xrightarrow{H_2O} AB_{(aq)} \longrightarrow A^+_{(aq)} + B^-_{(aq)}$$

(dissolution of salts)

Cationic Hydrolysis

$$A^+_{(aq)} + H_2O \rightleftharpoons AOH + H^+ \quad \text{[solution = acidic]}$$

Anionic Hydrolysis

$$B^-_{(aq)} + H_2O \rightleftharpoons BH + OH^- \quad \text{[solution = basic]}$$

Hydrolysis of Salt of Strong Acid + Strong Base (NaCl)

$$NaCl_{(s)} \longrightarrow NaCl_{(aq)} \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$$

$$Na^+_{(aq)} + H_2O \not\rightleftharpoons NaOH + H^+$$

$$Cl^-_{(aq)} + H_2O \not\rightleftharpoons HCl + OH^-$$

NaCN → Na^+ → NaOH (SB); Cl^- → HCl (WA)

Salt of (SA + SB) does not hydrolyse in water because anion & cation which are coming from SB + SA will form that acid & base when hydrolysed. Since S.B and S.A dissociates 100%, so no amount of cation and anion get hydrolysed.

Yeh bahut mazedaar hai

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pH of salt of (SA + SB) when dissolved in H_2O is decided by water only as there is no chance of Hydrolysis.

⇩

pH = 7 (neutral) at 25°

Hydrolysis of Strong Acid and Weak Base

Example: NH_4Cl

NH_4^+ → NH_4OH (W.B); Cl^- → HCl (S.A)

$$NH_4Cl_{(s)} \longrightarrow NH_4Cl_{(aq)} \longrightarrow NH^+_{4(aq)} + \cancel{Cl^-_{(aq)}}$$

$$NH^+_{4(aq)} + H_2O \rightleftharpoons \underset{\text{(weak base)}}{NH_4OH_{(aq)}} + H^+$$

Resultant solution should be acidic.

easy sa toh hai

pH calculation of a salt solution of weak base and strong acid

$$NH_4Cl_{(s)} \rightarrow NH_4Cl_{(aq)} \rightarrow NH^+_{4(aq)} + Cl^-_{aq}$$

$$NH^+_{4(aq)} + H_2O \overset{CM}{\rightleftharpoons} NH_4OH + H^+$$

	NH_4OH	$\rightleftharpoons$	NH_4^+	+	OH^-
	C		0		0
	C(1 - h)		Ch		Ch

Here, h= degree of hydrolysis

h= Moles hydrolysed/ Initial moles

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$K_h = \frac{[NH_4OH][H^+][OH^-]}{[NH_4^+][OH^-]} = \frac{K_w}{K_b} \simeq \frac{Ch \times Ch}{C(1-h)} = \frac{Ch^2}{1-h}$$

$Ch^2 [h \ll 1]$

$$h = \sqrt{\frac{K_w}{CK_b}}$$

$$[H^+] = Ch = C\sqrt{\frac{K_w}{CK_b}} = \sqrt{\frac{CK_w}{K_b}}$$

$$pH = -\log\sqrt{\frac{CK_w}{K_b}}$$

K_w = ionic product of water at 25°C is $K_w = 10^{-14}$
K_b = ionisation constant of weak base from which this cation is coming.
C = concentration of cation coming from salt.
Salt = ACl then conc of A^+ is equal to [salt].
Salt = A_2SO_4 then $[A^+] = 2[salt]$

mushkil kyu samajh rahe ho isko

45. 20 ml of 0.1 M NH_4OH is mixed with 40 mL of 0.05 M HCl. The pH of the mixture is nearest to: (Given K_b (NH_4OH) = 1×10^{-5}, log2 = 0.30, log3 = 0.48, log5 = 0.69, log7 = 0.84, log11 = 1.04) **[25 July, 2022 (Shift-1)]**

(A) 3.2 (B) 4.2 (C) 5.2 (D) 6.2

Sol. (C)

NH_4OH	+ HCl	→ NH_4Cl	+ H_2O
2 mmol	2 mmol	0	
0	0	2 mmol	

$$[NH_4^+] = \frac{2}{6}M$$

$$[H^+] = \sqrt{\frac{K_w C}{K_b}}$$

$$= \sqrt{\frac{10^{-14}}{10^{-5}} \times \frac{1}{30}} = \sqrt{\frac{10^{-9}}{30}} = \frac{10^{-5}}{\sqrt{3}}M$$

$$pH = -\log\left(\frac{10-5}{\sqrt{3}}\right) = 5 + \frac{1}{2}\log 3 = 5 + 0.24$$

$= 5.24$

aaya samajh ya pagla gaye?

46. The pH of a 0.02 M NH_4Cl solution will be [given K_b (NH_4OH) = 10^{-5} and log 2 = 0.301] **[10 Apr 2019]**

(A) 2.65 (B) 4.35 (C) 4.65 (D) 5.35

Sol. (D) For explanation, refer to QR code given on the last page.

Common Mistake

Students might incorrectly predict the nature of the salt.

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$$\frac{Ch^2}{1-h} = \frac{K_w}{K_b}$$

If $h = \sqrt{\frac{K_w}{CK_b}}$ is < 0.1 then apply all those formulas.

If $h \geq 0.1$ or absurd then solve $\frac{Ch^2}{1-h} = \frac{K_w}{K_b}$

$K_h = \frac{K_w}{K_b}$

$K_b \uparrow K_h \downarrow h \downarrow ch \downarrow [H^+] \downarrow pH \uparrow pOH \downarrow$

samajh aaye formulae ya phir dimag ghum gaya?

Important

Weaker the base from which that cation is coming more will be its hydrolysis more will be K_h value, more will be K_h value, more will be Ch value, lesser the pH value, more the pOH value.

yeh bahut important concept hai, isko zarur samajhana

47. Find out the decreasing order K_h, h, $[H^+]$, pH, pOH of a solution of different salts of weak base and strong acid.

Salt = ACl (CM), BCl (CM), CCl (CM), DCl (CM).
K_b of AOH = 2×10^{-5}, BOH = 10^{-5}
COH = 10^{-4}, DOH = 2×10^{-4}

Sol. Case of cationic hydrolysis

$A^+, B^+, C^+, D^+ \rightarrow$ hydrolysed

$K_h = \frac{K_w}{K_b}$ BCl > ACl > CCl > DCl → K_h

BCl > ACl > CCl > DCl → h
BCl > ACl > CCl > DCl → $[H^+]$
DCl > CCl > ACl > BCl → pH

48. 20 mL of 0.1 M NH_4OH is mixed with 40 mL of 0.05 M HCl . The pH of the mixture is nearest to : (Given K_b (NH_4OH) = 1×10^{-5}, log 2 = 0.30, log3 = 0.48, log 5 = 0.69, log7 = 0.84, log11 = 1.04) **[July, 2022 (Shift-I)]**

(A) 3.2 (B) 4.2
(C) 5.2 (D) 6.2

Sol. (C) $NH_4OH + HCl \rightleftharpoons NH_4Cl + H_2O$

$2.0 \times 0.1 = 2 \quad 0.05 \times 4 = 2$

$$\left[NH_4^+\right] = \frac{\text{Moles}}{V} = \frac{2}{60} = \frac{1}{30}$$

NH_4Cl is a salt of weak base and strong acid, so

$$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$$

$$= 7 - \frac{1}{2}\left(-\log\left(1\times10^{-5}\right)\right) - \frac{1}{2}\left(\log\frac{1}{30}\right)$$

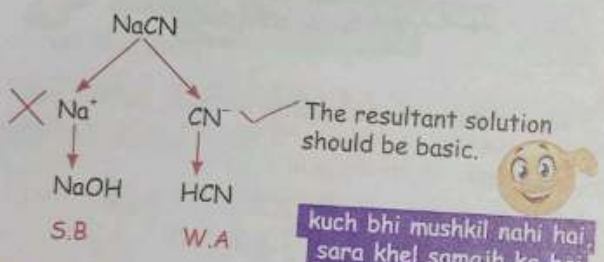
$$= 7 - 2.5 + 0.738 = 5.2$$

Hydrolysis of Strong Base and Weak Acid

Example - NaCN, CH_3COONa

$$NaCN_{(s)} \longrightarrow NaCN_{(aq)} \longrightarrow Na^+_{(aq)} + CN^-_{(aq)}$$

$$CN^-_{(aq)} + H_2O \rightleftharpoons HCN_{(aq)} + OH^-$$



The resultant solution should be basic.

kuch bhi mushkil nahi hai, sara khel samajh ka hai

pH calculation of a salt solution of weak acid and strong base

$$CH_3COONa_{(s)} \rightarrow \underset{CM}{CH_3COONa_{(aq)}} \rightarrow \underset{CM}{CH_3COO^-_{(aq)}} + \underset{CM}{Na^+_{(aq)}}$$

$$\underset{C}{CH_3COO^-_{(aq)}} + H_2O \rightleftharpoons \underset{O}{CH_3COOH_{(aq)}} + \underset{O}{OH^-_{(aq)}}$$

$$\underset{C(1-h)}{} \qquad \underset{Ch}{} \qquad \underset{Ch}{}$$

$$K_h = \frac{\left[CH_3COOH\right]\left[OH^-\right]\left[H^+\right]}{\left[CH_3COO^-\right]\left[H^+\right]} = \frac{K_w}{K_a} = \frac{Ch^2}{1-h} = Ch^2 \left[h<<1\right]$$

$$h = \sqrt{\frac{K_w}{Ck_a}}$$

$$\left[OH^-\right] = Ch = C\sqrt{\frac{K_w}{Ck_a}} = \sqrt{\frac{K_w}{Ck_a}}$$

$$\left[H^+\right] = \frac{K_w}{\left[OH^-\right]} = \frac{K_w}{\sqrt{\frac{CK_w}{K_a}}} = \sqrt{\frac{K_wK_a}{C}}$$

$$pH = -\log\sqrt{\frac{K_wK_a}{C}}$$

$$pH = -\frac{1}{2}\log\frac{K_wK_a}{C}$$

$$= -\frac{1}{2}\left[\log K_w + \log K_a - \log C\right]$$

$$= \frac{1}{2}\left[-\log K_w - \log K_a + \log C\right]$$

$$\boxed{pH = \frac{1}{2}\left[pK_w + pK_a + \log C\right]}$$

expression samajh aaya?

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$$K_h = \frac{K_w}{K_a}$$

$K_a \uparrow K_h \downarrow h \downarrow ch \downarrow \left[OH^-\right] \downarrow \left[H^+\right] \uparrow pH \downarrow pOH \uparrow$

Weaker the acid from which anion is coming, more will be the K_h value, more h value, more Ch value more $[OH^-]$, less $[H^+]$, more pH, less pOH value.

49. pH of K_2S solution at 25° C is

(A) 7 (B) less than 7

(C) more than 7 (D) 0

Sol. (A) It is formed from strong base KOH and weak acid H_2S.

It is a basic salt and hence, its pH will be more than 7.

50. Assuming that the degree of hydrolysis is small, the pH of 0.1 M sodium acetate solution ($K_a = 1.0 \times 10^{-5}$) will be -

(A) 5.0 (B) 6.0

(C) 8.0 (D) 9.0

Sol. (D) For explanation, refer to QR code given on the last page

kaafi mazedaar sawaal hai

51. What is the pH of 1 M CH_3COONa solution? K_a of acetic acid = 1.8×10^{-5}, $K_w = 10^{-14}$ $mol^2 L^{-2}$

(A) 2.4 (B) 3.6
(C) 4.8 (D) 9.4

Sol. (D) For explanation, refer to QR code given on the last page.

52. The dissociation constant of a substituted benzoic acid at 25° is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is.

[JEE 2009]

Sol. For a salt of weak acid and a strong base:

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

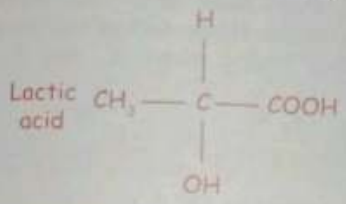
$$= 7 + \frac{1}{2}(-\log(1\times10^{-4})) + \frac{1}{2}\log 0.01 = 8$$

sawaal acha hai yeh, ignore mat karna

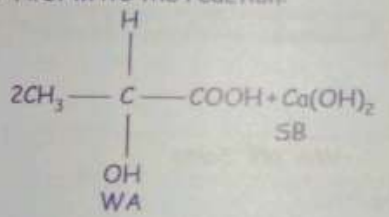
53. If the pK_a of lactic acid is 5, then the pH of 0.005 M calcium lactate solution at 25°C is ____ $\times 10^{-1}$ (Nearest integer)

[Jan. 2023 (Shift-II)]

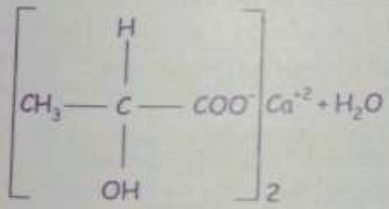
Lactic acid



Sol. First write the reaction:



WA SB



(calcium lactate) = (WA + SB)

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

$$= 7 + \frac{1}{2}(5) + \frac{1}{2}\log(0.005 \times 2) = 8.5$$

formula agar nahi pata toh yeh nahi hoga

54. HX is a weak acid [$K_a = 10^{-5}$]. It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is

[IIT JEE 2004]

(A) 0.01 % (B) 0.0001 %
(C) 0.1 % (D) 0.5%

Sol. (A) NaX is a weak acid – strong base salt.

$$\alpha = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{K_w}{K_a \times C}} = \sqrt{\frac{10^{-14}}{10^{-5}\times 0.1}} = 10^{-4}$$

So, % hydrolysis = 0.01

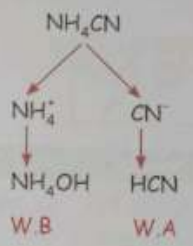
Hydrolysis of Weak Base and Weak Acid

Lets take the salt NH_4CN which consists of weak acid HCN and weak base NH_4OH.

$$NH_4CN_{(s)} \longrightarrow NH_4CN_{(aq)} \longrightarrow NH_{4(aq)}^{+} + CN_{(aq)}^{-}$$

$$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$$

$$CN_{(aq)}^{-} + H_2O \rightleftharpoons HCN + OH^-$$



resultant solution can be
- Acidic ($K_a > K_b$)
- Basic ($K_b > K_a$)
- Neutral ($K_a = K_b$)

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Common-ion effect will move the first eq^{bm} faster wrt the second equilibrium. So degree of hydrolysis of NH_4^+ ion increases faster as compared to CN^- ion. So the common degree of hydrolysis will be more than individual degree of hydrolysis.

kuch pada palle

pH calculation of a salt solution of WA + WB

$$NH_4CN_{(s)} \rightarrow \underset{CM}{NH_4CN_{(aq)}} \rightarrow NH^+_{4(aq)} + CN^-_{(aq)}$$

$$\underset{C}{NH^+_{4(aq)}} + \underset{C}{CN^-_{(aq)}} + H_2O \rightleftharpoons \underset{0}{NH_4OH_{(aq)}} + \underset{0}{HCN_{(aq)}}$$

$$\underset{C(1-h)}{} \quad \underset{C(1-h)}{} \qquad \underset{Ch}{} \qquad \underset{Ch}{}$$

$$K_h = \frac{[NH_4OH][OH^-][HCN][H^+]}{[NH_4^+][OH^-][CN^-][H^+]}$$

$$= \frac{K_w}{K_bK_a} = \frac{Ch.Ch}{C(1-h)C(1-h)} = \left(\frac{h}{(1-h)}\right)^2$$

$$\left(\frac{h}{1-h}\right)^2 = \frac{K_w}{K_aK_b} \cdot \left(\frac{h}{1-h}\right) = \sqrt{\frac{K_w}{K_aK_b}}$$

$$HCN_{(aq)} \rightleftharpoons CN^-_{(aq)} + H^+_{(aq)}$$

$$K_a = \frac{[CN^-][H^+]}{[HCN]} = \frac{C(1-h)[H^+]}{Ch} = \left(\frac{1-h}{h}\right)[H^+]$$

$$[H^+] = K_a\left(\frac{h}{1-h}\right) = K_a\sqrt{\frac{K_w}{K_aK_b}}$$

$$[H^+] = \sqrt{\frac{K_wK_a}{K_b}}$$

$$\boxed{pH = -\log\sqrt{\frac{K_wK_a}{K_b}}}$$

pura dhyan formulae par lagao bas

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- Degree of hydrolysis of a salt (WA + WB) is independent of the concentration of salt solution.
- pH is also independent of concentration of salt solution.

$[H^+] = \sqrt{\frac{K_wK_a}{K_b}}$ If $K_a > K_b$ [solution is acidic pH < 7 at 25° C]

If $K_b > K_a$ [solution is basic, pH > 7 at 25° C]

If $K_a = K_b$ [solution is neutral, pH = 7 at 25° C]

Important

Salt solution can be neutral because of two reasons:

1. No cationic and anionic hydrolysis
2. Hydrolysis of both cation and anion such that weak acid and weak base from which those cation and anions are coming, having same K_a and K_b value.

$$[H^+] = \sqrt{\frac{K_wK_a}{K_b}} = \sqrt{K_w} = \sqrt{10^{-14}} = 10^{-7}\ M$$

pH = neutral at 25° C.

kaafi mazedaar sawaal hai

Note

SA WB salts	WA SB Salts	WA WB Salts
Acidic	Basic	Almost neutral
Solution	Solution	Solution
Cationic hydrolysis	Anionic hydrolysis	Hydrolysis by both cation and anion
$K_h = \frac{K_w}{K_b}$	$K_h = \frac{K_w}{K_a}$	$K_h = \frac{K_w}{K_a \times K_b}$
$[H^+] = C \cdot h = \sqrt{\frac{K_w \cdot C}{K_b}}$	$[OH^-] = C \cdot h = \sqrt{\frac{K_w \cdot C}{K_a}}$	$[H^+] = K_a \cdot h = \sqrt{\frac{K_w \cdot K_a}{K_b}}$
$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$	$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$	$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$

55. The degree of hydrolysis of an anion B^- is 0.05. The degree of hydrolysis of a cation A^+ is 0.08. Find out the degree of hydrolysis of both the cation and anion when they hydrolysis in a common container.

(A) 0.05, 0.05 (B) 0.08, 0.08
(C) 0.05, 0.08 (D) 0.1, 0.1
(E) 0.1, 0.15

Sol. (D) 1. Common degree of hydrolysis should be greater than individual degree of hydrolysis.

2. Should be same.

56. pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is [JEE 2017]

(A) 7.2 (B) 6.9 (C) 7.0 (D) 1.0

Sol. (B) For a salt of weak acid and weak base, the formula to calculate pH is:

$$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

Substitute the values in the above expression:

$$pH = 7 + \frac{1}{2}(3.2) - \frac{1}{2}(3.4) = 7 + 1.6 - 1.7 = 6.9$$

sara khel formula ka hai or kuch hai hi nahi

57. Degree of hydrolysis of a salt of weak monobasic acid and weak monoacidic base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.24 M then its degree of hydrolysis is - [Mains - 2025]

(A) 100% (B) 50%
(C) 25% (D) None of these

Sol. (B) The degree of hydrolysis of a salt of weak acid and weak base do not depend on concentration of salt. Hence, the degree of hydrolysis remains 50%.

thoda tedha sawaal hai

Common Mistake

Students might calculate the degree of hydrolysis but in this, there is no need to calculate this.

Agar tumne calculate kia, tum lallu ho!

58. 0.01 M NaX solution is 1% hydrolysed. Find the ionization constant of weak acid HX and pH of NaX solution.

(A) 10^{-6}, 8 (B) 10^{-8}, 9
(C) 10^{-8}, 8.5 (D) 10^{-8}, 10

Sol. (D) For explanation, refer to QR code given on the last page.

sara khel samajh ka hai

59. The chloride salt of a certain weak monoacidic organic base is hydrolysed to an extent of 3% in its 0.1 M solution at 25°C. What is the dissociation constant of the base?

(A) 1×10^{-10} (B) 1×10^{-9}
(C) 3.33×10^{-9} (D) 3.33×10^{-10}

Sol. (A) For explanation, refer to QR code given on the last page.

60. Which salt solution is neutral?

(A) NaCl (B) CH_3COONa
(C) CH_3COONH_4 (D) NH_4Cl
(E) NH_4CN

Sol. (A,C) For explanation, refer to QR code given on the last page.

samajh aya ya nahi?

Hydrolysis of a Polyanion (Dianion or Trianion)

1. C M Na_2CO_3 Solution

K_{a_1} and K_{a_2} of H_2CO_3 are given

$H_2CO_3 \rightleftharpoons HCO_3^- + H^+ ; K_{a_1}$

$HCO_3^- \rightleftharpoons CO_3^{2-} + H^+ ; K_{a_2}$

$\Big\} K_{a_1} >> K_{a_2}$

$$Na_2CO_{3(s)} \rightarrow Na_2CO_{3(aq)} \rightarrow Na^+_{(aq)} + CO^{2-}_{3(aq)}$$

$$CO^{2-}_{3(aq)} + H_2O \rightleftharpoons HCO^-_{3(aq)} + OH^-_{(aq)} ; \quad K_{h_1} = \frac{K_w}{K_{a_2}}$$

$$HCO^-_{3(aq)} + H_2O \rightleftharpoons H_2CO_{3(aq)} + OH^-_{(aq)} ; \quad K_{h_2} = \frac{K_w}{K_{a_1}}$$

$K_{h_1} >>> K_{h_2}$, because $K_{a_2} << K_{a_1}$

Kyunki first equilibrium hi deciding factor or isme K_{a_2} k_a involvement hai, isliye anionic hydrolysis mei K_a ki jagah K_{a_2} ajaega.

$$\left[H^+\right] = \sqrt{\frac{K_w K_{a_2}}{C}}$$

The hydrolysis constant of second eq^{bm} is small as compare to first equilibrium. so it will face more common ion effect and OH^- coming from this eq^{bm} will be negligible as compared to first equilibrium.

yeh aaya samajh?

2. C M Na_3PO_4 solution

$$\left.\begin{aligned} H_3PO_4 &\rightleftharpoons H_2PO_4^- + H^+ : K_{a_1} \\ H_2PO_4^- &\rightleftharpoons HPO_4^{2-} + H^+ : K_{a_2} \\ HPO_4^{2-} &\rightleftharpoons PO_4^{3-} + H^+ : K_{a_3} \end{aligned}\right\} \quad K_{a_1} > K_{a_2} > K_{a_3}$$

$$Na_3PO_{4(s)} \rightarrow Na_3PO_{4(aq)} \rightarrow Na^+_{(aq)} + PO^{3-}_{4(aq)}$$

$$PO^{3-}_{4(aq)} + H_2O \rightleftharpoons HPO^{2-}_{4(aq)} + OH^-_{(aq)} ; K_{h_1} = \frac{K_w}{K_{a_3}}$$

$$HPO^{2-}_{4(aq)} + H_2O \rightleftharpoons H_2PO^-_{4(aq)} + OH^-_{(aq)} ; K_{h_2} = \frac{K_w}{K_{a_2}}$$

$$H_2PO^-_{4(aq)} + H_2O \rightleftharpoons H_3PO_{4(aq)} + OH^-_{(aq)} ; K_{h_3} = \frac{K_w}{K_{a_1}}$$

$$\left[H^+\right] = \sqrt{\frac{K_w K_{a_3}}{C}}$$

formulae par focus karo

Hydrolysis of Amphiprotic Anion

1. C M $NaHCO_3$

$$\left.\begin{aligned} H_2CO_3 &= HCO_3^- + H^+ : K_{a_1} \\ HCO_3^- &= CO_3^{2-} + H^+ : K_{a_2} \end{aligned}\right\} \quad K_{a_2} < K_{a_1}$$

$$NaHCO_{3(s)} \rightarrow NaHCO_{3(aq)} \rightarrow Na^+_{(aq)} + HCO^-_{3(aq)}$$

$$H_2CO_3 \xleftarrow{K_{a_1}} HCO_3^- \xrightarrow{K_{a_2}} CO_3^{2-}$$

$$\left[H^+\right] = \sqrt{K_{a_1} \times K_{a_2}}$$

$$pH = -\log\left[H^+\right] = -\log\sqrt{K_{a_1} \times K_{a_2}} = \frac{pK_{a_1} + pK_{a_2}}{2}$$

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

Kyu ghabra rahe ho bacho?

Note

pH is independent of the concentration of salt solution.

2. C M Na_2HPO_4

$$\left.\begin{aligned} H_3PO_4 &= H_2PO_4^- + H^+ : K_{a_1} \\ H_2PO_4^- &= HPO_4^{2-} + H^+ : K_{a_2} \\ HPO_4^{2-} &= PO_4^{3-} + H^+ : K_{a_3} \end{aligned}\right\} \quad K_{a_3} < K_{a_2} < K_{a_1}$$

$$Na_2HPO_4(s) \rightarrow Na_2HPO_{4(aq)} \rightarrow Na^+_{(aq)} + HPO^{2-}_{4(aq)}$$

$$H_2PO_4^- \xleftarrow{K_{a_2}} HPO_4^{2-} \xrightarrow{K_{a_3}} PO_4^{3-}$$

$$\left[H^+\right] = \sqrt{K_{a_2} \times K_{a_3}}$$

$$pH = \frac{pK_{a_2} + pK_{a_3}}{2}$$

Note

pH is independent of concentration of salt solution.

3. C M NaH_2PO_4

$$NaH_2PO_{4(s)} \rightarrow NaH_2PO_{4(aq)} \rightarrow Na^+_{(aq)} + H_2PO^-_{4(aq)}$$

$$H_3PO_4 \xleftarrow{Ka_1} H_2PO_4^- \xrightarrow{Ka_2} HPO_4^{2-}$$

$$\left[H^+\right] = \sqrt{Ka_1 \times Ka_2}$$

$$pH = \frac{pKa_1 + pKa_2}{2}$$

Note

pH is independent of concentration of salt solution.

araha hai maza?

4. C M NaH_2PO_3

$H_3PO_3 \rightarrow H^+ + H_2PO_3^-$: Ka_1
$H_2PO_3^- \rightarrow HPO_3^{2-} + H^+$: K_{a_2}
$\Big\}$ $K_{a_1} > K_{a_2}$

$NaH_2PO_{3(s)} \rightarrow NaH_2PO_{3(aq)}) \rightarrow Na^+_{(aq)} + H_2PO^-_{3(aq)}$

$H_3PO_3 \xleftarrow{K_{a_1}} H_2PO_3^- \xrightarrow{K_{a_2}} HPO_3^{2-}$

$[H^+] = \sqrt{K_{a_1} * K_{a_2}}$

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

Note

Independent of conc. of salt solution.

5. C M NaH_2PO_2

$H_3PO_2 = H_2PO_2^- + H^+$: K_a

$NaH_2PO_{2(s)} \rightarrow NaH_2PO_{2(aq)} \rightarrow Na^+_{(aq)} + H_2PO^-_{2(aq)}$

$H_3PO_2 \xleftarrow{K_a} H_2PO_2^- \xrightarrow{K_{a_2}} HPO_2^{2-}$

Kya bewkoofi kar rahe ho, H_3PO_2 toh monobasic acid hai isliye $H_2PO_2^-$ se further H^+ Nhi nikal sakta, isliye $H_2PO_2^-$ amphiprotic anion nhi hai isliye calculations normal mono anionic hydrolysis ke pH calculation ki tarah hongi.

$$[H^+] = \sqrt{\frac{K_w K_a}{C}}$$

Important

$H_2PO_2^-$ = anion of H_3PO_2 weak acid which is monobasic.

6. C M Na_2HPO_3

$H_3PO_3 \rightarrow H^+ + H_2PO_3^-$: K_{a_1}

$H_2PO_3^- \rightleftharpoons H^+ + HPO_3^{2-}$: K_{a_2}

H_3PO_3 = dibasic weak acid

$NaH_2PO_{3(s)} \rightarrow Na_2HPO_{3(aq)}) \rightarrow Na^+_{(aq)} + H_2PO^{2-}_{3(aq)}$

$HPO_3^{2-} + H_2O \rightleftharpoons H_2PO_3^- + OH^-$; $K_{h_1} = \frac{K_w}{K_{a_2}}$

$H_2PO_3^- + H_2O \rightleftharpoons H_3PO_3 + OH^-$; $K_{h_2} = \frac{K_w}{K_{a_1}}$

$$[H^+] = \sqrt{\frac{K_w K_{a_2}}{C}}$$

Formulae toh karne hi padenge, uske bina sawaal nahi niklega

61. The pH of ammonium phosphate solution, if pK_a of phosphoric acid and pK_b of ammonium hydroxide are 5.23 and 4.75 respectively, is _____. **[26 Feb, 2021 (Shift - II)]**

Sol. In this, both acid and base will undergo hydrolysis.

$$pH = 7 + \frac{pK_a}{2} - \frac{pK_b}{2} = 7 + \left(\frac{5.23 - 4.75}{2}\right) = 7.24$$

BUFFER SOLUTION

A buffer solution is a solution that resists changes in pH upon the addition of an acid or a base. It maintains a relatively stable pH level, making it crucial in various chemical processes and biological system.

K_b CH_3COO^-
K_a CH_3COOH
$\Big\}$ conjugate acid - base $\rightarrow K_w = K_a \times K_b$

For any conjugate acid - base pair

$$K_w = K_a \times K_b$$
$$14 = pK_a + pk_b$$

ab samajhte hai zabardast buffer kaunse hote hain?

Acidic Buffer

An acidic buffer solution is a solution that resists changes in pH when small amounts of acid or base are added to it. It typically consists of a weak acid and its conjugate base.

Addition of acid

When a strong acid like HCl is added to the buffer, the H^+ ions from the acid react with the acetate ions (CH_3COO^-) from the sodium acetate to form acetic acid. This reaction consumes the added H^+ ions, preventing a significant decrease in pH as the equilibrium constant of this reaction is $1/K_a$.

$$H^+(aq) + CH_3COO^-(aq) \rightarrow CH_3COOH(aq)$$

ddition of base

hen a strong base like NaOH is added, the OH^- ions om the base react with the acetic acid form water d acetate ions. This reaction consumes the added H^+ ions, preventing a significant increase in pH as e equilibrium constant of this reaction is K_a/K_w

$OH^-_{(aq)} + CH_3COOH(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$

Khoobsurat Baat

- Strong acid added to an acidic buffer solution will be consumed by conjugate base. If 'x' mole of strong acid is added then amount of conjugate base is decreased by 'x' mole and amount of weak acid is increased by 'x' mole.

Note

The added strong base is consumed by acid. If 'y' mole of strong base is added then amount of acid is reduced by 'y' and amount of conjugate base is increased by 'y' mole.

H calculation of acidic buffer

$H_3COOH \rightarrow CH_3COO^- + H^+$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}, \quad [H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]}$$

$$[H^+] = \frac{K_a[Acid]}{[C.B]}$$

$$pH = -\log\frac{K_a[Acid]}{[C.B]}$$

$$pH = pK_a + \log\frac{[C.B]}{[Acid]}$$ (general formula)

pagal toh nahi ho gaye?

Khoobsurat Baat

$= pK_a + \log\frac{[salt]}{[Acid]}$ assuming [C.B] = [salt]

salt is of 1:1 stoichiometry

62. For an acidic buffer containing (CH_3COO^- and CH_3COOH)

(A) $[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]}$

(B) $pH = pK_a + \log\frac{[\text{conjugate base}]}{[Acid]}$

(C) $pH = pK_a + \log\frac{[salt]}{[Acid]}$

(D) All of these

Sol. (D) Formulas given for an acidic buffer in all options are correct.

63. For acidic buffer $\Rightarrow CH_3COOH + (CH_3COO)_2Ca$

(A) $[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]}$

(B) $pH = pK_a + \log\frac{[salt]}{[Acid]}$

(C) $pH = pK_a + \log\frac{[C.B]}{[Acid]}$

(D) $pH = pK_a + \log 2\frac{[salt]}{[Acid]}$

Sol. (A,C,D) $(CH_3COO)_2Ca \rightarrow 2\,CH_3COO^- + Ca^{+2}$

$[CH_3COO^-] = 2[salt]$

So, except option (B), all are correct

64. The dissociation constant of acetic is $x \times 10^{-5}$. When 25 ml of 0.2 M CH_3COONa solution is mixed with 25 mL of 0.02 M CH_3COOH solution, the pH of the resultant solution is found to be equal to 5. The value of x is _____. [24 Jan 2023 Shift-I]

Sol. $pH = pK_a + \log\frac{[\text{conj base}]}{[Acid]}$

$5 = pK_a + \log\frac{0.1}{0.01}$, $pK_a = 4$

$K_a = 10^{-4} = 10 \times 10^{-5} = 10^{-4}$

65. A student needs to prepare a buffer solution of propanoic acid and its sodium salt with pH 4. The ratio of $\frac{[CH_3CH_2COOH^-]}{[CH_3CH_2COOH]}$ required to make buffer is _____.

Given: K_a (CH_3CH_2COOH) = 1.3×10^{-5}

[28 June, 2022 Shift-II]

(A) 0.03 (B) 0.13

(C) 0.23 (D) 0.33

Sol. (B) For explanation, refer to QR code given on the last page.

66. 50 ml of 0.1 M CH_3COOH is being titrated against 0.1 M NaOH. When 25 ml of NaOH has been added, the pH of the solution will be ____ $\times 10^{-2}$. (Nearest integer)

[26 June, 2022 (Shift-I)]

(Given : pK_a (CH_3COOH) = 4.76)

Log2 = 0.30 Log3 = 0.48

Log5 = 0.69 Log7 = 0.84

Log11 = 1.04

Sol. $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

5 mmol	2.5 m mol	0
2.5	0	2.5

$$pH = pK_a + \log \frac{[salt]}{[Acid]}$$

$$= 4.76 + \log \frac{2.5}{2.5} = 4.76 \text{ or } 476 \times 10^{-2}$$

Direct formula use karna hai

67. In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is _____ M. (Round off to the nearest integer).

[18 March, 2021(Shift-I)]

[Given : pK_a(acetic acid) = 4.74]

Sol. (10) For explanation, refer to QR code given on the last page.

tedha lagra hoga lekin hai nhi, bharosa karo

68. pH of a mixture containing 0.1 M X^- and 0.2 M HX is: [$pK_b(X^-) = 4$]

(A) 4 + log 2 (B) 4 - log 2

(C) 10 + log2 (D) 10 - log 2

Sol. (D) HX = weak acid, X^- = conj Base,

$K_a \times K_b = Kw$

$pK_a + pK_b = pK_w = 14$

$pK_a = 14 - pK_b = 10$

Acidic buffer:

$$pH = pK_a + \log \frac{[C.B]}{[Acid]}$$

$$= 10 + \log \frac{0.1}{0.2} = 10 - \log 2$$

69. K_a for HCN is 5×10^{-10}. For maintaining a constant pH 9, the volume of 5 M KCN solution required to be added to 10 ml of 2 M HCN solution is:

(A) 4 ml (B) 8 ml

(C) 2 ml (D) 10 ml

Sol. (C) Let the volume of KCN added = V ml

Total volume = (10 + V) ml

[HCN] = 2 M

$$[H^+] = K_a \frac{[HCN]}{[CN^-]}$$

$$10^{-9} = 5 \times 10^{-10} \frac{\left(\frac{2 \times 10}{10+V}\right)}{\left(\frac{5V}{10+V}\right)}$$

$$2 = \frac{2 \times 10}{5V}, \; V = 2 \text{ ml}$$

Yeh sawal acha hai isko dekho dhyan se or dimag lagao

70. A solution is prepared by mixing 0.01 mol each of H_2CO_3, $NaHCO_3$, Na_2CO_3 and NaOH in 100 ml of water. pH of the resulting solution is_____.

[JEE Adv. 2022]

[Given: pK_{a_1} and pK_{a_2} of H_2CO_3 are 6.37 and 10.32, respectively; log 2 = 0.30

Sol. $NaOH + H_2CO_3 \rightarrow NaHCO_3$

0.01	0.01	0
0	0	0.01 mol

Total $NaHCO_3$ = 0.01 + 0.01 = 0.02

Na_2CO_3 = 0.01

$$pH = pK_{a_2} + \log \frac{[CO_3^{2-}]}{[HCO_3^-]}$$

$$= 10.32 + \log \frac{0.01}{0.02} = 10.02$$

71. 50 ml of 0.2 M solution of an acid HA ($K_a = 10^{-5}$) and 50 ml of a NaA solution are given. What should be the concentration of NaA solution to make a buffer solution with pH = 4 upon mixing the two?

Sol. (0.02 M) For explanation, refer to QR code given on the last page.

direct formula lagaya hai bas

72. 20 ml of 0.1M NaOH is added to 50ml of 0.1M acetic acid solution. The pH of the resulting solution is $__ \times 10^{-2}$ (Nearest integer)

Given: $pK_a (CH_3COOH) = 4.76$

Log2 = 0.30

Log3 = 0.48 **[13 April, 2023 (Shift-II)]**

Sol. The neutralization reaction is given as,

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

Initially	5 mmol	2 mmol	0	0
After Rxn	3 mmol	0 mmol	2 mmol	0

The pH of resulting solution is calculated as,

$$pH = pK_a + \log_{10}\frac{[salt]}{[Acid]} \quad pH = 4.76 + \log_{10}\frac{2}{3}$$

$pH = 4.58 = 458 \times 10^{-2}$

NTA ans key → [448]

PW ans key →[458]

badhiya sawaal hai, chhod mat dena

Basic Buffer

$NH_4OH = NH_4^+ + OH^-$

$NH_4Cl \rightarrow NH_4^+ + Cl^-$

$$[NH_4^+]_{total} = [\text{conjugate acid}] = [NH_4^+]_{Base} + [NH_4^+]_{salt}$$

$$[NH_4^+]_{total} = [NH_4^+]_{salt} \qquad \left[[NH_4^+]_{salt} >> [NH_4^+]_{base}\right]$$

$$= \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$[OH^-] = \frac{K_b[NH_4OH]}{[NH_4^+]}$$

$$\boxed{pOH = pK_b + \log\frac{[\text{conj acid}]}{\text{Base}}}$$

73. pOH for this buffer ($NH_4OH + (NH_4)_2SO_4$)

(A) $pOH = pK_b + \log\frac{[NH_4^+]}{[NH_4OH]}$

(B) $pOH = pK_b + \log\frac{[salt]}{[Base]}$

(C) $pOH = pK_b + \log\frac{2[salt]}{[Base]}$

(D) none of these

Sol. (A,C) $(NH_4)_2SO_4 \rightarrow 2NH_4^+ + SO_4^{2-}$

$C \longrightarrow 2C$

Both options (A) and (C) are correct

Yeh sawal dimag ki batti jaga dega

74. a mole NH_4OH + b mole NH_4Cl

If x mole of HCl is added then

find out the difference b/w the initial and final pH value?

Sol. $NH_4OH = NH_4^+ + OH^-$

$NH_4Cl \rightarrow NH_4^+ + Cl^-$

If x mol of H^+ is added:

$$NH_4OH + H^+ = NH_4^+ + H_2O$$

	a	x	b
	a − x	0	b + x

$$pOH = pK_b + \log\frac{[C.A]}{[Base]}$$

$$(pOH) = pK_b + \log\frac{[b]}{[a]}$$

$$(pOH) = pK_b + \log\frac{[b+x]}{[a-x]}$$

prabhu ko yaad karo sawaal nikalne se pehle

75. A litre of buffer solution contains 0.1 mole of each of NH_3 and NH_4Cl. On the addition of 0.02 mole of HCl by dissolving gaseous HCl, the pH of the solution is found to be _____ $\times 10^{-3}$ (Nearest integer)

[25 Jan, 2023(Shift-1)]

[Given: $pK_b[NH_3] = 4.745$]
log 2 = 0.301 log 3 = 0.477
T = 298K]

Sol. $NH_4OH + H^+ \longrightarrow NH_4^+ + H_2O$

0.1	0.02	0.1
(0.1 - 0.02)	0	(0.1 + 0.02)
(0.08)		(0.12)

$$pOH = pK_b + \log\frac{[CA]}{[Base]}$$

$$= 4.745 + \log\left(\frac{0.12}{0.08}\right) = 4.745 + (\log 3 - \log 2)$$

$pOH = 4.745 + 0.18 = 4.925$

$pH = 14 - 4.925 = 9.075$

76. The OH^- concentration in a mixture of 5.0 mL of 0.0504 M NH_4Cl and 2 mL of 0.0210 M NH_3 solution is $x \times 10^{-9}$ M. The value of x is. (Nearest integer) **(26 Aug 2021)**

[Given $K_w = 1 \times 10^{-14}$ and $K_b = 1.8 \times 10^{-5}$]

Sol. First write the reaction as:

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

Calculate K_b as:

$$\text{So, } K_b = \frac{[NH_4^+][HO^-]}{[NH_3]}, \quad [HO^-] = \frac{K_b \times [NH_3]}{[NH_4^+]}$$

$$= 1.8 \times 10^{-5} \times \frac{2}{5} \times \frac{210}{504} = 3 \times 10^{-6}$$

asaan hai bacho, ghabrao mat

77. Class XII students were asked to prepare one litre of buffer solution of pH 8.26 by their Chemistry teacher. The amount of ammonium chloride to be dissolved by the student in 0.2 M ammonia solution to make one litre of the buffer is(Given : $pK_b(NH_3) = 4.74$)

[26 July, 2022 (Shift-II)]

Molar mass of NH_3 = 17 g mol^{-1} Molar mass of NH_4Cl = 53.5 g mol^{-1})

(A) 53.5g (B) 72.3g
(C) 107.0g (D) 126.0g

Sol. (C) For explanation, refer to QR code given on the last page.

Khoobsurat Baat

In acidic buffer if we add 'x' mole of acid then amount of (C.B) decreased by 'x' and amount of acid is increased by 'x' mole. In acidic buffer if we add 'y' mole of base then amount of acid is reduced by 'y mol' and amount of (CB) is increased by y mol.

And

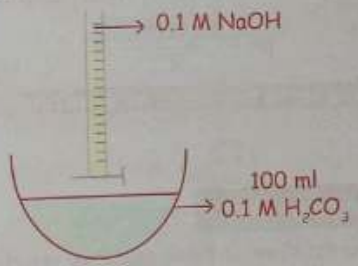
In basic buffer if we add 'x' mole of acid then amount of base is reduced by 'x' mole and amount of conjugate acid is increased by x mole. In basic buffer if we add 'y' mole of base then amount of conjugate acid is reduced by 'y' mole and amount of base is increased by 'y' mole.

sar ka pakadlo or prabhu ke gun gaao

78. Given: Ka_1 and Ka_2 values of H_2CO_3

Find out the pH of the solution when -

(i) 0 mL NaOH added
(ii) 50 mL NaOH added
(iii) 60 mL NaOH added
(iv) 100 mL NaOH added



Sol. (i) 0 ml NaOH added

Bartan says - I have a weak diprotic acid

$$[H^+] = \sqrt{K_{a_1} C}$$

$$pH = -\log\sqrt{K_{a_1} C}$$

(ii) 50 ml NaOH

$NaOH + H_2CO_3 \longrightarrow NaHCO_3 + H_2O$

$50 \times 0.1 = 5$ mol $100 \times 0.1 = 10$ mmol 0

0 5 5

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

Bartan says – I have 5 mml of H_2CO_3 + 5 mmol of $NaHCO_3$

$$pH = pK_{a_1} + \log \frac{[C \cdot B]}{[Acid]}$$

$$pH = pK_{a_1} + \log \frac{5}{5}, \quad pH = pK_{a_1}$$

(iii) $NaOH + H_2CO_3 \longrightarrow NaHCO_3 + H_2O$

6 mmol 10 mmol 0

0 4 6

Bartan says – (I have W.A = 4 m.mol + conj base = 6 mmol)

$$pH = pK_{a_1} + \log \frac{[C \cdot B]}{[Acid]}$$

$$pH = pK_{a_1} + \log \frac{6}{4}$$

(iv) $NaOH + H_2CO_3 \longrightarrow NaHCO_3 + H_2O$

10 mmol 10 mmol 0

0 0 10

Bartan says – I have only $NaHCO_3$ (salt)

Na^+ HCO_3^- (amphiprotic)

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

yeh sawaal kar lia toh matlab sab agaya samajh

Khoobsurat Baat

Jisko bartan ki awaaz sunna agaya, wo yeh chapter seekh gaya

Buffer Capacity

It is defined as the number of moles of strong acid (or strong base) added to one litre of a buffer solution to change its pH by one unit.

- Addition of strong acid or strong base.
- Per lt of Buffer solution
- Change in pH = 1 unit.

Let one lt of buffer solution having 'b' moles of salt and 'a' moles of weak acid.

$$(pH)_1 = pK_a + \log \frac{b}{a}$$

If we add 'x mole of strong acid -

$$(pH)_2 = pK_a + \log \frac{b-x}{a+x}$$

$$\Delta pH = (pH)_1 - (pH)_2 = \log \frac{b}{a} - \log \frac{b-x}{a+x}$$

$$d\frac{\Delta pH}{dx} = \frac{1}{2.303} \frac{a+b}{(a+x)(b-x)}$$

$$\text{Buffer capacity} = \frac{dx}{d\Delta pH} = \frac{2.303(a+x)(b-x)}{a+b}$$

x will be very much less than a and b

$$\frac{dx}{d\Delta pH} = 2.303\left(\frac{ab}{a+b}\right)$$

For maximum buffer capacity

$$\frac{d}{db}\left[\frac{dx}{d\Delta pH}\right] = 0$$

$a - b = 0, \quad a = b$

Note

For maximum Buffer capacity of acidic buffer [Acid] = (conjugate Base]

For maximum Buffer capacity of basic buffer [Base] = [conjugate acid]

iski language par mat jao iska bhaav samjho

79. Calculate the buffer capacity of 1lt of

(i) 0.1 M CH_3COOH and 0.1 M CH_3COONa

(ii) 0.2 M CH_3COOH and 0.2 M CH_3COONa

pK_a of CH_3COOH = 4. 74

which will you say a better buffer?

Sol. $B.C_1 = 2.303\left(\frac{ab}{a+b}\right) = 2.303\left(\frac{0.1 \times 0.1}{0.1+0.1}\right)$

$B \cdot C_2 = 2.303\left(\frac{a'b'}{a'+b'}\right) = 2.303\left(\frac{0.2 \times 0.2}{0.2+0.2}\right)$

$B \cdot C_2 > B \cdot C_1$

kaafi mazedaar sawaal hai

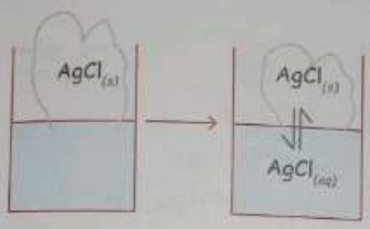
SOLUBILITY OF SALT

Solubility is the maximum amount of a substance that can be dissolved in a given amount of solvent at a specific temperature. When a salt dissolves in water, it dissociates into its constituent ions. This process reaches a state of equilibrium, known as **solubility equilibrium**.

In terms of gram, solubility is known as gram solubility

In terms of moles, solubility is known as molar **solublity**

Lets consider AgCl salt:



At any instant, $[Ag^+][Cl^-]$ = ionic product (Q)

$$AgCl_{(s)} \rightleftharpoons AgCl_{(aq)}$$

(Undissolved) (dissolved)

$$AgCl(s) \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$$

(dissolved salt exists as $(Ag^+(aq))$ and $(Cl^-(aq))$)

$K_{sp} = [Ag^+][Cl^-]$ = solubility product

Important

- If ionic product is less than K_{sp}, dissolution will occur.
- If ionic product is equal to K_{sp}, then saturation point has reached.
- If ionic product is greater than K_{sp}, then precipitation takes place.

Relation between K_{sp} and Solubility of Salts

1. **AB type salt**

Let the solubility of AB = S mol/Lt

$$AB_{(s)} \rightleftharpoons \underset{S}{A^+_{(aq)}} + \underset{S}{B^-_{(aq)}}$$

'S' mole घुल गये, जो घुल गये वो टूट गये

$$K_{sp} = [A^+][B^-] = S \times S$$

$$\boxed{S = \sqrt{K_{sp}}}$$

2. **AB_2 type salt**

Let the solubility of AB_2 = 'S' mol/Lt

$$AB_{2(s)} \rightleftharpoons \underset{S}{A^{2+}_{(aq)}} + \underset{2S}{2B^-_{(aq)}}$$

$$K_{sp} = [A][B]^2 = (S)(2S)^2$$

$$\boxed{K_{sp} = 4S^3}$$

3. **A_2B_3 Type salt**

Let the solubility of $A_2B_{3(s)}$ = 'S'mol/Lt

$$A_2B_{3(s)} \rightleftharpoons \underset{2S}{2A^{+3}_{(aq)}} + \underset{3S}{3B^{-2}_{(aq)}}$$

$$K_{sp} = [A]^2[B]^3 = (2S)^2(3S)^3$$

$$\boxed{K_{sp} = 108\ S^5}$$

4. **A_xB_y type salt**

Let the solubility of A_xB_y salt = 'S' mol/Lt

$$A_xB_{y(s)} \rightleftharpoons \underset{xS}{xA} + \underset{yS}{yB}$$

$$K_{sp} = [A]^x[B]^y = (xS)^x(yS)^y$$

$$k_{sp} = x^x y^y S^{x+y}$$

$$\boxed{S = \left[\frac{K_{sp}}{x^x y^y}\right]^{\frac{1}{x+y}}}$$

samjhe ke naahi?

80. A_3B_2 is a sparingly soluble salt of molar mass M(g mol^{-1}) and solubility x g L^{-1}. The solubility product satisfies $K_{sp} = a\left(\frac{x}{M}\right)^5$. The value of a is _____ (Integer answer)

[31 Aug, 2021 (Shift-I)]

Sol. $$A_3B_{2(S)} \rightleftharpoons \underset{3S}{3A^{2+}} + \underset{2S}{2B^{3-}}$$

Let the solubility = S mol/Lt

$$K_{sp} = [A^{2+}]^3[B^{3-}]^2 = (3S)^3(2S)^2$$

$$K_{sp} = 108\ S^5 = 108\left(\frac{x}{M}\right)^5$$

81. Relation between solubility S mole/Lt with K_{sp} of $Zr_3(PO_4)_4$ salt.

[(JEE Mains 2014, 8 April 2019)]

(A) $S = \left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$ (B) $S = \left(\frac{K_{sp}}{144}\right)^{\frac{1}{7}}$

(C) $S = \left(\frac{K_{sp}}{(6912)}\right)^{\frac{1}{7}}$ (D) $S = \left(\frac{K_{sp}}{6912}\right)^{7}$

Sol. (D) $A_3B_4 \rightarrow 3A^{4+} + 4B^{3-}$

$K_{sp} = [A^{4+}]^3[B^{3-}]^4 = (3S)^3(4S)^4$

$K_{sp} = 3^3 4^4 S^7 = S = \left(\frac{K_{sp}}{6912}\right)^{7}$

Common Mistake

It is important to write the correct dissociation reaction of given species otherwise it will lead to conceptual error.

dimag me kuch hui halchal?

82. The solubility of $Mg(OH)_2$ is 'S' mol/Lt. The Solubility product under same condition:

(A) $4S^3$ (B) $3S^4$

(C) $4S^2$ (D) S^3

[AIEEE-2002]

Sol. $Mg(OH)_2 \rightleftharpoons \underset{S}{Mg^{2+}} + \underset{2S}{2OH^-}$

$K_{sp} = [Mg^{+2}][OH^-]^2 = S(2S)^2 = 4S^3$

83. If the solubility of Lithium Sodium hexafluoridoaluminate $Li_3Na_3[AlF_6]_2$ is 's'mol/Lt then its solubility product -

(A) $18\ S^3$ (B) $186624\ S^8$

(C) $1458\ S^8$ (D) $2916\ S^8$

Sol. $3Li + 3Na + 2\ [AlF_6]$

$3S \quad 3S \quad 2S$

$k_{sp} = (3S)^3(3S)^3(2S)^2 = 2916\ S^8$

isme complex ka dissociation aana zaruri hai tabhi sawaal kr paoge

84. Let the solubility of an aqueous solution of $Mg(OH)_2$ be x, then its K_{sp} is [JEE 2002]

(A) $4x^3$ (B) $108\ x^5$

(C) $27x^4$ (D) $9x$

Sol. Consider the dissociation of $Mg(OH)_2$:

$Mg(OH)_2 \rightleftharpoons \underset{x}{Mg^{2+}} + \underset{2x}{2OH^-}$

$K_{sp} = x(2x)^2 = 4x^3$

85. The solubility in water of a sparingly soluble salt AB_2 is 1.0×10^{-5} mol L^{-1}. Its solubility product will be [JEE 2003]

(A) 4×10^{-10} (B) 1×10^{-15}

(C) 1×10^{-10} (D) 4×10^{-15}

Sol. (D) Consider the dissociation of AB_2 as:

$AB_2 \rightarrow A^{2+} + 2B^-$

$1 \quad S \quad 2S$

$K_{sp} = [A^{2+}][B^-]^2$

$= (S)(2S)^2 = 4S^3$

$= 4\ (1 \times 10^{-5})^3 = 4 \times 10^{-15}$

86. Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12}. The ratio of their molar solubilities [16 Mar 2021]

i.e. $\frac{S(A_2X)}{S(MX)}$ = (Round off to the Nearest Integer)

Sol. (50) For explanation, refer to QR code given on the last page.

dekh toh lo sawaal ko ki kya puch raha hai?

87. The solubility of $Ca(OH)_2$ is water is:
[Given : The solubility product of $Ca(OH)_2$ in water = 5.5×10^{-6}] [25 Feb 2019]

(A) 1.11×10^{-6} (B) 1.11×10^{-2}

(C) 1.77×10^{-6} (D) 1.77×10^{-2}

Sol. (B) Consider the dissociation of $Ca(OH)_2$ as:

$Ca(OH)_2 \rightleftharpoons \underset{S}{Ca^{2+}}(aq) + \underset{2S}{2OH^-}(aq)$

$K_{sp} = S(2S)^2 \Rightarrow 5.5 \times 10^{-6} = 4S^3$

Finally, calculate S as:

$S = \left(\frac{5.5}{4}\right)^{\frac{1}{3}} \times 10^{-2} = 1.11 \times 10^{-2}$

88. The molar solubility (in mol L^{-1}) of a sparingly soluble salt MX_4 is 's'. The corresponding solubility product is K_{sp}. 's' is given in term of K_{sp} by the relation : [JEE Adv.2004]

(A) $s = (256K_{sp})^{\frac{1}{5}}$ (B) $s = (128\,K_{sp})^{\frac{1}{4}}$

(C) $s = \left(\frac{K_{sp}}{128}\right)^{\frac{1}{4}}$ (D) $s = \left(\frac{K_{sp}}{256}\right)^{\frac{1}{5}}$

Sol. (D) For explanation, refer to QR code given on the last page.

89. Solubility of calcium phosphate (molecular mass, M) in water is W_g per 100 mL at 25° C. Its solubility product at 25°C will be approximately. **[01 Feb, 2024 (Shift-II)]**

(A) $10^7\left(\frac{W}{M}\right)^3$ (B) $10^7\left(\frac{W}{M}\right)^5$

(C) $10^3\left(\frac{W}{M}\right)^5$ (D) $10^5\left(\frac{W}{M}\right)^5$

Sol. (B) $S = \frac{W \times 10}{M}$

$$Ca_3(PO_4)_2(s) \rightleftharpoons \underset{3S}{3Ca^{2+}(aq.)} + \underset{2S}{2PO_4^{3-}(aq.)}$$

$$S = \frac{W \times 1000}{M \times 100} = \frac{W \times 10}{M}$$

$$K_{sp} = (3S)^3 (2S)^2 = 108\,S^5$$

$$= 108 \times 10^5 \times \left(\frac{W}{M}\right)^5$$

$$= 1.08 \times 10^7 \left(\frac{W}{M}\right)^5$$

90. The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-3}. The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be: **[9 Jan, 2020 (Shift-II)]**

(A) $(18 \times 10^{-31})^{\frac{1}{4}}$ (B) $(4.86 \times 10^{-29})^{\frac{1}{4}}$

(C) $(18 \times 10^{-31})^{\frac{1}{2}}$ (D) $(2.22 \times 10^{-31})^{\frac{1}{4}}$

($K_{sp} = 6.0 \times 10^{-31}$)

Sol. (A) For explanation, refer to QR code given on the last page.

91. Arrange the following in increasing order of solubility product : **(28 Jan, shift-II, 2025)**

$Ca(OH)_2$, AgBr, PbS, HgS

(A) PbS < HgS < $Ca(OH)_2$ < AgBr

(B) HgS < AgBr < PbS < $Ca(OH)_2$

(C) $Ca(OH)_2$ < AgBr < HgS < PbS

(D) HgS < PbS < AgBr < $Ca(OH)_2$

Sol. (D) Compounds with higher K_{sp} values are more soluble in water, while compounds with lower K_{sp} values are less soluble.

Based on the K_{sp} values, the increasing order of solubility product is:

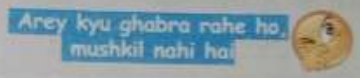
HgS < PbS < AgBr < $Ca(OH)_2$.

92. K_{sp} for $Cr(OH)_3$ is 1.6×10^{-30} What is the molar solubility of this salt in water?

(A) $\sqrt[4]{\frac{1.6 \times 10^{-30}}{27}}$ (B) $\sqrt[5]{1.8 \times 10^{-30}}$

(C) $\frac{1.8 \times 10^{-30}}{27}$ (D) $\sqrt[2]{1.6 \times 10^{-30}}$

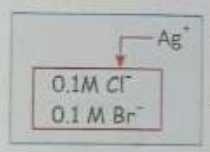
Sol. (A) For explanation, refer to QR code given on the last page.



Selective Precipitation

Selective precipitation is a technique used to separate ions from a solution by adding a reagent that precipitates one ion but not the other. This technique relies on the differences in the solubility product constants (K_{sp}) of different ionic compounds.

Case-1 :



$(K_{sp})_{AgCl} = 10^{-10}$, $(K_{sp})_{AgBr} = 10^{-13}$

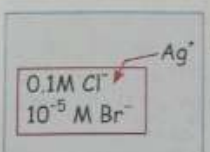
$$[Ag^+]_{AgCl} = \frac{(K_{sp})_{AgCl}}{[Cl^-]} = \frac{10^{-10}}{0.1} = 10^{-9}\,M$$

$$[Ag^+]_{AgBr} = \frac{(K_{sp})_{AgBr}}{[Br^-]} = \frac{10^{-13}}{0.1} = 10^{-12}\,M$$

Requirement of $[Ag^+]$ for the precipitation of AgBr is less than the requirement of $[Ag^+]$ for AgCl precipitation.

AgBr precipitates first.

Case-2 :



$(K_{sp})_{AgCl} = 10^{-10}$, $(K_{sp})_{AgBr} = 10^{-13}$

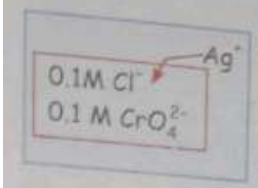
$$[Ag^+]_{AgCl} = \frac{10^{-10}}{0.1} = 10^{-9}\,M$$

$$[Ag^+]_{AgBr} = \frac{10^{-13}}{10^{-5}} = 10^{-8}\,M$$

Requirement of $[Ag^+]$ for the ppt^n of AgCl is less tha the requirement of $[Ag^+]$ for the ppt^n of AgBr.

AgCl precipitates first.

Case-3:



$(K_{sp})_{AgCl} = 10^{-10}$, $(K_{sp})_{Ag_2CrO_4} = 10^{-13}$

$$[Ag^+]_{AgCl} = \frac{10^{-10}}{0.1} = 10^{-9}\,M$$

$Ag_2CrO_{4(s)} \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

$K_{sp} = [Ag^+]^2\,[CrO_4^{2-}]$

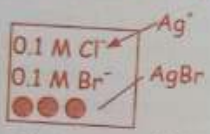
$$[Ag^+] = \sqrt{\frac{K_{sp}}{[CrO_4^{2-}]}} = \sqrt{\frac{10^{-13}}{0.1}} = 10^{-6}\,M$$

The requirement of $[Ag^+]$ for the ppt^n of AgCl is less than the ppt^n of Ag_2CrO_4.

AgCl precipitates first.

yeh concept khopdi ko hila dega pura

93. Let's consider two salts:



$(K_{sp})_{AgCl} = 10^{-10}$

$(K_{sp})_{AgBr} = 10^{-13}$

(i) Which salt precipitates first?

(ii) Find out the % Br^- left in the container when AgCl starts precipitating?

Sol. (i) $$[Ag^+]_{AgCl} = \frac{10^{-10}}{0.1} = 10^{-9}\,M$$

(AgBr precipitates first.)

$$[Ag^+]_{AgBr} = \frac{10^{-13}}{0.1} = 10^{-12}\,M$$

As the precipitation of AgBr proceeds the requirement of $[Ag^+]$ increases for the precipitation of AgBr only as $[Br^-]$ in solution phase decreases. As soon as $[Ag^+]$ required for AgBr precipitation meets the requirement for AgCl precipitation, AgCl starts precipitating also.

(ii) $(K_{sp})_{AgBr} = [Ag^+]\,[Br^-]$

$$[Br^-] = \frac{(K_{sp})_{AgBr}}{[Ag^+]} = \frac{10^{-13}}{10^{-9}} = 10^{-4}\,M$$

$$\%\,[Br^-]_{left} = \frac{10^{-4}}{0.1} \times 100 = 0.1\%$$

dimag ghum gaya kya ya sahi hai?

94. A solution is 0.1M in Cl^- and 0.001 M in CrO_4^{2-}. Solid $AgNO_3$ is gradually added to it. Assuming that the addition does not change in volume K_{sp} (AgCl) = $1.7 \times 10^{-10}\,M^2$ and K_{sp} (Ag_2CrO_4) = $1.9 \times 10^{-12}\,M^3$.

Select correct statement from the following: [20 July, 2021 (Shift-II)]

(A) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low.

(B) Ag_2CrO_4 precipitates first as its K_{sp} is low.

(C) AgCl will precipitate first as the amount of Ag^+ needed to precipitate is low.

(D) AgCl precipitates first because its K_{sp} is high.

Sol. (C) For explanation, refer to QR code given on the last page.

isko padhkar zarur mza aagya hoga!

95. An acidified solution of 0.05 M Zn^{2+} is saturated with 0.1 M H_2S. What the minimum molar concentration (M) is of H^+ required to prevent the precipitation of ZnS?

Use K_{sp} (ZnS) = 1.25×10^{-22} and overall dissociation constant of H_2S, $K_{net} = K_1 K_2 = 1 \times 10^{-21}$ [JEE Adv. 2020]

Sol. $K_{sp} = [Zn^{2+}]\,[S^{2-}]$

$$[S^{2-}] = \frac{K_{sp}}{[Zn^{2+}]} = \frac{1.25 \times 10^{-22}}{0.05} = 2.5 \times 10^{-21}\,M$$

$H_2S_{(aq.)} \rightleftharpoons 2H^+_{(aq)} + S^{2-}_{(aq)}$

$$K_1 K_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$$

$$[H^+]^2 = \frac{K_1 K_2\,[H_2S]}{[S^{2-}]} = \frac{10^{-21} \times 0.1}{2.5 \times 10^{-21}} = \frac{1}{25}$$

$$[H^+] = \frac{1}{5}\,M = 0.2\,M$$

SALT OF A STRONG ACID AND STRONG BASE (AgCl)

(i) In pure water

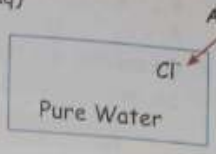
$AgCl_{(s)} \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$S_1 \quad S_1$

$K_{sp} = [Ag^+][Cl^-]$

$S_1 \times S_1$

$S_1 = \sqrt{K_{sp}}$

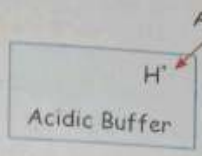


(ii) Acidic or basic buffer

$AgCl(s) \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

$S_2 \quad S_2$

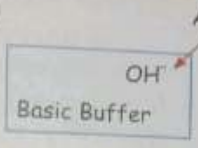
$S_2 = \sqrt{K_{sp}}$



$AgCl(s) = Ag^+(aq) + Cl^-(aq)$

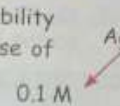
$S_3 \quad S_3$

$S_3 = \sqrt{K_{sp}}$



(iii) In a solution having common - ion

In presence of a common-ion solubility of Salt decreases. This is because of common-ion effect.



$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

$S_4 + 0.1 \quad S_4$

0.1M $AgNO_3$ = 0.1 M Ag^+ and 0.1 M NO_3^-

$K_{sp} = [Ag^+][Cl^-] = (S_4 + 0.1)\,S_4 \; [S_4 \ll 0.1] = 0.1\,S_4$

$S_4 = 10\,K_{sp}$

(iv) Simultaneous solubility

Simultaneous solubility refers to the solubility of two or more sparingly soluble salts in a common solvent, often water. When these salts share a common ion, their solubilities are influenced by each other.

Let's consider two salts: AgCl and AgBr.

$AgCl(s) \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

$S_5 + S$

$AgBr(s) \rightleftharpoons Ag^+_{(aq)} + Br^-_{(aq)}$

$S_5 + S$

Solubility of AgCl in presence of AgBr.

$S_5 < \sqrt{(K_{sp})_{AgCl}}$

$S_5 < \sqrt{(K_{sp})_{AgBr}}$

*Purpose: To find out the solubility of AgCl in Presence of AgBr

$(K_{sp})_{AgCl} = [Ag^+][Cl^-] = (S_5 + S)S_5$ ...(1)

$(K_{sp})_{AgBr} = [Ag^+][Br^-] = (S_5 + S)S$ ...(2)

$$\frac{S_5}{S} = \frac{(K_{sp})_{AgCl}}{(K_{sp})_{AgBr}}$$

Simultaneous solubility ko halke mei mat lo, yeh bahut important hai

(v) In a solution capable of forming complex ion with one component of salt

* In this case solubility increase

$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

$Ag^+_{(aq)} + 2NH_{3(aq)} \rightleftharpoons [Ag(NH_3)_2]^+_{(aq)}$

Let C M be the concentration of NH_3.

* In this case solubility increases drastically due to complex formation. The K_f or complex formation is very high.

96. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol /L) of Ag_2CrO_4 in a 0.1 M $AgNO_3$.

(A) 1.1×10^{-11} (B) 1.1×10^{-10}

(C) 1.1×10^{-12} (D) 1.1×10^{-9}

Sol. (B) $AgNO_3$ dissociates into Ag^+ and NO_3^-, so $[Ag^+] = 0.1$ M.

Now,

$Ag_2CrO_{4(S)} \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

$2S + 0.1 \quad S$

$K_{sp} = [Ag^+]^2\,[CrO_4^{-2}] = (2S + 0.1)^2 S$

$K_{sp} \approx 0.01S$

$S = 100\,K_{sp} = 100 \times 1.1 \times 10^{-12} = 1.1 \times 10^{-10}$

yeh sawaal dimag ko jagaega

97. The solubility of $CdSO_4$ in water is 8.0×10^{-4} mol L^{-1}. Its solubility in 0.01 M H_2SO_4 solution is _____ $\times 10^{-6}$ mol L^{-1}.

[18 March, 2021 (Shift-II)]

Sol. (64) For explanation, refer to QR code given on the last page.

arey common ion effect bhi agaya

98. The solubility product of $BaSO_4$ is 1×10^{-10} at 298 K. The solubility of $BaSO_4$ in 0.1 M K_2SO_4(aq) solution is _____ $\times 10^{-9}$ g L^{-1} (nearest integer).

Given: Molar mass of $BaSO_4$ is 233 g mol^{-1}

[18 April 2023 (Shift-II)]

Sol. $K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$

0.1M 0.2M 0.1M

$BaSO_4 \rightleftharpoons Ba^{+2} + SO_4^{2-}$

a –S S S + 0.1 = 0.1

$K_{sp} = S \times 10^{-1}$

$\Rightarrow 1 \times 10^{-10} = S \times 10^{-1}$

$\Rightarrow 1 \times 10^{-10} \times 10^{-1} = 10^{-9}$ mol L^{-1}

So, $S = 10^{-9} \times 233$ g L^{-1}

99. What is the molar solubility of $Al(OH)_3$ in 0.2 M NaOH solution? Given that, solubility product of $Al(OH)_3 = 2.4 \times 10^{-24}$:

[12 April, 2019 (Shift-I)]

(A) 12×10^{-23} (B) 12×10^{-21}

(B) 3×10^{-19} (D) 3×10^{-22}

Sol. (D) $Al(OH)_3 \rightleftharpoons Al^{+3} + 3\,OH^-$

$S' = 0.2 + 3(S') = 0.2$

$S' \times (0.2)^3 = K_{sp} = 2.4 \times 10^{-24}$

$(S') = 3 \times 10^{-22}$ M

bahut simple hai

100. If K_{sp} of Ag_2CO_3 is 8×10^{-12}, the molar solubility of Ag_2CO_3 in 0.1 M $AgNO_3$ is

[12 Jan 2019]

(A) 8×10^{-12} M (B) 8×10^{-11} M

(C) 8×10^{-10} M (D) 8×10^{-13} M

Sol. (C) For explanation, refer to QR code given on the last page.

101. The solubility product of PbI_2 is 8.0×10^{-9}. The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$ mol/ L. The value of x is_____. (Rounded off to the nearest integer) [Given $\sqrt{2} = 1.41$]

[24 Feb. 2021]

Sol. Lets consider the dissociation of $Pb(NO_3)_2$:

$Pb(NO_3)_2 \rightarrow Pb^{+2}_{(aq)} + 2\,NO_3^-$ (aq)

0.1 M – –

– 0.1M 0.2M

Similarly,

$PbI_2(s) \rightleftharpoons Pb^{+2}(aq) + 2I^-(aq)$

s 2s

$= s + 0.1 = 0.1$

Calculate K_{sp} as:

$K_{sp} = 8.0 \times 10^{-9} = [Pb^{+2}][I^-]^2$

$8 \times 10^{-8} = 4s^2$

On solving, $s = 141 \times 10^{-6}$ M

So, value of x = 141

dimag ki ghanti baji kya?

102. Which of the following happens when NH_4OH is added gradually to the solution containing 1M A^{2+} and 1M B^{3+} ions?

(23 Jan, shift-I, 2025)

Given: $K_{sp}[A(OH)_2] = 9 \times 10^{-10}$ and $K_{sp}[B(OH)_3] = 27 \times 10^{-18}$ at 298 K.

(A) Both $A(OH)_2$ and $B(OH)_3$ do not show precipitation with NH_4OH

(B) $B(OH)_3$ will precipitate before $A(OH)_2$

(C) $A(OH)_2$ will precipitate before $B(OH)_3$

(D) $A(OH)_2$ and $B(OH)_3$ will precipitate together

Sol. (B) For explanation, refer to QR code given on the last page.

103. In 1 L saturated solution of AgCl [K_{sp}(AgCl) $= 1.6 \times 10^{-10}$], 0.1 mol of CuCl [K_{sp} (CuCl) = 1.0 $\times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x}. The value of 'x' is

Sol. (7) For explanation, refer to QR code given on the last page.

104. The solubility of AgBr is max in –

(A) 0.1M $AgNO_3$ (B) 0.01M $AgNO_3$

(C) 0.001 M $AgNO_3$ (D) 0.1 NH_3

Sol. (D) For explanation, refer to QR code given on the last page.

yeh badiya sawaal hai

105. The solubility of AgCl will be maximum in which of the following?

[29 June, 2022 (Shift-I)]

(A) 0.01 M KCl (B) 0.01 M HCl

(C) 0.01 M $AgNO_3$ (D) De ionised water

Sol. (D) In de-ionized water, due to absence of any common ion, solubility will be maximum.

SALT OF WEAK ACID AND STRONG BASE

(i) Pure water

$CH_3COOAg_{(s)} \rightleftharpoons \underset{S_1 - x}{CH_3COO^-_{(aq)}} + Ag^+_{(aq)}$

$\underset{S_1 - x}{CH_3COO^-_{(aq)}} + H_2O \rightleftharpoons \underset{x}{CH_3COOH} + \underset{x}{OH^-}$

Due to the hydrolysis of CH_3COO^- the solubility increases slightly than $\sqrt{K_{sp}}$ value

Practically $S_1 \approx \sqrt{K_{sp}}$, Actually $S_1 > \sqrt{K_{sp}}$

$K_{sp} = (S_1 - x)S_1$ ...(1)

$\frac{K_w}{K_a} = \frac{x^2}{S_1 - x}$ ...(2)

On solving we get S_1

(ii) In acidic Buffer

$CH_3COOAg_{(s)} \rightleftharpoons \underset{S_2 - x}{CH_3COO^-_{(aq)}} + \underset{S_2}{Ag^+_{(aq)}}$

$\underset{S_2 - x}{CH_3COO^-} + H^+ \rightleftharpoons \underset{x}{CH_3COOH}$

$K_{sp} = (S_2 - x)\, S_x$ ...(1)

$\frac{1}{K_a} = \frac{x}{(S_2 - x)\left[H^+\right]_{buffer}}$ ...(2)

$S_2 = ?$

$S_2 \gg \sqrt{k_{sp}}$

Due to second eq^{bm} the solubility increases largely; as the equilibrium constant of second eq^{bm} $= \frac{1}{K_a}$ which is a large value.

mazedaar hai na

(iii) Basic Buffer

$CH_3COOAg_{(s)} \rightleftharpoons \underset{S_3}{CH_3COO^-_{(aq)}} + \underset{S_3}{Ag^+_{(aq)}}$

$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

$S_3 = \sqrt{K_{sp}}$, $S_3 \times S_3 = K_{sp}$

$S_3 = \sqrt{K_{sp}}$

There may be the chance of hydrolysis but due to common –ion effect of buffer OH^- this hydrolysis does not take place.

106. The solubility of AgCN in a buffer solution of pH = 3 is x.
The value of x is :
[Assume : No cyano complex is formed; K_{SP} (AgCN) = 2.2×10^{-16} and K_a(HCN) = 6.2×10^{-10}]
[25 Feb, 2021(Shift-I)]

(A) 1.6×10^{-6} (B) 0.625×10^{-6}
(C) 2.2×10^{-16} (D) 1.9×10^{-5}

Sol. (D) For explanation, refer to QR code given on the last page.

107. If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate of AY_2 at 300 K?
(Given K_{sp} (at 300 K) for $AY_2 = 5.2 \times 10^{-7}$)
[02 April, 2025 (Shift-I)]

(A) 3.6×10^{-3} M AB_2, 5.0×10^{-4} M XY
(B) 2.0×10^{-4} M AB_2, 0.8×10^{-3} M XY
(C) 2.0×10^{-2} M AB_2, 2.0×10^{-2} M XY
(D) 1.5×10^{-4} M AB_2, 1.5×10^{-3} M XY

Sol. (C) For explanation, refer to QR code given on the last page.

108. One litre buffer solution was prepared by adding 0.10 mol each of NH_3 and NH_4Cl in deionised water. The change in pH on addition of 0.05 mol of HCl to the above solution is _____ $\times 10^{-2}$, (Nearest integer) (Given : pK_b of NH_3 = 4.745 and $\log_{10}3$ = 0.477)
[07 April, 2025 (Shift-II)]

Sol. (48) For explanation, refer to QR code given on the last page.

pradhu ke gun gaao

Scan the QR code to get the explanations of unsolved questions.

6 Redox Reaction

INTRODUCTION

This chapter deals with the concept of oxidation, reduction, redox reactions, types of redox reactions followed by their balancing, titrations. Lets study about these topics in detail one by one. First we will study the old concept of oxidation and reduction.

Old Concept of Oxidation

(a) Oxidation is a chemical reaction in which oxygen is added

$$2HNO_2 + O_2 \rightarrow 2HNO_3;$$
$$CH_3CHO + O \rightarrow CH_3COOH$$

(b) Hydrogen is removed i.e. hydrogen becomes less

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2;$$
$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

(c) Electronegative element is added

$$2FeCl_2 + Cl_2 \rightarrow 2FeCl_3;$$
$$2Sb + 3Cl_2 \rightarrow 2SbCl_3$$

(d) Electropositive element is removed

$$2NaI + H_2O_2 \rightarrow 2NaOH + I_2;$$

(e) Valency of electropositive element increases

$$\underset{(+2)}{SnCl_2} + Cl_2 \rightarrow \underset{(+4)}{SnCl_4}$$

Old Concept of Reduction

(a) Hydrogen is added. For example

$$H_2 + Cl_2 \rightarrow 2HCl$$
$$N_2 + 3H_2 \rightarrow 2NH_3$$

(b) Oxygen is lost. For example

$$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$$
$$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$$

(c) Electropositive element is added. For example

$$2HgCl_2 + SnCl_2 \rightarrow Hg_2Cl_2 + SnCl_4$$
$$CuCl_2 + Cu \rightarrow Cu_2Cl_2$$

(d) Electronegative element is removed. For example

$$2FeCl_3 + H_2 \rightarrow 2FeCl_2 + 2HCl$$
$$PbS + H_2 \rightarrow Pb + H_2S$$

(e) Valency of electropositive element decreases. For example

$$\underset{(Cu^{+2})}{CuSO_4} + Fe \rightarrow FeSO_4 + \underset{(Cu^{0})}{Cu}$$
$$\underset{(Fe^{+3})}{FeCl_3} + H_2S \rightarrow \underset{(Fe^{+2})}{FeCl_2} + 2HCl + S$$

The old concept of oxidation and reduction, which was primarily based on the gain or loss of oxygen, failed due to several reasons:

purana concept dekh lia na ache se

Important

Limitations of the old concept

1. **Inability to explain reactions without oxygen:** The old concept couldn't explain reactions where oxidation or reduction occurred without the involvement of oxygen.
2. **Inadequate explanation of oxidation states:** The old concept didn't provide a clear understanding of oxidation states, leading to confusion and inconsistencies.
3. **Failure to account for electron transfer:** The old concept didn't recognize the importance of electron transfer in oxidation-reduction reactions.

Inability to explain oxidation-reduction reactions in non-aqueous solvents: The old concept was limited to aqueous solutions and couldn't explain oxidation-reduction reactions in non-aqueous solvents.

Due to these limitations, modern concepts of oxidation and reduction come into light. Lets study about these one by one.

samjh aaya kya Kuch

Modern Concept of Oxidation

(a) **Neutral atom:** When a neutral atom loses electron, it gets converted to a positive ion.

$$Na \rightarrow Na^{+1} + e^-$$
$$Al \rightarrow Al^{+3} + 3e^-$$

(b) **Cation:** When a cation loses electron, there is an increase in its positive charge.

$$Sn^{+2} \rightarrow Sn^{+4} + 2e^-$$
$$Hg^{+1} \rightarrow Hg^{+2} + e^-$$

(c) **Anion:** When an anion loses electron equal to its negative charge, it gets converted to a neutral atom.

$$2O^{-2} \rightarrow O_2 + 4e^-$$
$$2N^{-3} \rightarrow N_2 + 6e^-$$

(d) **Complex Anion:** When a complex anion loses electron, its negative charge decreases.

$$[Fe(CN)_6]^{-4} \rightarrow [Fe(CN)_6]^{-3} + e^-$$

(e) **Molecule:** When a molecule loses electrons, it breaks up into it constituents.

$$H_2O_2 \rightarrow 2H^{+1} + O_2 + 2e^-$$

Khoobsurat Baat

In oxidation reactions-

(i) Positive charge increases and negative charge decreases

(ii) Oxidation number increases

laga na mazedaar?

Modern Concept of Reduction

(a) **Neutral Atom:** When a neutral element or atom accepts electrons it gets converted into an anion.

$$N + 3e^- \rightarrow N^{-3}$$
$$S + 2e^- \rightarrow S^{-2}$$

(b) **Cation:** When a cation accepts electron equal to its charge, it gets converted into a neutral atom.

$$Mg^{+2} + 2e^- \rightarrow Mg^0$$
$$Al^{+3} + 3e^- \rightarrow Al^0$$

(c) Similarly, when a cation accepts less electrons than its charge, its positive charge decreases. For example

$$Cu^{+2} + e^- \rightarrow Cu^{+1}$$
$$Fe^{+3} + e^- \rightarrow Fe^{+2}$$

(d) **Anion:** When an anion accepts electron, its negative charge increases.

$$MnO_4^{-1} + e^- \rightarrow MnO_4^{-2}$$
$$[Fe(CN)_6]^{-3} + e^- \rightarrow [Fe(CN)_6]^{-4}$$

(e) **Molecule:** When a molecule accepts electron, it is a reduction reaction.

$$O_2 + 4e^- \rightarrow 2O^{-2}$$
$$I_2 + 2e^- \rightarrow 2I^{-1}$$

isko kyu ignore kar rahe ho?

Khoobsurat Baat

In reduction reactions-

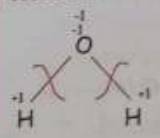
(i) Positive charge decreases and negative charge increases

(ii) Oxidation number decreases

OXIDATION STATE

"The oxidation state of an atom can be defined as the hypothetical charge that would be held by that atom if all of its bonds to other atoms were completely ionic in nature."

H_2O



O.S of both hydrogen atoms = +1

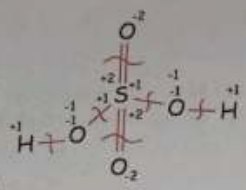
O.S of oxygan atom = -2

H_2SO_4

O.S of Sulphur atom = +6

O.S of all oxygen atoms = -2

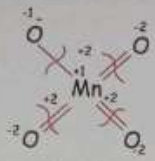
O.S of all hydrogen atoms = +1



MnO_4^-

O.S of manganese atom = +7

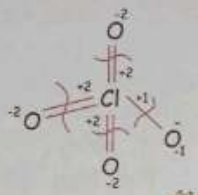
O.S of all oxygen atoms = -2



ClO_4^-

O.S of chlorine atom = +7

O.S of all oxygen atoms = -2

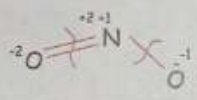


Structures to ab bnane aane chaiye

NO_2^-

O.S of nitrogen atom = +3

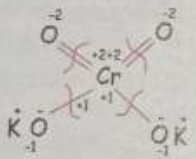
O.S of both oxygen atoms = -2



$K_2Cr_2O_4$

O.S of chromium atom = +6

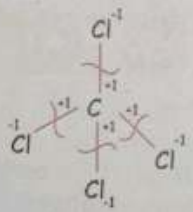
O.S of all oxygen atoms = -2



CCl_4

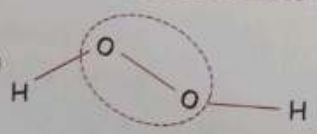
O.S of carbon atom = +4

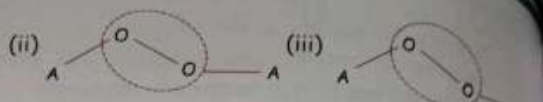
O.S of all chlorine atoms = -1



Compounds having Peroxy Linkage

Peroxy linkage ⇒ (i)



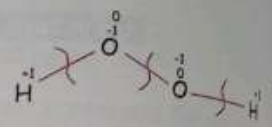


peroxy ko kyu halke mei le rahe ho?

H_2O_2

O.S of both the hydrogen atoms = +1

O.S of both the oxygen atoms = -1



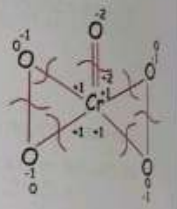
CrO_5

O.S of chromium atom = +6

O.S of double bonded oxygen atom = -2

O.S of rest all oxygen atom = -1

(Butterfly Structure)



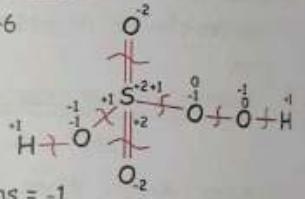
H_2SO_5 (Caro's Acid)

O.S of sulphur atom = +6

O.S of both hydrogen atoms = +1

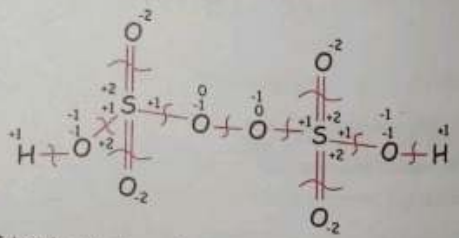
O.S of three oxygen atoms = -2

O.S of two oxygen atoms = -1



Darr toh nahi gaye?

$H_2S_2O_8$ (Marshal's Acid)



O.S of both sulphur atoms = +6

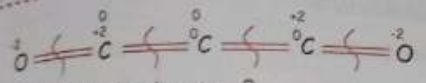
O.S of both hydrogen atoms = +1

O.S of 6 oxygen atoms = -2

O.S of 2 oxygen atoms = -1

SOME SPECIAL EXAMPLES

C_3O_2 (Carbon Suboxide)



O.S of both oxygen atoms = -2

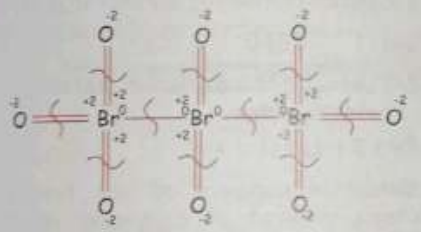
O.S of three carbon atoms = +2, 0, +2

lag raha hai na acha ?

Khoobsurat Baat

The average O.S. of carbon = $\frac{+2+0+2}{3} = +\frac{4}{3}$

Tribromooctaoxide (Br_3O_8)



O.S of three bromine atoms = +6, +4, +6

O.S of all the oxygen atoms = -2

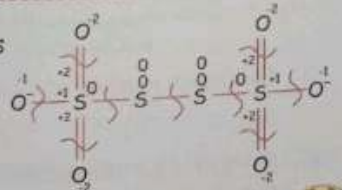
Khoobsurat Baat

The average O.S. of Br = $\frac{+6+4+6}{3} = +\frac{16}{3}$

$S_4O_6^{2-}$ (Tetrathionate Ion)

O.S. of sulphur atoms are = +5, 0, 0, +5

O.S. of all oxygen atoms = -2

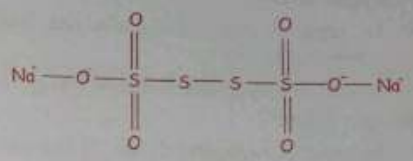


Gazab ka khatarnaak lag rha hoga

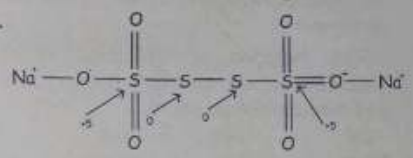
Khoobsurat Baat

The average O.S. of Sulphur = $\frac{+5+0+0+5}{4} = +\frac{5}{2}$

1. The difference in the oxidation states of two types of sulphur atoms in $Na_2S_4O_6$ is **[JEE Adv. 2011]**



Sol.



So there exist two types of O.S. of sulphur, +5 and 0. So difference is 5.

are structure pta hai toh sab pata hai

Note

The above method of finding O.S. of an element in a given compound can be useful only when we have an idea of the structure of the compound.

Kya saare structures yaad Karne honge

Khoobsurat Baat

Hum bina structure yaad kare bhi element ki oxidation state nikal salkte hain. Lekin uske liye kuch rules yaad karne padenge.

OXIDATION STATE RULES

(i) **Fluorine Atom**

* Fluorine is most electronegative element. It always has oxidation number equal to -1 in all its compounds.
* In case of F_2, it is zero.

(ii) **Oxygen Atom**

(a) In general as well as in its oxides = -2

(b) In peroxide (H_2O_2, Na_2O_2 etc) = -1

(c) In superoxide (KO_2 etc) = $-\frac{1}{2}$

(d) In ozonide (KO_3 etc) = $-\frac{1}{3}$

(e) In O_2F_2 and OF_2 = +1, + 2 respectively

(f) In Oxygen molecule (O_2) = 0

(iii) **Hydrogen Atom**

* In general hydrogen atom has oxidation number equal to +1
* In metallic hydrides (NaH, KH etc), it is -1

(iv) **Halogen Atom**

* In general, halogen (Cl, Br, I) has oxidation number = -1
* When halogen is attached with more electronegative element then it will show positive oxidation state.

$\overset{+5}{KClO_3}$ $\overset{+5}{HIO_3}$ $\overset{+5}{KBrO_3}$ $\overset{+7}{HClO_4}$

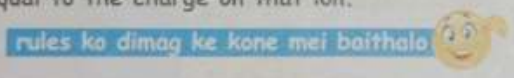
(v) **Metals**

* Alkali metals (Li, Na, K, Rb ----) = +1
* Alkaline Earth metals (Be, Mg, Ca, Sr ----) = +2
* Aluminium always has +3 oxidation number

(vi) **Oxidation number of an element in free state or in allotropic forms is always zero.**

(vii) **Sum of charges of elements in a molecule is zero.**

(viii) **Sum of charges of all elements in an ion is equal to the charge on that ion.**



List of Ions

Cl^-	Chloride	$C_2O_4^{2-}$	Oxalate
Br^-	Bromide	NO_3^-	Nitrate
F^-	Fluoride	N^{3-}	Nitride
I^-	Iodide	NO_2^-	Nitrite
CO_3^{2-}	Carbonate	ClO_4^-	Perchlorate
CN^-	Cyanide	ClO_3^-	Chlorate
NC^-	Isocyanide	ClO_2^-	Chlorite
SO_4^{2-}	Sulphate	ClO^-	Hypochlorite
SO_3^{2-}	Sulphite	CrO_4^{2-}	Chromate
$S_2O_3^{2-}$	Thiosulphate	$Cr_2O_7^{2-}$	Dichromate
S^{2-}	Sulphide	MnO_4^-	Permanganate
P^{3-}	Phosphide	PO_4^{3-}	Phosphate

Important

Ous (minimum)	ic (maximum)
Cuprous (Cu^+)	Cupric (cu^{2+})
Ferrous (Fe^{2+})	Ferric (Fe^{3+})
Stennous (Sn^{2+})	Stennic (Sn^{4+})
Mercurous (Hg_2^{2+})	Mercuric (Hg^{2+})

ous, ic ko samjhlo ki hota kya hai

2. Oxidation number of Sulphur in Na_2SO_4 is

(A) -2 (B) +6 (C) +2 (D) -6

Sol. (B) For explanation, refer to QR code given on the last page.

3. Oxidation number of Sulphur in S_2Cl_2 is

(A) +1 (B) 0 (C) -1 (D) + 6

Sol. (A) Let x be the oxidation state of S. Cl has -1 oxidation state.

$2x + 2(-1) = 0, x = +1$

4. Oxidation numbers of the two nitrogen atoms present in ammonium nitrate are respectively:

(A) +3 and +3 (B) 0 and 0

(C) -3 and +5 (D) -1 and -1

Sol. (C) Ammonium nitrate exists as $NH_4^+NO_3^-$

First calculate the oxidation state of N in NH_4^+:

$x + 4(+1) = +1, x = -3$

Now, calculate the N oxidation state in NO_3^-

$x + 3(-2) = -1, x = +5$

Ammonium nitrate nahi pta toh tum lallu ho

5. Oxidation number of Sulphur in $S_2O_2^{2-}$ is

(A) -2 (B) +1 (C) +6 (D) 0

Sol. (B) Let x be the oxidation state of sulphur. O has -2 oxidation state.

$2x + 2(-2) = -2, 2x - 4 = -2, x = +1$

6. N_2O_3, NO, NO_2, N_2O arrange these compounds of N in increasing order of oxidation states.

(A) N_2O, NO, N_2O_3, NO_2

(B) N_2O, NO_2, N_2O_3, NO

(C) NO_2, N_2O_3, N_2O, NO

(D) N_2O_3, NO, N_2O, NO_2

Sol. (A) For explanation, refer to QR code given on the last page.

7. In which of the following compound, oxidation number of Cl is +3?

(A) ICl (B) ClO_3 (C) ClF_3 (D) $HClO_4$

Sol. (C) Let x be the oxidation state of Cl.

F has -1 oxidation state.

$x - 3 = 0$, $x = +3$

8. If three electrons are lost by a metal ion M^{3+}, its final oxidation number should be

(A) 0 (B) +6 (C) +2 (D) +4

Sol. (B) If M^{3+} will lose three electrons, then it gets converted into +6.

$M^{3+} \rightarrow M^{6+} + 3e^-$

Hence, B is the correct option.

gazab ka assan hai bhayi

9. What should be the oxidation number of S in $H_2S_2O_7$?

(A) +5 (B) +6 (C) +4 (D) +7

Sol. (B) For explanation, refer to QR code given on the last page.

10. The oxidation number of cobalt in $[Co(CN)_6]^{3-}$ is-

(A) + 3 (B) – 3 (C) + 6 (D) – 6

Sol. (A) Let x be the oxidation state of Co.

CN has -1 charge.

$x - 6 = -3$, $x = +3$

So, A is the correct option.

are pagla mat jana

11. In which of the following compound, oxidation number of iron is not +3?

(A) Fe_3O_4 (B) Fe_2O_3

(C) $FeCl_3$ (D) $FePO_4$

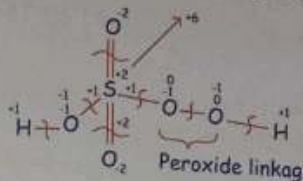
Sol. (A) Let the oxidation state of Fe be x. O has - 2 oxidation state.

$3x + 4(-2) = 0$, $x = +8/3$

12. Oxidation number of Sulphur in H_2SO_5 is

(A) +2 (B) +4 (C) +8 (D) +6

Sol. (D) In H_2SO_5, peroxide linkage is present.



Oxidation state of sulphur = +6

agr structure nhi aata toh beda garg

Common Mistake

$2(+1) + x + 5(-2) = 0$

$x = +8$ ✗

Khoobsurat Baat

- Maximum oxidation state for s, p block = valence electrons = (ns + np)
- For d block Maximum oxidation State = [ns+ (n - 1)d]

Whenever the oxidation state of any element in its combined state is coming more than its maximum value then maximum value is the O.S.

13. Calculate the oxidation state of S in $H_2S_2O_8$.

Sol. (+6) For explanation, refer to QR code given on the last page.

Common Mistake

$2(+1) + 2(x) + 8(-2) = 0$

$x = +7$ ✗

14. Determine the oxidation state of Cr in CrO_5.

Sol. (6) For explanation, refer to QR code given on the last page.

Common Mistake

CrO_5 = x + 5 (-2) = 0

x = +10

15. In which of the following compound, iodine is in its highest oxidation state

(A) KI (B) KIO_4 (C) KI_3 (D) IF_5

Sol. (B) For explanation, refer to QR code given on the last page.

16. The oxidation states of transition metal atoms in $K_2Cr_2O_7$, $KMnO_4$ and K_2FeO_4, respectively, are x, y and z. The sum of x, y and z is _____.

[JEE Main 2 Sep 2022 Shift-2]

Sol. $KMnO_4 \Rightarrow 1 + x - 8 = 0 \quad x = +7$

$K_2Cr_2O_7 \Rightarrow 2 + 2x - 14 = 0 \quad x = +6$

$K_2FeO_4 \Rightarrow 2 + x - 8 = 0 \quad x = +6$

+19

17. One mole of X_2H_4 releases 10 moles of electrons to form a compound Y. What should be the oxidation number of X in the compound Y?

(A) +3 (B) -3 (C) -6 (D) +1

Sol. (A) $X_2H_4 \xrightarrow{-2 \quad +n} X$

(Compound Y)

According to this equation, electron transfer is equal to 2(n + 2)

2 (n + 2) = 10; n = +3

equation likhni aati hai kya

18. Oxidation number of Fe in $K_3[Fe(CN)_6]$ is

(A) +2 (B) +3 (C) +1 (D) +4

Sol. (B) For explanation, refer to QR code given on the last page.

19. Phosphorus has the oxidation state of +3 in:

(A) Ortho phosphoric acid (H_3PO_4)

(B) Phosphorus acid (H_3PO_3)

(C) Meta phosphoric acid (HPO_3)

(D) Pyrophosphoric acid ($H_4P_2O_7$)

Sol. (B) In Phosphorous acid, H_3PO_3

3(+1) + x + 3(-2) = 0, x = +3

formula pta hai paglo

20. From the given list, the number of compounds with +4 oxidation state of Sulphur _____

SO_3, H_2SO_3, $SOCl_2$, SF_4, $BaSO_4$, $H_2S_2O_7$

[27 Jan, 2024 (Shift-I)]

Sol. (3) For explanation, refer to QR code given on the last page.

21. In ammonium-phosphomolybdate, the oxidation state of Mo is + (_____)

[6 April, 2023 (Shift-I)]

Sol. Formula of ammonium phosphomolybdate $(NH_4)_3PO_4.12MoO_3$

Let X be the oxidation state of Mo in MoO_3

X + (-2) × 3 = 0, X = + 6

isko kar dala toh balle balle

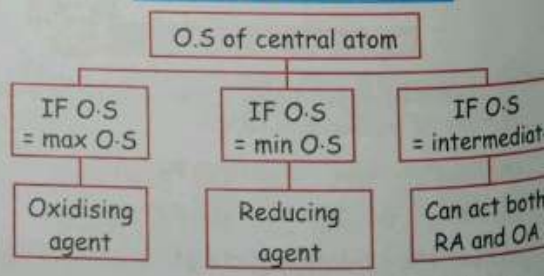
OXIDIZING AGENT AND REDUCING AGENT

An oxidizing agent is a substance that causes oxidat by accepting electrons from another substance. other words, it is a substance that itself gets reduc

A reducing agent is a substance that causes reduct by donating electrons to another substance. In ot words, it is a substance that itself gets oxidized.

Important

How to identify whether a particular substance is an OA or RA



(Can show Disproportionati

22. Which of the following reaction represents the oxidising behavior of H_2SO_4?

(A) $2PCl_5 + H_2SO_4 \rightarrow 2POCl_3 + 2HCl + SO_2Cl_2$

(B) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

(C) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

(D) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

Sol. (D) For explanation, refer to QR code given on the last page.

23. In which of the following reaction H_2O_2 acts as reducing agent?

(A) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$

(B) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$

(C) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$

(D) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

Sol. (B) $\overset{-1}{Cl_2} + \overset{-1}{H_2O_2} \rightarrow 2HCl + \overset{0}{O_2}$ (Oxidation)

Reducing nature ⇒ O.S. Increases

Oxidation agr pta chal gya toh yeh bhi kar hi loge

24. A sulphur containing species that can not be a reducing agent is

(A) SO_2 (B) SO_3^{-2}

(C) H_2SO_4 (D) $S_2O_3^{2-}$

Sol. (C) H_2SO_4 because sulphur has its maximum O.S. i.e. +6

25. Which one is the oxidising agent in the reaction given below?

$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{-2} + H_2O$

(A) H^+ (B) $Cr_2O_7^{-2}$

(C) Cr^{++} (D) none of these

Sol. (D) $2\overset{+6}{Cr}O_4^{2-} + 2H^+ \rightarrow \overset{+6}{Cr_2}O_7^{2-} + H_2O$

So there is no element that changes its O.S.

26. In the reaction: $C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$

HNO_3 acts as:

(A) An oxidising agent (B) An acid

(C) A reducing agent (D) None of these

Sol. (A) $\overset{0}{C} + 4\overset{+5}{H}\overset{+5}{N}O_3 \rightarrow \overset{+4}{C}O_2 + 2H_2O + 4\overset{+4}{N}O_2$ (Reduction)

HNO_3 ⇒ Oxidising agent

27. $H_2MoO_4 \rightarrow MoO_2^+$. In the process, H_2MoO_4

(A) Acts as a reducing agent.

(B) Acts as an oxidising agent.

(C) Acts both as a reducing and oxidising agent.

(D) None of these.

Sol. (B) $\overset{+6}{Mo} \rightarrow \overset{+5}{Mo}$

Mo is getting reduced from +6 to +5, so, it will act as an oxidizing agent.

ghabra kyu rahe ho bacho

28. Which of the following cannot function as an oxidising agent?

[27 Jan, 2024 (Shift-II)]

(A) N^{3-} (B) SO_4^{2-}

(C) BrO_3^- (D) MnO_4^-

Sol. (A) In N^{3-} ion 'N' is present in its lowest possible oxidation state, i.e., -3, therefore it cannot be reduced further because of which it cannot act as an oxidizing agent.

29. Thiosulphate reacts differently with iodine and bromine in the reaction given below:

$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

$S_2O_3^{2-} + 5Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 4Br^- + 10H^+$

Which of the following statement justifies the above dual behaviour of thiosulphate?

[08 April, 2024 (Shift-I)]

(A) Bromine undergoes oxidation and iodine undergoes reduction by iodine in these reactions

(B) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction

(C) Bromine is a stronger oxidant than iodine

(D) Bromine is a weaker oxidant than iodine

Sol. (C) For explanation, refer to QR code given on the last page.

30. In $C + H_2O \rightarrow CO + H_2$, H_2O acts as

(A) Oxidising agent (B) Reducing agent

(C) Both (D) None

Sol. (A) In H_2O, H has +1 oxidation state while in H_2, it has 0 oxidation state. It is getting reduced from +1 to 0, hence, acts as an oxidizing agent.

31. In which of the following reactions, hydrogen peroxide acts as an oxidizing agent?

[JEE Main 2017 Online]

(A) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$

(B) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(C) $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$

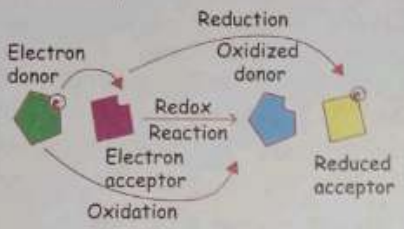
(D) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

Sol. (B) In PbS, S has -2 oxidation state while in $PbSO_4$, it has +6 oxidation state. H_2O_2 oxidizes S from -2 to +6 oxidation state.

agar yeh karlia matlab agaya samajh, balle balle

REDOX REACTION

A reaction in which reduction and oxidation takes place simultaneously.



isme oxidation, reduction sath sath chalengi

32. Which of the following reactions is an example of a redox reaction?

[JEE Main 2017 Offline]

(A) $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

(B) $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$

(C) $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$

(D) $XeF_2 + PF_5 \rightarrow [XeF]^+ PF_6^-$

Sol. (C) For explanation, refer to QR code given on the last page.

Types of Redox Reactions

(A) Combination Reaction

It is denoted as $A + B \rightarrow C$

Either or both A and B must be in the elemental form for such a reaction to be redox reaction. All combustion reactions which make use of elemental dioxygen as well as other electronegative elements are redox reactions.

$\overset{0}{C} + \overset{0}{O_2}(g) \xrightarrow{\Delta} \overset{+4\ -2}{CO_2}(g)$, $3\overset{0}{Mg}(s) + \overset{0}{N_2}(g) \xrightarrow{\Delta} \overset{+2\ -3}{Mg_3N_2}(s)$

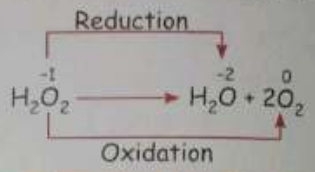
(B) Decomposition Reaction

These are opposite of combination reactions. It leads to the breakdown of a compound into two or more components, at least one of which must be in the elemental state.

$\overset{+1\ -2}{2H_2O}(l) \xrightarrow{\Delta} 2\overset{0}{H_2}(g) + \overset{0}{O_2}(g)$, $2\overset{+1\ -1}{NaH}(s) \xrightarrow{\Delta} 2\overset{0}{Na}(s) + \overset{0}{H_2}(g)$

(C) Disproportionation Reaction

Disproportionation reaction is when reduction and oxidation take place on same element of a compound.



Ek hi element ko reduced and oxidized kia jaega

(D) Displacement Reaction

These are of two types:

(i) Metal displacement reaction:

Eg: $\overset{+2+6-2}{CuSO_4} + \overset{0}{Zn} \xrightarrow{\Delta} \overset{0}{Cu} + \overset{+2\ +6-2}{ZnSO_4}$

(ii) Non-metal displacement reaction:

Eg: $2\overset{0}{Na} + 2\overset{+1\ -2}{H_2O} \xrightarrow{\Delta} \overset{+1\ -2+1}{2NaOH} + \overset{0}{H_2}$

33. Which of the following change represents a disproportionation reaction (s)?

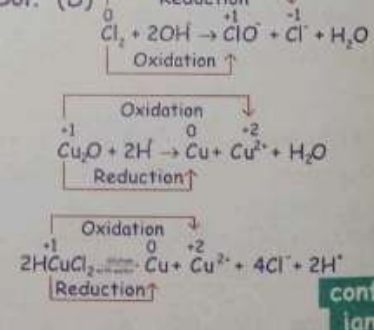
(A) $Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$

(B) $Cu_2O + 2H^+ \rightarrow Cu + Cu^{2+} + H_2O$

(C) $2HCuCl_2 \xrightarrow[\text{Water}]{\text{dilution with}} Cu + Cu^{2+} + 4Cl^- + 2H^+$

(D) All of the above

Sol. (D)



confusa mat jana bhayi

34. Match the LIST- I with LIST -II

	LIST-I (Redox Reaction)		LIST-II (Type of Redox Reaction)
A.	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	I	Disproportionation reaction
B.	$2NaH_{(s)} \xrightarrow{\Delta} 2Na(s) + H_2(g)$	II	Combinaton reaction
C.	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)} + 5CaO_{(s)}$	III	Decomposition reaction
D.	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	IV	Displacement reaction

Choose the correct answer from the options given below:

(A) A - III, B - IV, C - I, D - II
(B) A - II, B - III, C - IV, D - I
(C) A - IV, B - I, C - II, D - III
(D) A - II, B - III, C - I, D - IV

Sol. (B) For explanation, refer to QR code given on the last page.

kitna asaan hai, hai na, aaya maza

35. Which of the following options are correct for the reaction

$2[Au(CN)_2]^-_{(aq)} + Zn(s) \rightarrow 2Au(s) + [Zn(CN_4)]^{-2}_{(aq)}$?

[6 April, 2023 (Shift-I)]

A. Redox reaction
B. Displacement reaction
C. Decomposition reaction
D. Combination reaction

Choose the correct answer from the options given below:

(A) A and B only (B) A only
(C) C and D only (D) A and D only

Sol. (A) $2[\overset{+1}{Au}(CN)_2]^- + \overset{0}{Zn}(s) \rightarrow 2\overset{0}{Au} + [\overset{+2}{Zn}(CN)_4]^{-2}$

It is a redox reaction in which Zn displaced Au^+

Reduction and Oxidation both are taking place simultaneously.

assan hai yar kyu ghabra jate ho

36. Which of the given reactions is not an example of disproportionation reaction?

[26 July, 2022 (Shift-I)]

(A) $2H_2O_2 \rightarrow 2H_2O + O_2$
(B) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$
(C) $MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$
(D) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

Sol. (C) $MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$ is reduction process, while rest others are examples of disproportionation reaction.

37. The species which does not undergo disproportionation reaction is:

(A) ClO_2^- (B) ClO_3^-
(C) ClO^- (D) ClO_4^-

Sol. (D) In ClO_4^-, Cl is present in +7 oxidation state which is its maximum oxidation state and hence, it does not undergo disproportionation reaction.

LIMITATIONS OF MOLE CONCEPT

During the quantitative analysis of a chemical reaction in Mole Concept:

Step 1: We first find out the moles of the given reactants.

Step 2: With the help of a balanced chemical equation we find out the Limiting reagent.

Step 3: We get the result using the mole concept.

Ab Problem kya hai?

We don't have Chemical Equation now. So we can not use Mole Concept for quantitative analysis of a reaction.

Now comes Equivalent Concept

Moles → Equivalents (× n-factor)

Equivalents → Result (Using Law of Equivalence)

So we need to learn two things -

(i) How to calculate n-factor for different substances.

(ii) What is Law of equivalence and how do we apply it.

n-FACTOR CALCULATIONS FOR ACIDS

n-Factor = replaceable H^+ ion

(i) $HCl + NaOH \rightarrow NaCl + H_2O$

(n = 1)

(one H^+ ion is replaced by Na^+ ion)

(ii) $H_2SO_4 + NaOH \rightarrow Na_2SO_4 + H_2O$

(n = 2)

Two H^+ ions are replaced by Na^+ ions

(iii) $H_2SO_4 + NaOH \rightarrow NaHSO_4 + H_2O$

(n = 1)

Only one H^+ ion is replaced

(iv) $H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$

(n = 1)

(v) $H_3PO_4 + NaOH \rightarrow Na_2HPO_4 + H_2O$

(n = 2)

(vi) $H_3PO_4 + NaOH \rightarrow Na_3PO_4 + H_2O$

(n = 3)

Note

Basicity of H_3PO_4 is 3, but the n-factor can be 1, 2 or 3.

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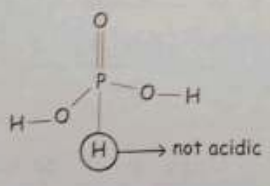
Basicity is the maximum n-factor of an acid

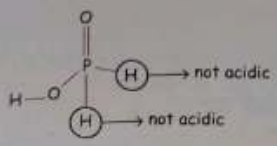
Some Special Examples

H_3PO_4 (orthophosphorous acid)

It is a dibasic acid as out of three hydrogen atoms, one hydrogen atom is directly attached with phosphorous atom, so this hydrogen atom is not acidic.



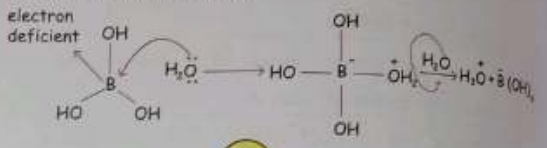
H_3PO_2 (Hypophosphorous acid)



It is a monobasic acid as out of three hydrogen atoms, two hydrogen atoms are directly attached with phosphorous atom so they are not acidic.

H_3BO_3 (Boric acid)

It is a monobasic acid.



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The n-factor of HCl is 1 only when it acts as an acid.

$HCl + NaOH \rightarrow NaCl + H_2O$

(acid) (base)

In the above reaction the n-factor of HCl is 1.

$K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + Cl_2 + H_2O$

(oxidising agent) (Reducing agent)

Here the n-factor of HCl is not 1 as it is behaving like the reducing agent.

The n-factor of HCl is $\frac{3}{7}$ here (will discuss later).

38. $K_xH_y(C_2O_4)_z \xrightarrow{NaOH}$

Find out the n-factor of this substance in this reaction?

Sol. (y) For explanation, refer to QR code given on the last page.

39. $KHC_2O_4 \cdot H_2C_2O_4 \cdot nH_2O \xrightarrow{NaOH}$

Find out the n-factor of the compound.

Sol. It has three H^+ ions so n-factor is three.

n-FACTOR CALCULATIONS FOR BASE

n-factor 1) replaceable OH^- ions

2) acceptable H^+ ions

$NaOH = 1$

$Ca(OH)_2 = 1, 2$

$Al(OH)_3 = 1, 2, 3$

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$ ($n = 1$)

$Na_2CO_3 + HCl \rightarrow H_2CO_3 + NaCl$ ($n = 2$)

$NaHCO_3 + HCl \rightarrow H_2CO_3 + NaCl$ ($n = 1$)

n-FACTOR CALCULATIONS FOR SALTS

Case-1

Salts which react such that no atom undergoes change in oxidation state

n-factor = Total cationic or total anionic charge.

Example:

$\overset{+2\ +4\ -2}{CaCO_3} \longrightarrow \overset{+2\ -2}{CaO} + \overset{+4\ -2}{CO_2}$

Ca^{+2} CO_3^{2-}

(n = 2)

hai na badhiya?

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moles x n-factor = Equivalents

$\frac{\text{given weight}}{\text{Equivalent weight}}$ = Equivalents

Equivalents wt = $\frac{\text{Molecular weight}}{\text{n-factor}}$

Since n-factor of a substance depends on chemical reaction, so Eq·wt also depends on reaction.

Example: $H_3PO_4 \rightarrow HPO_4^{2-}$ ($n = 2$) Eq·wt = $\frac{\text{mol wt}}{2}$

$H_3PO_4 \rightarrow PO_4^{3-}$ ($n = 3$) Eq wt = $\frac{\text{Mol wt}}{3}$

40. Find out the equivalent wt of $K_xH_y(C_2O_4)_z$ in the reaction given below

$K_xH_y(C_2O_4)_z \xrightarrow{NaOH}$

Sol. Since, NaOH is a base, $K_xH_y(C_2O_4)_z$ will act like an acid, so, its n-factor is equal to y.

Equivalent weight = $\frac{\text{Molecular weight}}{y}$

dimag ko khol lo ab

41. Find n-factor of reaction

$AgNO_3 + KI \rightarrow AgI + KNO_3$

Sol. (2) For explanation, refer to QR code given on the last page.

Case-2

Salts which react in a manner that only one atom undergoes change in oxidation state and goes in only one product

n-factor = moles of electron transfer by per mole of salt

K^+ MnO_4^-

$\overset{+7}{K}\overset{+7}{Mn}O_4 \xrightarrow[\text{medium}]{\text{acidic}} \overset{+2}{Mn}$ (n = 5)

$\overset{+7}{K}\overset{+7}{Mn}O_4 \xrightarrow{\text{Basic}} \overset{+6}{Mn}$ (n = 1)

$\overset{+7}{K}\overset{+7}{Mn}O_4 \xrightarrow[\text{weak basic}]{\text{neutral or}} \overset{+4}{Mn}$ (n = 3)

$\overset{+6}{K_2}\overset{+6}{Cr_2}O_7 \xrightarrow{\text{acidic}} Cr^{+3}$ (n = 6)

(+6 se +3, 3 ka change ek Cr ki wajah se, we have 2 Cr atoms, so, change is 3 × 2 = 6)

Cr is already in its max. O.S, so it con only be reduced. So always behaves like an Oxidising agent or oxidant.

Dimag ke kone mei bithalo isko

42. $O \cdot A + CuS \longrightarrow Cu^{+2} + SO_2$

Calculate the equivalent weight of CuS.

Sol. In CuS, oxidation state of S = −2

Oxidation state of S in SO_2 = +4

n-factor = 4 − (−2) = +6

Equivalent weight of CuS = M/6

are thoda toh try karo

Note

Cu^{+1} Cuprous	Cu^{+2} Cupric
Fe^{+2} Ferrous	Fe^{+3} Ferric
Sn^{+2} Stannous	Sn^{+4} Stannic
Hg_2^{+2} Mercurous	Hg^{2+} Mercuric

43. Find out the Eq wt of $Fe_2(C_2O_4)_3$ in this rx^n - $O.A + Fe_2(C_2O_4)_3 \rightarrow Fe^{3+} + CO_2$

Sol. $O.A + \overset{+3}{Fe_2}\overset{+3}{(C_2O_4)_3} \longrightarrow Fe^{+3} + \overset{+4}{CO_2}$

1×6

(+3 se +4, 1 ka change ek C ki wajah se, we have 6 C atoms, so, change is 1 × 6 = 6)

Eq wt = $\frac{M}{6}$

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Oxalate ion always converted into CO_2 after oxidation.

$\underset{+3}{C_2O_4^{2-}} \xrightarrow{[O]} \underset{+4}{CO_2}$

Case-3

Salts which react in a manner that only one atom undergoes change in oxidation state but goes in two products with different oxidation state (different than in the reactant) as a result of either oxidation or reduction

Changed

$3Mn^{+7} \longrightarrow 2Mn^{+2} + Mn^{+4}$

Changed

3↓

$2\overset{+7}{Mn} + \overset{+7}{Mn} \longrightarrow 2\overset{+2}{Mn} + \overset{+4}{Mn}$

$5 \times 2 = 10$ ↓

3 moles of Mn^{+7} transfer 13 moles of electron, per mole electron transfer is 13/3.

$\left(n = \frac{13}{3}\right)$

cases toh ab karne padenge, chhodoge toh pachhtaoge

Case-4

Salts which react in a fashion that only one atom undergoes change in oxidation state but goes in two products with different oxidation state (in one product with same oxidation state and in other with different oxidation state than in the reactant)

$\overset{+7}{Mn} + \overset{+7}{Mn} \longrightarrow \overset{+7}{Mn} + \overset{+2}{Mn}$

5e ↓

retained 5

$2Mn^{+7} \longrightarrow Mn^{+7} + Mn^{+2}$

changed

$n\text{-factor} = \frac{\text{electron transfer}}{\text{moles}} = \frac{5}{2}$

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$Mn^{+7} \longrightarrow Mn^{+2}$ (sirf change) $\longrightarrow n = 5$

$3Mn^{+7} \longrightarrow 2Mn^{+7} + Mn^{+2}$ $\longrightarrow n = \frac{5}{3}$

(Change nahi bhi hua and hua bhi)

$2Mn^{+7} \longrightarrow Mn^{+4} + Mn^{+2}$ $\longrightarrow n = \frac{8}{2} = 4$

(Do logo main change hua)

Khoobsurat baat ko kyu nahi dekh rhe lallu

Case-5

Salts that react in a manner that two type of atoms in the salt undergo change in oxidation state (both the atoms are either getting oxidised or reduced).

(1) All the atoms are getting oxidised

n-factor = Total moles of electrons lost by per mole of salt

oxidation → electron lost

Reduction → electron gain.

$\overset{+2}{FeC_2O_4} \xrightarrow{[O]} \overset{+3}{Fe} + CO_2$

(Ferrous oxalate or iron oxalate)

$C_2O_4^{2-}$ = oxalate ion

$\overset{+3}{C_2}\overset{2-}{O_4} \rightarrow \overset{+4}{CO_2}$

$\overset{+2}{Fe}\overset{+3}{C_2O_4} \xrightarrow{[O]} \overset{+3}{Fe} + \overset{+4}{CO_2}$

1×2

n-factor = $(2 \times 1) + 1 = 3$

dhyan toh lagao, thoda idhar

44. $Cu_2S \xrightarrow{[O]} \overset{+2}{Cu} + SO_2$

What is the Equivalent wt of Cu_2S in this reaction?

Sol. $\overset{+1}{Cu_2}\overset{-2}{S} \xrightarrow{[O]} \overset{+2}{Cu} + \overset{+4}{SO_2}$

2 | 6

(n = 8) 8

$E = \frac{M}{8}$

bdhiya sawaal hai

45. Find out the n-factor of SnC_2O_4 in the reaction with $KMnO_4$ in acidic medium?

Sol. (4) For explanation, refer to QR code given on the last page.

(2) When more than one atom are getting reduced

n-factor = Total moles of electrons gained by per mole salt

$\overset{+3}{Fe}(\overset{+5}{NO_3})_3 \longrightarrow Fe^{2+} + \overset{+2}{NO}$

1 | 3 × 3

(n = 10) 10

$(NO_3^- = x - 6 = -1 \quad x = +5)$

(3) Some are getting oxidised and some are getting reduced

If gain = loss, n-factor = gain or loss

If gain ≠ loss, n-factor = |gain-loss|

लेन = देन

n = 6

Eq. Wt = $\frac{\text{Mol.wt}}{6}$

Reduction ↓ 6

$\overset{+5}{K}\overset{-2}{ClO_3} \longrightarrow K\overset{+1}{Cl}\overset{-1}{} + \overset{0}{O_2}$

oxidation ↑ 6

46. $A + (NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + H_2O + \ldots$

Behaviour of A –

(i) O.A. (ii) R.A (iii) Both

(iv) None (v) Mujhe kya pata

6↓

Sol. (iv) $\overset{-3}{(NH_4)_2}\overset{+6}{Cr_2}\overset{-2}{O_7} \rightarrow \overset{0}{N_2} + \overset{+3}{Cr_2}\overset{-2}{O_3} + H_2O$

लेन = देन 6↑

n = 6

A ka kuch nhi de sakte or isse kuch e nhi sakte kyunki zarurat nhi hai

47. $[SCN]^- \xrightarrow{[O]} SO_4^{2-} + [CN]^-$

Calculate the n-Factor.

Sol. $\overset{-2\ +4\ -3}{[SCN]^-} \xrightarrow{[O]} \overset{+6\ -2}{SO_4^{2-}} + \overset{+2\ -3}{[CN]^-}$

8

$\overset{-1}{\overset{-1}{S}}\!\!-\!\!\overset{+1}{\overset{+3}{C}}\!\!\equiv\!\!\overset{-3}{N} \qquad \overset{-1}{\overset{+3}{C}}\!\!\equiv\!\!\overset{-3}{N}$

लेन ≠ देन

n = |लेन − देन| = |2 − 8| = 6

len den ka sara khel hai

48. Calculate the n-Factor

$Fe_{0.93}O \rightarrow Fe^{+3}$

Sol. (0.79) For explanation, refer to QR code given on the last page.

49. $Fe_{0.93}O \rightarrow Fe^{+2}$

find out the n-factor.

Sol. $Fe^{+3}_{0.14} \rightarrow Fe^{+2}$

1×0.14

(n = 0.14)

Note

To calculate the n-factor of a non stoichiometric compound we need not to calculate the moles of metal having different oxidations state. We can calculate the n-factor with the help of average oxidation state of metal.

$Fe_{0.93}O \longrightarrow Fe^{+2}$

$0.93x - 2 = 0,$

$x = \frac{2}{0.93}$

$Fe_{0.93}O \longrightarrow Fe^{+3}$

(n = 0.7a)

Change = $3 - \frac{2}{0.93} = \left(\frac{0.79}{0.93}\right)$

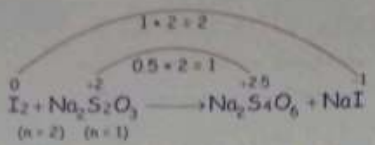
Net change = $\frac{0.79}{0.93} \times 0.93 = 0.79$

hai na kitna mazedaar?

50. $I_2 + Na_2S_2O_3 \longrightarrow Na_2S_4O_6 + NaI$

Find the equivalent weights of I_2 and $Na_2S_2O_3$ in terms of M_1 and M_2 respectively.

Sol.



$$E_1 = \frac{M_1}{2}, E_2 = \frac{M_2}{1}$$

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n-factor of HCl is always 1 (This is false).

Lets understand this by an example given below:

51. Find out the n-factor of both the species involved –

$K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + Cl_2 + H_2O$

Sol. $\overset{+6}{K_2}\overset{+6}{Cr_2}O_7 + \overset{-1}{H}\overset{-1}{Cl} \rightarrow \overset{-1}{K}\overset{-1}{Cl} + \overset{+3}{Cr}\overset{-1}{Cl_3} + \overset{0}{Cl_2} + H_2O$

(n = 6)

Reduction half: $6e^- + Cr_2O_7^{2-} \longrightarrow 2Cr^{+3}$

Oxidation half: $2Cl^- \rightarrow Cl_2 + 2e^- \times 3$

$1Cr_2O_7^{2-} + 6Cl^- \rightarrow 2Cr^{+3} + 3Cl_2$

$1K_2Cr_2O_7 + 14HCl \rightarrow 2CrCl_3 + 3Cl_2 + 2KCl + 7H_2O$

14 mol of HCl → 6 mol e^-

1 mol of HCl → $\frac{6}{14} = \frac{3}{7}$ mol e^-

$n = \frac{3}{7}$

Eq. wt of HCl in the previous slide

$$= \frac{M_{HCl}}{\frac{3}{7}} = \frac{7 \times 36.5}{3} = 85.2$$

Eq. of HCl as an acid = $\frac{M_{HCl}}{1} = \frac{36.5}{1} = 36.5$

Similarly, for

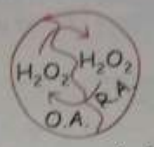
$KMnO_4 + HCl \rightarrow MnCl_2 + KCl + Cl_2 + H_2O$

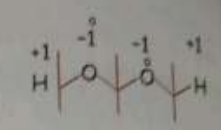
n-factor of HCl is $\frac{5}{8}$.

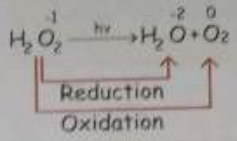
sar ko pakad lo zor se

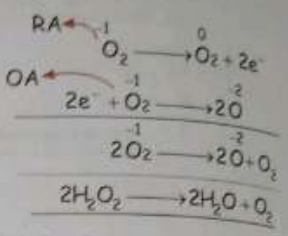
n-factor calculation in a disproportionation reaction

In a redox reaction a single substance is reduced and oxidised as well in the same reaction.









$H_2O_2 + H_2O_2 \rightarrow 2H_2O + O_2$

(n = 2) (n = 2)

2 mol H_2O_2 transfers 2 mol e^-

1 mol H_2O_2 transfers $\frac{2}{2}$ = 1 mol e^-

So, the n-factor of H_2O_2 in disproportionation reaction is 1.

Isko mat chhod dena, samjhe?

Important

1. n-factor of H_2O_2 as O.A = 2
2. n-factor of H_2O_2 as R.A = 2
3. Overall n-factor in this disproportionation reaction = 1

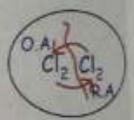
52. $Cl_2 \rightarrow Cl^- + ClO_3^-$

Calculate n-factor.

Sol. $\overset{0}{Cl_2} \longrightarrow \overset{-1}{Cl^-} + \overset{+5}{ClO_3^-}$

Reduction

Oxidation



$1\overset{0}{Cl_2} \longrightarrow 2\overset{+5}{ClO_3^-} + 10e^-$

$2e^- + \overset{0}{Cl_2} \longrightarrow 2Cl^- \times 5$

$6Cl_2 \longrightarrow 2ClO_3^- + 10Cl^-$

(i) As reducing agent = 10

(ii) As oxidising agent = 2

(iii) overall n-factor = $\frac{10}{6} = \frac{5}{3}$

isme dimag lagega

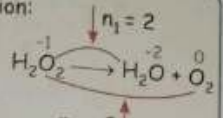
Jugaad!

In a disproportionation reaction:

$E_{net} = E_1 + E_2$

$\frac{M_{H_2O_2}}{n_{net}} = \frac{M_{H_2O_2}}{n_1} + \frac{M_{H_2O_2}}{n_2}$

$\frac{1}{n_{net}} = \frac{1}{n_1} + \frac{1}{n_2}$,



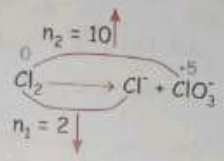
$\frac{1}{n_{net}} = \frac{1}{2} + \frac{1}{2}$, $n_{net} = 1$

Example:

$E_{net} = E_1 + E_2$

$\frac{M_{Cl_2}}{n_{net}} = \frac{M_{Cl_2}}{n_1} + \frac{M_{Cl_2}}{n_2}$

$\frac{1}{n_{net}} = \frac{1}{2} + \frac{1}{10}$, $n_{net} = \frac{5}{3}$



Khoobsurat Baat

For disproportionation reaction, the formula to calculate n-factor of the substance is:

$\frac{1}{n_{net}} = \frac{1}{n_1} + \frac{1}{n_2}$

n_1 = n-factor of that part of the substance getting reduced

n_2 = n-factor of that part of the substance getting oxidized

formula yaad karlo pehle

Note

- Equivalents = moles × n-factor
- Equivalents = $\frac{\text{Given mass}}{\text{Eq. mass}}$
- Equivalents mass = $\frac{\text{mol. wt}}{\text{n - factor.}}$
- Normality = $\frac{\text{no of equivalents of solute}}{\text{volume of solution in Lt.}}$
- Equivalent = Normality × volume

$N = \frac{\text{no. of eq. of solute}}{\text{vol. of solution in Lt}} = \text{n-factor} \times \left(\frac{\text{moles of solute}}{\text{vol. of solution in it.}}\right)$

N = n-factor × Molarity

sare formulae yaad karlo

LAW OF EQUIVALENCE

The law states that one equivalent of an element combine with one equivalent of the other, and in a chemical reaction same no. of equivalent (or mill equivalent) of reactants react to give same no. of equivalent (or milli equivalents) of products separately.

According:

(i) $aA + bB \rightarrow mM + nN$

m.eq of A = m.eq of B = m.eq of M = m.eq of N

(ii) In a compound M_xN_y

m.eq of M_xN_y = m.eq of M = m.eq of N

Important

- For Acid-base (Neutralization reaction) or redox reaction:

  $N_1V_1 = N_2V_2$ is always true

  But $M_1V_1 = M_2V_2$ (may or may not be true)

  $M_1 \times n_1 \times V_1 = M_2 \times n_2 \times V_2$ (always true where n terms represent n-factor)

isko kar daala toh life jhinga lalla

Khoobsurat Baat

Jo bhidte hain unke barabar hote hain

- Acid + base → Products
  Eq of Acid = Eq. of base
- O·A + R.A → Products
  Eq. of O.A = Eq. of R.A
- Acid 1 + Base + Acids 2 → Products
  Eq. of $Acid_1$ + Eq. of $Acid_2$ = Eq. of base
- $O{\cdot}A_1 + R{\cdot}A + O{\cdot}A_2$ → Products
  Eq. of OA_1 + Eq. of OA_2 = Eq. of RA

53. How many moles of SnC_2O_4 will be completely oxidized by 2 mol $KMnO_4$ in acidic medium?

$KMnO_4 + SnC_2O_4 \xrightarrow{H^+} Mn^{+2} + Sn^{+4} + CO_2$

Sol. $\overset{+2\ +3}{SnC_2O_4} \longrightarrow Sn^{+4} + \overset{+4}{CO_2}$

2↑ (n = 4) 2↑

$\overset{+7}{KMnO_4} + SnC_2O_4 \xrightarrow{H^+} Mn^{+2} + Sn^{+4} + CO_2$

2 mol $\quad \frac{10}{4} = 2.5$ mol.

×5 ÷4

10eq $\xrightarrow{LOE}$ 10eq

reactions likho phir sawaal karo

54. Find out the volume of 0.2 M $KMnO_4$ consumed to completely oxidised 100 ml 0.08 M SnC_2O_4 solution in acidic medium.

Sol. (32 ml) For explanation, refer to QR code given on the last page.

Khoobsurat Baat

Equivalents → n-factor × moles; → $\frac{\text{given mass}}{\text{Eq. mass}}$; → normality × volume; → n-factor × Molarity × volume

55. Metal reacts with chlorine to form chloride. Find out the eq wt of metal

Given: Weight of metal = 2.3 g, Weight of chloride = 3.1 g

Sol. (102.06 g) For explanation, refer to QR code given on the last page.

56. Metal reacts with nitrogen to give nitride

Find out the Eq wt of metal.

Given: Weight of metal =1.8 g, weight of nitride = 2.7 g

Sol: $\text{Metal} + \overset{0}{N_2} \longrightarrow \text{Metal nitride}^{-3}$

(n = 6)

LOE $\frac{1.8}{E} = \frac{0.9}{\frac{28}{6}}$

On solving, E = 9.3 g

57. In the ionic equation,

$2K^+BrO_3^- + 12H^+ + 10e^- \rightarrow Br_2 + 6H_2O + 2K^+$,

the equivalent weight of $KBrO_3$ will be (where M = molecular weight of $KBrO_3$):

(A) $\frac{M}{5}$ (B) $\frac{M}{2}$ (C) $\frac{M}{6}$ (D) $\frac{M}{4}$

Sol. (A) $\overset{+5}{BrO_3^-} \longrightarrow \overset{0}{Br_2}$

$\therefore (n_f)\ BrO_3^- = 5 \quad \therefore$ Eq wt = $\frac{M}{5}$

58. The equivalent weight of a metal is double that of oxygen. How many times is the weight of its oxide greater than weight of the metal?

(A) 1.5 (B) 2 (C) 0.5 (D) 3

Sol. (A) Given $E_{metal} = 2 \times 8 = 16$

$\frac{\text{Weight}_{oxide}}{\text{Weight}_{metal}} = ?$

$(n_{eq})_{metal} = (n_{eq})_{oxide}$

$\frac{w_{metal}}{16} = \frac{w_{oxide}}{16+8} \quad \therefore \frac{w_{oxide}}{w_{metal}} = \frac{24}{16} = \frac{3}{2} = 1.5$

dimag ki batti jali

59. If M represents molecular mass of Mn_3O_4 then what will be its equivalent mass if it undergoes disproportionation reaction as shown

$Mn_3O_4 \rightarrow MnO_4^- + Mn^{2+}$

(A) $\frac{M}{13}$ (B) $\frac{M}{2}$ (C) $\frac{15M}{26}$ (D) $\frac{26M}{15}$

Sol. (C) For explanation, refer to QR code given on the last page.

60. When a metal is burnt, its weight is increased by 24%. The equivalent weight of the metal will be

(A) 25 (B) 24 (C) 33.3 (D) 76

Sol. (C) For explanation, refer to QR code given on the last page.

61. The equivalent weight of Br_2 is 96 in the following disproportionation reaction.

$Br_2 + OH^- \rightarrow Br^- + H_2O + ?$ (Oxidized product)

The oxidation state of Br in the oxidized product is (Br = 80)

Sol. Given reaction

$Br_2 + OH^- \rightarrow Br^- + H_2O + P$

Eq. wt of $(Br_2) = \frac{\text{Molecular wt}}{\text{n-factor}}$

$\frac{1}{n_{net}} = \frac{1}{n_1} + \frac{1}{n_2}$, $\frac{3}{5} = \frac{1}{2} + \frac{1}{n_2}$

On solving, $n_2 = 10$

$Br_2 \xrightarrow{2 \times x = 2x} \overset{x}{Br^-}$

$2x = 10, x = +5$

62. Molecular masses of NH_3 and N_2 are x_1 and x_2, respectively. In the reaction

$N_2 + 3H_2 \rightarrow 2NH_3$

Their equivalent weights are y_1 and y_2. The $(y_1 - y_2)$ is

(A) $\left(\frac{2x_1 - x_2}{6}\right)$ (B) $(x_1 - x_2)$

(C) $(3x_1 - x_2)$ (D) $(x_1 - 3x_2)$

Sol. (A) Equivalent weight of $NH_3 = \frac{x_1}{3}$

Equivalent weight of $N_2 = \frac{x_2}{6}$

$y_1 - y_2 = \left(\frac{x_1}{3} - \frac{x_2}{6}\right) = \left(\frac{2x_1 - x_2}{6}\right)$

darr kyu rahe ho, asaan toh hai kitna

63. 0.7 gm of $Na_2CO_3.xH_2O$ is dissolved in 100 ml, 20 ml of which required 19.8 ml of 0.1 N HCl. The value of x is

(A) 4 (B) 3 (C) 2 (D) 1

Sol. (C) For explanation, refer to QR code given on the last page.

64. 0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation $5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the correct statements:

(a) 200 mL of KI solution reacts with 0.004 mol of KIO_3

(b) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4

(c) 0.5 L of KI solution produced 0.005 mol of I_2

(d) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$

Choose the correct answer from the options given below:

(A) (a) and (b) only (B) (a) and (d) only

(C) (b) and (c) only (D) (c) and (d) only

Sol. (B) For explanation, refer to QR code given on the last page.

SIMPLE TITRATIONS

The aim of simple titration is to find the concentration of an unknown solution with the help of the known concentration of another solution.

(a) Acid-base titrations (b) Redox titrations

$\text{Normality} = \frac{\text{Eq. of solute}}{\text{vol of sol}^n \text{ in Lt}}$

Eq of solute = Normality × volume (Lt)

m.eq of solute = Normality × vol of sol^n (ml)

m.eq of solute = $\underbrace{n \times \text{molarity}}_{\text{normality}} \times$ vol (ml)

(a) Acid-base Titration

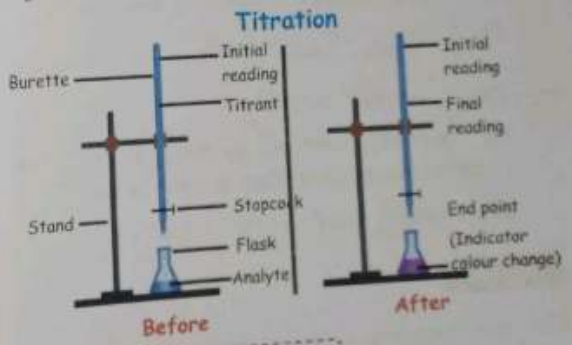
When acid reacts with base, then

Eq. of acid= Eq. of base

$N_1 \times V_1 = N_2 \times V_2$

N_1 = Normality of acid, V_1 = volume of acid,

N_2 = Normality of base, V_2 = volume of base



(b) Redox Titration

Titrations based on redox processes are called redox titrations. In these titrations titrate/titrant are oxidising/reducing agents. The calculations made in redox titrations are again based on law of chemical equivalence.

Titrate + Titrant → Products

Meq. of titrate = meq. of titrant (at equivalent point).

Or $N_1V_1 = N_2V_2$

asaan hai ghabra kyu rahe ho

Titration Using Acidified $KMnO_4$

Use of $KMnO_4$ as oxidising agent in redox titrations was first made for the titration of Fe^{2+} ions. It is readily available and colour of its solution is too intense that an indicator is not ordinarily required and MnO_4^- ions act as self indicator. Titrations of MnO_4^- are usually carried out in acidic medium.

Among the common mineral acids H_2SO_4, HCl and HNO_3, only H_2SO_4 is useful for providing acidic medium. HCl cannot be used since Cl^- are oxidised to Cl_2 by $KMnO_4$. On the other hand, HNO_3 itself acts as a strong oxidising agent.

$$2MnO_4^- + 10Cl^- + 16\,H^+ \rightarrow 2Mn^{2+} + 5Cl_2 + 8H_2O$$

Some important titrations using $KMnO_4$ are given below:

(a) **$KMnO_4$ vs. oxalic acid:**

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

End point: Light pink colour of MnO_4^- solution, $KMnO_4$ acts as self indicator.

At equivalence point:

(i) 2 mmol $KMnO_4$ reacts with 5 mmol oxalic acid.

(ii) meq. of $KMnO_4$ = meq. of oxalic acid

$$N_{KMnO_4} \times V_{KMnO_4} = N_{\text{oxalic acid}} \times V_{\text{oxalic acid}}$$

(iii) $$\frac{1}{2} \times M_{KMnO_4} \times V_{KMnO_4} = \frac{1}{5} \times M_{\text{oxalic acid}} \times V_{\text{Oxalic acid}}$$

(b) **$KMnO_4$ vs. Fe^{2+} ions:**

Fe^{2+} is readily oxidized by MnO_4^- to Fe^{3+} The reaction which takes place is given below:

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 8H_2O$$

End point: MnO_4^- acts as self indicator and light pink colour of solution indicates end point.

At equivalent point:

(i) 1 mol MnO_4^- reacts with 5 mol Fe^{2+}.

(ii) Meq. of MnO_4^- = Meq. of Fe^{2+}

Or $$N_{KMnO_4} \times V_{KMnO_4} = N_{Fe^{2+}} \times V_{Fe^{2+}}$$

Or $$5 \times M_{KMnO_4} \times V_{KMnO_4} = 1 \times M_{Fe^{2+}} \times V_{Fe^{2+}}$$

$$\left(\because N_{KMnO_4} = 5 \times M_{KMnO_4}, N_{Fe^{2+}} = M_{Fe^{2+}}\right)$$

samajh lo, phir yaad karo

(c) **Estimation of Fe^{2+} and Fe^{3+} ions in a solution:**

In this estimation, the solution is titrated twice. First without reduction in which only Fe^{2+} reacts and Fe^{3+} remains unreacted. In second step same volume of solution is taken, Fe^{3+} ions are reduced to Fe^{2+} and then titrated. Here

(i) In I step: Meq. of $KMnO_4$ (let x)

= Meq. of Fe^{2+}

(ii) In II step: Meq. $KMnO_4$ (y)

= Meq. of Fe^{2+} + Meq. of Fe^{3+}

(d) **Analysis of sample containing oxalic acid and sodium oxalate:**

Same volume of solution are titrated separately with NaOH and $KMnO_4$ solution

Meq. of NaOH = Meq. of oxalic acid

and Meq. of $KMnO_4$ = Meq. of oxalic acid + Meq. of sodium oxalate

samajhna toh padega pehle

Titration Using Acidified $K_2Cr_2O_7$

In volumetric analysis, $Cr_2O_7^{2-}$ is also commonly used as oxidant yet weaker than MnO_4^-, Ce^{4+} etc. In most of its applications, it is reduced to Cr^{3+}.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$E_{Cr_2O_7^{2-}} = \frac{M_{Cr_2O_7^{2-}}}{6} = \frac{296}{6} = 49$$

Note

In the laboratory, for practical purposes ferrous ammonium sulphate is taken as a source of Fe^{2+} ions. In the determination of Fe content in iron ores, the whole iron content present in a definite amount of its solution is first converted into Fe^{2+} by using suitable reducing agent like $SnCl_2$.

Important

These titrations are made in 1 - 2 N acid solution. In neutral or alkaline medium, it is not used. In alkaline medium, orange $Cr_2O_7^{2-}$ changes to yellow CrO_4^{2-}.

Iodometric and Iodimetric Titration

Iodine acts as mild oxidising agent

These titrations fall into two categories:

(a) **Iodimetry:** In these titrations, standard I_2 solution is used to titrate easily oxidisable substances. It includes the estimations of thiosulphates, sulphite arsenite etc.

$$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$$

$$SO_3^{2-} + I_2 + H_2O \rightarrow SO_4^{2-} + 2H^+ + 2I^-$$

$$AsO_3^{3-} + I_2 + H_2O \rightarrow AsO_4^{2-} + 2H^+ + 2I^-$$

$$H_2S + I_2 \rightarrow S + 2I^- + 2H^+$$

$$Sn^{2+} + I_2 \rightarrow Sn^{4+} + 2I^-$$

$$N_2H_4 + 2I_2 \rightarrow N_2 + 4H^+ + 4I^-$$

(b) **Iodometry:** In iodiometric titrations, an oxidising agent is allowed to react with excess of KI (or I^-) solution.

It includes the estimations of following species

$$2MnO_4^- + 10I^- + 16H^+ \rightleftharpoons Mn^{+2} + 5I_2 + 8H_2O$$

$$IO_3^- + 5I^- + 6H^+ \rightleftharpoons 3I_2 + 3H_2O$$

The I_2 liberated is titrated with hypo solution ($Na_2S_2O_3$).

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

I_2, though insoluble in water, remains in solution containing KI as KI_3.

iso mat chhodna, samjhe bacho

Note

On addition of starch after the liberation of iodine from the reaction of Cu^{2+} ions on iodide ions, an intense blue colour appears. This colour disappears as soon as the iodine is consumed by the thiosulphate ions. Thus, the end-point can easily be tracked and the rest is the stoichiometric calculation only.

65. Calculate the normality of a solution containing 50 mL of 5 M solution $K_2Cr_2O_7$ in acidic medium.

Sol. Normality (N) = Molarity × n-factor

= 5 × 6 = 30N

bilkul seedha sawaal hai

66. Find the number of moles of $KMnO_4$ needed to oxidise one mole Cu_2S in acidic medium.

The reaction is $KMnO_4 + Cu_2S \rightarrow Mn^{2+} + Cu^{2+} + SO_2$

Sol. From law of equivalence

Equivalents of Cu_2S = equivalents of $KMnO_4$

moles of Cu_2S × n-factor = moles of $KMnO_4$ × n-factor

1 × 8 = Moles of $KMnO_4$ × 5 (n-factor of Cu_2S = 8)

Moles of $KMnO_4 = \frac{8}{5} = 1.6$

67. When 30 ml of acid is neutralized by 15 ml of 0.2 N alkali, the acid concentration is _____

(A) 0.1 N (B) 0.2 N

(C) 0.3 N (D) 0.4 N

Sol. (A) Apply LOE as:

m. eq of acid = m. eq of alkali

$N \times 30 = 15 \times 0.2$, $N = 0.1$

lallu mat ban jana

68. If 83 mL of 0.45 M NaOH solution neutralizes a 235 mL HCl solution, then calculate the molarity of the HCl Solution.

Sol. (0.16 M) For explanation, refer to QR code given on the last page.

69. What volume of 0.2 M $KMnO_4$ solution is needed for complete reaction with 26.56 gm $Fe_{0.9}O_{1.0}$ in acidic medium? (Fe = 56)

(A) 280 ml (B) $\frac{280}{9}$ ml

(C) $\frac{2800}{9}$ ml (D) 560 ml

Sol. (A) $\overset{\frac{2}{0.9}}{Fe_{0.9}O} + \overset{+7}{KMnO_4} \rightarrow Fe^{+3} + \overset{2+}{Mn}$

$$\left(3 - \frac{2}{0.9}\right) \times 0.9 = 0.7$$

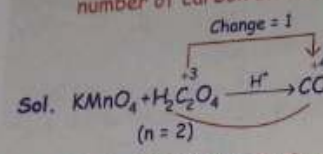
According to LOE

Eq of $Fe_{0.9}O$ = Eq of $KMnO_4$

$$\frac{26.56}{66.4} \times 0.7 = V \times 0.2 \times 5$$

V = 0.280 L or 280 ml.

70. In the titration of $KMnO_4$ and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is

Sol. 

71. 20 mL of 0.02 M $K_2Cr_2O_7$ solution is used for the titration of 10 mL of Fe^{2+} solution in the acidic medium. The molarity of Fe^{2+} solution is ____ $\times 10^{-2}$ M. (Nearest Integer)

[JEE Mains 2022]

Sol. $\overset{+6}{K_2Cr_2O_7} + Fe^{+2} \xrightarrow{H^+} Cr^{+3} + Fe^{+3}$
(n = 6) (n = 1)

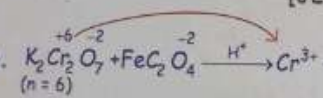
Apply LOE: M.eq of $K_2Cr_2O_7$ = M. eq of Fe^{+2}

$20 \times 0.02 \times 6 = 10 \times M \times 1$

$M = 0.24$

72. The volume, in ml, of 0.02 M $K_2Cr_2O_7$ solution required to react with 0.288 g of ferrous oxalate in acidic medium is ____
[Molar mass of Fe = 56mol^{-1}]

[JEE Mains 2020]

Sol. 

Apply LOE: Eq. of $K_2Cr_2O_7$ = Eq of FeC_2O_4

$V \times 0.02 \times 6 = \frac{0.288}{144} \times 3$

$V = \frac{2 \times 10^{-3}}{0.04}$ Lt or 50 mL

LOE toh laga loge na ya phir nahi haga?

73. If 50 mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is ____ g.

[29 Jan, 2024 (Shift-II)]

Sol. Equivalent of Oxalic acid = Equivalents of NaOH

$50 \times 0.5 \times 2 = 25 \times M \times 1$

$M_{NaOH} = 2M$

W_{NaOH} in 50ml = $2 \times 50 \times 40 \times 10^{-3}$ g = 4g

n-factor toh aata hai na nikalna

74. Only 2 mL of $KMnO_4$ solution of unknown molarity is required to reach the end point of a titration of 20 mL of oxalic acid (2 M) in acidic medium. The molarity of $KMnO_4$ solution should be ____ M.

[04 April, 2024 (Shift-2)]

Sol. As we know that,

eq. ($KMnO_4$) = eq. ($H_2C_2O_4$)

$(M \times n_f \times V)_{KMnO_4} = (M \times n_f \times V)_{H_2C_2O_4}$

$M \times 5 \times 2 = 2 \times 2 \times 20$

Hence, M = 8M

NTA ans → [50], PW ans → [8]

75. 20 mL of calcium hydroxide was consumed when it was reacted with 10 mL of unknown solution of H_2SO_4. Also 20 mL standard solution of 0.5 M HCl containing 2 drops of phenolphthalein was titrated with calcium hydroxide, the mixture showed pink colour when burette displayed the value of 35.5 mL whereas the burette showed 25.5 mL initially.

The concentration of H_2SO_4 is ____ M.
(Nearest integer)

[13 April, 2023 (Shift-I)]

Sol. (1M)For explanation, refer to QR code given on the last page.

76. 10.0 mL of Na_2CO_3 solution is titrated against 0.2 M HCl solution.

The following titre values were obtained in 5 readings: [18 March, 2021 (Shift-II)]

4.8 mL, 4.9 mL, 5.0 mL, 5.0 mL and 5.0 mL

Based on these readings, and convention of titrimetric estimation, the concentration of Na_2CO_3 solution is ____ mM. (Round off to the nearest integer)

Sol. $V_{Na_2CO_3}$ = 10 mL (Given)
Volume of HCl used will be 5 mL (average of titre values)

M_{eq} of HCl = m_{eq} of Na_2CO_3

$(M \times \text{n-factor} \times V)_{HCl} = (M \times \text{n-factor} \times V)_{Na_2CO_3}$

$0.2 \times 1 \times 5 = M \times 2 \times 10$

M = 0.05 M or 50 mM

Arey formula hi toh lagaya hai paglo

77. The exact volumes of 1 M NaOH solution required to neutralize 50 mL of 1 M H_3PO_3 solution and 100 mL of 2M H_3PO_2 solution respectively, are:

[16 March, 2021 (Shift-II)]

(A) 100 mL and 50 mL (B) 100 mL and 200 mL

(C) 50 mL and 50 mL (D) 100 mL and 100 mL

Sol. (B) According to question,

M_{eq} of Base = M_{eq} of Acid

For H_3PO_3: $1 \times V = 50 \times 1 \times 2$

$\therefore V = 100mL$ of NaOH

For H_3PO_2: $1 \times V = 100 \times 2 \times 1$

$\therefore V = 200mL$ of NaOH

badhiya sawaal hai bacho

78. For the given reaction:

$KMnO_4 + SnC_2O_4 + H^+ \rightarrow Mn^{2+} + Sn^{4+} + CO_2$

2 Mol. ?

Find out the moles of SnC_2O_4

Sol. $\overset{+7}{4}KMnO_4 + 5\overset{+2}{Sn}\overset{+3}{C_2}O_4 + H^+ \rightarrow Mn^{2+} + Sn^{4+} + \overset{+4}{CO_2}$

2Mol. ?

(n = 5) (n = 4)

Method: 1

Eq of $KMnO_4$ = Eq of SnC_2O_4

$2 \times 5 = x \times 4, \quad x = \frac{10}{4} = 2.5mol$

Method: 2

4 mol $KMnO_4 \rightarrow$ 5 mol SnC_2O_4

2mol $KMnO_4 \rightarrow \frac{5}{2}$ mol SnC_2O_4

dono methods hi asaan hai

79. A solution of 0.2 g of a compound containing Cu^{2+} and $C_2O_4^{2-}$ ions on titration with 0.02 M $KMnO_4$ in presence of H_2SO_4 consumes 22.6 ml of the oxidant. The resultant solution is neutralized with Na_2CO_3, acidified with dil. acetic acid and treated with excess KI. The liberated iodine requires 11.3 ml of 0.05M $Na_2S_2O_3$ solution for complete reduction. Find out the molar ratio of Cu^{2+} to $C_2O_4^{2-}$ in the compound. [IIT JEE Adv]

Sol. $\left(\frac{1}{2}\right)$ For explanation, refer to QR code given on the last page.

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Guru Mantra!

Don't be afraid after seeing the language

80. 2.480 g of $KClO_3$ are dissolved in conc. HCl and the solution was boiled. Chlorine gas evolved in the reaction was then passed through a solution of KI and liberated iodine was titrated with 100 ml of hypo. 12.3 ml of same hypo solution required 24.6 ml of 0.5 N iodine for complete neutralization. Calculate % purity of $KClO_3$ sample.

Sol. (82.34%) For explanation, refer to QR code given on the last page.

81. 20 mL of 0.02 M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}$ M [nearest integer]

[JEE Main 26 June 2022 Shift-2]

Given: $2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Sol. Meq of hypo solution = $20 \times 0.04 = 0.4$

Meq of Cu^{2+} in 10 mL = meq of $I_2 = 0.4$

Now, calculate mmol of Cu^{2+} ions in 10 mL = 0.4

Molarity = $\frac{0.4}{10} = 0.04$ M or 4×10^{-2} M

82. 0.3 g of an oxalate salt was dissolved in 100 mL solution. The solution required 90 mL of N/20 $KMnO_4$ for complete oxidation. The % of oxalate ion in salt is

(A) 33% (B) 66% (C) 70% (D) 40%

Sol. (B) $\frac{w}{\frac{88}{2}} \times 1000 = 90 \times \frac{1}{20}$

on solving, w = 0.198 g

% of oxalate ion = $\frac{0.198}{0.3} \times 100 = 66\%$

83. A mixture of pure $K_2Cr_2O_7$ and pure $KMnO_4$ weighing 0.561 g was treated with excess of KI in acidic medium. Iodine liberated required 100 ml of 0.15 M of sodium thiosulphate solution of exact oxidation. What is the percentage of each in the mixture?

[JEE Adv.]

Sol. (56.4%) For explanation, refer to QR code given on the last page.

84. An aqueous solution containing 0.10 g KIO_3 (formula weight = 214.0) was treated with an excess of KI solution. The solution was acidified with HCl. The liberated I_2 consumed 45.0 mL of thiosulphate solution to decolourise the blue starch-iodine complex. Calculate the molarity of the sodium thiosulphate solution.

[JEE Adv 1998]

Sol. $\overset{-5}{1KIO_3} + 5\overset{-1}{KI} \rightarrow 3\overset{0}{I_2}$

$(n = 5) \quad (n = 1)$

$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

Moles of I_2 formed $= 3\left(\frac{0.1}{214}\right)$ mol

Moles of hypo $= 2\left[3\left(\frac{0.1}{214}\right)\right]$ mol $= \frac{\text{Molarity} \times 45}{1000}$

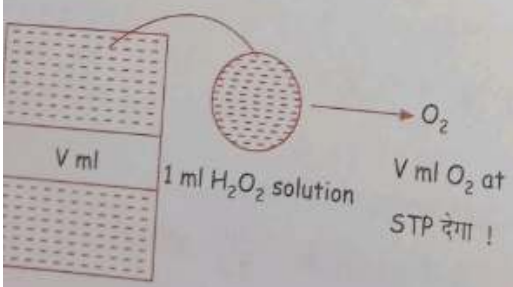
On solving, M = 0.0623 M

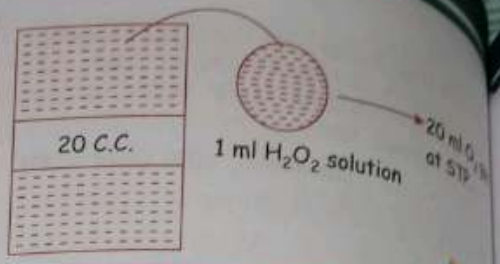
darr kyu rahe ho, koi baat hi nahi hai

VOLUME STRENGTH OF H_2O_2

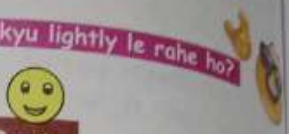
The volume strength of hydrogen peroxide (H_2O_2) is the number of volumes of oxygen gas produced by the decomposition of one volume of hydrogen peroxide.

$H_2O_2 \rightarrow H_2O + O_2$





iski strength ko kyu lightly le rahe ho?



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V.S = 5.6 × Normality or
= 11.2 × Molarity

85. The volume strength of 1.5 NH_2O_2 solution is
(A) 4.8 (B) 8.4 (C) 3.0 (D) 8.0

Sol. (B) V.S = 5.6 × Normality
= 5.6 × 1.5 = 8.40

86. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP.

[IIT JEE-1998]

Sol. $H_2O_2 + \overset{-1}{KI} \rightarrow \overset{0}{I_2}$
$(n = 2)$

Apply LOE: Eq. of H_2O_2 = Eq of KI = Eq of I_2

$\frac{N \times 5}{1000} = \frac{0.508}{254} \times 2$, V.S = 5.6 × 0.8 = 4.48

N = 0.8

dhyan lagao idhar

PERCENTAGE LABELLING OF OLEUM

The percentage labelling of oleum is based on the percentage of free sulfur trioxide (SO_3) present in the solution.

Or

Amount of total H_2SO_4 obtained on dilution of 100 g of oleum sample is also called percentage labelling of that sample.

Example:

$\xrightarrow{\text{dilute}}$ 109 gm total H_2SO_4 (मिलेगा)

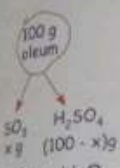
Oleum sample $\rightarrow H_2S_2O_7$

forming $H_2SO_4 = (H_2SO_4 + SO_3)$

87. Find out the percentage of free SO_3 in 109% oleum sample.

Sol. Oleum

$\xrightarrow{+H_2O}$ H_2SO_4 (Total) (109 gm) $\rightarrow$ H_2SO_4 (initial) + H_2SO_4 (formed) } 109 g



$1SO_3 + H_2O \longrightarrow \underset{\text{(Formed)}}{H_2SO_4}$

$\frac{x}{80}$ mol $SO_3 \longrightarrow \frac{x}{80}$ mol $H_2SO_4 \longrightarrow \frac{x}{80} \times 98$ gm.

$x = 40$, so, free SO_3 is 40% $\leftarrow (100 - x) + \frac{x}{80} \times 98 = 109$

jugaadu tarika hai kaam aaega bahut

88. If % free SO_3 in oleum = 40% then what is its labelling?

Sol. (109%) For explanation, refer to QR code given on the last page.

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Oleum $\xrightarrow{+H_2O}$ Total H_2SO_4

If labelling = 109%

100gm $\xrightarrow{+H_2O}$ 109 gm H_2SO_4

पानी डाला = 9gm $\rightarrow \frac{1}{2}$ mol $H_2O \rightarrow \frac{1}{2}$ mol SO_3

$\downarrow$

$80 \times \frac{1}{2} = 40$g

89. If labelling is 104.5%, then find out free SO_3?

Sol. $100 + 4.5 \rightarrow 104.5$ Total H_2SO_4

Oleum $H_2O \rightarrow \frac{1}{4}$ mol $H_2O \rightarrow \frac{1}{4}$ mol $SO_3 \rightarrow$

20 g $SO_3 \rightarrow$ 20% free SO_3

Note

For minimum labelling of oleum sample

oleum sample (100 g) + $H_2O \rightarrow H_2SO_4$ (Total)

For minimum labelling of oleum sample, water added should be minimum so that we get minimum amount of H_2SO_4 (Total). For this assuming minimum amount of SO_3 present in the oleum sample (just greater than zero).

So, minimum labelling should be just greater than 100%.

For maximum labelling of oleum sample

SO_3 should be maximum in 100 g oleum sample so that we add maximum water to get maximum H_2SO_4 (Total).

$SO_3 + H_2O \rightarrow H_2SO_4$

Assuming free SO_3 is almost 100 g. So H_2SO_4 formed after dilution is $\frac{100}{80} \times 98 = 122.5$ gm

So, maximum labelling should be just less than 122.5 %

dekho kitna asaan bna dia isko, araha hai na maza?

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100% < labelling < 122.5 %

BALANCING OF REDOX REACTION

1. First write oxidation half and reduction half separately.
2. Balance all the atoms other than O and H.
3. Find change in oxidation number in reduction and oxidation and equate them.
4. Balance charge on both sides by adding H^+ ions in case of acidic medium and in case of basic medium, add OH^- ions.
5. Balance O and H with water.

90. $Fe^{2+} + Cr_2O_7^{2-} \rightarrow Fe^{3+} + Cr^{3+}$

Balance in acidic medium.

Sol. $(6Fe^{3+} + 2Cr^{3+} + 7H_2O)$

For explanation, refer to QR code given on the last page.

91. In the following redox equation

$xUO^{2+} + Cr_2O_7^{2-} + yH^+ \rightarrow aUO_2^{2+} + zCr^{3+} + bH_2O$

The values of coefficients x, y and z respectively.

(A) 3, 8, 2 (B) 3, 8, 7

(C) 3, 2, 4 (D) 3, 1, 8

Sol. (A) $3H_2O + 3\overset{+4}{U}O^{2+} \rightarrow 3\overset{+6}{U}O_2^{2+} + 6e^- + 6H^+$

$14H^+ + 6e^- + \overset{+6}{Cr}_2O_7^{2-} \rightarrow 2\overset{+3}{Cr}^{+3} + 7H_2O$

$3UO^{2+} + Cr_2O_7^{2-} + 8H^+ \rightarrow 3UO_2^{2+} + 2Cr^{3+} + 4H_2O$

92. For the redox reaction

$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

the correct coefficient of the reactants for the balanced equation are

Sol. $5e^- + \overset{+7}{Mn}O_4^- \rightarrow Mn^{2+} \times 2$

$\overset{+3}{C}_2O_4^{2-} \rightarrow 2\overset{+4}{C}O_2 + 2e^-) \times 5$

$16H^+ + 10e^- + 2MnO_4^- \rightarrow 2Mn^{2+} + 8H_2O$

$5C_2O_4^{2-} \rightarrow 10CO_2 + 10e^-$

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

93. $2KMnO_4 + 5H_2S + 6H^+ \rightarrow 2Mn^{2+} + 2K^+ + 5S + 8H_2O$. In the above reaction, how many moles of electrons would be involved in the oxidation of 1 mole of reductant?

(A) Two (B) Five (C) Ten (D) One

Sol. (A) For explanation, refer to QR code given on the last page.

94. In redox reaction:

$Fe + HNO_3 \rightarrow Fe(NO_3)_2 + NH_4NO_3 + H_2O$

The coefficient of HNO_3, $Fe(NO_3)_2$, NH_4NO_3 is:

(A) 1 : 10 : 4 (B) 10 : 4 : 1

(C) 4 : 10 : 1 (D) 10 : 1 : 4

Sol. (B) For explanation, refer to QR code given on the last page.

95. In the reaction $xHI + yHNO_3 \rightarrow NO + H_2O$, upon balancing with whole number coefficients:

(A) x = 3, y = 2 (B) x = 2, y = 3

(C) x = 6, y = 2 (D) x = 6, y = 1

Sol. (C) Valency factor ratio is inversely related to molar ratio.

$(n_f)HI : (n_f)HNO_3 = 1 : 3 = 2 : 6$

$\therefore$ Molar ratio = 6 : 2

are balance hi toh karna hai or kya karna hai

96. Balance the following equation and choose the quantity which is the sum of the coefficients of reactants and products:

$KMnO_4 + H_2O_2 + \ldots\ldots H_2SO_4 \rightarrow \ldots\ldots MnSO_4 + \ldots\ldots O_2 \ldots\ldots + H_2O + \ldots\ldots K_2SO_4$

(A) 26 (B) 23 (C) 28 (D) 22

Sol. (A) Balance reaction is

$2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \longrightarrow 2MnSO_4 + 5O_2 + 8H_2O + K_2SO_4$

$\therefore$ Sum of stoichiometric coefficients = 2 + 5 + 3 + 2 + 5 + 8 + 1 = 26

pehle balance toh karo tabhi toh pata chalega coefficients ka

97. H_2S (5 moles) reacts completely with acidified aqueous potassium permanganate solution. In this reaction, the number of moles of water produced is x, and the number of moles of electrons involved is y. The value of (x + y) is _____

[JEE Adv. 2023]

Sol. First write the balanced equation:

$2KMnO_4 + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + 2MnSO_4 + 5S + 8H_2O$

According to the balanced equation, x = 8

Each molecule of H_2S loses 2 electrons, so, for every mole of H_2S, 2 moles of electrons are involved.

For 5 moles of H_2S, the number of electrons involved is 10 so, y = 10

x + y = 8 + 10 = 18

Agar reaction nahi likh paye toh nhi kr paoge paaglo

98. $MnO_4^- + I^- \rightarrow MnO_2 + IO_3^-$

Balance in Basic medium.

Sol.

$$3e^- + \overset{+7}{Mn}O_4^- \rightarrow \overset{+4}{Mn}O_2 \times 2$$

$$\overset{-1}{I} \rightarrow \overset{+5}{I}O_3^- + 6e^-$$

$$4H_2O + 6e^- + 2MnO_4^- \rightarrow 2MnO_2 + 8OH^-$$

$$6OH^- + I^- \rightarrow IO_3^- + 6e^- + 3H_2O$$

$$2MnO_4^- + H_2O + I^- \rightarrow IO_3^- + 2MnO_2 + 2OH^-$$

99. $Br_2 \rightarrow Br^- + BrO_3^-$

Balance in Basic medium.

Sol.

$$2e^- + \overset{0}{Br_2} \rightarrow 2\overset{-1}{Br} \times 5$$

$$\overset{0}{Br_2} \rightarrow \overset{+5}{Br}O_3^- + 10e^-$$

$$10e^- + 5Br_2 \rightarrow 10Br^-$$

$$12OH^- + Br_2 \rightarrow 2BrO_3^- + 10e^- + 6H_2O$$

$$12OH^- + 6Br_2 \rightarrow 2BrO_3^- + 10Br^- + 6H_2O$$

100. For the redox reaction $Zn + NO_3^- \rightarrow Zn^{2+} + NH_4^+$ in basic medium, coefficients of Zn, NO_3^- and OH^- in the balanced equation respectively are:

(A) 4, 1, 7 (B) 7, 4, 1

(C) 4,1,10 (D) 1,4,10

Sol. (C) Balanced reaction is

$$4Zn + NO_3^- + 7H_2O \rightarrow 4Zn^{2+} + NH_4^+ + 10OH^-$$

mazedaar hai, is par dhyan lagao ache se

101. Chlorine undergoes disproportionation in alkaline medium as shown below:

$$aCl_2(g) + bOH^-(aq) \rightarrow cClO^-(aq) + dCl^-(aq) + eH_2O(l)$$

The value of a, b, c and d in a balanced redox reaction are respectively:

[29 Jan, 2024 (Shift-I)]

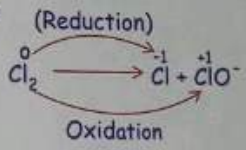
(A) 1, 2, 1 and 1

(B) 2, 2, 1 and 3

(C) 3, 4, 4 and 2

(D) 2, 4, 1 and 3

Sol. (A)



Balanced redox reaction is:-

$$1Cl_2 + 2OH^- \rightarrow 1Cl^- + 1ClO^- + 1H_2O$$

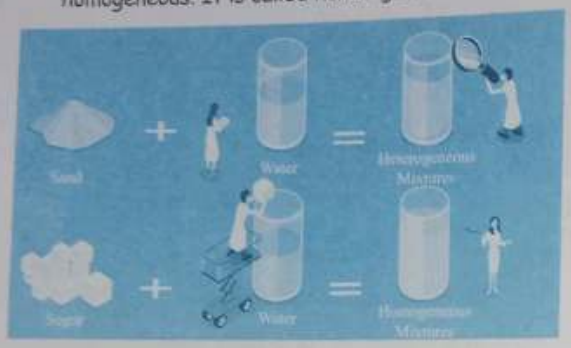
Scan the QR code to get the explanations of unsolved questions.

7 Solutions

INTRODUCTION

- Solutions are homogeneous mixtures of two or more than two components.
- Homogeneous mixture means composition and properties remain uniform throughout the mixture.

Example: Mixture of sugar in water is a homogeneous system. But mixture of sand in water is not homogeneous. It is called heterogeneous mixture.



Solutions

Solvent

- Component present in the largest quantity.
- Decides the physical state of solution.

Solute

- Component present in the solution other than solvent.

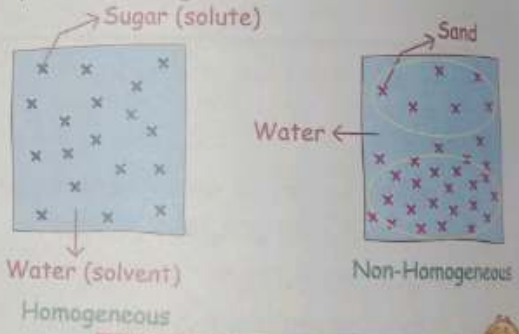
Important

Type of Solution	Solute	Solvent	Common Examples
Gaseous solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

yaad karna pdega

CONCENTRATION TERMS

A mixture of two or more substances can be a solution. We can also say that a solution is a homogeneous mixture of two or more substances. Homogeneous means uniform composition throughout. Thus, a homogeneous mixture i.e a solution will have uniform composition throughout.



Yeh toh bhut asaan hai yaar

Molarity

Molarity (M) is defined as the number of moles of solute present per litre of solution. The formula for molarity is:

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution in liters}}$$

Molality (Molal or m)

It is defined as the number of moles of solute dissolved in 1 kg or 1000 gm solvent

$$\text{Molality} = \frac{\text{moles of solute}}{\text{weight of solvent in Kg}}$$

Mole Fraction

It is defined as the ratio of the number of moles of a specific component to the total number of moles of all components in the mixture.

$$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{n_1}{n_1 + n_2}$$

$$X_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}} = \frac{n_2}{n_1 + n_2}$$

$$X_{solute} + X_{solvent} = \frac{n_1}{n_1 + n_2} + \frac{n_2}{n_1 + n_2} = 1$$

%w/w

It is defined as the "amount of solute dissolved (in gm) in 100 gm of solution".

$$\%w/w = \frac{\text{wt. of solute in gm}}{\text{wt. of solution in gm}} \times 100$$

%w/V

It is defined as the mass of solute (in gm) dissolved in 100 ml solution.

$$\%w/V = \frac{\text{wt. of solute in gm}}{\text{volume of solution in ml}} \times 100$$

%V/V

It is defined as the volume of solute (in ml) dissolved in 100 ml solution

$$\%V/V = \frac{\text{volume of solute in ml}}{\text{volume of solution in ml}} \times 100$$

Ppm

It is defined as the number of gm of solute present in 10^6 gm of solution.

$$ppm = \frac{\text{mass of solute (gm)}}{\text{mass of solution (gm)}} \times 10^6$$

Ppb

It is defined as the number of gm of solute dissolved in 10^9 gm of solution

$$Ppb = \frac{\text{mass of solute (gm)}}{\text{mass of solution (gm)}} \times 10^9$$

Formality

It is defined as the number of gram formula masses of the ionic solute dissolved per lt of solution.

- It is defined for ionic solids.

$$\text{Formality} = \frac{\text{no. of gm formula mass of solute}}{\text{volume of solution in lt}} = \frac{\text{mass of solute}}{\text{formula mass of solute} \times \text{vol. of solution in lt}}$$

Strength of Solution

It is defined as the mass of solute (in gm) present in 1 lt of solution

$$\text{Strength} = \frac{\text{mass of solute in gm}}{\text{volume of solution in lt}}$$

itne sare formulae, baap re baap!

Note

We have provided practice questions for these concentration terms in some basic concepts of chemistry. Kindly refer to that chapter for practice and for detailed theory.

VAPOUR PRESSURE OF PURE LIQUID

The vapour pressure of a pure liquid is the pressure exerted by the vapour molecules above the liquid surface when the liquid is in equilibrium with its vapour.

Molecules of solvent can be present in bulk as well as on the surface. Their respective characteristics are:

Molecules of solvent present in bulk

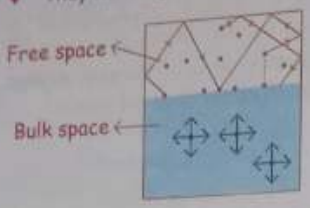
- They are in equilibrium
- They are stable
- They have less energy

$F_{net} = 0$

Molecules of solvent present on surface

- They are not at equilibrium
- They are not stable
- They have high energy

$F_{net} \neq 0$



samjho-samjho ek ek shabd ko

Important

Initially, the rate of evaporation is maximum and the rate of condensation is zero.

At equilibrium, rate of evaporation = rate of condensation

Saturated vapours (Maximum moles which can be present under given T and free volume)

Vapour Pressure

The pressure exerted by the saturated vapour on the surface of liquid or on the walls of container is called V.P of that liquid at that particular temperature.

$$liq \rightleftharpoons vapour$$

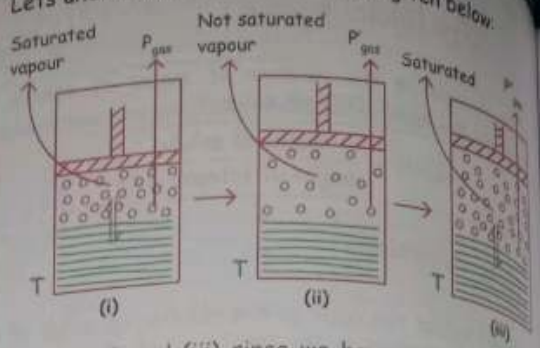
The equilibrium constant K_p for this equilibrium is

$$K_p = P_{vap}$$

Note

Since K_p is only temperature dependent so vapour pressure of a pure liquid is also temperature dependent.

Lets understand this with example given below:



In case (i) and (iii) since we have an equilibrium in between liq and vapours and temperature is constant so K_p is fixed, so vapour pressure is also fixed (same)

$P_{gas} = P''_{gas}$ and $P'_{gas} < P''_{gas}$; $P'_{gas} < P_{gas}$

dimag lagana padega tabhi samajh aaega

Important

K_p (equilibrium const) is only only temperature dependent.

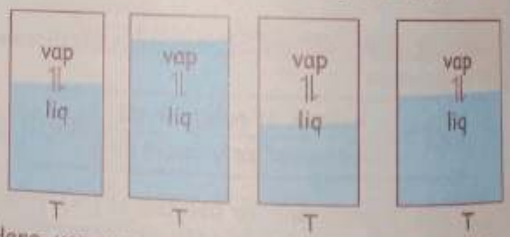
Means P_{vap} of pure liquid is only only only temperature dependent.

Note

Irrespective of the available volume for the vapours above the liquid surface, the vapour pressure is constant at constant T.

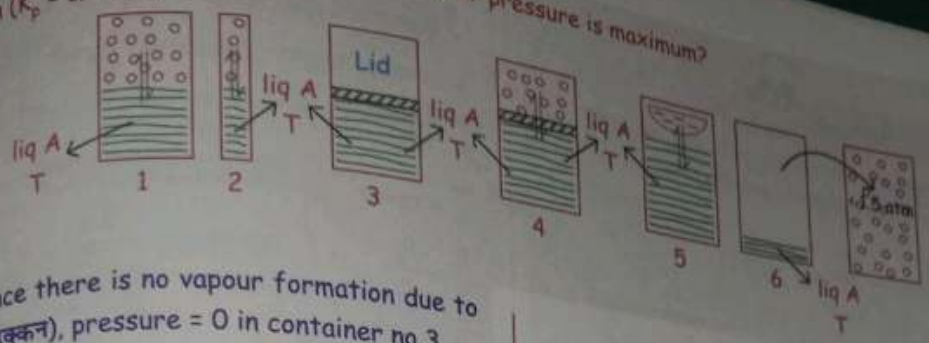
Khoobsurat Baat

Lets consider four containers as given below:



Here, vapour pressure will remain same in all four containers because at constant temperature, K_p is fixed and so vapour pressure.

1. Find out for which of these containers vapour pressure is maximum? Given (K_p = 1.5 atm)



Sol. • Since there is no vapour formation due to lid (ढक्कन), pressure = 0 in container no 3.

• In containers 1, 2, 3, 4, 5 since there is an equilibrium between vapour and liquid, the vapour pressure is same.

• Aisa lagta hai ki container 6 has not enough liquid to maintain the equilibrium, so press < 1.5.

yeh sawaal dimag dauda dega

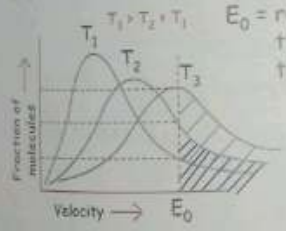
Khoobsurat Baat

Irrespective of the amount of liquid and shape of container the vapour pressure of the liquid is constant at a particular temperature. Provided the liquid is sufficient enough to make that liq ⇌ vap equilibrium.

Important

Explanation of temperature dependence of vapour pressure

1. Lets consider the graph given below:



E_0 = required energy for the molecules to leave the surface of liquid.

On increasing the temperature, the fraction of molecules having energy greater than E_0 increases. More particles leave the surface of liquid and go into vapour phase and V.P increases.

2. Consider an ideal gas equation:

$PV = nRT$, $P \alpha T$

Vapour pressure increases with the increase in temperature.

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अब आएगा मजा

Statement I: Vapour pressure increases on increasing the temperature.

Statement II: Vapour pressure linearly depends on temperature.

Statement I is correct but Statement II is wrong.

Clausius-Clapeyron Equation

Let consider the liquid-vapor equilibrium:

$$liq \rightleftharpoons vap \qquad K_p = V.P = P$$

$$\log\frac{(K_p)_{T_2}}{(K_p)_{T_1}} = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

K_p depends on temperature exponentially.

$$\log\frac{P_{T_2}}{P_{T_1}} = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Vapour pressure depends on temperature exponentially not linearly.

ab aaya samajh?

$(K_p)_{T_2}$ = Equilibrium constant at T_2 temperature

$(K_p)_{T_1}$ = Equilibrium constant at T_1 temperature

ΔH_{vap} = Heat of vapourisation of pure liquid

P_{T_2} = V.P at T_2 temperature

P_{T_1} = V.P at T_1 temperature

Equation ko yaad karlena kaam ban jaega

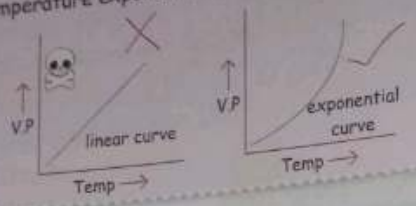
Note

Vapour Pressure and Temperature

The vapour pressure is directly proportional to temperature. As the temperature of liquid or solid increases, its vapour pressure also increases.

Common Mistake

Vapour pressure increases with the increase in temperature exponentially not linearly.



Note

- Vapour pressure of pure liquid is only temperature dependent.
- Irrespective of the amount of liquid and shape of container the vapour pressure is constant at a particular temperature.
- Liquid should be sufficient enough to make liq $\rightleftharpoons$ vap equilibrium.
- Vapour pressure depends on temperature exponentially.
- $\log\frac{P_{T_2}}{P_{T_1}} = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

(Temperature is in kelvin)

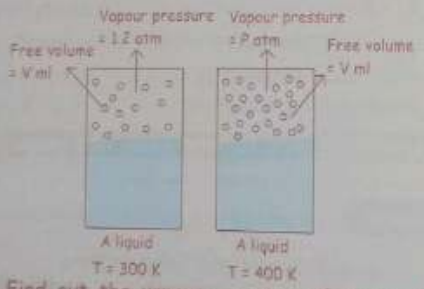
2. Vapour pressure of a pure liquid is 1.5 atm at 350 K. If its vapour pressure at 500 K is P atm then

(A) P = 1.0 atm (B) P > 1.5 atm

(C) P < 1.5 atm (D) can't say

Sol. (B) As temperature is increasing from 350 K to 500 K, vapor pressure will also increase. So, P > 1.5 atm

3.



Find out the vapour pressure of liquid A at 400 K if its vapour pressure at 300 K is 1.2 atm.

Sol. As temperature is increasing from 300 K to 400 K, vapor pressure will also increase. So, P > 1.2 atm

diagram ka khel hai sara, samjho

Common Mistake

Students might calculate vapour pressure at 400 K as:

$P \propto T$

$\frac{P}{T} = \text{const}$

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$\frac{1.2}{300} = \frac{P_2}{400}$

$P_2 = 1.6$ atm

But this is wrong as V.P exponentially depends on temperature.

hahaha

4. Calculate the vapour pressure of pure water at 60°C. Vapour pressure at 100°C is 1.0 atm and ΔH_{vap} of water = 9800 cal/mol

Sol. (0.6851 atm) For explanation, refer to QR code given on the last page.

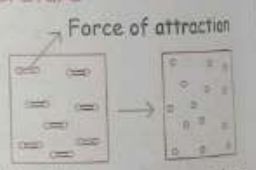
5. Vapour pressure of liquid Nickel at 1606°C is 0.100 torr whereas at 1805°C, its vapour pressure is 1.000 torr. At what temperature does the liquid have vapour pressure 2.500 torr?

Sol. (2169 K) For explanation, refer to QR code given on the last page.

Vapour Pressure of Two Liquids at Same Temperature

$(V.P)_A \neq (V.P)_B$ at same temperature

For different liquids at same temp since equilibrium constants are different, vapour pressure also different. Vapour pressure of different liquids has different values at same temperature because of different molecular interactions.



Formula dekhlo dhyan lagakar, samjhe

Important

$$\text{Vapour pressure} \propto \frac{1}{\text{Force of attraction}}$$

Force of attraction↑ vapours↓ vapour pressure↓

yeh mazedaar hai

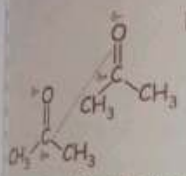
Khoobsurat Baat

The order of forces of attraction between molecules is:

Ionic-force > ion-dipole > Hydrogen-bond

$Na^+ \equiv Br^-$ $Na^+—H_2O$ $H_2O—H_2O$

> dipole-dipole > dipole-induced dipole > dispersion force or Vander Waal force



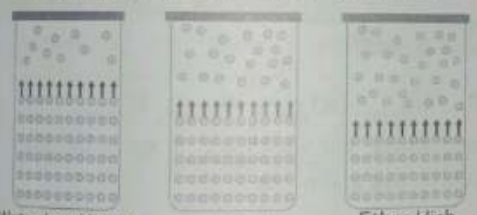
H_2O Br—Br Br—Br Br—Br

- For Vander Waal forces, more the molecular mass, more will be the force of attraction.

$Cl_2 < Br_2 < I_2$

Important

Vapour Pressure of Different Solvent



Water Low Vapour Pressure | Alcohol | Ether High Vapour Pressure

Liquids having small intermolecular attractive forces have a high tendency to evaporate and have high vapour pressure. Liquids having large intermolecular attractive forces have a small tendency to evaporate and have low vapour pressure. (at same temperature)

Khoobsurat Baat

Force of attraction:
Polar molecules > non-polar molecules
Vapour pressure:
Polar molecules < non-polar molecules

badia concept hai, chhod mat dena

6. Compare the vapour pressure of $H_2O(l)$ and $CHCl_3(l)$ at temperature T.

Sol. Force of attraction increases, vapour pressure decreases.

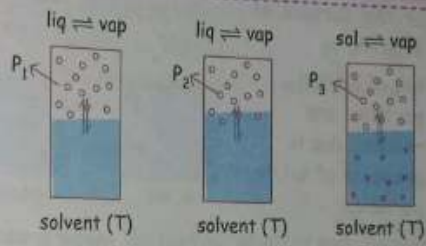
In H_2O, hydrogen bond is present while in $CHCl_3$, dipole-dipole forces are present.

Strength of H-bond> strength of dipole-dipole forces

$(V.P)_{H_2O} < (V.P)_{CHCl_3}$ at temperature T

VAPOR PRESSURE FOR SOLUTIONS

1. Solution of a non-volatile Solute in a Volatile Solvent



- Solute is non volatile
- Solvent is volatile
- In vapour phase of a solution, there must be solvent only

Note

volatile - उड़ता है।
$P^\circ \neq 0$

Non-volatile - नहीं उड़ता है।
$P^\circ = 0$

Common Mistake

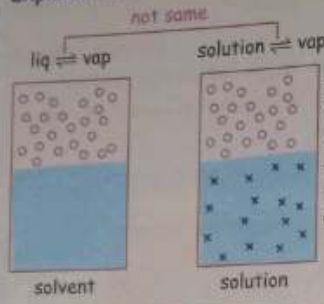
Arey temperature same hai to vapour pressure bhi same hoga jaise pahle kiya tha.

So, $P_1 = P_2 = P_3$

Lekin aisa nahi hai?

Khoobsurat Baat

Explanation:

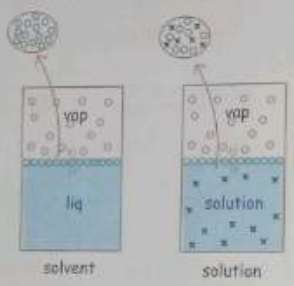


Since the equilibria are different so equilibrium constant are different at temperature T that is is why vapour pressures are different.

samjhe ab ki sahi reason kya hai

Now consider two different equilibria as:

- There exist solute-solvent interaction at the surface of solution.
- No. of solvent particles are reduced due to presence of solute particles.



Conclusion: Due to these two reasons, vapour pressure of solution decreases.

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- Due to these two reasons, vapour pressure of solution (P_s) is less than the vapour pressure of pure solvent (P°).

$$P_s < P^\circ$$

- As the amount of solute increases, vapour pressure further decreases.

balle balle

RAOULT'S LAW

This law states that the partial vapour pressure of each component in a solution is proportional to its mole fraction in the solution.

Derivation of Raoult's Law

$(rate)_{evaporation} \propto X_{solvent} = K_1 X_{solvent}$

$(rate)_{condensation} \propto P_S = K_2 P_S$

$(rate)_{evaporation} = (rate)_{condensation}$

$K_1 X_{solvent} = K_2 P_S, \quad P_S = \frac{K_1}{K_2} X_{solvent}$

if $X_{solvent} = 1$ then $P_S = P^\circ, \quad P^\circ = \frac{K_1}{K_2}(1)$

$P_S = P^\circ X_{solvent}, \quad P_S = P^\circ X_{solvent}$

P_S = vapour pressure of solution

P° = vapour pressure of solvent

$X_{solvent}$ = mole fraction of solvent

$P_s = P^\circ (1 - X_{solute}), \quad P^\circ - P_S = P^\circ (X_{solute})$

$$\frac{P^\circ - P_S}{P^\circ} = X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

if $n_{solute} << n_{solvent}$ (diluted solution)

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solvent}}$$

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{(w)_{solute} / M_{solute}}{\frac{(w)_{solute}}{M_{solute}} + \frac{(w)_{solvent}}{M_{solvent}}}$$

Formula dekhlo dhyan lagakar

Important

$P^\circ, P_S, (w)_{solute}, (w)_{solvent}, M_{solute}, M_{solvent}$

Among these six variables, 5 must be given to calculate the sixth one.

7. When a certain amount of solid A is dissolved in 100 g of water at 25°C to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is 23.75 mmHg. The number of moles of solute A added is _____ (Nearest Integer)

[27 July, 2022 (Shift-II)]

Sol. Apply Raoult's law:

$$\frac{P^\circ - P_S}{P_S} = \frac{n_{solute}}{n_{solvent}}, \quad \frac{P^\circ - \frac{P^\circ}{2}}{\frac{P^\circ}{2}} = \frac{n_{solute}}{n_{solvent}}$$

Finally,

$n_{solute} = n_{solvent} = \frac{100}{18} = 5.55$ mol

are mushkil nahi hai

8. The vapour pressure of solvent is 20 torr, while that of its dilute solution is 17 torr, the mole-fraction of the solvent is

(A) 0.6 (B) 0.85 (C) 0.5 (D) 0.7

Sol. (B) For explanation, refer to QR code given on the last page.

9. Two solutions A and B of cane sugar in water are prepared. In solution A, mole fraction of cane sugar is 0.1. Solution B is 1 molal. Which solution has a higher vapour pressure?

Sol. (B) For explanation, refer to QR code given on the last page.

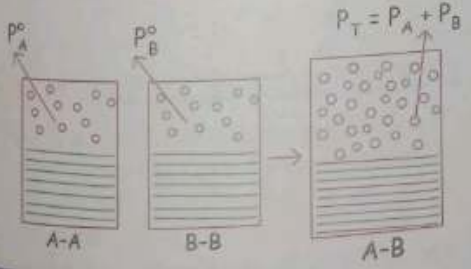
10. Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 gm of glucose is dissolved in 450 gm of water.

Sol. (17.438 mm Hg) For explanation, refer to QR code given on the last page.

Common Mistake

Applying incorrect formulas will lead to incorrect answers. So, check the formula carefully before proceeding further. Direct formula lagega

2. Solution of Volatile Solute in a Volatile Solvent where Both are Miscible



Note

solute = उड़ता है $p^\circ \neq 0$

solvent = उड़ता है $p^\circ \neq 0$

1. **Calculation of total vapour pressure of solution**

$P_A = P^\circ_A X_A$; $P_B = P^\circ_B X_A$

$P_T = P_A + P_B$ $P_T = P^\circ_A X_A + P^\circ_B X_B$

X_A and X_B are the mole fractions of A and B in liquid phase.

$P_T = P^\circ_A X_A + P^\circ_B (1 - X_A)$

$$P_T = P^\circ_B + (P^\circ_A - P^\circ_B) X_A$$

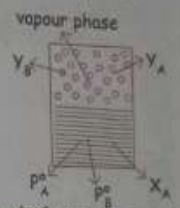
Khoobsurat Baat

To calculate the vapour pressure of solution we need mole fractions of A and B in liquid phase.

2. **Calculation of composition of vapour phase**

X_A = mole fraction of A in liquid phase.

Y_A = mole fraction of A in vapour phase.



According to Dalton's law:

Partial pressure of gas A = mole fraction × total pressure

$P_A = P_{Total} Y_A$

$$Y_A = \frac{P_A}{P_T} = \frac{P^\circ_A X_A}{P^\circ_A X_A + P^\circ_B X_B}, \quad Y_B = \frac{P_B}{P_T} = \frac{P^\circ_B X_B}{P^\circ_A X_A + P^\circ_B X_B}$$

$Y_A + Y_B = 1$

dhyan lagao idhar

Note

If X_A, X_B, P°_A, P°_B are known then we can easily calculate Y_A and Y_B.

If Y_A, Y_B, P°_A, P°_B are known then we can easily calculate X_A and X_B.

11. $P^\circ_A = 300$ torr

$P^\circ_B = 400$ torr, $X_A = 0.4$

Calculate-

(i) vapour pressure of A in vapour phase

(ii) vapour pressure of B in vapour phase

(iii) Total vapour pressure

(iv) Y_A and Y_B

Sol. (i) $P_A = P^\circ_A X_A = 300 \times 0.4 = 120$ torr

(ii) $P_B = P^\circ_B X_B = 400 \times 0.6 = 240$ torr

(iii) $P_T = P_A + P_B = 120 + 240 = 360$ torr

(iv) $Y_A = \frac{P_A}{P_T} = \frac{120}{360} = \frac{1}{3} = 0.33$

$$Y_B = 1 - Y_A = 1 - \frac{1}{3} = \frac{2}{3} = 0.67$$

yeh sawaal seedhe tarike se hoga, jada dimag nhi lgane ka

12. Equal moles of liq P and liq Q are mixed. What is the ratio of their moles in vapour phase?

Given: ($P^\circ_P = 2P^\circ_Q$)

Sol. (2:1) For explanation, refer to QR code given on the last page.

13. If the total vapour pressure of the liquid mixture A and B is given by the equation:

$P = 180\ X_A + 90$ then the ratio of the vapour pressure of the pure liquids A and B is given by:

(A) 3 : 2 (B) 4 : 1 (C) 3 : 1 (D) 6 : 2

Sol. (C) $P_T = P^\circ_A X_A + P^\circ_B X_B$

$P_T = P^\circ_A X_A + P^\circ_B (1 - X_A)$, $P_T = P^\circ_B + X_A (P^\circ_A - P^\circ_B)$

$P^\circ_B = 90$ $P^\circ_A - P^\circ_B = 180$

$P^\circ_A = 180 + P^\circ_B = 180 + 90 = 270$

$$\frac{P^\circ_A}{P^\circ_B} = \frac{270}{90} = 3:1$$

14. Liquids A and B form ideal solution for all compositions of A and B at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of A have the total vapor pressures of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid B in bar?

[JEE Adv.-2020]

Sol. For an ideal solution,

$P_T = P^\circ_A X_A + P^\circ_B X_B$

Substitute the values in the above equation:

$0.3 = P^\circ_A \times 0.25 + P^\circ_B \times 0.75$ ...(i)

Also, $0.4 = P^\circ_A \times 0.5 + P^\circ_B \times 0.5$

$0.8 = P^\circ_A + P^\circ_B$ ...(ii)

From equations (i) and (ii):

$P^\circ_A = 0.6$ bar, $P^\circ_B = 0.2$ bar

unit likhna mat bhul jana

15. The Total pressure observed by mixing two liquid A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively. The total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is mm Hg. (Nearest Integer) Consider the liquid and solution behave ideally. [24 Jan, 2023 (Shift-II)]

Sol. (434 mm Hg) For explanation, refer to QR code given on the last page.

16. The vapour pressure of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is_____. [28 June, 2022 (Shift-II)]

Sol. $Y_B = \frac{P^\circ_B X_B}{P^\circ_A X_A + P^\circ_B X_B}$

$$= \frac{100 \times 0.7}{(50 \times 0.3) + (100 \times 0.7)} = \frac{70}{85} = \frac{14}{17}$$

17. A gaseous mixture of two substances A and B, under a total pressure of 0.8 atm is in equilibrium with an ideal liquid solution. The mole fraction of substance A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is ____ atm. (Nearest Integer)

[28 July, 2022 (Shift-II)]

Sol. $P_A = Y_A \times P_T$

Substitute the values in the above expression:

$P_A = 0.5 \times 0.8 = 0.4$

$$P_A = X_A \times P^\circ_A \Rightarrow P^\circ_A = \frac{0.4}{0.2} = 2 \text{ atm}$$

are kitna asaan hai waah-waah

18. The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is _____ (Nearest Integer)

[20 July, 2021 (Shift-II)]

Sol. (1) For explanation, refer to QR code given on the last page.

19. Two liquids A and B form ideal solution. At 300 K, the vapour pressure of a solution containing 1 mol of A and 3 mol of B is 550 mm of Hg. At the same temperature one more mol of B is added to this solution, the vapour pressure of solution increases by 10 mm of Hg. Determine the vapour pressure of A and B in their pure state.

[Advanced Level]

Sol. (P°_B = 600 mm of Hg, P°_A = 400 mm of Hg)

For explanation, refer to QR code given on the last page.

20. The vapour pressure of pure benzene and methyl benzene at 27°C is given as 80 Torr and 24 Torr, respectively. The mole fraction of methyl benzene in vapour phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is_____ $\times 10^{-2}$ (nearest integer)

[09 April, 2024 (Shift-II)]

Sol. In vapour phase,

$$Y_{\text{Methylbenzene}} = \frac{P_{\text{Methylbenzene}}}{P_{\text{Total}}}$$

on substituting values,

$$Y_{\text{Methylbenzene}} = \frac{0.5 \times 24}{0.5 \times 80 + 0.5 \times 24}$$

$$= \frac{12}{40+12} = 0.23 = 23 \times 10^{-2}$$

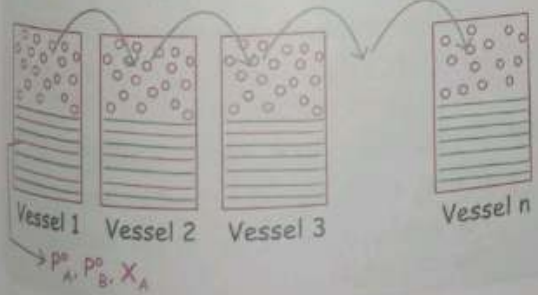
Common Mistake

When 1 more mole of B is added, mole fraction of A becomes $\frac{1}{5}$ instead of $\frac{1}{4}$.

Important

Calculation of Y_A and Y_B for n^{th} vessel during distillation process

Lets consider the 'n' no. of vessels:



Important

Step 1: Calculation of Y_A, Y_B of vessel 1

$$Y_{A,1} = \frac{P^\circ_A X_{A,1}}{P^\circ_A X_{A,1} + P^\circ_B (1 - X_{A,1})}$$

$$1 - Y_{A,1} = \frac{P^\circ_B (1 - X_{A,1})}{P^\circ_A X_{A,1} + P^\circ_B (1 - X_{A,1})}$$

$$\boxed{\frac{Y_{A,1}}{1 - Y_{A,1}} = \frac{P^\circ_A}{P^\circ_B}\left(\frac{X_{A,1}}{1 - X_{A,1}}\right)}$$

$X_{A,1}$ = mole fraction of A in liquid phase of vessel 1.

$Y_{A,1}$ = mole fraction of A in vapour phase of vessel 1.

Step 2: Calculation of mole fraction in vapour phase of vessel 2

Y_A and Y_B of vessel 1 = X_A and X_B for vessel 2

$$\updownarrow \begin{matrix} Y_{A,1} = X_{A,2} \\ Y_{B,1} = X_{B,2} \end{matrix} \updownarrow$$

$$Y_{A,2} = \frac{P^\circ_A X_{A,2}}{P^\circ_A X_{A,2} + P^\circ_B (1 - X_{A,2})} = \frac{P^\circ_A Y_{A,1}}{P^\circ_A Y_{A,1} + P^\circ_B (1 - Y_{A,1})}$$

$$Y_{B,2} = 1 - Y_{A,2} = \frac{P^\circ_B (1 - Y_{A,1})}{P^\circ_A Y_{A,1} + P^\circ_B (1 - Y_{A,1})}$$

$$\frac{Y_{A,2}}{1 - Y_{A,2}} = \frac{P^\circ_A}{P^\circ_B}\left(\frac{Y_{A,1}}{1 - Y_{A,1}}\right), \quad \frac{Y_{A,2}}{1 - Y_{A,2}} = \frac{P^\circ_A}{P^\circ_B} \cdot \frac{P^\circ_A}{P^\circ_B}\left(\frac{X_{A,1}}{1 - X_{A,1}}\right)$$

$$\boxed{\frac{Y_{A,2}}{1 - Y_{A,2}} = \left(\frac{P^\circ_A}{P^\circ_B}\right)^2 \left(\frac{X_{A,1}}{1 - X_{A,1}}\right)}$$

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$$\frac{Y_{A,n}}{1 - Y_{A,n}} = \left(\frac{P^\circ_A}{P^\circ_B}\right)^n \frac{X_{A,1}}{1 - X_{A,1}}$$

where $Y_{A,n}$ = mole fraction of A in the vapour phase of n^{th} vessel.

21. A vessel labelled 1 contains 3 mol of A and 1 mol of liquid B forming an ideal solution. The solution is allowed to reach equilibrium with its vapours. The vapours are then quickly taken out and condensed in vessel labelled 2. Here again the solution is allowed to reach an equilibrium with its vapours. The vapours are again quickly taken out and condensed in vessel labelled 3. This process is continued several times.

If P°_A = 700 mm Hg, P°_B = 350 mm Hg **[Advanced level]**

(i) composition of vapour in vessel 1

(ii) composition of vapour in vessel 2

(iii) composition of vapour in vessel 100

yeh bdia sawal hai, dimag ko pura jaga dega

Sol. (i) $Y_{A,1} = \frac{6}{7}; Y_{B,1} = \frac{1}{7}$ (ii) $Y_{A,2} = \frac{12}{13}; Y_{B,2} = \frac{1}{13}$

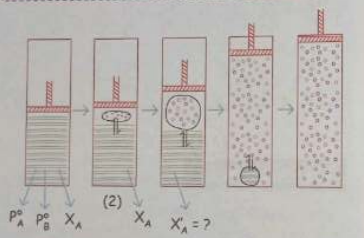
(iii) $Y_{A,100} = 1,\ Y_{B,100} = 0$

For explanation, refer to QR code given on the last page.

22. 100 gm of liquid A (molar mass 140 g/mol) was dissolved in 1000 gm of liquid B (molar mass 180 g/mol). The vapour pressure of pure liquid B was found to be 500 torr. Calculate the vapour pressure of pure liquid A and its vapour pressure in the solution if the total vapour pressure of solution is 475 torr.

Sol. (280.5 torr) For explanation, refer to QR code given on the last page.

CASE OF EVAPORATION



Dheere dheere piston uthate ja rhe hain, vapors bante ja rhe hain

23. Find out the vapour pressure of the liquid when first drop of vapour formed.

Sol. $P_T = P^\circ_A X_A + P^\circ_B X_B$ **बेईमानी**

Assuming the X_A and X_B values in first container is equal to the X_A and X_B in this container as the amount of A and B in the first drop of vapour is negligible.

24. Find out the composition of this first drop of vapour formed.

Sol. $Y_A = \frac{P^\circ_A X_A}{P^\circ_A X_A + P^\circ_B (1 - X_A)}$ or $\frac{Y_A}{1 - Y_A} = \left(\frac{P^\circ_A}{P^\circ_B}\right)\left(\frac{X_A}{1 - X_A}\right)$

25. Find out the composition of vapours when last drop of liquid remained. **बेईमानी**

Sol. Assuming all the liquid is now in vapour phase so the initial X_A and X_B will be now Y_A and Y_B of this container as there is negligible amount of A and B left in the last drop of liquid remained.

$X_A = Y_A,\ X_B = Y_B$

Thodi se beyimaani krni pdegi, varna dikkat aaegi

26. What will be the composition of the last drop of liquid remained?

Sol. $\frac{Y_A}{1 - Y_A} = \frac{P''_A}{P''_B}\frac{X'_A}{1 - X'_A}$ $X'_A = ?$

27. What will be the vapour pressure of last drop of liquid remained?

Sol. $P_T = P^\circ_A X'_A + P^\circ_B X'_B$

28. The vapour pressure of two pure liquids, A and B, which form an ideal solution are 300 and 800 torr respectively, at temperature T. A liquid solution of A and B for which the mole fraction of A is 0.60 is contained in a cylinder closed by a piston on which the pressure can be varied. The solution is slowly vaporized at temperature T by decreasing the applied pressure, starting with a pressure of about 1 atm. Calculate

(A) The pressure at which the first bubble of vapour is formed

(B) The composition of the vapour in this bubble

(C) The composition of the last droplet

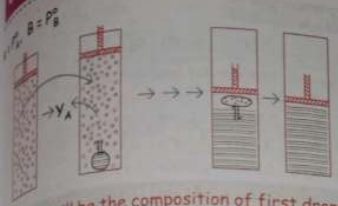
(D) The pressure when only this last droplet of liquid remains

Sol. (A) 500 torr (B) $Y_A = 0.36,\ Y_B = 0.64$

(C) $X'_A = 0.8,\ X'_B = 0.2$ (D) 400 torr

For explanation, refer to QR code given on the last page.

CASE OF CONDENSATION



29. What will be the composition of first droplet of liquid formed?

Sol. Assuming the vapour phase ↑composition remains same as that of first container because the amount of A and B in first droplet of liquid is negligible

$$Y_A = \frac{P_A^\circ X_A}{P_A^\circ X_A + P_B^\circ (1 - X_A)} \text{ or } \frac{Y_A}{1 - Y_A} = \left(\frac{P_A^\circ}{P_B^\circ}\right)\frac{X_A}{1 - X_A}$$

$X_A = ?$

30. Find out the vapour pressure of the first droplet of liquid formed.

Sol. $P_T = P_A^\circ X_A + P_B^\circ X_B$

31. What is the composition of the liquid phase when last drop of vapour remained?

Sol. Assuming all the vapour are now in liquid phase as the amount left in the last drop of vapour is negligible.

$Y_A = X'_A$, $Y_B = X'_B$

32. What will be the vapour pressure of the liquid when last drop of vapour remained?

Sol. $P_T = P_A^\circ X'_A + P_B^\circ X'_B$, $P_T = P_A^\circ Y_A + P_B^\circ Y_B$

33. What will be the composition of the last drop of vapour remained?

Sol. $$\frac{Y_A}{1 - Y_A} = \frac{P_A^\circ}{P_B^\circ}\left(\frac{X_A}{1 - X_A}\right) = \frac{P_A^\circ}{P_B^\circ}\left(\frac{Y_A}{1 - Y_A}\right)$$

34. The vapour pressure of two pure liquids, A and B that form an ideal solution are 300 and 800 torr respectively, at temperature T. A mixture of the vapours of A and B for which the mole fraction of A is 0.25 is slowly compressed at temperature T. Calculate

(1) The composition of the first drop of the condensate

(2) The total pressure when this drop is formed

(3) The composition of the solution whose normal boiling point is T

(4) The pressure when only the last bubble of vapour remains, and

(5) The composition of the last bubble

Sol. (1) 0.47 (2) 565 torr (3) 0.08
(4) 675 torr (5) 0.11
For explanation, refer to QR code given on the last page.

35. Two liquids, A and B have the same molecular weights and form an ideal solution. The solution of composition x_A has the vapour pressure 700 mm Hg at 80°C. The above solution is distilled without reflux till 3/4 of the solution is collected as condensate. The composition of the condensate is $x'_A = 0.75$ and that of residue is $x_A = 0.3$. If the vapour pressure of the residue at 80°C is 600 mm Hg, calculate x_A, P_A° and P_B°.

Sol. ($x_A = 0.63$, $P_A^\circ = 809$ mm Hg ; $P_B^\circ = 510$ mm Hg)
For explanation, refer to QR code given on the last page.

36. Vapour pressure of methylalcohol and ethyl alcohol solution is represented by $P = 115\,X_A + 140$ where X_A = mole fraction of methyl alcohol. The value of $\lim_{x_A \to 0} \frac{P_B^\circ}{X_B}$ is-

(A) 225 (B) 115 (C) 140 (D) 135

Sol. (C) $P_T = P_A^\circ X_A + P_B^\circ (1 - X_A)$

$P_T = P_B^\circ + X_A (P_A^\circ - P_B^\circ)$

$P_B^\circ = 140$, $P_A^\circ - P_B^\circ = 115$, $P_A^\circ = 255$

37. Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mm Hg, respectively, at the same temperature. Then, correct statement is
(X_M = Mole fraction of 'M' in solution;
X_N = Mole fraction of 'N' in solution;
y_M = Mole fraction of 'M' in vapour phase;
y_N = Mole fraction of 'N' in vapour phase)
[9 April, 2019 (Shift-I)]

(A) $\frac{X_M}{X_N} = \frac{y_M}{y_N}$ (B) $\frac{X_M}{X_N} > \frac{y_M}{y_N}$

(C) $\frac{X_M}{X_N} < \frac{y_M}{y_N}$ (D) $(X_M - y_M) < (X_N - y_N)$

Sol. (B) $\because P_N^\circ > P_M^\circ$

$\therefore y_N > X_N$ and $X_M > y_M$

On multiplying, we get, $y_N X_M > X_N y_M$

dimag ki batti jali?

3. Mixture of two volatile liquid where both are immiscible

[Since the liquid immiscible]

$P_T = P^\circ_A + P^\circ_B$

$Y_A = \frac{P^\circ_A}{P^\circ_A + P^\circ_B}$, $Y_B = \frac{P^\circ_B}{P^\circ_A + P^\circ_B}$

$Y_A = \frac{P^\circ_A}{P^\circ_A + P^\circ_B}$ $\quad \frac{n_A}{(n_A + n_B)} = \frac{P^\circ_A}{P^\circ_B}$ $\quad \frac{W_A / M_A}{W_B / M_B} = \frac{P^\circ_A}{P^\circ_B}$

$Y_B = \frac{P^\circ_B}{P^\circ_A + P^\circ_B}$ $\quad \frac{n_B}{(n_A + n_B)}$

$\frac{Y_A}{Y_B} = \frac{P^\circ_A}{P^\circ_B} \rightarrow \frac{n_A}{n_B} = \frac{P^\circ_A}{P^\circ_B} \rightarrow \frac{W_A}{W_B} = \frac{P^\circ_A \cdot M_A}{P^\circ_B \cdot M_B}$

n_A = moles of A in vapour phase
n_B = mole of B in vapour phase
W_A = weight of A in vapour phase
W_B = weight of B in vapour phase

Khatarnaak toh nahi laga

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Agar hume vapour phase mein 'A' ka amount (W_A) zyada chahiye to $P^\circ_B M_B$ ki value chhoti honi chahiye.

38. Which liquid is best for the purification of a liquid which is immiscible?

(A) $P^\circ = a$; $M = b$ (B) $P^\circ = \frac{a}{2}$; $M = 3b$

(C) $P^\circ = 2a$; $M = 2b$ (D) $P^\circ = \frac{a}{2}$; $M = b$

Sol. (D) $\frac{W_A}{W_B} = \frac{P^\circ_A M_A}{P^\circ_B M_B}$

For W_A to be more, $P^\circ_B M_B$ should be less.

39. If liq A (P°_A = 300 torr) immiscible with liq B (P°_B = 500 torr) are mixed then find out the mole ratio of A and B in vapour phase.

Sol. $\frac{Y_A}{Y_B} = \frac{P^\circ_A}{P^\circ_B} \rightarrow \frac{n_A}{n_B} = \frac{P^\circ_A}{P^\circ_B} = \frac{300}{500} = \frac{3}{5}$

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Maze Ki Baat

- When V.P liquid becomes equal to atmospheric pressure, boiling starts.

Important

	(A)	(H_2O)	(Mixture)	
at t = 40°C	0.05 atm	0.1 atm	0.15 atm	Na A boil karega, Na H_2O boil karega but Haa hi mixture boil karega.
t = 45°C	0.08 atm	0.2 atm	0.28 atm	
t = 70°C	0.2 atm	0.5 atm	0.7 atm	
t = 90°C	0.3 atm	0.65 atm	0.95 atm	A and H_2O to boil Nahi karege but mixture boil karega.
t = 98°C	0.32 atm	0.68 atm	1.0 atm	

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The boiling point of this mixture is less than that of boiling point of either liquid.

40. If the boiling point of liquid A = 100°C and liquid B is 140°C and both are immiscible then find out the B.P of this liquid mixture-

(A) 100°C (B) 120°C

(C) 140°C (D) 98°C

Sol. (D) Boiling point of mixture of two volatile immiscible liquids is less than the boiling point of either of the liquids.

41. An unknown compound is immiscible with water. It is steam distilled at 98°C and P_T = 737 torr, $P^\circ_{H_2O}$ = 707 torr at 98°C. This distillate was 75% by weight water, Calculate the molecular weight of the unknown.

Sol. (141.4 g/mol) For explanation, refer to QR code given on the last page.

IDEAL SOLUTIONS

Solutions which follow Raoult's law on entire range of composition are called ideal solutions.

For ideal solutions-

A-A and B-B interactions = A-B interaction

(i) $\Delta H_{mix} = 0$ (ii) $\Delta V_{mix} = 0$

(iii) $\Delta G_{mix} < 0$ (iv) $\Delta S_{mix} > 0$

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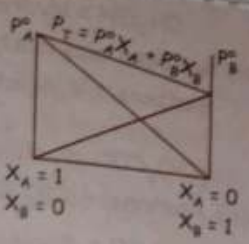
Whenever particular interactions are replaced by similar type of interactions then the resultant solution is ideal solution.



graph dekh kar hosh toh nahi udd gaye?

Example:

(i) Benzene + Toluene

(ii) Hexane + Heptane

(iii) Ethyl bromide + Ethyl iodide

(iv) Chlorobenzene + Bromobenzene

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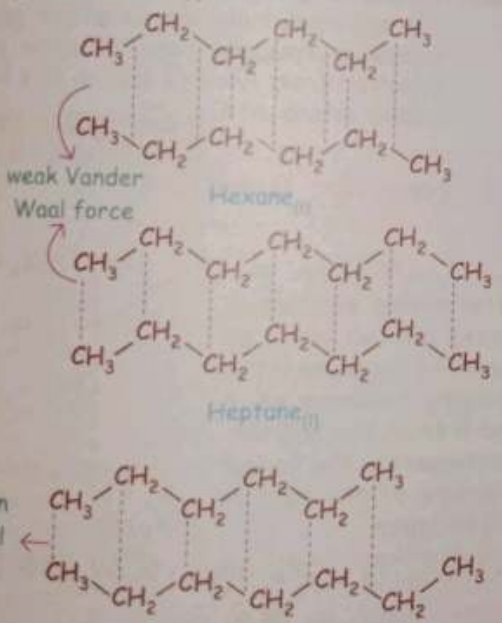
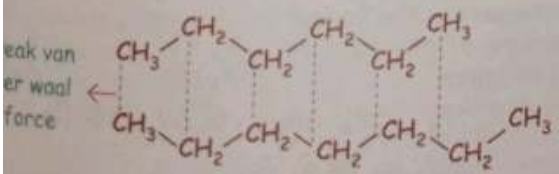
$Hexane_{(l)}$ and $Heptane_{(l)}$





Van der waal interaction is replaced by van der Waal interaction, so, solution is ideal.

NON-IDEAL SOLUTIONS

1. Positive Deviation

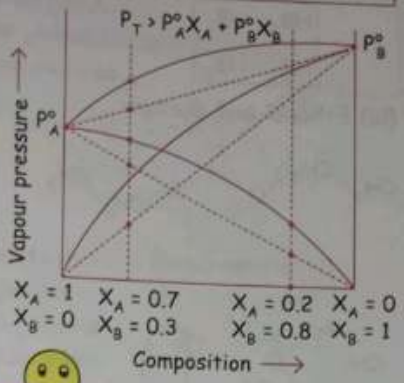
A-B interaction < A-A interaction or B-B interaction

(i) $\Delta H_{mix} > 0$

(ii) $\Delta V_{mix} > 0$

(iii) $\Delta G_{mix} < 0$

(iv) $\Delta S_{mix} > 0$



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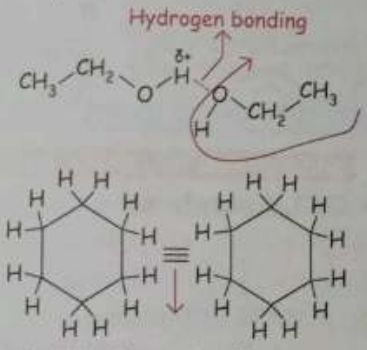
Whenever stronger interactions are replaced by weaker interactions then we observe positive deviated solution.

Hydrogen Bonding > dipole-dipole > vander Waal forces

Examples of positive deviation

(i) Ethanol + Cyclohexane



- Cyclohexane will reduce the extent of hydrogen bonding leads to (+) ve deviated solution.

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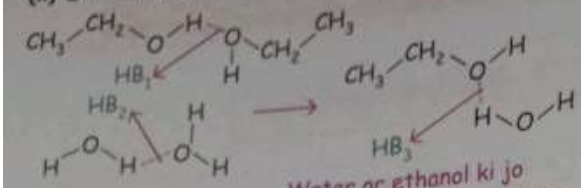
More the M wt more the vander Waal interaction

bhut mazedaar hai

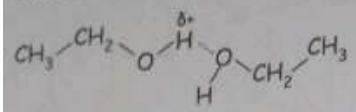
(ii) Ethanol and water or methanol and water



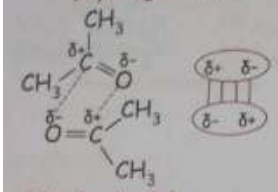
$HB_3 > HB_1$
$HB_3 < HB_2$

Water or ethanol ki jo H-bonding hai wo water-water ki H-bonding se kam hai, isliye positive deviation hoga

(iii) Ethanol and acetone



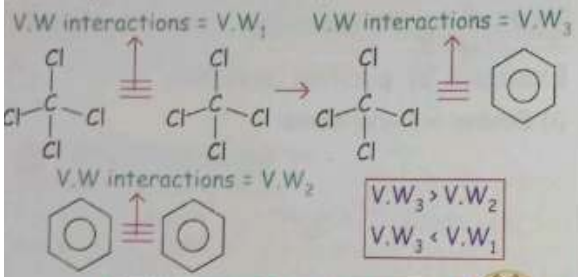
(Hydrogen bond)



(dipole-dipole)

Acetone reduces the extent of hydrogen bonding between ethanol molecules, leads to (+) ve deviated solution

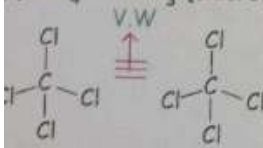
(iv) CCl_4 and Benzene or CCl_4 and Toluene

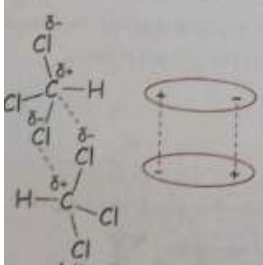


$V.W_3 > V.W_2$
$V.W_3 < V.W_1$

badia sawaal hai, dhyan lga lo

(v) CCl_4 + $CHCl_3$ (chloroform)





(dipole-dipole)

V.W interactions reduces the extent of dipole-dipole interaction, leads to (+) ve deviated solution.

(vi) CCl_4 and methanol

CCl_4 = V.W interactions → CH_3OH = Hydrogen bond

V.W interactions reduces the extent of H-Bonding

(vii) Benzene and acetone

Benzene = V.W interactions → $(CH)_2CO$ = dipole-dipole

V.W interactions reduces the extent of dipole-dipole

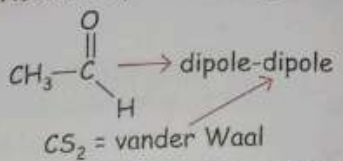
(viii) Acetone + CS_2 (carbon disulphide)

$(CH)_2CO$ = dipole-dipole

CS_2 = non polar → Vander Waal interaction

V.W interactions reduces the extent of dipole-dipole

(ix) Acetaldehyde and CS_2



V.W interactions reduces the extent of dipole-dipole

sare examples dekhlo ache se, varna pagla jaoge

42. Identify the mixture that shows positive deviation. **[Jan-2024]**

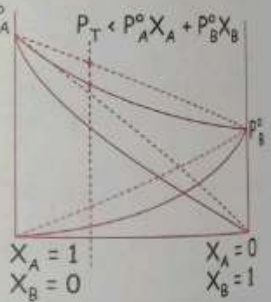
(A) $(CH_3)_2CO + C_6H_5NH_2$ (B) $CHCl_3 + C_6H_6$
(C) $CHCl_3 + (CH_3)_2CO$ (D) $(CH_3)_2CO + CS_2$

Sol. (D) Acetone is a polar molecule while CS_2 is a non-polar molecule. Polar and non-polar molecules typically tend to show positive deviation from Raoult's law as this leads to weaker interaction.

2. Negative Deviation

$F_{A-B} > F_{A-A}$ and F_{B-B}

After mixing, interaction forces are increasing so there is a decrease in the escaping tendency of A and B from the solution. Consequently, the vapour pressure of the solution will be lower.



$P_A > P^\circ_A X_A$; $P_B > P^\circ_B X_B$; $\Delta H_{mix} < 0$, $\Delta V_{mix} < 0$

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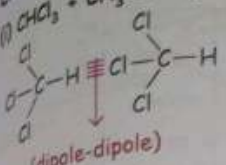
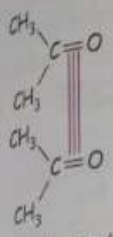
Whenever weaker interactions are replaced by stronger interactions, solution shows (-) ve deviation.

Examples of negative deviation

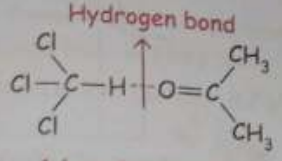
(i) $CHCl_3 + CH_3COCH_3$



(dipole-dipole)



(dipole-dipole)

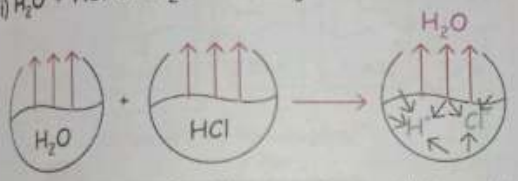


1 (-ve deviation)

Here, dp-dp and dp-dp(weaker) interactions are replaced by H-bonding(stronger)

structures toh pta hai na lallu

(ii) $H_2O + HCl$ or $H_2O + HNO_3$



In case of solution of acid and water, acid dissociates into ions. Ions do no evaporate so vapour pressure decreased leading to negative deviation.

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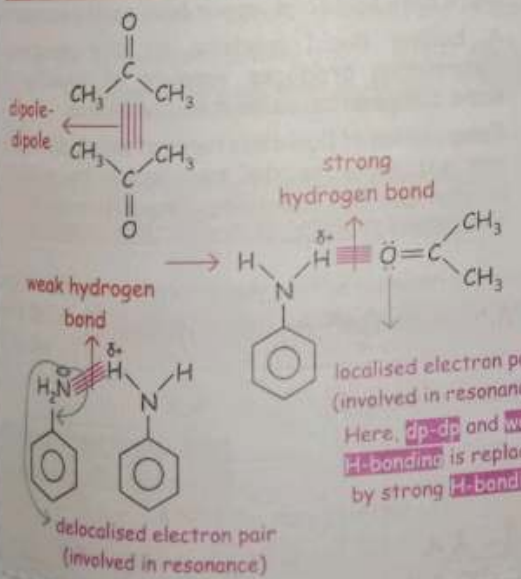
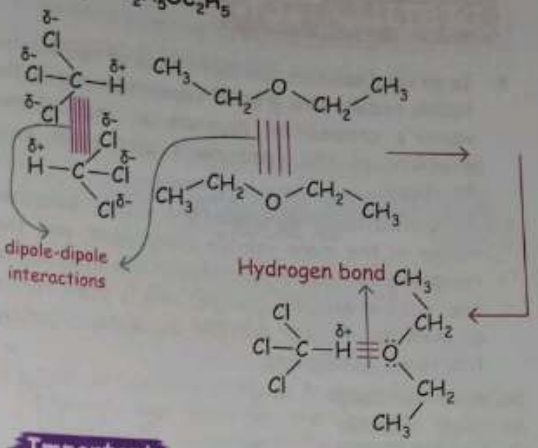


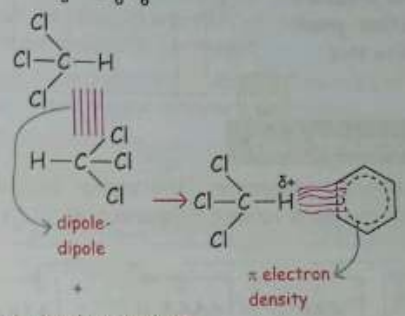
Here, dp-dp and weak H-bonding is replaced by strong H-bonding

(iii) $CHCl_3 + C_2H_5OC_2H_5$



Important

(iv) $CHCl_3 + C_6H_6$



Vander Waal interactions

(non polar molecule)

Due to interaction of H and π electron density this system shows negative deviation.

isko karke dekho kitab mei baar-baar

43. Solution of phenol and aniline shows-
(A) Positive deviation (B) Negative deviation
(C) Zero deviation (D) Can't predict

Sol. (B) For explanation, refer to QR code given on the last page.

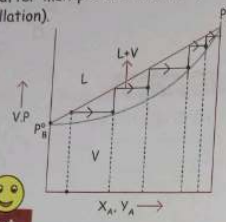
44. A solution of two miscible liquids showing negative deviation from Raoult's law will have [Jan-2024]
(A) increased V.P, increased B.P
(B) increased V.P, decreased B.P
(C) decreased V.P, decreased B.P
(D) decreased V.P, increased B.P

Sol. (D) For explanation, refer to QR code given on the last page.

DISTILLATION

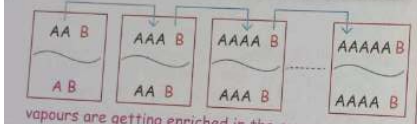
- In an ideal solution, the distillation of two volatile liquids results in a gradual separation where the vapour's composition depends on the relative volatility of the components and their mole fractions.
- As distillation progresses, the vapour becomes richer in the more volatile component, and the remaining liquid becomes richer in the less volatile one, until eventually, you get nearly pure forms of each liquid after multiple distillations (or by fractional distillation).

Suppose we have an ideal solution of A and B liquids (A is more volatile) then the pressure composition graph will be like this



Khoobsurat Baat

As distillation progresses, the vapour becomes richer in the more volatile component 'A'.



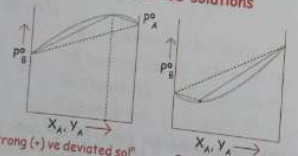
vapours are getting enriched in the component which is more volatile.

Lekin ab kamal dekho!

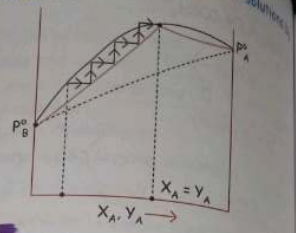
For largely (+) ve or (–) ve deviated solutions we can not separate the two components after a particular composition is achieved.

Note

Pressure-composition curve for (+) ve deviated or (–) ve deviated solutions

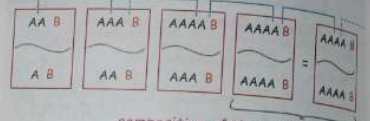


Now let's try to separate these kind of solutions by distillation method:-



Note

After few steps of distillation the vapour produced during boiling has the same composition as the liquid mixture, which makes it impossible to separate the components by simple distillation.



composition of the solution is fixed

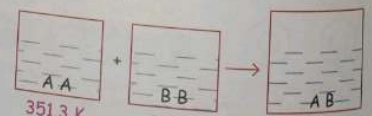
yeh tedha hai thoda

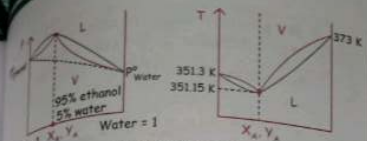
AZEOTROPIC MIXTURE

- A binary solution containing two volatile liquids in which both liquid and vapour have same composition.
- A boiling liquid mixture at the azeotropic composition produces vapours of exactly the same composition as that of the liquid.
- Composition of liquid mixtures at which distillation can not separate the two liquids because the condensate has the same composition as that of the azeotropic liquid.

Minimum Boiling Azeotropes

(Ethanol + water)



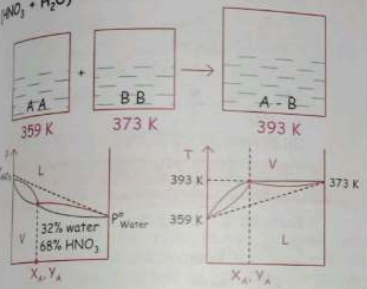


- These azeotropes are formed by the solutions having large (+) ve deviation from Raoult's law.
- Boiling point of the solution is less than the B.P of either component.

Ho kya raha hai, koi toh rok lo

Maximum Boiling Azeotropes

($HNO_3 + H_2O$)



- These are formed by the solutions which show (-) ve deviation from Raoult's law.
- Boiling point of this azeotropic mixture is more than the either component.

Ho kya raha hai, koi toh rok lo

COLLIGATIVE PROPERTIES

Properties which depend only on the number of solute particles and are independent on the nature of solute particles.

Khoobsurat Baat

Solute should be non-volatile

Solution should be dilute

There are four colligative properties:

1. Relative lowering of vapour pressure
2. Elevation in boiling point
3. Depression in freezing point
4. Osmotic pressure

Let's discuss these properties one by one in detail.

sahab sirf number dekhti hai ya nature nahi

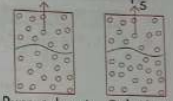
1. Relative Lowering of Vapour Pressure (RLVP)

- Relative Lowering of Vapour Pressure is a colligative property that describes the decrease in vapour pressure of a solvent due to the presence of a solute.
- It is defined as the ratio of the decrease in vapour pressure of a solvent to its original vapour pressure.
- Due to the interaction of solute and solvent, vapour pressure decreases.
- Due to decrease in per unit surface area of solvent particles, vapour pressure decreases.
- Mathematically,

Here, $P_s < P^\circ$

$P_S = P^\circ X_{solvent}$

$P_S = P^\circ(1 - X_{solute})$



$$\frac{P^\circ - P_S}{P^\circ} = X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

So $\frac{P^\circ - P_S}{P^\circ}$ depends on the number of solute particles, that's why it is a colligative property.

samjhe kyu ise colligative bola hai?

Important

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{\frac{W_{solute}}{M_{solute}}}{\frac{W_{solute}}{M_{solute}} + \frac{W_{solvent}}{M_{solvent}}}$$

We have six variable $\Rightarrow P^\circ$, P_S, W_{solute}, $W_{solvent}$, M_{solute}, $M_{solvent}$. Any 5 can be given and sixth will be asked.

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{n_{solute}}{n_{solvent}} \text{ (if } n_{solute} \ll n_{solvent})$$

Note

$$\frac{P^\circ}{P^\circ - P_S} = 1 + \frac{n_{solvent}}{n_{solute}} \cdot \frac{P_S}{P^\circ - P_S} = \frac{n_{solvent}}{n_{solute}}$$

$$\frac{P^\circ - P_S}{P_S} = \frac{n_{solute}}{n_{solvent}}$$

Khoobsurat Baat

$$\frac{P^\circ - P_S}{P^\circ} = \frac{\frac{W_1}{M_1}}{\frac{W_1}{M_1} + \frac{W_2}{M_2}} \Rightarrow \text{Relative lowering of vapour pressure}$$

$$\frac{P^\circ - P_S}{P_S} = \frac{\frac{W_1}{M_1}}{\frac{W_2}{M_2}} \Rightarrow \text{Not relative lowering of vapour pressure}$$

Important

Ab dhyan se sunna kaam ki baat hai

- If RLVP is given or asked then we use

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

If in the problem we are given that solution is highly diluted then we can conclude

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solvent}}$$

- If P° and P_S is given then better to use

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solvent}}$$

45. When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg. The mole fraction of the solute in the solution is 0.2. What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg?

[23 Jan, 2025 (Shift-II)]

(A) 0.2 (B) 0.4 (C) 0.8 (D) 0.6

Sol. (D) By relative lowering in vapour pressure.

$$\frac{P^0 - P_s}{P^0} = x_{solute}, \quad \frac{10}{P^0} = x_{solute} \quad ...(I)$$

$$\frac{P^0 - P_s}{P^0} = x_{solute}, \quad \frac{20}{P^0} = x_{solute}$$

By equating (I) and (II) ...(II)

$$\frac{\frac{10}{P^0}}{\frac{20}{P^0}} = \frac{0.2}{x_{solute}} \Rightarrow x_{solute} = 0.4, x_{solvent} = 1 - 0.4 = 0.6$$

46. The vapour pressure of benzene at 80°C is lowered by 10 mm by dissolving 2g of a non-volatile substance in 78g of benzene. The vapour pressure of pure benzene at 80°C is 750 mm. The molecular weight of the substance will be :

(A) 15 (B) 150 (C) 1500 (D) 148

Sol. (D) $P^\circ - P_S = 10$

Bachcha-1 Beimani krega dekhna

$$\frac{P^\circ - P_S}{P^\circ} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{n_{solute}}{n_{solvent}}$$

$$\frac{750 - 740}{750} = \frac{\frac{2}{M}}{1}$$

On Solving, M = 150 g/mol

Jab yeh dia hi ni ki solute solvent se bhut chhota hai toh apne ap kyu kar dia, gazab lallu ho

Bachcha-2- Samjhdar hai

$$\frac{P^\circ - P_S}{P_S} = \frac{n_{solute}}{n_{solvent}}$$

Let M be the molecular weight of substance.

Substitute the values in the above expression

$$\Rightarrow \frac{750 - (750 - 10)}{750} = \frac{\frac{2}{M}}{\frac{2}{M} + \frac{78}{78}}$$

On solving, M = 148 g/mol

jo bacha beimani karega wo lallu hai, jo samjhdari dikhaega, uski balle balle hogi

47. A set of solutions is prepared using 180 g of water as a solvent and 10 g of different non-volatile solutes A, B and C. The relative lowering of vapour pressure in the presence of the solutes are in the order [Given, molar mass of A = 100 g mol^{-1}; B = 200 g mol^{-1}; C = 10,000 g mol^{-1}]

[06 Sept, 2020 (Shift-II)]

(A) A > C > B (B) C > B > A

(C) A > B > C (D) B > C > A

Sol. (C) Calculate the mole fraction of A, B and C. Higher the mole fraction of solute, higher will the relative lowering of vapor pressure. The correct order is: A > B > C

Common Mistake

Assuming that relative lowering of vapour pressure inversely varies with mole fraction.

48. 224 mL of $SO_2(g)$ at 298 K and 1 atm is passed through 100 mL of 0.1 M NaOH solution. The non-volatile solute produced is dissolved in 36 g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) ($P^\circ_{(H_2O)}$ = 24 mm of Hg) is $x \times 10^{-2}$ mm of Hg, the value of x is _____. (Integer answer)

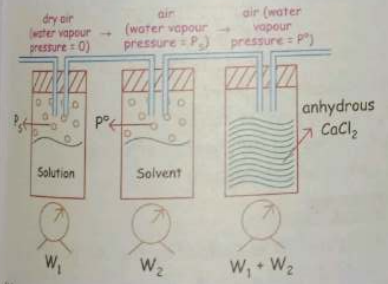
[26 Feb, 2021 (Shift-I)]

Sol. (18) For explanation, refer to QR code given on the last page.

OSTWALT-WALKER EXPERIMENT

It demonstrates the relationship between the vapour pressure of a solution and the concentration of the solute. The objective of the Ostwald-Walker experiment is to measure the vapour pressure of a solution as a function of the concentration of the solute.

This experiment shows that the vapour pressure of a solution decreases with increasing concentration of the solute.



W_1 = wt. loss in solution container

W_2 = wt. loss in solvent container

$W_1 + W_2$ = wt. gain in $CaCl_2$ container

$$W_1 = \frac{P_s VM}{RT}$$

$$W_2 = \frac{(P^\circ - P_s)VM}{RT} \quad ...(i)$$

$$W_1 + W_2 = \frac{P^\circ VM}{RT} \quad ...(ii)$$

$$\frac{2}{3} = \frac{P^\circ - P_s}{P^\circ} = \frac{W_2}{W_1 + W_2} \quad ...(iii)$$

$$\frac{P^\circ - P_s}{P^\circ} = \frac{\text{wt. loss in solvent container}}{\text{wt. loss in solvent + wt. loss in solution container}}$$

$$\frac{P^\circ - P_s}{P^\circ} = \frac{\text{wt. loss in solvent container}}{\text{wt. gain in } CaCl_2 \text{ container}}$$

$$\frac{P^\circ - P_s}{P^\circ} = \frac{\text{wt. loss in solvent container}}{\text{wt. loss in solution container}}$$

formula baitha lo dimag ke kone mei

49. Dry air was passed through a solution of 5 gm of a solute in 80 gm of water and then it passed through pure water, loss in weight of solution was 2.50 gm and that of pure solvent was 0.04 gm. Calculate molecular weight of solute.

Sol. Here, $w_2 = 0.04$ g, $w_1 = 2.50$ g

$$n_{solute} = \frac{5}{M}, \; n_{solvent} = \frac{80}{18}$$

$$\frac{P^\circ - P_s}{P_s} = \frac{\frac{5}{M}}{\frac{80}{18}} = \frac{0.04}{2.5}$$

On solving, M = 31.25 g/mol

ghabra kyu rahe ho

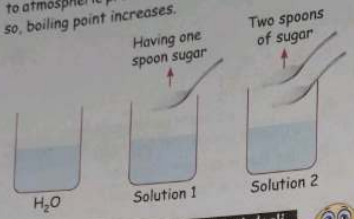
50. Dry air is passed through a solution containing 20 g. of an organic non-volatile solute in 250 ml of water. Then the air was passed through pure water and then through a U-tube containing anhydrous $CaCl_2$. The mass lost in solution is 26 g and the mass gained in the U-tube is 26.48 g. Calculate the molecular mass of the organic solute.

Sol. (78 g/mole) For explanation, refer to QR code given on the last page.

2. Elevation in Boiling Point

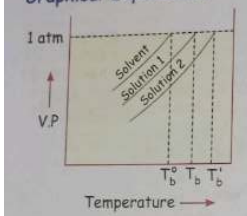
It is known that boiling point of solution is inversely proportional to vapour pressure.

Vapour pressure of solution is less than the vapour of pure solvent so to attain the vapour pressure equals to atmospheric pressure we need more and more heat, so, boiling point increases.



dusre box mei kam cheeni daali, teesre mei jada

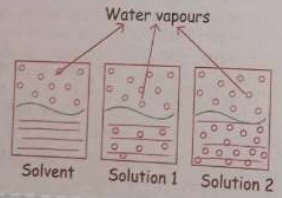
Graphical Explanation



$T_b' > T_b > T_b^\circ$

It is evident from the graph that on increasing the spoons of sugar, boiling point increases so increase in boiling point is a colligative property.

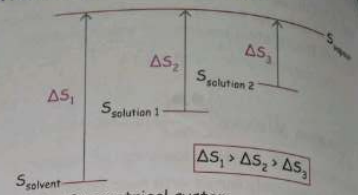
Khoobsurat Baat



Important

Vapour phase always contains molecules of solvent (H_2O) only as solute is non-volatile.

Thermodynamic Explanation



$\Delta S_1 > \Delta S_2 > \Delta S_3$

solvent = Symmetrical system
solution 1 = Unsymmetrical system
solution 2 = more unsymmetrical system

Unsymmetrical results into more entropy

$\Delta G = \Delta H - T\Delta S$ (at boiling point $\Delta G = 0$ and $T = T_b$)

$$T_b = \frac{\Delta H_{vap}}{\Delta S}$$

as $\Delta S \uparrow$ $T_b \downarrow$
$\Delta S \downarrow$ $T_b \uparrow$

as we add sugar to the solvent, ΔS decreases and boiling point increases.

isne dimag ghuma dia hoga pakka

Calculation of ΔT_b

$\Delta T_b \propto$ molality $\quad \Delta T_b = K_b m$

K_b = ebullioscopic constant
or molal elevation of boiling point
Unit of K_b = K kg mol^{-1} or K $molal^{-1}$

- K_b is a property of solvent only irrespective of the solute. If solvent is same, K_b remains same.
- It does not depend on the nature or the number of solute particles.

$$K_b = \frac{MRT_b^2}{\Delta H_{vap} \times 1000}$$

M = molecular weight of solvent
T_b = Boiling point of solvent
ΔH_{vap} = Heat of vaporisation of solvent
R = universal gas constant

Note

- Unit of 'R' is decided by the unit of ΔH_{vap}
  If ΔH_{vap} = in Joule
  then R = 8.314 J/K-mole
  ΔH_{vap} = in cal
  then R = 1.98 = 2.0 cal/K-mole

51. A solution containing 2 g of a non-volatile solute in 20 g of water boils at 373.52 K. The molecular mass of solute is _____ g/mole.

Given: T_b of water 373 K and K_b of H_2O = 0.52 K kg mole [JEE Mains-2023]

Sol. First calculate ΔT_b as:

$\Delta T_b = T_b^\circ - T_b = 373.52 - 373 = 0.52$

$\Delta T_b = K_b m$

Substitute the values in the above expression:

$$0.52 = 0.52 \times \frac{2}{\text{Molar Mass}} \times \frac{1}{20 \times 10^{-3}}$$

On solving, M = 100 g/mol

(Seedha formula lagega or kuch-nahi karna)

Common Mistake

Take the mass of solvent in g which is already given. It will lead to a conceptually wrong answer.

52. A solution containing 2.5×10^{-3} kg of solute dissolved in 75×10^{-3} kg of water, boils at 373.535 K. The molar mass of solute is _____ g/mol.

Given: T_b of water = 373.15 K, K_b of H_2O = 0.52 K kg mol^{-1} [JEE Mains-2022]

Sol. First calculate ΔT_b as:

$\Delta T_b = T_b^\circ - T_b = 373.52 - 373 = 0.52$

Now, $\Delta T_b = K_b m$

Substitute the values in the above expression:

$$M_{solute} = \frac{0.52 \times 10^3 \times 2.5 \times 10^{-3}}{75 \times 10^{-3} \times 0.385} = 45.02$$

53. Boiling point of water at 750 mm Hg is 99.63°C. How much sucrose should be added to 500 g of water such that it boils at 100°C?

Sol. (121.67 g) For explanation, refer to QR code given on the last page.

54. Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous of a non-volatile solute B. The relation between molecular weights of A and B is- [27 July, 2022 (Shift-I)]

(A) $M_A = 4M_B$ (B) $M_B = 4M_A$

(C) $M_A = 8M_B$ (D) $M_B = 8M_A$

Sol. (B) For explanation, refer to QR code given on the last page.

55. 2 g of a non-volatile non-electrolyte solute is dissolved in 200 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 8. The elevation in boiling points A and B are in the ratio of x/y (x : y). The value of y is _____ (Nearest Integer). [27 June, 2022 (Shift-I)]

Sol. wt. of solute = 2 g

wt. of $(solvent)_1$ = 200 g, wt. of $(solvent)_2$ = 200 g

$(K_b)_1 : (K_b)_2 = 1 : 8$

$$\frac{(\Delta T_b)_1}{(\Delta T_b)_2} = \frac{(K_b)_1\left(\frac{2}{M} \times \frac{200}{1000}\right)}{(K_b)_2\left(\frac{2}{M} \times \frac{200}{1000}\right)} = \frac{(K_b)_1}{(K_b)_2} = \frac{1}{8} = \frac{x}{y}$$

56. Arrange the following solutions in order of their increasing boiling points. [22 Jan, 2025 (Shift-I)]

(i) 10^{-4} M NaCl (ii) 10^{-4} M Urea

(iii) 10^{-3} M NaCl (iv) 10^{-2} M NaCl

(A) (ii) < (i) ≅ (iii) < (iv) (B) (iv) < (iii) < (i) < (ii)

(C) (ii) < (i) < (iii) < (iv) (D) (i) < (ii) < (iii) < (iv)

Sol. (C) As, $\Delta T_b = i \cdot K_b \cdot m$

∴ More the value of (i × m), more will be the elevation in boiling point and hence, more will be the boiling point.

(i) $\Delta T_b = 2 \times K_b \times 10^{-4}$ (For NaCl, i = 2)

(ii) $\Delta T_b = 1 \times K_b \times 10^{-4}$ (For urea, i = 1)

(iii) $\Delta T_b = 2 \times K_b \times 10^{-3}$ (For NaCl, i = 2)

(iv) $\Delta T_b = 2 \times K_b \times 10^{-2}$ (For NaCl, i = 2)

Therefore, correct order of increasing boiling points:

(ii) < (i) < (iii) < (iv)

kitna asaan hai, maze hi maze

57. When 3.0 g of a substance 'X' is dissolved in 100 g of CCl_4, it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _____ g mol^{-1}. (Nearest Integer)

[Given: K_b for CCl_4 is 5.0 K kg mol^{-1}] [06 Sept. 2020 (Shift-II)]

Sol. $\Delta T_b = K_b \times$ molality

Molality = Moles of solute/ mass of solvent in kg

Substitute the values in the above expression:

$0.60 = 5 \times (3 / M) / (100 / 100)$

On solving, M = 250 g/mol

58. If the boiling points of two solvents X and Y (having same molecular weights) are in the ratio 2 : 1 and their enthalpy of vaporization are in the ratio 1 : 2, then the boiling point elevation constant of X is 'm' times the boiling point elevation constant of Y. The value of m is _____ (Nearest Integer)

[08 April, 2023 (Shift-II)]

Sol. (8) For explanation, refer to QR code given on the last page.

59. 1 g of non-volatile non-electrolyte solute is dissolved in 100 g of two different solvents A and B whose ebulliopscopic constant are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$ is:

[10 April, 2019 (Shift-II)]

(A) 5 : 1 (B) 10 : 1

(C) 1 : 5 (D) 1 : 0.2

Sol. (C) The formula of elevation in boiling point is:

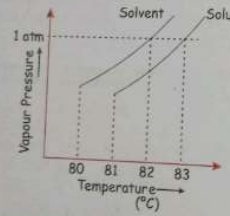
$\Delta T_b = K_b m$

According to the given information:

$\therefore \frac{\Delta T_{b(A)}}{\Delta T_{b(B)}} = \frac{K_{b(A)}}{K_{b(B)}}$ as $m_A = m_B$

$\therefore \frac{\Delta T_{b(A)}}{\Delta T_{b(B)}} = \frac{1}{5}$

60. The vapour pressure vs. temperature curve for a solution solvent system is shown below.



The boiling point of the solvent is °C

[8 April, 2023 (Shift-I)]

Sol. Boiling point of solvent is 82°C

Boiling point of solution is 83°C

graph nahi aata toh beda garg

3. Depression in Freezing Point

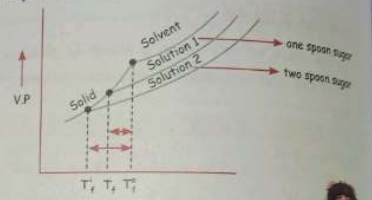
- Freezing point is the point where first crystal of ice separates.
- At freezing point of substance, the solid phase is in dynamic equilibrium with liquid phase

$$H_2O(s) \rightleftharpoons H_2O(l)$$

- Freezing of solvent occurs only.
- Freezing point is the point where vapour pressure of liquid phase is equal to the vapour pressure of solid phase.
- The difference in the freezing point of pure solvent T_f° and that of solution (T_f) is called depression in freezing point.

$$\boxed{\Delta T_f = T_f^\circ - T_f}$$

Graphical Explanation



(samjhe kuch ya pagla gaye)

As we increase the spoons of sugar (quantity of solute) depression in F.P also increases. So depression in F.P is a colligative property.

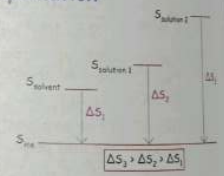
Thermodynamic Explanation

at F.P liq $\rightleftharpoons$ solid

$\Delta G = 0$

$\Delta G = \Delta H - T\Delta S$

$$\boxed{T_F = \frac{\Delta H_{Fusion}}{\Delta S}}$$



as ΔS increases, freezing point decreases

$\boxed{(T_F)_{solution\ 2} < (T_F)_{solution\ 1} < (T_F)_{solvent}}$

$\boxed{(\Delta T_F)_{solution\ 2} < (\Delta T_F)_{solution\ 1}}$

Freezing point sabse kam hoga to depression in freezing point maximum hoga.

(yeh samajhne mei halaat tight hogi)

Calculation of freezing point

$\Delta T_F \propto$ molality

$$\Delta T_F = K_F\, m$$

K_F = cryoscopic constant or molal depression of F.P

Unit of K_F = K molal^{-1} or K kg mol^{-1}

- K_F is a property of solvent
- It does not depend on the nature or number of solute particles

$$K_F = \frac{MRT_F^2}{\Delta H_{Fusion} \times 1000}$$

M = molar mass of solvent

T_f = F.P of solvent

ΔH_{fusion} = Heat of fusion solvent

R = universal gas constant

61. What happens to freezing point of benzene when small quantity of napthalene is added to benzene? [30 Jan, 2024 (Shift-I)]

(A) Increases

(B) Remains unchanged

(C) First decreases and then increases

(D) Decreases

Sol. (D) Adding naphthalene to benzene lowers its freezing point.

62. The solution from the following with highest depression in freezing point/lowest freezing point is [30 Jan, 2024 (Shift-II)]

(A) 180 g of acetic acid dissolved in water

(B) 180 g of acetic acid dissolved in benzene

(C) 180 g of benzoic acid dissolved in benzene

(D) 180 g of glucose dissolved in water

Sol. (A) For explanation, refer to QR code given on the last page.

63. The amount of urea to be dissolved in 500 cc of water (K_F = 1.86 K kg mol^{-1}) to produce a depression of 0.186°C in the freezing point

Sol. (3 g) For explanation, refer to QR code given on the last page.

64. The elevation in boiling point for 1 molal solution of non-volatile solute A is 3K. The depression in freezing point for 2 molal solution of A in the same solvent is 6K. The ratio of K_b and K_f i.e. K_b/K_f is 1 : X. The value of X is _____. [Nearest Integer]

[26 July, 2022 (Shift-II)]

Sol.

molality = 1 m	molality = 2 m	$\frac{K_b}{K_F} = \frac{3}{3} = \frac{1}{1}$
$\Delta T_b = 3$ K	$\Delta T_F = 6$ K	
$K_b = \frac{\Delta T_b}{M} = \frac{3}{1}$	$K_F = \frac{\Delta T_F}{M} = \frac{6}{2}$	
= 3K Kg mol^{-1}	= 3K Kg mol^{-1}	

Khoobsurat Baat

Case of ice separation

Suppose the F.P of a solution = -2°C

Means at -2°C first crystal of ice separates. As some amount of water is converted into ice, the molality of the remaining solution increases and depression in F.P also increases and F.P decreases.

if solution is kept at -5°C then-

Assuming the freezing point of the left solution to be -5°C we calculate the molality of the remaining solution ($5 = K_f m$) then we calculate the wt of solvent remained in the solution. On subtracting this wt from original solvent wt we get the mass of ice.

(samjhe baraf kaise alag hokar banti hai)

65. A solution contains 62 gm ethylene glycol ($CH_2OH - CH_2OH$) in 250 gm H_2O is cooled upto -10°C. If K_F = 1.86 K kg mol^{-1}, then amount of water (in gm) separated as ice is- [JEE-Mains 2019]

(A) 32 (B) 48 (C) 64 (D) 16

Sol. (C) $\Delta T_f = K_f m$

$$0 - (-10) = 186 \times \frac{62 \times 1000}{62 \times w_A}$$

$$w_A = \frac{186 \times 62 \times 1000}{10 \times 62} = 186\,g$$

Now, calculate amount of water separated as ice = 250 - 186 = 64 g

Common Mistake

Students might forget to subtract the calculated weight from given weight which means they can mark 186 g as the correct answer.

66. Given below are two statements:

[29 Jan, 2025 (Shift-II)]

Statement (I): NaCl is added to the ice at 0°C, present in the ice cream box to prevent the melting of ice cream.

Statement (II): On addition of NaCl to ice at 0°C, there is a depression in freezing point.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Sol. (B) When NaCl (sodium chloride, or table salt) is added to ice at 0°C in an ice cream box, it lowers the freezing point of the water, effectively preventing the ice cream from melting by creating a colder mixture around the container; this phenomenon is called "freezing point depression"

67. 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest Integer)

[Given: K_f (H_2O) = 1.86 K kg mol^{-1}]

[27 Aug, 2021 (Shift-I)]

67. (518.31 gm) For explanation, refer to QR code given on the last page.

68. On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, vapour pressur decreases from 650 mm Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is _____

(Given data: Molar maas and the molal freezing point depression constant of benzene are 78 g mol^{-1} and 5.12 K kg mol^{-1}, respectively)

[JEE Adv. 2019]

Sol. Relative lowering of vapor pressure is calculated as:

$$\because \frac{p^\circ - p}{p^\circ} = x_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

$$\Rightarrow \frac{10}{650} = \frac{n_{solute}}{n_{solute} + 0.5}$$

$$n_{solute} + 0.5 = 65\, n_{solute} \Rightarrow n_{solute} = \frac{0.5}{64} = \frac{1}{2\times 64}$$

Calculate depression in freezing point as:

$$\Delta T_f = K_f m = 5.12 \times 1000 \times \frac{1}{2\times 64\times 39} = 1.02\,K$$

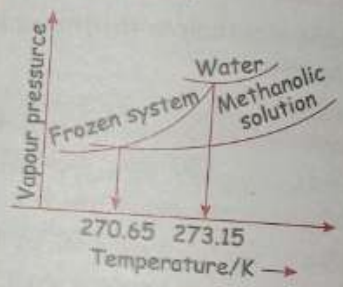
do do properties agayi, halat khrab ho gayi kya

69. 150 g of acetic acid contaminated with 10.2 g ascorbic acid ($C_6H_8O_6$) to lower down its freezing point by ($x \times 10^{-1}$)°C The value of x is _____ (Nearest Integer)

(Given K_f = 3.9 K mol^{-1}; molar mass of ascorbic acid = 176 g mol^{-1}) [28 July, 2022 (Shift-I)]

Sol. (15.06) For explanation, refer to QR code given on the last page.

70. When 'x' × 10^{-2} mL methanol (molar mass = 32 g; density = 0.792 g/cm^3) is added to 100 mL water (density =1 g/cm^3), the following diagram is obtained.



x = ... (nearest integer)

[Given: Molal freezing point depression constant of water at 273.15K is 1.86 K kg mol^{-1}]

[06 April, 2024 (Shift-II)]

Sol. $\Delta T_f = T_{f\,(solvent)} - T_{f\,(solution)}$

$= 273.15 - 270.65 = 2.5\,K$

$\Delta T_f = K_f m \Rightarrow 2.5 = 1.86 \times \frac{n}{0.1}$ (Given K_f = 1.86)

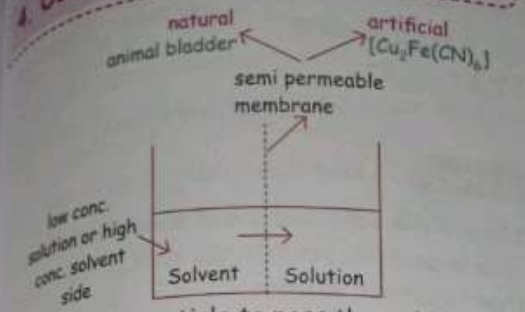
n = 0.1344 moles, w = 0.1344 × 32 = 4.3 g

(molar mass of methanol = 32g/mol)

Hence, Volume = $\frac{4.3}{0.792} = 5.43\,ml$

$= 543 \times 10^{-2}$ ml

4. Osmosis and Osmotic Pressure

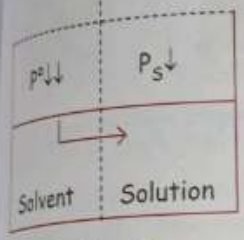


(It allows solvent particle to pass through)

Observation: The movement (resultant) of solvent particles is observed from low conc. solution to high conc. solution or from high conc. solvent side to low conc. solvent side.

(Osmosis)

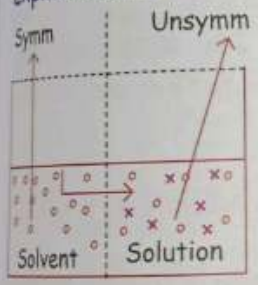
Explanation 1



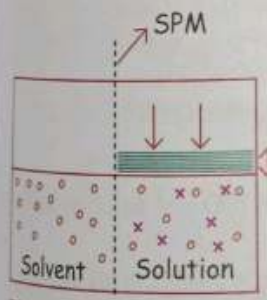
$P^\circ > P_S$

solvent particles feel more pressure to move through spm towards high concentrated solution side.

Explanation 2

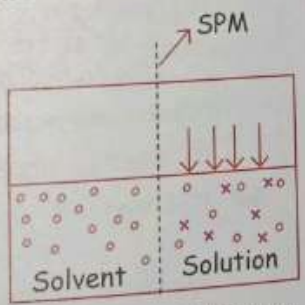


The entropy of solvent particles in solvent side is less than the entropy of solvent particles in solution side. That's why the resultant flow of solvent is observed from solvent to solution side.



This extra hydrostatic pressure which stops the osmosis is called Osmotic pressure.

$\pi = hpg$



The extra pressure applied to high conc. solution side to prevent the osmosis from the starting also called Osmotic Pressure.

explanation ko samjh lo acche se, varna dikkat aaegi

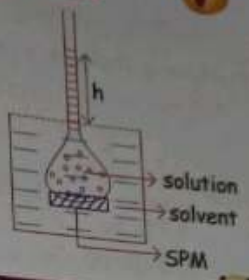
$\pi = hpg$

$\pi = CRT$

C = concentration of solution (molarity)

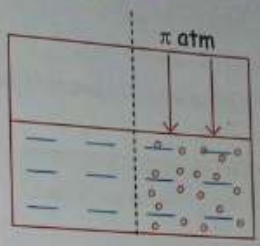
T = temperature in K.

R = 0.0821 L-atm/K-mol

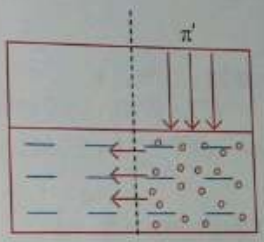


(yeh property jada mazedaar hai)

Important



If we apply π atm pressure on high conc. solution side then osmosis can be prevented



$\pi' > \pi$

If π' (applied pressure) is greater than osmotic pressure the resultant flow of solvent is observed from high conc. solution side to low conc. solution dide. The process is 'reverse osmosis'

71. The relationship between Osmotic pressure at 273 K when 10 gm of glucose (P_1), 10 gm urea (P_2) and 10 gm sucrose (P_3) are dissolved in 250 ml of water is

(A) $P_1 > P_2 > P_3$ (B) $P_3 > P_1 > P_2$

(C) $P_2 > P_1 > P_3$ (D) $P_2 > P_3 > P_1$

Sol. (C) Osmotic pressure = CRT

$\pi \propto C$

For glucose, Moles = 10/180

For urea: Moles = 10/60

For sucrose, Moles = 10/342

Lower the molar mass, higher will be the osmotic pressure.

Common Mistake

Assuming that osmotic pressure varies directly with molar mass.

72. What will be the concentration of sucrose solution which develops an osmotic pressure of 1.5 atm at 300 K.

Sol. (0.0609 M) For explanation, refer to QR code given on the last page.

73. Calculate the osmotic pressure of these solutions.

(i) 5% w/V glucose solution at 27°C

(ii) 0.1 M urea solution at 300 K

Sol. (i) 5% w/V means 5 g of glucose is present in 100 ml of solution.

Osmotic pressure = CRT

$C = n/V$

$= \frac{\left(\frac{5}{180}\right)}{0.1} = 0.28$ M

Osmotic pressure = 0.28 × 0.0821 × (27 + 273) = 6.90 atm

(ii) Osmotic pressure = CRT = 0.1 × 0.0821 × 300 = 2.46 atm

(waah waah, bhut hi asaan)

74. Consider the dissociation of the weak acid HX as given below

$HX(aq) \rightleftharpoons H^+(aq) + X^-(aq)$, $K_a = 1.2 \times 10^{-5}$

[K_a: dissociation constant]

The osmotic pressure of 0.03M aqueous solution of HX at 300 K is _____ $\times 10^{-2}$ bar (nearest integer).

[Given: R = 0.083 L bar Mol^{-1} K^{-1}]

[06 April, 2024 (Shift-I)]

Sol. (76) For explanation, refer to QR code given on the last page.

Hypertonic and hypotonic solutions

- Solution which has higher osmotic pressure is hypertonic solution with respect to the solution having lower osmotic pressure (Hypotonic).

$C_1T_1 > C_2T_2$ ← $\pi_1 > \pi_2$

T_1 = temp of sol^{n}1

T_2 = temp of sol^{n}2

Hypertonic, Hypotonic

$C_1RT > C_2RT$

$C_1 > C_2$ [Temp = same]

Note

If both the solutions are at same temperature then hypertonic solution has more concentration than hypotonic solution.

Isotonic solutions

Solution having same osmotic pressure are isotonic

$\pi_1 = \pi_2$ → $C_1 = C_2$ (Temp same)

$\pi_1 = \pi_2$ → $C_1T_1 = C_2T_2$ (Temp different)

Hypertonic | Isotonic | Hypotonic

(diagram dekhkr toh ho gaya hoga all clear)

75. Assume a living cell with 0.09%(w/w) of glucose solution (aqueous). This cell is immersed in another solution having equal mole fraction of glucose and water.

[28 Jan, 2025 (Shift-II)]

(Consider the data up to first decimal place only)

The cell will:

(A) show no change in volume since solution is 0.9%(w/w)

(B) shrink since solution is 0.45%(w/w) as a result of association of glucose molecules (due to hydrogen bonding)

(C) shrink since solution is 0.5%(w/w)

(D) swell up since solution is 1%(w/w)

Sol. (*) Living cell = 0.9 g in 100 g of solution %w/w = 0.9

Weight of solution = 0.5 × 180 + 0.5 × 18 = 99g

% w/w is approximately 90%

Concentrated solution = Cell will shrink

None of the given option is correct (Bonus Ques)

76. A 5% w/V cane sugar solution is isotonic with 0.877% w/V solution of substance 'X'. Calculate the molecular mass of 'X'. (Mol wt of cane sugar = 342 g/mol).

Sol. For isotonic solutions,

Osmotic pressure of cane sugar = osmotic pressure of X

Let M be the molecular mass of X.

Substitute the given values in the above expression:

$$\left(\frac{5}{\frac{342}{0.1}}\right) = \left(\frac{0.877}{\frac{M}{0.1}}\right)$$

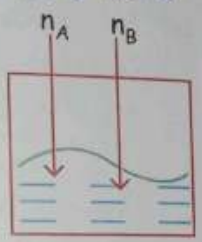
On solving, M = 60 g/mol

(Solution isotonic hai lallu, dhyan lagao)

Important

Osmotic pressure of solution having two or more solutes

The osmotic pressure of a solution having more than one solute is sum of partial osmotic pressures.

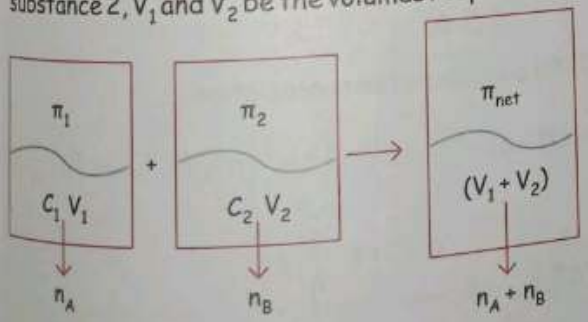


$$\pi_{net} = \pi_A + \pi_B = \left(\frac{n_A}{V}\right)RT + \left(\frac{n_B}{V}\right)RT$$

$$\boxed{\pi_{net} = (n_A + n_B)\frac{RT}{V}}$$

Osmotic pressure of a resultant solution when two solutions of same substance of different osmotic pressure are mixed.

Let π_1 be the osmotic pressure of substance 1 and π_2 be the osmotic pressure of substance 2, C_1, C_2 be the concentrations of substance 1 and substance 2, V_1 and V_2 be the volumes respectively.



$$\pi_1 = C_1RT = \frac{n_A}{V_1}RT \longrightarrow n_A = \frac{\pi_1 V_1}{RT}$$

$$\pi_2 = C_2RT = \frac{n_B}{V_1}RT \longrightarrow n_B = \frac{\pi_2 V_2}{RT}$$

$$\pi_{net} = \left(\frac{n_A + n_B}{V_1 + V_2}\right)RT$$

$$n_A + n_B = \frac{\pi_{net}(V_1 + V_2)}{RT}$$

$$\frac{\pi_1 V_1}{RT} + \frac{\pi_2 V_2}{RT} = \frac{\pi_{net}(V_1 + V_2)}{RT}$$

formula mushkil nahi hai agar dimag lagaoge toh

Khoobsurat Baat

$$\pi_{net} = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2} \nearrow \frac{\Sigma \pi_i V_i}{\Sigma V_i}$$

→ Mixing of two solution of different osmotic pressure

$$\pi_{net} = (n_A + n_B)\frac{RT}{V} \nearrow$$ when more than one solute is added

$$(\Sigma n_i)\frac{RT}{V}$$

77. Two solutions of glucose have osmotic pressures 1.5 and 2.5 atm. 1 Lt of first solution is mixed with 3 lt of second solution. Calculate the osmotic pressure of resultant solution.

Sol. $\pi_{net} = \frac{\Sigma \pi_i V_i}{\Sigma V_i} = \frac{(1.5 \times 1) + (2.5 \times 3)}{4} = 2.25$ atm

Gazab asaan hai bacho

78. 9 gm glucose and 12 gm urea are dissolved in 1 lt aqueous solution at 27°C. Calculate the osmotic pressure of the solution.

Sol. (6.155 atm) For explanation, refer to QR code given on the last page.

VAN'T HOFF FACTOR

The Van't Hoff factor (i) is a measure of the extent to which a solute dissociates or associates in a solution. It is defined as the ratio of the actual number of particles in a solution to the number of particles that would be present if the solute did not dissociate or associate.

$$\frac{(\Delta T_b)_{observed}}{(\Delta T_b)_{theoretical}} = \text{van't Hoff factor} = i$$

$$\boxed{\frac{(\text{colligative prop})_{observed}}{(\text{colligative prop})_{theoretical}} = i}$$

$$\text{van't Haff factor} = \frac{\text{Observed colligative property}}{\text{Theoretical willigative property}}$$

$$= \frac{\text{actual moles of solute after association or dissociation}}{\text{initial moles of solute}}$$

(yeh kya hua, kaise hua, kab hua)

Khoobsurat Baat

$(\text{coll. property})_{observed} = (\text{coll. prop})_{actual}$

$= (i \times \text{coll. prop})_{theoretical}$

Important

Colligative properties in terms of Van't Hoff factor can be represented as:

1. $\frac{P^\circ - P_S}{P^\circ} = \frac{in_{solute}}{in_{solute} + n_{solvent}}$

   For diluted solution $n_{solute} << n_{solvent}$

   $\frac{P^\circ - P_S}{P^\circ} = i\left(\frac{n_{solute}}{n_{solvent}}\right)$

2. $\Delta T_b = i K_b m$
3. $\Delta T_f = i K_f m$
4. $\pi = i CRT$

Calculation of Van't Hoff factor during dissociation of solute

A_n	$\rightleftharpoons$	nA
C mol		0
$C(1-\alpha)$		$nc\alpha$

Total moles after dissociation $= C(1-\alpha) + nc\alpha$

$= C[1 + (n-1)\alpha]$

α = degree of dissociation

$(\Delta T_F)_{th} = K_f\left(\frac{C}{W}\right)$, $(\Delta T_F)_{obs} = K_f\frac{C[1+(n-1)\alpha]}{W}$

$i = \frac{(\Delta T_F)_{obs}}{(\Delta T_b)_{th}} = 1 + (n-1)\alpha$

$\boxed{i = 1 + (n-1)\alpha}$

(pagla toh nahi gaye?)

Khoobsurat Baat

$1 + (n-1)\alpha = i$ (during dissociation of a solute having degree of dissociation α)

α = D.OD

n = Jitne logo mein toota

79. Find out the Van't Hoff factor for AB_3 if degree of dissociation = 0.4

Sol. For dissociation, $i > 1$

Lets consider the dissociation of AB_3:

$AB_3 \rightleftharpoons A + 3B$

$4 = n$

$i = 1 + (n-1)\alpha$

$= 1 + (4-1)\,0.4 = 2.2$

$(\text{coll. prop})_{obs} = 2.2\ (\text{Coll. prop})_{Th}$

$\Delta T_b = 2.2\ K_b.m$

Calculation of Van't Hoff factor during association

nA	$\rightleftharpoons$	A_n
C		0
$C(1-\beta)$		$\frac{C\beta}{n}$

Total particles after association

$= C(1-\beta) + \frac{C\beta}{n} = C\left[1 - \left(1 - \frac{1}{n}\right)\beta\right]$, $(\Delta T_b)_{th} = K_b\left(\frac{C}{W}\right)$

$(\Delta T_b)_{obs} = K_b\frac{C\left[1 - \left(1 - \frac{1}{n}\right)\beta\right]}{W}$

$\frac{(\Delta T_b)_{obs}}{(\Delta T_b)_{th}} = 1-\left(1-\frac{1}{n}\right)\beta$ $\boxed{i = 1-\left(1-\frac{1}{n}\right)\beta}$

(yeh toh bhut khatarnaak lagra hai)

Khoobsurat Baat

β = degree of association
n = Jitne log jud rahe hain
$i = 1+(n-1)\alpha$
α = degreee of dissociation
n = Jitne logo mein toota
$(coll\ prop)_{obs} = i\ (coll\ prop)_{theoretical}$

$\boxed{i = 1-\left(1-\frac{1}{n}\right)\beta}$

80. 0.004 M K_2SO_4 solution is isotonic with 0.01 M glucose solution. Percentage dissociation of K_2SO_4 is _____ (Nearest Integer)

[11 April, 2023 (Shift-I)]

Sol. For isotonic solutions:

Osmotic pressure of K_2SO_4 = Osmotic pressure of glucose

$i \times 0.004 \times RT = 0.01 \times RT$

On solving, $i = 2.5$

Lets consider the dissociation of K_2SO_4:

$K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$

For dissociation,

$i = 1 + (n-1)\alpha$

Rearrange the above equation for α:

$$(\alpha) = \frac{i-1}{n-1} = \frac{2.5-1}{3-1} = 0.75$$

So, the degree of dissociation in percent is 75%.

Common Mistake

Assuming that value of Van't Hoff factor is 3 but here n will be equal to 3.

81. The relationship between the values of osmotic pressure of 0.1 M solution of KNO_3 (P_1) and CH_3COOH (P_2) is

(A) $P_1 > P_2$ (B) $P_2 > P_1$

(C) $P_1 = P_2$ (D) $P_1/(P_1+P_2) = P_2/(P_1+P_2)$

Sol. (A) For explanation, refer to QR code given on the last page.

82. Maximum boiling point will be for the solution

(A) 0.1 M NaCl (B) 0.1 M $Al_2(SO_4)_3$

(C) 0.1 M urea (D) 0.1 M K_2SO_4

Sol. (B) The elevation in boiling point is calculated as:

$\Delta T_b = i\ K_b m$

For NaCl, $i = 2$

For $Al_2(SO_4)_3$, $i = 5$

For urea, $i = 1$

For K_2SO_4, $i = 3$

Higher the value of i, higher will be elevation in boiling point. Also, $\Delta T_b \uparrow$ B.P↑

are 'i' ko mat bhul jana varna sab galat

Khoobsurat Baat

More the van't Hoff factor more will be the observed C.P.

83. Maximum freezing point depression for same concentrated solution-

(A) solute = NaCl ($\alpha = 0.9$), solvent ($K_f = 0.3$ K Kg mol^{-1})

(B) solute = K_2SO_4 ($\alpha = 0.7$), solvent ($K_f = 0.3$ K Kg mol^{-1})

(C) solute = $CaCl_2$ ($\alpha = 0.8$), solvent ($K_f = 0.2$ K Kg mol^{-1})

(D) solute = urea ($\alpha = 0$), solvent ($K_f = 0.4$ K Kg mol^{-1})

Sol. (B) For explanation, refer to QR code given on the last page.

84. 1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C

$K_b(H_2O) = 0.52$ K kg mol^{-1}

Which of the following is correct?

[29 Jan, 2021 (Shift-I)]

(A) AX_2 and AY_2 (both) are completely unionised.

(B) AX_2 is fully ionised while AY_2 is completely unionised.

(C) AX_2 and AY_2 (both) are ionised.

(D) AX_2 is completely unionised while AY_2 is fully ionised.

Sol. (B) For explanation, refer to QR code given on the last page.

85. The observed and normal molar masses of compound MX_2 are 65.6 and 164 respectively. The percent degree of ionisation MX_2 of is _____ %. (Nearest integer)

[24 Jan, 2025 (Shift-II)]

Sol. [75] First calculate van't hoff factor (i):

i = Normal molar mass/ observed molar mass
= 164/65.6 = 2.5

Consider the dissociation of MX_2:

$MX_2 \rightarrow M^{+2} + 2X^-$

$i = 1 + (n-1)\alpha$

For dissociation:

Here, n = 3

$1.5 = 2\alpha$

$2.5 = 1 + (3-1)\alpha$

$\alpha = 0.75$, % Ionization = 75%

reaction toh likh loge na

86. The Van't Hoff factor for a dilute aqueous solution of glucose is

(A) Zero (B) 1.0

(C) 1.5 (D) 2.0

Sol. (B) Glucose is a non-electrolyte and hence, its van't hoff factor is 1.

Common Mistake

Assuming that glucose will dissociate but it is a non-electrolyte for which i = 1.

87. Find K_a, the ionization constant of a monobasic acid HA if a 0.025 molal aqueous solution of acid freezes at -0.060°C (K_f = 1.86 K Kg mol^{-1} and molality = molarity)

Sol. (2.1×10^{-3}) For explanation, refer to QR code given on the last page.

88. When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be -0.93°C [$K_f(H_2O)$ = 1.86 K kg mol^{-1}]. The number (n) of benzoic acid molecules associated (assuming 100% association) is _____

[26 Feb, 2021 (Shift-II)]

Sol. The formula to calculate depression in freezing point is:

$\Delta T_f = iK_f m$

Substitute the values in the above expression:

$0 - (-0.93) = i \times 1.86 \times \frac{12.2}{122 \times 100} \times 1000$

On solving, i = 0.5

Now, $i = 1 + \left(\frac{1}{n} - 1\right)\alpha$, $\frac{1}{2} = 1 + \left(\frac{1}{n} - 1\right) \times 1$

On solving, n = 2

Common Mistake

Incorrect relation of i and α will give incorrect value of n.

89. The number of pairs of the solution having the same value of the osmotic pressure from the following is _____.

(Assume 100% ionization)

[25 Jan, 2023 (Shift-II)]

(A) 0.500 M C_2H_5OH and 0.25 M KBr (aq)

(B) 0.100 M $K_4[Fe(CN)_6]$ (aq) and 0.100 M $FeSO_4(NH_4)_2\ SO_4$(aq)

(C) 0.05 M $K_4[Fe(CN)_6]$ (aq) and 0.25 M NaCl (aq)

(D) 0.15 M NaCl (aq) and 0.1 M $BaCl_2$ (aq)

(E) 0.02 M $KCl.MgCl_2.6H_2O$ (aq) and 0.05 M KCl (aq)

Sol. $\pi = iCRT$, $\pi \propto iC$

where, i is Van't Hoff factor and C is the concentration.

Hence, A, B, D and E have same value of osmotic pressure.

kyu confusa rhe ho baalako

90. Sea water contains 29.25% NaCl and 19% $MgCl_2$ by weight of solution. The normal boiling point of the sea water is °C (Nearest Integer)

Assume 100% ionization for both NaCl and $MgCl_2$

Given: K_b (H_2O) = 0.52 K kg mol^{-1}

Molar mass of NaCl and $MgCl_2$ is 58.5 and 95 g mol^{-1} respectively.

[13 April, 2023 (Shift-II)]

Sol. (116.07°C) For explanation, refer to QR code given on the last page.

Q1. The percentage dissociation of a salt (MX_3) solution at given temperature (van't Hoff factor $i = 2$) is.....% (Nearest integer)

[07 April, 2025 (Shift-I)]

Sol. [33]

The dissociation of MX_3 is given by:

$MX_3 \rightarrow M^{3+} + 3X^-$

The van't Hoff factor (i) is the total number of particles in solution, which is:

$i = 1 + (n - 1)\alpha$

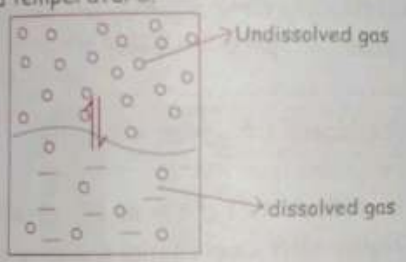
$i = 1 + (4 - 1)\alpha = 2$

$\alpha = \frac{1}{3} = 33.33\% \approx 33\%$

percentage nikalni hai toh 100 se toh karoge multiply

SOLUBILITY

It is defined as a maximum amount that can be dissolved in a specified amount of solvent at a specified temperature.



Note

There will be an equilibrium between the dissolved and undissolved gas.

Undissolved gas $\rightleftharpoons$ dissolved gas

- Factors which move this equilibrium toward forward direction will increase the solubility of the gas.
- Factors which move this equilibrium towards backward direction will decrease the solubility of the gas.

Factors Affecting Solubility of Gas

1. Temperature:

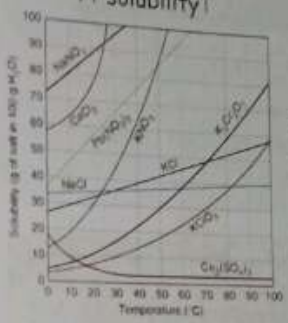
(a) **Solubility of solid in liquid:**

- For an endothermic reaction, solubility increases with an increase in temperature.
- For an exothermic reaction, solubility decreases with an increase in temperature.

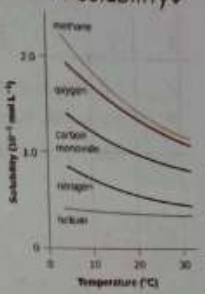
(b) **Solubility of gas in liquid:**

- On increasing temperature of gas, solubility decreases.
- Graphical representations of solubility of gas in liquid and solid in liquid are given below as:

T↑ solubility↑



T↑ solubility↓



sar toh nahi chakra gaya

Khoobsurat Baat

Dissolution of a gas is some sort of condensation process, which is exothermic in nature.

2. Effect of pressure:

(a) **Solubility of solids in liquids**

- Pressure does not have any significant effect on solubility of solids in liquids. It is so because solids and liquids are highly incompressible and practically remain unaffected by changes in pressure.

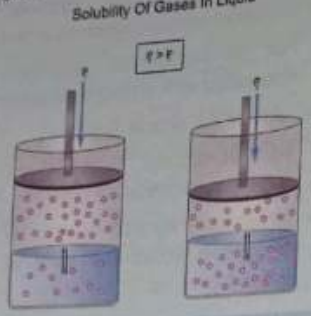
(b) **Solubility of gas in liquid**

- Effect of pressure on solubility of gas in liquid is defined by Henry's law.

Important

Henry's Law

This law states that "solubility of the gas in a particular solvent is directly proportional to the partial pressure of the gas present above the liquid."



As pressure increases, the solubility of gases in liquid also increases.

$P_{gas} \propto X_{gas}$ $\quad \boxed{P_{gas} = K_H X_{gas}}$

P_{gas} = P.P of gas above liquid

X_{gas} = mole fraction of gas in the liquid

K_H = Henry's constant

92. CO_2 gas is bubbled through water during a soft drink manufacturing process at 298 K, if CO_2 exerts a partial pressure of 0.835 bar, then x mmol of CO_2 would dissolve in 0.9 L of water. The value of x is _____. (Nearest Integer) (Henry's law constant for CO_2 at 298 K is 1.67×10^3 bar)

Sol. (25) For explanation, refer to QR code given on the last page.

93. If O_2 gas is bubbled through water at 303 K, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____.

(Given: Henry's Law constant for O_2 at 303 K is 46.82 k bar and partial pressure of O_2 = 0.920 bar)

(Assume solubility of O_2 in water is too small, nearly negligible)

Sol. Use Henry's law: $P_{gas} = K_H \cdot X_{gas}$

Substitute values in above equation:

$$0.920 = 46.82 \times 10^3 \text{ bar} \times \frac{\text{mol of } O_2}{\text{mol of } H_2O}$$

$$0.920 = 46.82 \times 10^3 \times \frac{\text{mol of } O_2}{1000}$$

$$P = \frac{0.920}{46.82 \times 18} = n_{O_2}$$

On solving, $1.09 \times 10^{-3} = n_{O_2}$

mmol of O_2 = 1

direct hi toh hai yaar

94. The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is x _____ 10^{-5} mol dm^{-3}.

(Round off to the Nearest Integer).

[Given: Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O_2 Density of water with dissolved oxygen = 1.0 kg dm^{-3}]

Sol. The formula to calculate solubility is:

$P = K_H \times$ Solubility

Rearrange the above equation for solubility:

Solubility = P/K_H

$$\text{Solubility} = \frac{20 \times 10^3}{8.0 \times 10^7} = 2.5 \times 10^{-4} = 25 \times 10^{-5}$$

3. Nature of the gas

It is based on the rule 'like dissolves like.'

In water (polar solvent), gas which is polar dissolves more as compare to non-polar gas.

Example: HCl, CO_2, NH_3, Cl_2, SO_2

polar polar polar

Applications of Henry's Law

1. It is difficult to breath at higher altitudes

At higher altitudes, partial pressure of oxygen is less as compared to ground level which leads to decrease of oxygen in blood , as a result, it becomes difficult to breathe at high altitudes.

upar jane ka nhi, khatra hai

2. Scuba divers come up slowly

When scuba divers go down in water, pressure increases which increases the solubility of N_2 in blood.

When divers come towards the surface, pressure gradually decreases which decreases the solubility of N_2 in blood that leads to painful conditions known as bends which are painful and dangerous to life.

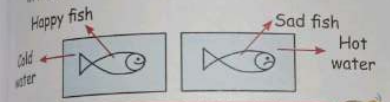
To avoid bend situation

He add with (N_2 + O_2) — come up slowly

DECOMPRESSION SICKNESS

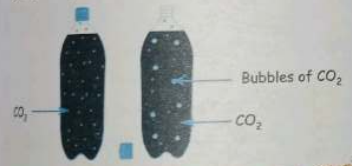
3. **It is difficult to survive in hot water for aquatic life**

As temperature increases, solubility of O_2 in water decreases. As a result, aquatic species are more comfortable in cold waters rather than in warm waters.



(thande pani mei fish khush hai or garam pani mei dukhi)

4. To increase the solubility of CO_2 in soft drinks and soda water, the bottle is sealed under high pressure.



is samjhe kyu bubbles aate hai bottle khulte hue, wo jaadu nhi hai

Henry's Constant

$P_{gas} = K_H X_{gas}$

K_H = henry's constant which does not dependent on X_{gas} and P_{gas}

K_H → Nature of gas

K_H → Temperature

If pressure is const for different gases

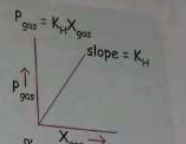
$$K_H \propto \frac{1}{X_{gas}}$$

Note

More the solubility of gas, lesser will be the K_H value.

Important

The graphical representation of pressure with mole fraction can be represented as:



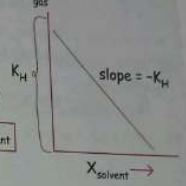
The more the slope of this curve, more will be the value of K_H.

$P_{gas} = K_H X_{gas}$,

$P_{gas} = K_H (1 - X_{solvent})$

$P_{gas} = K_H - K_H X_{solvent}$

$Y = C + MX$



Khoobsurat Baat

Gases having more force of attraction are more soluble and having less K_H value.

Polar gas has more force of attraction as compare to non-polar gas. So the K_H value is less for polar gases as compare to non-polar gas and for non-polar gases, the force of attraction factor is more for the gas having more Molecular weight.

Limitations

- It is applicable at low pressure and high temperature as the solubility of the gas should be low.
- It is not applicable for the gases which react with water.

95. Which gas has higher K_H value-SO_2 or H_2?

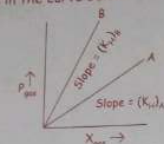
Sol. Higher the force of attraction, lower will be the K_H value.

$S_{SO_2} > S_{H_2}$, $(K_H)_{SO_2} < (K_H)_{H_2}$

Common Mistake

Relates K_H value directly with the force of attraction.

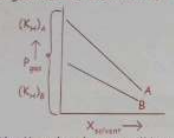
96. Compare K_H and solubility of gases A and B given in the curve below:



Sol. More the slope of the curve, more will be the value of K_H.

$(K_H)_B > (K_H)_A$. $X_B < X_A$

97. Which gas has more solubility- A or B?



Sol. More the K_H value, lesser will be the solubility.

Also, more the slope of the curve, more will be the value of K_H.

$(K_H)_A > (K_H)_B$

$(solubility)_A < (solubility)_B$

kuch kuch ho rha hoga dimag mei

98. Henry's constant (in kbar) for four gases α, β, γ and δ in water at 298 K is given below:

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3 kg m^{-3} at 298 K)

This table implies that:

(A) α has the highest solubility in water at a given pressure

(B) solubility of γ at 308 K is lower than at 298 K.

(C) The pressure of a 55.5 molal solution of δ is 250 bar.

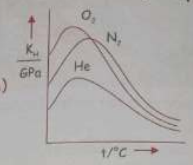
(D) The pressure of a 55.5 molal solution of γ is 1 bar. [3 Sept, 2020 (Shift-I)]

98. (C) For explanation, refer to QR code given on the last page.

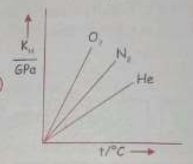
99. Which of the following graph correctly represents the plots of K_H at 1 bar gases in water versus temperature?

[02 April, 2025 (Shift-I)]

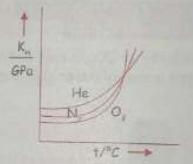
(A)



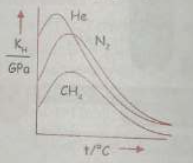
(B)



(C)



(D)



Sol. (D)

The solubility of gases first decreases and then increases with rise in temperature. Consequently, Henry's law constant (K_H) initially increases and then decreases. Also, the value of K_H at moderate temperatures follow the order He > N_2 > CH_4.

Scan the QR code to get the explanations of unsolved questions.